Marcelo Sampaio de Alencar
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Digital Communications

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ISBN 978-3-031-92396-8 ISBN 978-3-031-92397-5 (eBook)
https://doi.org/10.1007/978-3-031-92397-5

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Preface

This book is intended to serve as a textbook for courses dealing with Digital Communications, Cellular Mobile Systems, or Modulation Theory. The modulation theory is dealt with using stochastic processes, which remains a novel approach for undergraduate texts. The book is suitable for the undergraduate as well as the initial graduate levels of Electrical Engineering courses.

Chapter 1 presents a guided tour through communications history, covering important communications and historical facts around the world up to the beginning of the Internet. Chapter 2 covers signals and linear systems, time signals, discrete-time signals, power, and autocorrelation.

Chapter 3 introduces the concepts of system response and convolution, including discrete impulse response, discrete convolution, algorithm for discrete deconvolution, as well as continuous impulse response and continuous convolution.

Spectral analysis is treated in Chap. 4 including Fourier analysis, Fourier transform and its properties, the sampling theorem, Parseval's theorem, average power and autocorrelation, Hilbert transform and its properties, and useful relations involving Hilbert transforms.

Chapter 5 presents random signal and noise, including the autocorrelation function, stationarity, the power spectral density, linear systems, expected value of the output signal, the response of linear systems to random signals, phase information, analysis of a digital signal, signal fading, Rayleigh fading, Nakagami distribution, Rice distribution, slow and fast fading, and Jakes' model.

Analog-to-digital conversion is the subject of Chap. 6, including signal coding, signal compression systems, features of speech signals, pulse code modulation, the uniform quantizer, noise spectrum for the uniform quantizer, the nonuniform quantizer, vector quantization, linear predictive coding parameters, overview of speech coding, waveform coding, and parametric and hybrid coding.

Chapter 7 explains modulation theory, covering amplitude modulation, amplitude modulation by random signals, suppressed carrier amplitude modulation and its spectrum, quadrature amplitude modulation, quadrature modulation with random signals, single sideband modulation, independent sideband modulation,

angle modulation, angle modulation using random signals, and frequency and phase modulation.

Digital modulation theory is the subject of Chap. 8, presenting signal space, digital modulation schemes, differential coding, offset phase modulation, constant envelope modulation, the rotated constellation, computing the spectra of digital schemes, calculation of error probability, and orthogonal frequency division multiplexing.

Chapter 9 addresses information theory, covering uncertainty measure, the information unit, entropy, conditional entropy, mutual information, discrete memoryless information source, extension of a discrete memoryless source, coding for a discrete information source, types of source codes, prefix-free codes, instantaneous codes, Kraft inequality, Huffman code, joint information measurement, noiseless channel, discrete noisy channel, and channel capacity.

Chapter 10 presents basic concepts of error-correcting codes, addressing block codes, cyclic codes, decoding of cyclic codes, algebraic decoding, soft decision decoding, and convolutional codes.

Computer networks are presented in Chap. 11, addressing data flow in networks, queue models, local area networks, network protocols and architecture, layer architecture, TCP/IP protocol history, internet coordination, types of networks, transmission protocols, interconnection equipment, interconnection protocols, the IP protocol, the TCP/IP protocol, and security protocols for TCP/IP.

The book has six appendices. Appendix A covers Fourier series and Fourier transform and their properties, as well as Fourier transform in two dimensions. A table of Fourier transforms is included. Appendix B presents discrete Fourier transforms and brings a table of discrete Fourier transforms. Appendix C presents a table of Hilbert transform formulas and some important inequalities. Appendix D presents some important formulas used in the text. Appendix E presents a review of probability theory covering set theory, functions and measure probability theory, and random variables. It also includes Bayes' theorem, moments, variance, the characteristic function and function of a random variable, and the concept of joint random variables. Glossary presents the usual acronyms and the terminology of digital communications.

As an important part for learning the contents, many examples and problems are provided throughout the book.

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Acknowledgments

The authors are grateful to Professor Elvino S. Sousa (University of Toronto, Canada), Professor Paddy Farrell, Emeritus Professor Garik Markarian, and Emeritus Professor Bahram Honary (in memoriam) (Lancaster University, UK) for technical communications and useful discussions related to digital communication systems.

The authors are also grateful to all the members of their respective communications research groups, certified by the Brazilian National Council for Scientific and Technological Development (CNPq), at Federal University of Rio Grande do Norte, Federal University of Bahia, Federal University of Campina Grande, Senai Cimatex (Salvador, Bahia), Federal University of Rio Grande do Norte, Federal University of Paraíba, and Federal University of Pernambuco, for their collaboration in many ways, helpful discussions and friendship, as well as our colleagues at the Institute for Advanced Studies in Communications (Iecom).

The authors wish also to acknowledge the contribution of Prof. Maria de Lourdes Alcoforado (State University of Pernambuco) to the chapter on error-correcting codes, and the support of Prof. José Sampaio Lemos-Neto (Federal University of Pernambuco), who helped with the reviewing process and producing part of the figures in the text.

The authors also acknowledge the support of Prof. Francisco Madeiro (State University of Pernambuco) and Prof. Waslon T. A. Lopes (Federal University of Paraíba), who contributed to the chapter on speech coding, and Prof. Wamberto J. L Queiroz (Federal University of Campina Grande), who contributed to the chapter on digital modulation.

The authors are indebted to their families for their patience and support during the course of the preparation of this book.

Finally, the authors are thankful to Mary E. James, from Springer, who strongly supported this project from the beginning and helped with the reviewing process.

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About the Authors



Marcelo Sampaio de Alencar was born in Serrita, Brazil, in 1957. He received his bachelor degree in Electrical Engineering, from Universidade Federal de Pernambuco (UFPE), Brazil, 1980; his master degree in Electrical Engineering, from Universidade Federal da Paraiba (UFPB), Brazil, 1988; and his PhD from University of Waterloo, Department of Electrical and Computer Engineering, Canada, 1993. Marcelo S. Alencar has more than 44 years of engineering experience, and he is currently IEEE Life Senior Member. Between 1982 and 1984, he worked for the State University of Santa Catarina (UDESC). From 1984 to 2002, he worked for the Department of Electrical Engineering, Federal University of Paraiba, Campina Grande, Brazil. From 2002 to 2017, he was Chair Professor at the Department of Electrical Engineering, Federal University of Campina Grande, Brazil. He spent some time working for MCI-Embratel and for the University of Toronto, as Visiting Professor. Since 2017, he has spent some time as Visiting Professor at the Federal University of Bahia, Salvador, at Cimatec Salvador, and at the Federal University of Rio Grande do Norte, Natal, Brazil. Marcelo Alencar supervised seven post-doctoral fellows, 23 PhDs, 36 master's, and several undergraduate students.

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He published over 570 papers in scientific journals and conferences. He wrote chapters for 13 books. He also published 31 books: Set, Measure and Probability Theory, Cryptography and Network Security, Economic Theory, Linear Electronics, Music Science, Modulation Theory, Spectrum Sensing and Applications, Scientific Style in English, Cellular Network Planning, by River Publishers; Digital Television Systems, by Cambridge; Informação, Codificação e Segurança de Redes, by Elsevier; Communication Systems, Third Edition, Nanotechnology-Based Smart Remote Sensing Networks for Disaster Prevention, IoT Based Control Networks and Intelligent Systems, by Springer; Communication, Management and Information Technology: International Conference on Communication, Management and Information Technology, by CRC Press, Taylor & Francis Group; Probability Theory, Information Theory, by Momentum Press; Televisão Digital, Second Edition, Telefonia Celular Digital, Third Edition, Telefonia Digital, Fifth Edição, Teoria de Conjuntos, Medida e Probabilidade, Engenharia de Redes de Computadores, Probabilidade e Processos Estocásticos, Sistemas de Comunicações, Ondas Eletromagnéticas e Teoria de Antenas, by Editora Érica Ltda; História, Tecnologia e Legislação de Telecomunicações, Divulgação Científica, História da Comunicação no Brasil, Soluços d'Alma, Sexo Conexo by Gráfica e Editora Epgraf; Princípios de Comunicação, by Editora Universitária, UFPB.

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Valdemar Cardoso da Rocha, Jr. born in Jaboatão, Pernambuco, Brazil, on August 27, 1947, is a former chair of the Department of Electronics and Systems at the Federal University of Pernambuco (UFPE), Brazil. His work focuses on coding theory, information theory, and cryptography. He received the BS degree in electrical and electronics engineering from Escola Politécnica, Recife, Brazil, 1970. He began graduate studies in 1972 at the University of Kent, concentrating on coding theory and was awarded a PhD in 1976, with Paddy Farrell as his supervisor. He returned to Brazil and became a Faculty Member at UFPE, at the Department of Electrical Engineering (1976–1979) and the

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He has been a consultant to the Brazilian Ministry of Education, the Ministry of Science and Technology, and the Ministry of Defense on postgraduate education and research in electrical engineering. He is a founding member, former President, and Emeritus Member of the Brazilian Telecommunications Society; a Life Senior Member of the IEEE Communications Society and the IEEE Information Theory Society; was elected a Fellow of the Institute of Mathematics and its Applications in 1992; and has taught at the Swiss Federal Institute of Technology-Zürich, the University of Leeds, and Lancaster University. Published over 100 engineering and scientific papers, including journal and conference papers, and the books Communication Systems, Springer, 3rd edition, 2023, Principles of Applied Digital Information Theory (in Portuguese), Interciência, 2018, and Elements of Algebraic Coding Systems, Momentum Press, 2014.

Chapter 1
Communications History



A man has died and his body turned into sand.
All his relatives disintegrated into dust.
It is by his writings that he will be remembered.
Egyptian scribe, four thousand years ago (Fischer 2007)

1.1 The Beginning of Communications in America

The history of communications in the United States spans two centuries and encompasses a wide range of technologies and innovations. This chapter presents the main milestones and developments in the history of American communications.

The first known attempt at long-distance communication in the United States occurred in 1639, when a system of signal fires was set up between Boston and Salem. In the late 1700s, the telegraph was invented in Europe, and by the mid-1800s, it had been introduced to the United States (Beauchamp 2001).

The United States Postal Service was established in 1775, with Benjamin Franklin serving as the first postmaster general. The postal service played a critical role in early American communications, helping to connect people across vast distances.

The telegraph was introduced in the United States by Samuel Morse, who developed a printing telegraph system in 1835. The first commercial telegraph line between Baltimore and Washington, DC, was established in 1844 (Bray 1995).

In 1844, Samuel Morse sent the first telegraph message from Washington, DC, to Baltimore, Maryland, using a code he had developed. The telegraph allowed for fast and reliable long-distance communication and was widely used by businesses and governments. Morse sent the first telegraph message, which said "What hath God wrought," from Washington, DC, to Baltimore.

The telegraph was a system for transmitting messages over long distances using electrical signals. The telegraph revolutionized communications by allowing people to send and receive information almost instantaneously and laid the groundwork for future innovations like the telephone and the Internet.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_1

Alexander Graham Bell patented the first telephone in 1876. The first commercial telephone service was established in 1878. The telephone quickly became a ubiquitous form of communication in the United States, connecting people across the country and around the world.

The telephone made it possible for people to communicate with each other from afar and transformed the way businesses and individuals conducted their affairs. By the early 1900s, the Bell Telephone Company had become the dominant telephone provider in the United States.

1.2 The Early Radio Broadcasting

Radio broadcasting began in the United States in the early 1900s and became a popular medium for entertainment and news. The first transoceanic radio transmission was sent by Guglielmo Marconi across the Atlantic Ocean in 1901 (Lewis 1991).

The first commercial radio station, KDKA, in Pittsburgh began broadcasting in 1920. Radio broadcasting emerged as a new form of mass communication. Initially used mainly for entertainment, radio soon became a key means of disseminating news and information to a broad audience.

The Communications Act of 1934 established the Federal Communications Commission (FCC) to regulate interstate and international communications by radio, television, wire, satellite, and cable. The FCC was established to regulate the use of the radio spectrum and ensure fair competition in the industry. The FCC is responsible to assigning to each radio station in the United States a precise carrier frequency (Frost 2010). During World War II, radio played a crucial role in keeping people informed about the progress of the conflict.

The first experimental television broadcast began in the late 1920s. This broadcast was made, in 1927, by Philo Farnsworth, though it took several years for it to become commercially available. The television development was halted during World War II. After the war, television quickly became the dominant form of entertainment in the United States for decades.

Television broadcasting began in the United States in the late 1930s. The first TV stations began broadcasting in 1941, and by the early 1950s, TV had become a major source of entertainment and news. But it was not until the 1950s that the technology became widely available to American consumers. Television quickly became the dominant form of mass media and had a profound impact on American culture and society. By the end of the decade, television was also available to consumers in Brazil (Alencar et al. 2010).

1.3 The Empire of Communications

The beginning of communications in the United States and Brazil had a common patron who, despite not having had access to a formal education in communications, became one of the greatest enthusiasts of the area. D. Pedro II was educated at home by a tutor, José Bonifácio de Andrada e Silva, one of the first scientists to have a doctoral degree in Brazil (Alencar 2003a).

José Bonifácio de Andrada e Silva (1763–1838), a diplomat, professor, jurist, and Brazilian politician, graduated in Law, Mathematics, and Natural Philosophy, in Portugal and obtained a doctorate in Mineralogy in France. He also wrote for "*O Patriota*" and was a supporter of the installation of blast furnaces in Brazil, a thesis also embraced by the editor of the newspaper "*Correio Braziliense*."

The Patriarch of Independence, as Andrada e Silva was called, in addition to promoting science, was also a member of the Brazilian Historical and Geographical Institute, the Geographical Societies of Rio de Janeiro and Lisbon, the Geographical Institute of Paraíba do Norte, and the Geographical Institute of Pará, corresponding member of the Ateneu Ibero-Americano, and permanent honorary member of the Brazilian Institute of Culture.

With the guidance of José Bonifácio, D. Pedro II developed scientific curiosity and absorbed the state-of-the-art knowledge of the time and a little of the revolutionary spirit. Therefore, technological innovations always attracted the Emperor's attention. The organization of postal services in Brazil dates back to 1829. In 1843, just over 2 years after D. Pedro II assumed power in Brazil, the first Brazilian postage stamp, the well-known "Olho de Boi," was launched. Brazil was the third country in the world to use the stamp.

The first submarine cables in Brazil were inaugurated by D. Pedro II in 1874, connecting Rio-Salvador-Recife-Belém. The Recife-João Pessoa-Natal line was established in 1875. The first Brazilian international cable connection was made in the same year, with Portugal, having been completed through a contract with the British Eastern Telegraph Company.

Less than a year later, in 1877, D. Pedro II inaugurates the telephone in Brazil. He had met the telephone at the Philadelphia Exposition (Alencar 2003a).

To get an idea of the evolution of events at that time, the first telephone exchange in Paris was activated in 1879. In the same year, D. Pedro II gave permission to install the one in Rio de Janeiro. Companhia Telefonica do Brasil was created on November 15, 1879, with initial capital of 1,500,000$000 réis, divided into 7,500 shares distributed by the Western Telegraph Company, the first telephone concessionaire in Brazil.

The Rio de Janeiro exchange is among the first telephone exchanges in the world, inaugurated in 1881, at the same time that the Anglo-Portuguese Telephone Company installed the one in Lisbon. In April 1885, Brazil had seven exchanges in operation, with 3335 subscribers. Comparing with what was happening in the world at that time, the United States had 137,570 subscribers in the same year, Germany with 14,732, Italy with 4,346, France with 7,175, and Sweden with 5,705.

Excluding the United States, Brazil did not do poorly in telephony in the last century—taking into account that the country's per capita income was very small due to the huge number of slaves still existing. Brazil would start the Republic as a member of the International Postal Union and as part of all international agreements that regulated telegraphy, submarine cables, and maritime signaling.

1.4 The Emperor and the Teacher of the Deaf and Dumb

Alexander Graham Bell arrived in Boston in 1868 as a teacher for the deaf and dumb from Edinburgh and Montreal, becoming a naturalized citizen of the United States. In the year of the Philadelphia Centennial Exposition, 1876, Bell decided to make an exhibition of his inventions to Sir William Thomson (Lord Kelvin), including a demonstration of the telephone, which had been under development since the autumn of 1875. The device was developed between the fall of 1875 and the summer of 1877. The official date for the invention of the telephone is 1876 (Shiers 1977).

D. Pedro II, who was on a visit to the United States, discovered the telephone at the exhibition marking the first centenary of the country's independence. He accepted the invitation of the President of the United States, Ulysses Grant, to attend the inauguration and, together with him, to start the machines in the Machinery Hall. About 250,000 people attended the ceremony.

It is said that the Emperor, when trying the telephone, was surprised and said "Holy God! This speaks…." Enthusiastic about the invention, he offered Bell a sum of money to develop it, with the condition that Brazil would be the first country to use it. The inventor fulfilled his promise and sent, in the following year, the first telephone sets for the installation of a telephone line between the palace of *São Cristóvão* and the *Santa Cruz* farm, of the imperial family—long before the telephone was commercialized in the world (Lopes 1997).

The Emperor's main interest was to meet the poet Henry Wadsworth Longfellow and General William T. Sherman but ended up knowing the inventions of Thomas Edison and Alexander Graham Bell. Less than a year later, in 1877, D. Pedro II inaugurated the telephone in Brazil, installed by the American company Western and Brazilian Telegraph Company. Brazil joined the International Telegraphy Convention in the same year (Hill 1947).

In 1879, Charles Paul MacKie obtained the first concession to establish a telephone network in the country. The company was incorporated in Boston under the name of Telephone Company of Brazil with an initial capital of US$ 300,000. The first three lines were built in May 1881, with a cable that totaled 1,600 meters length (Bandeira 2007).

The Bell Telephone Company was established in 1878 in New York City, with 5,000 shares issued and distributed among 7 members of Graham Bell family and friends as shareholders. AT&T was created on February 28, 1885, in the State of

New York, with the initial mission to build and operate long-distance telephone circuits. In 1899, AT&T became the holding of the Bell Company (Chapuis 1982).

In 1879, D. Pedro II granted Thomas Edison the privilege of introducing his devices for transmitting electric light in Brazil. The lighting contract for the capital was with a company linked to the Baron of Mauá, Rio de Janeiro Gás Co., an English consortium. The first attempt to implement electric lighting was made in 1883, but it was only implemented in 1905, through the Canadian consortium Rio de Janeiro Tramway, Light and Power Company, organized by Alexander Mackenzie and Fred Stark Pearson, inspired by the American capitalist Percival Farquhar (Bandeira 2007).

1.5 The Inception of Digital Communications in America

Claude Elwood Shannon (1916–2001), an American mathematician, electrical engineer, computer scientist, and cryptographer, is considered the Father of Information Theory. In 1948, he published a seminal paper on the mathematical concept of information, which has remained the most cited for decades. From that article, information left the field of Journalism to occupy the more formal field of Probability Theory (Shannon 1948a).

Shannon's development was also based on the work of Harry Nyquist (Harry Theodor Nyqvist, 1889–1976), who determined the necessary sampling rate, as a function of the maximum frequency of a continuous signal, so that this signal could be reproduced completely only with a set of discrete samples (Widrow 1956).

Independently, Andrey Nikolaevich Kolmogorov (1903–1987), a Soviet mathematician, developed his Complexity Theory in the 1960s. It was a new theory of information based on the length of the algorithm designed to describe a given sequence of data. He used Alan Turing's machine model for this new definition. Under certain conditions, Shannon's and Kolmogorov's definitions are equivalent.

The idea of relating the number of states of a system to a physical measure, however, had been circulating since the nineteenth century. Rudolf Julius Emanuel Clausius (1822–1888), a German physicist and mathematician, proposed the term entropy for this measure in 1895.

Entropy comes from the Greek word for transformation, and in Physics, it is related to the logarithm of the ratio between the final and the initial temperature of a system or with the ratio between the change in heat and the temperature of the same system.

Shannon defined the entropy of an alphabet as the negative mean value of the symbol probability logarithm. Thus, when the symbols are equiprobable, that is, they have the same probability, this definition of entropy is equivalent to that created by Nyquist.

However, as it is more generic, Shannon's entropy also applies to the calculation of the capacity of communication channels. Much of the work of researchers in the

field is devoted to calculating the capacity or developing error correcting codes to achieve this capacity.

Shannon died on February 24, 2001, a victim of the disease named after the physician Alois Alzheimer (1864–1915), a German psychiatrist and neuropathologist. According to his wife, he lived calmly but without the possibility of retaining new information—a bad joke of nature on the genius that reinvented information.

1.6 Coding in Space

The history of coding, the branch of Information Theory that deals with the correction of transmission errors and made possible the development of the Internet, had a special chapter with the launch of the Voyager 1 and 2 spacecraft three decades ago.

The probes began their journey on August 20, 1977, with the launch of Voyager 2. Voyager 1 was launched on September 5 of the same year. Currently, Voyager 2 is 15.5 billion kilometers from the Sun and Voyage 1 12.5 billion. They travel 1.6 million kilometers a day.

The probes' original mission was to obtain data from Jupiter and Saturn, which would take just 4 years. However, the technology employed proved to be much more resistant and efficient. In the first 12 years, they conducted explorations of Jupiter, Saturn, and the planets' moons and the first expeditions to Uranus and Neptune, sending back never-before-seen images and scientific data that resulted in fundamental discoveries about the Solar System's outer planets.

In December 2004, Voyager 2 reached the Solar System's final frontier, a turbulent region that begins approximately 14 billion kilometers from the Sun and in which the solar wind is reduced as it encounters the interstellar medium, the rarefied gas that fills space. among the stars.

Each spacecraft carries five functional science instruments, used to study the solar wind, energetic particles, magnetic fields, and other types of waves as they make their way through deep space hitherto unexplored by humans.

Voyagers use radioisotope thermoelectric generators to power the instruments. With a consumption of less than 300 watts per probe, equivalent to a medium-sized digital television, the equipment should operate by 2020.

The probes carry a 12-inch copper and gold disk that contains data on the diversity of life, society, and culture on Earth. 117 images and various sounds obtained in nature were recorded, in addition to musical selections from different cultures and times and greetings in 54 languages.

Communication between the Voyager 1 and 2 probes and National Aeronautics and Space Administration (NASA) takes place through the Deep Space Network, a network of antennas installed in several countries. Commands sent from Earth take 12 hours to reach Voyager 1 and 14 hours to be received by Voyager 2. Signals sent by spacecraft take the same time to reach Earth and arrive very weakly,

heavily corrupted by noise. Therefore, the probes are equipped with error correction encoders.

The ships used the convolutional code (2, 1, 6), developed by Peter Elias in 1955, concatenated with the Reed-Solomon code (255, 223), known as RS, created by Irving Stoy Reed (1923–2012), an American mathematician and engineer, and Gustave Solomon (1930–1996), an American mathematician and electrical engineer, in 1960. This code concatenation scheme can achieve a rate of only one error per million transmitted bits, with signal power just under twice the noise power (van Lint and Springer 1987).

The convolutional code used produces two output bits for each input bit and has order 6, related to the memory used. The RS code can correct 16 symbol errors per transmitted block. This code uses 32 eight-bit symbols, or octets, for parity checking, which added to the 223 information symbols gives the 255 transmitted symbols.

When the Voyager space probes were launched, there were no adequate Reed-Solomon decoders. Only in 1986, when Voyager 2 was approaching Uranus, did technology and the evolution of algorithms finally make it possible to build an operational RS (255, 223) decoder.

Elwyn Ralph Berlekamp (1940–2019), a professor of Mathematics and Computer Science at the University of California, Berkeley, was the inventor of the efficient decoding algorithm for the RS code. The algorithm for decoding the convolutional code was invented by Andrew James Viterbi (1935–), born Andrea Giacomo Viterbi, an Italian Jewish-American electrical engineer and businessman who co-founded Qualcomm Inc. and invented the Viterbi algorithm (Berlekampe and Tong 1985).

In 1969, Viterbi, along with Irvin Mark Jacobs (1935–), an American electrical engineer and businessman, and Leonard Kleinrock (1934–), an American computer scientist and Internet pioneer, founded Linkabit. They were professors at the University of California at Los Angeles and also consultants for NASA for projects in the field of Digital Transmission (Viterbi and Omura 1979).

The company was acquired by the company Microwave Associates Communications in 1980. In 1985, Viterbi and Jacobs left the company to found Qualcomm, which became a giant in the field of Cellular Mobile Communications. Kleinrock, who invented packet switching and was one of the creators of ARPANET, the network that gave rise to the Internet, remained in academia.

1.7 The Beginning of the Internet

Incredible as it may seem, there was a time when there was no Internet. At that time, when you wanted to set up a meeting, it was necessary to place a phone call. To announce a birthday or birth, it was common use to send a telegram. People would even leave their rooms, in companies, to talk to colleagues about a work topic or just to make small talk (Alencar 2008a).

Of course, some companies started out in the business of exchanging posts. In the 1970s, there was the Telex service, provided by the European Post, Telephone and Telegraph (PTT) companies, the American Telephone and Telegraph (AT&T) Corporation, the Brazilian Telecommunications Company (Embratel), and other public companies. They formed an international system of written communications, which prevailed until the 1990s, and consisted of a worldwide network with a numeric addressing plan, with terminals that could send written messages to other terminals.

The Telex had some limitations, as it did not have all the characters used in the most languages, such as accents, and all the letters were capitalized. The older machines had no random access memory (RAM). Therefore, the messages were prerecorded by mechanical means, typically with perforated paper tape, and then transmitted. Incoming messages were printed on continuous rolls of paper (Alencar 2011a).

The service worked in some countries, until the past decade, despite the number of falling subscribers due to the introduction of electronic mail. The terminals resembled electric typewriters, and there was a guarantee of immediate delivery, with terminal authentication, which was important for the companies that hired the service (Alencar 2011b).

In Brazil, Embratel already had an information packet exchange service in the early 1980s. It was known as Projeto Ciranda and served only its employees. Then Cirandão was launched to the public access, but few people took notice. The marketers of company changed its name to STM-400, but it did not have much effect on the market. The company missed the target audience. The company should have bet on the university market.

Regarding the use of data communication networks, a significant event was the decision of the National Science Foundation (NSF) of the United States in 1985 to invest in setting up networks to serve the academic and research community. In 1986, articles were published describing the networks used by the academic and research community in the United States, with emphasis on the Bitnet network, which has been in operation since 1981, and for the National Science network Foundation (NSFNET), created in 1986.

Bitnet, also known as Because It's Time Network, was a large computer network, which carried electronic mail messages. It used technology developed by IBM, mainly the protocol Network Job Entry (NJE), and gave rise to the Listserv program for maintenance of lists of debates. Its appeal was its simplicity of adhesion and operation, mainly for institutions that had an IBM computer (Alencar 2008b).

Bitnet was managed by the Corporation for Research and Educational Networking (CREN), from Washington, USA, and was used to provide electronic mail and file transfer between large computers in educational and research institutions in North America, South America, Europe, and Japan. It reached more than 2,500 universities and institutes worldwide.

The network NSFNET, which would be part of the Internet, used the family of protocols TCP/IP, developed from Defense Advanced Research Projects Agency (DARPA). This protocol would allow several applications via the network, espe-

cially the interactive use of remote computers (Telnet); the transfer of files, known as File Transfer Protocol (ftp); and, from the decade the 1990s, interactive consultation of information bases on the World Wide Web (www), in addition to electronic mail.

Considering the importance for the academic community of the use of computer networks, several projects in this direction took place in the 1980s at the National Scientific Computing Laboratory (LNCC) of CNPq, at the São Paulo State Research Support Foundation (FAPESP), and at the Federal University of Rio de Janeiro (UFRJ).

The first meeting to discuss the establishment of a national network for researchers, with access to international networks, was held at the University of São Paulo (USP) in October 1987. The meeting was attended by research institutions, development agencies, and Embratel. The network aimed at allowing broad access for members of the research community to Bitnet, using the facilities of the LNCC, through dial-up, or through the Embratel's National Network of Packages (Renpac).

The message exchange service for university students in Brazil started in mid-1987, with the arrival of Bitnet. After 8 months of connection, the network reached 110 nodes in January 1988, when the decision was taken to use FAPESP to implement an international connection to state universities.

Embratel, concerned with the monopoly it had in the area, only allowed the transport of third-party traffic through community networks research and academic research in October 1988, a month after the establishment of of the first international connection. Thus, the first connection established with Bitnet, with a rate of 9,600 baud or symbols per second, was made between the LNCC, in Rio de Janeiro, and the University of Maryland, in the United States.

Then, negotiations began with the Central Committee of the Bitnet (Educom). The entry point was placed at Fermi National Laboratory (Fermilab), which had a Vax platform (Vax 750) and a cooperation contact signed with the Institute of Physics of the University of São Paulo (IFUSP).

In September 1988, Educom changed its rules, no longer accepting us at Bitnet in foreign countries. It was then necessary to structure a cooperation network, which changed the long-term plans. Negotiation finals were closed with Fermilab in October 1988 in Batavia, Illinois, and then with Educom. Thus, a cooperation network was established with the Bitnet, which became known as Academic Network at São Paulo (ANSP) (Alencar 2008c).

The network's second international connection, which initially operated at a of 4,800 baud, was installed in November 1988 between FAPESP and Fermilab in Chicago. This connection provided for the service of the university system and research center in São Paulo and used DECnet technology, which allowed access to *High-Energy Physics Network* (HEPnet) and Bitnet.

In February 1989, the Embratel connection from São Paulo, Brazil, to Batavia, USA, was installed, using HEPnet (DECnet), with the address user@FPSP.HEPNET. It became the IFUSP DECnet node.

A third independent connection to Bitnet, also at 4,800 baud, was installed in May 1989 between UFRJ and the University of California, Los Angeles (UCLA).

The Federal University of Rio Grande do Sul (UFRGS) entered the network in July 1989, using the Phonenet protocol over X.25, a set of protocols standardized by the ITU for wide area networks, using the telephone system or the Integrated Services Digital Network (ISDN) as a means of transmission (Alencar 2003b).

In December 1989, a Vax6330 and an mVax3600 were purchased for FAPESP, with financial support from the Department of Science and Technology of the State, leaving the mVax3600 exclusively for network management. The international line has increased from 4800 to 9600 baud in September 1990.

At the end of the 1980s, Embratel offered the following communication services: private lines, with transmission rates between 300 and 9600 baud; a switched data service called Renpac, with X.25 and X.28 access, at rates up to 9,600 baud; and satellite service, using technology Very Small Aperture Terminal (VSAT).

In May 1989, the country had three islands of access to Bitnet. Two in the city of Rio de Janeiro and one in São Paulo. Communications between these islands were through the international Bitnet network. The Federal University of Paraíba was interconnected to the network in 1989, with a direct connection with Embratel in Recife.

After the elimination, by Embratel, of the restriction on traffic of third parties, it was possible to establish a national network to share the access to international networks. This was accomplished in 1991, with the interconnection between islands and with the extension of connectivity to other research centers in the country.

The adoption of a final form of addressing by the National Network of Research (RNP), from CNPq, only took place in December 1990. The addresses in the form user@maquina.departamento.instucional.BR started to be used by institutions that had the capacity to manage their own sub-domain (Alencar 2008d).

Addresses such as user@maquina.departamento.instucional.ANxx.BR still remained for some of the other institutions. Pseudo-addresses related to Bitnet nodes, such as user@node.ANxx.BR, were formally eliminated, although their use has continued in practice.

Brazil's access to the Internet became possible in February of 1991, when FAPESP increased the transmission rate to 9,600 baud, for its connection to Fermilab and installed the Multinet program from TGV, to carry traffic that used the Internet Protocol (IP), in addition to DECnet, which was a proprietary network technology from Digital Equipment Corporation (DEC), and also Bitnet traffic (Tolhurst and Pike 1994).

IP connectivity has been extended to a small number of institutions in São Paulo, Rio de Janeiro, Rio Grande do Sul, and Minas Gerais, with low transmission rate private lines, between 2,400 and 9,600 baud, or through Renpac.

The components of the second-generation network were installed in 1992 and included RNP and the state networks of Rio de Janeiro and São Paulo, which were financed by FAPERJ and FAPESP. State networks were installed for the United Nations Conference on Environment and Development (UNCED-92 or Rio-92), held in June 1992, in Rio de Janeiro, and both the networks used new international connections of 64 kbit/s. They served to support the Global Forum, a meeting of nongovernmental organizations (NGOs), carried out in parallel to UNCED-92.

Bitnet was important for worldwide connectivity until the beginning of the 1990s, when it was supplanted by the Internet. The main application of Bitnet was the maintenance of distribution lists. The most visible difference between Bitnet and the Internet was the addresses of the servers. The addresses of Bitnet had no points to separate server names from domains.

From the beginning of the 1990s, RNP started to provide access to approximately 600 teaching and research institutions, serving a wide range of community of about 65 thousand users.

Chapter 2
Signals and Linear Systems



Pure mathematics is, in its way, the poetry of logical ideas.
– Albert Einstein

Signals are physical processes that are expressed as functions of independent variables, usually time, that contain information about the behavior and characteristics of given physical phenomena. For example, the voice signal is the result of the sound pressure caused by the vibration of the vocal chords over time (Lourtie 2007). Other examples of signals include the temperature, the pressure, the stock prices, and the transmission of messages using the Internet Mandal and Asif (2007).

Most signals are usually processed to extract, digest, or remove information. They are used to excite electronic circuits, such as control or robotic systems, video and image systems, communication equipment, and a large variety of information processing systems.

These systems are complex in nature and involve analog or digital devices. The modeling of a complex system is difficult and requires that it could be analyzed, or separated, into simpler and mathematically tractable parts.

For most systems, restrictions on the frequency or power range, as well as limits on operating lifetime, are established to render them amenable to mathematical analysis. It is important to obtain solutions that reflect the behavior of the original equipment or system, but real life is always very complex.

2.1 Usual Time Signals

As mentioned, the mathematical model of a signal is a functional relation in which the argument, also known as the independent variable, is the time. A typical signal is the voltage across the terminals of a circuit or the current flowing in that circuit (Baskakov 1986).

Signals can be one-dimensional, when described by one-time function, $x(t)$, or they can be multidimensional, $\mathbf{x}(t) = [x_1(t), x_2(t), \cdots, x_N(t)]$, as seen in Fig. 2.1. In this case, $\mathbf{x}(t)$ is the multidimensional input signal, and $\mathbf{y}(t) =$

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_2

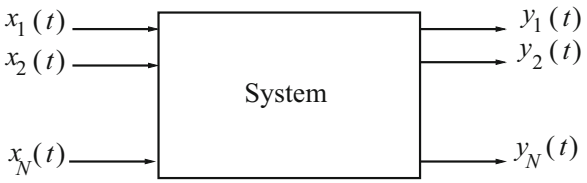


Fig. 2.1 A system with multiple terminals

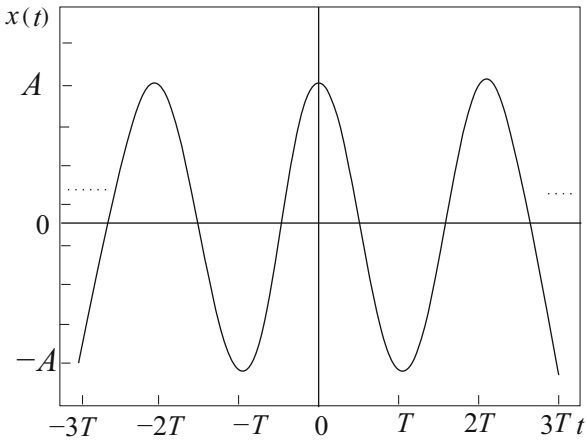


Fig. 2.2 A deterministic signal

$[y_1(t), y_2(t), \cdots, y_N(t)]$ is the output signal. The number N is the dimensionality of the multidimensional signal. The set of messages that flow through a mobile cellular communication system is an example of a multidimensional signal.

It is possible to characterize a system as analog or digital. A system that has an analog input and an analog output is defined as an analog system. On the other hand, a system that has a digital input and a digital output is considered a digital system.

Systems can have mixed-type attributes too. A system with analog input and digital output is an analog-to-digital converter. By the same token, a system with digital input and analog output is referred to as digital-to-analog converter. Of course, those systems can include several parts, which are either analog or digital.

Signals, on the other hand, can be deterministic or random in nature. A deterministic signal is usually described by an equation, such as

$$x(t) = A\cos(\omega t + \phi), \tag{2.1}$$

in which A represents the signal amplitude, $\omega = 2\pi f$ is the angular frequency, f is the frequency in hertz (Hz), and ϕ is its initial phase.

Sometimes, a signal is described by a drawing, as seen in Fig. 2.2, that characterizes the sinusoidal signal for the time interval of interest $[-3T, 3T]$. In this case, the sinusoidal function has a repetition period, of just period, of $2T$. This means that two peaks, or valleys, of the cosine function are separated by $2T$.

Regarding the nature of the signals, they can be described as continuous or discrete in time, as well as continuous or discrete in amplitude. Signals that are

continuous in time and continuous in amplitude are known as analog signals, in this case $x(t) \in \mathbb{R}, t \in \mathbb{R}$. Signals that are continuous in time and discrete in amplitude are called discrete-amplitude signals, that is, $x(t) \in \mathbb{R}, t \in \mathbb{Q}$. Signals that are discrete in time and continuous in amplitude are discrete-time signals, that is, $x(t) \in \mathbb{Q}, t \in \mathbb{R}$. Signals that are discrete in time and discrete in value are digital signals, that is, $x(t) \in \mathbb{Q}, t \in \mathbb{Q}$. Sometimes, the set of integers $\mathbb{Z}$ is used in place of the set of rationals $\mathbb{Q}$ to characterize the discrete support or represent the discrete amplitude.

2.1.1 Stochastic Signals

Deterministic signals are hardly found in real life, because nature is random and its outcomes are usually stochastic and, sometimes, unpredictable. But it is sometimes convenient to think of signals as being deterministic, because this expedient facilitates the analysis.

One of the convenient properties of deterministic signals states that if the signal is known at a certain time, it can be inferred for the whole time interval. This is clear, for instance, in the case of the sinusoidal function.

A random, or stochastic, signal can be described by one of its samples or realizations, as depicted in Fig. 2.3, or by some of its probabilistic properties or statistical means, such as its probability density function, as shown in Fig. 2.4.

The important stochastic signals are studied in the chapter on random signals and noise. A review of probability, which discusses the fundamental concepts of set and measure, as well as random variables and their moments, is presented in Appendix E.

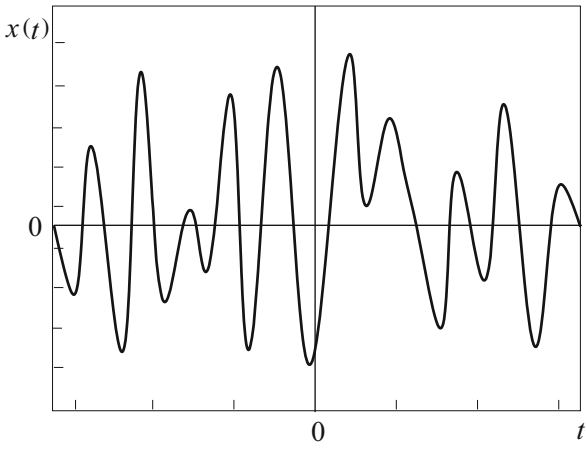


Fig. 2.3 A random, or stochastic, signal

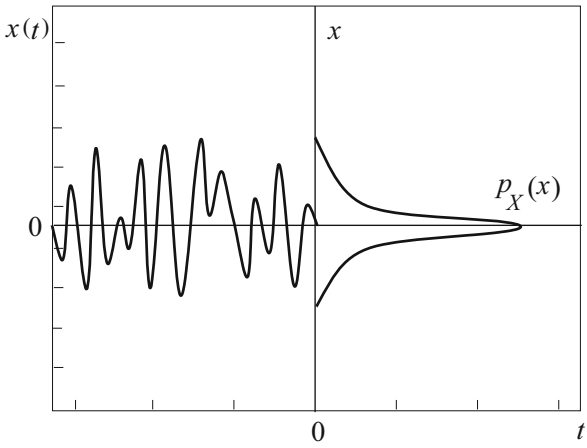


Fig. 2.4 A random signal and its probability density function

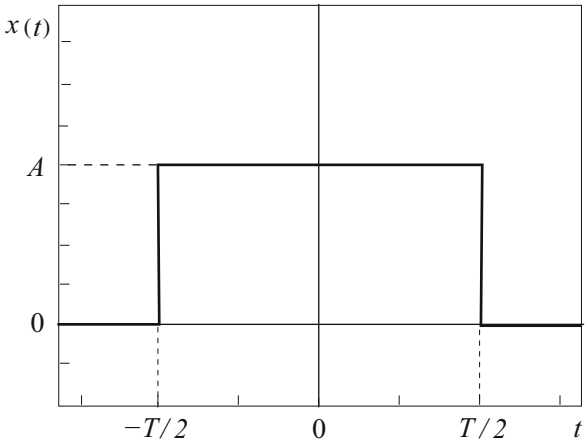


Fig. 2.5 The gate function

Examples of random functions are abundant and include voice signals, television signals, telephone traffic, digital video, Internet traffic, and communication signals in general.

2.1.2 The Gate and the Unit Step Functions

The gate function, $g(t)$, plays an important role in the study of signals and systems, because it can be used to represent a pulse signal, the response of a linear system to a given signal, or the time a certain circuit is in operation, for example. Figure 2.5 illustrates the gate function.

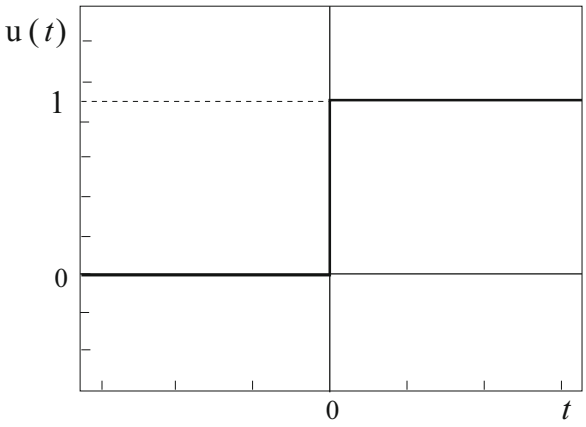


Fig. 2.6 A unit step function

The gate function, or signal, can be obtained as the composition of two displaced unit step functions,

$$g(t) = A[\mathrm{u}(t + T/2) - \mathrm{u}(t - T/2)], \tag{2.2}$$

in which the unit step function is depicted in Fig. 2.6 and is defined as

$$\mathrm{u}(t) = \begin{cases} 0 \text{ if } t \in (-\infty, 0) \\ 1 \text{ if } t \in (0, \infty). \end{cases} \tag{2.3}$$

The value of the unit step function at zero is usually undefined. Some authors prefer to define $\mathrm{u}(0) = 1/2$ for reasons of expediency.

The unit step function lasts from minus infinity to plus infinity, which spans the whole real line $\mathbb{R}$. It can indicate the exact time moment a certain phenomenon occurs, for instance, when a switch is closed or a signal is turned on.

Example The unit step function can be used to generate a digital waveform from a sinusoidal function, using a composite function. Consider that the input signal is $x(t) = \cos(\omega t)$. Then, the required formula for the output signal can be written as

$$y(t) = \mathrm{u}[x(t)] = \mathrm{u}[\cos(\omega t)] = \begin{cases} 1, \text{ if } x(t) \in (0, \infty) \\ 0, \text{ if } x(t) \in (-\infty, 0), \end{cases} \tag{2.4}$$

and the output signal is depicted in Fig. 2.7. Note that the signal is a square wave, a periodic signal that can only assume the values 0 and 1.

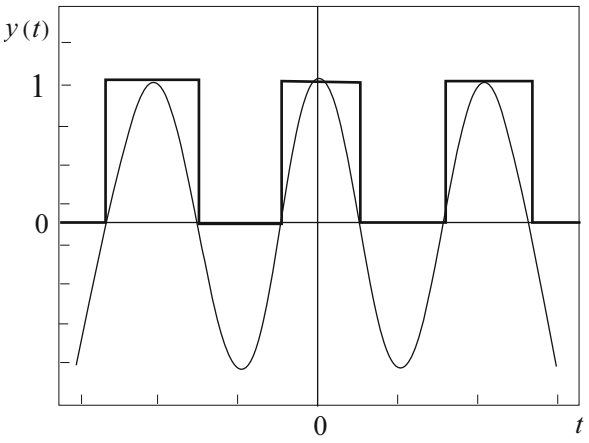


Fig. 2.7 A square wave produced from a cosine wave, using the unit step function

2.1.3 Properties of the Unit Step Function

The unit step function is a very useful one for applications in Mathematics and Engineering, because it can be used to delimit other functions. Some of its interesting properties are presented in the sequence.

For instance, the following property illustrates a typical application of the unit step function to delimit the range of integration of a given signal:

$$\int_{-\infty}^{\infty} x(t)\mathrm{u}(t)\mathrm{d}t = \int_{0}^{\infty} x(t)\mathrm{d}t \tag{2.5}$$

Note, in the following property, that the unit step function has been inverted, regarding the abscissa, and shifted to the position t on the τ axis. Thus, $\mathrm{u}(t-\tau)$ is equal to one from $\tau = -\infty$ to $\tau = t$ and zero elsewhere:

$$\int_{-\infty}^{\infty} x(\tau)\mathrm{u}(t-\tau)\mathrm{d}\tau = \int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \tag{2.6}$$

2.1.4 The Sign Function

The signum, or sign, function attributes one to positive values of the independent variable and minus one to the negative values as follows:

$$\mathrm{sgn}(t) = \begin{cases} 1, & \text{if } x(t) \in (0, \infty) \\ -1, & \text{if } x(t) \in (-\infty, 0) \end{cases} \tag{2.7}$$

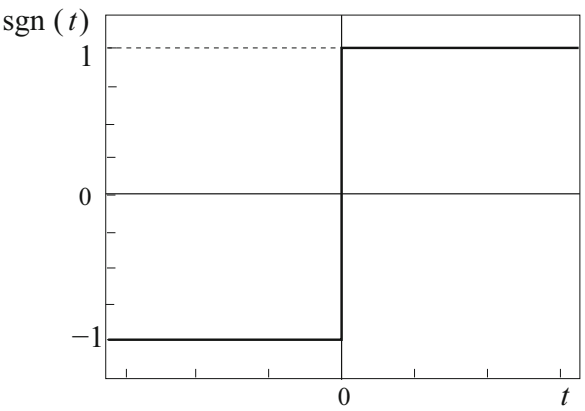


Fig. 2.8 The signum function

The sign function, depicted in Fig. 2.8, is usually not defined for $t = 0$, but the value zero can be attributed in some cases to facilitate limiting operations.

The sign function can be defined with the help of the unit step function as

$$\operatorname{sgn}(t) = \mathrm{u}(t) - \mathrm{u}(-t). \tag{2.8}$$

Example A formula for an alternating square wave can be written using the $\operatorname{sgn}(t)$. It is only necessary to write the composite function $f(t) = \operatorname{sgn}[\sin(t)]$.

2.1.5 The Impulse Function

The impulse function, also called Dirac function or Dirac delta distribution, after the British theoretical physicist Paul Adrien Maurice Dirac (1902–1984), is an important mathematical construct that helps explain several phenomena. An impulse function can be seen in Fig. 2.9.

The impulse function, which does not meet the requirements of a typical function but can be considered as a generalized function, is defined as follows:

$$\delta(t) = \begin{cases} \infty, & \text{if } t = 0 \\ 0, & \text{if } t \neq 0 \end{cases} \tag{2.9}$$

The value of the impulse function at zero is infinity. This can cause problems, sometimes, and therefore some authors prefer to define the impulse function as a limit. For instance, it can be defined as the limit of the Laplace function as the parameter α goes to infinity:

$$\delta(t) = \lim_{\alpha \to \infty} \frac{\alpha}{2} e^{-\alpha |x|} \tag{2.10}$$

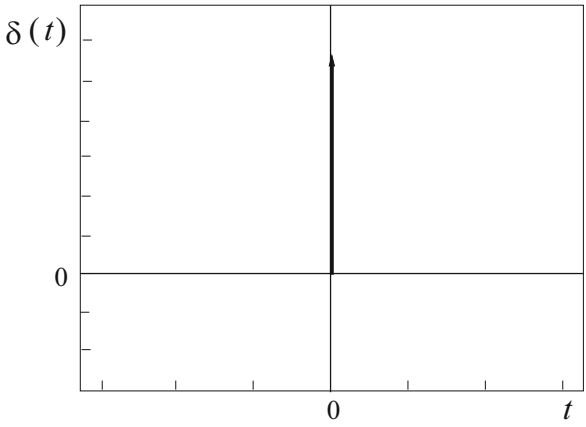


Fig. 2.9 The impulse function

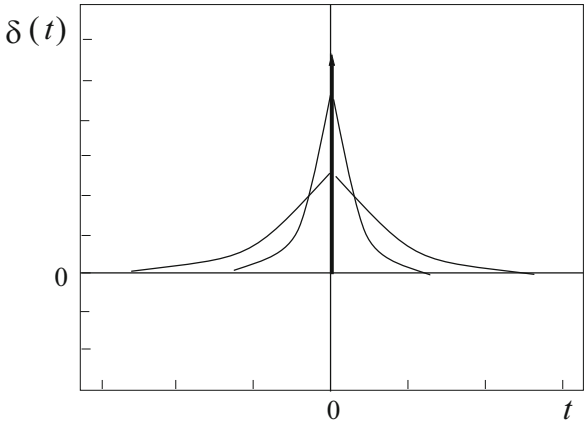


Fig. 2.10 The limiting definition of the impulse function

This can be observed in Fig. 2.10. Note that the Laplace function is squeezed, as $\alpha \to \infty$, but the area under the curve remains the same.

The Laplace function was named after Pierre-Simon, Marquis de Laplace (1749–1827), a French scholar and polymath with important contributions to the development of Engineering, Mathematics, Statistics, Physics, Astronomy, and Philosophy.

Note that the Laplace function is normalized, as can be verified, which means that it can be used as a probability distribution function:

$$\int_{-\infty}^{\infty} \frac{\alpha}{2} e^{-\alpha|x|} \mathrm{d}x = \int_{-\infty}^{0} \frac{\alpha}{2} e^{\alpha x} \mathrm{d}x + \int_{0}^{\infty} \frac{\alpha}{2} e^{-\alpha x} \mathrm{d}x$$
$$= \frac{\alpha}{2} \left[\frac{e^{\alpha x}}{\alpha} \right]_{-\infty}^{0} + \frac{\alpha}{2} \left[-\frac{e^{-\alpha x}}{\alpha} \right]_{0}^{\infty} = 1 \qquad (2.11)$$

The same result for the impulse can also be obtained from the limit of a normalized gate function, considering the interval $(0, \tau)$, which one defines as $g_\tau(t)$,

$$g_\tau(t) = \frac{1}{\tau}[\mathrm{u}(t) - \mathrm{u}(t-\tau)], \tag{2.12}$$

as the parameter $\tau \to 0$:

$$\delta(t) = \lim_{\tau \to 0} g_\tau(t) = \lim_{\tau \to 0} \frac{\mathrm{u}(t) - \mathrm{u}(t-\tau)}{\tau} \tag{2.13}$$

The operational definition of the impulse function is therefore related to the unit step function, as follows,

$$\frac{\mathrm{du}(t)}{\mathrm{d}t} = \delta(t), \tag{2.14}$$

that is, the impulse function is the derivative of the unit step function. This is a useful relation, and several examples are given to emphasize its importance.

Therefore, because of the fundamental theorem of Calculus,

$$\int_{-\infty}^{t} \delta(\tau)\mathrm{d}\tau = \mathrm{u}(t). \tag{2.15}$$

The impulse function has other important properties, for example,

$$\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = 1, \tag{2.16}$$

the area of the impulse function is one, which means that the impulse function is normalized.

The impulse function can also extract a sample of a given signal, as can be observed in the following,

$$\int_{-\infty}^{\infty} x(t)\delta(t)\mathrm{d}t = x(0), \tag{2.17}$$

and as displayed in Fig. 2.11,

$$\int_{-\infty}^{\infty} x(t)\delta(t-\tau)\mathrm{d}t = x(\tau). \tag{2.18}$$

The impulse function can be used to model voltage, current, or power peaks. For example, a lightning can be modeled as a voltage impulse, because it occurs during a short time interval and has an amplitude that can reach millions of volts.

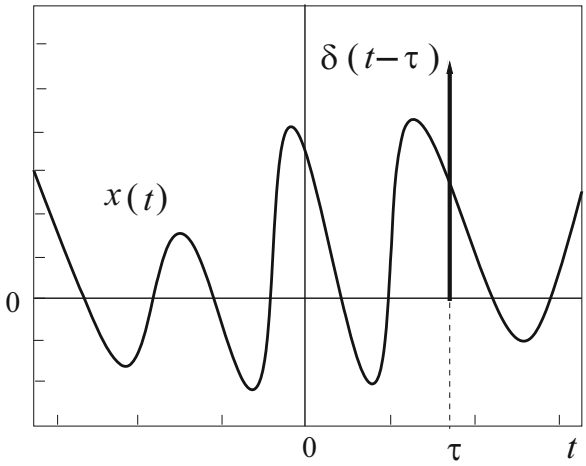


Fig. 2.11 The impulse function shifted to τ to sample the signal $x(t)$

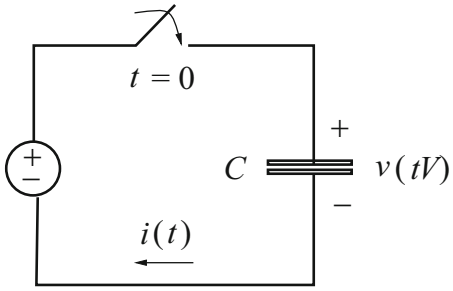


Fig. 2.12 Capacitive circuit subject to a sudden voltage change

Example The derivative of the sign function can be computed using the properties of the unit step function,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{sgn}(t) = \frac{\mathrm{d}}{\mathrm{d}t}[\mathrm{u}(t) - \mathrm{u}(-t)] = \delta(t) + \delta(-t) = 2\delta(t),$$

which is true, because the impulse is an even function.

Example The circuit depicted in Fig. 2.12 is closed at $t = 0$. Determine the voltage across the capacitor and the current that flows in the circuit.

Using the relationship between the voltage and the current for a capacitor of capacitance C, one can write

$$i(t) = C\frac{\mathrm{d}v(t)}{\mathrm{d}t}. \tag{2.19}$$

The voltage across the capacitor terminals is $v(t) = V\mathrm{u}(t)$ for all time. Therefore, the current that flows in the circuit is given by

$$i(t) = C\frac{\mathrm{d}[V\,\mathrm{u}(t)]}{\mathrm{d}t} = CV\delta(t). \tag{2.20}$$

That is, the current is an impulse whose area is proportional to the capacitance times the source voltage. This happens because there is no resistance to limit the current that flows in the circuit.

Example A series of alternating impulses can be created from the formula for the sinusoidal wave, as a composite function, $y(t) = \mathrm{u}[x(t)] = \mathrm{u}[\cos(\omega t + \phi)]$.

The derivative of $y(t)$ can be obtained using the formula for the derivative of a composite function, that is,

$$z(t) = \frac{\mathrm{d}y(t)}{\mathrm{d}t} = \frac{\mathrm{d}u(t)}{\mathrm{d}x(t)} \times \frac{\mathrm{d}x(t)}{\mathrm{d}t}, \tag{2.21}$$

$$z(t) = \frac{\mathrm{d}\,\mathrm{u}[\cos(\omega t + \phi)]}{\mathrm{d}t} = \delta[\cos(\omega t + \phi)] \times [-\omega \sin(\omega t + \phi)], \tag{2.22}$$

$$z(t) = -\omega\delta[\cos(\omega t + \phi)] \times \sin(\omega t + \phi), \tag{2.23}$$

which is the formula for the series of alternating impulses depicted in Fig. 2.13.

2.1.6 Properties of the Impulse Function

Some important properties of the impulse generalized function, which are useful to model linear systems, are listed in the following Hsu (1973):

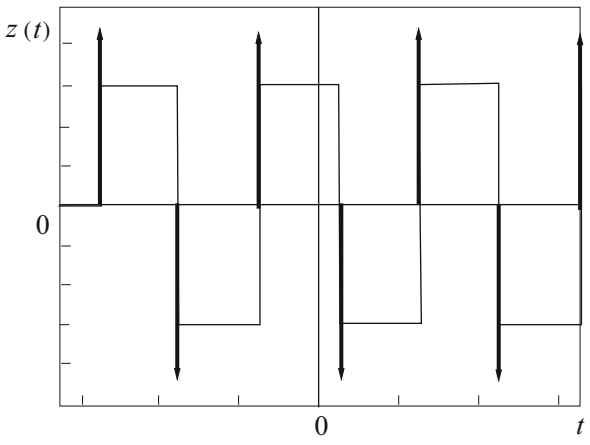


Fig. 2.13 Periodic series of alternating impulses

First, recall that $\delta(t) = \mathrm{u}'(t)$. Then, from the properties of the unit step function,

$$\mathrm{u}(\alpha t) = \begin{cases} \mathrm{u}(t) & \text{if } \alpha \text{ is positive, } \textit{and} \\ \mathrm{u}(-t) & \text{if } \alpha \text{ is negative.} \end{cases} \tag{2.24}$$

Taking the derivative of the unit step function $\mathrm{u}(\alpha t)$, one obtains

$$\frac{\mathrm{du}(\alpha t)}{\mathrm{d}t} = \begin{cases} \alpha\delta(t) & \text{if } \alpha \text{ is positive, } \textit{and} \\ -\alpha\delta(t) & \text{if } \alpha \text{ is negative.} \end{cases} \tag{2.25}$$

Therefore, combining both results, one obtains

$$\delta(\alpha t) = \frac{1}{|\alpha|}\delta(t). \tag{2.26}$$

Because it is possible to write $\mathrm{u}(-t) = 1 - \mathrm{u}(t)$, then

$$\mathrm{u}'(-t) = -\delta(-t) = \frac{\mathrm{d}[1 - \mathrm{u}(t)]}{\mathrm{d}t} = -\delta(t);$$

thus,

$$\delta(-t) = \delta(t). \tag{2.27}$$

By convention, one has the following property:

$$t\delta(t) = 0 \tag{2.28}$$

The following property is a result of the sampling property, because the impulse is different from zero only at $t = 0$. It means that the area of the impulse takes the value of the signal at $t = 0$,

$$x(t)\delta(t) = x(0)\delta(t) \tag{2.29}$$

if $x(t)$ is continuous at $t = 0$.

The last property can be generalized for any time $t = \tau$,

$$x(t)\delta(t - \tau) = x(\tau)\delta(t - \tau), \tag{2.30}$$

and, for any continuous-time signal $x(t)$,

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)\mathrm{d}\tau. \tag{2.31}$$

The equation that defines the derivatives of the impulse function is

$$\int f(t)\delta^{(n)}(t)\mathrm{d}t = -\int f'(t)\delta^{(n-1)}(t)\mathrm{d}t. \tag{2.32}$$

2.1.7 Composite Function with the Impulse

The composite function that results by combining the impulse with a function of time $f(t)$, which has N roots $t_i, i = 1, 2, \ldots, N$, is given by

$$\delta[f(t)] = \sum_{i=1}^{N} \frac{\delta(t - t_i)}{|f'(t_i)|}. \tag{2.33}$$

This is an important result, which has several applications, mainly regarding the identification of zeros of a given function. It is demonstrated for a polynomial but can be generalized for any function that has zeros on the real line.

To demonstrate this last property of the impulse function for a polynomial, consider that the function $f(t)$ can be put in the following form, with no loss of generality,

$$f(t) = (t - t_1)(t - t_2)\cdots(t - t_N) = \prod_{i=1}^{N}(t - t_i),$$

in which t_i are the ordered $(t_1 < t_2 \cdots < t_N)$ roots of the polynomial $f(t)$.

Figure 2.14 illustrates the composite function $\mathrm{u}[f(t)]$. Observe that this composite function is the sum of gate functions of the form

$$\mathrm{u}[f(t)] = \sum_i g(t - t_i) = \sum_i [\mathrm{u}(t - t_i) - \mathrm{u}(t - t_{i+1})]. \tag{2.34}$$

Note also, by the chain rule, that

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{u}[f(t)] = \delta[f(t)]f'(t). \tag{2.35}$$

Taking the derivative of the composite function in Eq. 2.34 and equating to 2.35 yield

$$\delta[f(t)]f'(t) = \sum_i [\delta(t - t_i) - \delta(t - t_{i+1})], \tag{2.36}$$

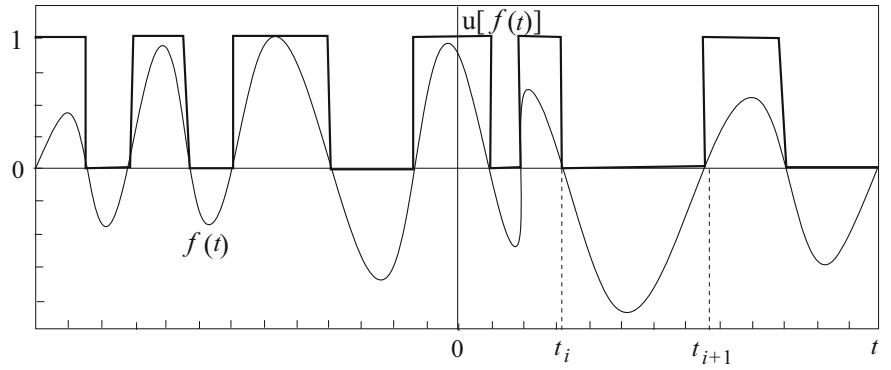


Fig. 2.14 The function u[$f(t)$]

which can be put into the following form:

$$\delta[f(t)] = \sum_i \frac{[\delta(t - t_i) - \delta(t - t_{i+1})]}{f'(t_i)} \tag{2.37}$$

This represents a series of alternating impulses that occur at the roots of the polynomial. Because the sign of each impulse coincides with the sign of the polynomial derivative at each point, it is possible to put the equation in its final form:

$$\delta[f(t)] = \sum_i \frac{[\delta(t - t_i)]}{|f'(t_i)|} \tag{2.38}$$

Example Consider that $f(t) = t^2 - \alpha^2$, and determine the formula for $\delta[f(t)]$. This is an equation for a parabola that crosses the abscissa axis, as sketched in Fig. 2.15.

Substituting the expression for $f(t)$ into the formula, one obtains

$$\delta[t^2 - \alpha^2] = \sum_{i=1}^{2} \frac{\delta(t - t_i)}{2|t_i|}$$

for $t_1 = -\alpha$ and $t_2 = \alpha$. Thus,

$$\delta[t^2 - \alpha^2] = \frac{1}{2|\alpha|} [\delta(t + \alpha) + \delta(t - \alpha)],$$

as can be observed in the figure.

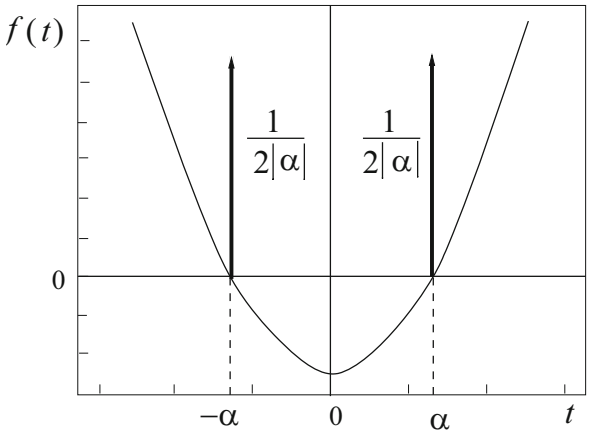


Fig. 2.15 Parabola that crosses the abscissa at two points

Example Consider now that $f(t) = t^2 + 2t - 3$, and again, determine the formula for $\delta[f(t)]$. For this polynomial, after determining the roots, one obtains $f(t) = (t-1)(t+3)$. Thus, $f'(t) = 2t + 2$, $f'(1) = 4$, and $f'(-3) = -4$.

Substituting the expression for $f(t)$, and the root values, into the formula, one obtains

$$\delta[t^2 + 2t - 3] = \sum_{i=1}^{2} \frac{\delta(t - t_i)}{4};$$

therefore,

$$\delta[t^2 + 2t - 3] = \frac{1}{4}\left[\delta(t-1) + \delta(t+3)\right].$$

2.1.8 Doublet Generalized Function

The doublet generalized function $\delta'(t)$ is obtained as the derivative of the impulse function. It is shown in Fig. 2.16:

$$\delta'(t) = \frac{\mathrm{d}\delta(t)}{\mathrm{d}t} \tag{2.39}$$

The doublet function has the following fundamental property,

$$\int_{-\infty}^{\infty} x(t)\delta'(t)\mathrm{d}t = -\int_{-\infty}^{\infty} x'(t)\delta(t)\mathrm{d}t = -x'(0), \tag{2.40}$$

which can be obtained using integration by parts for a function $x(t)$ which is continuous at $t = 0$ and vanishes outside some fixed interval.

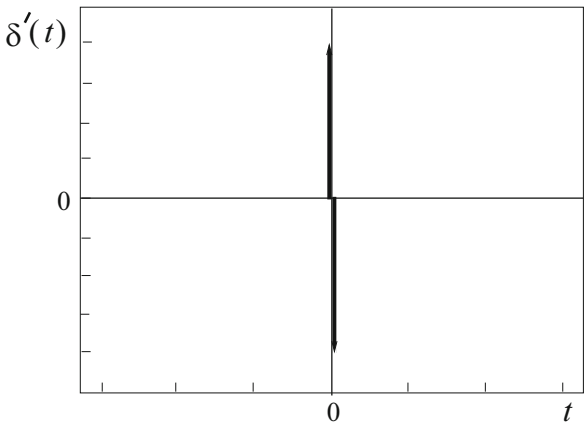


Fig. 2.16 Doublet generalized function

Defining the n-th derivative of the impulse function as

$$\delta^{(n)}(t) = \frac{\mathrm{d}^n \delta(t)}{\mathrm{d}t^n}, \tag{2.41}$$

one obtains

$$\int_{-\infty}^{\infty} x(t)\delta^{(n)}(t)\mathrm{d}t = (-1)^n x^{(n)}(0). \tag{2.42}$$

Finally, observe that putting $x(t) = \tau\delta(\tau - t)$ and $n = 1$ into Eq. 2.41 and integrating in τ, one obtains

$$\int_{-\infty}^{\infty} \tau\delta(\tau - t)\delta'(\tau)\mathrm{d}\tau = t\delta'(t); \tag{2.43}$$

thus,

$$t\delta'(t) = -\delta(t). \tag{2.44}$$

Also, putting $x(t) = t^n\delta(t)$ into Eq. 2.40 yields

$$\int_{-\infty}^{\infty} t^n\delta(t)\delta'(t)\mathrm{d}t = -\int_{-\infty}^{\infty} nt^{n-1}\delta'(t)\delta(t)\mathrm{d}t. \tag{2.45}$$

Integrating again the second term $n - 1$ times, one obtains

$$\int_{-\infty}^{\infty} t^n\delta(t)\delta'(t)\mathrm{d}t = \int_{-\infty}^{\infty} (-1)^n n!\delta'(t)\delta(t)\mathrm{d}t; \tag{2.46}$$

therefore, observing the integrand, the following result is obtained:

$$t^n \delta^{(n)}(t) = (-1)^n n! \delta(t) \tag{2.47}$$

Finally,

$$\delta'(-t) = -\delta'(t), \tag{2.48}$$

which means that the doublet generalized function is antisymmetric.

2.1.9 The Ramp Function

The ramp function, $\mathrm{r}(t)$, is obtained as the integral of the unit step function, as sketched in Fig. 2.17:

$$\mathrm{r}(t) = \int_{-\infty}^{t} \mathrm{u}(\tau) \mathrm{d}\tau \tag{2.49}$$

It can also be written as $\mathrm{r}(t) = t\,\mathrm{u}(t)$.

Of course, by the fundamental theorem of Calculus, the derivative of the ramp function gives the unit step function,

$$\frac{\mathrm{d}\,\mathrm{r}(t)}{\mathrm{d}t} = \mathrm{u}(t). \tag{2.50}$$

As can be seen, the ramp function grows without limit as time goes by. This function can be used to model different signals, including the sawtooth wave, which is used to synchronize the time base in oscilloscopes. The ramp function can also be

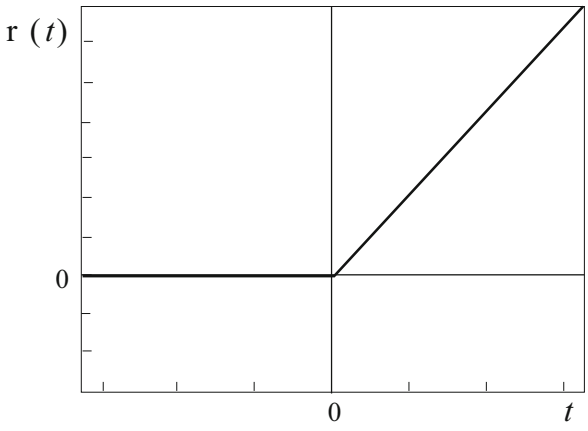


Fig. 2.17 Ramp function

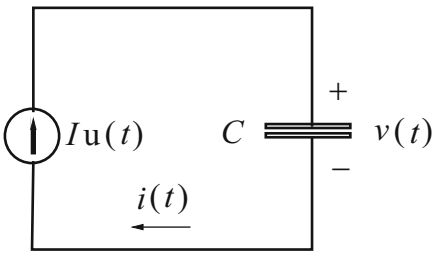


Fig. 2.18 A capacitive circuit excited by a current source

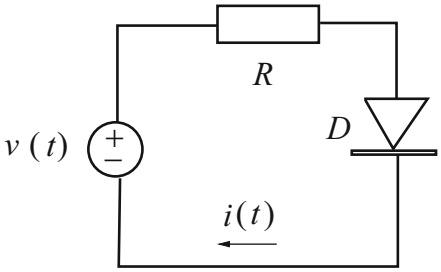


Fig. 2.19 Rectifier using a semiconductor diode

used to produce composite signals, such as the half-wave signal that is produced as the result of the signal rectification by a semiconductor diode.

Example The capacitive circuit, depicted in Fig. 2.18, is excited by a current source $i(t) = I\mathrm{u}(t)$. Compute the voltage across the capacitor.

Using the relation between the current and the voltage across the terminals of a capacitor,

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\mathrm{d}\tau. \tag{2.51}$$

Substituting the equation for the current source and computing the integral, one obtains

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} I\mathrm{u}(\tau)\mathrm{d}\tau = \frac{I}{C}\mathrm{r}(\tau). \tag{2.52}$$

Therefore, the voltage across the terminals of the capacitor will linearly increase without limit, until the device is disrupted.

Example The circuit depicted in Fig. 2.19 has an ideal semiconductor diode. Write an equation for the current that flows through the resistor using the ramp function, considering that the input voltage is

$$v(t) = V\cos(\omega t). \tag{2.53}$$

The current only flows through the circuit during the positive circle of the input signal; therefore, using Kirchhoff's law, a relation derived by the German physicist Gustav Robert Kirchhoff (1824–1887), one obtains

$$i(t) = \begin{cases} \frac{V}{R}\cos(\omega t) & \text{if the cosine is positive, } and \\ 0 & \text{if the cosine is negative.} \end{cases} \tag{2.54}$$

The previous result can be written in a concise manner, using the ramp function,

$$i(t) = \frac{V}{R}\mathrm{r}[\cos(\omega t)]. \tag{2.55}$$

2.1.10 The Exponential Function

The exponential function, depicted in Fig. 2.20, is expressed as (Oppenheim et al. 2002)

$$x(t) = e^{\alpha t}, \tag{2.56}$$

in which α is a constant parameter. For $\alpha > 0$, the function grows without limit, and for $\alpha < 0$, it decreases to zero as times goes by.

This function has large application in Electrical Engineering, as well as several other scientific areas. It is usually combined with the unit step function to delimit its support. For instance, the decreasing exponential

$$y(t) = e^{-\alpha t}\mathrm{u}(t), \tag{2.57}$$

for α positive, has its domain in the interval $[0, \infty]$.

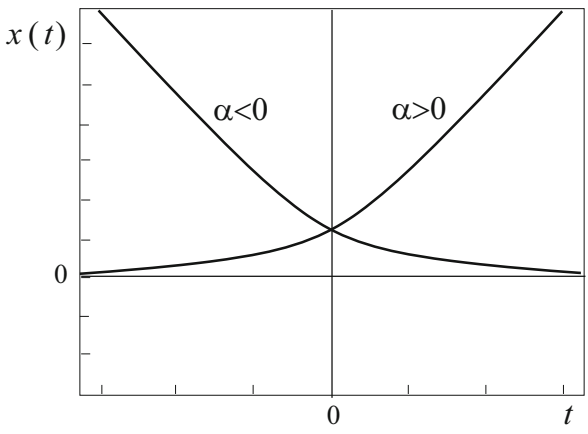


Fig. 2.20 The exponential function

A diode is a nonlinear device, and the relationship between the voltage across its terminals and the current that flows through it is better modeled by the Shockley diode equation, or the diode law, named after the Nobel Prize winner and transistor co-inventor William Bradford Shockley Jr. (1910–1989).

The diode equation gives a formula for the current through a diode, $i(t)$ as a function of voltage, $v(t)$, as in the following,

$$i(t) = I_0 \left[e^{\frac{qv(t)}{kT}} - 1 \right], \tag{2.58}$$

in which I_0 is the diode leakage current density in the absence of light (dark saturation current), q is the absolute value of the electron charge, k is the Boltzmann's constant, and T is the absolute temperature in kelvin (K). At 300K, the thermal voltage $kT/q = 25.85$ mV.

Figure 2.21 sketches the diode equation. The negative part is emphasized to facilitate the observation.

Systems composed of nonlinear devices, such as the semiconductor diode, are difficult to solve or require complex mathematical tools to produce results. Anyway, the obtained solutions are usually of a particular type.

That is one of the reasons the linear systems are studied. They provide simple, analytical, results based on strong mathematical foundations. Even better, linear systems also produce general solutions, which can be used in different cases.

2.1.11 The Growing and Decaying Sinusoidal Functions

The growing sinusoidal function is defined as (Oppenheim et al. 2002)

$$y(t) = e^{\alpha t} \cos(\omega t + \phi), \tag{2.59}$$

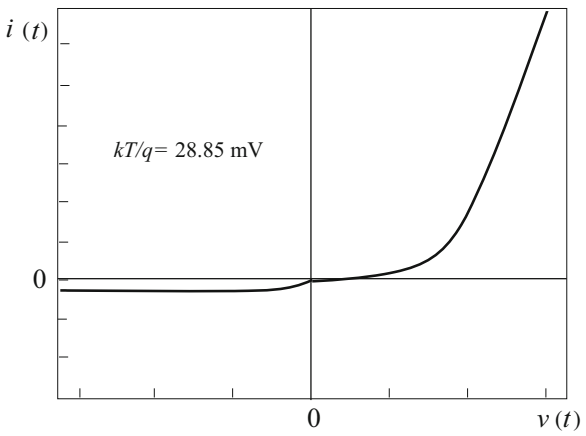


Fig. 2.21 Model for a semiconductor diode

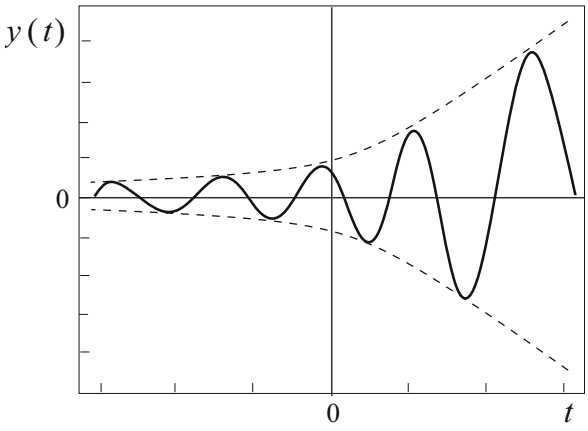


Fig. 2.22 Growing sinusoidal function

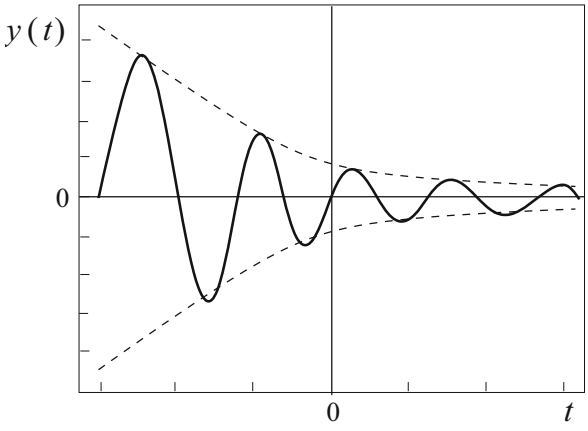


Fig. 2.23 Decaying sinusoidal function

in which $\alpha > 0$ is a real number, ϕ is the initial phase, and ω is the angular frequency. This function is depicted in Fig. 2.22.

The decaying sinusoidal function, which frequently appears as a response of systems with elements that store energy, is defined as

$$y(t) = e^{-\alpha t} \cos(\omega t + \phi). \tag{2.60}$$

This function is shown in Fig. 2.23.

The decaying sinusoidal function is usually referred to as a dumped sinusoid. This type of signal appears in the response of *RLC* circuits and in excited mechanical systems, composed of mass and spring, for example, because the nonconservative damping force removes energy from the system.

The energy is removed from the systems, usually, in the form of thermal energy. A guitar string stops oscillating a few seconds after being plucked, because it is also subject to dumping, and part of the energy is released by means of a mechanical

wave, called sound, that propagates along or through a medium by particle-to-particle interaction.

2.1.12 The Complex Exponential Function

The complex exponential function is defined as (Oppenheim et al. 2002)

$$x(t) = e^{j\omega t} = \cos(\omega t) + j\,\sin(\omega t), \tag{2.61}$$

in which $\omega = 2\pi f$ is the angular frequency and f is the frequency in hertz [Hz]. The complex exponential function is depicted in Fig. 2.24.

An important property of the complex exponential function is that it is periodic, with period $T = 1/f$ given in seconds [s]. To verify this, consider a shifted version of the complex exponential

$$x(t+T) = e^{j\omega(t+T)} = e^{j\omega t} \cdot e^{j\omega T}, \tag{2.62}$$

in which, from the preceding definition, $T = 2\pi/\omega$. Substituting this value into the previous equation, one obtains

$$x(t+T) = e^{j\omega t} \cdot e^{j2\pi}. \tag{2.63}$$

But

$$e^{j2\pi} = \cos(2\pi) + j\,\sin(2\pi) = 1. \tag{2.64}$$

Therefore,

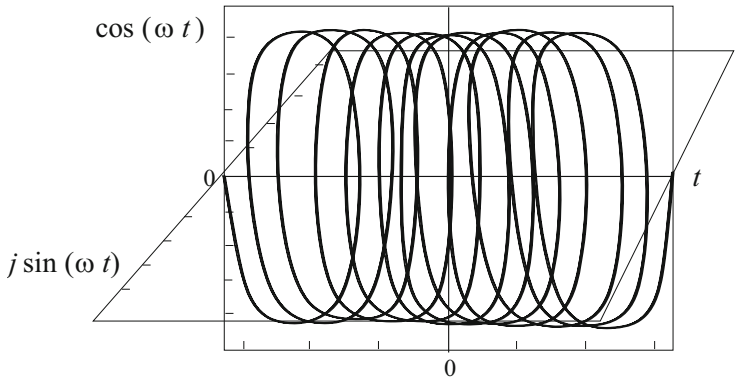


Fig. 2.24 Complex exponential function

$$x(t+T) = e^{j\omega t}, \tag{2.65}$$

as expected.

2.2 Discrete-Time Signals

Mathematical modeling is the art of establishing the conditions to make the devices and systems amenable to general solutions. For example, if a device is linear only for a certain frequency range or for a given maximum power, it is the job of the designer to make sure these limits are not exceeded to maintain the system under control.

Several phenomena are modeled as discrete in time, for example, the number of objects produced by a factory per hour, the number of people that enter a bank agency at a given date, or the output of a digital circuit.

Discrete-time systems are characterized by difference equations, which express the relationship between discrete-time functions or sequences. A sequence of numbers $\{x_k\}$ is an ordered collection, which is indexed by a set of integers $\{k\}$. Thus, if $k \in \mathbb{Z}$, the set of integers, then $x_k \in \{\ldots, x_{-2}, x_{-1}, x_0, x_1, x_2, \ldots\}$ (Gabel and Roberts 1973).

2.2.1 Discrete Unit Step Function

The discrete unit step function $\mathrm{u}(k)$ is the infinite sequence represented by $\{\ldots, 0, 0, 0, 1, 1, 1, \ldots\}$, in which the first one occurs at $k = 0$, as shown in Fig. 2.25.

The discrete step function can be defined as

$$\mathrm{u}(k) = \begin{cases} 0 \text{ if } k < 0 \\ 1 \text{ if } k \geq 0 \end{cases} \tag{2.66}$$

for $k \in \mathbb{Z}$.

2.2.2 Discrete Impulse Function

The discrete unit step function can be used to define the discrete impulse function, also known as Kronecker impulse, $\delta(k)$, after the German mathematician Leopold Kronecker (1823–1891), who made contributions to the fields of Algebra and Continuity of Functions.

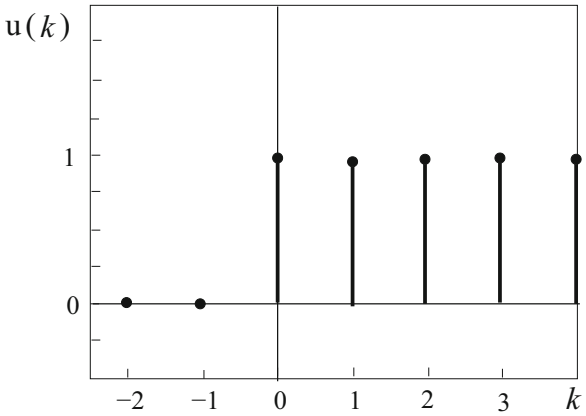


Fig. 2.25 The discrete version of the unit step function

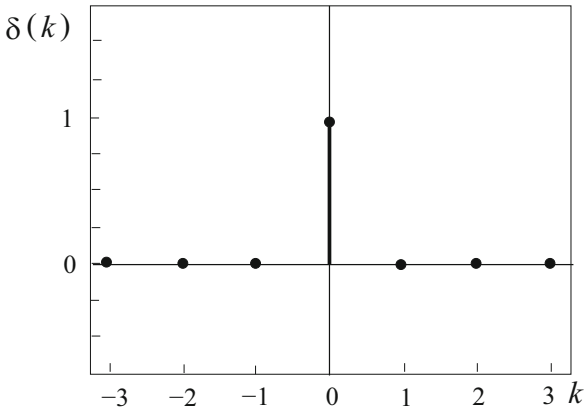


Fig. 2.26 The discrete version of the impulse function

The discrete impulse function is defined as

$$\delta(k) = \begin{cases} 0 \text{ if } k \neq 0 \\ 1 \text{ if } k = 0 \end{cases} \tag{2.67}$$

for $k \in \mathbb{Z}$. Figure 2.26 sketches the discrete impulse function.

If one takes the difference between the discrete unit step function $\mathrm{u}(k)$ and a delayed version of it $\mathrm{u}(k-1)$, the result is the Kronecker impulse function, as follows:

$$\delta(k) = \mathrm{u}(k) - \mathrm{u}(k-1) \tag{2.68}$$

By the same token, the discrete unit step function can be obtained as

$$\mathrm{u}(k) = \sum_{n=-\infty}^{k} \delta(n), \tag{2.69}$$

in which the summation is done over the set of integers $\mathbb{Z}$.

2.2.3 Discrete Ramp Function

The discrete-time ramp, shown in Fig. 2.27, can be written as the summation of the discrete unit step function, in the following manner,

$$\mathrm{r}(k+1) = \sum_{n=-\infty}^{k} \mathrm{u}(n), \tag{2.70}$$

in which the summation is done over the set of integers $\mathbb{Z}$. The discrete ramp function can also be expressed as $\mathrm{r}(k) = k\mathrm{u}(k)$ for all $k \in \mathbb{Z}$.

On the other hand, the discrete unit step function can be obtained from the discrete ramp function, as indicated:

$$\mathrm{u}(k) = \mathrm{r}(k+1) - \mathrm{u}(k) \tag{2.71}$$

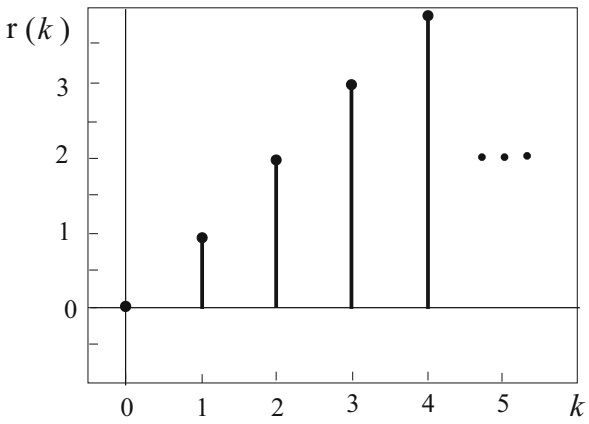


Fig. 2.27 The discrete version of the ramp function

In fact, any arbitrary sequence can be represented as a linear combination of time-shifted Kronecker functions, as follows, Mandal and Asif (2007)

$$x(k) = \sum_{n=-\infty}^{\infty} x(n)\delta(k-n), \tag{2.72}$$

in which, once more, the summation is done over the set of integers $\mathbb{Z}$.

Example The window function, $\mathrm{w}(k)$, is used to filter a certain set of samples from a given signal. It is defined as

$$\mathrm{w}(k) = \begin{cases} 0 \text{ if } k < -n, k > n \text{ and} \\ 1 \text{ if } -n \leq k \leq n \end{cases} \tag{2.73}$$

for $k, n \in \mathbb{Z}$. Write this function using the discrete unit step function.

From definition of the unit step function, as sketched in Fig. 2.28 for $n = 2$, one can write the window function as

$$\mathrm{w}(k) = \mathrm{u}(k+n) - \mathrm{u}(k-n). \tag{2.74}$$

Example A decaying discrete-time exponential may be written as

$$x(k) = e^{-k}\mathrm{u}(k), \tag{2.75}$$

while a discrete-time complex exponential may be written as

$$y(k) = e^{j\omega k}, \tag{2.76}$$

which has the following property:

$$y(k) = e^{j(\omega+2\pi)k} = e^{j\omega k}e^{j2\pi k} = e^{j\omega k} \tag{2.77}$$

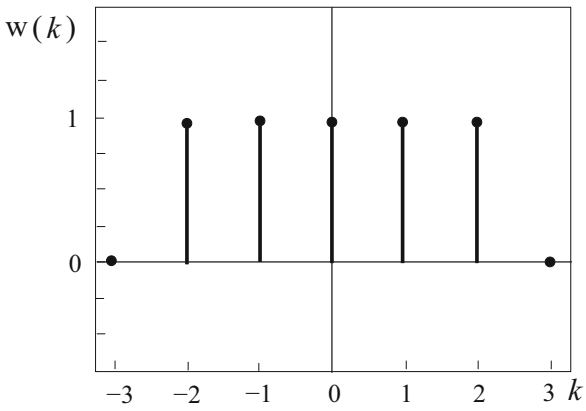


Fig. 2.28 The discrete version of the window function for $n = 2$

2.3 Linear Systems

The word system comes from the ancient Greek $\sigma\acute{\nu}\sigma\tau\eta\mu\alpha$, which is used to refer to a musical scale, an organized body or a whole compounded of parts or members. A system can be continuous or discrete in time or amplitude. A true linear system is hardly found in practice, but the concept is useful because there are several mathematical tools to solve linear systems.

2.3.1 Continuous-Time and Discrete-Time Systems

If the input and output of a system are capable of changing at any instant of time, it is called a continuous-time system. The continuous-time systems are usually modeled by the use of differential equations. The domain of the continuous-time systems is the real line $\mathbb{R}$.

When the signals associated with the system change only at discrete instants, the system is considered discrete-time. Difference equations are used to model discrete-time systems. The domain of the discrete-time systems is the set of integers $\mathbb{Z}$. Consider the following example:

Example A discrete-time circuit is shown in Fig. 2.29, in which $\{x_k\}$ represents the sequence of inputs $\{x_1, x_2, \ldots, x_k, \ldots\}$ and $\{y_k\}$ represents the output values $\{y_1, y_2, \ldots, y_k, \ldots\}$. The system can be modeled by the following equation, taken at time $t = k$,

$$y_k = \alpha x_{k-1} + \beta x_k, \tag{2.78}$$

in which α and β are constant parameters, representing the gains associated with each arm of the system, and Δ represents a unit delay. It is not difficult to verify that this system is linear.

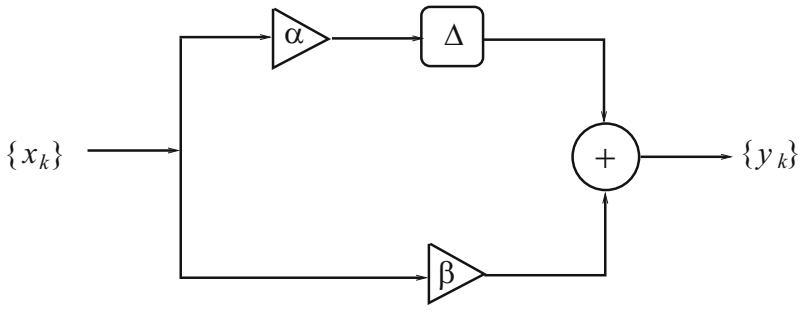


Fig. 2.29 Discrete system

Example In the previous example, if one puts $\alpha = \beta = 1/2$, the discrete-time circuit is known as the moving average filter,

$$y_k = \frac{x_{k-1} + x_k}{2}. \tag{2.79}$$

A moving average filter is used with time signals and series data to smooth out short-term fluctuations and highlight longer-term trends or cycles. It also reduces random noise while retaining a sharp step response.

Example The general formula for the response of a moving average filter, which takes the mean of the previous N samples, is given by

$$y_k = \frac{1}{N} \sum_{i=0}^{N-1} x_{k-i}. \tag{2.80}$$

The moving average is a finite impulse response (FIR) filter that takes N input points, computes their average, and produces a single output point. As the filter length expands, that is, as the parameter N increases, the smoothness of the output signal increases, whereas the sharp transitions in the data are made progressively flat.

A discrete-time signal $x[k]$ may represent a phenomenon for which the independent variable is inherently discrete, such as the number of coronavirus cases that occur per day, or it may be obtained by sampling a continuous-time signal $x(t)$, such as when a voice signal is digitized to be transmitted over the Internet (Hsu 2011).

2.3.2 *Time-Invariant and Time-Varying Systems*

The system parameters can vary in time, and the system is therefore considered time-varying, or the system parameters do not change with time, that is, the system is time-invariant.

The time-varying systems are modeled by differential equations or difference equations with time-varying coefficients. The time-invariant systems are modeled by differential equations or difference equations with constant coefficients.

The shifting property is useful to characterize a system as time-invariant. For the case of a linear system, if, for any values of t and τ,

$$y(t - \tau) = L[x(t - \tau)], \tag{2.81}$$

then the system is time-invariant.

Regarding the discrete-time system, in analogy to the precedent discussion,

$$y(n - k) = L[x(n - k)], \tag{2.82}$$

and the system response is independent of the time origin and depends only on the shape of the input.

Example A general moving average system is described by the following equation,

$$y(k) = \frac{1}{1 + N_I + N_S} \sum_{n=-N_I}^{N_S} x(k-n), \tag{2.83}$$

in which N_I and N_S refer to the limits of the window that are used to take $1+N_I+N_S$ samples of $x(k)$ around n to be averaged. The moving average system is used as a filter, which has a smoothing effect on the input signal.

2.3.3 Memory and Memoryless Systems

The systems can also be characterized as memory or memoryless. If a system output at time t, say $y(t)$, depends only on the input value at the same instant, $x(t)$, it is considered memoryless.

Memoryless systems do not depend on any past inputs and are also independent of future inputs. Therefore, the impulse response of a memoryless system is an impulse whose area depends on the system features. Causal systems, on the other hand, do not depend on future inputs.

On the other hand, if the output at the present time depends on past values of the input, for example, the interval $[t-\tau, t]$, then the system is said to have memory of length τ.

Example The system that takes the square of the input signal is memoryless, as follows:

$$y(k) = [x(k)]^2 \tag{2.84}$$

Example A system that has an inductor, with the voltage $v(t)$ as the input and the current $i(t)$ as the output,

$$i(t) = \frac{1}{L} \int_{-\infty}^{t} v(\sigma) \mathrm{d}\sigma, \tag{2.85}$$

has a memory of infinite duration, because the output depends on the input for the whole interval $(-\infty, t]$.

Analog systems with memory may be also classified as lumped or distributed. If the number of state variables necessary to describe a future output is finite, the system is lumped, but if it is infinite, the system is distributed.

2.3.4 Lumped and Distributed Systems

Most systems are composed of several elements, devices, or even sub-systems, which are interconnected to perform in a certain manner or execute a given procedure or function. Sometimes, because of the size of the system, as in the case of an electrical transmission line, the signal response is not instantaneous. It takes a finite, nonzero, amount of time for the signal to propagate through the system. This is an example of a distributed parameter system.

On the other hand, if the stimulus is propagated instantaneously throughout the system, it is characterized as a lumped parameter system. Ordinary differential equations are used to model the lumped system, while the distributed systems are modeled by partial differential equations.

2.3.5 Conditions for a Continuous Linear System

A continuous linear system has its output in some sense proportional to the input. This means that if $x(t)$ produces an output $y(t)$, that is Gabel and Roberts (1973),

$$x(t) \rightarrow y(t), \tag{2.86}$$

then it is expected that

$$\alpha x(t) \rightarrow \alpha y(t). \tag{2.87}$$

This property, shared by every linear system, is called homogeneity. But linear systems must also possess the superposition property, that is, if

$$\begin{aligned} x_1(t) &\rightarrow y_1(t) \\ x_2(t) &\rightarrow y_2(t), \end{aligned} \tag{2.88}$$

then

$$x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t). \tag{2.89}$$

This can be summarized in a single relation, that is, a system is linear if and only if

$$\alpha x_1(t) + \beta x_2(t) \rightarrow \alpha y_1(t) + \beta y_2(t), \tag{2.90}$$

in which α and β are constants.

It is usually convenient to use a compact notation to represent the transformation implied by the previous relations

$$y(t) = L[x(t)], \tag{2.91}$$

and L is considered a linear transformation if and only if

$$L[\alpha x_1(t) + \beta x_2(t)] = \alpha L[x_1(t)] + \beta L[x_2(t)]. \tag{2.92}$$

2.3.6 Conditions for a Discrete Linear System

A discrete linear system also has its output proportional to the input. This means that if $x(n)$ produces an output $y(n)$, that is Gabel and Roberts (1973),

$$x(n) \rightarrow y(n), \tag{2.93}$$

then one expects that

$$\alpha x(n) \rightarrow \alpha y(n). \tag{2.94}$$

A linear system must satisfy the property of homogeneity and also possess the superposition property; this means that

$$\begin{aligned} x_1(n) &\rightarrow y_1(n) \\ x_2(n) &\rightarrow y_2(n) \end{aligned} \tag{2.95}$$

and

$$x_1(n) + x_2(n) \rightarrow y_1(n) + y_2(n), \tag{2.96}$$

for a discrete system can be also summarized in a single relation, implying that a system is linear if and only if

$$\alpha x_1(n) + \beta x_2(n) \rightarrow \alpha y_1(n) + \beta y_2(n), \tag{2.97}$$

in which α and β are constant parameters.

The compact notation is used to represent the transformation implied by the previous relations

$$y(n) = L[x(n)], \tag{2.98}$$

and L is considered a linear transformation if and only if

$$L[\alpha x_1(n) + \beta x_2(n)] = \alpha L[x_1(n)] + \beta L[x_2(n)]. \tag{2.99}$$

2.3.7 Deterministic or Stochastic Systems

A system that always produces the same output for a given input is said to be deterministic. This means that the system parameters do not change with time, by chance, or other conditions. A system whose parameters are subject to random behavior is said to be stochastic. A stochastic system generates different outputs for a given input.

Most systems are stochastic by nature. This can be caused by aging components, by electric faults, by chance, or by the randomness of the quantum particles that interact within the devices. Some systems are designed to be stochastic, such as the random noise generator used in the laboratories.

The signal processing and communication systems are particularly stochastic. The vocal tract, for example, is a signal processing system that produces voice when excited, depending on many random parameters, which include the humor and health of the speaker, the subject of the speech, the ambient temperature, the memories associated with the matter at hand, and so on.

Communication systems are subject to random noise, stochastic fading, random delays, interference from multiple signals, scattering, and many other problems, besides components aging and design errors, to name a few.

However, the inclusion of randomness in the analysis of linear systems greatly increases the complexity. Therefore, most authors leave this question to research papers and theses. This book is only about systems with deterministic parameters.

2.4 Power and Autocorrelation

The average value of a real signal $x(t)$ is given by the normalized integral of the signal over the time interval. The normalization parameter is the time interval T:

$$\overline{x}(t) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)\mathrm{d}t \tag{2.100}$$

The instantaneous power of $x(t)$ is given by

$$p_X(t) = x^2(t). \tag{2.101}$$

This is a simplification, which usually assumes a voltage applied to a resistor of one ohm [ω].

The instantaneous power, in fact, is the product of voltage and current and is measured in joules per second [J/s] or watts [W]. It is, of course, a function of time (Scott 1960),

$$p(t) = v(t) \cdot i(t), \tag{2.102}$$

the voltage is measured in volts [V], and the current is in amperes [A].

If the signal $x(t)$ exists for the whole interval $(-\infty, +\infty)$, the total average power $\overline{P}_X$ is defined for a real signal $x(t)$ as the power dissipated in a 1-ohm resistor, when a voltage $x(t)$ is applied to this resistor (or a current $x(t)$ flows through the resistor) (Lathi 1989). Thus,

$$\overline{P}_X = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x^2(t)\mathrm{d}t. \tag{2.103}$$

From the previous definition, the unit to measure $\overline{P}_X$ corresponds to the square of the units of the signal $x(t)$ (either volt^2 or ampère^2, depending on the use of voltage or current). These units are commonly converted to watt, using a normalization by units of impedance (ohm).

It is common use to express the power in decibels (dBm), relative to the reference power of 1 mW. The power in dBm is given by the following formula (Gagliardi 1988):

$$\overline{P}_X = 10\log\left[\frac{\overline{P}_X}{1\text{ mW}}\right] \text{ dBm} \tag{2.104}$$

The total power ($\overline{P}_X$) contains two components: one constant component, because of the nonzero average value of the signal $x(t)$ ($\overline{P}_{DC}$), and an alternating component ($\overline{P}_{AC}$). The DC power of the signal is given by

$$\overline{P}_{DC} = [\overline{x}(t)]^2 \text{ W}. \tag{2.105}$$

Therefore, the AC power can be determined by removing the DC power from the total power, that is,

$$\overline{P}_{AC} = \overline{P}_X - \overline{P}_{DC} \text{ W}. \tag{2.106}$$

2.4.1 Time Autocorrelation of a Continuous Signal

The average time autocorrelation $\overline{R}_X(\tau)$, or simply autocorrelation, of a real signal $x(t)$ is defined as follows:

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)x(t+\tau)\mathrm{d}t \tag{2.107}$$

A simple change of variable allows Eq. 2.107 to be written as

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)x(t-\tau)\mathrm{d}t. \tag{2.108}$$

Example Compute the autocorrelation function for the sinusoidal signal

$$x(t) = V\sin(\omega t + \phi),$$

in which ω is the angular frequency and ϕ is the initial phase of the signal.

From the definition 2.108,

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} V\sin[\omega t + \phi]V\sin[\omega(t-\tau)+\phi]\mathrm{d}t,$$

which gives, using a trigonometric property,

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{V^2}{2T} \int_{\frac{-T}{2}}^{\frac{T}{2}} [\cos(\omega\tau) - \cos(2\omega t - \tau + 2\phi)]\,\mathrm{d}t.$$

Separating the terms and integrating

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{V^2}{2T}\cos(\omega\tau) \int_{\frac{-T}{2}}^{\frac{T}{2}} \mathrm{d}t + \lim_{T\to\infty} \frac{V^2}{2T} \int_{\frac{-T}{2}}^{\frac{T}{2}} \cos(2\omega t - \tau + 2\phi)\mathrm{d}t,$$

one obtains

$$\overline{R}_X(\tau) = \frac{V^2}{2}\cos(\omega\tau),$$

because the mean value of the cosine is zero.

From Eqs. 2.107 and 2.108, it follows that $\overline{R}_X(\tau)$ is an even function of τ, and thus Lathi (1989)

$$\overline{R}_X(-\tau) = \overline{R}_X(\tau). \tag{2.109}$$

From the definition of autocorrelation and power, one obtains

$$\overline{P}_X = \overline{R}_X(0) \tag{2.110}$$

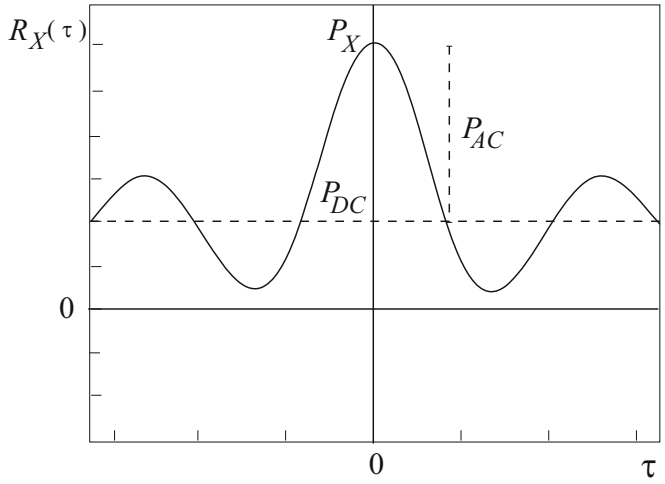


Fig. 2.30 Autocorrelation and its parameters

and

$$\overline{P}_{\mathrm{DC}} = \overline{R}_X(\infty), \tag{2.111}$$

that is, from its autocorrelation function, it is possible to obtain information about the power of a signal. The AC power can be obtained as

$$P_{\mathrm{AC}} = \overline{P}_X - \overline{P}_{\mathrm{DC}} = \overline{R}_X(0) - \overline{R}_X(\infty). \tag{2.112}$$

The parameters derived from the autocorrelation can be seen in Fig. 2.30.

Example Considering that the DC power is zero for a sinusoidal signal, the AC power can be computed from the autocorrelation function as follows:

$$P_{\mathrm{AC}} = \overline{R}_X(0) = \frac{V^2}{2}\cos(\omega \cdot 0) = \frac{V^2}{2}$$

2.5 Problems

1. Draw the graphs for the following discrete signals:
   (a) $x(k) = ku(k)$.
   (b) $x(k) = \alpha^k u(k)$ for α larger, smaller, or equal to zero.
   (c) $x(k) = \alpha^{-k} u(k)$ for α larger, smaller, or equal to zero.

(d) $x(k) = (-\alpha)^k u(k)$.
(e) $x(k) = ku(k) - (k-1)u(k-1)$.
(f) $x(k) = \cos \omega k$.

2. Sketch the graphs for the following signals:

(a) $x(t) = \frac{\mathrm{d}}{\mathrm{d}t}[tu(t)]$.
(b) $x(t) = e^{-\alpha t}u(t) - e^{-\beta t}u(t)$ for diverse values of α and β.
(c) $x(t) = u(\cos \pi t)$.
(d) $x(t) = \delta(\sin \pi t)$.
(e) $x(t) = tu(t) - (t-T)u(t-T)$.
(f) $x(t) = \frac{\mathrm{d}}{\mathrm{d}t}[u(\cos \pi t)]$.

3. Using the following property of the step function, demonstrate the fundamental property of the impulse function:

$$\int_{-\infty}^{\infty} x(\tau)\mathrm{u}(t-\tau)\mathrm{d}\tau = \int_{0}^{t} x(\tau)\mathrm{d}\tau$$

4. Compute the derivative of the gate function. Draw the graphic:

$$g(t) = \mathrm{u}(t + T/2) - \mathrm{u}(t - T/2)$$

5. Determine the voltage across the terminals of an inductor that is fed by a current source $i(t) = I\mathrm{u}(t)$.
6. Determine the voltage across the terminals of an inductor that is fed by a current source $i(t) = I\delta(t)$.
7. Demonstrate the fundamental property of the doublet function, Formula 2.40, using integral by parts.
8. Determine the voltage $v(t)$ across the terminals of the semiconductor diode, using the following equation for the current $i(t)$:

$$i(t) = I_0\left[e^{\frac{qv(t)}{kT}} - 1\right]$$

9. Write the equation for the growing sinusoidal function assuming that the signal starts at time $t = T$ seconds. Use the unit step function:

$$y(t) = e^{\alpha t}\cos(\omega t + \phi)$$

10. Obtain the derivative of the following version of the decaying sinusoidal function:

$$y(t) = e^{-\alpha t}\cos(\omega t + \phi)\,\mathrm{u}(t)$$

11. Demonstrate the fundamental property of the doublet function,

$$\int_{-\infty}^{\infty} x(t)\delta'(t)\mathrm{d}t = -\int_{-\infty}^{\infty} x'(t)\delta(t)\mathrm{d}t = -x'(0).$$

12. Demonstrate the following result, using the definition of the impulse function:

$$\delta(\alpha t) = \frac{1}{|\alpha|}\delta(t).$$

13. Using the previous result, demonstrate the following property of the impulse function:

$$\delta(-t) = \delta(t)$$

14. Show that the doublet generalized function is antisymmetric, that is,

$$\delta'(-t) = -\delta'(t).$$

15. Consider that $f(t) = t^2 + t - 2$, and determine the formulas for $\mathrm{u}[f(t)]$ and $\delta[f(t)]$.

Chapter 3
System Response and Convolution



Mathematics is the most beautiful and most powerful creation of the human spirit.

– Stefan Banach

Convolution is an important concept in Engineering, because it can be used to determine the output of a system for a given input signal, to compute the probability distribution of the sum of random variables, and to determine the effect of channel noise in the probability of error of a received signal, among other important applications.

3.1 Discrete Impulse Response

Figure 3.1 illustrates a discrete linear system that has an impulse response $\{h_n\}$. The system is excited by the input sequence $\{x_n\}$ to produce the output sequence $\{y_n\}$. The impulse response implies that $\delta(n) \rightarrow h(n)$, that is, the application of a Kronecker impulse at the input of the system gives as output the sequence $h(n)$.

The discrete impulse function can be used to test discrete systems, such as digital filters, to verify if the response corresponds to the design specifications.

3.2 Discrete Convolution

The discrete convolution process can be seen graphically in the set of Figs. 3.2, 3.3, and 3.4. The procedure is also equivalent to a computer algorithm that can be written directly in any mathematical language, such as Python, Fortran, C++, or Maple.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_3

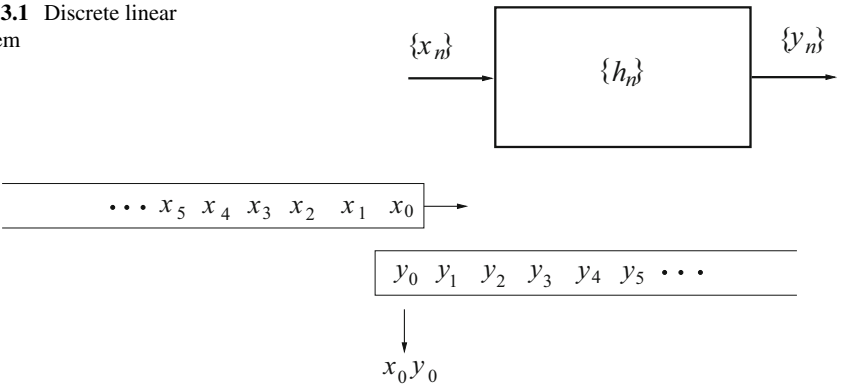


Fig. 3.1 Discrete linear system

Fig. 3.2 Convolution as a computer algorithm, $n = 0$

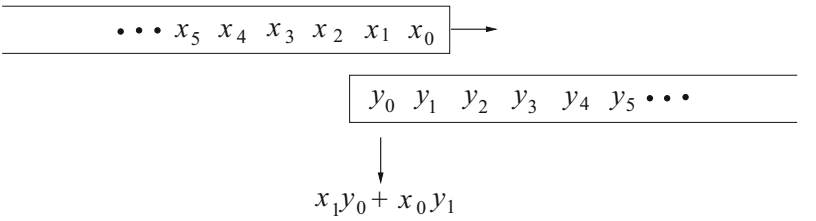


Fig. 3.3 Convolution as a computer algorithm, $n = 1$

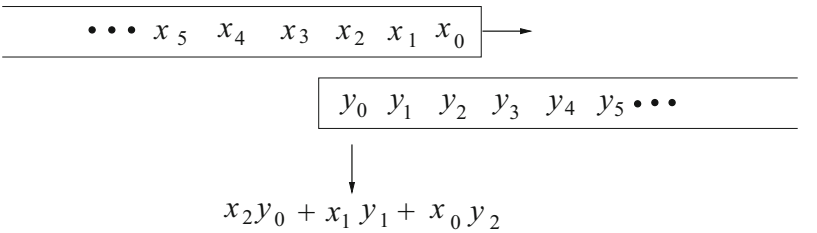


Fig. 3.4 Convolution as a computer algorithm, $n = 2$

The sequence $\{x_n\} = \{x_0, x_1, x_2, \ldots, x_n, \ldots\}$ that can be expressed as

$$\{x_n\} = \sum_{i=0}^{\infty} x_i \delta_{n-i} \tag{3.1}$$

is convolved with the sequence $\{y_n\} = \{y_0, y_1, x_2, \ldots, y_n, \ldots\}$, which can be written as

$$\{y_n\} = \sum_{i=0}^{\infty} y_i \delta_{n-i}, \tag{3.2}$$

to produce the sequence $\{z_n\} = \{z_0, z_1, z_2, \ldots, z_n, \ldots\}$ that can be expressed as

$$\{z_n\} = \sum_{i=0}^{\infty} z_i \delta_{n-i}. \tag{3.3}$$

In the previous equations, $\delta_n = \delta(n)$ is the discrete impulse or Kronecker impulse function. Observe that the impulses inside the summations merely indicate the positions where the signals have values different from zero.

As can be deducted by inductive reasoning, the convolution between $x(n)$ and $y(n)$, $z(n) = x(n) * y(n)$ gives the following sequence of values:

$$\begin{aligned} z(0) &= x(0)y(0) \\ z(1) &= x(1)y(0) + x(0)y(1) \\ z(2) &= x(2)y(0) + x(1)y(1) + x(0)y(2) \\ &\vdots \\ z(n) &= \sum_{k=0}^{\infty} x(n-k)y(k) \end{aligned} \tag{3.4}$$

The preceding formula can be generalized to include the whole set of integers $\mathbb{Z}$. Therefore, the discrete convolution of two functions $x(n)$ and $y(n)$ is defined by the following summation,

$$z(n) = x(n) * y(n) = \sum_{k=-\infty}^{\infty} x(k)y(n-k), \tag{3.5}$$

which is equivalent to the following result, as can be verified by a change of variables:

$$z(n) = y(n) * x(n) = \sum_{k=-\infty}^{\infty} x(n-k)y(k) \tag{3.6}$$

To understand the convolution operation, it is necessary to plot each function. Then, one of the functions has its plot reflected across the $k = 0$ axis. This function is shifted right by n, as $n \in \mathbb{Z}$ increases. The point-wise product of the resulting plots is computed, and all of the terms are summed.

3.2.1 Discrete Convolution Properties

The discrete convolution is a linear operation. It is also a commutative operation, because

$$z(n) = x(n) * y(n) = y(n) * x(n), \tag{3.7}$$

as can be verified with a change of variables. The convolution operation is also associative, that is,

$$z(n) * [x(n) * y(n)] = [z(n) * y(n)] * x(n). \tag{3.8}$$

The operation of convolution is distributive over the addition operation, that is,

$$z(n) * [x(n) + y(n)] = [z(n) * y(n)] + [z(n) * x(n)]. \tag{3.9}$$

If the set $\{x_n\}$ has N elements and $\{y_n\}$ has M elements, the set that results from the convolution operation, $\{z_n\}$, will have $N + M - 1$ elements.

The difference operator, $D[x(n)] = x(n+1) - x(n)$, satisfies the relationship

$$D[z(n)] = D[x(n) * y(n)] = D[x(n)] * y(n) = x(n) * D[y(n)]. \tag{3.10}$$

The discrete convolution formula can be simplified, regarding the summation limits if one takes into consideration the causality of $x(n)$ and $y(n)$. The following properties are useful to simplify the computation.

If $x(n)$ is general and $y(n)$ is general, then

$$z(n) = x(n) * y(n) = \sum_{k=-\infty}^{\infty} x(k)y(n-k). \tag{3.11}$$

If $x(n)$ is causal and $y(n)$ is general, then

$$z(n) = x(n) * y(n) = \sum_{k=0}^{\infty} x(k)y(n-k), x(k) = 0, \text{ for } k < 0. \tag{3.12}$$

If $x(n)$ is general and $y(n)$ is causal, then

$$z(n) = x(n) * y(n) = \sum_{k=-\infty}^{K} x(k)y(n-k), y(k) = 0, \text{ for } k < 0. \tag{3.13}$$

Finally, if $x(n)$ is causal and $y(n)$ is causal, then

$$z(n) = x(n) * y(n) = \sum_{k=0}^{K} x(k)y(n-k), x(k) = 0, y(k) = 0, \text{ for } k < 0. \quad (3.14)$$

The convolution of a signal $x(n)$ with the unit step function $\mathrm{u}(n)$ produces the summation of the signal samples, because the inverted and shifted discrete unit step function $\mathrm{u}(n-k)$ is equal to one from minus infinity to n, as follows:

$$\begin{aligned} x(n) * \mathrm{u}(n) &= \sum_{k=-\infty}^{\infty} x(k)\mathrm{u}(n-k) \\ &= \sum_{k=-\infty}^{n} x(k) \end{aligned} \quad (3.15)$$

The convolution of a signal $x(n)$ with the discrete impulse function $\delta(n)$ produces the same signal:

$$\begin{aligned} x(n) * \delta(n) &= \sum_{k=-\infty}^{\infty} x(k)\delta(n-k) \\ &= x(n) \sum_{k=-\infty}^{\infty} \delta(n-k) \\ &= x(n) \end{aligned} \quad (3.16)$$

Also, it is possible to verify that $x(n)*\alpha\delta(n) = \alpha x(n)$ and $x(n)*\delta(n-k) = x(n-k)$.

Example Suppose that the impulse response for a discrete-time system with parameter α is given by

$$h(n) = \alpha^n \, \mathrm{u}(n), \quad (3.17)$$

in which $0 < \alpha < 1$, and consider the response of this linear system to an input signal that is also a geometrical function,

$$x(n) = \beta^n \, \mathrm{u}(n), \quad (3.18)$$

in which $0 < \beta < 1$.

The output of the linear system for this input is given by the convolution of the impulse response with the input signal, as expressed in the following formula:

$$y(n) = h(n) * x(n) \quad (3.19)$$

To compute this convolution operation, one uses the convolution definition from Eq. 3.5, substituting the appropriate terms in the formula

$$y(n) = \sum_{k=-\infty}^{\infty} \alpha^k \, \mathrm{u}(k) \beta^{n-k} \, \mathrm{u}(n-k). \tag{3.20}$$

Appendix D contains a set of formulas and inequalities that can help compute summations.

The properties of the unit step function can be used to simplify this sum, giving $y(n) = 0$ for $n < 0$ and

$$y(n) = \sum_{k=0}^{n} (\alpha\beta)^k \text{ for } n \geq 0. \tag{3.21}$$

Therefore, provided that $\alpha\beta \neq 0$, one obtains

$$y(n) = \begin{cases} 0 & \text{if } n < 0 \\ \frac{1-(\alpha\beta)^{n+1}}{1-\alpha\beta} & \textit{and if } n \geq 0. \end{cases} \tag{3.22}$$

It is possible to visualize the convolution computation as a graphical process in the sequence of Figs. 3.5 and 3.6, as the reversed version of the input signal shifts to the right, with increasing n.

The final result of the convolution process, which produces the output $y(n)$, is seen in Fig. 3.7. Note that the resulting function reflects the behavior of the convolution product, which increases at first, because large terms are multiplied and summed, and then decreases, as the sum of the product of the two discrete functions reduces.

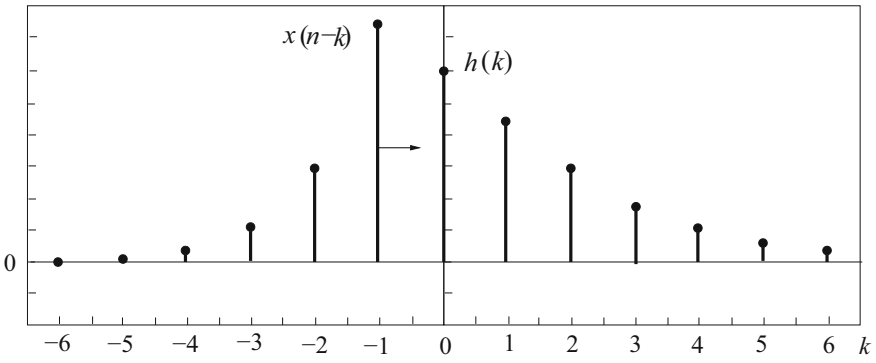


Fig. 3.5 Digital convolution as a graphical process, $n = -1$

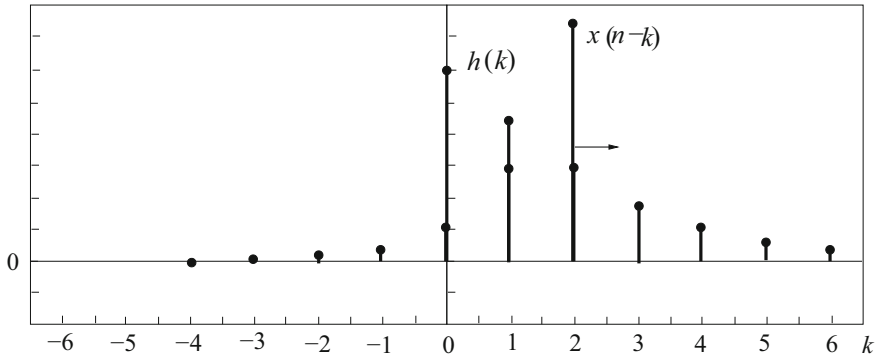


Fig. 3.6 Digital convolution as a graphical process, $n = 2$

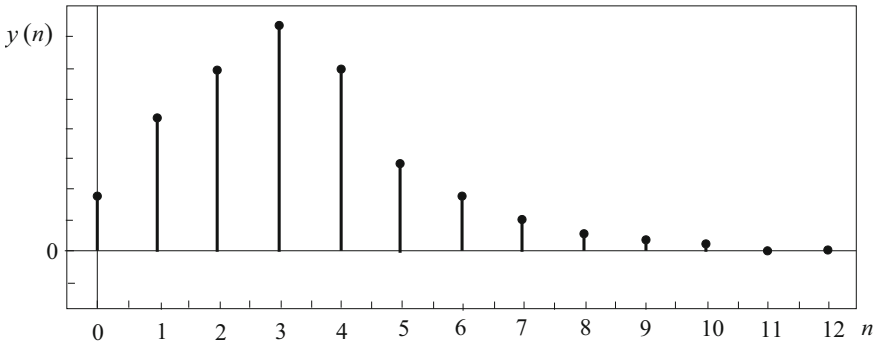


Fig. 3.7 Final result of the digital convolution process

Example The response of a discrete linear system, with discrete impulse response $h(n) = \mathrm{u}(n) - \mathrm{u}(n - N)$, to the input $x(n) = \alpha^n \mathrm{u}(n)$. This system represents a discrete-time window with a width of N samples (Oppenheim and Schafer 1989).

The response of the discrete system is given by the convolution

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k),$$

which is the sum over all k of the product $x(k)h(n-k)$. This yields

$$y(n) = \sum_{k=-\infty}^{\infty} \alpha^k \mathrm{u}(k)[\mathrm{u}(n-k) - \mathrm{u}(n-k-N)],$$

which is different from zero only for $k \geq 0$; thus,

$$y(n) = \sum_{k=0}^{\infty} \alpha^k [\mathrm{u}(n-k) - \mathrm{u}(n-k-N)]. \tag{3.23}$$

This summation can be divided into two parts: First, for $0 \leq n \leq N-1$, Eq. 3.23 is the sum of $n+1$ terms of a geometric series, which can be expressed in closed form as

$$y(n) = \sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}.$$

This result can be obtained from the list of formulas presented in Appendix D.

On the other hand, for the remaining interval, $1+n-N < k \leq n$, for $N-1 < n$, the summation yields

$$y(n) = \sum_{k=1+n-N}^{n} \alpha^k = \frac{\alpha^{1+n-N} - \alpha^{n+1}}{1-\alpha} = \alpha^{1+n-N} \left[\frac{1-\alpha^N}{1-\alpha} \right].$$

The intermediary steps toward the convolution are shown in Figs. 3.8 and 3.9, as the inverted version of the impulse response $h(n-k)$ is shifted to the right and passes through the input signal $x(k)$.

Finally, the complete response $y(n)$ can be written as

$$y(n) = \begin{cases} 0 & \text{if } n < 0, \\ \frac{1-\alpha^{n+1}}{1-\alpha} & \text{if } 0 \leq n \leq N-1, \\ \alpha^{1+n-N} \left[\frac{1-\alpha^N}{1-\alpha} \right] & N-1 \leq n. \end{cases}$$

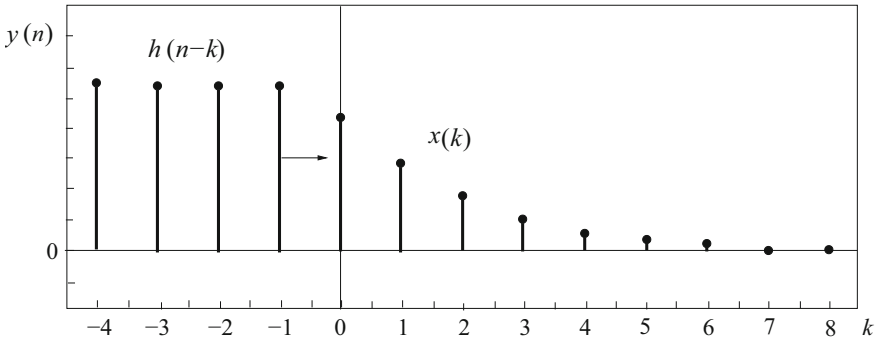


Fig. 3.8 Response of a discrete system, $n < 0$

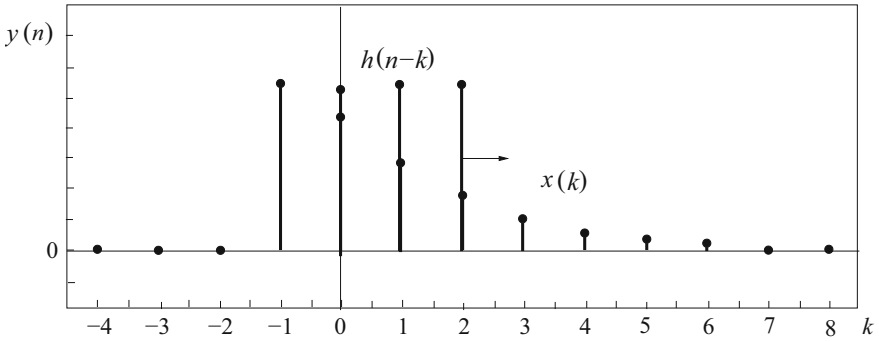


Fig. 3.9 Response of a discrete system, $0 \leq n \leq N-1$

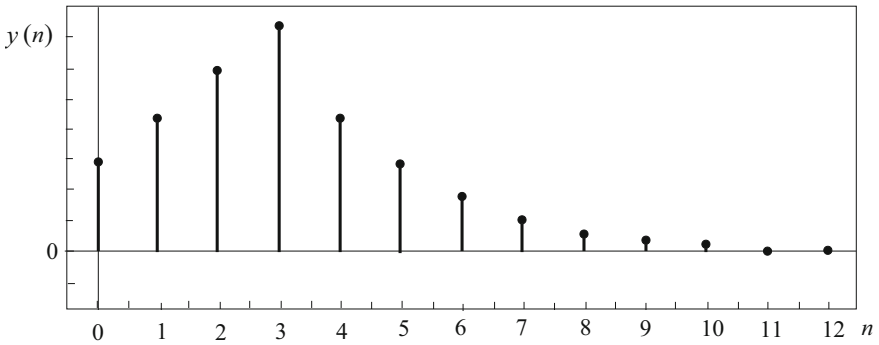


Fig. 3.10 Response of a discrete system

The final result is seen in Fig. 3.10. Note that the response $y(n)$ is spread and occupies a number of points that are the sum of the points of $x(n)$ and $h(n)$, minus one point.

This example illustrates the analytical convolution computation when both the input signal and the impulse response are given by known formulas. In this case, it is possible to resort to tabulated geometric series. The example also draws attention to the importance of sketching the functions as careful as possible to correctly determine the limits on the summation.

3.3 Algorithm for Discrete Deconvolution

Sometimes, it is necessary to obtain the input sequence, given the output sequence, for a certain transfer function. It is also interesting, and useful in the semiconductor industry, to discover the transfer function of a determined digital circuit, given the set of outputs in response to a given input sequence.

This can be computed using the set of algebraic Eqs. 3.4. Consider that $y(n)$ is the output of a digital filter defined by the transfer function $h(n)$, given the input $x(n)$, as follows:

$$\begin{aligned} y(0) &= x(0)h(0) \\ y(1) &= x(1)h(0) + x(0)h(1) \\ y(2) &= x(2)h(0) + x(1)h(1) + x(0)h(2) \\ &\vdots \\ y(n) &= \sum_{k=0}^{\infty} x(n-k)h(k) \end{aligned} \tag{3.24}$$

Suppose one is interested in discovering the transfer function of the filter. The procedure is the following: First, invert the term $n = 0$ from the set of Eqs. 3.25 to obtain

$$h(0) = \frac{y(0)}{x(0)}.$$

Then, solve the second equation of the set for $h(1)$ to get

$$h(1) = \frac{y(1) - x(1)h(0)}{x(0)},$$

and substitute the value of $h(0)$ obtained in the first step of the process to achieve

$$h(1) = \frac{y(1) - \frac{y(0)x(1)}{x(0)}}{x(0)}.$$

Continuing the iterative process, it is possible to obtain all the terms of the sequence $h(n)$ which defines the transfer function of the digital filter. This is all that is required to design the required circuit.

3.3.1 Moving Average Filter

As can be observed, the window $h(n)$ has the property of smoothing the original signal $x(n)$. This is a useful feature to soften the borders of a given signal, for example, to filter a video signal that is affected by impulsive noise.

In fact, a scaled version of this impulse response,

$$l(n) = \frac{h(n)}{N} = \frac{\mathrm{u}(n) - \mathrm{u}(n-N)}{N}, \tag{3.25}$$

is a finite impulse response (FIR) filter, called moving average filter, that performs the following functions:

- The filter takes N input points, computes their average, and produces a single output point.
- Because of the computation involved, the filter introduces a finite delay.
- The equation works as a low-pass filter.

The difference equation for an N-point discrete-time moving average filter with input represented by $x(k)$ and the averaged output by the signal $y(k)$ is the following:

$$y(n) = \frac{1}{N} \sum_{i=0}^{N-1} x(n-i) \tag{3.26}$$

Example The unit step response of the moving average filter is given by

$$y(n) = \mathrm{u}(n) * l(n) = \mathrm{u}(n) * \left[\frac{\mathrm{u}(n) - \mathrm{u}(n-N)}{N} \right],$$

which produces

$$y(n) = \frac{\mathrm{r}(n) - \mathrm{r}(n-N)}{N}, \tag{3.27}$$

which is equivalent to

$$y(n) = \begin{cases} 0 & \text{if } n < 0, \\ \frac{n+1}{N} & \text{if } 0 \leq n < N, \\ 1 & \text{if } n \geq N. \end{cases} \tag{3.28}$$

Figure 3.11 sketches the unit step response of an FIR filter, when a Heaviside step function is applied to the input, for $N = 5$. The function is named after Oliver Heaviside (1850–1925), an English mathematician and physicist that applied complex numbers to circuit analysis, invented a technique equivalent to the Laplace transform for solving differential equations, developed the Vector Calculus, and expressed the equations derived by James Clerk Maxwell (1831–1879) in the way they are used today (Alencar 2018).

3.4 Continuous Impulse Response

It is possible to obtain an analogous solution for the response of a continuous linear system using the convolution operation, as the one obtained for the discrete system.

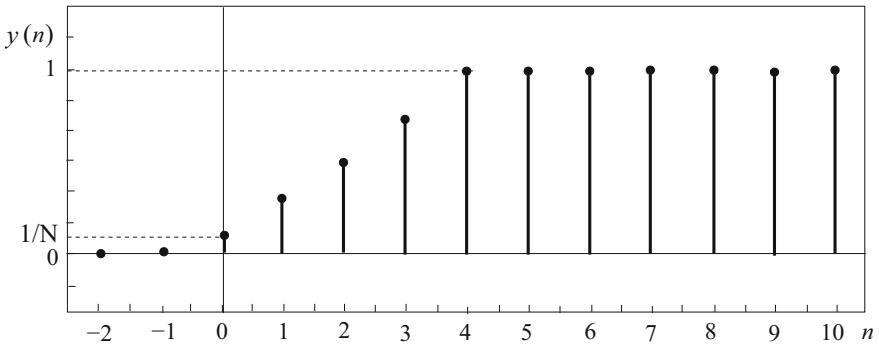


Fig. 3.11 Unit step response of a moving average FIR filter

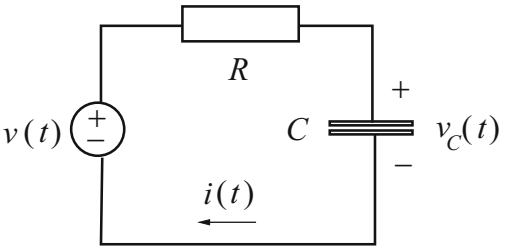


Fig. 3.12 Impulse response of an RC circuit

The convolution operation, in this case, involves functions defined in the set of real numbers $\mathbb{R}$, and it is important to take care with the definition of integration limits. It is important to clarify the concept of impulse response with an example.

Example Consider again the circuit presented in Fig. 3.12, which is reproduced here for convenience, in which the input signal is the voltage $v(t) = V\mathrm{u}(t)$ and the output is the voltage across the capacitor terminals $v_C(t)$.

The voltage step $v_C(t)$ response is given by

$$v_C(t) = V\left[1 - e^{-\frac{t}{RC}}\right]\mathrm{u}(t). \tag{3.29}$$

Because this is the response to a voltage step and, by the properties of a linear circuit the derivative of the input, $v'(t) = V\delta(t)$, produces the derivative of the output, the impulse response of the circuit is given by

$$\frac{\mathrm{d}v_C(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left\{V\left[1 - e^{-\frac{t}{RC}}\right]\mathrm{u}(t)\right\}, \tag{3.30}$$

which, using the multiplication rule for the derivative, results in

$$\frac{\mathrm{d}v_C(t)}{\mathrm{d}t} = V\left[\frac{1}{RC}e^{-\frac{t}{RC}}\right]\mathrm{u}(t) + V\left[1 - e^{-\frac{t}{RC}}\right]\delta(t), \tag{3.31}$$

but by the properties of the impulse function, the second term of the equation is null, and one finally obtains the impulse response of this system in terms of voltage,

$$h(t) = \frac{\mathrm{d}v_C(t)}{\mathrm{d}t} = \frac{V}{RC} e^{-\frac{t}{RC}} \mathrm{u}(t). \tag{3.32}$$

3.4.1 Impulse Response Using Convolution

Consider the general analog linear system shown in Fig. 3.13, which has impulse response $h(t)$ and provides a response $y(t)$ to the input $x(t)$.

The impulse response $h(t)$ is defined as the output that results from the application of an impulse $\delta(t)$ to the system. The impulse excites the energy-storing devices of the system and produces a response

$$h(t) = L[\delta(t)], \tag{3.33}$$

in which $L[\cdot]$ represents the linear operation executed by the system.

For a linear time-invariant system, the application of an impulse of area $x(\tau)$ at time $t = \tau$, that is, $x(\tau)\delta(t - \tau)$, produces

$$x(\tau)h(t - \tau) = L[x(\tau)\delta(t - \tau)].$$

The complete response $y(t)$ is the sum of all the infinitesimal contributions, which gives, in the limit,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)\mathrm{d}\tau, \tag{3.34}$$

which is called the continuous convolution of $x(t)$ and $h(t)$. The notation

$$y(t) = x(t) * h(t) \tag{3.35}$$

is a compact way of expressing the same idea.

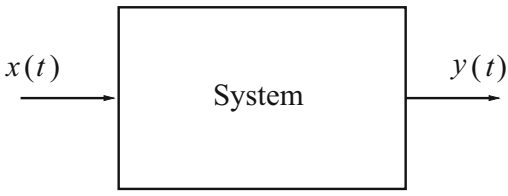


Fig. 3.13 Continuous linear system

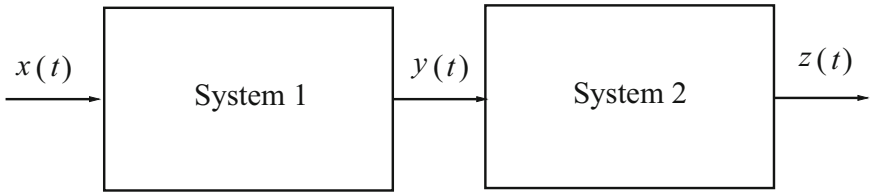


Fig. 3.14 Cascade of continuous linear systems

Example Suppose two linear systems are mounted in cascade, as shown in Fig. 3.14, in which the first system has impulse response $h(t)$ and the second system has impulse response $g(t)$. The overall response of the composite system is given by

$$z(t) = y(t) * g(t) = [x(t) * h(t)] * g(t) = x(t) * [h(t) * g(t)], \tag{3.36}$$

because the convolution operation is associative. Therefore, it is equivalent to a linear system whose impulse response is the convolution of the responses of both systems.

Example Compute the unit step response of the linear system, supposing that the impulse response of the system is given by the following formula:

$$h(t) = \frac{V}{RC} e^{-\frac{t}{RC}} \mathrm{u}(t) \tag{3.37}$$

From Formula 3.34, the unit step response is given by

$$y(t) = \int_{-\infty}^{\infty} \mathrm{u}(t-\tau) \frac{V}{RC} e^{-\frac{\tau}{RC}} \mathrm{u}(\tau) \mathrm{d}\tau. \tag{3.38}$$

Because the product of the unit step functions $\mathrm{u}(t-\tau)\mathrm{u}(\tau)$ is one only in the interval $0 \leq \tau \leq t$, it is possible to simplify the equation to

$$y(t) = \int_0^t \frac{V}{RC} e^{-\frac{\tau}{RC}} \mathrm{d}\tau = \frac{V}{RC} \left[-RC e^{-\frac{\tau}{RC}} \right]_0^t = V \left[1 - e^{-\frac{1}{RC}} \right], \tag{3.39}$$

which is the unit step response, obtained from the impulse response of the circuit.

3.5 Continuous Convolution

As discussed, the behavior of a linear, continuous-time, time-invariant system, defined by the impulse response $h(t)$, with input signal $x(t)$ and output signal $y(t)$

is described by the convolution integral,

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(t-\tau) \cdot h(\tau)\, \mathrm{d}\tau = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, \mathrm{d}\tau. \tag{3.40}$$

The system response, $y(t)$, is proportional to a weighted average of the input signal $x(\tau)$. This process is made clear with an example.

Example Suppose the signal $x(t) = \mathrm{u}(t)$ is the input to a system with impulse response given by $h(t) = \mathrm{u}(t)$. For this case, the output is

$$y(t) = \mathrm{u}(t) * \mathrm{u}(t) = \int_{-\infty}^{\infty} \mathrm{u}(t-\tau) \cdot \mathrm{u}(\tau)\, \mathrm{d}\tau.$$

As mentioned, when computing the convolution, it is important to observe carefully the definition of the limits. In this case, the limits are explicitly defined by the unit step functions.

Because the product $\mathrm{u}(t-\tau) \cdot \mathrm{u}(\tau)$ is zero for $t < 0$ and equals one for $t \geq 0$, as seen in Figs. 3.15 and 3.16, the integral can be put into the following form,

$$y(t) = \int_{0}^{t} \mathrm{u}(\tau)\, \mathrm{d}\tau = \mathrm{r}(t),$$

by the definition of the ramp function. Therefore, the integrator has impulse response $h(t) = \mathrm{u}(t)$.

This result can be observed in Fig. 3.17. Note that this result is of a general nature, that is, the convolution of any signal with the unit step function gives the integral of

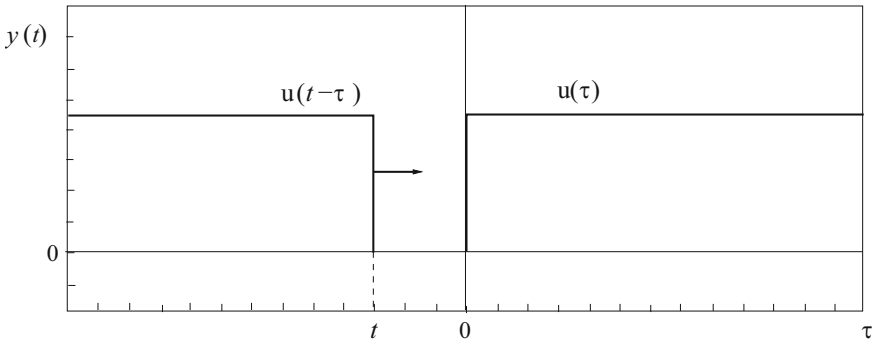


Fig. 3.15 Graphical convolution for unit step functions, $t < 0$

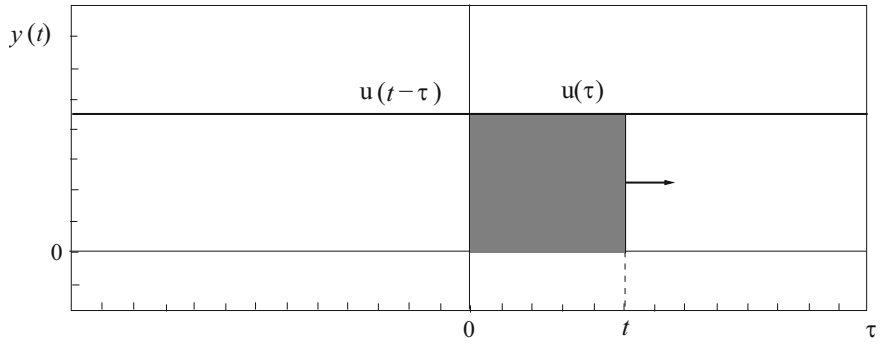


Fig. 3.16 Graphical convolution for unit step functions, $t \geq 0$

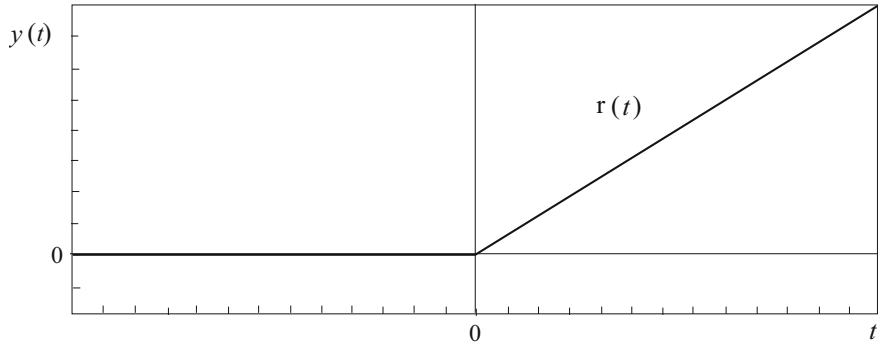


Fig. 3.17 Final result of the convolution of unit step functions

that signal:

$$x(t) * \mathrm{u}(t) = \int_{-\infty}^{\infty} x(\tau)\mathrm{u}(t-\tau)\,\mathrm{d}\tau = \int_{0}^{t} x(\tau)\,\mathrm{d}\tau \tag{3.41}$$

Example A differentiator has impulse response $h(t) = \delta'(t)$, as can be demonstrated in the following,

$$y(t) = x(t) * \delta'(t) = \int_{-\infty}^{\infty} x(\tau)\cdot\delta'(t-\tau)\,\mathrm{d}\tau \tag{3.42}$$

$$= \int_{-\infty}^{\infty} x'(\tau)\cdot\delta(t-\tau)\,\mathrm{d}\tau = x'(t),$$

taking into account the fundamental property of the impulse function.

3.5.1 Linear System with an Exponential Impulse Response

As another application of the convolution operation, consider that a time-invariant linear system, with impulse response $h(t) = e^{-\alpha t}\mathrm{u}(t)$, is excited with an input signal $x(t) = e^{-\beta t}\mathrm{u}(t)$. This type of impulse response is typical of linear systems that have memory, or energy-storing devices.

The weighting function is an inverted version of the impulse response, $h(\tau)$, shifted by the amount of time t, as shown in Fig. 3.18. It is important to consider the delimiting effect of the unit step functions to define the region of integration.

As depicted in Fig. 3.19, as time goes by, the weighting function is displaced and emphasizes different parts of the input function.

The output signal is then obtained by computing the convolution integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\,\mathrm{d}\tau, \tag{3.43}$$

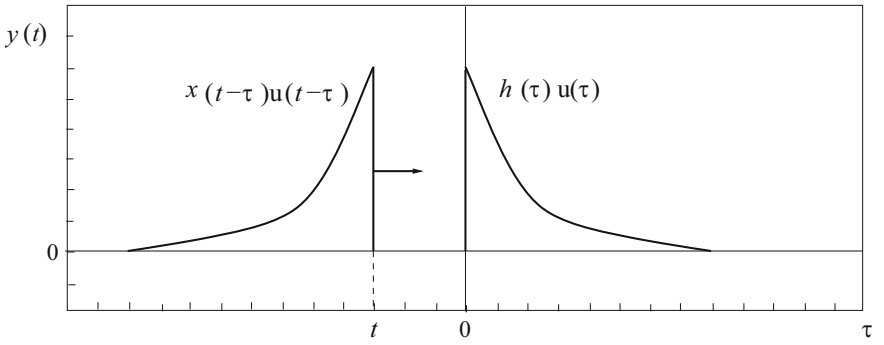


Fig. 3.18 Convolution as a graphical process, $t < 0$

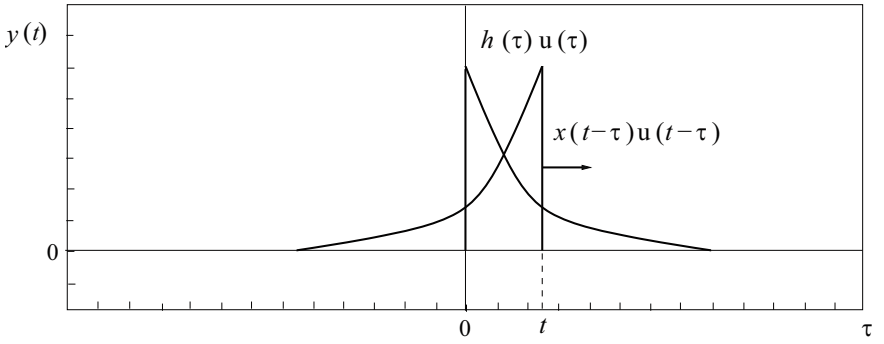


Fig. 3.19 Convolution as a graphical process, $t > 0$

with the limits defined by the unit step functions, as follows,

$$y(t) = \int_{-\infty}^{\infty} e^{-\beta\tau} \mathrm{u}(\tau) e^{-\alpha(t-\tau)} \mathrm{u}(t-\tau)\, \mathrm{d}\tau,$$

in which the product $\mathrm{u}(\tau)\mathrm{u}(t-\tau) = 1$ for $0 \leq \tau \leq t$ and zero otherwise. Therefore,

$$y(t) = \int_0^t e^{-\beta\tau} e^{-\alpha(t-\tau)}\, \mathrm{d}\tau = e^{-\alpha t} \int_0^t e^{(\alpha-\beta)\tau}\, \mathrm{d}\tau.$$

The integral can be obtained, as follows, for $\alpha \neq \beta$

$$y(t) = e^{-\alpha t} \left[\frac{e^{(\alpha-\beta)\tau}}{\alpha-\beta} \right]_0^t = e^{-\alpha t} \left[\frac{e^{(\alpha-\beta)t}}{\alpha-\beta} - \frac{1}{\alpha-\beta} \right].$$

Finally,

$$y(t) = \frac{e^{-\beta t} - e^{-\alpha t}}{\alpha - \beta}.$$

If $\alpha = \beta$, the integral

$$y(t) = e^{-\alpha t} \int_0^t e^{(\alpha-\beta)\tau}\, \mathrm{d}\tau = e^{-\alpha t} \int_0^t \mathrm{d}\tau = t e^{-\alpha t}.$$

The final result is observed in Fig. 3.20. Note that the output signal first increases from zero to a maximum value, which can be determined deriving the resulting

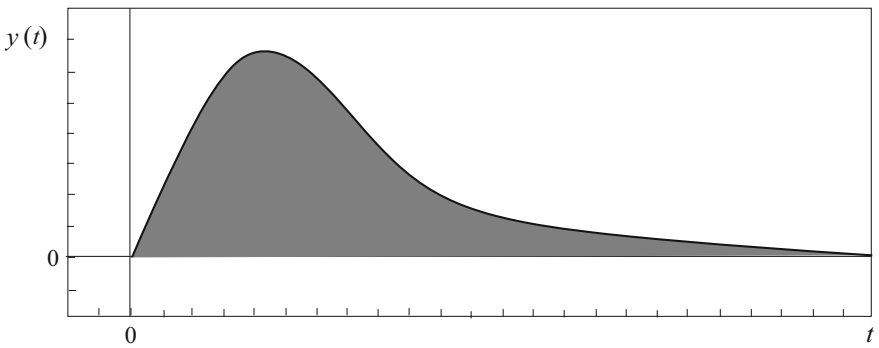


Fig. 3.20 Final result of the convolution

formula and equating the derivative to zero. The second derivative is negative, indicating that there is a maximum value at that point.

For example, considering the case in which $\alpha = \beta$, one obtains

$$y'(t) = e^{-\alpha t} - \alpha t e^{-\alpha t} = 0,$$

which gives

$$t = \frac{1}{\alpha}.$$

3.5.2 Continuous Convolution Properties

The continuous convolution formula can be simplified, regarding the integration limits taking into consideration the causality of the operands $x(t)$ and $y(t)$. The following properties are useful to simplify the calculation:

If $x(t)$ is general and $y(t)$ is general, then their convolution $z(t) = x(t) * y(t)$ is

$$z(t) = \int_{-\infty}^{\infty} x(\tau) \cdot y(t - \tau)\, \mathrm{d}\tau. \tag{3.44}$$

If $x(t)$ is causal and $y(t)$ is general, then

$$z(t) = \int_{0}^{\infty} x(\tau) \cdot y(t - \tau)\, \mathrm{d}\tau. \tag{3.45}$$

If $x(t)$ is general and $y(t)$ is causal, then

$$z(t) = \int_{-\infty}^{t} x(\tau) \cdot y(t - \tau)\, \mathrm{d}\tau. \tag{3.46}$$

Finally, if $x(t)$ is causal and $y(t)$ is causal, then

$$z(t) = \int_{0}^{t} x(\tau) \cdot y(t - \tau)\, \mathrm{d}\tau. \tag{3.47}$$

Example The convolution of a signal $x(t)$ with the unit step function $\mathrm{u}(t)$ produces the integral of the signal. Making $y(t) = \mathrm{u}(t)$ in the convolution operation gives

$$z(t) = x(t) * \mathrm{u}(t) = \int_{-\infty}^{\infty} x(\tau) \cdot \mathrm{u}(t - \tau)\, \mathrm{d}\tau,$$

which, by Property 3.46, results in

$$z(t) = \int_{-\infty}^{t} x(\tau)\, \mathrm{d}\tau.$$

Example The convolution of a signal $x(t)$ with the impulse function $\delta(t)$ produces the same signal, as follows,

$$z(t) = x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \cdot \delta(t - \tau)\, \mathrm{d}\tau = x(t),$$

by the integration property of the impulse function.

3.6 Problems

1. The unit step response of a linear system is given by $y(t) = [\cos \omega t]\mathrm{u}(t)$. Find the impulse response.
2. Compute the response of a linear system described by $h(t) = e^{-\beta}u(t)$, $\beta > 0$, to the input signals $u(t)$ and $r(t)$. Draw the graphics.
3. An integrator is described by the equation $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$. Find the impulse response of the integrator and verify if the system is stable.
4. For a periodic input $x(t)$ with period N, show that the output of a discrete linear time-invariant system $y(t)$ is also periodic, with the same period.
5. Verify if the system with an impulse response $h(n) = \alpha^n u(n)$ is causal and for what values of α it is stable.
6. For a system described by the equation $y(n) - \frac{1}{2}y(n-1) = x(n)$, com $y(-1) = 0$, find the response $y(n)$ for the following inputs. Draw the graphics:

   (a) $x(n) = (\frac{1}{2})^n u(n)$.
   (b) $x(n) = (\frac{1}{3})^n u(n)$.

7. Design a linear system described by the equation $y(n) - \alpha y(n-1) = x(n)$. Obtain the system impulse response.

8. Make the drawing of a linear system described by the equation $y(n) - \alpha y(n - 1) - \beta y(n - 2) = x(n)$. Determine the impulse response of the system.
9. Compute the response of a linear system that is described by the equation $h(t) = e^{-\beta}u(t)$, $\beta > 0$, to the input signals $u(t)$ e $r(t)$. Make the graphics.
10. For the system described by the equation $y(n)-y(n-1) = x(n)$, with $y(-1) = 0$, find the response $y(n)$ for the following inputs. Sketch the plots:

    (a) $x(n) = (\frac{1}{2})^n u(n)$.
    (b) $x(n) = (\frac{1}{3})^n u(n)$.

11. Compute the output signal for a circuit with impulse response $h(t) = (1 - t)[u(t) - u(t - 1)]$ for the input $x(t) = e^{-t}\mathrm{u}(t)$.
12. Compute the output signal of a circuit with impulse response $h(t) = t[\mathrm{u}(t) - \mathrm{u}(t - 1)]$ for the input $x(t) = e^{-t}\mathrm{u}(t)$.

Chapter 4
Fourier and Hilbert Transforms



Mathematics compares the most diverse phenomena and discovers the secret analogies that unite them.

—Joseph Fourier

4.1 Introduction

The objective of this chapter is to provide the reader with the necessary mathematical basis for understanding communication systems in conjunction with probability theory and stochastic processes. The reader will become familiar with concepts and equations involving Fourier series, which have a significant historical relevance for the theory of communications. Furthermore, both the theory and properties of the Fourier and Hilbert transforms are presented, which constitute powerful tools for spectral and temporal analyses (Alencar 1999).

4.2 Fourier Analysis

The basic Fourier theory establishes fundamental conditions for the representation of a fairly arbitrary function defined in a finite interval as a sum of sinusoids. In fact, this is just an instance of the more general Fourier representation of signals in which a periodic signal $f(t)$, under fairly general conditions, can be represented by a complete set of orthogonal functions. By a complete set $\mathcal{S}$ of orthogonal functions, it is understood that, except for those orthogonal functions already in $\mathcal{S}$, there are no other orthogonal functions not belonging to $\mathcal{S}$ to be considered.

In the following, it is assumed that a periodic signal $f(t)$ satisfies the Dirichlet conditions, i.e., that $f(t)$ is a bounded function which in any one period has at most a finite number of local maxima and minima and a finite number of points of discontinuity (Wylie 1966). The representation of signals by orthogonal functions has very often an error, which diminishes as the number of component terms in the corresponding series is increased.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_4

The fact that a periodic signal $f(t)$ can in general be expanded as a sum of mutually orthogonal functions demands for a closer look at the concepts of periodicity and orthogonality.

Periodicity relates to the repetitive character of the function. A function $f(t)$ is defined to be a periodic function of period T if and only if T is the smallest positive number for which $f(t+T) = f(t)$. In other words, $f(t)$ is periodic if its domain contains $t+T$ whenever it contains t, and $f(t+T) = f(t)$. It follows from the definition of a periodic function that if T represents the period of $f(t)$ then $f(t) = f(t+nT)$ for $n = 1, 2, \ldots$, i.e., $f(t)$ will repeat its values when integer multiples of T (Wozencraft and Jacobs 1965) are added to its argument. Figure 4.1 represents a general periodic signal, $x(t)$, with period $2T$ and amplitude B.

If $f(t)$ and $g(t)$ are two periodic functions with the same period T, then their sum $f(t)+g(t)$ will also be a periodic function with period T. We prove this result by making $h(t) = f(t)+g(t)$ and noticing that $h(t+T) = f(t+T)+g(t+T) = f(t)+g(t) = h(t)$.

We shall now investigate the concept of orthogonality. Orthogonality provides the tool for introducing the concept of a basis, i.e., of a minimum set of functions that can be used to generate other functions. However, orthogonality by itself does not guarantee that a complete vector space is generated.

Two real functions $u(t)$ and $v(t)$, defined in the interval $\alpha \leq t \leq \beta$, are orthogonal if their inner product is null, that is, if

$$(u(t), v(t)) = \int_{\alpha}^{\beta} u(t)v(t)\mathrm{d}t = 0. \tag{4.1}$$

The set of functions $f_n(t)$, $n = 1, 2, 3, \ldots$, as illustrated in Fig. 4.2, can be used for representing signals in the time domain. This set of functions constitutes an orthogonal set in the interval $(-3T, 3T)$.

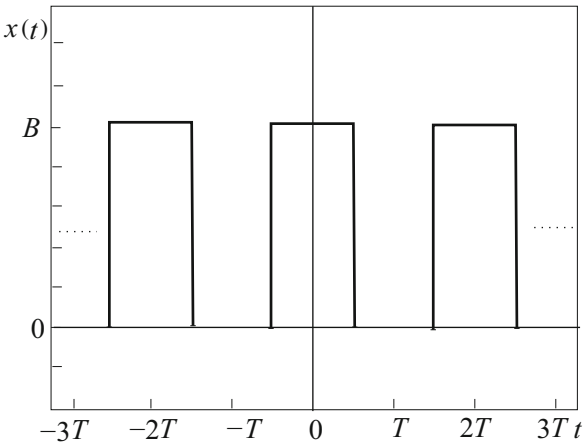


Fig. 4.1 Example of a periodic signal

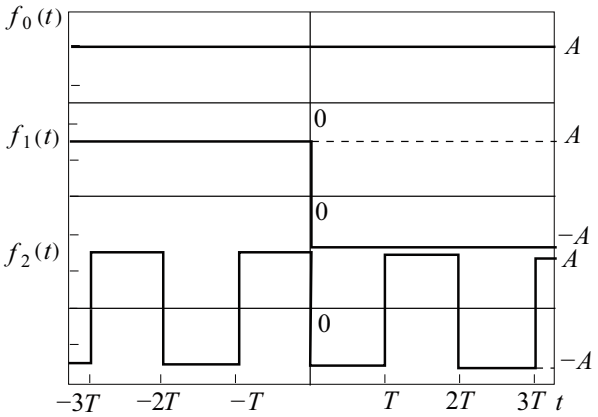


Fig. 4.2 Set of orthogonal functions

4.2.1 The Trigonometric Fourier Series

The trigonometric Fourier series representation of a signal $f(t)$ can be written as

$$f(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)], \tag{4.2}$$

in which the term a_0 (the average value of the function $f(t)$) indicates whether or not the signal contains a DC value and the terms a_n and b_n are denominated the Fourier series coefficients, in which n is a positive integer.

The equality sign holds in (4.2) for all values of t only when $f(t)$ is periodic. However, the Fourier series representation is a useful tool for any type of signal as long as that signal representation is required only in the $[0, T]$ interval. Outside that interval, the Fourier series representation will always be periodic, even if the signal $f(t)$ is not periodic (Knopp 1990).

The sine and cosine functions are examples of orthogonal functions because they satisfy the following equations for integer values of n and m, denominated orthogonality relations:

$$\int_0^T \cos(n\omega_o t)\ \sin(m\omega_o t)\mathrm{d}t = 0 \text{ for all integers } n \text{ and } m, \tag{4.3}$$

$$\int_0^T \cos(n\omega_o t)\cos(m\omega_o t)\mathrm{d}t = \begin{cases} 0 & \text{if } n \neq m \\ \frac{T}{2} & \text{if } n = m \end{cases} \tag{4.4}$$

$$\int_0^T \sin(n\omega_o t)\ \sin(m\omega_o t)\mathrm{d}t = \begin{cases} 0 & \text{if } n \neq m \\ \frac{T}{2} & \text{if } n = m \end{cases} \tag{4.5}$$

in which $\omega_0 = 2\pi/T$.

As a consequence of the orthogonality conditions, explicit expressions for the coefficients a_n and b_n of the Fourier trigonometric series can be computed. By integrating both sides in expression (4.2) in the interval $[0, T]$, it follows that (Oberhettinger 1990)

$$\int_0^T f(t)\mathrm{d}t = \int_0^T a_o \mathrm{d}t + \sum_{n=1}^{\infty} \int_0^T a_n \cos(n\omega_o t)\mathrm{d}t + \sum_{n=1}^{\infty} \int_0^T b_n \sin(n\omega_o t)\mathrm{d}t,$$

and since

$$\int_0^T a_n \cos(n\omega_o t)\mathrm{d}t = \int_0^T b_n \sin(n\omega_o t)\mathrm{d}t = 0,$$

it follows that

$$a_o = \frac{1}{T}\int_0^T f(t)\mathrm{d}t. \tag{4.6}$$

Now, by multiplying both sides in expression (4.2) by $\cos(m\omega_o t)$ and integrating in the interval $[0, T]$, it follows that

$$\begin{aligned}\int_0^T f(t)\cos(m\omega_o t)\mathrm{d}t = &\int_0^T a_o \cos(m\omega_o t)\mathrm{d}t \\ &+ \sum_{n=1}^{\infty}\int_o^T a_n \cos(n\omega_o t)\cos(m\omega_o t)\mathrm{d}t \\ &+ \sum_{n=1}^{\infty}\int_o^T b_n \cos(m\omega_0 t)\sin(n\omega_o t)\mathrm{d}t,\end{aligned} \tag{4.7}$$

which after simplification produces

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_o t)\mathrm{d}t, \text{ for } n = 1, 2, 3, \ldots. \tag{4.8}$$

In a similar manner, b_n is found by multiplying both sides in expression (4.2) by $\sin(n\omega_o t)$ and integrating in the interval $[0, T]$, i.e.,

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega_o t)\mathrm{d}t \tag{4.9}$$

for $n = 1, 2, 3, \ldots$.

4.2.2 Even Functions and Odd Functions

A function is called an odd function if it is antisymmetric with respect to the ordinate axis, i.e., if $f(-t) = -f(t)$, in which $-t$ and t are assumed to belong to the function domain. Examples of odd functions are provided by the functions t, t^3, $\sin t$ and $t^{|2n+1|}$.

Similarly, a function is called an even function if it is symmetric with respect to the ordinate axis, i.e., if $f(-t) = f(t)$, in which t and $-t$ are assumed to belong to the function domain. Examples of even functions are provided by the functions 1, t^2, $\cos t$, $|t|$, $\exp(-|t|)$, and $t^{|2n|}$.

Some Elementary Properties

(a) The sum (difference) and the product (quotient) of two even functions are an even function.
(b) The sum (difference) of two odd functions is an odd function.
(c) The product (quotient) of two odd functions is an even function.
(d) The sum (difference) of an even function and an odd function is neither an even function nor an odd function.
(e) The product (quotient) between an even function and an odd function is an odd function.

Two other important properties are the following:

(f) If $f(t)$ is an even periodic function of period T, then

$$\int_{-T/2}^{T/2} f(t)\mathrm{d}t = 2\int_{0}^{T/2} f(t)\mathrm{d}t. \tag{4.10}$$

(g) If $f(t)$ is an odd periodic function of period T, then

$$\int_{-T/2}^{T/2} f(t)\mathrm{d}t = 0. \tag{4.11}$$

Properties (4.2.2) and (4.2.2) allow for a considerable simplification when computing coefficients of a trigonometric Fourier series:

(h) If $f(t)$ is an even function then $b_n = 0$, and

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_o t)\mathrm{d}t \quad \text{for } n = 1, 2, 3, \ldots. \tag{4.12}$$

(i) If $f(t)$ is an odd function, then $a_n = 0$ and

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega_o t)\mathrm{d}t \text{ for } n = 1, 2, 3, \ldots. \tag{4.13}$$

Example Compute the coefficients of the trigonometric Fourier series for the waveform $f(t) = A[u(t+\tau) - u(t-\tau)]$, which repeats itself with period T, in which $u(t)$ denotes the unit step function and $2\tau \leq T$.

Solution Since the given signal is symmetric with respect to the ordinate axis, it follows that $f(t) = f(-t)$ and the function is even. Therefore, $b_n = 0$, and all that is left for computing is a_o and a_n for $n = 1, 2, \ldots$. The expression for computing the average value a_0 is given by

$$a_o = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\mathrm{d}t = \frac{1}{T}\int_{-\tau}^{\tau} A\mathrm{d}t = \frac{2A\tau}{T}.$$

In the previous equation, the maximum value of τ is $T/2$. The coefficients a_n for $n = 1, 2, \ldots$ are computed as

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_o t)\mathrm{d}t = \frac{2}{T}\int_{-\tau}^{\tau} A\cos(n\omega_o t)\mathrm{d}t,$$

$$a_n = \frac{4A}{T}\int_0^{\tau}\cos(n\omega_o t)\mathrm{d}t = \frac{4A}{Tn\omega_o}\sin(n\omega_o t)\Big|_0^{\tau} = (4A\tau/T)\frac{\sin(n\omega_o\tau)}{n\omega_0\tau}.$$

The signal $f(t)$ is then represented by the following trigonometric Fourier series:

$$f(t) = \frac{2A\tau}{T} + \left(\frac{4A\tau}{T}\right)\sum_{n=1}^{\infty}\frac{\sin(n\omega_o\tau)}{n\omega_0\tau}\cos(n\omega_o t).$$

4.2.3 The Compact Fourier Series

It is also possible to represent the Fourier series in a form known as the *compact* Fourier series as follows:

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n\cos(n\omega_o t + \theta_n) \tag{4.14}$$

By expanding the expression $C_n\cos(n\omega_o t + \theta)$ as $C_n\cos(n\omega_o t)\cos\theta_n - C_n\sin(n\omega_o t)\sin\theta_n$ and comparing this result with (4.2), it follows that $a_o = C_o$, $a_n = C_n\cos\theta_n$, and $b_n = -C_n\sin\theta_n$. It is now possible to compute C_n as a function of a_n and b_n. For that purpose, it is sufficient to square a_n and b_n and add the result, i.e.,

$$a_n^2 + b_n^2 = C_n^2\cos^2\theta_n + C_n^2\sin^2\theta_n = C_n^2. \tag{4.15}$$

From Eq. (4.15), the modulus of C_n can be written as

$$C_n = \sqrt{a_n^2 + b_n^2}. \tag{4.16}$$

In order to determine θ_n, it suffices to divide b_n by a_n, i.e.,

$$\frac{b_n}{a_n} = -\frac{\sin\,\theta_n}{\cos\theta_n} = -\tan\theta_n, \tag{4.17}$$

which when solved for θ_n produces

$$\theta_n = -\arctan\left(\frac{b_n}{a_n}\right). \tag{4.18}$$

4.2.4 The Exponential Fourier Series

Since the set of exponential functions $e^{jn\omega_o t}$, $n = 0, \pm 1, \pm 2, \ldots$, is a complete set of orthogonal functions in an interval of magnitude T, in which $T = 2\pi/\omega_o$, then it is possible to represent a function $f(t)$ by a linear combination of exponential functions in an interval T,

$$f(t) = \sum_{-\infty}^{\infty} F_n e^{jn\omega_0 t} \tag{4.19}$$

in which

$$F_n = \frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} f(t)e^{-jn\omega_0 t}\mathrm{d}t. \tag{4.20}$$

Equation (4.19) represents the exponential Fourier series expansion of $f(t)$, and Eq. (4.20) is the expression to compute the associated series coefficients. The exponential Fourier series is also known as the *complex Fourier series*. It is immediate to show that Eq. (4.19) is just another way of expressing the Fourier series as given in (4.2). Replacing $\cos(n\omega_o t) + j\sin(n\omega_0 t)$ for $e^{n\omega_0 t}$ (Euler's identity) in (4.19), it follows that

$$f(t) = F_o + \sum_{n=-\infty}^{-1} F_n[\cos(n\omega_o t) + j\sin(n\omega_o t)]$$
$$+ \sum_{n=1}^{\infty} F_n[\cos(n\omega_o t) + j\sin(n\omega_o t)]$$

or

$$f(t) = F_o + \sum_{n=1}^{\infty} F_n[\cos(n\omega_o t) + j\sin(n\omega_o t)] + F_{-n}[\cos(n\omega_o t) - j\sin(n\omega_o t)].$$

Grouping the coefficients of the sine and cosine terms, it follows that

$$f(t) = F_o + \sum_{n=1}^{\infty}(F_n + F_{-n})\cos(n\omega_o t) + j(F_n - F_{-n})\sin(n\omega_o t). \tag{4.21}$$

Comparing the above expression with (4.2), it follows that

$$a_o = F_o,\ a_n = (F_n + F_{-n}) \text{ and } b_n = j(F_n - F_{-n}) \tag{4.22}$$

and that

$$F_o = a_o, \tag{4.23}$$

$$F_n = \frac{a_n - jb_n}{2} \tag{4.24}$$

and

$$F_{-n} = \frac{a_n + jb_n}{2}. \tag{4.25}$$

In case the function $f(t)$ is even, i.e., if $b_n = 0$, then

$$a_o = F_o,\ F_n = \frac{a_n}{2} \text{ and } F_{-n} = \frac{a_n}{2}. \tag{4.26}$$

Example Compute the exponential Fourier series for the train of impulses

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$

Solution The complex coefficients are given by

$$F_n = \frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} \delta_T(t)e^{-jn\omega_o t}\mathrm{d}t = \frac{1}{T}, \tag{4.27}$$

since

$$\int_{-\infty}^{\infty} \delta(t - t_o)f(t)\mathrm{d}t = f(t_o) \text{ (impulse filtering)}. \tag{4.28}$$

It follows that $f(t)$ can be written as

$$f(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{-jn\omega_o t}. \tag{4.29}$$

In practice, in order to obtain an approximation of an impulse train, it is sufficient to pass a binary digital signal through a differentiator circuit and then pass the resulting waveform through a half-wave rectifier.

The Fourier series expansion of a periodic function is equivalent to its decomposition in frequency components. In general, a periodic function with period T has frequency components $0, \pm\omega_o, \pm 2\omega_o, \pm 3\omega_o, \ldots, \pm n\omega_o$, in which $\omega_o = 2\pi/T$ is the fundamental frequency and the multiples of ω_0 are called harmonics. Notice that the spectrum exists only for discrete values of ω and that the spectral components are spaced by at least ω_o.

4.3 Fourier Transform

It was shown earlier that an arbitrary function can be represented in terms of an exponential (or trigonometric) Fourier series in a finite interval. If such a function is periodic, this representation can be extended for the entire interval $(-\infty, \infty)$.

However, it is interesting to observe the spectral behavior of a function in general, periodic or not, in the entire interval $(-\infty, \infty)$. In order to do that, the function $f(t)$ is truncated in the interval $[-T/2, T/2]$, obtaining $f_T(t)$. It is possible then to represent this function as a sum of exponentials in the entire interval $(-\infty, \infty)$ by making T approach infinity. In other words,

$$\lim_{T\to\infty} f_T(t) = f(t).$$

The $f_T(t)$ signal can be represented by the exponential Fourier series as

$$f_T(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_o t}, \tag{4.30}$$

in which $\omega_o = 2\pi/T$ and

$$F_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f_T(t) e^{-jn\omega_o t} \mathrm{d}t. \tag{4.31}$$

F_n represents the spectral amplitude associated with a component of frequency $n\omega_o$. As T increases, the amplitudes diminish, but the spectrum shape is not altered. This increase in T forces ω_o to diminish and the spectrum to become denser.

In the limit, as $T \to \infty$, ω_o becomes infinitesimally small, being represented by $\mathrm{d}\omega$. On the other hand, there are now infinitely many components, and the spectrum is no longer a discrete one, becoming a continuous spectrum in the limit.

For convenience, write $TF_n = F(\omega)$, that is, the product TF_n becomes a function of the variable ω, since $n\omega_o \to \omega$. Replacing $\frac{F(\omega)}{T}$ for F_n in 4.30, one obtains

$$f_T(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} F(\omega) e^{j\omega t}. \tag{4.32}$$

Replacing $\omega_0/2\pi$ for $1/T$, it follows that

$$f_T(t) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} F(\omega) e^{j\omega t} \omega_0. \tag{4.33}$$

In the limit, as T approaches infinity, one has

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} \mathrm{d}\omega \tag{4.34}$$

which is known as the inverse Fourier transform.

Similarly, from 4.31, as T approaches infinity, one obtains

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} \mathrm{d}t \tag{4.35}$$

which is known as the direct Fourier transform, sometimes denoted in the literature as $F(\omega) = \mathcal{F}[f(t)]$. A Fourier transform pair is often denoted as $f(t) \longleftrightarrow F(\omega)$.

In the sequel, some important Fourier transforms are presented (Haykin 1988).

4.3.1 Bilateral Exponential Signal

If $f(t) = e^{-a|t|}$, it follows from (4.35) that

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} \mathrm{d}t \\ &= \int_{-\infty}^{0} e^{at} e^{-j\omega t} \mathrm{d}t + \int_{0}^{\infty} e^{-at} e^{-j\omega t} \mathrm{d}t \\ &= \frac{1}{a - j\omega} + \frac{1}{a + j\omega}, \end{aligned} \tag{4.36}$$

$$F(\omega) = \frac{2a}{a^2 + \omega^2}. \tag{4.37}$$

4.3.2 Gate Function

The gate function is defined by the expression $p_T(t) = A[u(t + T/2) - u(t - T/2)]$ or

$$p_T(t) = \begin{cases} A \text{ if } |t| \leq T/2 \\ 0 \text{ if } |t| > T/2 \end{cases} \tag{4.38}$$

in which $u(t)$ denotes the *unit step* function, defined as

$$u(t) = \begin{cases} 1 \text{ if } t \geq 0 \\ 0 \text{ if } t < 0 \end{cases} \tag{4.39}$$

and represented in Fig. 4.3.

The gate function is illustrated in Fig. 4.4. The Fourier transform of the gate function can be calculated as

$$\begin{aligned} F(\omega) &= \int_{-\frac{T}{2}}^{\frac{T}{2}} Ae^{-j\omega t} \mathrm{d}t \\ &= \frac{A}{j\omega}\left(e^{j\omega\frac{T}{2}} - e^{-j\omega\frac{T}{2}}\right) \\ &= \frac{A}{j\omega} 2j \sin(\omega T/2), \end{aligned} \tag{4.40}$$

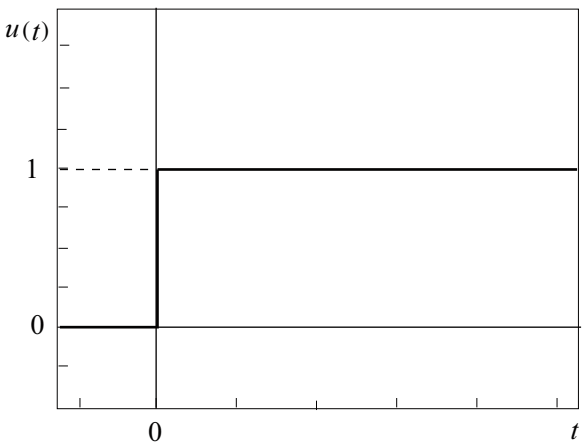


Fig. 4.3 The unit step function

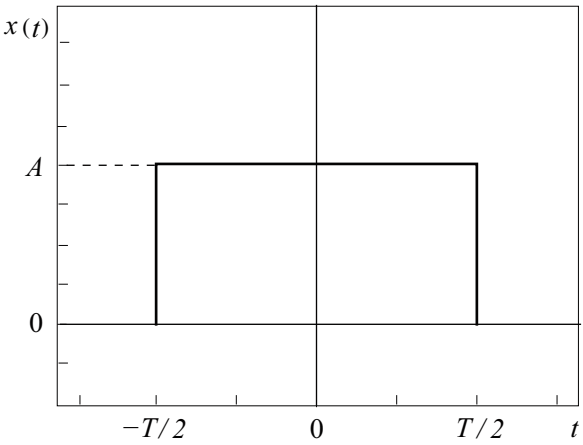


Fig. 4.4 The gate function

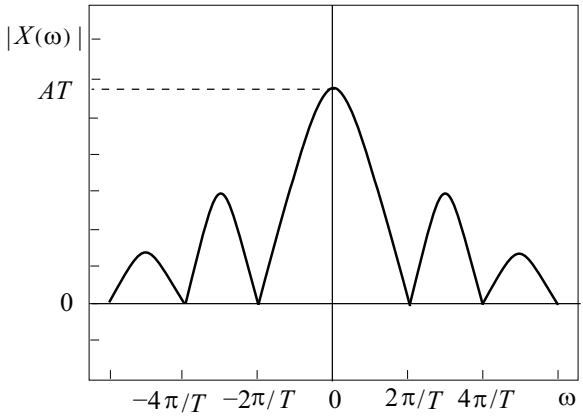


Fig. 4.5 Magnitude plot of the Fourier transform of the gate function

which can be rearranged as

$$F(\omega) = AT\left(\frac{\sin(\omega T/2)}{\omega T/2}\right),$$

and finally

$$F(\omega) = AT\,\mathrm{Sa}\left(\frac{\omega T}{2}\right), \tag{4.41}$$

in which $\mathrm{Sa}(x) = \frac{\sin x}{x}$ is the *sampling* function. This function converges to one, as x goes to zero. The sampling function, the magnitude of which is illustrated in Fig. 4.5, is of great relevance in communication theory.

The *sampling* function obeys the following important relationship:

$$\int_{-\infty}^{\infty} \frac{k}{\pi} \mathrm{Sa}\,(kt)\mathrm{d}t = 1 \tag{4.42}$$

The area under this curve is equal to 1. As k increases, the amplitude of the sampling function increases, the spacing between zero crossings diminishes, and most of the signal energy concentrates near the origin. For $k \to \infty$, the function converges to an impulse function, i.e.,

$$\delta(t) = \lim_{k\to\infty} \frac{k}{\pi} \mathrm{Sa}\,(kt). \tag{4.43}$$

In this manner, in the limit, it is true that $\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = 1$. Since the function concentrates its nonzero values near the origin, it follows that $\delta(t) = 0$ for $t \neq 0$. Therefore,

$$\int_{-\infty}^{\infty} f(t)\delta(t)\mathrm{d}t = f(0)\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = f(0). \tag{4.44}$$

In general, it is possible to write (4.44) as

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_o)\mathrm{d}t = f(t_o). \tag{4.45}$$

This important relationship, mentioned earlier in (4.28), is known as the filtering property of the impulse function.

4.3.3 Impulse Function or Dirac's Delta Function

By making $f(t) = \delta(t)$ in (4.35), it follows that

$$F(\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}\mathrm{d}t. \tag{4.46}$$

Using the impulse filtering property, it follows that $F(\omega) = 1$. Therefore, the impulse function contains a continuum of equal amplitude spectral components.

Alternatively, by making $F(\omega) = 1$ in 4.34 and simplifying, the impulse function can be written as

$$\frac{1}{\pi}\int_{0}^{\infty} \cos \omega t\, d\omega.$$

4.3.4 The Constant Function

If $f(t)$ is a constant function, then its Fourier transform in principle would not exist since this function does not satisfy the absolute integrability criterion. In general, $F(\omega),$, the Fourier transform of $f(t)$, is expected to be finite, i.e.,

$$|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)||e^{-j\omega t}|\mathrm{d}t < \infty, \tag{4.47}$$

since $|e^{-j\omega t}| = 1$; then,

$$\int_{-\infty}^{\infty} |f(t)|\mathrm{d}t < \infty. \tag{4.48}$$

However, that is just a sufficiency condition and not a necessary condition for the existence of the Fourier transform, since there exist functions that, although do not satisfy the condition of absolute integrability, in the limit have a Fourier transform (Carlson 1975). This is a very important observation since this approach is often used in the computation of Fourier transforms of many functions. Returning to the constant function, it can be approximated by a gate function with amplitude A and width τ and then making τ approach very large values,

$$\mathcal{F}[A] = \lim_{\tau\to\infty} A\tau \mathrm{Sa}\left(\frac{\omega\tau}{2}\right) \tag{4.49}$$

$$= 2\pi A \lim_{\tau\to\infty} \frac{\tau}{2\pi} \mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$\mathcal{F}[A] = 2\pi A\delta(\omega). \tag{4.50}$$

This result is not only a very interesting one but also somehow intuitive since a constant function in time represents a DC level and, as was to be expected, contains no spectral component except for the one at $\omega = 0$.

4.3.5 Fourier Transform of Sine and Cosine

Since both the sine and the cosine functions are periodic functions, they do not satisfy the condition of absolute integrability. However, their respective Fourier transforms exist in the limit when τ goes to infinity. Assuming the function to exist only in the interval $(\frac{-\tau}{2}, \frac{\tau}{2})$ and to be zero outside this interval and considering the limit of the expression when τ goes to infinity,

$$\mathcal{F}(\sin\omega_0 t) = \lim_{\tau\to\infty} \int_{\frac{-\tau}{2}}^{\frac{\tau}{2}} \sin\omega_0 t \; e^{-j\omega t}\mathrm{d}t \tag{4.51}$$

$$= \lim_{\tau \to \infty} \int_{\frac{-\tau}{2}}^{\frac{\tau}{2}} \frac{e^{-j(\omega-\omega_0)t}}{2j} - \frac{e^{-j(\omega+\omega_0)t}}{2j} \mathrm{d}t$$

$$= \lim_{\tau \to \infty} \left[\frac{j\tau \sin(\omega+\omega_0)\frac{\tau}{2}}{2(\omega+\omega_0)\frac{\tau}{2}} - \frac{j\tau \sin(\omega-\omega_0)\frac{\tau}{2}}{2(\omega-\omega_0)\frac{\tau}{2}} \right]$$

$$= \lim_{\tau \to \infty} \left\{ j\frac{\tau}{2}\mathrm{Sa}\left[\frac{(\omega+\omega_0)}{2}\right] - j\frac{\tau}{2}\mathrm{Sa}\left[\frac{\tau(\omega+\omega_0)}{2}\right] \right\}.$$

Therefore,

$$\mathcal{F}(\sin\omega_0 t) = j\pi[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)].$$

Applying a similar reasoning, it follows that

$$\mathcal{F}(\cos\omega_0 t) = \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)], \tag{4.52}$$

in which $\omega_0 = 2\pi/T$. Figure 4.6 illustrates the cosine function, and Fig. 4.7 depicts the magnitude of the Fourier transform of the cosine function.

4.3.6 The Fourier Transform of $e^{j\omega_0 t}$

Using Euler's identity, $e^{j\omega_0 t} = \cos\omega_0 t + j\sin\omega_0 t$, it follows that

$$\mathcal{F}[e^{j\omega_0 t}] = \mathcal{F}[\cos\omega_0 t + j\sin\omega_0 t]. \tag{4.53}$$

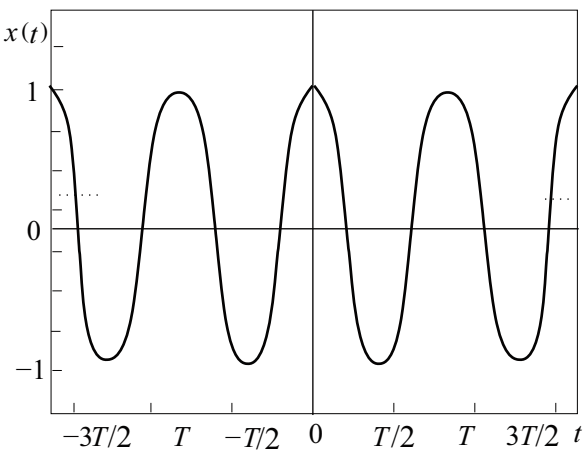


Fig. 4.6 The cosine function

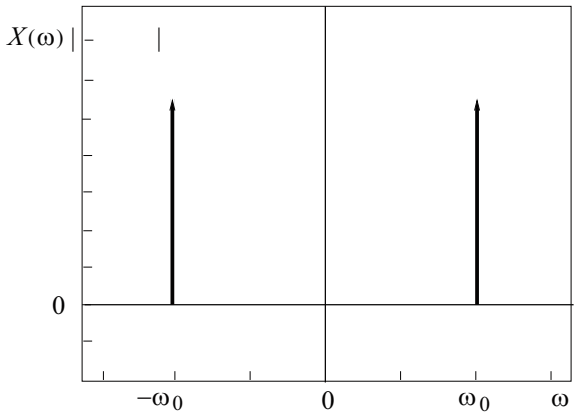


Fig. 4.7 Magnitude plot of the Fourier transform of the cosine function

Substituting in (4.53) the Fourier transforms of the sine and of the cosine functions, respectively, it follows that

$$\mathcal{F}[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0). \tag{4.54}$$

4.3.7 The Fourier Transform of a Periodic Function

We consider next the exponential Fourier series representation of a periodic function $f_T(t)$ of period T:

$$f_T(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t} \tag{4.55}$$

Applying the Fourier transform to both sides in (4.55), it follows that

$$\mathcal{F}[f_T(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}\right] \tag{4.56}$$

$$= \sum_{n=-\infty}^{\infty} F_n \mathcal{F}[e^{jn\omega_0 t}]. \tag{4.57}$$

Now, applying in (4.57) the result from (4.54), it follows that

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0). \tag{4.58}$$

4.4 Some Properties of the Fourier Transform

4.4.1 Linearity

Linearity is an important property when studying communication systems. A system is defined to be a linear system if satisfies the properties of homogeneity and additivity:

1 Homogeneity—If the application of the signal $x(t)$ at the system input produces $y(t)$ at the system output, then the application of the input $\alpha x(t)$, in which α is a constant, produces $\alpha y(t)$ at the output.
2 Additivity—If the application of the signals $x_1(t)$ and $x_2(t)$ at the system input produces, respectively, $y_1(t)$ and $y_2(t)$ at the system output, then the application of the input $x_1(t) + x_2(t)$ produces $y_1(t) + y_2(t)$ at the output.

By applying the tests for homogeneity and additivity, it is immediate to check that the process that generates the signal $s(t) = A\cos(\omega_c t + \Delta m(t) + \theta)$ from an input signal $m(t)$ is nonlinear. By applying the same test to the signal $r(t) = m(t)\cos(\omega_c t + \theta)$, it is immediate to show that the process generating $r(t)$ is linear.

The Fourier transform is a linear operator, i.e., if a function can be written as a linear combination of other (well behaved) functions, the corresponding Fourier transform will be given by a linear combination of the corresponding Fourier transforms of each one of the functions involved in the linear combination (Gagliardi 1988).

If $f(t) \longleftrightarrow F(\omega)$ and $g(t) \longleftrightarrow G(\omega)$, then it follows that

$$\alpha f(t) + \beta g(t) \longleftrightarrow \alpha F(\omega) + \beta G(\omega). \tag{4.59}$$

Proof Letting $h(t) = \alpha f(t) + \beta g(t) \rightarrow$, it then follows that

$$\begin{aligned} H(\omega) &= \int_{-\infty}^{\infty} h(t) e^{-j\omega t} \mathrm{d}t \\ &= \alpha \int_{-\infty}^{\infty} f(t) e^{-j\omega t} \mathrm{d}t + \beta \int_{-\infty}^{\infty} g(t) e^{-j\omega t} \mathrm{d}t \end{aligned}$$

and finally

$$H(\omega) = \alpha F(\omega) + \beta G(\omega). \tag{4.60}$$

4.4.2 Scaling

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} f(at)e^{-j\omega t}\mathrm{d}t. \tag{4.61}$$

Initially, let us consider $a > 0$ in (4.61). By letting $u = at$, it follows that $\mathrm{d}t = (1/a)\mathrm{d}u$. Replacing u for at in (4.61), it follows that

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} \frac{f(u)}{a} e^{-j\frac{\omega}{a}u}\mathrm{d}u$$

which simplifies to

$$\mathcal{F}[f(at)] = \frac{1}{a}F\left(\frac{\omega}{a}\right). \tag{4.62}$$

Consider now the case in which $a < 0$. By a similar procedure, it follows that

$$\mathcal{F}[f(at)] = -\frac{1}{a}F\left(\frac{\omega}{a}\right). \tag{4.63}$$

Finally, Eqs. 4.62 and 4.63 can be combined and written as

$$\mathcal{F}[f(at)] = \frac{1}{|a|}F\left(\frac{\omega}{a}\right). \tag{4.64}$$

This result points to the fact that if a signal is compressed in the time domain by a factor a, then its frequency spectrum will expand in the frequency domain by that same factor.

4.4.3 Symmetry

This is an interesting property which can be fully observed in even functions. The symmetry property states that if

$$f(t) \longleftrightarrow F(\omega), \tag{4.65}$$

then it follows that

$$F(t) \longleftrightarrow 2\pi f(-\omega). \tag{4.66}$$

Proof By definition,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} \mathrm{d}\omega,$$

which after multiplication of both sides by 2π becomes

$$2\pi f(t) = \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} \mathrm{d}\omega.$$

By letting $u = -t$, it follows that

$$2\pi f(-u) = \int_{-\infty}^{+\infty} F(\omega) e^{-j\omega u} \mathrm{d}\omega,$$

and now by making $t = \omega$, one obtains

$$2\pi f(-u) = \int_{-\infty}^{+\infty} F(t) e^{-jtu} \mathrm{d}t.$$

Finally, by letting $u = \omega$, it follows that

$$2\pi f(-\omega) = \int_{-\infty}^{+\infty} F(t) e^{-j\omega t} \mathrm{d}t. \tag{4.67}$$

Example The Fourier transform of a constant function can be easily derived by the use of the symmetry property. Since

$$A\delta(t) \longleftrightarrow A,$$

it follows that

$$A \longleftrightarrow 2\pi A\delta(-\omega) = 2\pi A\delta(\omega).$$

4.4.4 Time-Domain Shift

Given that $f(t) \longleftrightarrow F(\omega)$, it then follows that $f(t - t_0) \longleftrightarrow F(\omega)e^{-j\omega t_0}$. Let $g(t) = f(t - t_0)$. In this case, it follows that

$$G(\omega) = \mathcal{F}[g(t)] = \int_{-\infty}^{\infty} f(t - t_0) e^{-j\omega t} \mathrm{d}t. \tag{4.68}$$

By making $\tau = t - t_0$, it follows that

$$G(\omega) = \int_{-\infty}^{\infty} f(\tau) e^{-j\omega(\tau+t_0)} \mathrm{d}\tau \tag{4.69}$$

$$= \int_{-\infty}^{\infty} f(\tau) e^{-j\omega\tau} e^{-j\omega t_0} \mathrm{d}\tau \tag{4.70}$$

and finally

$$G(\omega) = e^{-j\omega t_0} F(\omega). \tag{4.71}$$

This result shows that whenever a function is shifted in time its frequency domain amplitude spectrum remains unaltered. However, the corresponding phase spectrum experiences a rotation proportional to ωt_0.

4.4.5 Frequency Domain Shift

Given that $f(t) \longleftrightarrow F(\omega)$, it then follows that $f(t)e^{j\omega_0 t} \longleftrightarrow F(\omega - \omega_0)$:

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = \int_{-\infty}^{\infty} f(t) e^{j\omega_0 t} e^{-j\omega t} \mathrm{d}t \tag{4.72}$$

$$= \int_{-\infty}^{\infty} f(t) e^{-j(\omega-\omega_0)t} \mathrm{d}t,$$

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0) \tag{4.73}$$

4.4.6 Differentiation in the Time Domain

Given the following transformation

$$f(t) \longleftrightarrow F(\omega), \tag{4.74}$$

it then follows that

$$\frac{\mathrm{d}f(t)}{\mathrm{d}t} \longleftrightarrow j\omega F(\omega); \tag{4.75}$$

notice that if $f(t)$ has a constant term then it will disappear after the derivative.

Proof Let us consider the expression for the inverse Fourier transform:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\mathrm{d}\omega \tag{4.76}$$

Differentiating in time, it follows that

$$\begin{aligned}\frac{\mathrm{d}f(t)}{\mathrm{d}t} &= \frac{1}{2\pi}\frac{\mathrm{d}}{\mathrm{d}t}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\mathrm{d}\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}F(\omega)e^{j\omega t}\mathrm{d}\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} j\omega F(\omega)e^{j\omega t}\mathrm{d}\omega\end{aligned}$$

and then

$$\frac{\mathrm{d}f(t)}{\mathrm{d}t} \longleftrightarrow j\omega F(\omega). \tag{4.77}$$

In general, it follows that

$$\frac{d^n f(t)}{\mathrm{d}t} \longleftrightarrow (j\omega)^n f(\omega). \tag{4.78}$$

By computing the Fourier transform of the signal $f(t) = \delta(t) - \alpha e^{-\alpha t}u(t)$, it is immediate to show that, by applying the property of differentiation in time, this signal is the time derivative of the signal $g(t) = e^{-\alpha t}u(t)$.

4.4.7 Integration in the Time Domain

Let $f(t)$ be a signal with zero average value, i.e., let $\int_{-\infty}^{\infty} f(t)\mathrm{d}t = 0$. By defining

$$g(t) = \int_{-\infty}^{t} f(\tau)\mathrm{d}\tau, \tag{4.79}$$

it follows that

$$\frac{\mathrm{d}g(t)}{\mathrm{d}t} = f(t),$$

and since

$$g(t) \longleftrightarrow G(\omega), \tag{4.80}$$

then

$$f(t) \longleftrightarrow j\omega G(\omega),$$

and

$$G(\omega) = \frac{F(\omega)}{j\omega}. \tag{4.81}$$

In this manner, it follows that for a signal with zero average value

$$f(t) \longleftrightarrow F(\omega)$$

$$\int_{-\infty}^{t} f(\tau)\mathrm{d}\tau \longleftrightarrow \frac{F(\omega)}{j\omega}. \tag{4.82}$$

Generalizing, for the case in which $f(t)$ has a nonzero average value, it follows that

$$\int_{-\infty}^{t} f(\tau)\mathrm{d}\tau \longleftrightarrow \frac{F(\omega)}{j\omega} + \pi\delta(\omega)F(0). \tag{4.83}$$

4.4.8 The Convolution Theorem

The convolution theorem is a powerful tool for analyzing the frequency contents of a signal, allowing obtention of many relevant results. One instance of the use of the convolution theorem, of fundamental importance in communication theory, is the sampling theorem which will be the subject of the next section.

The convolution between two time functions $f(t)$ and $g(t)$ is defined by the following integral,

$$\int_{-\infty}^{\infty} f(\tau)g(t-\tau)\mathrm{d}\tau, \tag{4.84}$$

which is often denoted as $f(t) * g(t)$.

Let $h(t) = f(t) * g(t)$ and let $h(t) \longleftrightarrow H(\omega)$. It follows that

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}\mathrm{d}t = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\tau)g(t-\tau)e^{-j\omega t}\mathrm{d}t\,d\tau, \tag{4.85}$$

$$H(\omega) = \int_{-\infty}^{\infty} f(\tau)\int_{-\infty}^{\infty} g(t-\tau)e^{-j\omega t}\mathrm{d}t\,d\tau, \tag{4.86}$$

$$H(\omega) = \int_{-\infty}^{\infty} f(\tau)G(\omega)e^{-j\omega\tau}\mathrm{d}\tau, \tag{4.87}$$

and, finally,

$$H(\omega) = F(\omega)G(\omega). \tag{4.88}$$

The convolution of two time functions is equivalent in the frequency domain to the product of their respective Fourier transforms. For the case in which $h(t) = f(t) \cdot g(t)$, proceeding in a similar manner, one obtains

$$H(\omega) = \frac{1}{2\pi}[F(\omega) * G(\omega)]. \tag{4.89}$$

In other words, the product of two time functions has a Fourier transform given by the convolution of their respective Fourier transforms. The convolution operation is often used when computing the response of a linear circuit, given its impulse response and an input signal.

Example The circuit in Fig. 4.8 has the impulse response $h(t)$ given by

$$h(t) = \frac{1}{RC}e^{-\frac{t}{RC}}u(t).$$

The application of the unit impulse $x(t) = \delta(t)$ as the input to this circuit causes an output $y(t) = h(t) * x(t)$. In the frequency domain, by the convolution theorem, it follows that $Y(\omega) = H(\omega)X(\omega) = H(\omega)$, i.e., the Fourier transform of the impulse response of a linear system is the system transfer function.

Using the frequency domain convolution theorem, it can be shown that

$$\cos(\omega_c t)u(t) \longleftrightarrow \frac{\pi}{2}[\delta(\omega + \omega_c) + \delta(\omega - \omega_c)] + j\frac{\omega}{\omega_c^2 - \omega^2}.$$

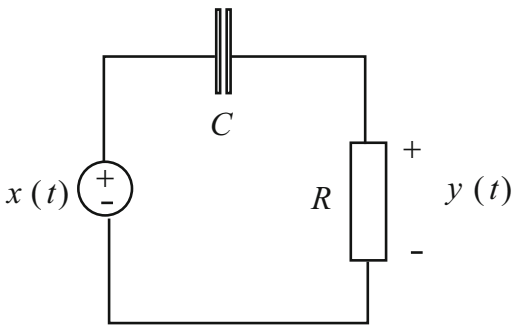


Fig. 4.8 An RC circuit

4.5 Parseval's Theorem

For a real signal $f(t)$ of finite energy, often called simply a real energy signal, the energy E associated with $f(t)$ is given by

$$E = \int_{-\infty}^{\infty} f^2(t)\mathrm{d}t$$

and can equivalently be calculated by the formula

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2 \mathrm{d}.\omega.$$

It follows that

$$\int_{-\infty}^{\infty} f^2(t)\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2 \mathrm{d}\omega. \tag{4.90}$$

The relationship given in (4.90) is known as Parseval's theorem or Parseval's identity. For a real signal $x(t)$ with energy E, it can be shown, by using Parseval's identity, that the signals $x(t)$ and $y(t) = x(t-\tau)$ have the same energy E.

Another way of expressing Parseval's identity is as follows:

$$\int_{-\infty}^{\infty} f(x)G(x)\mathrm{d}x = \int_{-\infty}^{\infty} F(x)g(x)\mathrm{d}x \tag{4.91}$$

4.6 Average Value, Power, and Autocorrelation

As mentioned earlier, the average value of a real signal $x(t)$ is given by

$$\overline{x}(t) = \lim_{T\to\infty} \frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)\mathrm{d}t. \tag{4.92}$$

The instantaneous power of $x(t)$ is given by

$$p_X(t) = x^2(t). \tag{4.93}$$

If the signal $x(t)$ exists for the whole interval $(-\infty, +\infty)$, the total power $\overline{P}_X$ is defined for a real signal $x(t)$ as the power dissipated in a 1-ohm resistor, when a voltage $x(t)$ is applied to this resistor (or a current $x(t)$ flows through the

resistor) (Lathi 1989). Thus,

$$\overline{P}_X = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x^2(t)\mathrm{d}t. \tag{4.94}$$

From the previous definition, the unit to measure $\overline{P}_X$ corresponds to the square of the units of the signal $x(t)$ ($volt^2$, amp^2). These units will only be converted to watts if they are normalized by units of impedance (ohm). It is common use to express the power in decibel (dB). The power in decibel is given by the expression (Gagliardi 1988)

$$\overline{P}_{X,dB} = 10 \log \overline{P}_X. \tag{4.95}$$

The total power ($\overline{P}_X$) contains two components: one DC component, due to a nonzero average value of the signal $x(t)$ ($\overline{P}_{DC}$), and an AC component ($\overline{P}_{AC}$). The DC power of the signal is given by

$$\overline{P}_{DC} = (\overline{x}(t))^2. \tag{4.96}$$

It follows that the AC power can be determined by removing the DC power from the total power, i.e.,

$$\overline{P}_{AC} = \overline{P}_X - \overline{P}_{DC}. \tag{4.97}$$

4.6.1 Time Autocorrelation of Signals

The average time autocorrelation $\overline{R}_X(\tau)$, or simply autocorrelation, of a real signal $x(t)$ is defined as follows:

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)x(t+\tau)\mathrm{d}t \tag{4.98}$$

The change of variable $y = t + \tau$ allows Eq. (2.107) to be written as

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)x(t-\tau)\mathrm{d}t. \tag{4.99}$$

From Eqs. (4.98) and (4.99), it follows that $\overline{R}_X(\tau)$ is an even function of τ, and thus (Lathi 1989)

$$\overline{R}_X(-\tau) = \overline{R}_X(\tau). \tag{4.100}$$

From the definition of autocorrelation and power, it follows that

$$\overline{P}_X = \overline{R}_X(0) \tag{4.101}$$

and

$$\overline{P}_{DC} = \overline{R}_X(\infty), \tag{4.102}$$

i.e., from its autocorrelation function, it is possible to obtain information about the power of a signal. The autocorrelation function can also be considered in the frequency domain by taking its Fourier transform, i.e.,

$$\mathcal{F}\{\overline{R}_X(\tau)\} = \int_{-\infty}^{+\infty} \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)x(t+\tau)e^{-j\omega\tau}\mathrm{d}t\,\mathrm{d}\tau = \tag{4.103}$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t) \int_{-\infty}^{+\infty} x(t+\tau)\mathrm{d}\tau\mathrm{d}t$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)X(\omega)e^{j\omega t}\mathrm{d}t$$

$$= X(\omega) \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)e^{j\omega t}\mathrm{d}t$$

$$= \lim_{T\to\infty} \frac{X(\omega)X(-\omega)}{T}$$

$$= \lim_{T\to\infty} \frac{|X(\omega)|^2}{T}. \tag{4.104}$$

The power spectral density $\overline{S}_X$ of a signal $x(t)$ is defined as the Fourier transform of the autocorrelation function $\overline{R}_X(\tau)$ of $x(t)$, i.e., as

$$\overline{S}_X = \int_{-\infty}^{\infty} \overline{R}_X(\tau)e^{-j\omega\tau}\mathrm{d}\tau. \tag{4.105}$$

Example Find the power spectral density of the sinusoidal signal $x(t) = A\cos(\omega_0 t + \theta)$, which is illustrated in Fig. 4.9.

Solution

$$\overline{R}_X(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} A^2 \cos(\omega_0 t + \theta)\cos\left[\omega_0(t+\tau) + \theta\right]\mathrm{d}t$$

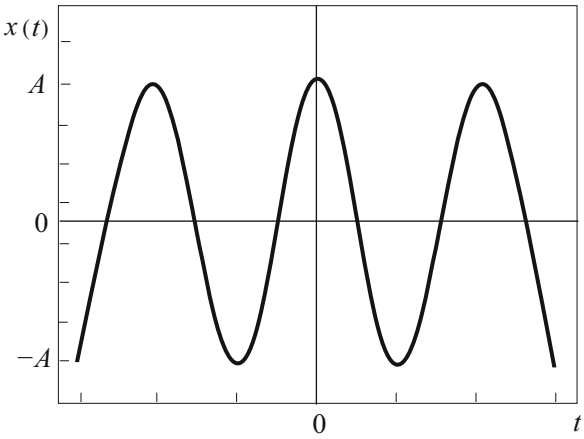


Fig. 4.9 Sinusoidal signal

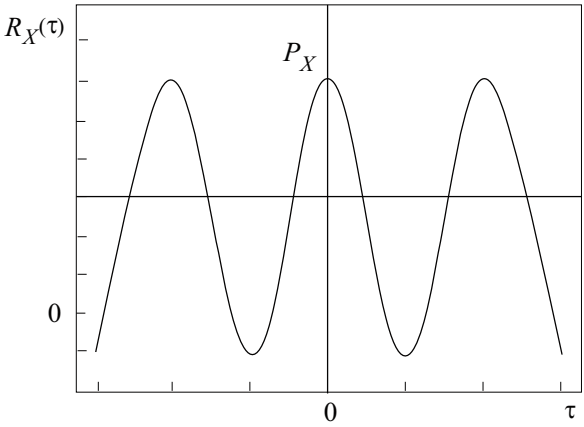


Fig. 4.10 Autocorrelation of the sinusoidal signal

$$= \frac{A^2}{2} \lim_{T\to\infty} \frac{1}{T} \left[\int_{\frac{-T}{2}}^{\frac{T}{2}} \cos \omega_0 \tau \, \mathrm{d}t + \int_{\frac{-T}{2}}^{\frac{T}{2}} \cos\left(2\omega_0 t + \omega_0 \tau + 2\theta\right) \mathrm{d}t \right]$$

$$= \frac{A^2}{2} \cos \omega_0 \tau .$$

Notice that the autocorrelation function, depicted in Fig. 4.10, is independent of the phase θ. The power spectral density, shown in Fig. 4.11, is given by

$$\overline{S}_X(\omega) = \mathcal{F}\left[R_X(\tau)\right]$$

$$\overline{S}_X(\omega) = \frac{\pi A^2}{2} \left[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)\right].$$

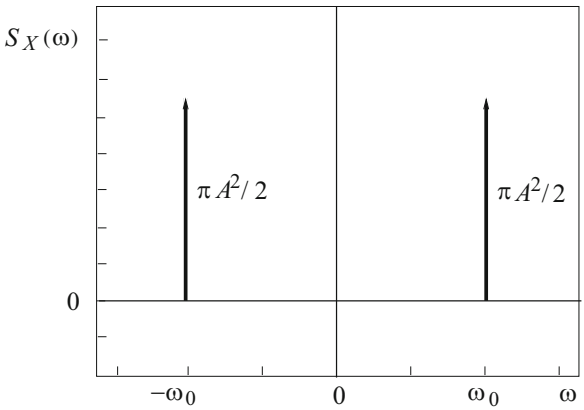


Fig. 4.11 Power spectral density of the sinusoidal signal

Finally, the power, or mean square average, of $x(t)$ is given by

$$\overline{P}_X = \overline{R}_X(0) = \frac{A^2}{2}.$$

4.7 The Hilbert Transform

The Hilbert transform is a time-domain-to-time-domain transformation which shifts the phase of a signal. The application of a Hilbert transform to a signal, twice in succession, shifts the phases of all of the components of this signal by 180^0 and therefore produces the negative of the original signal. It is named after the German mathematician David Hilbert (1862–1943) (Hilbert 1912).

The Hilbert transform of a time signal $f(t)$ is denoted by $\mathcal{H}[f(t)]$, or $\hat{f}(t)$, and defined by the following integral (Gagliardi 1978):

$$\mathcal{H}[f(t)] = \hat{f}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f(\tau)}{t-\tau}\mathrm{d}\tau \tag{4.106}$$

The functions $f(t)$ and $\hat{f}(t)$ form a pair of Hilbert transforms. It is important to mention that the transform is actually the Cauchy principal value of the integral.

The Hilbert transform is a linear operation, and its inverse is given by

$$\mathcal{H}^{-1}[\hat{f}(t)] = f(t) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\hat{f}(t)}{t-\tau}\mathrm{d}\tau. \tag{4.107}$$

From the definition, it is possible to interpret $\hat{f}(t)$ as the convolution of $f(t)$ with the function $\frac{1}{\pi t}$. From a known property of the Fourier transform, the convolution

in time is equivalent to the multiplication in the frequency domain. Therefore,

$$G(\omega) = \mathcal{F}\left[f(t) * \frac{1}{\pi t}\right] = \mathcal{F}[f(t)] \cdot \mathcal{F}\left[\frac{1}{\pi t}\right]. \tag{4.108}$$

4.8 Linearity of the Hilbert Transform

The Hilbert transform is also a linear operation. Consider the transform of the linear combination of functions $h(t) = \alpha f(t) + \beta g(t)$,

$$\begin{aligned} \mathcal{H}[g(t)] &= \mathcal{H}[\alpha f(t) + \beta g(t)] \\ &= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\alpha f(\tau) + \beta g(\tau)}{t-\tau} \mathrm{d}\tau \\ &= \frac{\alpha}{\pi}\int_{-\infty}^{\infty} \frac{f(\tau)}{t-\tau} \mathrm{d}\tau + \frac{\beta}{\pi}\int_{-\infty}^{\infty} \frac{g(\tau)}{t-\tau} \mathrm{d}\tau \\ &= \alpha\mathcal{H}[f(t)] + \beta\mathcal{H}[g(t)]. \end{aligned} \tag{4.109}$$

Example The Hilbert transform for a constant function $f(t) = c$ is easy to calculate using the definition

$$\mathcal{H}[c] = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{c}{t-\tau} \mathrm{d}\tau = \frac{c}{\pi}\int_{-\infty}^{\infty} \frac{1}{t-\tau} \mathrm{d}\tau = 0. \tag{4.110}$$

The last equality is a result of the integrand $1/(t-\tau)$ being an odd function over a symmetric interval around $t = \tau$. Hence, $\mathcal{H}[c] = 0$ for any constant c.

It is not always easy and straightforward to compute the Hilbert transform for a general function. For more advanced functions, it is necessary to use techniques from complex analysis to handle the integral. These techniques include contour integrals in the complex plane and the residue theorem. The singularity in $t = \tau$ on the real line has to be taken care of in this case.

4.9 Multiple Hilbert Transforms

The Hilbert transform applied twice on a real function produces the same real function but changes its sign. That is,

$$\mathcal{H}^2[f(t)] = \mathcal{H}[\mathcal{H}[f(t)]] = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\hat{f}(\tau)}{t-\tau} \mathrm{d}\tau = -f(t). \tag{4.111}$$

From this property, it is possible to derive the following,

$$\mathcal{H}^4[f(t)] = f(t), \tag{4.112}$$

that is, it is possible to use the multiple Hilbert transform to calculate the inverse Hilbert transform,

$$\mathcal{H}^{-1}[\hat{f}(t)] = \mathcal{H}^3[f(t)]. \tag{4.113}$$

By multiplying the Hilbert transform operator by itself, one obtains a practical method to do multiple Hilbert transforms:

$$\mathcal{H}^{-1}[\hat{f}(t)] = \mathcal{H}^3[f(t)] \tag{4.114}$$

Hilbert transform can be applied in the time domain by using the former definition. In the frequency domain, it is only necessary to multiply the Hilbert transform operator $-j\text{sgn}(\omega)$ by the Fourier transform $F(\omega)$ of the function $f(t)$,

$$\mathcal{H}^n[f(t)] \leftrightarrow [-j\text{sgn}\,(\omega)]^n F(\omega), \tag{4.115}$$

in which n is the number of time that the Hilbert transform is applied.

4.10 The Fourier Transform of $\frac{1}{\pi t}$

To obtain the Fourier transform of the hyperbolic function, $g(t) = \frac{1}{\pi t}$, shown in Fig. 4.12, it is possible to use the following reasoning:

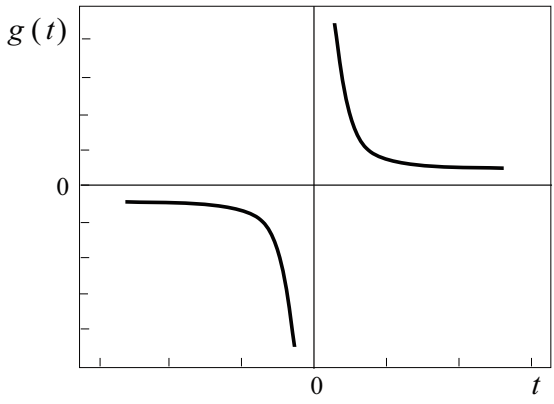


Fig. 4.12 Signal function

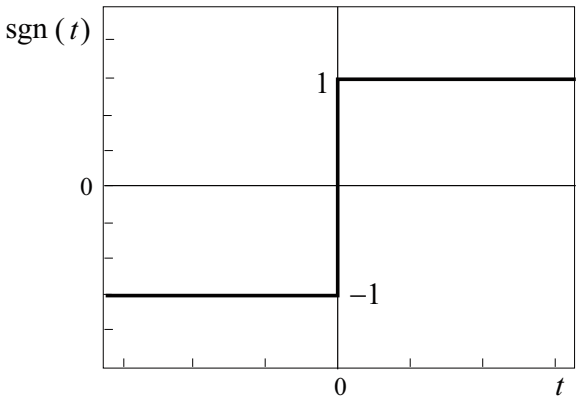


Fig. 4.13 Signal function

- Consider the signal function presented in Fig. 4.13.
- The derivative of this function results in an impulse centered at the origin of the axis, with total area equal to 2, and the Fourier transform of the impulse is equal to 2.
- Because the impulse has been obtained from the signal function, then, its Fourier transform is obtained dividing the previous result by $j\omega$. Therefore,

$$\text{sgn}(t) = \text{u}(t) - \text{u}(-t) \leftrightarrow \frac{2}{j\omega}. \tag{4.116}$$

- Finally, using the symmetry property, one obtains

$$\frac{1}{\pi t} \leftrightarrow j[\text{u}(-\omega) - \text{u}(\omega)] = -j\text{sgn}(\omega).$$

Therefore, the Fourier transform of the hyperbolic function is the imaginary signal function, as follows:

$$G(\omega) = -j\text{sgn}(\omega) \tag{4.117}$$

This property helps computing the Hilbert transform because it is possible to use the available tables of Fourier transform with this objective.

4.11 Hilbert Transform of the Signal Derivative

The Hilbert transform of the derivative of a signal is the derivative of the Hilbert transform of that signal. This can be verified using a property of the doublet function $\delta'(t)$.

Recall that

$$\delta'(t) = \frac{\mathrm{d}}{\mathrm{d}t}\delta(t) = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathrm{u}(t), \tag{4.118}$$

in which $\delta(t)$ is the impulse function and $\mathrm{u}(t)$ is the unit step function.

Because, by a property of the doublet function, $g(t) * \delta'(t) = g'(t)$, then

$$\mathcal{H}[g'(t)] = [g(t) * \delta'(t)] * \frac{1}{\pi t} = \left[g(t) * \frac{1}{\pi t}\right] * \delta'(t) = \frac{\mathrm{d}}{\mathrm{d}t}\hat{g}(t). \tag{4.119}$$

Iterating this identity, one obtains

$$\mathcal{H}[g^{(n)}(t)] = \frac{\mathrm{d}^n}{\mathrm{d}t^n}\hat{g}(t). \tag{4.120}$$

Example Calculate the Hilbert transform of the delta function $\delta(t)$ and its derivatives. Consider the Hilbert transform of the delta function,

$$\mathcal{H}[\delta(t)] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\delta(\tau)}{t-\tau}\mathrm{d}\tau = \frac{1}{\pi t}, \tag{4.121}$$

by the fundamental property of the convolution with an impulse function.

Using the property of the Hilbert transform derivative, one obtains

$$\mathcal{H}[\delta'(t)] = -\frac{1}{\pi t^2}. \tag{4.122}$$

By the same token,

$$\mathcal{H}[\delta''(t)] = \frac{2}{\pi t^3}. \tag{4.123}$$

4.12 Hilbert Transform of Trigonometric Functions

The Hilbert transform of the cosine function can be computed directly from the definition. Considering that $f(t) = \cos(\omega t)$, then

$$\begin{aligned}\mathcal{H}[\cos(\omega t)] &= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\cos(\omega\tau)}{t-\tau}\mathrm{d}\tau, \text{ change variable } \tau = \sigma + t \\ &= -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\cos(\omega(\sigma+t))}{\sigma}\mathrm{d}\sigma\end{aligned} \tag{4.124}$$

$$
\begin{aligned}
&= -\frac{1}{\pi}\left[\int_{-\infty}^{\infty}\cos(\omega t)\frac{\cos(\omega\sigma)}{\sigma}\mathrm{d}\sigma - \int_{-\infty}^{\infty}\sin(\omega t)\frac{\sin(\omega\sigma)}{\sigma}\mathrm{d}\sigma\right]\\
&= -\frac{1}{\pi}\left[\cos(\omega t)\int_{-\infty}^{\infty}\frac{\cos(\omega\sigma)}{\sigma}\mathrm{d}\sigma - \sin(\omega t)\int_{-\infty}^{\infty}\frac{\sin(\omega\sigma)}{\sigma}\mathrm{d}\sigma\right]\\
&= \sin(\omega t).
\end{aligned}
$$

The result comes from the fact that $\cos(\omega\sigma)/\sigma$ is an odd function and

$$\int_{-\infty}^{\infty}\frac{\sin(\omega\sigma)}{\sigma}\mathrm{d}\sigma = \pi. \tag{4.125}$$

Following the same reasoning, one obtains

$$
\begin{aligned}
\mathcal{H}[\sin(\omega t)] &= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin(\omega\tau)}{t-\tau}\mathrm{d}\tau,\ \text{change variable } \tau = \sigma + t \qquad (4.126)\\
&= -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin(\omega(\sigma+t))}{\sigma}\mathrm{d}\sigma\\
&= -\frac{1}{\pi}\left[\int_{-\infty}^{\infty}\cos(\omega t)\frac{\sin(\omega\sigma)}{\sigma}\mathrm{d}\sigma - \int_{-\infty}^{\infty}\sin(\omega t)\frac{\cos(\omega\sigma)}{\sigma}\mathrm{d}\sigma\right]\\
&= -\frac{1}{\pi}\left[\cos(\omega t)\int_{-\infty}^{\infty}\frac{\sin(\omega\sigma)}{\sigma}\mathrm{d}\sigma - \sin(\omega t)\int_{-\infty}^{\infty}\frac{\cos(\omega\sigma)}{\sigma}\mathrm{d}\sigma\right]\\
&= -\cos(\omega t).
\end{aligned}
$$

4.13 Hilbert Transform of the Complex Exponential Functions

The Hilbert transform of the complex exponential function $e^{j\omega t}$ can be computed in the following manner,

$$
\begin{aligned}
\mathcal{H}[e^{j\omega t}] &= \mathcal{H}[\cos(\omega t) + j\sin(\omega t)] = \sin(\omega t) - j\cos(\omega t) \qquad (4.127)\\
&= -j\,\mathrm{sgn}\,(\omega)e^{j\omega t} = \mathrm{sgn}\,(\omega)e^{j(\omega t-\pi/2)},
\end{aligned}
$$

in which the operator $-j\,\mathrm{sgn}\,(\omega)$ produces a phase lag of $\pi/2$ for all positive frequencies and a phase lead of $\pi/2$ for the negative frequencies.

By the same token, the Hilbert transform of the complex exponential function $e^{-j\omega t}$ can be computed as

$$
\begin{aligned}
\mathcal{H}[e^{-j\omega t}] &= \mathcal{H}[\cos(\omega t) - j\sin(\omega t)] = \sin(\omega t) + j\cos(\omega t) \qquad (4.128)\\
&= j\,\mathrm{sgn}\,(\omega)e^{-j\omega t} = \mathrm{sgn}\,(\omega)e^{-j(\omega t-\pi/2)}.
\end{aligned}
$$

4.14 Hilbert Transform of Periodic Functions

A periodic function can be written in trigonometric form as

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t + \phi_n), \tag{4.129}$$

in which $\omega_0 = 2\pi/T$ and T is the period of the waveform.

Applying the Hilbert transform to the previous signal, using the discussed properties, one obtains

$$\hat{f}(t) = \mathcal{H}[f(t)] = \sum_{n=1}^{\infty} a_n \sin(n\omega_0 t + \phi_n), \tag{4.130}$$

because the Hilbert transform of a constant is zero.

A periodic signal can also be expressed in complex form, using the complex exponential formula, also known as Euler's formula, $e^{jn\omega_0 t} = \cos(n\omega_0 t) + j\sin(n\omega_0 t)$. This formula was originally derived by Leonhard Paul Euler (1707–1783), a Swiss mathematician, physicist, astronomer, geographer, logician, and engineer, who founded the studies of graph theory and topology and made pioneering discoveries in many other branches of Mathematics, including analytic number theory, complex analysis, and infinitesimal calculus:

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{n\omega_0 t} \tag{4.131}$$

Applying the Hilbert transform signal, one obtains

$$\hat{f}(t) = \mathcal{H}[f(t)] = \sum_{n=-\infty}^{\infty} C_n \mathcal{H}[e^{n\omega_0 t}] = -\sum_{n=-\infty}^{\infty} C_n j\,\mathrm{sgn}(n)\, e^{n\omega_0 t}. \tag{4.132}$$

4.15 Properties of the Hilbert Transform

The Hilbert transform has the following properties (Johansson 1999):

- The Hilbert transform is a linear operation; thus,

$$\mathcal{H}[\alpha f(t) + \beta g(t)] = \alpha\mathcal{H}[f(t)] + \beta\mathcal{H}[g(t)]. \tag{4.133}$$

- The Hilbert transform of a constant is zero, as follows,

$$\mathcal{H}[c] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{c}{t-\tau}\mathrm{d}\tau = \frac{c}{\pi}\int_{-\infty}^{\infty}\frac{1}{\sigma}\mathrm{d}\sigma = 0, \tag{4.134}$$

in which the change of variables $\sigma = t - \tau$ and $\mathrm{d}\tau = -\mathrm{d}\sigma$ was performed.

- The time shift property implies that

$$\mathcal{H}[f(t-\alpha)] = \hat{f}(t-\alpha). \tag{4.135}$$

- The scaling property implies that

$$\mathcal{H}[f(\alpha t)] = \hat{f}(\alpha t), \alpha > 0. \tag{4.136}$$

- The time reversal property implies that

$$\mathcal{H}[f(-\alpha t)] = -\hat{f}(-\alpha t), \alpha > 0. \tag{4.137}$$

- The derivative property implies that

$$\mathcal{H}[f'(t)] = \hat{f}'(t). \tag{4.138}$$

In this case, computing the derivative of the Hilbert transform, one obtains

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}[f(t)] = \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f(\tau)}{t-\tau}\mathrm{d}\tau\right] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\partial}{\partial t}\left[\frac{f(\tau)}{t-\tau}\right]\mathrm{d}\tau,$$

which, by a change of variables, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}[f(t)] = \mathcal{H}\left[\frac{\mathrm{d}}{\mathrm{d}t}f(t)\right].$$

- The Hilbert transform of the convolution of functions $f(t)$ and $g(t)$ yields

$$\mathcal{H}[f(t) * g(t)] = \mathcal{H}[f(t)] * g(t) = f(t) * \mathcal{H}[g(t)]. \tag{4.139}$$

- Applying the Hilbert transform of the convolution of functions $f(t)$ and $g(t)$ in Formula 4.139 gives

$$\mathcal{H}[\mathcal{H}[f(t) * g(t)]] = -\mathcal{H}[f(t)] * \mathcal{H}[g(t)]. \tag{4.140}$$

- The signal $f(t)$ and its transform $\hat{f}(t)$ have the same power spectral density.
- The signal $f(t)$ and its transform $\hat{f}(t)$ have the same autocorrelation function.
- The signal $f(t)$ and its transform $\hat{f}(t)$ are orthogonal.
- If $\hat{f}(t)$ is the Hilbert transform of $f(t)$, then the Hilbert transform of $\hat{f}(t)$ is $-f(t)$.

For a function $f(t)$ that is either even or odd and has a well-defined Hilbert transform, the following holds:

- If $f(t)$ is even, its Hilbert transform is an odd function.
- If $f(t)$ is odd, its Hilbert transform is an even function.

It is important to remember that a signal $f(t)$ and its Hilbert transform $\hat{f}(t)$ have the same amplitude spectrum and the same autocorrelation function. Also, the power spectral density is the same for both $f(t)$ and $\hat{f}(t)$. Finally, if the Fourier transform of $f(t)$ exists, then the Hilbert transform also exists for energy and power signals.

Example Using the corresponding properties, it is possible to calculate the Hilbert transforms of the basic trigonometric functions. Supposing that $\omega > 0$, then

$$\begin{aligned}\mathcal{H}[\cos(\omega t)] &= \mathcal{H}\left[\frac{e^{j\omega t}+e^{-j\omega t}}{2}\right] = \frac{\mathcal{H}[e^{j\omega t}]+\mathcal{H}[e^{-j\omega t}]}{2} \\ &= \frac{-j\,\mathrm{sgn}(\omega)e^{j\omega t} - j\,\mathrm{sgn}(-\omega)e^{-j\omega t}}{2} \\ &= \frac{-je^{j\omega t}+je^{-j\omega t}}{2} = \frac{e^{j\omega t}-e^{-j\omega t}}{2j} = \sin(\omega t).\end{aligned} \tag{4.141}$$

By the same token,

$$\mathcal{H}[\sin(\omega t)] = \mathcal{H}\left[\cos\left(\omega t - \frac{\pi}{2}\right)\right] = \sin(\omega t - \frac{\pi}{2}) = -\cos(\omega t). \tag{4.142}$$

Concluding, if $\omega > 0$, then $\mathcal{H}[\cos(\omega t)] = \sin(\omega t)$, and $\mathcal{H}[\sin(\omega t)] = -\cos(\omega t)$. Using the time reversal property, if $\omega < 0$, then $\mathcal{H}[\cos(\omega t)] = -\sin(\omega t)$, and $\mathcal{H}[\sin(\omega t)] = -[-\cos(\omega t)] = \cos(\omega t)$.

It is possible to write a compact set of formulas, using the signum function, for all $\omega \in \mathbb{R}$,

$$\begin{aligned}\mathcal{H}[\cos(\omega t)] &= \mathrm{sgn}(\omega)\sin(\omega t) \\ \mathcal{H}[\sin(\omega t)] &= -\mathrm{sgn}(\omega)\cos(\omega t),\end{aligned} \tag{4.143}$$

which are the Hilbert transforms for the cosine and sine functions.

4.16 Orthogonality of the Hilbert Transform

If $f(t)$ is a real-valued energy signal, then $f(t)$ and $\hat{f}(t)$ are orthogonal. To prove this, recall that a complex function is called Hermitian if its real part is even and its imaginary part is odd. Therefore, the Fourier transform $F(\omega)$ of a real function $f(t)$ is Hermitian:

$$\begin{aligned}\langle f(t), \hat{f}(t)\rangle &= \int_{-\infty}^{\infty} f(t)\hat{f}^*(t)\mathrm{d}t \\ &= \int_{-\infty}^{\infty} F(\omega)\hat{F}^*(\omega)\mathrm{d}\omega \\ &= \int_{-\infty}^{\infty} F(\omega)[-j\,\mathrm{sgn}(\omega)\hat{F}(\omega)]^*\mathrm{d}\omega \\ &= \int_{-\infty}^{\infty} j|F(\omega)|^2\mathrm{sgn}(\omega)\mathrm{d}\omega \\ &= 0,\end{aligned} \tag{4.144}$$

in which the use has been made of the property that since $|G(\omega)|^2$ is an even function, then $|G(\omega)|^2\mathrm{sgn}(\omega)$ is an odd function of ω and hence the value of the integral is zero.

4.17 Energy of a Hilbert Transform of a Signal

The energy of a function $f(t)$ is related to the energy of its Fourier transform $F(\omega)$ by Parseval's theorem,

$$E_f = \int_{-\infty}^{\infty} |f(t)|^2\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\mathrm{d}\omega, \tag{4.145}$$

assuming that the energy is finite.

The same theorem can be used to define the energy of the Hilbert transform of $f(t)$, that is,

$$\begin{aligned}E_{\hat{f}} = \int_{-\infty}^{\infty} |\hat{f}(t)|^2\mathrm{d}t &= \frac{1}{2\pi}\int_{-\infty}^{\infty} |-j\,\mathrm{sgn}\,(\omega)F(\omega)|^2\mathrm{d}\omega, \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} |-j\,\mathrm{sgn}\,(\omega)|^2|F(\omega)|^2\mathrm{d}\omega.\end{aligned} \tag{4.146}$$

It can be seen that $|-j\,\mathrm{sgn}\,(\omega)|^2 = |-j|^2|\mathrm{sgn}\,(\omega)|^2 = 1$, except for $\omega = 0$. But since $F(\omega)$ does not contain any impulses at the origin, one concludes that $E_f = E_{\hat{f}}$.

Example Compute the Hilbert transform of $tf(t)$. In order to solve this, consider the original definition of the Hilbert transform,

$$\mathcal{H}[tf(t)] = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\tau f(\tau)}{t-\tau}\mathrm{d}\tau,$$

and insert the new variable $\sigma = t - \tau$ to obtain

$$\begin{aligned}\mathcal{H}[tf(t)] &= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{(t-\sigma)f(t-\sigma)}{\sigma}\mathrm{d}\sigma \\ &= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{tf(t-\sigma)}{\sigma}\mathrm{d}\sigma - \frac{1}{\pi}\int_{-\infty}^{\infty}f(t-\sigma)\mathrm{d}\sigma \\ &= t\mathcal{H}[f(t)] - \frac{1}{\pi}\int_{-\infty}^{\infty}f(t-\sigma)\mathrm{d}\sigma.\end{aligned} \tag{4.147}$$

4.18 Hilbert Transform of a Low-Pass Signal

Let $f(t)$ be signal whose Fourier transform satisfies $F(\omega) = 0$ for $|\omega| \geq W$, and let $h(t)$ be a signal with $H(\omega) = 0$ for $|f| < W$. Then (Kschischang 2015),

$$\mathcal{H}[f(t)h(t)] = f(t)\hat{h}(t), \tag{4.148}$$

that is, it is only necessary to transform the high-pass signal to compute the Hilbert transform of the product of a low-pass signal with a high-pass signal.

To prove this property, let $g(t) = f(t)h(t)$. Then,

$$G(\omega) = F(\omega) * H(\omega) = F(\omega) * [H(\omega)\mathrm{u}(\omega) + H(\omega)\mathrm{u}(-\omega)],$$

in which $\mathrm{u}(\omega)$ is the unit step function.

It is possible to note that $F(\omega) * [H(\omega)\mathrm{u}(\omega)]$ is zero if $\omega < 0$ and that $F(\omega) * [H(\omega)\mathrm{u}(-\omega)]$ is zero if $\omega > 0$. Therefore,

$$\begin{aligned}\hat{G}(\omega) &= -j\mathrm{sgn}(\omega)G(\omega) \\ &= -jF(\omega) * [H(\omega)\mathrm{u}(\omega)] + jF(\omega)[H(\omega)\mathrm{u}(-\omega)] \\ &= F(\omega) * [-jH(\omega)\mathrm{u}(\omega) + jH(\omega)\mathrm{u}(-\omega)] \\ &= F(\omega) * [-j\mathrm{sgn}(\omega)H(\omega)] \\ &= F(\omega) * \hat{H}(\omega).\end{aligned} \tag{4.149}$$

An important special case of this property appears in the case of quadrature amplitude modulation (QAM) modulation. Assuming that $m_I(t)$, the in-phase component, and $m_Q(t)$, the quadrature component of the modulating signal, are bandlimited to W Hz, then if $\omega_c > W$, it can be shown that

$$\mathcal{H}[m_I(t)\cos(\omega_c t) + m_Q(t)\sin(\omega_c t)] = m_I(t)\sin(\omega_c t) - m_Q(t)\cos(\omega_c t), \tag{4.150}$$

in which ω_c is the carrier frequency.

4.19 Hilbert Transform of an Amplitude-Modulated Signal

The Hilbert transform of an amplitude-modulated signal is given by Kschischang (2015)

$$\mathcal{H}[m(t)\cos(\omega_c t+\phi)] = \left[m(t) * \frac{\cos(\omega_c t)}{\pi t}\right]\cos(\omega_c t+\phi) + \left[m(t) * \frac{\sin(\omega_c t)}{\pi t}\right]\sin(\omega_c t+\phi). \tag{4.151}$$

In order to obtain this result, it is necessary to write $m(t) = m_L(t) + m_H(t)$, in which $m_L(t)$ is the low-pass component of the modulating signal $m(t)$, given by

$$m_L(t) = m(t) * \frac{\sin(\omega_c t)}{\pi t},$$

and $m_H(t)$ is the high-pass component

$$m_H(t) = m(t) * \left[\delta(t) - \frac{\sin(\omega_c t)}{\pi t}\right].$$

Note that $m_L(t)$ contains all frequency components of $m(t)$ from zero up to frequency ω_c, while $m_H(t)$ contains the remaining components at higher frequencies. Therefore,

$$\begin{aligned}\mathcal{H}[m(t)\cos(\omega_c t+\phi)] &= \mathcal{H}[m_L(t)\cos(\omega_c t+\phi)] \\ &+ \mathcal{H}[m_H(t)\cos(\omega_c t+\phi)] \\ &= m_L(t)\mathcal{H}[\cos(\omega_c t+\phi)] \\ &+ m_H(t)\mathcal{H}[\cos(\omega_c t+\phi)],\end{aligned} \tag{4.152}$$

in which the second equality follows from application of the previous results on low-pass and high-pass products.

From the properties of the Hilbert transform, $\mathcal{H}[\cos(\omega_c t+\phi)] = \sin(\omega_c t+\phi)$. The remaining term can be computed as follows,

$$\begin{aligned}\mathcal{H}[m_H(t)] &= m(t) * \left[\delta(t) - \frac{\sin(\omega_c t)}{\pi t}\right] * \frac{1}{\pi t} \\ &= m(t) * \left[\frac{1}{\pi t} - \frac{\sin(\omega_c t)}{\pi t} * \frac{1}{\pi t}\right] \\ &= m(t) * \left[\frac{1}{\pi t} - \frac{\sin(\omega_c t)}{\pi t} * \frac{1}{\pi t}\right]\end{aligned} \tag{4.153}$$

$$= m(t) * \left[\frac{1}{\pi t} [1 - 2\sin^2(\omega_c t/2)] \right]$$
$$= m(t) * \frac{\cos(\omega_c t)}{\pi t},$$

which leads to the main result. In the derivation, one used the fact that $\mathcal{H}[\text{sinc}(t)] = \text{sinc}(t)\sin(\pi t/2)$.

4.20 Problems

1. Consider the signal

$$x(t) = \begin{cases} 1, & 0 \leq t < \pi \\ -1, & \pi \leq t \leq 2\pi \end{cases}$$

which is approximated as $\tilde{x}(t) = \frac{4}{\pi}\sin(t)$ in the time interval considered.

   (a) Show that the error in the approximation is orthogonal to the function $\tilde{x}(t)$.
   (b) Show that the energy of $x(t)$ is the sum of the energy in the error signal with the energy of the signal $\tilde{x}(t)$.

2. Calculate the instantaneous power and the average power of the following signals:

   (a) $x(t) = \cos(2\pi t)$
   (b) $y(t) = \sin(2\pi t)$
   (c) $z(t) = x(t) + y(t)$

3. Determine the constant A such that $f_1(t)$ and $f_2(t)$ are orthogonal for all t, in which $f_1(t) = e^{-|t|}$ and $f_2(t) = 1 - Ae^{-2|t|}$.
4. Given the set of functions $f_n(t)$, as illustrated in Fig. 4.2, show the following:

   (a) This set of functions constitutes an orthogonal set in the interval (0, 1). For an orthonormal set, the integral of the product of the functions is one or zero. Is the set orthonormal?
   (b) Represent a given signal $f(t) = 2t$ in the interval (0, 1), using this set of orthogonal functions.
   (c) Plot the function $f(t)$ and its approximate representation $\tilde{f}(t)$ in the same graph.
   (d) Determine the energy of the error signal resulting from the approximation.

5. Represent the gate function, and its complement, using the unit step.
6. Represent analytically the graph of a series of triangular functions, using generalized functions.
7. Calculate the following integrals:

(a) $\int_{-\infty}^{\infty} e^{-\alpha t} u(t)\mathrm{d}t$
(b) $\int_{-\infty}^{\infty} e^{-\alpha t} \delta(t)\mathrm{d}t$
(c) $\int_{-\infty}^{\infty} e^{-\alpha t} r(t)\mathrm{d}t$ for $r(t) = \int_{-\infty}^{t} u(\tau)d\tau$

8. Calculate the Fourier transform of the impulse function assuming the Fourier transform of the unit step function is known.
9. Calculate the inverse Fourier transform of the function

$$F(\omega) = A[u(\omega + \omega_0) - u(\omega - \omega_0)].$$

10. Calculate the Fourier transform of the function $f(t) = Ae^{-\alpha t}u(t)$, and plot the corresponding magnitude and phase diagrams.
11. Plot the magnitude and phase diagrams of the Fourier transform of the function $\delta(t + t_0)$.
12. For a circuit with impulse response $h(t)$,

$$h(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} u(t),$$

find the response for the excitation $x(t)$ given by

$$x(t) = te^{-\frac{t}{\tau}} u(t).$$

13. Find the Fourier transform of the function $g(t) = f(t)\cos(\omega_c t)$, given the Fourier transform of $f(t)$.
14. Show that for a function $f(t)$ in general,

$$\int_{-\infty}^{t} f(t)\mathrm{d}t \longleftrightarrow \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega).$$

15. Prove that, for a real energy signal $f(t)$, the energy associated with $f(t)$,

$$\int_{-\infty}^{\infty} f^2(t)\mathrm{d}t,$$

can be calculated by the formula

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega.$$

16. Use the property of the convolution in the frequency domain to show that

$$\cos(\omega_c t)u(t) \longleftrightarrow \frac{\pi}{2}[\delta(\omega + \omega_c) + \delta(\omega - \omega_c)] + j\frac{\omega}{\omega_c^2 - \omega^2}.$$

17. A signal $x(t)$ has the exponential Fourier series expansion as given:

$$x(t) = -\frac{2A}{\pi} \sum_{n=-\infty}^{\infty} \frac{1}{4n^2 - 1} e^{j2\pi nt}$$

Find its corresponding trigonometric Fourier series expansion.

18. By defining the cutoff frequency as the smallest frequency for which the first spectral zero occurs, determine the cutoff frequency (ω_0) of the signal $x(t)$ in Fig. 4.14.
19. Calculate the Fourier transform of the signals represented in Figs. 4.15 and 4.16.
20. Calculate the frequency response of a linear system the transfer function of which is given, when the input is the pulse $x(t) = A[u(t + T/2) - u(t - T/2)]$. Plot the corresponding magnitude and phase diagrams of the frequency

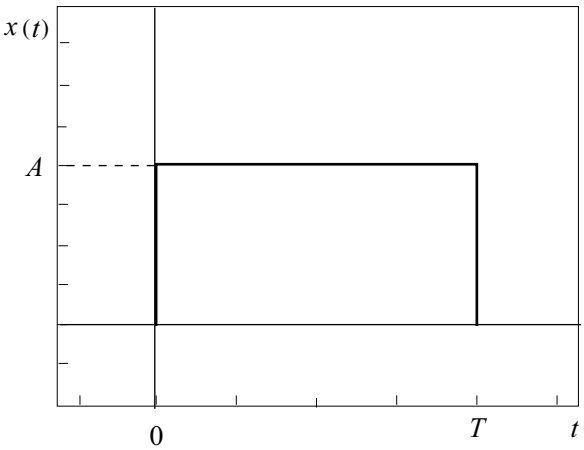


Fig. 4.14 Shifted gate function

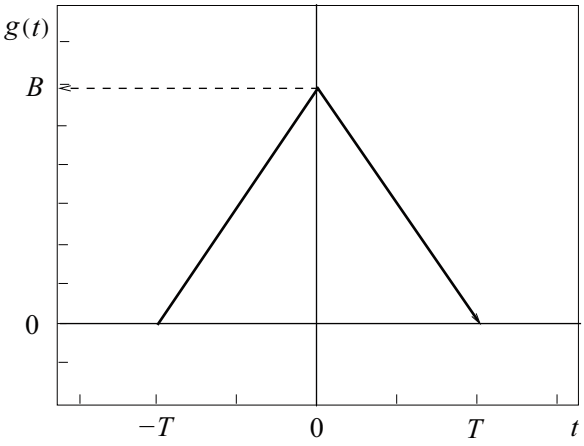


Fig. 4.15 Triangular waveform

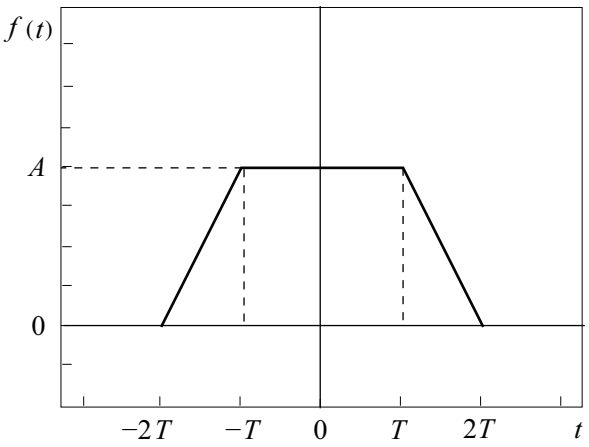


Fig. 4.16 Trapezoidal waveform

response:

$$H(\omega) = ju(-\omega) - ju(\omega)$$

21. A signal $x(t)$ is given by the expression

$$x(t) = \frac{\sin(At)}{\pi t}.$$

Determine the Nyquist frequency for sampling this signal.

22. What is the least sampling rate that is required to sample the signal $f(t) = \sin^3(\omega_0 t)$? Show graphically the effect caused by a reduction of the sampling rate, falling below the Nyquist rate.
23. Calculate the Fourier transform of the signal

$$g(t) = Ae^{-t}u(t),$$

and then apply the property of integration in the time domain to obtain the Fourier transform of $f(t) = A(1 - e^{-t})u(t)$.

24. Determine the magnitude spectrum and phase function of the signal $f(t) = te^{-at}u(t)$.
25. A voltage signal

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n)$$

is applied to the input of a circuit, producing the current

$$i(t) = I_0 + \sum_{m=1}^{\infty} I_m \cos(m\omega_0 t + \phi_m).$$

Using the orthogonality concept, calculate the power (P) absorbed by the circuit, considering

$$P = \frac{1}{T}\int_{-T/2}^{T/2} v(t)i(t)\mathrm{d}t.$$

What is the power for the case in which $\theta_n = \phi_n$?

26. Define an ideal low-pass filter, and explain why it is not physically realizable. Indicate the corresponding filter transfer function and the filter impulse response.
27. Given the linear system shown in Fig. 4.17, in which T represents a constant delay, determine

    (a) The system transfer function $H(\omega)$
    (b) The system impulse response $h(t)$

28. Let $f(t)$ be the signal with spectrum $F(\omega)$ as follows,

$$F(\omega) = \frac{AT}{1 + jwT}, \text{ in which } T = 0.5\,\mu\text{s}, \ A = 5\,\text{V}.$$

    (a) Calculate and plot the magnitude of the Fourier transform, $|F(\omega)|$.
    (b) Calculate the frequency for which $|F(\omega)|$ corresponds to a value 3dB below the maximum amplitude value in the spectrum.
    (c) Calculate the energy of the signal in time, $f(t)$.

29. Represent the following signals using the unit step:

    (a) The ramp function, $r(t + T)$
    (b) The echo function, $\delta(t - T) + \delta(t + T)$
    (c) A periodic sawtooth waveform with period T and peak amplitude given by A

    Plot the corresponding graphs.

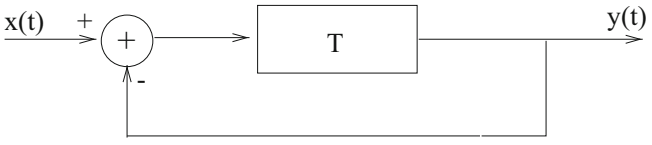


Fig. 4.17 Linear system with feedback

30. Calculate the Fourier transform of each one of the following signals:

    (a) $x(t) = e^{-t+t_o}u(t - t_o)$
    (b) $y(t) = tu(t)$
    (c) $z(t) = y'(t)$

    Draw the time signals, as well as the associated magnitude spectra.
31. Verify, by applying the properties of homogeneity and additivity, whether the process generating the signal

$$s(t) = A\cos(\omega_c t + \Delta m(t) + \theta)$$

    from the input signal $m(t)$ is linear. Perform the same test for the signal

$$r(t) = m(t)\cos(\omega_c t + \theta).$$

32. A linear system has impulse response $h(t) = 2[u(t) - u(t - T)]$. Using the convolution theorem, determine the system response to the input signal $x(t) = u(t) - u(t - T)$.
33. A digital signal $x(t)$ has autocorrelation function

$$\bar{R}_X(\tau) = A^2\left[1 - \frac{|\tau|}{T_b}\right][u(\tau + T_b) - u(\tau - T_b)],$$

    in which T_b is the bit duration. Determine the total power, the AC power, and the DC power of the given signal. Calculate the signal power spectral density. Plot the diagrams representing these functions.
34. Calculate the Fourier transform of a periodic signal $x(t)$, represented analytically as

$$x(t) = \sum_{-\infty}^{\infty} F_n e^{jn\omega_0 t}.$$

35. Prove the following property of the Fourier transform:

$$x(\alpha t) \longleftrightarrow \frac{1}{|\alpha|}X\left(\frac{\omega}{\alpha}\right)$$

36. Calculate the average value and the power of the signal

$$x(t) = Vu(\cos t).$$

37. For a given real signal $x(t)$, prove the following Parseval identity:

$$E = \int_{-\infty}^{\infty} x^2(t)\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Consider the signals $x(t)$ and $y(t) = x(t - \tau)$, and show, by using Parseval's identity, that both signals have the same energy E.

38. A signal $y(t)$ is given by the following expression,

$$y(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau,$$

in which the signal $x(t)$ has a Fourier transform $X(\omega)$. By using properties of the Fourier transform, determine the Fourier transform of $y(t)$.

39. Given that the Fourier transform of the signal $f(t) = \cos(\omega_o t)$ is $F(\omega) = \pi[\delta(w+\omega_o)+\delta(w-\omega_o)]$, determine the Fourier transform of the signal $g(t) = \sin(\omega_o t - \phi)$, in which ϕ is a phase constant. Sketch the magnitude and phase graphs of the Fourier transform of this signal.

40. Calculate the Fourier transform of the radio frequency pulse

$$f(t) = \cos(\omega_o t)[u(t + T) - u(t - T)],$$

considering that $\omega_o \gg \frac{2\pi}{T}$. Sketch the magnitude and phase graphs of the Fourier transform of this signal.

41. Calculate the Fourier transform of the signal $f(t) = \delta(t) - \alpha e^{-\alpha t} u(t)$, and show, by using the property of the derivative in the time domain, that this signal is the derivative of the signal $g(t) = e^{-\alpha t} u(t)$. Sketch the respective time and frequency domain graphs of the signals, specifying the magnitude and the phase spectra of each signal.

42. By making use of properties of the Fourier transform, show that the derivative of the signal $h(t) = f(t) * g(t)$ can be expressed as

$$h'(t) = f'(t) * g(t), \quad \text{or} \quad h'(t) = f(t) * g'(t).$$

43. Determine the Nyquist frequency for which the following signal can be recovered without distortion:

$$f(t) = \frac{\sin \alpha t \cdot \sin \beta t}{t^2}, \quad \alpha > \beta.$$

Sketch the signal spectrum, and give a graphical description of the procedure.

44. Using the Fourier transform, show that the unit impulse function can be written as

$$\frac{1}{\pi} \int_0^{\infty} \cos(\omega t) d\omega.$$

45. Using properties of the Fourier transform, determine the Fourier transform of the function $|t|$. Plot the corresponding magnitude and phase spectrum of that transform.

46. Show that $u(t) * u(t) = r(t)$, in which $u(t)$ represents the unit step function and $r(t)$ denotes the ramp with slope 1.
47. Determine the Fourier transform for each one of the following functions: $u(t - T)$, t, $te^{-at}u(t)$, $\frac{1}{t}$, $\frac{1}{t^2}$. Plot the corresponding time-domain diagrams and the respective magnitude and phase spectrum of the associated transforms.
48. Calculate the Fourier transform $P(\omega)$ of the signal $p(t) = v^2(t)$ representing the instantaneous power in a 1- Ω resistor, as a function of the Fourier transform $V(\omega)$ of $v(t)$. Using the expression obtained for $P(\omega)$, plot the instantaneous power spectrum for a sinusoidal input signal $v(t) = A\cos(\omega_o t)$.
49. Find the complex Fourier series for the signal, its Fourier transform, and plot the corresponding magnitude spectrum:

$$f(t) = \cos(\omega_o t) + \sin^2(\omega_o t)$$

50. Show that if $x(t)$ is a bandlimited signal, i.e., $X(\omega) = 0$ for $|\omega| > \omega_M$, then

$$x(t) * \frac{\sin(at)}{\pi t} = x(t), \text{ if } a > \omega_M.$$

Plot the corresponding graphs to illustrate the proof.
51. Prove the following Parseval equation,

$$\int_{-\infty}^{\infty} f(x)G(x)dx = \int_{-\infty}^{\infty} F(x)g(x)\mathrm{d}x,$$

in which $f(\cdot)$ and $g(\cdot)$ are the original functions and $F(\cdot)$ and $G(\cdot)$ are their Fourier transforms.
52. Find the Fourier transform of the current through a diode, represented by the expression $i(t) = I_o[e^{\alpha v(t)} - 1]$, given the voltage $v(t)$ applied to the diode and its Fourier transform $V(\omega)$, in which α is a diode parameter and I_o is the reverse current. Plot the magnitude spectrum of the Fourier transform.
53. Give a physical interpretation for the Hilbert transform of a constant.
54. Obtain a general formula for the iteration of the Hilbert transform $\mathcal{H}^n[f(t)]$.
55. Compute the Hilbert transform of the function $f(t) = \mathrm{sgn}\,(t)$, and sketch the graph of the transform.
56. Use the formula for the Hilbert transform of the n-ary derivative of a function, $\mathcal{H}[f^{(n)}(t)]$, to obtain a general formula in the case $f(t) = \sin(\alpha t)$.
57. Obtain a general formula for the Hilbert transform of the derivative of the function $f(t) = \delta(\alpha t)$, using the property associated with the Hilbert transform of the function derivative, $\mathcal{H}[f^{(n)}(t)]$.
58. Prove that

$$\int_{-\infty}^{\infty} \frac{\sin(\omega\sigma)}{\sigma} d\sigma = \pi.$$

59. Calculate the Hilbert transform of the complex exponential function $f(t) = e^{j\omega t+\phi}$, using Euler's formula $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$.
60. Demonstrate the time reversal property of the Hilbert transform, $\mathcal{H}[f(-\alpha t)] = -\hat{f}(-\alpha t), \alpha > 0$.
61. Demonstrate that

$$\int_{-\infty}^{\infty} f(t-\sigma)d\sigma = \pi[t\mathcal{H}[f(t)] - \mathcal{H}[tf(t)]].$$

62. Considering that $f(t)$ is a signal whose Fourier transform satisfies $F(\omega) = 0$ for $|\omega| \geq W$ and that $h(t)$ is a signal with $H(\omega) = 0$ for $|f| < W$, prove that

$$\mathcal{H}[f(t)h(t)] = f(t)\hat{h}(t).$$

63. Explain how to obtain a single-sideband modulated signal, in which a sideband is suppressed, using some Hilbert transform properties.

Chapter 5
Random Signals and Noise



In mathematics the art of proposing a question must be held of higher value than solving it.

—Georg Cantor

Signals that convey information are by nature random. A random signal, also known generically as a stochastic process, is an extension of the concept of a random variable, involving a time variable. In fact, a stochastic process is a function of a random variable, or of a combination of random variables, and time. Figure 5.1 illustrates a random signal, and Fig. 5.2 depicts its associated pdf.

A random process $X(t)$ defines a random variable for each point on the time axis. A stochastic process is said to be stationary in the strict sense if the probability densities associated with the process do not change with time.

5.1 The Autocorrelation Function

The autocorrelation function is an important joint moment of the random process $X(t)$, because it reveals certain characteristics of the signal, that could not be apparent at first sight. It is used to detect signals in noise but has many other applications:

$$R_X(\xi, \eta) = \mathrm{E}[X(\xi)X(\eta)], \tag{5.1}$$

in which

$$\mathrm{E}[X(\xi)X(\eta)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(\xi)x(\eta)p_{X(\xi)X(\eta)}(x(\xi)x(\eta))\mathrm{d}x(\xi)\mathrm{d}y(\eta) \tag{5.2}$$

denotes the joint moment of the random variable. $X(t)$ at $t = \xi$ and at $t = \eta$.

The random process is called wide-sense stationary if its autocorrelation depends only on the interval of time separating $X(\xi)$ and $X(\eta)$, that is, it depends only on

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_5

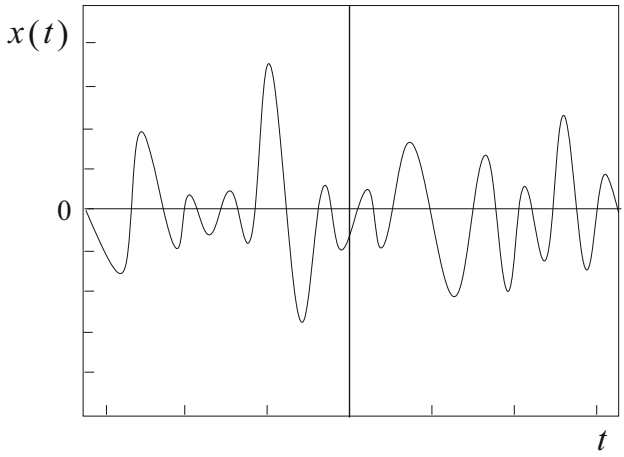


Fig. 5.1 Signal representing a sample of a random process

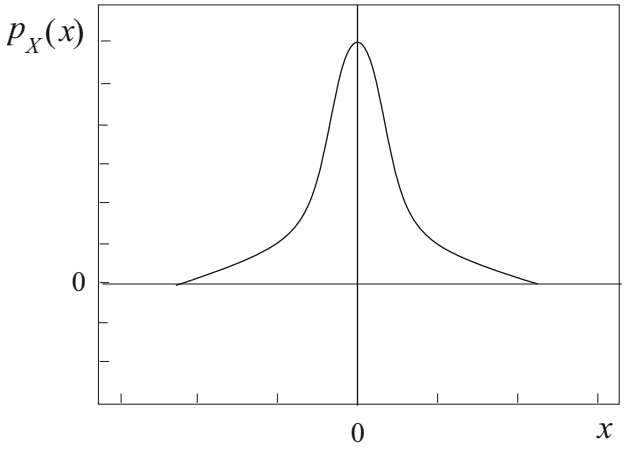


Fig. 5.2 Probability density function for the random process to which the sample in Fig. 5.1 belongs

$\tau = \xi - \eta$. Equation 5.1 in this case can be written as

$$R_X(\tau) = \mathrm{E}[X(t)X(t+\tau)]. \tag{5.3}$$

5.2 Stationarity

In general, the statistical mean of a random signal is a function of time, that is, it changes with time. Thus, the mean value

$$\mathrm{E}[X(t)] = m_X(t),$$

the power

$$\mathrm{E}[X^2(t)] = P_X(t),$$

and the autocorrelation

$$R_X(\tau, t) = \mathrm{E}[X(t)X(t+\tau)],$$

are, in general, time-dependent. However, there exists a set of time signals the mean value of which is time-independent. These signals are called stationary signals, as illustrated in the following example:

Example Consider the digital signal shown in Fig. 5.3, with equiprobable amplitudes A and $-A$.

The pdf of the digital signal, as shown in Fig. 5.4, is given by

$$p_X(x) = \frac{1}{2}\left[\delta(x+A) + \delta(x-A)\right].$$

Applying the definition of the mean to $\mathrm{E}[X(t)]$, it follows that

$$\mathrm{E}[X(t)] = 0.$$

The power is given as

$$\mathrm{E}[X^2(t)] = \frac{1}{2}A^2 + \frac{1}{2}(-A)^2 = A^2.$$

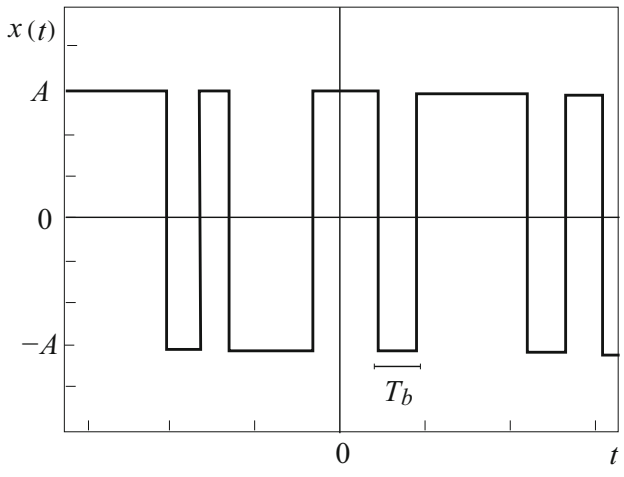


Fig. 5.3 A digital signal

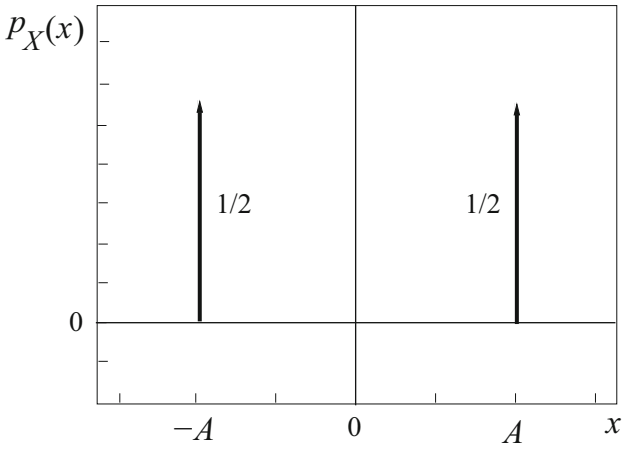


Fig. 5.4 Probability density function for the digital signal of Fig. 5.3

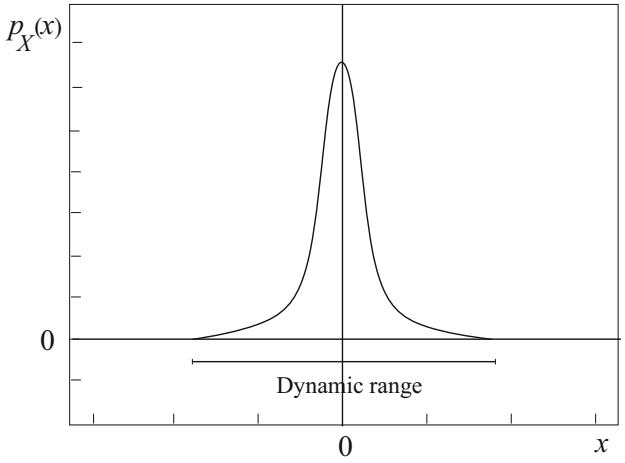


Fig. 5.5 The dynamic range of a signal

Finally, the variance and the standard deviation are calculated as

$$\sigma_X^2 = \mathrm{E}[X^2(t)] = A^2 \;\Rightarrow\; \sigma_X = A.$$

Example The dynamic range of a signal, from a probabilistic point of view, is illustrated in Fig. 5.5. As can be seen, the dynamic range depends on the standard deviation, or RMS voltage, being usually specified for $2\sigma_X$ or $4\sigma_X$. For a signal with a Gaussian probability distribution of amplitudes, this corresponds to a range encompassing, respectively, 97% and 99.7% of all signal amplitudes.

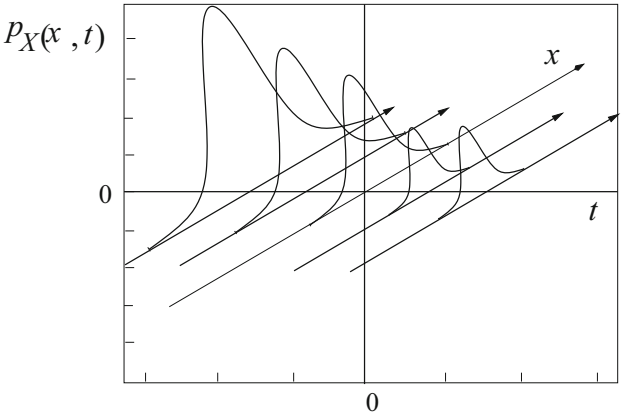


Fig. 5.6 A time-varying probability density function

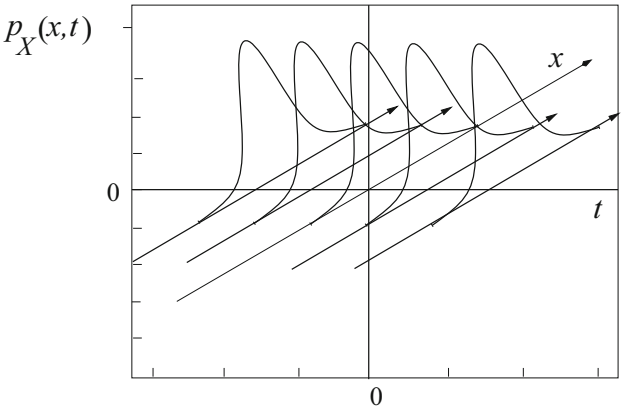


Fig. 5.7 A time-invariant probability density function

However, since the signal is time-varying, its statistical mean can also change with time, as illustrated in Fig. 5.6. In the example considered, the variance is initially diminishing with time, and later it is growing with time. In this case, an adjustment in the signal variance, by means of an automatic gain control mechanism, can remove the pdf dependency on time.

A signal is stationary whenever its pdf is time-independent, that is, whenever $p_X(x,t) = p_X(x)$, as illustrated in Fig. 5.7.

Stationarity may occur in various instances:

1. The signal is stationary in the mean when the mean is independent of time, $m_X(t) = m_X$.
2. The signal is stationary in power when the power is independent of time, $P_X(t) = P_X$.

3. First-order stationarity implies that the first-order moment of the signal is independent of time.
4. Second-order stationarity implies that the first- and second-order moments of the signal are independent of time.
5. Strict sense stationarity implies that the signal is stationary for all orders, that is, $p_{X_1 \cdots X_M}(x_1, \cdots, x_M; t) = p_{X_1 \cdots X_M}(x_1, \cdots, x_M)$.

A real stochastic signal is seldom strictly stationary for the whole time. But most signals can be considered stationary, in the mean and power, if some conditions are met, for instance, if the time interval is restricted or if the audio equipment or the communication receiver has automatic gain control.

5.2.1 Wide-Sense Stationarity

However, there is a type of stationarity that is useful to most purposes but does not require the signal to fulfill many statistical tests. The following conditions are necessary to guarantee that a stochastic process is wide-sense stationary:

1. The signal mean and the power are constant.
2. The autocorrelation depends only on the time difference between the measurements,

$$R_X(t_1, t_2) = R_X(t_2 - t_1) = R_X(\tau),$$

that is, the autocorrelation does not depend on the origin of the time interval.

5.2.2 Ergodic Signals

Ergodicity is another important characteristic of random processes. For simplicity, a signal is ergodic in the mean if the time expected value of the signal coincides with its statistical mean. The same is valid for other signal statistics. Therefore, an ergodic signal has a typical waveform that represents the whole ensemble.

Ergodicity can occur on the mean, on the power, on the autocorrelation, or with respect to other statistics. Ergodicity of the mean implies that the time average is equivalent to the statistical signal average. Therefore,

1. Ergodicity of the mean—$\overline{X(t)} \sim \mathrm{E}[X(t)]$.
2. Ergodicity of the power—$\overline{X^2(t)} \sim \overline{R_X}(\tau) \sim R_X(\tau)$.
3. Ergodicity of the autocorrelation—$\overline{R_X}(\tau) \sim R_X(\tau)$.

A strictly stationary stochastic process has a joint pdf that is independent of time. A wide-sense stationary process has the first- and second-order means constant and the autocorrelation depending only on the measuring time interval.

Therefore, a stochastic process is ergodic whenever its statistical means, which are functions of time, can be approximated by their corresponding time averages, which are random processes, with a standard deviation which is close to zero. The ergodicity may appear only on the mean value of the process, in which case the process is said to be ergodic in the mean.

5.2.3 Properties of the Autocorrelation

It is not necessary to compute the autocorrelation using the definition every time. The autocorrelation function has some important properties, presented in the following, that help solve most of the problems:

1. The total power of a signal is obtained by computing the autocorrelation at the origin,

$$R_X(0) = \mathrm{E}[X^2(t)] = P_X. \tag{5.4}$$

2. The average power or DC power level is the limit of the autocorrelation, as the time difference goes to infinity,

$$R_X(\infty) = \lim_{\tau\to\infty} R_X(\tau) = \lim_{\tau\to\infty} \mathrm{E}[X(t+\tau)X(t)] = \mathrm{E}^2[X(t)]. \tag{5.5}$$

3. The signal mean value is the square root of the autocorrelation evaluated at the infinity,

$$\mathrm{E}[X(t)] = \sqrt{R_X(\infty)}. \tag{5.6}$$

4. The autocovariance is the unbiased autocorrelation,

$$\begin{aligned} C_X(\tau) &= \mathrm{E}[(X(t+\tau) - m_X)(X(t) - m_X)] \\ &= R_X(\tau) - \mathrm{E}^2[X(t)]. \end{aligned} \tag{5.7}$$

5. The variance, or AC power, is the unbiased total power,

$$\begin{aligned} \mathrm{V}[X(t)] &= \mathrm{E}[(X(t) - \mathrm{E}[X(t)])^2] \\ &= \mathrm{E}[X^2(t)] - \mathrm{E}^2[X(t)] \\ &= R_X(0) - R_X(\infty) = P_{AC}. \end{aligned} \tag{5.8}$$

6. The autocorrelation has a maximum at the origin,

$$R_X(0) \geq |R_X(\tau)|. \tag{5.9}$$

This property is demonstrated by considering the following mathematical tautology, which is always true for real signals,

$$\mathrm{E}[(X(t) - X(t+\tau))^2] \geq 0.$$

Thus,

$$\mathrm{E}[X^2(t) - 2X(t)X(t+\tau)] + \mathrm{E}[X^2(t+\tau)] \geq 0,$$

that is,

$$2R_X(0) - 2RX(\tau) \geq 0 \;\Rightarrow\; R_X(0) \geq R_X(\tau).$$

7. The autocorrelation is a symmetric function,

$$R_X(\tau) = R_X(-\tau). \tag{5.10}$$

In order to prove this property, it is sufficient to use the definition

$$R_X(-\tau) = \mathrm{E}[X(t)X(t-\tau)].$$

Considering a change of variables, $t - \tau = \sigma \;\Rightarrow\; t = \sigma + \tau$, one obtains

$$R_X(-\tau) = \mathrm{E}[X(\sigma+\tau) \cdot X(\sigma)] = R_X(\tau).$$

The relationship between the autocorrelation function and various other power measures is illustrated in Fig. 5.8.

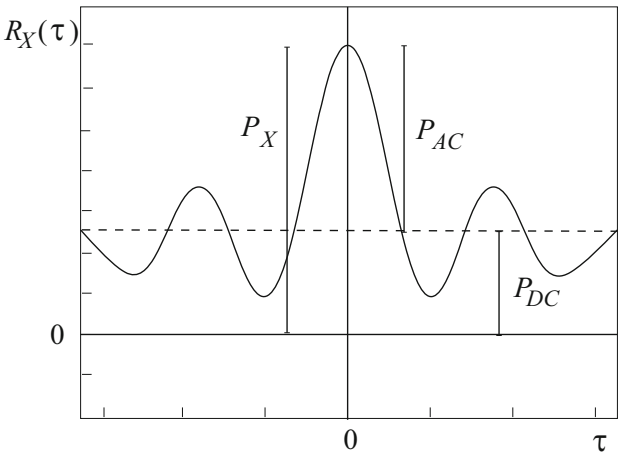


Fig. 5.8 Relationship between the autocorrelation and various power measures

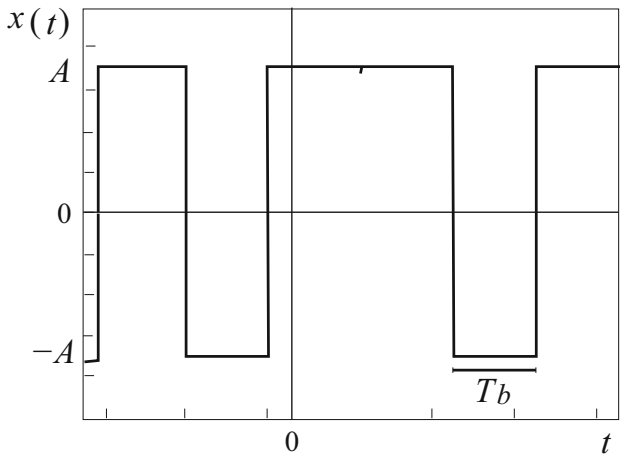


Fig. 5.9 Digital random signal

Example The autocorrelation of a digital signal $X(t)$, with equiprobable amplitude levels A and $-A$, and a random initial transition, as seen in Fig. 5.9, with uniform distribution, can be computed in the following manner:

A baseband random binary signal assumes only two states, $-A$ and A, and the transition between the states can take place every T_b seconds. The autocorrelation can be computed by Lathi (1968)

$$R_X(\tau) = \sum_{X_1} \sum_{X_2} x_1 x_2 P_{X_1 X_2}(x_1, x_2) \tag{5.11}$$

in which x_1 and x_2 represent the signal amplitudes at t and $t+\tau$ and the summations are over the sets $X_1 = X_2 = \{-A, A\}$. This can be written as

$$\begin{aligned} R_X(\tau) = A^2 \big[& P_{X_1 X_2}(A, A) + P_{X_1 X_2}(-A, -A) \\ & - P_{X_1 X_2}(-A, A) - P_{X_1 X_2}(A, -A) \big]. \end{aligned} \tag{5.12}$$

But

$$P_{X_1 X_2}(A, A) = P_{X_1}(A) P_{X_2}(A | x_1 = A),$$

and, since the signal states are equiprobable,

$$P_{X_1}(A) = P_{X_1}(-A) = 0.5.$$

Therefore, it is possible to write

$$\begin{aligned} R_X(\tau) = & A^2 P_{X_1}(A) P_{X_2}(A|x_1 = A) \\ & + A^2 P_{X_1}(-A) P_{X_2}(-A|x_1 = -A) \\ & - A^2 P_{X_1}(-A) P_{X_2}(A|x_1 = -A) \\ & - A^2 P_{X_1}(A) P_{X_2}(-A|x_1 = A). \end{aligned} \tag{5.13}$$

On the other hand,

$$P_{X_2}(A|x_1 = A) = 1 - P_{X_2}(-A|x_1 = A),$$

in which $P_{X_2}(A|x_1 = A)$ is the probability of observing $x_2 = A$ given that $x_1 = A$.

First, consider the condition $\tau < T_b$. In this case, $x_2 = -A$, when $x_1 = A$ only if a transition occurs in the interval $[t, t + \tau]$, and the state changes at this point.

But because of the assumption of a uniform distribution for the initial transition, the probability that a transition lies in the interval $[t, t + \tau]$ is proportional to the size of the interval, that is, τ / T_b. Hence,

$$P_{X_2}(-A|x_1 = A) = \frac{\tau}{2T_b},\ 0 < \tau < T_b,$$

and

$$P_{X_2}(A|x_1 = A) = 1 - \frac{\tau}{2T_b},\ 0 < \tau < T_b.$$

In a similar way,

$$P_{X_2}(A|x_1 = -A) = \frac{\tau}{2T_b},\ 0 < \tau < T_b,$$

and

$$P_{X_2}(-A|x_1 = -A) = 1 - \frac{\tau}{2T_b},\ 0 < \tau < T_b.$$

Substituting into Eq. 5.13 and making the simplifications, one obtains

$$R_X(\tau) = 1 - \frac{\tau}{T_b},\ 0 < \tau < T_b.$$

Therefore, considering that the autocorrelation is an even function and that the signal pulses are uncorrelated outside the symbol interval T_b, the digital signal $X(t)$

has the following autocorrelation function for any time difference,

$$R_X(\tau) = A^2 \left[1 - \frac{|\tau|}{T_b}\right] [u(\tau + T_b) - u(\tau - T_b)], \tag{5.14}$$

in which T_b is the pulse duration (Fig. 5.10).

Example A telegraphic signal $X(t)$, with equiprobable amplitude levels A and $-A$, is illustrated in Fig. 5.11. For this signal, the transitions can occur at any time, and they are usually modeled as a Poisson stochastic process.

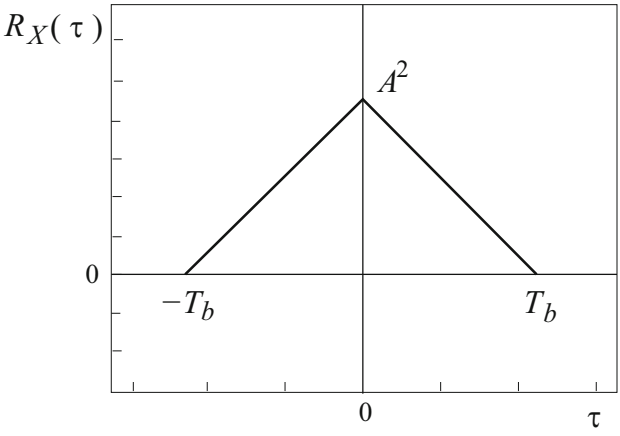


Fig. 5.10 Autocorrelation of a digital random signal

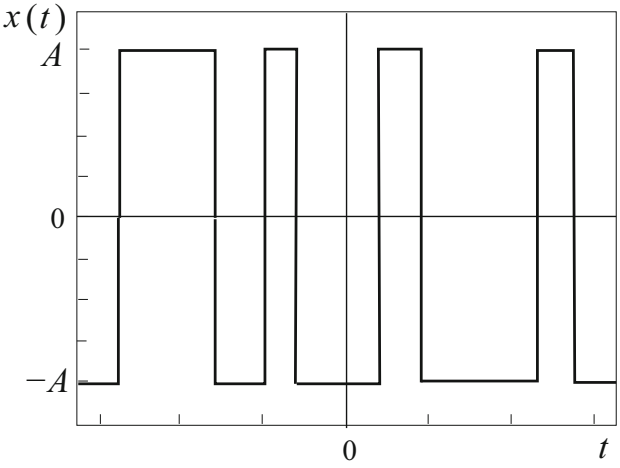


Fig. 5.11 Telegraphic random signal

For the Poisson process, with an arriving rate λ, the probability that the system is in state k, at time τ, is given by the following formula:

$$p_k(\tau) = \frac{(\lambda\tau)^k}{k!} e^{-\lambda\tau} \tag{5.15}$$

Using 5.15, the probability of no transition in the interval is

$$p_0(\tau) = \frac{(\lambda\tau)^0}{0!} e^{-\lambda\tau} = e^{-\lambda\tau}, \tag{5.16}$$

and the probability of, at least, a transition occurring is

$$p_T(\tau) = 1 - e^{-\lambda\tau}. \tag{5.17}$$

The autocorrelation of the telegraphic signal can be computed in the following manner. As before,

$$R_X(\tau) = \sum_{X_1} \sum_{X_2} x_1 x_2 P_{X_1 X_2}(x_1, x_2) \tag{5.18}$$

in which x_1 and x_2 represent the signal amplitudes at t and $t+\tau$ and the summations are over the sets $X_1 = X_2 = \{-A, A\}$.

This can again be written as

$$\begin{aligned} R_X(\tau) = A^2 \big[& P_{X_1 X_2}(A, A) + P_{X_1 X_2}(-A, -A) \\ & - P_{X_1 X_2}(-A, A) - P_{X_1 X_2}(A, -A) \big]. \end{aligned} \tag{5.19}$$

Substituting the transition probabilities, from the Poisson formula, and simplifying the equation give

$$R_X(\tau) = A^2 e^{-\lambda|\tau|}. \tag{5.20}$$

The autocorrelation function for the telegraphic signal is shown in Fig. 5.12. One can verify that the signal power is $P_X = R_X(0) = A^2$.

5.3 The Power Spectral Density

The autocorrelation function makes it possible to define the following Fourier transform pair, known as the Wiener-Khinchin theorem:

$$S_X(w) = \int_{-\infty}^{+\infty} R_X(\tau) e^{-jw\tau} \mathrm{d}\tau \tag{5.21}$$

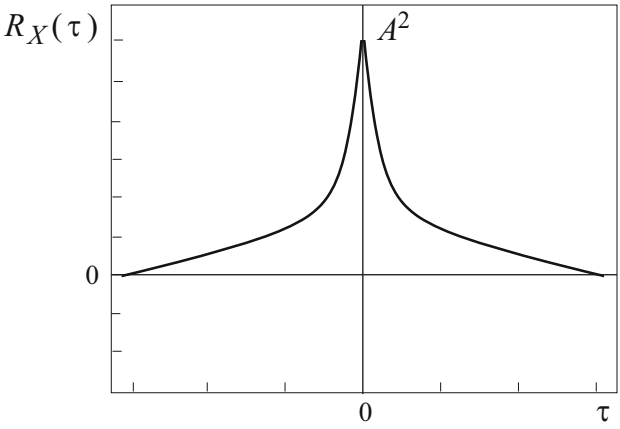


Fig. 5.12 Autocorrelation of a telegraphic random signal

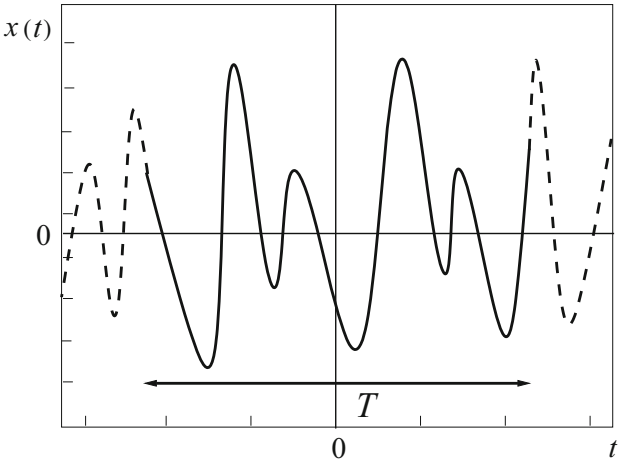


Fig. 5.13 Random signal truncated in an interval T

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_X(w) e^{jw\tau} \mathrm{d}w \tag{5.22}$$

The function $S_X(w)$ is called the power spectral density (PSD) of the random process.

The Wiener-Khinchin theorem relates the autocorrelation function with the PSD, that is, it plays the role of a bridge between the time domain and the frequency domain for random signals. This theorem will be proved in the sequel. Figure 5.13 shows a random signal truncated in an interval T.

The Fourier transform for the signal $x(t)$ given in Fig. 5.13 is given by $\mathcal{F}[x_T(t)] = X_T(\omega)$. The time, or deterministic, PSD of $x(t)$ is calculated as follows:

$$\lim_{T\to\infty} \frac{1}{T} |X_T(\omega)|^2 = \overline{S_X}(\omega)$$

The result obtained is obviously a random quantity, and it is possible to compute its statistical mean to obtain the PSD:

$$S_X(\omega) = \mathrm{E}[\overline{S_X}(\omega)] \tag{5.23}$$

Recall that

$$|X_T(\omega)|^2 = X_T(\omega) \cdot X_T^*(\omega) = X_T(\omega) \cdot X_T(-\omega)$$

for a real $X(t)$, in which

$$X_T(\omega) = \int_{-T/2}^{T/2} X_T(t) e^{-j\omega t}\, dt.$$

Substituting the last transform into Eq. 5.23, with an adequate variable change, gives

$$S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} E[|X_T(\omega)|^2] = \lim_{T\to\infty} \frac{1}{T} E[X_T(\omega) \cdot X_T(-\omega)] =$$

$$= \lim_{T\to\infty} E\left[\int_{-T/2}^{T/2} X_T(\sigma) e^{-j\omega\sigma}\, \mathrm{d}\sigma \cdot \int_{-T/2}^{T/2} X_T(\rho) e^{j\omega\rho}\, \mathrm{d}\rho\right] =$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} E[X_T(\sigma) X_T(\rho)] e^{-j(\sigma-\rho)\omega}\, \mathrm{d}\sigma\, \mathrm{d}\rho =$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} R_{X_T}(\sigma - \rho) e^{-j(\sigma-\rho)\omega}\, \mathrm{d}\sigma\, \mathrm{d}\rho.$$

Figure 5.14 shows the integration region. The real plane is covered as $T \to \infty$.

Making the variable changes $\tau = \sigma - \rho$ and $t = \sigma$, the power spectral density is given by

$$S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \int_{-T/2-t}^{T/2-t} R_{X_T}(\tau) e^{-j\tau\omega}\, \mathrm{d}\tau\, \mathrm{d}t.$$

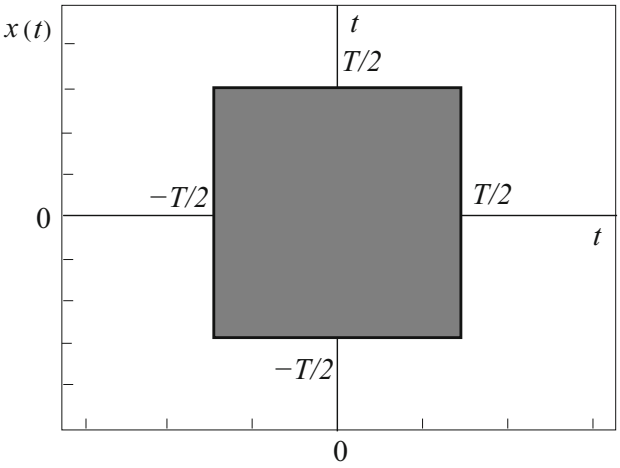


Fig. 5.14 Rectangular integration region

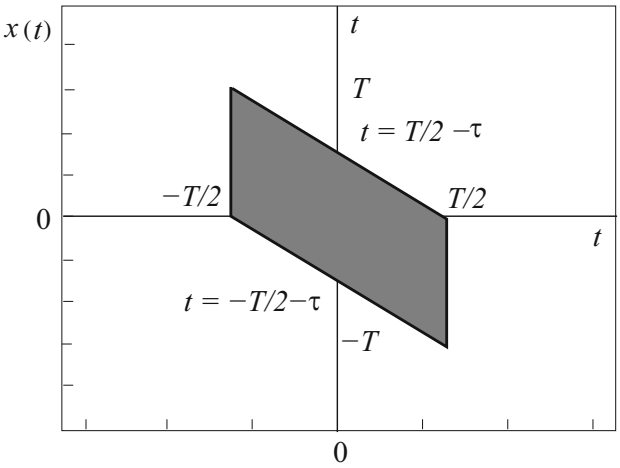


Fig. 5.15 Integration region obtained after the variable change

Figure 5.15 shows the new integration region, after the variable changes. In order to compute the integral, with the indicated limits, it is useful to separate the integration region into two parts, the first one for $\tau \geq 0$ and the other for $\tau < 0$.

Therefore, the expression is given by

$$S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} \left[\int_0^T \int_{-T/2}^{T/2-\tau} R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau \, \mathrm{d}t \right.$$

$$+ \int_{-T}^{0} \int_{-T/2-\tau}^{T/2} R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau \, \mathrm{d}t \Bigg].$$

Integration of the equation gives

$$S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} \left[\int_0^T (T-\tau) R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau + \int_{-T}^{0} (T+\tau) R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau \right].$$

Combining the integrals, one obtains

$$S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} \left[\int_{-T}^{T} R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau - 2 \int_0^T \tau \cos(\omega\tau) R_{X_T}(\tau) e^{-j\tau\omega} \, \mathrm{d}\tau \right].$$

Finally, taking the limit and computing the integral in the second term of the expression lead to the Wiener-Khinchin theorem

$$S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} \, \mathrm{d}\tau. \tag{5.24}$$

Therefore, $S_X(\omega) = \mathcal{F}[R_X(\tau)]$, given that τR_X is absolutely integrable, which implies that

$$\int_{-\infty}^{\infty} |\tau \, R_X(\tau)| \, \mathrm{d}\tau < \infty.$$

The function $S_X(\omega)$ represents the power spectral density of the random signal $X(t)$, which measures power per unit frequency. The corresponding inverse Fourier transform is the autocorrelation function

$$R_X(\tau) = \mathcal{F}^{-1}[S_X(\omega)],$$

that is,

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) e^{j\omega\tau} \, \mathrm{d}\omega. \tag{5.25}$$

As an example, Fig. 5.16 illustrates the power spectral density of a bandlimited signal. This is a common type of signal, because the filtered noise at the input of a receiver is usually considered a bandlimited white noise.

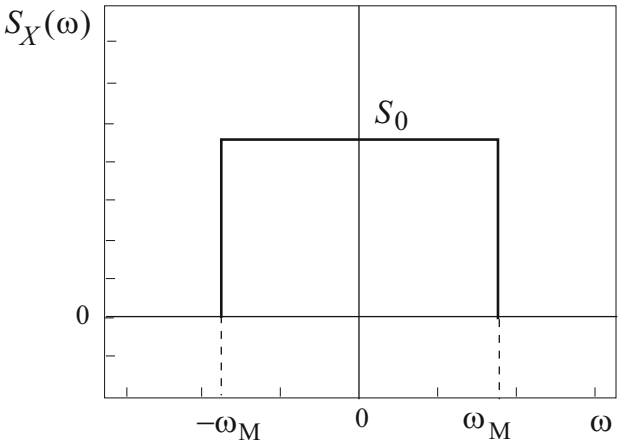


Fig. 5.16 Power spectral density for a bandlimited signal

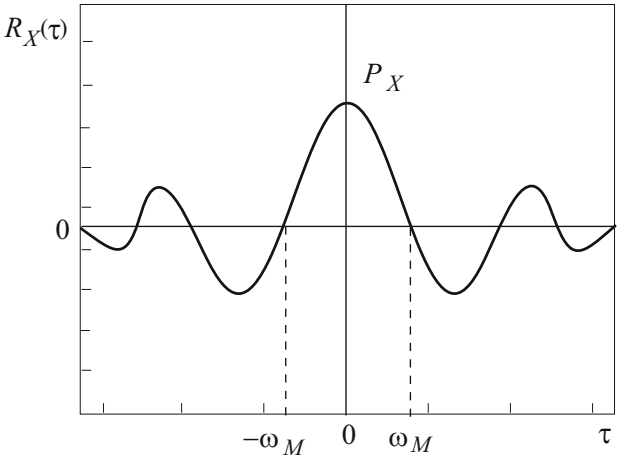


Fig. 5.17 Autocorrelation function for a bandlimited signal

The autocorrelation function for the bandlimited signal is depicted in Fig. 5.17. Note that the autocorrelation has several zeros along the abscissa, which indicate the points at which the correlation is null. For the case, the signal total power is $P_X = \omega_M S_0/\pi$.

Figure 5.18 illustrates the power spectral density for a white noise. Note that the spectrum indicates that the noise has a constant power density associated with every frequency.

The corresponding autocorrelation function for white noise is shown in Fig. 5.19.

As mentioned, the power spectral density $S_X(\omega) = S_0$, which indicates a uniform distribution for the power spectral density along the whole spectrum, and $R_X(\tau) =$

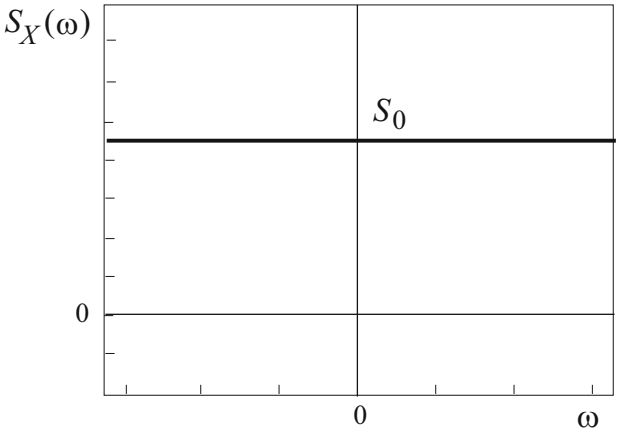


Fig. 5.18 Power spectral density for white noise

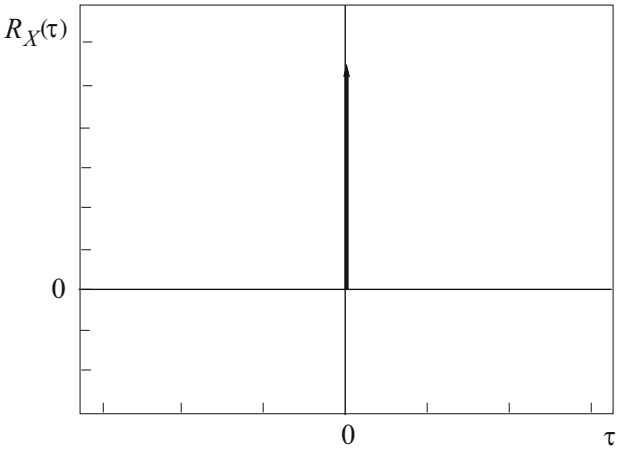


Fig. 5.19 Autocorrelation function for white noise

$S_0\delta(\tau)$, which indicates that the white noise is the most uncorrelated or most random signal. Correlation is nonzero for this signal only at $\tau = 0$.

On the other hand, the PSD for a constant signal is an impulse at the origin, and its autocorrelation is a constant for all τ, which renders the constant function as the most predictable among all signals.

Example The power spectral density for the telegraphic signal can be computed using the Wiener-Khinchin theorem, due to Norbert Wiener (1894–1964), an American mathematician and philosopher, and Aleksandr Yakovlevich Khinchin

(1894–1959), a Soviet mathematician. It states that

$$S_X(\omega) = A^2 \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau}\, e^{-\lambda|\tau|} \mathrm{d}\tau.$$

Splitting into two integrals,

$$S_X(\omega) = A^2 \int_0^{\infty} R_X(\tau) e^{-(j\omega+\lambda)\tau} \mathrm{d}\tau + A^2 \int_{-\infty}^{0} R_X(\tau) e^{-(j\omega-\lambda)\tau} \mathrm{d}\tau.$$

Changing the variables and the order of integration and computing the integrals,

$$S_X(\omega) = \frac{A^2}{\lambda + j\omega} + \frac{A^2}{\lambda - j\omega}.$$

Computing the common divisor and adding the fractions, one obtains

$$S_X(\omega) = \frac{2\lambda A^2}{\lambda^2 + \omega^2}. \tag{5.26}$$

The power spectral density of a random telegraphic signal is shown in Fig. 5.20.

5.3.1 *Properties of the Power Spectral Density*

Some properties of the PSD function are listed in the following:

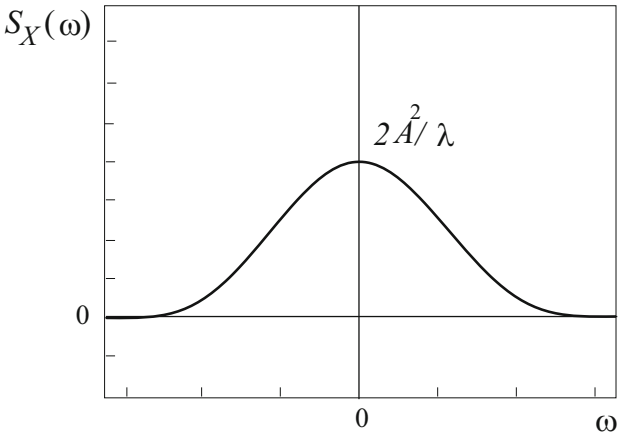


Fig. 5.20 Power spectral density of a random telegraphic signal

1. The area under the curve of the PSD is equal to the total power of the random process, that is,

$$P_X = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X(\omega)\mathrm{d}\omega. \tag{5.27}$$

This fact can be verified directly as

$$\begin{aligned} P_X = R_X(0) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega)e^{j\omega 0}\,\mathrm{d}\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega)\,\mathrm{d}\omega = \int_{-\infty}^{\infty} S_X(f)\,\mathrm{d}f, \end{aligned}$$

in which $\omega = 2\pi f$.

2. The area under the autocorrelation function is PSD computed at the origin,

$$S_X(0) = \int_{-\infty}^{\infty} R_X(\tau)\,\mathrm{d}\tau. \tag{5.28}$$

The proof is similar to the previous property.

3. If $R_X(\tau)$ is real and even, then

$$\begin{aligned} S_X(\omega) &= \int_{-\infty}^{\infty} R_X(\tau)[\cos\omega\tau - j\sin\omega\tau]\,\mathrm{d}\tau, \\ &= \int_{-\infty}^{\infty} R_X(\tau)\cos\omega\tau\,\mathrm{d}\tau, \end{aligned} \tag{5.29}$$

that is, $S_X(\omega)$ is real and even.

4. $S_X(\omega) \geq 0$, since the density reflects a power measure.
5. The following identities hold:

$$\int_{-\infty}^{+\infty} R_X(\tau)R_Y(\tau)\mathrm{d}\tau = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X(\omega)S_Y(w)\mathrm{d}\omega \tag{5.30}$$

$$\int_{-\infty}^{+\infty} R_X^2(\tau)\mathrm{d}\tau = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X^2(\omega)\mathrm{d}\omega. \tag{5.31}$$

Finally, the cross-correlation, or just correlation, between two random processes $X(t)$ and $Y(t)$ is defined as

$$R_{XY}(\tau) = \mathrm{E}[X(t)Y(t+\tau)]. \tag{5.32}$$

The cross-correlation has some important properties that are useful in deriving the quadrature modulation theory:

1. For any two stochastic processes,

$$R_{XY}(-\tau) = \mathrm{E}[X(t)Y(t-\tau)] = \mathrm{E}[X(t+\sigma)Y(t)] = R_{YX}(\tau). \tag{5.33}$$

2. The correlation is bounded by

$$|R_{XY}(\tau)| \leq \sqrt{R_X(0)R_Y(0)}. \tag{5.34}$$

3. If the two stochastic processes $x(t)$ or $y(t)$ are independent, then

$$R_{XY}(\tau) = R_{YX}(\tau) = \mathrm{E}[X(t)]\mathrm{E}[Y(t)]. \tag{5.35}$$

The Fourier transform of the correlation leads to the definition of the cross-power spectral density $S_{XY}(\omega)$:

$$S_{XY}(\omega) = \int_{-\infty}^{+\infty} R_{XY}(\tau)e^{-j\omega\tau}d\tau \tag{5.36}$$

Example By knowing that $\hat{m}(t) = \frac{1}{\pi t} * m(t)$ is the Hilbert transform of $m(t)$ and using properties of the autocorrelation and of the cross-correlation, it can be shown that

$$\mathrm{E}[m(t)^2] = \mathrm{E}[\hat{m}(t)^2]$$

and that

$$\mathrm{E}[m(t)\hat{m}(t)] = 0.$$

The following property is commonly used in communications to detect a carrier or a pilot signal. If two stationary processes, $X(t)$ and $Y(t)$, are added to form a new process $Z(t) = X(t) + Y(t)$, then the autocorrelation function of the new process is given by

$$\begin{aligned} R_Z(\tau) &= \mathrm{E}[Z(t) \cdot Z(t+\tau)] \\ &= \mathrm{E}[(x(t) + y(t))(x(t+\tau) + y(t+\tau))], \end{aligned}$$

which implies

$$R_Z(\tau) = \mathrm{E}[x(t)x(t+\tau) + y(t)y(t+\tau) + x(t)y(t+\tau) + x(t+\tau)y(t)].$$

By applying properties of the expected value to the above expression, it follows that

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau). \tag{5.37}$$

If the stochastic processes $X(t)$ and $Y(t)$ can be considered uncorrelated, then $R_{XY}(\tau) = R_{YX}(\tau) = 0$, and $R_Z(\tau)$ is written as

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau). \tag{5.38}$$

This equation is useful to detect a periodic signal in the presence of noise by correlation.

Example A radar signal is a periodic waveform that is, for example, transmitted to detect the presence of an object, usually an enemy plane. Of course, the enemy does not want to be detected and generates an electromagnetic noise signal to counterattack and hide the plane.

Suppose, for the sake of simplicity, that the transmitted radar signal is a sinusoidal waveform, $X(t) = A\cos(\omega_0 + \phi)$, and the thermal noise $N(t)$ is produced by a device operating at a high temperature.

Then, the received signal is

$$Z(t) = A\cos(\omega_0 + \phi) + N(t),$$

and the autocorrelation is given by

$$R_Z(\tau) = \frac{A^2}{2}\cos(\omega_0\tau) + \alpha kTGe^{-\alpha|\tau|},$$

in which α is a parameter that depends on the medium characteristics, the parameter $k = 1.38 \times 10^{-23}$ is the Boltzmann constant, G is the conductance of the device, and T is the absolute temperature in kelvin.

Figure 5.21 shows that, for large values of τ, the thermal noise fades down, and the autocorrelation of the received signal shows a periodic behavior of the same period as that of $X(t)$. This permits the detection of the periodic signal.

The decorrelation time, which is used to estimate how large τ should be to extract the periodic signal, is given by Lévine (1973)

$$\tau_0 = \frac{1}{2}\int_{-\infty}^{\infty} |\rho(\tau)| d\tau, \tag{5.39}$$

in which the correlation coefficient is defined as

$$\rho(\tau) = \frac{R(\tau)}{R(0)}, \ -1 \leq \rho \leq 1.$$

If, in the formula for the noise autocorrelation, the parameter $\alpha \to \infty$, the function becomes an impulse, and the interfering signal is called additive white Gaussian noise (AWGN). In this case, for any value of τ, the noise autocorrelation fades down immediately, and the periodic signal can be easily detected.

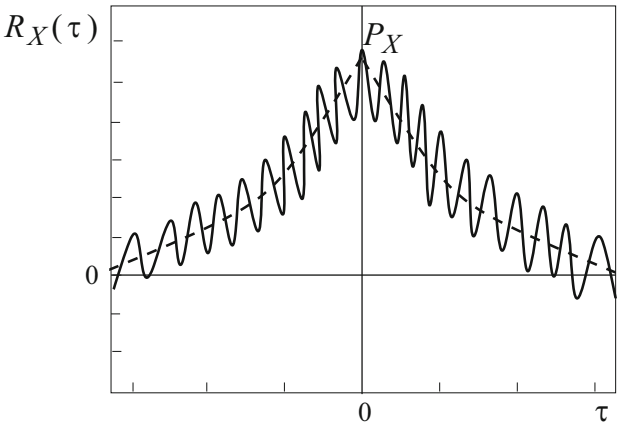


Fig. 5.21 Autocorrelation at the output of a noisy channel with a periodic input signal

The resulting detected power can be written as

$$P_Z = R_Z(0) = P_X + P_Y$$

and for the example

$$P_Z = \frac{A^2}{2} + \alpha kTG.$$

The corresponding PSD is given by

$$S_Z(\omega) = S_X(\omega) + S_Y(\omega), \tag{5.40}$$

which results in

$$S_Z(\omega) = \frac{\pi A^2}{2}\left[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)\right] + \frac{2\alpha^2 kTG}{\alpha^2 + \omega^2},$$

that is shown in Fig. 5.22.

5.4 Linear Systems

Linear systems can be analyzed using the theory of stochastic processes. This provides more general and more interesting solutions than those resulting from classical analysis. This section deals with the response of linear systems to a random input $X(t)$.

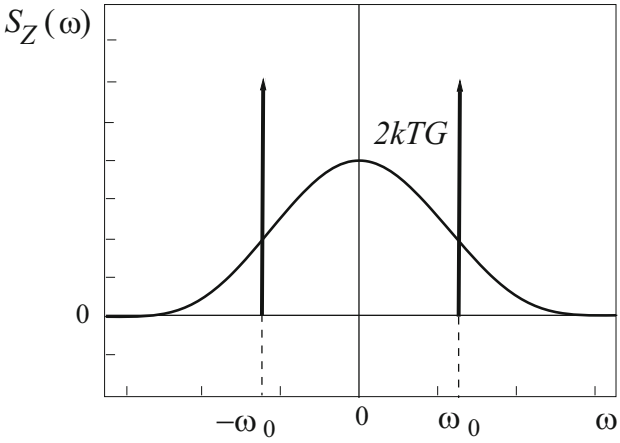
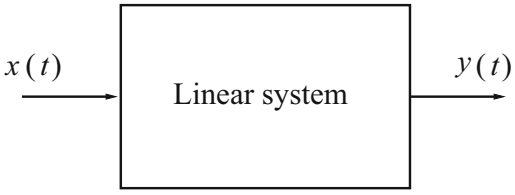


Fig. 5.22 Spectrum at the output of a noisy channel with a periodic input signal

Fig. 5.23 A random signal at the input of a linear system



For a linear system, as illustrated in Fig. 5.23, the Fourier transform of its impulse response $h(t)$ is given by

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} \, \mathrm{d}t. \tag{5.41}$$

The linear system response $Y(t)$ is obtained by means of the convolution of the input signal with the impulse response as follows:

$$Y(t) = X(t) * h(t) \;\Rightarrow\; Y(t) = \int_{-\infty}^{\infty} X(t-\alpha)\, h(\alpha)\, \mathrm{d}\alpha$$

$$= \int_{-\infty}^{\infty} X(\alpha)\, h(t-\alpha)\, \mathrm{d}\alpha$$

5.5 Expected Value of the Output Signal

The mean value of the random signal at the output of a linear system is calculated as follows:

$$\mathrm{E}[Y(t)] = \mathrm{E}\left[\int_{-\infty}^{\infty} X(t-\alpha)\, h(\alpha)\, d\alpha\right] = \int_{-\infty}^{\infty} E[X(t-\alpha)]\, h(\alpha)\, \mathrm{d}\alpha$$

Considering the random signal $X(t)$ to be narrow-sense stationary, it follows that $\mathrm{E}[X(t-\alpha)] = \mathrm{E}[X(t)] = m_X$, and thus

$$\mathrm{E}[Y(t)] = m_X \int_{-\infty}^{\infty} h(\alpha)\, \mathrm{d}\alpha = m_X H(0),$$

in which $H(0) = \int_{-\infty}^{\infty} h(\alpha)\, d\alpha$ follows from (5.41) computed at $\omega = 0$. Therefore, the mean value of the output signal depends only on the mean value of the input signal and on the value assumed by the transfer function at $\omega = 0$.

5.6 The Response of Linear Systems to Random Signals

The computation of the autocorrelation of the output signal, given the autocorrelation of the input signal to a linear system, can be performed as follows:

The relationship between the input and the output of a linear system was shown earlier to be given by

$$Y(t) = \int_{-\infty}^{\infty} X(\rho)\, h(t-\rho)\, \mathrm{d}\rho = \int_{-\infty}^{\infty} X(t-\rho)\, h(\rho)\, \mathrm{d}\rho = X(t) * h(t).$$

The output autocorrelation function can be calculated directly from its definition as

$$\begin{aligned} R_Y(\tau) &= \mathrm{E}[Y(t)Y(t+\tau)] \\ &= \mathrm{E}\left[\int_{-\infty}^{\infty} X(t-\rho)\, h(\rho)\, \mathrm{d}\rho \cdot \int_{-\infty}^{\infty} X(t+\tau-\sigma)\, h(\sigma)\, \mathrm{d}\sigma\right] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{E}[X(t-\rho)X(t+\tau-\sigma)] \cdot h(\rho) \cdot h(\sigma)\, \mathrm{d}\rho\, \mathrm{d}\sigma \\ &= \int_{=\infty}^{\infty}\int_{-\infty}^{\infty} R_{XX}(\tau+\rho-\sigma)\, h(\rho)\, h(\sigma)\, \mathrm{d}\rho\, \mathrm{d}\sigma. \end{aligned}$$

Example Suppose that white noise with autocorrelation $R_X(\tau) = \delta(\tau)$ is the input signal to a linear system. The corresponding autocorrelation function of the output signal is given by

$$R_Y(\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \delta(\tau+\rho-\sigma)\, h(\rho)\, h(\sigma)\, \mathrm{d}\rho\, \mathrm{d}\sigma$$

$$= \int_{-\infty}^{\infty} h(\sigma - \tau) \cdot h(\sigma)\, \mathrm{d}\sigma$$
$$= h(-\tau) * h(\tau).$$

The Fourier transform of $R_Y(\tau)$ leads to the following result,

$$R_Y(\tau) = h(-t) * h(t) \iff S_Y(\omega) = H(-\omega) \cdot H(\omega),$$

and for $h(\tau)$ a real function of τ, it follows that $H(-\omega) = H^*(\omega)$ and consequently

$$S_Y(\omega) = H(-\omega) \cdot H(\omega) = H^*(\omega) \cdot H(\omega) = |H(\omega)|^2.$$

Summarizing, the output PSD is $S_Y(\omega) = |H(\omega)|^2$ when white noise is the input to a linear system.

In general, the output spectrum can be computed by applying the Wiener-Khinchin theorem $S_Y(\omega) = \mathcal{F}[R_Y(\tau)]$:

$$S_Y(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} R_X(\tau + \rho - \sigma)\, h(\rho)\, h(\sigma) \cdot e^{-j\omega\tau}\, \mathrm{d}\rho\, \mathrm{d}\sigma\, \mathrm{d}\tau$$

Integrating on the variable τ, it follows that

$$S_Y(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} S_X(\omega) e^{j\omega(\rho-\sigma)}\, h(\rho)\, h(\sigma)\, \mathrm{d}\rho\, \mathrm{d}\sigma.$$

Finally, removing $S_X(\omega)$ from the double integral and then separating the two variables in this double integral it follows that

$$S_Y(\omega) = S_X(\omega) \int_{-\infty}^{\infty} h(\rho) e^{j\omega\rho}\, \mathrm{d}\rho \int_{-\infty}^{\infty} h(\sigma) e^{-j\omega\sigma}\, \mathrm{d}\sigma$$
$$= S_X(\omega) \cdot H(-\omega) \cdot H(\omega).$$

Therefore, $S_Y(\omega) = S_X(\omega) \cdot |H(\omega)|^2$ will result whenever the system impulse response is a real function.

Example Consider again white noise with autocorrelation function $R_X(\tau) = \delta(\tau)$ applied to a linear system. The white noise spectrum is calculated as follows,

$$S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau}\, \mathrm{d}\tau = \int_{-\infty}^{\infty} \delta(\tau) e^{-j\omega\tau}\, \mathrm{d}\tau = 1,$$

from which it follows that

$$S_Y(\omega) = |H(\omega)|^2,$$

similar to the previous example.

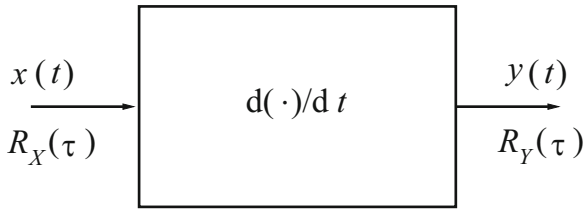


Fig. 5.24 A differentiator circuit

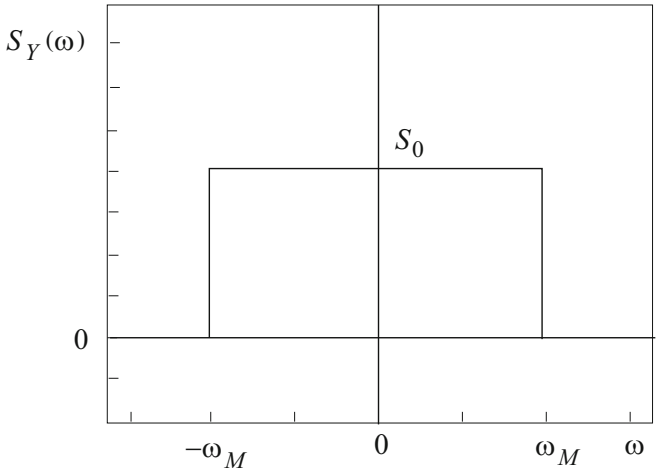


Fig. 5.25 Spectrum of the low-pass noise

Example The linear system shown in Fig. 5.24 is a differentiator, used in control systems or demodulator/detector for frequency-modulated signals.

The output PSD for this circuit (or its frequency response) is equal to

$$S_Y(\omega) = |j\omega|^2 \cdot S_X(\omega) = \omega^2 S_X(\omega).$$

It is thus noticed that, for frequency-modulated signals, the noise at the detector output follows a square law, that is, the output PSD grows with the square of the frequency. In this manner, in a frequency division multiplexing of frequency-modulated channels, the noise will affect more intensely those channels occupying the higher-frequency region of the spectrum.

Figure 5.25 shows, as an illustration of what has been discussed so far about square noise, the spectrum of a low-pass flat noise (obtained by passing white noise through an ideal low-pass filter). This filtered white noise is applied to the differentiator circuit of the example, which in turn produces at the output the square law noise shown in Fig. 5.26 (Alencar and da Rocha Jr. 2022).

Observation Preemphasis circuits are used in FM modulators to compensate for the quadratic noise and other effects.

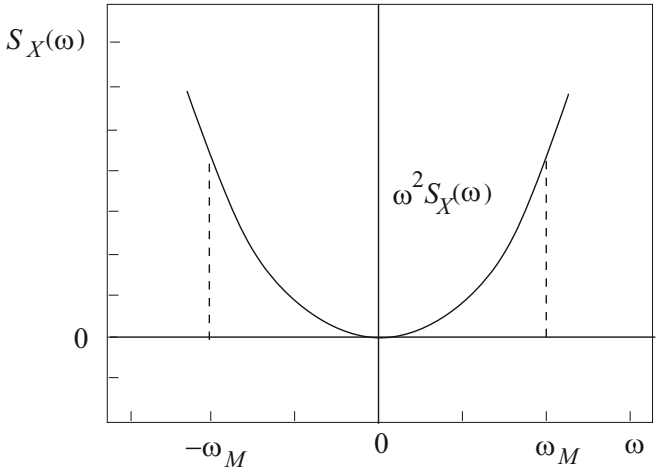


Fig. 5.26 Spectrum of the quadratic noise

Other relationships among different correlation functions can be derived. The correlation measures between input and output (input-output cross-correlation) and correlation between output and input can also be calculated from the input signal autocorrelation.

The correlation between the input and the output can be calculated with the formula

$$R_{XY}(\tau) = \mathrm{E}[X(t)Y(t+\tau)],$$

and in an analogous manner, the correlation between the output and the input can be calculated as

$$R_{YX}(\tau) = \mathrm{E}[Y(t)X(t+\tau)].$$

For a linear system, the correlation between output and input is given by

$$R_{YX}(\tau) = \mathrm{E}\left[\int_{-\infty}^{\infty} X(t-\rho)\,h(\rho)\,\mathrm{d}\rho \cdot X(t+\tau)\right].$$

Exchanging the order of the expected value and integral computations, due to their linearity, it follows that

$$R_{YX}(\tau) = \int_{-\infty}^{\infty} \mathrm{E}[X(t-\rho)X(t+\tau)]\,h(\rho)\,\mathrm{d}\rho = \int_{-\infty}^{\infty} R_X(\tau+\rho)\,h(\rho)\,\mathrm{d}\rho.$$

In a similar manner, the correlation between the input and the output is calculated as

$$\begin{aligned} R_{XY}(\tau) &= \mathrm{E}\left[X(t) \cdot \int_{-\infty}^{\infty} X(t+\tau-\rho)\, h(\rho)\, \mathrm{d}\rho \right] \\ &= \int_{-\infty}^{\infty} \mathrm{E}[X(t) X(t+\tau-\rho)]\, h(\rho)\, \mathrm{d}\rho, \end{aligned}$$

and, finally,

$$R_{XY}(\tau) \;=\; \int_{-\infty}^{\infty} R_X(\tau-\rho)\, h(\rho)\, \mathrm{d}\rho.$$

Therefore,

$$R_{XY}(\tau) \;=\; R_X(\tau) * h(\tau)$$

and

$$R_{YX}(\tau) \;=\; R_X(\tau) * h(-\tau).$$

The resulting cross-power spectral density (CPSD) between input and output is given by

$$S_{XY}(\tau) \;=\; S_X(\omega) \cdot H(\omega)$$

and between output and input by

$$S_{YX}(\tau) \;=\; S_X(\omega) \cdot H^*(\omega).$$

By assuming $S_Y(\omega) \;=\; |H(\omega)|^2 S_X(\omega)$, the following relationships are immediate:

$$\begin{aligned} S_Y(\omega) &= H^*(\omega) S_{XY}(\omega) \\ &= H(\omega) S_{YX}(\omega). \end{aligned}$$

It is usually easier to deal with PSD rather than autocorrelations, and the determination of $S_Y(\omega)$ from $S_X(\omega)$ is a rather direct operation for linear systems. The CPSDs are calculated afterward, and the correlations are then obtained by

inverse Fourier transformation. This is indicated in the following:

$$\begin{array}{ccccccc} R_X(\tau) & \longleftrightarrow & S_X(\omega) & & & & \\ & & \downarrow & & & & \\ R_Y(\tau) & \longleftrightarrow & S_Y(\omega) & \longrightarrow & S_{XY}(\omega) & \longleftrightarrow & R_{XY}(\tau) \\ & & \downarrow & & & & \\ & & S_{YX}(\omega) & & & & \\ & & \updownarrow & & & & \\ & & R_{YX}(\tau) & & & & \end{array}$$

5.7 Phase Information

The autocorrelation is a special measure of average behavior of a signal. Consequently, it is not always possible to recover a signal from its autocorrelation. Since the PSD is a function of the autocorrelation, it also follows that signal recovery from its PSD is not always possible because phase information about the signal has been lost in the averaging operation involved.

On the other hand, the CPSDs, relating input-output and output-input, preserve signal phase information and can be used to recover the phase function explicitly.

The transfer function of a linear system can be written as

$$H(\omega) = |H(\omega)|e^{j\theta(\omega)},$$

in which the modulus $|H(\omega)|$ and the phase $\theta(\omega)$ are clearly separated. The complex conjugate of the transfer function is

$$H^*(\omega) = |H(\omega)|e^{-j\theta(\omega)}.$$

Since

$$S_Y(\omega) = H^*(\omega)S_{XY}(\omega) = H(\omega) \cdot S_{YX}(\omega),$$

it follows for a real $h(t)$ that

$$|H(\omega)|e^{-j\theta\omega} \cdot S_{XY}(\omega) = |H(\omega)| \cdot e^{j\theta\omega} \cdot S_{YX}(\omega)$$

and, finally,

$$e^{2j\theta(\omega)} = \frac{S_{XY}(\omega)}{S_{YX}(\omega)}.$$

The function $\theta(\omega)$ can then be extracted, thus giving

$$\theta(\omega) = \frac{1}{2j} \ln \frac{S_{XY}(\omega)}{S_{YX}(\omega)},$$

which is the desired signal phase information.

Thus, the previous diagram can be completed with the information on how to compute the phase function of the signal, as follows:

$$\begin{array}{ccccccc} R_X(\tau) & \longleftrightarrow & S_X(\omega) & & & & \\ & & \downarrow & & & & \\ R_Y(\tau) & \longleftrightarrow & S_Y(\omega) & \longrightarrow & S_{XY}(\omega) & \longleftrightarrow & R_{XY}(\tau) \\ & & \downarrow & & \downarrow & & \\ & & S_{YX}(\omega) & \longrightarrow & \theta(\omega) & & \\ & & \updownarrow & & & & \\ & & R_{YX}(\tau) & & & & \end{array}$$

Example The Hilbert transform provides an example of how to apply the preceding theory. A representation of the filter impulse response for the Hilbert transform has been shown previously. The frequency domain representation of the Hilbert transform is shown in Fig. 5.27, and its phase function is depicted in Fig. 5.28.

Since for the Hilbert transform $H(\omega) = j\,[u(-\omega) - u(\omega)]$, it follows that $|H(\omega)|^2 = 1$, and from the important result

$$S_Y(\omega) = |H(\omega)|^2 \cdot S_X(\omega),$$

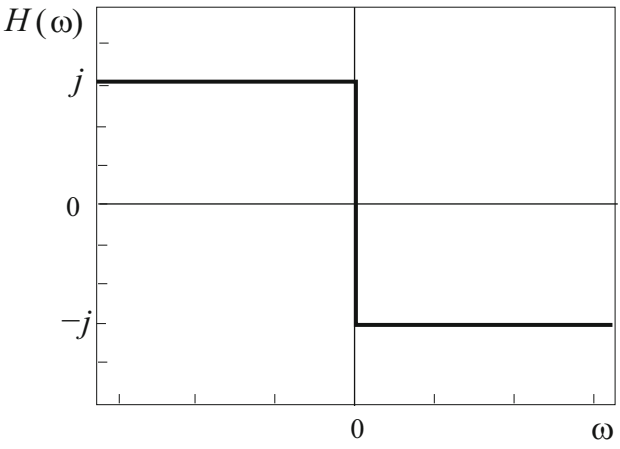


Fig. 5.27 Transfer function of the Hilbert transform

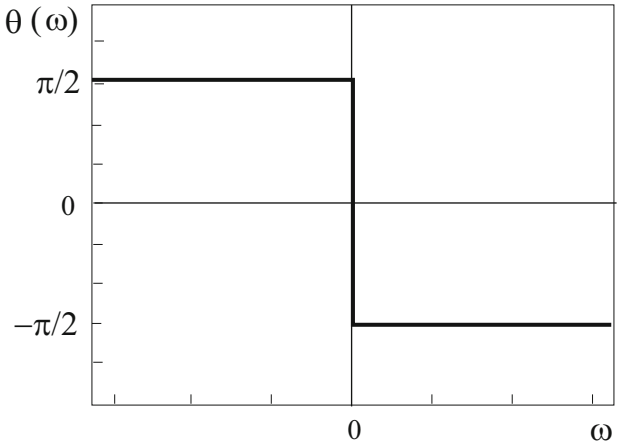


Fig. 5.28 The phase function of the Hilbert transform in the frequency domain

it follows that

$$S_Y(\omega) = S_X(\omega).$$

Therefore, the PSD of a signal at the output of a Hilbert filter is exactly the same as the input. The fact that $S_Y(\omega) = S_X(\omega)$ is expected, since the Hilbert transform operates only on the signal phase and the PSD does not contain phase information.

5.8 Analysis of a Digital Signal

Most modulation techniques use digital signals as inputs, and this is the only reason they are called digital modulation schemes. Then, digital signal analysis is important to obtain the spectra of digitally modulated carriers.

This section presents a mathematical formulation for the digital signal, which includes the computation of its autocorrelation and power density spectrum.

The digital signal can be produced by the digitization of an audio or a video signal or can be directly generated by a digital equipment or by a computer connected to a network. A possible mathematical expression of the digital signal is as follows,

$$m(t) = \sum_{k=-\infty}^{\infty} m_k p(t - kT_b), \tag{5.42}$$

in which m_k represents the k-th randomly generated symbol from a discrete alphabet $\mathbb{M}$, $p(t)$ is the pulse function that shapes the transmitted signal, and T_b is the bit interval.

5.8.1 Autocorrelation of a Digital Signal

As previously discussed, the autocorrelation function for signal $m(t)$, which can be nonstationary, is computed by the formula

$$R_M(\tau, t) = \mathrm{E}[m(t)m(t+\tau)]. \tag{5.43}$$

Substituting the expression for $m(t)$ into Formula 5.43 gives

$$R_M(\tau, t) = \mathrm{E}\left[\sum_{k=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} m_k\, p(t-kT_b) m_j\, p(t+\tau-iT_b)\right]. \tag{5.44}$$

Because of the linearity property, the expected value operator applies directly to the random signals,

$$R_M(\tau, t) = \sum_{k=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} \mathrm{E}\left[m_k m_j\right] p(t-kT_b)\, p(t+\tau-iT_b). \tag{5.45}$$

Equation 5.45 is averaged in time to eliminate the time dependency, producing

$$R_M(\tau) = \frac{1}{T_b}\int_0^{T_b} R_M(\tau, t)\mathrm{d}t, \tag{5.46}$$

or, equivalently,

$$R_M(\tau) = \frac{1}{T_b}\int_0^{T_b} \sum_{k=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} \mathrm{E}\left[m_k m_j\right] p(t-kT_b)\, p(t+\tau-iT_b)\mathrm{d}t. \tag{5.47}$$

The integral and summation operations can be changed, which gives

$$R_M(\tau) = \frac{1}{T_b}\sum_{k=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} \mathrm{E}\left[m_k m_j\right] \int_0^{T_b} p(t-kT_b)\, p(t+\tau-iT_b)\mathrm{d}t. \tag{5.48}$$

In the last equation, it is possible to define the discrete autocorrelation, to simplify the expression, as

$$R(k-i) = \mathrm{E}\left[m_k m_j\right], \tag{5.49}$$

and the signal autocorrelation can be written as

$$R_M(\tau) = \frac{1}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) \int_0^{T_b} p(t - kT_b)\, p(t + \tau - iT_b)\mathrm{d}t, \tag{5.50}$$

which is the general formula for the autocorrelation of a digital signal.

Example For the particular case of a signal, shaped as a rectangular pulse defined in the interval $0 \leq t \leq T_b$, with independent and equiprobable symbols from the set $\mathbb{M} = \{-A, A\}$, the autocorrelation function is given by

$$R_M(\tau) = A^2[1 - \frac{|\tau|}{T_b}][u(\tau + T_b) - u(\tau - T_b)], \tag{5.51}$$

in which T_b is the bit interval and A represents the pulse amplitude.

Figure 5.29 shows that the autocorrelation function has a triangular shape. Its maximum, the signal power, occurs at the origin and is equal to A^2. The autocorrelation decreases linearly with the time interval and reaches zero at time T_b.

5.8.2 *Power Spectral Density for the Digital Signal*

The Fourier transform of the autocorrelation function, Eq. 5.50, is computed to obtain the PSD for the digital signal:

$$S_M(\omega) = \int_{-\infty}^{\infty} R_M(\tau) e^{-j\omega\tau} \mathrm{d}\tau \tag{5.52}$$

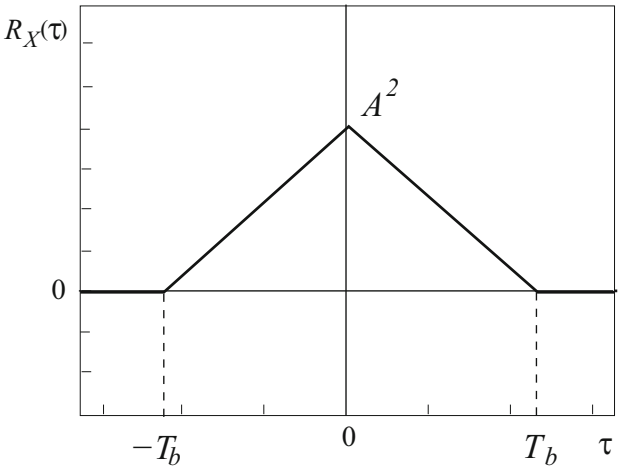


Fig. 5.29 Autocorrelation for the digital signal

Substituting the expression for the autocorrelation,

$$S_M(\omega) = \frac{1}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) \cdot \int_{-\infty}^{\infty} \int_{0}^{T_b} p(t-kT_b)\, p(t+\tau-iT_b) e^{-j\omega\tau} \mathrm{d}t \mathrm{d}\tau. \tag{5.53}$$

The order of integration is changed to compute the Fourier integral of the shifted pulse. This can be written as

$$S_M(\omega) = \frac{1}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) \int_{0}^{T_b} p(t-kT_b) P(\omega) e^{-j\omega(kT_b-t)} \mathrm{d}t. \tag{5.54}$$

Because the term $P(\omega)e^{-j\omega kT_b}$ is independent of time, it can be taken out of the integral, that is,

$$S_M(\omega) = \frac{1}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) P(\omega) e^{-j\omega kT_b} \int_{0}^{T_b} p(t-kT_b) e^{j\omega t} \mathrm{d}t. \tag{5.55}$$

The integral in 5.55 is then evaluated to give

$$S_M(\omega) = \frac{1}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) P(\omega) P(-\omega) e^{-j\omega(k-j)T_b}. \tag{5.56}$$

The shape of the spectrum for the random digital signal depends on the pulse shape, defined by $P(\omega)$, and also on the manner the symbols relate to each other, specified by the discrete autocorrelation function $R(k-i)$.

Therefore, the signal design, necessary to produce an adequate modulation scheme, involves pulse shaping as well as the control of the correlation between the transmitted symbols. It can be obtained by signal processing.

For a real pulse, $P(-\omega) = P^*(\omega)$, and the PSD can be written as

$$S_M(\omega) = \frac{|P(\omega)|^2}{T_b} \sum_{k=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} R(k-i) e^{-j\omega(k-j)T_b}, \tag{5.57}$$

which can be simplified to

$$S_M(\omega) = \frac{|P(\omega)|^2}{T_b} S(\omega); \tag{5.58}$$

if one puts $l = k - i$, the summations can be simplified, and the PSD for the discrete sequence of symbols is given by

$$S(\omega) = \sum_{l=-\infty}^{\infty} R(l) e^{-j\omega l T_b}. \tag{5.59}$$

For the example of Eq. 5.51, the corresponding spectrum is

$$S_M(\omega) = A^2 T_b \frac{\sin^2(\omega T_b/2)}{(\omega T_b/2)^2}, \tag{5.60}$$

which is the squared sampling function and shows that the random digital signal has a continuous spectrum that occupies a large portion of the spectrum. The function is sketched in Fig. 5.30. The first null is a usual measure of the bandwidth and is given by $\omega_M = 2\pi/T_b$.

Example Suppose the digital signal has an expected value different from zero, as is the case illustrated in Fig. 5.31.

For this case, the signal is shaped as a rectangular pulse defined in the interval $0 \leq t \leq T_b$, with independent and equiprobable symbols from the set $\mathbb{M} = \{0, A\}$.

The autocorrelation function is given by

$$R_M(\tau) = \frac{A^2}{4}\left[1 - \frac{|\tau|}{T_b}\right][u(\tau + T_b) - u(\tau - T_b)] + \frac{A^2}{4}, \tag{5.61}$$

in which T_b is the bit interval, and shown in Fig. 5.32.

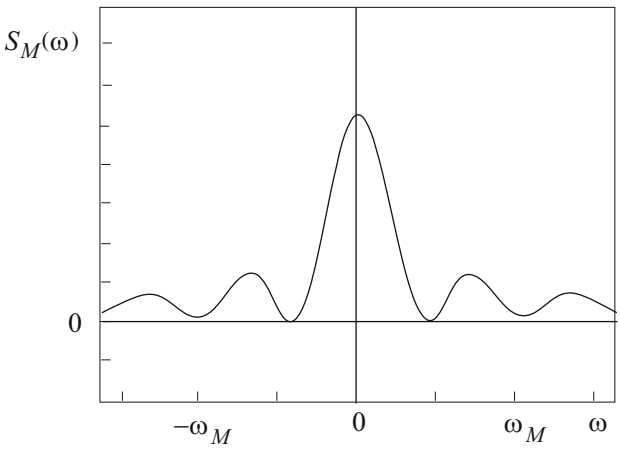


Fig. 5.30 Power spectral density for the random digital signal

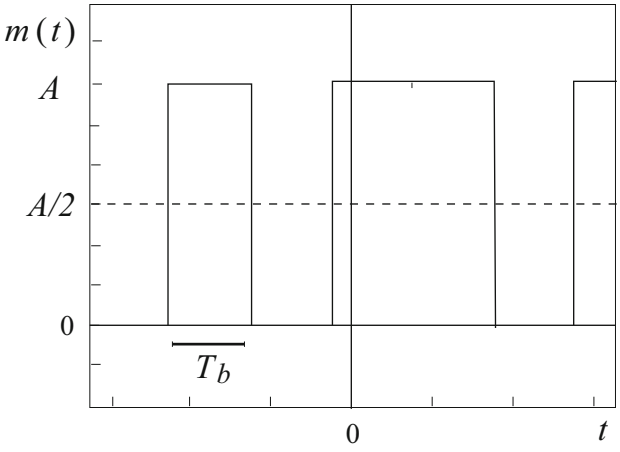


Fig. 5.31 Random digital signal with mean different from zero

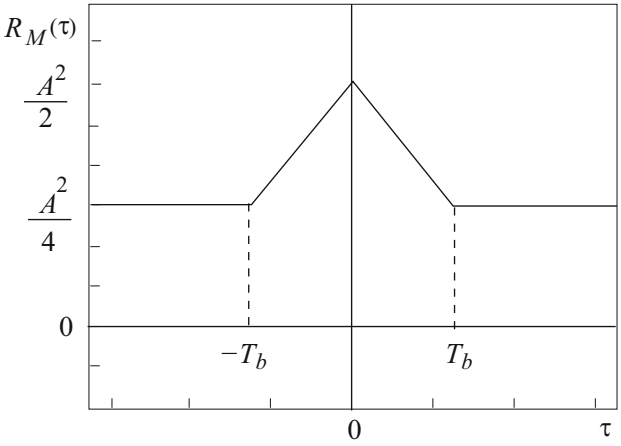


Fig. 5.32 Autocorrelation function of a random digital signal with mean different from zero

The power spectral density is then obtained by Fourier transforming the autocorrelation function,

$$S_M(\omega) = A^2 T_b \frac{\sin^2(\omega T_b/2)}{(\omega T_b/2)^2}, \tag{5.62}$$

which is the squared sampling function, and shows an impulse at the origin. The first null is again a measure of the bandwidth, given by $\omega_M = 2\pi/T_b$.

The spectrum is illustrated in Fig. 5.33. Notice that an impulse appeared at zero, indicating the digital signal has an expected value different from zero.

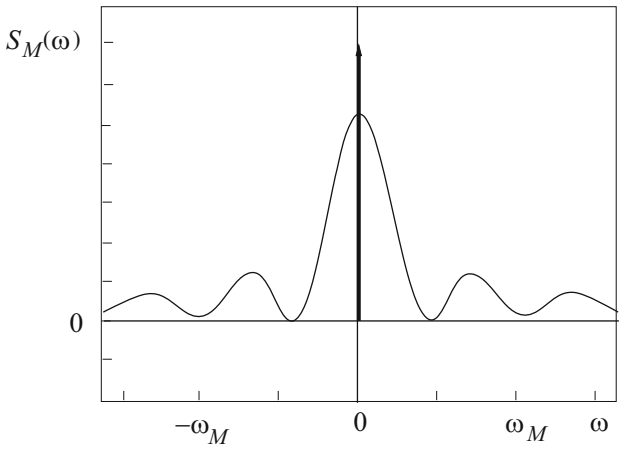


Fig. 5.33 Spectrum of a random digital signal with mean different from zero

5.8.3 Spectrum for Non-equiprobable Levels

The results of the previous subsection have been obtained for a digital signal with equiprobable levels. If level A has probability p and level $-A$ has probability $1-p$, then the PSD can be computed and gives (Gagliardi 1988)

$$S_M(\omega) = [4p(1-p)]\, A^2 T_b \frac{\sin^2(\omega T_b/2)}{(\omega T_b/2)^2} + 2\pi A^2 (2p-1)^2 \delta(\omega). \tag{5.63}$$

This represents a similar spectrum, with the amplitude adjusted by the factor $[4p(1-p)]$ and with an additional impulse at the origin, indicating a possible DC level, depending on the symbol probability.

It is also possible to shape the signal spectrum by controlling the symbol sequence statistics. First, substitute Eq. 5.59 into 5.58:

$$S_M(\omega) = \frac{|P(\omega)|^2}{T_b} \sum_{l=-\infty}^{\infty} R(l) e^{-j\omega l T_b}. \tag{5.64}$$

The summation in Eq. 5.64, which shapes the pulse spectrum, is the discrete Fourier transform of the discrete autocorrelation $R(l)$. Therefore, as the properties of the binary signal depend on the characteristics of the information sequence, it is possible to change the signal spectrum by the introduction of correlation between the sequence bits.

Example The input bit sequence, m_k, is equally likely and independent, as assumed previously. Suppose a new discrete sequence is produced as

$$b_k = \frac{m_k + m_{k-1}}{2}. \tag{5.65}$$

Equation 5.65 represents a moving average operation, in which symbol b_k is obtained by averaging the current symbol with the previous one.

The discrete autocorrelation function is then

$$R(l) = \delta(l) + \frac{\delta(l+1) + \delta(l-1)}{2}, \tag{5.66}$$

in which $\delta(\cdot)$ is the Kronecker delta function, after the German mathematician Leopold Kronecker (1823–1891), defined as $\delta(l) = 1$, for $l = 0$ and zero otherwise.

And the discrete Fourier transform is

$$\begin{aligned} S(\omega) &= \sum_{l=-\infty}^{\infty} R(l) e^{-j\omega l T_b} = 1 + \frac{e^{-j\omega T_b} + e^{j\omega T_b}}{2} \\ &= 1 + \cos \omega T_b \\ &= 2 \cos^2 \frac{\omega T_b}{2}. \end{aligned}$$

Finally, the signal spectrum is given by

$$S_M(\omega) = \frac{2|P(\omega)|^2}{T_b} \cos^2 \frac{\omega T_b}{2}. \tag{5.67}$$

It is possible to note that the pulse spectrum has been low-pass filtered by the sequence spectrum, as a result of the introduction of correlation between the symbols, and also there are no discrete lines in the resulting spectrum.

5.8.4 The Digital Signal Bandwidth

The signal bandwidth can be defined in several ways. In industry, the most usual definition is the half-power bandwidth (ω_{3dB}). This bandwidth is computed by dividing the maximum value of the function by two and finding the frequency for which this particular value occurs.

The root mean square (RMS) bandwidth is computed using the frequency deviation around the carrier, if the signal is modulated, or around the origin, for a baseband signal. The frequency deviation, or RMS bandwidth, for a baseband

signal is given by

$$\omega_{RMS} = \sqrt{\frac{\int_{-\infty}^{\infty} \omega^2 S_M(\omega)\mathrm{d}\omega}{\int_{-\infty}^{\infty} S_M(\omega)\mathrm{d}\omega}}, \tag{5.68}$$

and the RMS bandwidth is $2\omega_{RMS}$.

The previous formula is equivalent to

$$\omega_{RMS} = \sqrt{\frac{-R''_M(0)}{R_M(0)}}. \tag{5.69}$$

The RMS bandwidth for a modulated signal, around the carrier frequency ω_c, is given by

$$\omega_{RMS} = \sqrt{\frac{\int_{-\infty}^{\infty} (\omega - \omega_c)^2 S_M(\omega - \omega_c)\mathrm{d}\omega}{\int_{-\infty}^{\infty} S_M(\omega - \omega_c)\mathrm{d}\omega}}. \tag{5.70}$$

The white noise bandwidth can be computed by equating the maximum of the signal power spectrum density to the noise power spectrum density $S_N = \max S_M(\omega)$. After that, the power for both signal and noise are equated and ω_N is obtained. The noise power is $P_N = 2\omega_N S_N$, and the signal power, P_M, is given by the formula

$$P_M = R_M(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_M(\omega)\mathrm{d}\omega. \tag{5.71}$$

There are signals that exhibit a finite bandwidth, when the spectrum vanishes after a certain frequency. They exhibit a real bandwidth. Finally, the percent bandwidth is computed by finding the frequency range that includes a known percentage, for instance, 90%, of the signal power.

5.9 Signal Fading

In wireless communications, fading is a variation induced on the attenuation of a transmitted signal as a result of various channel effects. These effects are caused by variations in time, geographical position, and radio frequency.

Fading is frequently modeled as a random process, and a fading channel is a communication channel that experiences this kind of attenuation. In wireless systems, fading may either be due to multipath propagation, referred to as multipath-induced fading (caused by multiple reflections, refractions, or diffractions), weather

(e.g., dust or rain), or shadowing from obstacles affecting the signal propagation, which is called shadow fading.

The superposition of multiple copies of the transmitted signal, traversing different paths, in which each signal copy experiences different attenuation, delay, and phase shift while traveling from the source to the receiver, results in either constructive or destructive interference, amplifying or attenuating the signal amplitude at the receiver. Strong destructive interference is referred to as a deep fade and may result in temporary failure of communication due to a severe drop in the channel signal-to-noise ratio.

5.10 Rayleigh Fading

Rayleigh fading is a statistical model for the effect of a propagation environment on a radio signal, such as that used by wireless devices. Rayleigh fading models assume that the magnitude of a received signal varies randomly, or fades, according to a Rayleigh distribution, which represents the radial component of the sum of two uncorrelated Gaussian random variables.

This type of fading was named after John William Strutt, 3rd Baron Rayleigh, (1842–1919), an English mathematician and physicist who made extensive contributions to science. Lord Rayleigh spent his academic career at the University of Cambridge, served as president of the Royal Society, and received the 1904 Nobel Prize in Physics.

Rayleigh fading is an adequate model for tropospheric and ionospheric signal propagation, as well as the effect of heavily built-up urban environments on radio signals. This kind of fading is usual when the number of multiple reflective paths is large and there is no dominant propagation along a line of sight between the transmitter and the receiver. If there is a dominant line of sight, Rician fading may be more adequate.

In probability theory and statistics, the Rayleigh distribution is a continuous probability distribution for nonnegative-valued random variables. The probability density function of the Rayleigh distribution is

$$p_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0, \tag{5.72}$$

in which σ is the scale parameter of the distribution. An illustration of the Rayleigh distribution is presented in Fig. 5.34.

The cumulative distribution function is

$$P_R(r) = 1 - e^{-r^2/(2\sigma^2)} \tag{5.73}$$

for $r \in [0, \infty)$.

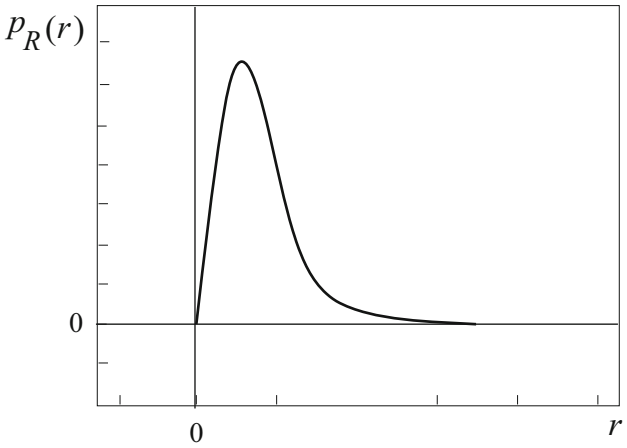


Fig. 5.34 Envelope of the Rayleigh distribution

Consider the two-dimensional vector $Y = (U, V)$, which has components that are bivariate normally distributed, centered at zero, and independent. Then U and V have density functions

$$p_U(u) = \frac{e^{-u^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \tag{5.74}$$

and

$$p_V(v) = \frac{e^{-v^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}. \tag{5.75}$$

Let R be the length of the radius. That is, $R = \sqrt{U^2 + V^2}$. Then R has cumulative distribution function

$$P_R(r) = \iint_{D_R} p_U(u)\, p_V(v)\, \mathrm{d}A, \tag{5.76}$$

in which $\mathrm{d}A = \mathrm{d}u\mathrm{d}v$ and D_R is the disk

$$D_R = \left\{(u, v) : \sqrt{u^2 + v^2} \leq r\right\}.$$

Writing the double integral in polar coordinates, one obtains

$$P_R(r) = \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_0^r x e^{-x^2/(2\sigma^2)}\, \mathrm{d}x\, \mathrm{d}\theta = \frac{1}{\sigma^2} \int_0^r x e^{-x^2/(2\sigma^2)}\, \mathrm{d}x. \tag{5.77}$$

Finally, the probability density function for X is the derivative of its cumulative distribution function, which by the fundamental theorem of calculus is

$$p_R(r) = \frac{\mathrm{d}}{\mathrm{d}r} P_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \tag{5.78}$$

which is the Rayleigh distribution. The statistical mean is

$$\mathrm{E}[R] = \sigma \sqrt{\frac{\pi}{2}},$$

and the variance is given by

$$\mathrm{V}[R] = \frac{(4-\pi)\sigma^2}{2}.$$

5.11 Nakagami Distribution

The Nakagami distribution is a recent probability distribution, which first appeared in 1960. It generalizes the modeling of small-scale fading for dense signal scatters and is one of the most common distributions for modeling right-skewed, positive data sets. Applications include modeling wireless signal and radio wave propagation, the characterization of breast tumors using ultrasound imaging, and in meteorology (Nakagami 1960).

The Nakagami or the Nakagami-m probability distribution, named after professor Minoru Nakagami, from the Kobe University, Japan, is related to the gamma distribution. The family of Nakagami distributions has two parameters: a shape parameter $m \geq 1/2$ and the parameter that controls the distribution spreading $\Omega > 0$.

It is known that the Nakagami-m distribution approximates both the Hoyt distribution, for $m < 1$, and the Rice distribution, for $m > 1$. In fact, for the Hoyt parameter $b = 0$, the Rice parameter $k = 0$, and the Nakagami parameter $m = 1$, the three distributions converge to the Rayleigh, one for which the uniform phase condition applies. But it is not the same for the Nakagami distribution.

The Nakagami-m phase-envelope joint distribution is described as follows: Let R and Θ be random variables representing, respectively, the envelope and phase of the Nakagami-m distribution. The corresponding joint probability density function is given by Yacoub et al. (2005)

$$p_{R\Theta}(r,\theta) = \frac{m^m |\sin(2\theta)|^{m-1}}{2^{m-1}\Gamma^2(m/2)\Omega^m} r^{2m-1} \exp\left(-\frac{m}{\Omega} r^2\right), \ \forall r \geq 0, \tag{5.79}$$

in which $m \geq 1/2$ and $\Omega > 0$.

The envelope pdf is given by the formula

$$p_R(r) = \frac{2m^m}{\Gamma(m)\Omega^m} r^{2m-1} \exp\left(-\frac{m}{\Omega} r^2\right), \forall r \geq 0. \tag{5.80}$$

The phase pdf is given by

$$p_\Theta(\theta) = \frac{\Gamma(m)|\sin(2\theta)|^{m-1}}{2^m \Gamma^2(m/2)}. \tag{5.81}$$

It is not difficult to show that

$$p_{R\Theta}(r, \theta) = p_R(r) \times p_\Theta(\theta),$$

and the envelope and phase are independent random variables.

The cumulative distribution function of the envelope can be computed to give

$$P(r) = G\left(m, \frac{mr^2}{\Omega}\right), \tag{5.82}$$

in which G is the regularized (lower) incomplete gamma function.

The parameters m and Ω are given by

$$m = \frac{\left(\mathrm{E}\left[X^2\right]\right)^2}{\mathrm{V}\left[X^2\right]} \tag{5.83}$$

and

$$\Omega = \mathrm{E}\left[X^2\right]. \tag{5.84}$$

5.12 Rice Distribution

In probability theory, the Rice or Rician distribution is the probability distribution of the magnitude of a circularly symmetric bivariate normal random variable, usually with a mean different from zero. It was named after Stephen Oswald Rice (1907–1986), an electrical engineer who pioneered in the fields of Information Theory, Communications Theory, and Telecommunications.

Consider again the two-dimensional vector $Y = (U, V)$, which has components that are bivariate normally distributed and independent. But let one of the components to reflect the possibility of a direct (line-of-sight) path, with amplitude A.

Then, U and V have density functions

$$p_U(u) = \frac{e^{-(u+A)^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \tag{5.85}$$

and

$$p_V(v) = \frac{e^{-v^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}. \tag{5.86}$$

Then, the resulting combination of random components gives

$$|R|^2 = (U + A)^2 + V^2 = U^2 + V^2 + 2AU + A^2, \tag{5.87}$$

and the Jacobian of the transformation is $J(R, \Theta) = R$.

Therefore, the joint probability distribution can be obtained,

$$p_{R,\Theta}(r, \theta) = \frac{r}{2\pi\sigma^2} \exp\left[\frac{-(u^2 + v^2 + 2Au + A^2)}{2\sigma^2}\right]. \tag{5.88}$$

This last formula can be integrated in θ to obtain the marginal probability density function of R,

$$p_R(r) = \int_0^{2\pi} \frac{r}{2\pi\sigma^2} \exp\left[\frac{-(u^2 + v^2 + 2Au + A^2)}{2\sigma^2}\right] \mathrm{d}\theta. \tag{5.89}$$

Taking out of the integrand the part that do not depend on θ, one obtains

$$p_R(r) = \frac{r}{2\pi\sigma^2} e^{\frac{-A^2}{2\sigma^2}} \int_0^{2\pi} \exp\left[\frac{-(u^2 + v^2 + 2Au)}{2\sigma^2}\right] \mathrm{d}\theta. \tag{5.90}$$

Recalling that $u = r\cos\theta$, one obtains

$$p_R(r) = \frac{r}{2\pi\sigma^2} e^{\frac{-A^2}{2\sigma^2}} \int_0^{2\pi} \exp\left[\frac{-(u^2 + v^2 + 2Ar\cos\theta)}{2\sigma^2}\right] \mathrm{d}\theta, \tag{5.91}$$

which can be written as

$$p_R(r) = e^{\frac{-A^2}{2\sigma^2}} \frac{r}{2\pi\sigma^2} \int_0^{2\pi} \exp\left[\frac{-(r^2 + 2Ar\cos\theta)}{2\sigma^2}\right] \mathrm{d}\theta, \tag{5.92}$$

which can be adjusted to

$$p_R(r) = \frac{r}{\sigma^2} \exp\left[\frac{-(r^2 + A^2)}{2\sigma^2}\right] \frac{1}{2\pi} \int_0^{2\pi} \exp\left[\frac{-Ar\cos\theta}{\sigma^2}\right] \mathrm{d}\theta. \tag{5.93}$$

Considering the definition of the modified Bessel function,

$$I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{x \cos\theta} \mathrm{d}\theta, \tag{5.94}$$

the required probability density function is then obtained,

$$p_R(r) = \frac{r}{\sigma^2} \exp\left[\frac{-(r^2 + A^2)}{2\sigma^2}\right] I_0\left(\frac{rA}{\sigma^2}\right), \tag{5.95}$$

in which $I_0(z)$ is the modified Bessel function of the first kind with order zero.

The pdf is depicted in Fig. 5.35, for some values of the parameter A, considering σ fixed. Note that the Rice distribution approaches the Gaussian distribution, as the parameter A increases.

In the context of Rician fading, the distribution is often also rewritten using the shape parameter

$$K = \frac{A^2}{2\sigma^2},$$

defined as the ratio of the power contributions by the line-of-sight path, represented by A^2, to the remaining multipaths, and the scale parameter

$$\Omega = A^2 + 2\sigma^2,$$

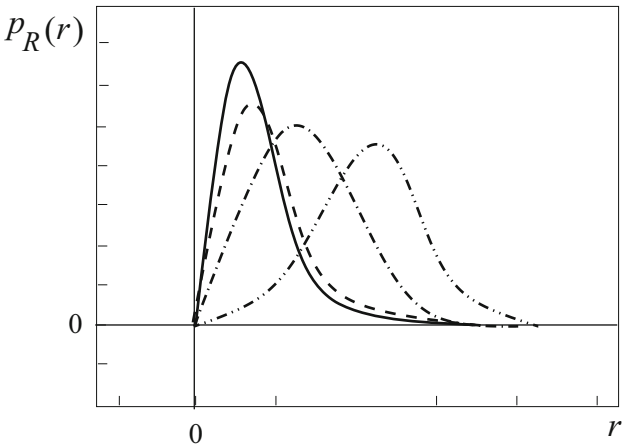


Fig. 5.35 Envelope of the Rice distribution

defined as the total power received in all paths. The mean of the Rice distribution is

$$\mathrm{E}[X] = \sigma\sqrt{\frac{\pi}{2}}L_{1/2}\left(-\frac{A^2}{2\sigma^2}\right), \tag{5.96}$$

and the variance is

$$\mathrm{V}[X] = 2\sigma^2 + A^2\frac{\pi\sigma^2}{2}L_{1/2}^2\left(-\frac{A^2}{2\sigma^2}\right), \tag{5.97}$$

in which $L_n(x)$ denotes the Laguerre polynomials, named after Edmond Nicolas Laguerre (1834–1886), a French mathematician, that may be defined by the Rodrigues formula, which was independently introduced by Benjamin Olinde Rodrigues (1795–1851), a French banker, mathematician, and social reformer; by Sir James Ivory, (1765–1842), a British mathematician; and by Carl Gustav Jacob Jacobi (1804–1851), a German mathematician who made fundamental contributions to elliptic functions, dynamics, differential equations, determinants, and number theory,

$$L_n(x) = \frac{e^x}{n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(e^{-x}x^n) = \frac{1}{n!}\left(\frac{\mathrm{d}}{\mathrm{d}x} - 1\right)^n x^n. \tag{5.98}$$

The formula can also be written as

$$L_n(x) = L_n^{(0)}(x) = M(-n, 1, x) = {}_1F_1(-n; 1; x), \tag{5.99}$$

in which

$$M(a, b, z) = {}_1F_1(a; b; z) \tag{5.100}$$

is the confluent hypergeometric function of the first kind.

5.13 Slow and Fast Fading

The terms slow and fast fading refer to the rate at which the magnitude and phase change imposed by the channel on the signal changes. The coherence time is a measure of the minimum time required for the magnitude change or phase change of the channel to become uncorrelated from its previous value.

Slow fading arises when the coherence time of the channel is large relative to the delay requirement of the application. In this regime, the amplitude and phase change imposed by the channel can be considered roughly constant over the period of use. Slow fading can be caused by events such as shadowing, in which a large obstruction such as a hill or large building obscures the main signal path between

the transmitter and the receiver. The received power change caused by shadowing is often modeled using a log-normal distribution with a standard deviation according to the log-distance path loss model.

Fast fading occurs when the coherence time of the channel is small relative to the delay requirement of the application. In this case, the amplitude and phase change imposed by the channel varies considerably over the period of use.

5.14 Jakes' Model

The Doppler power spectral density of a fading channel describes how much spectral broadening the relative movement between the antennas causes. This means that a sinusoid signal, which Fourier transforms into an impulse in the frequency domain, is spread out across frequency when it passes through the communication channel. The Doppler power spectral density is the Fourier transform of the corresponding autocorrelation function in the time domain.

Christian Doppler (1803–1853) was director of the Physical Institute and professor of Experimental Physics of the University of Vienna. He was an Austrian physicist who described how the observed frequency of light and sound waves is affected by the relative motion of the source and the detector.

The Doppler effect is defined as the increase or decrease in the frequency of a transmitted signal as the transmitter and receiver move toward or away from each other,

$$\Delta f(t) = f_c \left(\frac{v}{c}\right) \cos\theta(t), \tag{5.101}$$

in which $\Delta f(t)$ is the frequency shift, f_c is the carrier frequency in Hz, v is the mobile velocity, c is the speed of light, and $\theta(t)$ is the angle function between the mobile velocity vector and the imaginary line that extends from the transmitting antenna and the mobile.

It is important to point out that this angle is a stochastic process, because one considers that the mobile changes direction at random. Because of this, the frequency deviation is also a function of time.

If one multiplies Eq. 5.101 by 2π, one obtains the frequency in radians per second [rd/s], as follows

$$\omega(t) = 2\pi\,\Delta f(t) = 2\pi f_c \left(\frac{v}{c}\right) \cos\theta(t) = \omega_c \left(\frac{v}{c}\right) \cos\theta(t), \tag{5.102}$$

which is more convenient to work with.

Consider that an unmodulated signal $s(t)$ is transmitted,

$$s(t) = A\cos[\omega_c t + \phi], \tag{5.103}$$

in which ϕ is a random phase, uniformly distributed in the interval $[0, 2\pi]$.

The received signal, by the mobile, is given by

$$s(t) = A\cos\left[\omega_c t + \omega_c\left(\frac{v}{c}\right)\int \cos\theta(t)\,\mathrm{d}t + \phi)\right]. \tag{5.104}$$

Equation 5.104 is equivalent to a frequency-modulated signal, with frequency deviation index given by

$$\Delta_{FM} = \omega_c\left(\frac{v}{c}\right). \tag{5.105}$$

Therefore, the power spectral density can be obtained as follows if one considers a large frequency deviation (Alencar and da Rocha Jr. 2022),

$$S_S(\omega) = \frac{\pi A^2}{2\Delta_{FM}}\left[p_M\left(\frac{\omega+\omega_c}{\Delta_{FM}}\right) + p_M\left(\frac{\omega-\omega_c}{\Delta_{FM}}\right)\right], \tag{5.106}$$

in which $p_M(\cdot)$ is the probability distribution function of the random process $m(t) = \cos\theta(t)$.

Considering that the distribution of $\theta(t)$ is known, it is possible to obtain the probability density function of $m(t)$ by a transformation of probability density function, as follows:

$$p_M(m) = \frac{p_\Theta(\theta)}{\left|\frac{\mathrm{d}m}{\mathrm{d}\theta}\right|},\ \theta = f^{-1}(m) \tag{5.107}$$

Taking into account that

$$\frac{\mathrm{d}m}{\mathrm{d}\theta} = -\sin\theta = -\sqrt{1-\cos^2\theta} = -\sqrt{1-m^2},$$

one obtains

$$p_M(m) = \frac{1}{\pi\sqrt{1-m^2}}. \tag{5.108}$$

Substituting 5.108 into 5.106 and cancelling the appropriate constants, one obtains

$$S_S(\omega) = \frac{A^2}{2}\left[\frac{1}{\sqrt{\Delta_{FM}^2-(\omega+\omega_c)^2}} + \frac{1}{\sqrt{\Delta_{FM}^2-(\omega-\omega_c)^2}}\right]. \tag{5.109}$$

Considering the definition of Δ_{FM} from Formula 5.105, the final formula is obtained for the power spectral density induced by the Doppler effect caused by

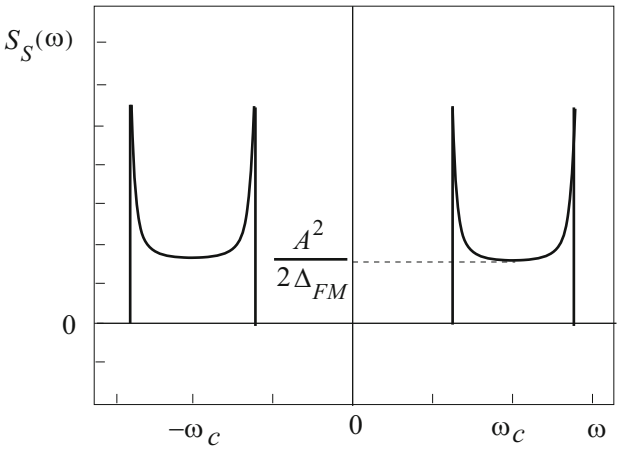


Fig. 5.36 Spectrum of the received signal subject to the Doppler effect

the movement of the mobile station,

$$S_S(\omega) = \frac{A^2}{2}\left[\frac{1}{\sqrt{\left(\frac{\omega_c v}{c}\right)^2 - (\omega+\omega_c)^2}} + \frac{1}{\sqrt{\left(\frac{\omega_c v}{c}\right)^2 - (\omega-\omega_c)^2}}\right]. \tag{5.110}$$

Note that the power spectral density broadens as the mobile velocity increases toward the base station and narrows as the mobile moves away from the base station with increased speed. The spectrum is illustrated in Fig. 5.36.

On the other hand, consider that the Fourier transform of the probability density function $p_M(m)$ is the characteristic function $P_M(\omega)$, given by

$$P_M(\omega) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{j\omega\sin\tau}\mathrm{d}\tau = J_0(\omega). \tag{5.111}$$

Therefore, the associated autocorrelation function for the received signal affected by the random frequency shift, caused by the Doppler effect, is given by (Fig. 5.37)

$$R_S(\tau) = \frac{A^2 c}{4\omega_c v} J_0\left(\frac{c\tau}{\omega_c v}\right)\cos(\omega_c\tau). \tag{5.112}$$

5.15 Problems

1. For the following joint probability distribution, determine the first and second moments of the random variable X. Draw the graphics for the distribution,

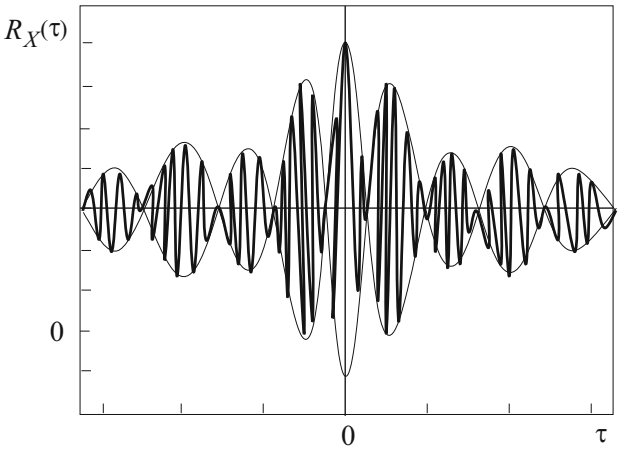


Fig. 5.37 Autocorrelation of the received signal subject to the Doppler effect

considering $r = 0$ and $r = 1$:

$$p_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-r^2}}\, e^{-\frac{1}{2(1-r^2)}\left(\frac{(x-m)^2}{\sigma_X^2} - \frac{2rxy}{\sigma_X\sigma_Y} + \frac{y^2}{\sigma_Y^2}\right)}.$$

2. A signal $x(t)$, with Gauss distribution, pass through a system with response $Y = X^2$. Compute the probability distribution of the output signal $y(t)$. Draw the graphics:

$$p_X(x) = \frac{1}{\sigma_X\sqrt{2\pi}} e^{-x^2/2\sigma_X^2}.$$

3. Determine and sketch the autocorrelation and the power spectrum density for the radar signal, given by

$$Y(t) = V\cos(\omega_0 t + \phi) + N(t),$$

which is subject to thermal noise, with autocorrelation given in the following, ω_0 is constant and ϕ is a random phase, with uniform distribution in the interval $[-\pi, \pi]$. Compute the decorrelation time to be used in the detection process:

$$R_N(\tau) = A^2 e^{-\lambda|\tau|}.$$

4. Compute the correlation between input and output signals for the following filter, given that the input has autocorrelation $R_X(\tau) = \delta(\tau)$. Relate the points in which the input and output are orthogonal. Are the signals decorrelated at

those points? Explain and draw the graphics:

$$H(\omega) = [u(\omega + \omega_M) - u(\omega - \omega_M)].$$

5. Project an equipment to compute the modulus and the phase of a complex Gaussian signal $Z(t) = X(t) + jY(t)$. What is the probability distribution of the signal amplitude at the output of the device?
6. Compute the correlations and the phase function for a channel that has the following cross-power densities. Sketch the correlations and the phase function for a transmitted signal that has the same characteristics of a white noise:

$$S_{XY}(\omega) = j[u(-\omega) - u(\omega)]S_X(\omega),$$

$$S_{YX}(\omega) = j[u(\omega) - u(-\omega)]S_X(\omega).$$

7. Compute and sketch the phase response of a signal at the output of the following system, considering a wide-sense random stationary process, $X(t)$,

$$h(t) = e^{-\alpha(t-\sigma)}u(t - \sigma).$$

8. Determine and sketch the power spectral density of the signal at the output of the following system, in which $X(t)$ is a wide-sense random stationary process,

$$Y(t) = \alpha X(t) - \beta X(t - \sigma)$$

for the following input autocorrelation,

$$R_X(\tau) = A^2[1 - \frac{|\tau|}{T}],\ -T \leq \tau \leq T.$$

9. A stochastic process is given by the following formula

$$Y(t) = V\cos[\omega_0 t + \alpha X(t) + \phi],$$

in which ω_0 is constant and ϕ is a random phase with uniform distribution in the interval $[-\pi, \pi]$. Consider that the parameter α has a high value, and determine and sketch the power spectral density for a Gaussian signal $X(t)$, with power spectral density

$$S_X(\omega) = S_0[u(\omega + \omega_M) - u(\omega - \omega_M)].$$

10. The model for the telephone traffic measured at the busy hour, considering infinite traffic sources, is characterized by the Poisson distribution, given in the following. Determine the characteristic function, and compute the average

traffic and the standard deviation in relation to the average traffic,

$$p_X(x) = e^{-b} \sum_{k=0}^{\infty} \frac{b^k}{k!} \delta(x-k).$$

11. Find the distribution of the random variable $Z = X + Y$, considering that X and Y are independent random variables with exponential distribution. Draw the graphics:

$$p_X(x) = \frac{\alpha}{2} e^{-\alpha x} u(x),$$

$$p_Y(y) = \frac{\beta}{2} e^{-\beta y} u(y).$$

12. A signal $x(t)$, with Gamma probability distribution, in the interval $[0, \pi]$, passes through a circuit with response $Y = X^2 \cdot u(X)$. Compute the signal distribution at the output of the circuit $y(t)$, and sketch the graphics:

$$p_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

13. The stochastic process $Z(t) = X(t)X'(t)$ is obtained from a Gaussian process $X(t)$, which has zero mean and power spectral density $S_X(\omega)$. Compute the power spectral density and the autocorrelation of $Z(t)$, considering that for a Gaussian process

$$S_{X^2}(\omega) = 2 \int_{-\infty}^{\infty} S_X(\omega - \phi) S_X(\phi) \mathrm{d}\phi.$$

Taking into account that the signal $X(t)$ has a uniform power spectral density $S_X(\omega) = S_o$, between $-\omega_m$ and ω_M, determine the autocorrelation and the power spectral density of $Z(t)$.

Chapter 6
Analog-to-Digital Conversion*



Without mathematics, there's nothing you can do. Everything around you is mathematics. Everything around you is numbers.

– Shakuntala Devi

6.1 Introduction

The fundamental purpose of signal compression techniques is to reduce the number of bits required to represent a signal (e.g., speech, image, video, audio) while maintaining an acceptable signal quality. Signal compression is essential for applications which need the minimization of the storage and/or transmission requirements, such as multimedia systems, integrated services digital networks (ISDN), video conference, voice response systems, voice mail, music broadcast, high-resolution facsimile, high-definition television (HDTV), mobile telephony, storage of medical images, archiving of fingerprints, and transmission of remote sensing images obtained from satellites.

Although some systems do not present severe bandwidth limitations, such as optical fiber networks, and although the technological evolution is continuously leading to memories with high storage capacity, signal compression plays an important role, because of many aspects, such as (Jayant and Noll 1984; Gersho and Gray 1992):

- The wide use of multimedia systems has led to an increasing demand concerning the storage of speech, audio, image, and video in compressed form;
- A larger number of communication channels may be multiplexed in broadband systems, using compression techniques to reduce the bandwidth requirements of each signal to be multiplexed;
- In mobile telephony, the bandwidth is severely limited, which has motivated research and development in speech coding.

* This chapter received contributions from Francisco Madeiro and Waslon Terllizzie Araújo Lopes.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_6

6.2 Signal Sampling

A bandlimited signal $f(t)$, having no frequency components above $\omega_M = 2\pi f_M$, can be reconstructed from its samples, collected at uniform time intervals $T_s = 1/f_s$, i.e., at a sampling rate f_s, in which $f_s \geq 2f_M$.

By a bandlimited signal $f(t) \longleftrightarrow F(\omega)$, it is understood that there is a frequency ω_M above which $F(\omega) = 0$, i.e., that $F(\omega) = 0$ for $|\omega| > \omega_M$. Nyquist concluded that all the information about $f(t)$, as illustrated in Fig. 6.1, is contained in the samples of this signal, collected at regular time intervals T_s.

In this manner the signal can be completely recovered from its samples. For a bandlimited signal $f(t)$, i.e., such that $F(\omega) = 0$ for $|\omega| > \omega_M$, as shown in Fig. 6.2, it follows that

$$f(t) * \frac{\sin(at)}{\pi t} = f(t), \text{ if } a > \omega_M,$$

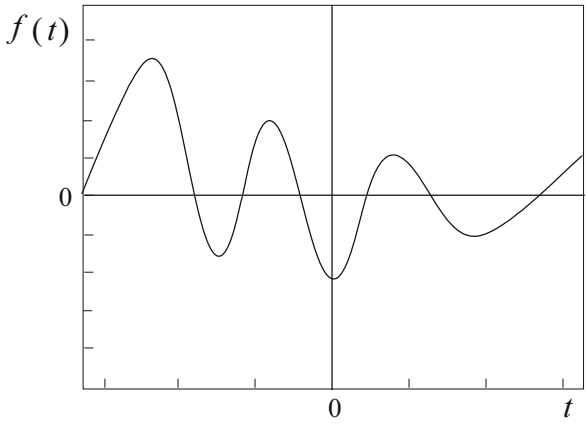


Fig. 6.1 Bandlimited signal $f(t)$

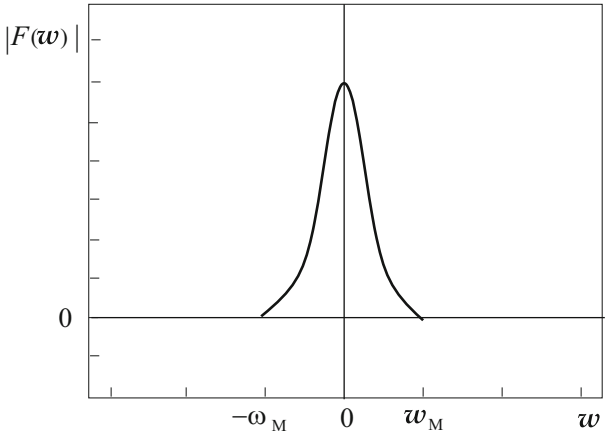


Fig. 6.2 Spectrum of a bandlimited signal $F(\omega)$

because in the frequency domain this corresponds to the product of $F(\omega)$ by a gate function of width greater than $2\omega_M$.

The function $f(t)$ is sampled once every T_s seconds or, equivalently, sampled with a sampling frequency f_s, in which $f_s = 1/T_s \geq 2f_M$.

Consider the signal $f_s(t) = f(t)\delta_T(t)$, in which

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT). \tag{6.1}$$

Computing the Fourier transform of the series of impulses, one obtains

$$\Delta_T(\omega) = \omega_o \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_o). \tag{6.2}$$

The signal $\delta_T(t)$ is illustrated in Fig. 6.3, and its Fourier transform is depicted in Fig. 6.4.

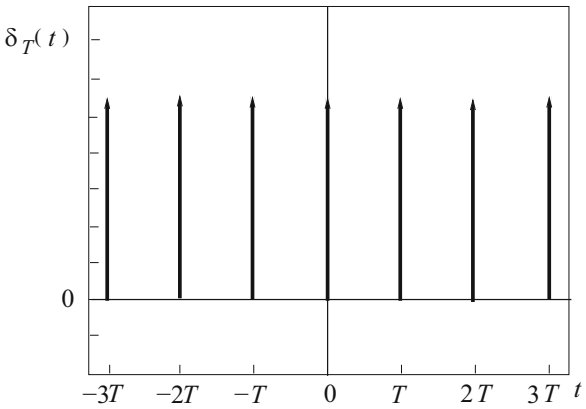


Fig. 6.3 Impulse train used for sampling

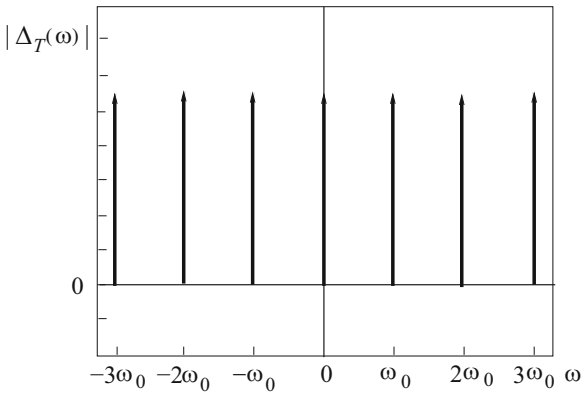


Fig. 6.4 Fourier transform of the impulse train

The signal $f_s(t)$ represents $f(t)$ sampled at uniform time intervals T_s seconds. From the frequency convolution theorem, it follows that the Fourier transform of the product of two functions in the time domain is given by the convolution of their respective Fourier transforms. It now follows that

$$f_s(t) \longleftrightarrow \frac{1}{2\pi}[F(\omega) * \Delta_T(\omega)] \tag{6.3}$$

and thus

$$f_s(t) \longleftrightarrow \frac{1}{T} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_o). \tag{6.4}$$

Figure 6.5 shows the sampled signal. It can be observed from Fig. 6.6 that if the sampling frequency ω_s is less than $2\omega_M$, there will be an overlap of spectral components. This will cause a loss of information because the original signal can

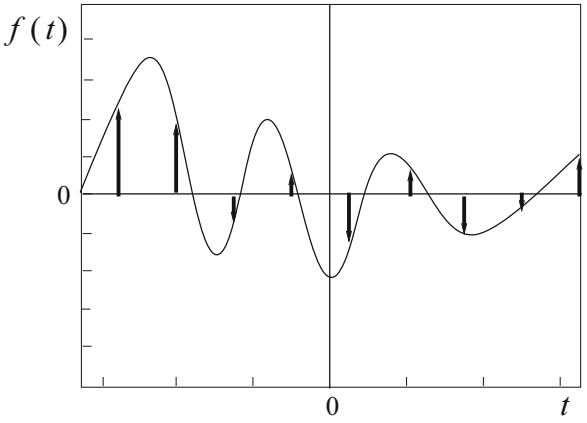


Fig. 6.5 Sampled signal

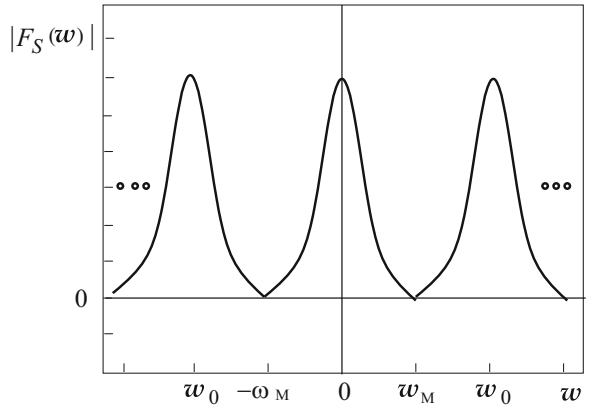


Fig. 6.6 Spectrum of a sampled signal

no longer be fully recovered from its samples. As ω_s becomes smaller than $2\omega_M$, the sampling rate diminishes causing a partial loss of information.

Therefore the minimum sampling frequency that allows perfect recovery of the signal is $\omega_s = 2\omega_M$ and is known as the Nyquist sampling rate. In order to recover the original spectrum $F(\omega)$, it is enough to pass the sampled signal through a low-pass filter with cutoff frequency ω_M.

For applications in telephony, the sampling frequency is $f_S = 8000$ samples per second or 8 k samples/s. Then the speech signal is quantized, as will be discussed later, for 256 distinct levels. Each level corresponds to an 8 bit code ($2^8 = 256$).

After encoding, the signal is transmitted at a rate of 8000 samples/s $\times$ 8 bits/sample = 64 kbits/s and occupies a bandwidth of approximately 64 kHz.

If the sampling frequency w_S is lower than $2\pi B$, there will be spectra overlap and, as a consequence, information loss. Therefore, the sampling frequency for a baseband signal to be recovered without loss is $w_S = 2\pi B$, known as the Nyquist sampling frequency.

As just mentioned, if the sampling frequency is lower than the Nyquist frequency, the signal will not be completely recovered, since there will be spectral superposition, leading to distortion in the highest frequencies. This phenomenon is known as *aliasing*. On the other hand, increasing the sampling frequency for a value higher than the Nyquist frequency leads to spectra separation higher than the minimum necessary to recover the signal.

6.3 Signal Coding

Signal coding is the process of representing an information signal with the purpose of reaching a communication target, such as analog to digital conversion, low bit rate transmission, or message encryption. In the literature, the terms source coding, signal compression, data compression, digital coding, and bandwidth compression are all employed when referring to the techniques to obtain a compact digital representation of a signal (Jayant 1992; Jayant et al. 1993).

6.4 The Performance of a Signal Compression System

The purpose of signal compression is to reduce the bit rate in the digital representation of a signal while maintaining the required levels of signal quality, implementation complexity, and communication delay. Each one of those aspects will be discussed in the following.

6.4.1 Quality of the Reconstructed Signals

It is important, in digital coding of signals, to develop methods for assessing the quality of reconstructed signals, usually obtained from the application of compression techniques. The measures used for assessing the quality of signals may be classified into two general groups: subjective quality measures and objective quality measures.

The subjective measures are based on comparisons (performed by means of listening or visualization tests) between the original signal and the processed signal, performed by a group of human observers which subjectively rank the quality of the processed signal according to a predetermined scale. The objective measures are based on a direct mathematical comparison between the original and processed signals (Deller Jr. et al. 1993).

To be useful, the objective quality measures must present two main attributes. They should have a subjective significance, in the sense that low and high changes in the objective measures correspond, respectively, to low and high changes in the subjective quality of the reconstructed signals. Hence, they should present a strong correlation, positive or negative, with the subjective evaluation results. Second, they should be mathematically tractable and be easily implementable.

Subjective quality measures are used to definitely assess the quality of algorithms and techniques of signal coding. However, subjective tests are difficult to implement, since they require personnel to participate in the evaluation, including lay men, specialists, and possible users of the signal coding system, involve a large volume of processed signal, and require the availability of laboratories with conditions suitable for the subjective evaluations.

Therefore, subjective tests are very time-consuming. For those reasons, objective quality measures, which are less time-consuming, play an important role in the process of assessing the quality of reconstructed signals and are very useful for the task of adjusting the parameters of algorithms and techniques of compression. The assessment of speech and image quality has been a research theme for a long time, being addressed by many researchers (Eskicioglu and Fischer 1995; Dimolitsas 1991).

In the following, an important subjective measure, the mean opinion score (MOS), is discussed. Three objective quality measures are addressed: the signal to noise ratio (SNR), the segmental signal to noise ratio (SNRseg), and the spectral distortion (SD).

Mean Opinion Score

A subjective quality measure widely used for assessing the performance of speech compression systems is the mean opinion score (MOS). Each listener rates the quality of the reconstructed signal according to a predetermined scale, shown in Table 6.1 (Jayant and Noll 1984; Deller Jr. et al. 1993). Then the scores are averaged

Table 6.1 Mean Opinion Score (MOS) five-point scale

Score (s)	Quality
5	Excellent
4	Good
3	Fair
2	Poor
1	Unsatisfactory

to determine the final value of the evaluation, that is,

$$MOS = \frac{1}{L}\sum_{l=1}^{L} s_l, \tag{6.5}$$

in which L is the number of listeners in the test and s_l is the score assigned by the l-th listener.

In speech coding, assessments by means of MOS are well accepted and sometimes complemented by intelligibility measures, such as MRT (modified rhyme test) and DRT (diagnostic rhyme test) (Deller Jr. et al. 1993). Other measures have been used to assess the quality of speech signals, such as IAJ (isometric absolute judgment), QUART (quality acceptance rating test), and DAM (diagnostic acceptability measure) (Deller Jr. et al. 1993).

Signal-to-Noise Ratio (SNR)

Let $x(n)$ be the original signal, $y(n)$ the processed signal, and $e(n) = x(n) - y(n)$ the error signal at time n.

The energy contained in the original signal is

$$E_x = \sum_n x^2(n). \tag{6.6}$$

The energy contained in the error signal is

$$E_e = \sum_n e^2(n) = \sum_n [x(n) - y(n)]^2. \tag{6.7}$$

The resulting SNR, in dB, is given by Deller Jr. et al. (1993)

$$\text{SNR} = 10\log_{10}\left(\frac{E_x}{E_e}\right) = 10\log_{10}\left[\frac{\sum_n x^2(n)}{\sum_n [x(n) - y(n)]^2}\right]. \tag{6.8}$$

Segmental Signal-to-Noise Ratio (SNRseg)

In spite of its mathematical simplicity, the SNR measure has a limitation: it weights all time domain errors equally. This is the reason why a high SNR measure, with undesirable results, can be obtained if the speech utterance presents high concentration of voiced segments (high energy segments), since noise has a greater perceptual effect in low-energy segments, such as unvoiced fricatives (Deller Jr. et al. 1993).

An improved quality measure may be obtained if SNR is measured over short time intervals and the results averaged. The frame-based measure is called the segmental signal to noise ratio (SNRseg) and is expressed as

$$\text{SNRseg} = E[\text{SNR}(j)], \tag{6.9}$$

in which $\text{SNR}(j)$ denotes the conventional SNR for the j-th frame (time interval) of the signal.

The SNRseg measure is formulated as Deller Jr. et al. (1993)

$$\text{SNRseg} = \frac{1}{J}\sum_{j=0}^{J-1} 10\log_{10}\left[\frac{\sum_{n=m_j-N_A-1}^{m_j} x^2(n)}{\sum_{n=m_j-N_A-1}^{m_j}[x(n)-y(n)]^2}\right], \tag{6.10}$$

in which $m_0, m_1, \ldots, m_{J-1}$ are the end-times for the J frames, each of which is length N_A samples, typically 15–25 milliseconds.

Other objective quality measures may be pointed out, such as LAR (log-area ratio), Itakura log-likelihood measure (Deller Jr. et al. 1993), and the spectral distortion.

Spectral Distortion

For applications that require low-rate speech coding, it is essential to quantize suitably the line spectral frequency (LSF) parameters, using a number of bits as low as possible. The development of LSF coding methods has been an intense research area for many years (Paliwal and Atal 1993; LeBlanc et al. 1993; Eriksson et al. 1999).

The quality of the LSF quantization is measured by the spectral distortion (SD), formulated as

$$\text{SD} = \left[\frac{1}{F_S}\int_0^{F_S}[10\log_{10} S(f) - 10\log_{10}\hat{S}(f)]^2 \mathrm{d}f\right]^{1/2}, \tag{6.11}$$

in which $S(f)$ and $\hat{S}(f)$ denote, respectively, the original and quantized envelope.

Table 6.2 Digital audio formats (Jayant 1992)

Format	Sampling rate (kHz)	Bandwidth (kHz)	Frequency range (Hz)
Telephony	8	3.2	200–3400
Teleconferencing	16	7	50–7000
Compact Disc (CD)	44.1	20	20–20,000
Digital Audio Tape (DAT)	48	20	20–20,000

6.4.2 Bit Rate

The bit rate of a digital representation may be measured in bits per sample, bits per pixel (bpp), or bits per second, depending on the scenario. The rate in bits per second is simply the product of the sampling rate by the number of bits per sample. The sampling rate is usually slightly higher than twice the signal bandwidth, as stated by the Nyquist sampling theorem (Jayant and Noll 1984; Lathi 1988).

Table 6.2 shows some formats usually used for audio (Jayant 1992; Jayant et al. 1993). Typical sampling rates are 8 kHz for telephone speech, 16 kHz for AM radio-grade audio, 32 kHz for FM audio, and 44.1 kHz or 48 kHz for CD (compact disc) audio or DAT (digital audio tape) audio. Notice that the bandwidths are lower than half the corresponding sampling rates, in accordance to Nyquist sampling principle. It is worth mentioning that a PCM signal has a band limited by a low-pass filter. The telephony band usually transmitted is 300 Hz to 3400 Hz in Europe and Latin America and 200 Hz to 3400 Hz in United States and Japan.

6.4.3 Complexity

The complexity of a coding algorithm is related to the computational effort required to implement the encoding and decoding processes. Therefore, it concerns to arithmetic capability and memory requirements. The complexity is usually measured in millions of instructions per second (MIPS). Other measures related to the complexity are the physical size of the encoder, decoder, or codec (encoder plus decoder), the cost, and the power consumption (measured, for instance, in milliwatt, mW), which, in turn, is a criterion particularly important for handheld systems (Jayant 1992; Jayant et al. 1993).

6.4.4 Communication Delay

The complexity increase of a coding algorithm is generally associated with a processing delay increase in encoder and decoder. The importance of delay in a communication system depends on the application. Depending on the commu-

nication environment, the tolerated overall delay may be severely limited, as in network telephony (Jayant 1992; Jayant et al. 1993). Thus, the delay produced by a *codec* imposes some practical constraints concerning the use in communication systems, since the delay must not be higher than a certain limit. However, for some applications the communication delay in irrelevant.

6.5 Features of Speech Signals

The knowledge of the features of speech signals has been efficiently used in techniques for speech coding and speech synthesis, as well as in speech and speaker recognition systems.

The mechanism of speech production, as occurs in any physical system, presents limited frequency response. The limit, which varies from person to person, is about 10 kHz (Aguiar Neto 1995). However, in telephony systems, the speech signal is limited to the range 200–3400 Hz, without great damage to speech quality. In other applications, as teleconference, the speech signal is limited to 7 kHz, which is known as broadband transmission.

Speech sounds may be classified in three distinct classes: voiced sounds (such as /a/ and /i/), unvoiced sounds (such as /sh/), and plosive sounds (such as /p/, /t/ and /k/) (Rabiner and Schafer 1978).

The voiced sounds are quasi-periodic in time domain and harmonically structured in frequency, while the unvoiced sounds have a random nature and broadband. The energy of the voiced sounds is generally higher than that of unvoiced frames. Voiced sounds are produced by quasi-periodic pressure waves exciting the vocal tract, which, acting as a resonator, produces resonance frequencies called formants, which characterize the different voiced sounds. In general there are three to five formants below 5 kHz.

The amplitudes and the location of the first three formants are very important to synthesis and perception of speech (Spanias 1994). The fundamental frequency of the voiced sounds is typically in the range of 80 Hz (for men) to 350 Hz (for children), with 240 Hz being a typical value for women (Aguiar Neto 1995). In generation of plosive sounds, the air is completely oriented to the mouth, which is completely closed. With the increase of the pressure, the occlusion is suddenly broken.

One can also cite the mixed excitation sounds, such as unvoiced fricative sounds (such as /j/, /v/ and /z/), which are produced by combining the vocal chords vibration with a turbulent excitation, and the occlusive (or plosive) voiced sounds (such as /d/ and /b/).

Speech signals are non-stationary. However, they may be considered as quasi-stationary in short frames (short time intervals), typically 5–20 ms (Spanias 1994). In this sense, the speech signals can be considered approximately ergodic.

One of the simplest descriptions of a waveform is given by a graphical representation of amplitude versus time, as shown in Fig. 6.7. In the waveform, one can

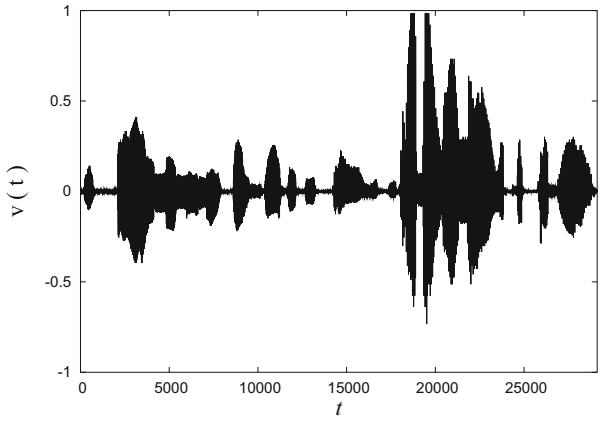


Fig. 6.7 Voice waveform corresponding to the Portuguese sentence "*O sol ilumina a fachada de tarde. Trabalhou mais do que podia*.", with 29,120 samples, 3.64 s duration, 8 kHz sampling rate

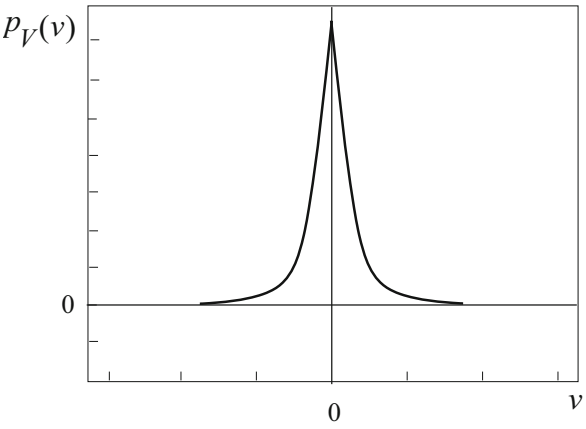


Fig. 6.8 Model for the probability density function of a speech signal

identify high energy frames, low energy frames, and inter-syllabic pause frames. These and the frames of pause between words correspond to about 50 to 60 % of the total duration of the speech signal (Aguiar Neto 1995).

Speech signals present a high variation of signal amplitude. The amplitude variation, which corresponds to about 50 dB, is called the dynamic range of the signal. It is worth mentioning that the acquisition (8.0 bit/sample, 8 kHz sampling rate) of the speech signal shown in Fig. 6.7, as well as in Fig. 6.8, was performed by using a Sun® workstation, equipped with audio processing tools.

Regarding the probability density function for the amplitudes of the speech signal, approximations by the Laplacian model, for long voice sequences (hundreds of milliseconds); bilateral Gamma model, for short periods (tens of milliseconds); and Gaussian model, for very short periods of time (hundreds of microseconds), have been used (Aguiar Neto 1995).

Concerning the energy of the speech signal, a concentration in the lowest frequencies of the spectrum is observed, mainly in the range 500 to 800 Hz. However, although presenting low values of energy, the highest frequency components are important since they determine most of the speech intelligibility. The spectrum decays about 8–10 dB per octave.

Frequencies below 500 Hz have a minor contribution for the speech understanding, but they have a crucial role in the naturalness of the speech (Aguiar Neto 1995).

The probability density function, $p_V(v)$ of a speech signal, $v(t)$, is presented in Fig. 6.8, modeled after a result from Alcaim et al. (1992), produced by ten different speakers, five males and five females. A typical feature of the speech signals is that it can be modeled by a Laplace probability density function, because of the predominance of small amplitude samples.

6.6 Pulse Code Modulation

Pulse code modulation (PCM) provides a method for accomplishing the digital transmission or storage of an original analog signal. In this method of signal coding, the analog message signal is sampled, and the amplitude of each sample is mapped (approximated, rounded off) to the nearest one of a finite set of discrete levels (allowed quantization levels), so that both time and amplitude are represented in discrete form. This allows the message to be transmitted by means of a digital (coded) waveform.

The fundamental operations in the transmitter of a PCM system are sampling, quantizing, and encoding.

The incoming message waveform is sampled using a train of narrow rectangular pulses so as to closely approximate the instantaneous sampling process. In the sampling process, a continuous-time signal is transformed to a discrete-time signal. The sampling, or Nyquist theorem, previously discussed, states that a signal $f(t)$, bandlimited to B Hz (i.e., $F(w) = 0$ for $|w| > 2\pi B$) may be reconstructed exactly (without error) from its samples taken at a rate $R_S > 2B$ Hz (samples per second). In other words, the minimum sampling frequency for a baseband signal to be recovered without distortion is $f_S = 2B$ Hz (Lathi 1988).

In practice, a low-pass pre-alias filter is used at the front end of the sampler in order to exclude frequencies greater than B, before sampling.

After sampling, the sample values are still analog because they lie in a continuous range. By means of the quantization process, each sample is approximated to the nearest quantization level. Each sample is approximated to the midpoint of the interval in which the sample value falls. The information is thus digitized.

The quantization process introduces a certain amount of error or distortion in the signal samples. This error, known as quantization noise or quantization error, is intrinsic to the process of analog to digital conversion (Sripad and Snyder 1977). It is minimized by establishing a large number (L) of quantization intervals. To

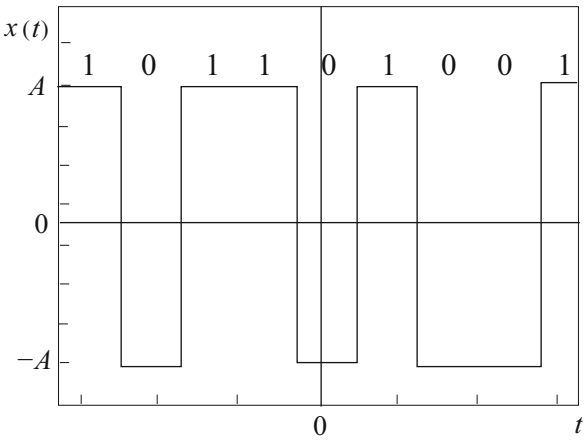


Fig. 6.9 Binary bipolar waveform

assure the intelligibility of voice signals, for example, $L = 8$ or 16 is sufficient. For commercial use, $L = 32$ is a minimum, and for telephone communication, $L = 128$ or 256 is commonly used (Lathi 1988).

During each sampling interval, one quantized sample is transmitted, which takes one of the L possible quantized values. To exploit the advantages of sampling and quantizing, it is required the use of an encoding process to translate the discrete set of sample values to a binary signal, because of its simplicity and ease of detection. The quantization process performed by using $L = 8$ quantization levels requires the use of codewords with $\log_2 8 = 3$ bits to represent each quantization level.

Several formats (waveforms) can be used to represent the binary sequences produced by the analog-to-digital conversion. Figure 6.9 depicts an example of a format, by which the binary symbol "1" is represented by a positive pulse of constant amplitude and duration of one bit and symbol "0" is represented by a negative pulse for the same duration. This format is called bipolar signal.

The scheme to transmit or store the digitized data using pulses is known as pulse code modulation (PCM) (Shannon 1948b).

The quantization noise can be minimized by increasing the number of quantization levels. This leads to the use codewords with a higher number of bits for representing each quantization level, which increases the transmission rate.

6.6.1 Uniform Quantization

Quantization can be uniform or nonuniform. In uniform quantization, the quantization levels are uniformly spaced. Otherwise, the quantization is nonuniform. The quantizer characteristic can be of midtread or midriser type. Fig. 6.10a shows the input-output characteristic of a uniform quantizer of the midtread type, which is so called because the origin lies in the middle of a tread of the staircase-like figure.

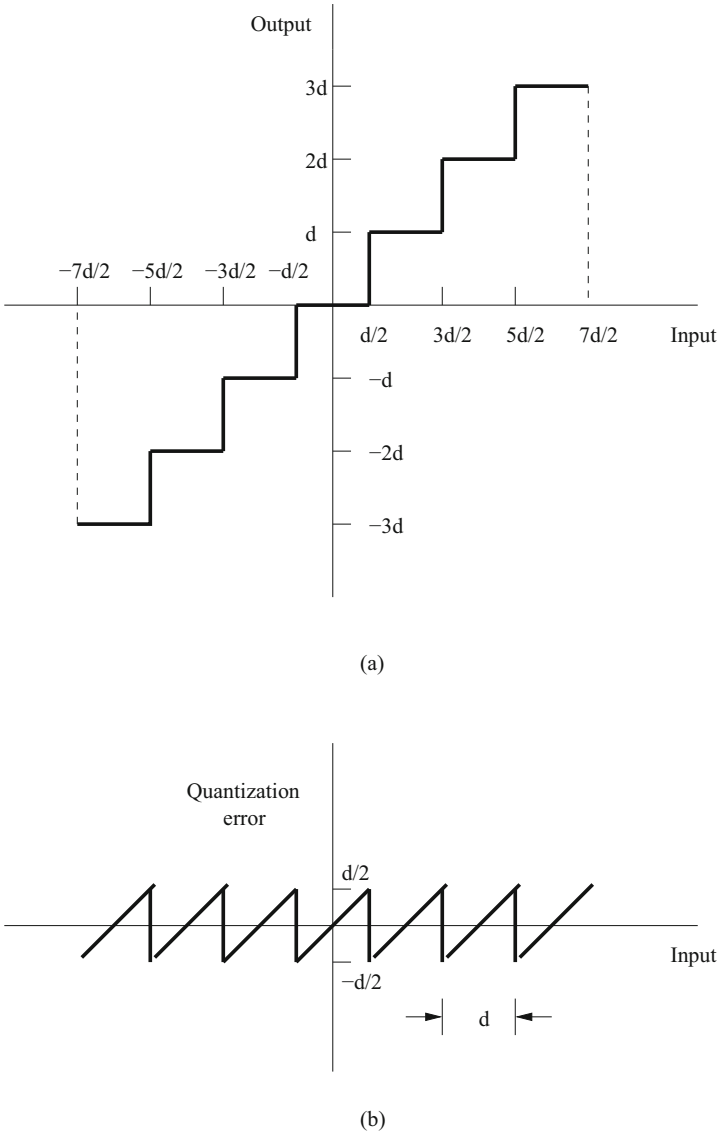


Fig. 6.10 (**a**) Characteristic of the midtread quantizer. (**b**) Variation of the corresponding quantization error with input

Figure 6.11a shows the input-output characteristic of a uniform quantizer of the midriser type, in which the origin lies in the middle of a rising part of the staircase like diagram.

The staircase-like characteristic of the midtread quantizer in Fig. 6.10a shows that the decision thresholds of the quantizer are located at $\pm d/2$, $\pm 3d/2$, $\pm 5d/2$,

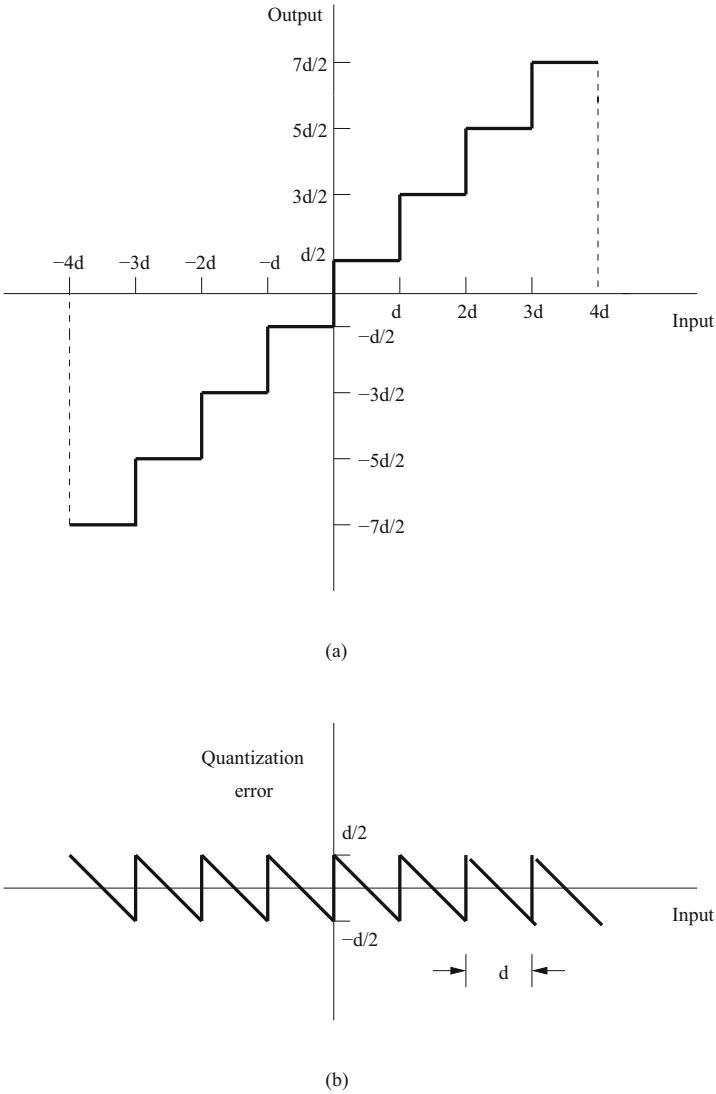


Fig. 6.11 (**a**) Characteristic of the midriser quantizer. (**b**) Variation of the corresponding quantization error with input

$\ldots$, and the representation levels (quantization levels) are located at $0, \pm d, \pm 2d, \ldots$, in which d is the stepsize. The staircase-like characteristic of the midriser quantizer in Fig. 6.11a shows that the decision thresholds of the quantizer are located at $0, \pm d, \pm 2d, \ldots$, and the representation levels (quantization levels) are located at $\pm d/2, \pm 3d/2, \pm 5d/2, \ldots$, in which d is the stepsize.

Figures 6.10b and 6.11b show that the maximum quantization error (which corresponds to the difference between the output and input values of the quantizer) for a sample is half one stepsize. It is also observed that the total range of variation of the quantization noise is from $-d/2$ to $d/2$.

6.6.2 *Quantization Noise*

The error or quantization noise consists of the difference between the signal at the quantizer input and the signal at the output, $n = x - y$, in which $y = q(x)$ and $q(\cdot)$ represents the quantization function. The performance of coding or processing systems is limited by the level of the quantization noise. The channel capacity itself is limited by that noise. As a consequence, the figure of merit which is most used in comparative analysis is the signal to quantization noise ratio (SQNR).

Throughout this chapter, the notation SNR (signal to noise ratio) is also used to denote the signal to quantization noise ratio (Paez and Glisson 1972). The mean square error, for a uniform quantizer, is given by $d^2/12$ (Bennett 1948), supposing a uniform distribution for the quantization noise, in which d represents the quantization step. This result is shown in the sequel.

Assuming a uniform probability distribution for the noise in the interval $[-d/2, d/2]$,

$$p_N(n) = \frac{1}{d}\left[\mathrm{u}(n + d/2) - \mathrm{u}(n - d/2)\right], \tag{6.12}$$

the power of the quantization noise is determined by the expression

$$P_N = \int_{-\infty}^{\infty} n^2 p_N(n) dn. \tag{6.13}$$

Taken into account the probability distribution assumed,

$$P_N = \frac{1}{d}\int_{-d/2}^{d/2} n^2 dn = \frac{1}{d}\frac{d^3}{12} = \frac{d^2}{12}. \tag{6.14}$$

This result was obtained by the first time by Claude E. Shannon, in 1948 (Shannon 1948b).

In what follows it is shown that each additional bit increases the SQNR by about 6 dB. The SQNR is given by

$$\mathrm{SQNR} = 10 \log\left(\frac{P_X}{P_N}\right), \tag{6.15}$$

in which P_X denotes the power of the signal to be quantized and $P_N = d^2/12$ is the power of the quantization noise.

For a signal with dynamic range (i.e., the interval corresponding to the signal amplitude variation) that is equal to $2\sqrt{P_X}$ and quantization with $N = 2^m$ levels, in which m denotes the number of coding bits, the quantization step is given by

$$d = \frac{2\sqrt{P_X}}{N} = \frac{2\sqrt{P_X}}{2^m}. \tag{6.16}$$

As a consequence, the power of the quantization noise is given by

$$P_N = \frac{d^2}{12} = \frac{4P_X}{12N^2} = \frac{P_X}{3 \cdot 4^m}. \tag{6.17}$$

Substituting Eq. 6.17 in 6.15, it follows that

$$\text{SQNR} = 10 \log 3 \cdot 4^m = 10 \log 3 + 10m \log 4 \approx 5 + 6m \text{ dB}. \tag{6.18}$$

Hence, from Eq. 6.18 it is observed that the SNQR increase is about 6 dB for each additional bit, considering uniform quantization.

6.7 The Uniform Quantizer

The need for representing signals by a finite number of bits implies that quantization noise is present in almost all digital signal processing systems and inherently occurs in the analog-to-digital conversion process. The quantization process is a nonlinear mapping from the domain of continuous amplitude signals to one of a countable number of possible output levels (Sripad and Snyder 1977).

The distortion error, or quantization noise, consists of the difference between the input to the quantizer and the discrete output signal. The figure of merit most commonly used in comparative analyses is the signal to quantization noise ratio (SQNR) (Paez and Glisson 1972). The mean square error for a uniform quantizer is approximately given by $d^2/12$, under a supposition of uniform probability density function for the quantization noise, where d represents the stepsize (Bennett 1948).

The usual optimization techniques for the scalar quantizer are centered on properties of the probability density function (pdf) of the input signal (Paez and Glisson 1972; Lloyd 1982; Max 1960). In fact, there seems to be a tendency of the proposed schemes to obtain an output pdf that more closely resembles the uniform type (Jayant and Noll 1984). Usually, the information on the noise power is sufficient to approach a given problem. Sometimes, as in the case of matched filters design, the shape of the noise spectrum plays a more important role.

Some theoretical models have been proposed to explain the power spectral density of the quantization noise, and conditions have been given for the quantization noise to have a white spectrum (Sripad and Snyder 1977). In particular, a characteristic function has been found to account for the generated noise (Bennett

1948). The idea of using the Fourier series decomposition of the characteristic function is due to Lévine (1973).

An interesting model was derived for deterministic signals that made use of Fourier series analysis (Claasen and Jongepier 1981). It can be seen as the first attempt to use Woodward's theorem to approach the problem, but the results were stated without a formal proof and required considerations on the convergence of the model introduced. An interesting model for the quantization noise autocorrelation, for the uniform quantizer, was introduced in a recent article. The result is based on the two-dimensional characteristic function of the input signal (Gray 1990).

The main body of this work is divided into five parts. First, the mathematical model for evaluating the power spectral density of the quantization noise is reviewed. Second, the rationale behind the minimization of the noise is discussed. This is followed by a model for a two-level quantizer. Then, an optimum method, in the sense of the mean square error, for filtering out the quantizing noise is presented. Finally, a view of quantization as a sampling process is discussed.

It is shown that the spectrum of the quantization noise is quite independent of the spectrum of the applied signal and remarkably related to the probability density function of the signal derivative. It is also shown, for Gaussian signals, that the noise is statistically uncorrelated, and thus independent, of the input signal.

6.8 Noise Spectrum for the Uniform Quantizer

Quantization noise can be thought as the result of the application of the signal $x(t)$ to a circuit with characteristic $f(x)$, as shown in Fig. 6.12 (Alencar 1998a).

The input signal is assumed stationary in the wide sense, but no restriction is made concerning its probability density function (Alencar 1993a). A small quantization step and a uniform quantization scheme are considered in the model. The function $f(x)$ is periodic, with period d; therefore, it can be written as

$$f(x) = x - md$$
$$\left(m - \frac{1}{2}\right) d < x \leq \left(m + \frac{1}{2}\right) d, \; m = 0, \pm 1, \pm 2, \cdots \tag{6.19}$$

A Fourier series representation for the preceding function can be obtained in terms of the input signal in which d represents the stepsize, $x(t)$ is the input signal, and X_M is the quantizer threshold, given by Lévine (1973)

$$f(x(t)) = \frac{d}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin\left[\frac{2\pi n x(t)}{d}\right], \; |x(t)| \leq X_M. \tag{6.20}$$

The signal is not allowed to exceed the threshold level, and so there is no overload noise in the model. Equation 6.20 represents a sum of phase modulated signals,

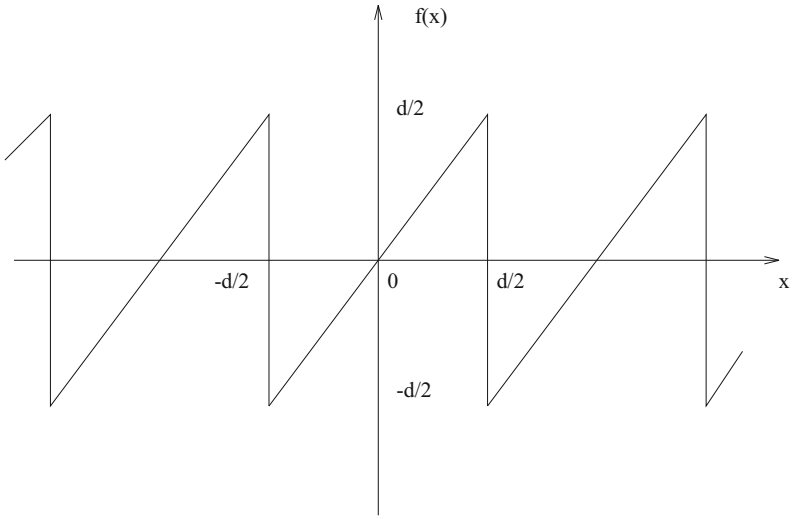


Fig. 6.12 Characteristic function of the quantizer

with phase deviation $2\pi n/d$ and amplitude $d/\pi n$. The autocorrelation function of Eq. 6.20 can be computed, as described in Alencar (1993a), giving

$$R_N(\tau) = \frac{d^2}{\pi^2} \sum_{n=1}^{\infty} \sum_{n=1}^{\infty} \frac{(-1)^{n+m}}{n.m} \cdot E\left[\sin\left(\frac{2\pi n x(t)}{d}\right) \sin\left(\frac{2\pi n x(t+\tau)}{d}\right)\right]. \tag{6.21}$$

The phase modulated signals are assumed to form an orthogonal set, which implies that the cross-correlations are zero. Equation 6.21 can be written as

$$R_N(\tau) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} E\left[\cos\left(\frac{2\pi n}{d}(x(t+\tau) - x(t))\right)\right]. \tag{6.22}$$

Equation 6.22 can be re-written, using the exponential formula for the cosine, to emphasize the role of the expectancy operator in that equation

$$R_N(\tau) = \frac{d^2}{4\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} E[e^{\pm j \frac{2\pi n}{d}(x(t+\tau)-x(t))}]. \tag{6.23}$$

The linear mean square estimator can be used in the last expression to account for the expression $x(t+\tau) - x(t)$. This estimator can be shown to be unbiased and

consistent (Alencar 1998b). In fact, the mean square error of the estimator is always reduced as $\tau \to 0$.

Finally, the autocorrelation function of Eq. 6.20 is evaluated

$$R_N(\tau) = \frac{d^2}{4\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} E[e^{-j\frac{2\pi n}{d}\tau x'(t)} + e^{+j\frac{2\pi n}{d}\tau x'(t)}]. \tag{6.24}$$

The previous formula can be easily verified by evaluation at the origin, which gives the well-known value for the total noise power

$$P_N = R_N(0) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{d^2}{2\pi^2} \cdot \frac{\pi^2}{6} = \frac{d^2}{12}. \tag{6.25}$$

The actual noise power, inside the signal bandwidth, can be much smaller. It is worth noting that the last result was obtained without any constraints on the shape of the noise probability density function. This means that the probability distribution for the quantization noise does not have to be uniform in order to verify Eq. 6.25.

The power spectrum density of the quantization noise can be obtained by using the Wiener-Khinchin theorem in Eq. 6.24 (Alencar and Neto 1991)

$$S_N(w) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^3} p_{X'}\left(\frac{wd}{2\pi n}\right), \tag{6.26}$$

in which $p_{X'}(\cdot)$ is the probability density function of the derivative of the input signal.

Equation 6.26 demonstrates that the power spectral density of the quantization noise is related to the probability density function of the derivative of the input signal. The convergence of the noise spectrum to Eq. 6.26, as the stepsize decreases, is a result of a previous work Alencar (1993b). In fact, the noise spectrum reflects an infinite sum of contributions, each one with the shape of the probability density function, but with decreasing intensity and increasing bandwidth.

In order to give an example of how to use Formula 6.26, suppose the input signal is zero mean Gaussian, with variance $\sigma_X^2 = P_X$

$$p_X(x) = \frac{1}{\sqrt{2\pi P_X}} e^{-x^2/2P_X}, \quad P_X = R_X(0). \tag{6.27}$$

For this type of signal it is possible to compute the probability density function of its derivative, in terms of the second derivative of the autocorrelation function $R_X(\tau)$

$$p_{X'}(x) = \frac{1}{\sqrt{2\pi P_{X'}}} e^{-x^2/2P_{X'}}, \quad P_{X'} = -R_X''(0). \tag{6.28}$$

Substituting $p_{X'}(x)$ from Eq. 6.28 into Formula 6.26 yields the power spectrum density of the quantization noise, when the input to the quantizer is a stationary Gaussian signal:

$$S_N(w) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^3} \frac{1}{\sqrt{-2\pi R_X''(0)}} e^{\frac{(wd)^2}{8\pi^2 n^2 R_X''(0)}}. \tag{6.29}$$

Figure 6.13 illustrates the application of the formula for a Gaussian input signal, showing the spectrum of the quantization noise for two values of the stepsize. It is noted that the spectrum broadens for a smaller stepsize.

Another example can be given, to clarify the use of Formula 6.26. Suppose a sine waveform, $x(t) = Asin(\omega_c t + \phi)$, is applied to the input of the quantizer. The pdf for a sine waveform is Papoulis (1981b)

$$p_X(x) = \frac{1}{\pi\sqrt{2P_X - x^2}}, \; P_X = R_X(0) = \frac{A^2}{2}. \tag{6.30}$$

Again, for a sinusoidal signal, it is easy to compute the pdf of its derivative, in terms of the second derivative of the autocorrelation function $R_X(\tau)$

$$p_{X'}(x) = \frac{1}{\pi\sqrt{(\omega_c A)^2 - x^2}}. \tag{6.31}$$

Substituting this equation into Formula 6.26 yields the power spectrum density of the quantization noise, when the input to the quantizer is a sine waveform

$$S_N(w) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^3} \frac{1}{\pi\sqrt{(\omega_c A)^2 - (\frac{wd}{2\pi n})^2}}. \tag{6.32}$$

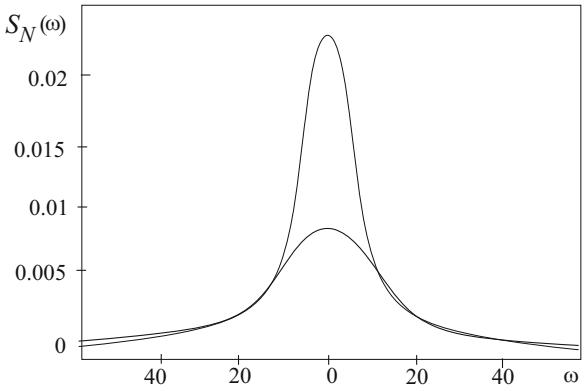


Fig. 6.13 Spectrum of the quantization noise for a Gaussian input signal, for two values of the stepsize

The method for voice waveform uniform coding, called pulse code modulation (PCM), is defined in ITU-T G.711 and AT&T 43801. Basically, the signal is sampled at a rate 8000 times per second and uniformly coded using 8 bits per sample.

6.8.1 Nonuniform Quantization

In uniform quantization, the signal to quantization noise ratio is given by Eq. 6.15, and the noise power depends only on the magnitude of the quantization intervals, d, as shown in Eq. 6.14. Thus, SQNR depends directly on the power of the input signal. As the dynamic range of speech signals is about 50 dB, the SQNR varies significantly, decreasing with the decrease of the power of the input signal.

The signal quality may deteriorate, for instance, in time intervals when the person speaks very softly. Statistically, it is observed a predominance of low amplitude speech samples, meaning that the SQNR is low most of the time.

However, it is ideally desirable to obtain an SQNR that is as constant (same quality) as possible for all the values of power of the message signal (input signal). The use of a nonuniform quantizer attempts to assure the SQNR constancy. In this case, the quantization step is not constant; it varies as a function of the signal amplitude. For lower levels of the input signal, the quantization step is small. The width of the quantization step is logarithmically increased as the level of the message signal increases.

Alternatively, the logarithm of the input signal can be uniformly quantized rather than the input signal itself. Indeed, at the transmitter, the encoder expands the number of levels at low amplitudes and compresses those at high amplitudes, as depicted in Fig. 6.14. At the receiver, this procedure is reversed to bring the relative amplitude levels of the signal to their original value. The process of first compressing and then expanding a signal is referred to as *companding*. Figure 6.15 illustrates a block diagram of a communication system based on companding (Alencar and da Rocha Jr. 2022).

Concerning nonuniform quantization, the term companding is used to address the processes of compression and expansion of the signal to be coded.

The compression is necessary for increasing the weaker levels of the signal, making the coder more efficient. The expansion is carried out at the receiver, with a function which is the inverse of the compression function.

It is possible to obtain an ideal model for a voice compandor, to minimize the quantizatio noise, using the transformation of probability density function, as shown in the following example.

Example Assume that the voice signal, $v(t)$, can be modeled by a Laplace probability density function, as shown in Fig. 6.16.

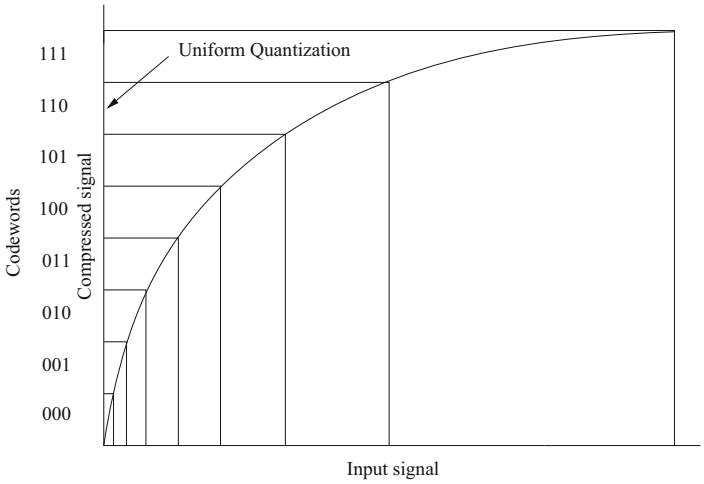


Fig. 6.14 Typical compression curve for nonuniform quantization

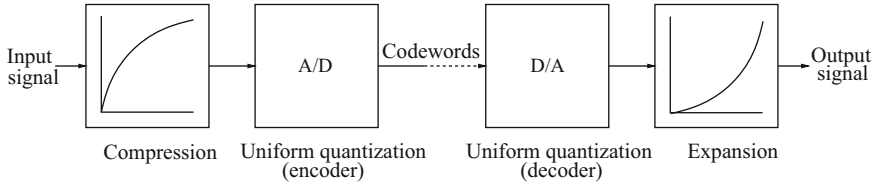
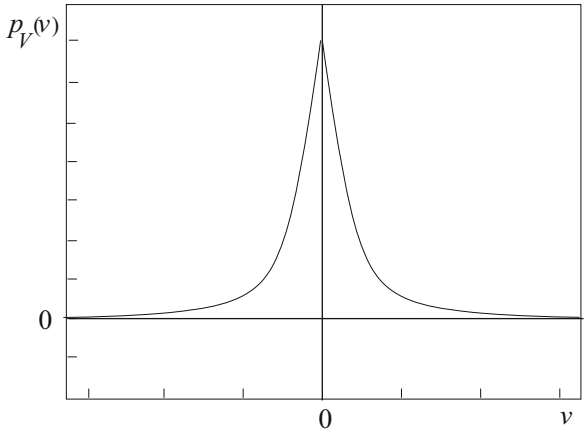


Fig. 6.15 Block diagram of a communication system based on *companding*

Fig. 6.16 Laplace probability density function



The equation that describes the Laplace distribution is

$$p_V(v) = \frac{\alpha}{2} e^{-\alpha|v|}, \tag{6.33}$$

which has zero mean, as expected, and the variance is given by $\frac{2}{\alpha^2}$. This is a usual model for the voice signal, because it is simple and relatively accurate.

As discussed, because the compandor function is invertible, it does not distort the voice signal in the process. Also, the Nyquist criterion garantees that the signal spectrum can be completely recovered. Therefore, in order to minimize the quantization noise, it is necessary to optimize the nonuniform quantizer to match the signal probability distribution.

Because the nonuniform quantizer can be decomposed into two parts, a nonlinear function followed by a uniform quantizer, the objective is to convert the Laplacian probability density function into a uniform distribution, which is the optimal one, in the sense that it produces the minimum quantization error for this case.

The transformation of a general pdf, $p_V(v)$, into a uniform pdf, $p_Y(y)$, in the interval $0 \leq y \leq 1$, is explained in the following. In this case, the output distribution is given by

$$p_y(y) = \frac{p_V(v)}{|\frac{\mathrm{d}y}{\mathrm{d}v}|}, \quad v = f^{-1}(y), \quad \frac{\mathrm{d}y}{\mathrm{d}v} \geq 0,$$

$$p_y(y) = \frac{p_V(v)}{\mathrm{d}y/\mathrm{d}v} \quad \Rightarrow \quad \frac{p_V(v)}{\mathrm{d}y/\mathrm{d}v} = 1\,, \quad 0 \leq y \leq 1.$$

Therefore,

$$p_V(v) = \frac{\mathrm{d}y}{\mathrm{d}v} \quad \Rightarrow \quad \mathrm{d}y = p_V(v)\,\mathrm{d}v.$$

Then, integrating the differential in y, the desired function is obtained

$$y(v) = \int_{-\infty}^{y} dy = \int_{-\infty}^{v} p_V(v)\,\mathrm{d}v = P_V(v). \tag{6.34}$$

Therefore, the compandor function must take the format of the signal cumulative probability function in the defined interval.

Suppose the uniform quantizer operates in the interval $[-A, A]$, then the output of the compressor must be a signal with a uniform probability distribution, as follows:

$$p_Y(y) = \frac{1}{2A}\left[\mathrm{u}(y + A) - \mathrm{u}(y - A)\right]. \tag{6.35}$$

This function is depicted in Fig. 6.17.

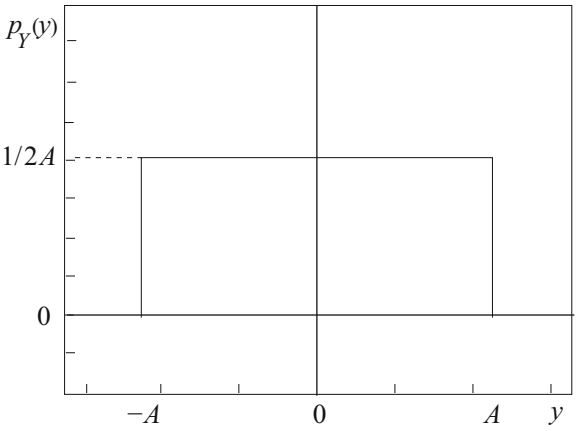


Fig. 6.17 Uniform probability density function

Thus, for the right-hand side of the input signal domain,

$$y(v) = \int_0^v \frac{\alpha}{2} e^{-\alpha|v|}\,\mathrm{d}v,\ 0 \le y \le A.$$

In this case,

$$\frac{p_V(v)}{\mathrm{d}y/\mathrm{d}v} = \frac{1}{2A},\ 0 \le y \le A.$$

Therefore,

$$p_V(v) = \frac{1}{2A}\frac{\mathrm{d}y}{\mathrm{d}v} \Rightarrow \mathrm{d}y = 2Ap_V(v)\,\mathrm{d}v.$$

Integrating the differential in y, one obtains

$$y(v) = 2A\int_{-\infty}^{v} p_V(v)\,\mathrm{d}v, \tag{6.36}$$

which gives

$$y(v) = 2A\frac{\alpha}{2}\int_0^v e^{-\alpha v}\,\mathrm{d}v = A\left[1 - e^{-\alpha v}\right],\ 0 \le y \le A,\ 0 \le v < \infty.$$

Because the input signal is symmetric, the compressor function is anti-symmetric. Therefore, the function for the left-hand side of the input domain is obtained by the following transformation $f(v) = -y(-v)$. Figure 6.18 shows the compressor function.

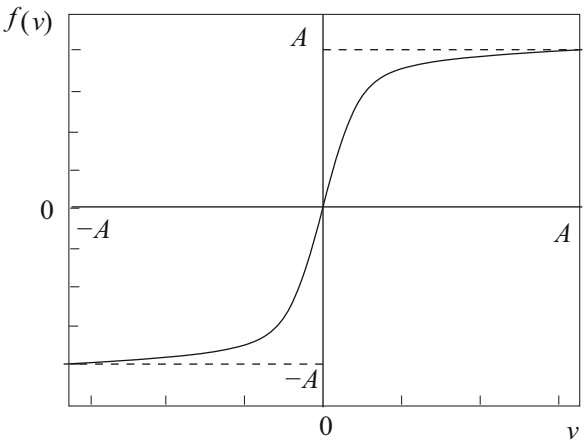


Fig. 6.18 Optimal compressor function for the Laplace distribution

6.9 Compression Laws

For non-uniform coding, two compression laws are recommended by the International Telecommunication Union (ITU-T), formerly known as ITU-T: the μ-law used in North America and Japan, and the A-law used in the Europe and most of the rest of the world. The A-law is also used in international routes.

The corresponding expressions for the previous compression laws are given next. Note that the maximum amplitudes of both input and output signals are V.

- μ-law

$$y = \begin{cases} C(x) = \dfrac{V \ln(1 + \mu x/V)}{\ln(1 + \mu)}, & x > 0 \\ C(x) = -C(-x), & x \leq 0 \end{cases} \tag{6.37}$$

- A-law

$$y = \begin{cases} C(x) = \dfrac{Ax}{1 + \ln A}, & 0 \leq x \leq V/A \\ C(x) = \dfrac{V(1 + \ln(Ax/V))}{1 + \ln A}, & V/A \leq x \leq V \\ C(x) = -C(-x), & x \leq 0. \end{cases} \tag{6.38}$$

The corresponding compression curves are shown, respectively, in Figs. 6.19 and 6.20. The curves show the effect of changing the parameters μ and A. An increase on the respective parameter leads to an increase in the curve nonlinearity. For $A = 1$ and $\mu = 0$, the compression curves are linear, i.e., no compression is

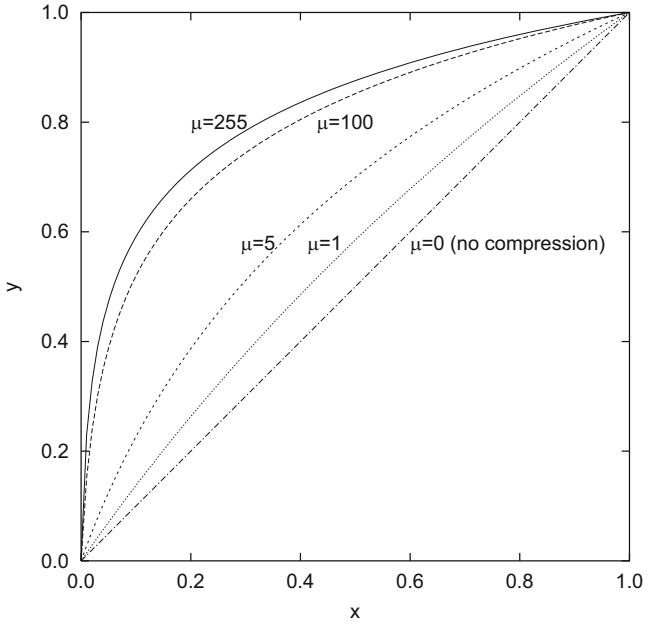


Fig. 6.19 Compression curves for μ-law

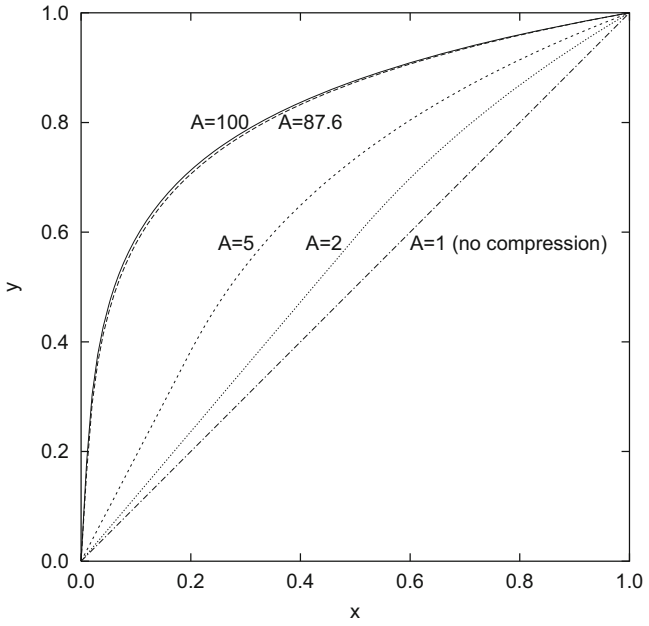


Fig. 6.20 Compression curves for A-law

obtained. The typical values, obtained from subjective tests (*mean opinion score*, MOS), are $\mu = 255$ and $A = 87.6$. The G.711 PCM A-law or μ-law has an MOS of about 4.2.

The curve of signal to quantization noise, as a function of the inverse of the signal amplitude, for nonuniform quantizers is flatter when compared to the uniform quantizer. This way, the input lower levels are preserved.

For the nonuniform case, one can assume that the signal is transformed by a nonlinear function $g(\cdot)$ prior to the quantization process. This gives

$$S_N(w) = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^3} p_{g'(x)} \left(\frac{wd}{2\pi n} \right), \tag{6.39}$$

in which $g'(x)$ is the derivative of the compression function. Its probability density function is usually difficult to compute. For example, using a sine waveform, with amplitude V, as input and the nonlinear $A - Law$ scheme, the resulting pdf is given by

$$p_{g'(x)}(\xi) = \begin{cases} [\pi \sqrt{(\frac{A.V}{1+\ln A})^2 - \xi^2}]^{-1} & |\xi| < \frac{A.V}{1+\ln A} \\ \frac{\xi^2 (1+\ln A)}{\pi X_M V} & |\xi| \geq \frac{A.V}{1+\ln A} \end{cases}. \tag{6.40}$$

Careful selection of $g(x)$ can minimize the following expression and maximize the signal to quantization noise ratio. The compression function must be chosen in order to displace the peak of the quantization noise spectrum far outside the signal bandwidth:

$$P'_N = \int_{-\omega_M}^{\omega_M} S_N(\omega) \mathrm{d}\omega = \frac{d^2}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^3} \int_{-\omega_M}^{\omega_M} p_{g'(x)} \left(\frac{\omega d}{2\pi n} \right) \mathrm{d}\omega, \tag{6.41}$$

in which P'_N represents the noise power that falls inside the signal bandwidth. This quantity can be made quite small, compared to the total noise P_N in 6.25.

6.10 Vector Quantization

Vector quantization (Gersho and Gray 1992; Gray 1984) may be seen as an extension of scalar quantization to a multidimensional space and is supported by Shannon's rate distortion theory, which states that a better performance is achievable by coding blocks of samples (vectors) instead of individual samples (scalars). Mathematically, vector quantization may be defined as a mapping Q of a vector $\boldsymbol{x}$ belonging to the K-dimensional Euclidean space, $\mathbb{R}^K$, into a vector belonging to a finite subset W of $\mathbb{R}^K$, that is,

$$Q : \mathbb{R}^K \rightarrow W. \tag{6.42}$$

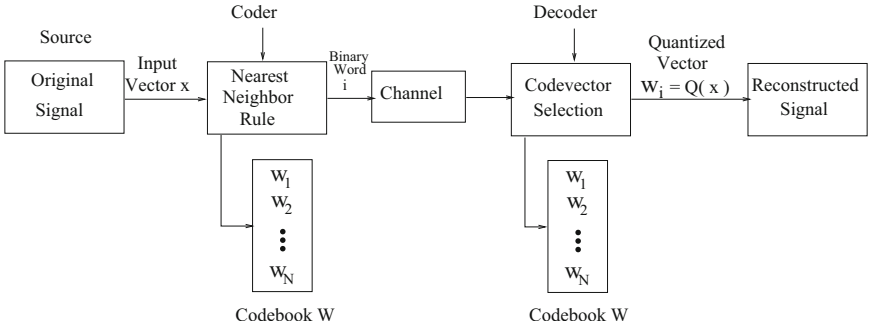


Fig. 6.21 Coding system based on vector quantization

The codebook $W = \{\boldsymbol{w}_i;\ i = 1,\ 2,\ \ldots,\ N\}$ is the set of the K-dimensional codevectors (reconstruction vectors). The index of codevector $\boldsymbol{w}_i$ is denoted by $\boldsymbol{i}$. Each index $\boldsymbol{i} \in \{0, 1\}^b$ represents a binary word of b bits. The code rate of a vector quantizer, measuring the number of bits per vector component, is $R = \frac{1}{K} \log_2 N = \frac{b}{K}$. In voice waveform coding, R is expressed in bits/sample. In image coding, R is expressed in bits per pixel (bpp).

In a signal coding system based on vector quantization, shown in Fig. 6.21 (Alencar and da Rocha Jr. 2022), the coder and the decoder operate as follows. Given a vector $\boldsymbol{x} \in \mathbb{R}^K$ from the signal to be coded, the coder determines the distortion $d(\boldsymbol{x}, \boldsymbol{w}_i)$ between that vector and each codevector $\boldsymbol{w}_i$, $i = 1,\ 2,\ \ldots,\ N$ from the codebook W. The optimum coding rule is the nearest neighbor rule, by which a binary word $\boldsymbol{i}$ is transmitted to the decoder if codevector $\boldsymbol{w}_i$ corresponds to the minimum distortion, that is, if $\boldsymbol{w}_i$ is the vector with greatest similarity to $\boldsymbol{x}$ among all codevectors of the codebook.

In other words, the coder employs the coding rule $C(\boldsymbol{x}) = \boldsymbol{i}$ if $d(\boldsymbol{x}, \boldsymbol{w}_i) < d(\boldsymbol{x}, \boldsymbol{w}_j), \forall j \neq i$. The task of the decoder is very simple: upon receiving the index $\boldsymbol{i}$ of b bits, the decoder simply looks for the vector $\boldsymbol{w}_i$ in its copy of the codebook W and outputs $\boldsymbol{w}_i$ as the reproduction (reconstruction) of $\boldsymbol{x}$. Hence, the decoder uses the decoding rule $D(\boldsymbol{i}) = \boldsymbol{w}_i$. The mapping of $\boldsymbol{x}$ in $\boldsymbol{w}_i$ is usually expressed as $\boldsymbol{w}_i = Q(\boldsymbol{x})$.

Thus, in the scenario of digital coding of signals, vector quantization is a lossy compression technique, since the reconstructed signal is a distorted version of the original signal. The quantization error, introduced when representing the input signal by its corresponding quantized version, is called quantizer distortion. A key question in vector quantizer design is the trade-off between rate and distortion. The target is an optimum codebook, which minimizes, for a given code rate, the average distortion introduced by approximating the input vectors by the corresponding codevectors.

The Linde-Buzo-Gray (LGB) algorithm (Linde et al. 1980) is a widely used technique for codebook design. Other methods have been applied for codebook design, such as Kohonen learning algorithm (Kohonen 1990) and other unsuper-

vised learning algorithms (Krishnamurthy et al. 1990; Chen et al. 1994); stochastic relaxation (Zeger et al. 1992); fuzzy algorithms (Karayiannis and Pai 1995), and genetic algorithm (Pan et al. 1995).

Codebook design plays an important role for the performance of digital processing systems based on vector quantization (VQ) (Gray 1984; Gersho and Gray 1992). In speech and image coding based on VQ (Abut et al. 1982; Ramamurthi and Gersho 1986), the quality of the reconstructed signals depends on the designed codebook. In speaker identification systems based on VQ (Soong et al. 1987; Fechine 2000), the recognition rates depend on the codebook of acoustic parameters designed for each speaker registered in the system.

The mapping Q leads to a partitioning of $\mathbb{R}^K$ into N subspaces (cells, called Voronoi regions) $S_i, i = 1,\ 2,\ \ldots,\ N$, for which

$$\bigcup_{i=1}^{N} S_i = \mathbb{R}^K \text{ and } S_i \cap S_j = \emptyset \text{ if } i \neq j. \tag{6.43}$$

Each Voronoi region S_i is defined as

$$S_i = \{\boldsymbol{x} :\ Q(\boldsymbol{x}) = \boldsymbol{w}_i\} = \{\boldsymbol{x} :\ C(\boldsymbol{x}) = \boldsymbol{i}\}. \tag{6.44}$$

Codevector $\boldsymbol{w}_i$ is the representative vector of all input vectors belonging to cell S_i, as shown in Fig. 6.22.

6.10.1 LBG Algorithm

Let the iteration step of the LBG algorithm be denoted by n. Given K, N and a distortion threshold $\epsilon \geq 0$, the LBG algorithm (Linde et al. 1980) consists of the following steps:

- *Step (1)* initialization: given an initial codebook W_0 and a training set $\boldsymbol{X} = \{\boldsymbol{x}_m;\ m = 1,\ 2,\ \ldots,\ M\}$, set $n = 0$ and $D_{-1} = \infty$.
- *Step (2)* partitioning: given W_n (codebook at the n-th iteration), assign each training vector (input vector) in the corresponding class (Voronoi cell) according to the nearest neighbor rule; determine the distortion

$$D_n = \sum_{i=1}^{N} \sum_{\boldsymbol{x}_m \in S_i} d(\boldsymbol{x}_m, \boldsymbol{w}_i). \tag{6.45}$$

- *Step (3)* convergence test (stop criterion): if $(D_{n-1} - D_n)/D_n \leq \epsilon$ then stop, with W_n representing the final codebook (designed codebook); else, continue.
- *Step (4)* codebook updating: calculate the new codevectors as the centroids of the classes; set $W_{n+1} \leftarrow W_n$; set $n \leftarrow n + 1$ and go to *Step (2)*.

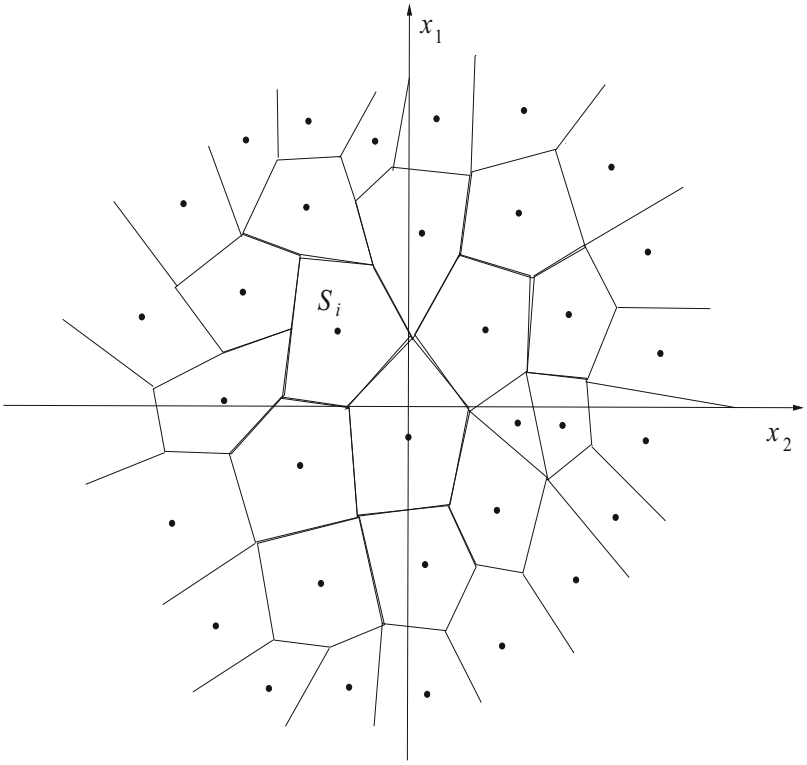


Fig. 6.22 Partitioning of the Euclidean space $\mathbb{R}^2$, introduced by mapping input vectors $\boldsymbol{x}$ into codevectors $\boldsymbol{w}_i$. Coordinates x_1 and x_2 represent the first and the second components of vector $\boldsymbol{x} \in \mathbb{R}^2$, respectively

The distortion decreases monotonically in the LBG algorithm, since the codebook is iteratively updated attempting to satisfy the centroid condition and nearest neighbor condition. In the LBG algorithm the distortion introduced by representing the training vectors by the corresponding codevectors (centroids) is monitored at each iteration. The stopping rule (convergence test) is based on that monitored distortion. The codebook training stops when $(D_{n-1} - D_n)/D_n \leq \epsilon$. The convergence speed of the LBG algorithm depends on the initial codebook.

6.11 LPC Parameters

Most current speech coding algorithms use the source-filter model of speech production. The voice is modeled as the response of a linear time-variant synthesis filter to an input signal called excitation. Examples of speech coders based on that model include the numerous variations of the LPC (linear predictive

coding) vocoder and the wide family of analysis by synthesis coders based on linear prediction (LPAS, linear-prediction-based analysis-by-synthesis)—including CELP (code-excited linear prediction). The synthesis filter, which determines the short-term spectral envelope of the synthesized speech, is characterized by the linear prediction coefficients, obtained from a linear prediction (LP) analysis of the input speech signal. Those coefficients are usually called LPC coefficients and are related generically to each of the many different (but equivalent) sets of parameters that specify the synthesis filter.

Much research has been developed in LPC quantization (Paliwal and Atal 1993; Kleijn and Paliwal 1995; Atal et al. 1993). The main purpose of LPC quantization for speech coding is to avoid the introduction of any audible distortion in the coded speech, maintaining the bit rate as low as possible. If this purpose is reached, then we have *transparent quality* or *transparent quantization*.

Due to the cost and difficulty to perform subjective quality tests, the researchers have used objective measures to assess the distortion in spectral envelope caused by the LPC quantization. Specifically, the spectral distortion or spectral distance has been used as a standard performance criterion. Associated with that measure, an objective criterion for transparent LPC quantization has been proposed by Paliwal and Atal (1993), based on the role of the *outliers*.

The first study on vector quantization applied to LPC parameters is due to Buzo et al. in 1980 (Buzo et al. 1980). Recently, many papers regarding sophisticated schemes of vector quantization have been presented. Paliwal and Atal (1993) have efficiently applied split VQ (split vector quantization) for obtaining transparent quality at 24 bits per frame. That work has been a reference usually used as a benchmark for comparing other results. Other researchers have obtained similar or better results.

6.11.1 LPC Quantization

In a speech coder based on the source-filter model, the LPC coefficients $\{a_i\}$ are obtained by using linear predictive analysis (Markel and Gray 1976) in each speech frame. The coefficients are used to form a synthesis filter given by $H(z) = 1/A(z)$, in which $A(z)$ is the inverse filter, given by

$$A(z) = 1 + a_1 z^{-1} + \cdots + a_M z^{-M}, \tag{6.46}$$

in which M is typically a number between 10 and 16 called order of the predictor. Due to the quasi stationary nature of the speech, that filter is updated at every frame, whose typical size is 20 ms. This leads to a rate of 50 frames per second. A description of that synthesis filter may be communicated to the receiver at each frame. The quantization of the filter into a finite number of bits per frame is known as LPC spectrum quantization.

The purpose of LPC quantization is to efficiently code the LPC parameters without introducing audible distortion in the coded speech. As stated previously, the difficulties associated with subjective evaluation tests have led to performance assessments of quantization schemes by using the spectral distortion, SD, which is expressed in dB and determined to a speech frame according to

$$\mathrm{SD}^2 = \frac{2}{F_S}\int_0^{F_S} \{20\log_{10}|H(e^{j\pi f/F_S})| - 20\log_{10}|\hat{H}(e^{j\pi f/F_S})|\}^2 \mathrm{d}f, \qquad (6.47)$$

in which F_S is the sampling frequency and $\hat{H}(z)$ is the quantized synthesis filter transfer function.

Thus, SD^2 is the mean square error related to the log magnitudes of the synthesis filter frequency responses, quantized and non-quantized, considering that the averaging is taken in the frequency domain. For evaluating the performance of a LPC quantization scheme, the average spectral distortion (average value of SD considering all speech frames) is determined and the percentage of outliers is determined. The outliers are frames whose SD exceeds a determined threshold value. In Paliwal and Atal (1993) three conditions based on SD for transparent quantization are presented:

1. The average spectral distortion is lower than 1 dB.
2. There are no outliers with SD larger than 4 dB.
3. The percentage of outliers with SD in the range 2–4 dB is smaller than 2%.

For quantization of the LPC parameters, the prediction coefficients $\{a_i\}$ are mapped into an equivalent representation, with good quantization properties in terms of distribution, stability, and spectral sensitivity. Representations such as log-area ratios, arcsines of reflection coefficients. and line spectrum frequencies (also called line spectrum pairs, LSPs) have been studied. They give better quantization efficiency and better stability properties when compared to the LPC coefficients. Since the 1980s, the LSF representation has been the dominant form for the purpose of LPC spectrum quantization. According to Soong and Juang (1993), the formal result in terms of DRT evaluation of an 800 bit/s vocoder LSP is only 0.7 worse than the 2400 bit/s residual excited vocoder LPC. At a rate 4800 bit/s, the DRT score of the vocoder LSP in only 0.7 worse than the 9600 bit/s residual excited vocoder LPC.

It is worth mentioning that in speech coding algorithms using linear prediction, the transmission of the LPC parameters, usually transformed to the LSF representation, consumes most of the total bit rate of the coder.

In order to define the LSF parameters, the inverse filter polynomial is used to construct two polynomials:

$$P(z) = A(z) + z^{-(M+1)}A(z^{-1}) \qquad (6.48)$$

and

$$Q(z) = A(z) - z^{-(M+1)}A(z^{-1}). \tag{6.49}$$

The roots of polynomials $P(z)$ and $Q(z)$ are called LSFs. The polynomials $P(z)$ and $Q(z)$ have the following properties: 1) all zeros of $P(z)$ and $Q(z)$ lie on the unit circle and 2) zeros of $P(z)$ and $Q(z)$ are interlaced with each other, that is, the LSFs w_i are in ascending order in $(0, \pi)$, in the form (Kim and Oh 1999)

$$0 < w_1 < w_2 < \ldots < w_p < \pi. \tag{6.50}$$

An important question in LSF coding is that the ordering relation is required for assuring the stability of the synthesis filter (Sugamura and Farvardin 1988). The properties (1) and (2) previously presented are useful for determining the LSF parameters from $P(z)$ and $Q(z)$.

For a long time, almost all coding schemes have used some form of scalar quantization (Eriksson et al. 1999). Complexity questions (number of operations performed to compare an input vector to each codevector of the codebook) and memory requirements were viewed as limiting factors to the effective use of vector quantization. The first work incorporating VQ was described in Buzo et al. (1980), but a performance far from the acceptable was obtained with VQ at 10 bits per frame.

Hence, due to the prohibitive size of the training set, to the high computational cost of the codebook design, and to the computational requirements prohibitive to full search vector quantization, hybrid schemes (scalar and vectorial) (Grass and Kabal 1991; Laroia et al. 1991) and methods for reducing the computational complexity have been investigated. As an example, one can cite the method proposed by Paliwal and Atal in Paliwal and Atal (1993), by which the vector of LSF parameters is divided into two vectors, each one quantized by a different codebook. This procedure is known as split VQ.

In memoryless quantization, each vector of LSF parameters is quantized independently of the past LSF vectors. However, this is not the most efficient method for coding the LSF vectors, which present significant interframe correlation (correlation regarding successive frames). As a consequence, performance gains in coding may be obtained by exploring the interframe correlation, as occurs in predictive vector quantization (Shoham 1987) and finite-state vector quantization.

Besides the type of quantization (scalar, vectorial, hybrid) and the inclusion or exclusion of memory in the quantization process, many aspects, generally related, affect the performance of an LSF quantizer, such as Ramachandran et al. (1995): the distortion measure used for the quantizer design, the codebook design, the complexity of the search, the number of bits, the memory requirements for codebook storage, and the robustness to channel errors.

Code excited linear prediction (CELP) (Schoroeder and Atal 1985) is a class of speech coders which present a good strategy for high-quality digital speech transmission at low bit rates. An important technique belonging to that class of

coders is the vector sum excited linear prediction (VSELP) (Electronic Industries Association (EIA), EIA).

6.12 Overview of Speech Coding

Speech compression has been an intense research area for decades. Almost all works in speech compression regard to lossy compression, by which the numerical representation of the signal samples is never recovered exactly after decoding. There exists a wide range of compromises between the bit rate and the quality of the reconstructed signal that are of practical interest for telephone speech coding. The cell phone users, for instance, are accustomed to varied levels of signal degradation.

The speech coding algorithms may be divided into two main groups: waveform coders and vocoders (Gersho 1994).

Historically, the term vocoder originated from the contraction of voice encoder and decoder. In waveform coders, the data transmitted from the coder to the decoder specify a speech representation as a waveform of amplitude *versus* time, such that the reconstructed signal approximates the original waveform and, consequently, provides an approximated recreation of the original sound.

On the other hand, vocoders do not produce an approximation of the original waveform. In vocoders, parameters which characterize the segments of sounds are specified and transmitted to the decoder, which reconstructs a new waveform, with a similar sound. Vocoders are also called parametric coders by an obvious reason. Frequently those parameters characterize the short-term spectrum of a sound.

The parameters specify the mathematical model of human speech production. But, the parameters do not provide sufficient information to produce a good approximation of the original waveform, but the information is sufficient for the decoder to synthesize a sound which is perceptually similar to the voice signal. Vocoders operate at rates lower than the ones of waveform coders, but the quality of the reconstructed speech, although intelligible, suffers from loss of naturalness, and some unique features which identify a speaker may be damaged.

For waveform coders the quantization is performed directly on the signal waveform, while for parametric coders, the quantization is performed on the parameters of the model under consideration. Hybrid coders are based on the models of speech production, but they use an excitation for the synthesizer which is more accurate than that one used in parametric coders

Most of the research in speech coding is based on the speech allocated in the telephony bandwidth, that is, limited to 3.2 kHz, which corresponds to the range 200 Hz to 3.4 kHz, sampled at 8 ksamples/s. Speech coding in broadband has gained attention and concerns 7 kHz signals, sampled at 16 ksamples/s.

A popular coding technique is the code-excited linear prediction (CELP). Some other coding methods are adaptive delta modulation (ADM), adaptive differential pulse code modulation (ADPCM), adaptive predictive coding (APC), multipulse linear predictive coding (MP-LPC), and regular pulse excitation (RPE).

The coders MP-LPC, RPE, and CELP belong to the family of algorithms of analysis by synthesis, which can be seen as hybrid coders, since they combine some features of vocoders and waveform coders. A well-known analysis by synthesis speech coder is the full rate GSM RPE-LTP 13 kbit/s, standardized by ETSI, in 1988, for the digital cellular mobile system (Hersent et al. 2002).

The linear predictive coding (LPC) vocoder is widely used in secure telephone speech. It is also possible to cite the vocoder sinusoidal coding technique, whose corresponding versions are the sinusoidal transform coding (STC) and the multiband excitation (MBE).

6.13 Waveform Coding

For waveform coders the quantization is performed directly on the voice waveform. The purpose is to reproduce the waveform, sample by sample, as efficiently as possible. In general the coders have low implementation complexity, present low delay, and are suitable for use in telephone networks at bit rates $R \geq 16$ kbit/s. An exception is the VQ-based technique, which code a set of samples.

Regarding the complexity, usually the coders may be classified as low, medium, and high complexity. Low-complexity waveform coders present good speech quality at rates of 56 kbit/s minimum. Lower rates, up to 32 kbit/s, are obtained with medium complexity coders and lower rates, up to 16 kbit/s, with high-complexity coders.

Differential Pulse Code Modulation

Differential coding exploits the fact that the speech signal presents significant correlation between successive samples. This means that the speech signal is very redundant. The purpose of differential pulse code modulation (DPCM) is to reduce the redundancy of the speech signal, by quantizing the difference between amplitudes of adjacent samples. As the difference presents lower variance when compared to the original signal, fewer bits may be used for its representation.

Figure 6.23 shows a block diagram of a DPCM encoder. It can be seen that the previous input value is reconstructed by a feedback loop that integrates the encoded sample differences. The advantage of the feedback implementation is that

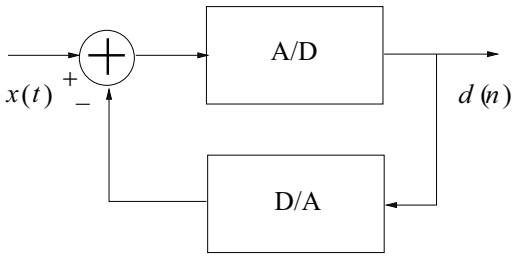


Fig. 6.23 Block diagram of differential PCM

quantization errors do not accumulate indefinitely (Bellamy 1991). As in PCM systems, the analog-to-digital conversion process can be uniform or companded. By using DPCM, the transmission rate may be decreased to 56 kbit/s with comparable quality to that of PCM.

Adaptive Differential Pulse Code Modulation

It is interesting to notice that a DPCM coder may increase its performance gain if the quantization and/or prediction is carried out adaptively. In such case the coder is called adaptive differential pulse code modulation (ADPCM).

The adaptive techniques for speech coding try to solve the trade-off between a stepsize large enough to accommodate the maximum peak-to-peak range of the signal and small enough to minimize the quantization noise. So, adaptive quantization consists on adjusting the quantization steps as a function of the signal level. A nonuniform quantization is performed, by which the width of the quantization levels is not predetermined.

Adaptive prediction consists on the dynamic adjustment of the predictor coefficients, according to the variations of the speech signal. This way, the non-stationary nature of the speech signal is taken into consideration in the coding process. ADPCM coders present good speech quality for rates between 24 and 48 kbit/s.

Delta Modulation

Delta modulation (DM) is a special case of DPCM. The variation of amplitude from sample to sample is quantized, by using only two quantization levels. Thus, DM uses only one bit per sample of the difference signal.

The output of the two-level quantizer is related to the input by the expression $y = 2du(x) - d$, in which $u(\cdot)$ is the unit step function, d denotes the quantization step, and x is the difference signal between successive samples of the original signal (Alencar 2002). The single bit is used to increment or decrement, by a constant stepsize, the coded signal when the difference signal is positive or negative, respectively. Thus, the coded signal resembles a staircase, as illustrated in Fig. 6.24.

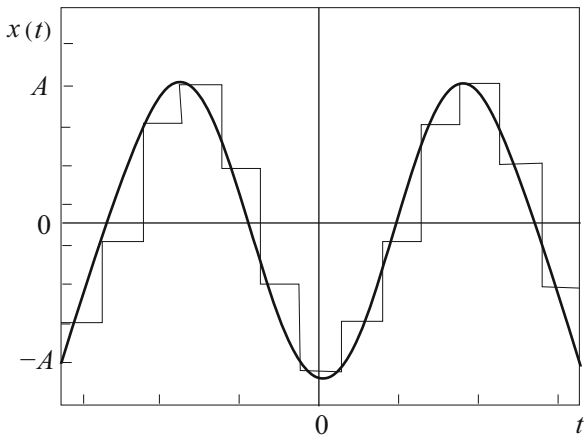


Fig. 6.24 Example of the delta modulation

The practical advantages of using a small number of quantization levels, including quantization with only one bit, have been confirmed by the popularity of *sigma-delta* ($\Sigma - \Delta$) coding schemes. Those schemes have wide acceptance due to their robustness to circuitry imperfections and suitability to VLSI (very large scale integration) implementation (Gray 1987; Zamir and Feder 1995).

The quantization noise, caused by the quantization stage in the $\Sigma - \Delta$ modulator, has been analyzed by many techniques (Gray 1989; Galton 1993). The use of pseudo-random noise (*dither*) and the sampling at high rates have been studied, demonstrating the viability of recovering a signal with two levels (Chou and Gray 1991; Shamai 1994).

6.14 Parametric and Hybrid Coding

Parametric coders, or vocoders, are based on the model of speech production. This model is represented by a set of parameters periodically updated. For determining those parameters, the signal is segmented at periodic intervals, called frames. The parameters are usually updated at each frame. The rate required by the vocoders is low (lower than 4.8 kbit/s), but the delay and the complexity are high and the signal sounds synthetic.

Different from waveform coders, in parametric coders the quantization is carried out on the parameters of the speech production model, which are used in speech synthesis. The parameters of the speech model are determined over short frames, in which the speech signal may be considered stationary, and transmitted to the synthesizer in the receiver. Parametric coders do not provide speech quality required by the telephone network. They are more used in military applications.

Hybrid coders combine the quality of waveform coders with the efficiency of parametric coders. Hybrid coders, more sophisticated than vocoders, are based on models of speech production and use a better excitation to the synthesizer. The excitation improvement is responsible for the improvement in the quality of the synthesized speech, which is more intelligible than the speech from conventional vocoders. The improvement is due to quantization and coding of the parameters that define the excitation as well as to parameters of the synthesis filter. A process known as analysis by synthesis is responsible for obtaining the parameters used in short frames. Hybrid coders are usually complex and lead, for rates from 4 to 16 kbit/s, to a better quality when compared to the one obtained from waveform coding with higher rates.

Channel Vocoder

Most of the coding processes of the channel coders involve the determination of the sampled spectrum of the speech signal as a time function. In channel vocoder, a bank of bandpass filters is used to separate the energy of the speech signal in sub-bands, which are completely rectified and filtered for determining the relative power levels. The individual power levels are coded and transmitted to the destination.

In addition to measuring the signal spectrum, modern channel vocoders also determine the nature of the speech excitation.

Most of the difficulties encountered in the great majority of vocoder implementations regard the determination of the speech harmonics. Without an accurate information of excitation, the quality in the coder output is poor, and many times it depends on the person who talks. Some channel vocoders have produced high intelligibility, in spite of producing a synthetic sound at a rate 2.4 kbit/s (Bayless et al. 1973).

Formant Vocoder

A formant vocoder determines the localization and the amplitude of the formants (in which the energy of the speech signal is concentrated) and transmits that information instead of transmitting the whole spectral envelope. Hence, the formant vocoder produces a low bit rate by coding only the most important pulses of the speech spectrum (Alencar 2002).

The most important point in formant vocoder is the accurate tracking of the formants. Once that requirement is satisfied, the formant vocoder may provide speech intelligibility at a rate lower than 1 kbit/s (Flanagan et al. 1979).

Linear Predictive Coder

Linear predictive coder (LPC) is a widely used vocoder. It extracts the perceptually important features of the speech directly from the waveform in the time domain. The effect is better than that one obtained from the frequency spectrum, such as in channel vocoder and formant vocoder. Fundamentally, an LPC analyses the voice waveform for producing a model of the time-varying vocal tract and for producing the transfer function of the vocal model. A synthesizer at the receiver terminal recreates the speech signal by using a mathematical model of the vocal tract. By the periodic updating of the parameters of the model and the specification of the excitation, the synthesizer is adapted to the changes performed. Figure 6.25 presents

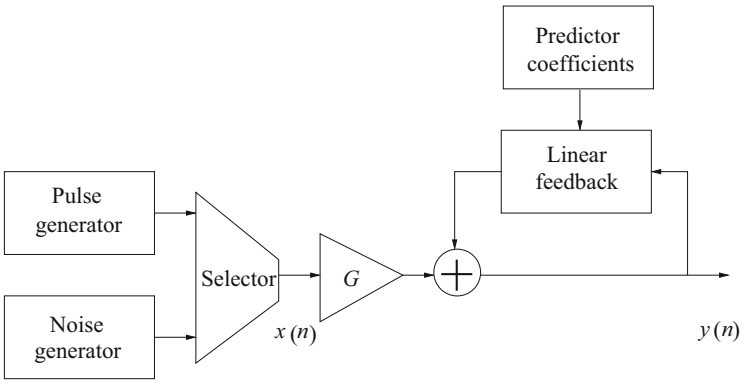


Fig. 6.25 Model of speech generation in linear predictive coding

the basic model of speech generation of the linear predictive coding. The figure is also a model of an LPC coder/synthesizer.

The equation of the model of the vocal tract is defined by

$$y(n) = \sum_{k=1}^{p} a_k y(n-k) + Gx(n), \tag{6.51}$$

in which $y(n)$ denotes de n-th output sample, a_k denotes the k-th predictor coefficient, G is the gain factor, $x(n)$ denotes the input sampled at time n, and p is the order of the model.

The speech output in Eq. 6.51 is represented as the actual input of the system, summed to a linear combination of the predicted outputs of the vocal tract. The model is adaptive and the coder determines, periodically, a new set of parameters corresponding to the successive frames of speech. A basic LPC does not perform the measurement and coding of the differences of waveforms. Instead, the error signals are minimized in a mean square sense, when the predictor coefficients are determined (Alencar 2002).

The information that the LPC coder/analyzer determines and transmits to the decoder/synthesizer consists of:

1. Nature of excitation, if voiced or not.
2. Counting of the period for the excitation of the speech.
3. Gain factor.
4. Predictor coefficients, that is, parameters of the vocal tract model.

An LPC coder is capable of performing gradual changes in the spectral envelope. As a final result, a more natural speech representation is obtained when compared to that of vocoders based purely in the frequency domain (Bayless et al. 1973). Most LPCs have concentrated speech coding in the range 1.4 to 2.4 kbit/s.

Code-excited Linear Prediction: CELP

The basic LPC algorithm synthesizes the speech in the decoder by using a very simple excitation model, which requires only 10% of the data rate (Alencar 2002). The simplicity of the model leads to speech that sounds synthetic. To overcome this problem, many techniques have been developed to improve the excitation. Three methods may be cited: multipulse excitation LPC (MLPC), residual excitation LPC (RELP), and code-excited linear prediction (CELP).

The CELP coders look like the multipulse coders. The synthesis filter and the predictors are the same. The difference resides in the fact that in CELP coders the sequence of excitation pulses is selected from a set of random vectors, previously stored, forming a kind of codebook. The vectors stored in the codebook have Gaussian distribution and zero mean, as an attempt to match the short-term characteristics of the speech signal (Aguiar Neto 1995).

The choice of the ideal sequence to be used as the excitation of the synthesis filter at the decoder is performed by a search in a codebook, using the analysis by synthesis technique. As in MLPC, a short-term correlation analysis and a long-term

correlation analysis are carried out, leading to the spectral envelope (formants) and the periodicity (pitch) of the speech signal. By using those parameters, the speech signal is synthesized.

The CELP coders lead to a good speech quality at rates ranging from 4.8 kbit/s to 16 kbit/s and present a better performance in terms of quality versus bit rate when compared to other coders. However, the CELP coding demands a high computational effort, which was seen as an obstacle, for a long time, to real-time implementation.

Simplifications in the basic structure of CELP coder (introduction of efficient search methods in the codebook or use of algebraic codebooks) and the increase of MIPS in digital signal processors (DSPs) made the coding method feasible. It is worth mentioning that the decoder is much simpler than the coder (since there is no analysis by synthesis search), and a posterior filtering may be optionally performed (Hersent et al. 2002).

A modification introduced in the CELP coder, such that the delay was lowered from about 20–40 ms to about 2 ms, has originated the LD-CELP (low delay CELP).

Many international standards in the range 4.8 kbit/s to 16 kbit/s are CELP coders or CELP based coders, such as ITU-T G.729, multimedia standard ITU-T G.723.1, full-rate GSM ETSI, and half-rate GSM.

6.15 Speech Coder Attributes

Speech coders have attributes classified into four groups: bit rate, quality, complexity, and delay. For a given application, some attributes are pre-specified, while trade-offs can be made among the others. As an example, the communication channel may establish a limit in the bit rate; cost constraints may set limits in the complexity.

Quality can be generally improved by increasing the bit rate or the complexity and sometimes by increasing the delay. Requirements or goals may be set for all of these speech coders attributes. For the purposes of standards, two additional attributes are the method of specification and conformance validation and the schedule in which the work plan is to be accomplished (Cox 1995).

Bit Rate

As mentioned earlier, telephone bandwidth speech signals have a bandwidth of about 300–3400 Hz and are generally sampled at 8000 Hz. The speech coders that have been standardized in recent years have bit rates ranging from 800 bit/s to 16 kbit/s (Cox 1995).

Some of these coders, notably the ones that have been standardized for cellular telephony, also have a channel coder associated with them. In such case, the bit rate increases (for instance, to 22.8 kpbs). The lower bit rates (800–4800 bit/s) are

associated mainly to secure telephony, while cellular telephony speech coding rates range from 3.3 kbit/s to 14 kbit/s, for regular voice services (Cox 1995).

Wideband speech signals have a bandwidth of 50–7000 Hz and are sampled at 16 kHz. Due to the wider bandwidth, intelligibility and naturalness are improved, and listener fatigue during long conversations is lessened. Among the wideband telephony standards (7 kHz), one can cite G.722, with rates 64, 56, and 48 kbit/s.

Delay

Low-rate speech coders can be considered block coders. They encode a block (frame) of speech at a time. Depending on the application, the total speech coding delay is some multiple of the frame size. The minimum delay is generally about three to four times the frame size. For purposes of standardization, the delay is a factor in real-time conversational systems (Cox 1995).

Complexity

Regarding speech coder implementation, speed and random access memory (RAM) usage are the two most important aspects to complexity. The faster the chip (DSP) or the greater the chip size, the greater the cost. The same attributes also influence the power consumption, which is a critical attribute to handheld applications.

Hence, complexity is determinant for both cost and power consumption. For the perspective of standardization, complexity is determined by the application (Cox 1995). Speech coders that require less than 15 MIPS are considered low-complexity coders. The ones that require 30 MIPS or more are considered high-complexity coders (Rao et al. 2002).

Quality

The Speech Quality Experts Group (SQEG) of the International Telecommunications Union (ITU) has a very strict view of what constitutes network toll quality.

The first digital speech coding standard was ITU-T Recommendation G.711 for 64 kbit/s PCM speech. The distortion introduced by a G.711 codec is considered one QDU (quantization distortion unit). SQEG uses QDU for network planning purposes. The second digital speech coding standard was G.721 32 kbit/s ADPCM. This coder was standardized to be used in combination with a G.711 codec as its input and output. The resulting distortion is considered to be 3.5 QDU (Cox 1995).

SQEG network planning guidelines call for a maximum of 14 QDU for an end-to-end international connection and less than 4 QDU for a domestic connection (Cox 1995). G.728 16 kbit/s LD-CELP codec is considered to have the same QDU of G.721. Both are ITU coders which present toll quality.

Regarding quality, some aspects must be considered, such as intelligibility, naturalness, and coder performance when the input contains other signals (such as background noise or music) besides speech. Speech quality can be measured by subjective tests.

Other dimensions of speech quality can be considered. The channel error sensitivity can be regarded as one aspect of quality. In the case of digital cellular standards, additional bits for channel coding are provisioned to protect the information bearing bits.

6.16 Problems

1. Explain signal compression (definition, types, importance, purpose, applications).
2. Discuss the function of source coding, channel coding, and modulation, in a typical digital communication system.
3. Enunciate the sampling theorem, presenting a brief mathematical analysis of the sampling process. Comment on the following: if the sampling frequency is lower than the Nyquist frequency, then the signal cannot be completely recovered.
4. Discuss the aspects used for assessing the performance of a signal compression system.
5. Present the main features of speech signals. Some features are efficiently explored in speech coding systems. Explain.
6. Pulse code modulation (PCM) transforms an analog signal into a series of binary pulses. What is the minimum sampling frequency for a signal with bandwidth of 3.4 kHz?
7. The minimum square error of a uniform scalar quantizer is given by $d^2/12$, in which d is the quantization step. What are the reasons by which this result is obtained?
8. The scalar quantization noise leads to a loss in signal quality. What is the improvement, in terms of signal to quantization noise ratio (SQNR), obtained by using two additional bits in the coding process? Explain.
9. Define mathematically vector quantization. Discuss its importance in the scenario of signal compression.
10. Regarding LPC quantization, what is transparent quantization?
11. Describe what is meant by waveform coders, parametric coders, and hybrid coders. Present a comparison concerning reconstructed signal quality *versus* bit rate.
12. Discuss the following attributes of speech coders: bit rate, delay, complexity and quality.

Chapter 7
Modulation Theory



Go down deep enough into anything and you will find mathematics.
– Dean Schlicter

7.1 Introduction

The signals that carry information are transmitted from one point to another by means of electromagnetic fields, using a given propagation media, as illustrated in block diagram, as devised in Fig. 7.1 by Claude Elwood Shannon (1916–2001), an American mathematician, electrical engineer, and cryptographer.

In Fig. 7.1, $m(t)$ is the message, or modulating signal, $s(t)$ is the modulated signal that is transmitted through the channel, and $r(t) = s(t) + n(t)$ represents the received signal, in which $n(t)$ is the additive noise, representing a combination of all sources of random interference in the channel. Finally, $\tilde{m}(t)$ is the demodulated signal, which incorporates noise and other channel effects, such as fading.

The efficient radiation of radio signals employs antennas, the dimensions of which are compatible with the wavelength of the electromagnetic field and with the media characteristics. Such waves provide a support for carrying information, which is also represented by an electrical signal, called modulating signal, used to modify one or more parameters of the carrier (Schwartz 1970).

As briefly mentioned earlier, modulation consists of the variation of one or more characteristics of the carrier waveform as a function of the modulating signal. A sinusoidal waveform is used as the carrier, and the modulation is performed in three main ways, as represented in the following:

$$s(t) = a(t)\cos[\omega_c t + \theta(t) + \phi], \tag{7.1}$$

in which $s(t)$ is the modulated carrier, $a(t)$ is the amplitude modulating signal, $\theta(t)$ represents the angle modulating signal, and ϕ is a carrier random phase:

- Amplitude modulation (AM), when the carrier amplitude is varied according to the variation of the modulating signal;

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_7

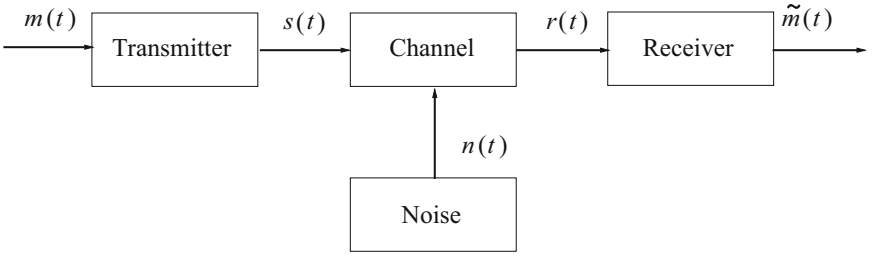


Fig. 7.1 Elementary model for and information transmission system

- Angle modulation, when either the phase (PM) or the frequency (FM) of the carrier is varied as a function of the modulating signal;
- Quadrature amplitude modulation (QAM), when both the amplitude and the phase of the carrier are varied simultaneously as a function of the modulating signal.

7.2 Amplitude Modulation

Amplitude modulation is a transmission system in which the instantaneous amplitude of the carrier electromagnetic wave varies according to the modulating signal (Alencar 1999). It is also known as double-sideband amplitude modulation (AM-DSB) and is by far the most widely used modulation system, because it has been adopted for commercial broadcast. This is a consequence of some practical advantages of the technique, regarding economy, receiver design simplicity, and easy maintenance. The AM carrier is a sinusoid represented as

$$c(t) = A\cos(\omega_c t + \phi), \tag{7.2}$$

in which A denotes the carrier amplitude, ω_c denotes the angular frequency (measured in radians per second, rad/s), and ϕ denotes the carrier phase, measured in radians (rad).

Figure 7.2 illustrates an AM modulator, which includes the signal source, such as a microphone, the modulator, the oscillator that produces the carrier, the carrier amplifier, and the transmitting antenna.

The message signal, or modulating signal, usually written as $m(t)$, must have its largest frequency well below the carrier frequency ω_c. The available modulation bandwidth is restricted by the linear region of operation of the transmitter amplifiers and varies from 0.1% to 1% of the carrier frequency (Gagliardi 1978).

The instantaneous carrier amplitude needs to have the following form, so that the carrier amplitude varies according to $m(t)$:

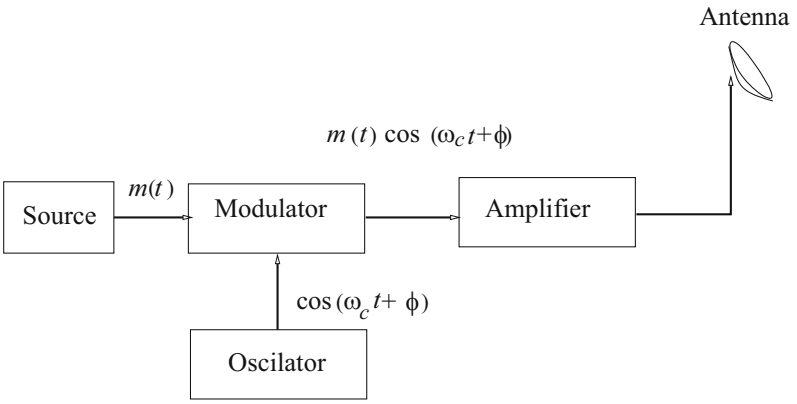


Fig. 7.2 Amplitude modulation technique

$$a(t) = A + Bm(t) = A[1 + \Delta_{AM} m(t)], \tag{7.3}$$

in which $\Delta_{AM} = B/A$ is called the AM modulation index. The instantaneous amplitude of the message signal produces the modulated waveform as follows:

$$s(t) = a(t)\cos(\omega_c t + \phi), \tag{7.4}$$

This can be also written as

$$s(t) = A[1 + \Delta_{AM} m(t)]\cos(\omega_c t + \phi), \tag{7.5}$$

in which ω_c is the unmodulated carrier frequency, in rad/s, and ϕ is the carrier phase, which is usually random.

Figure 7.3 shows the modulating waveform, which alters the carrier amplitude in order to produce the modulated waveforms seen in Figs. 7.4, 7.5, and 7.6, for various values of the modulation index.

It is noticed in Fig. 7.4 that the envelope of the modulated waveform has the same format as the modulating signal. This follows because $A + Bm(t) > 0$, that is, $\Delta_{AM} = B/A < 1$. In Fig. 7.5 the modulation index is equal to one ($\Delta_{AM} = 1$) and is usually referred as a 100% modulated carrier. In Fig. 7.6 the modulation index is greater than one ($\Delta_{AM} > 1$) and causes phase inversion or phase rotation, in the modulated carrier, which is said to be over modulated.

The modulation index indicates how strong the modulating signal is with respect to the carrier. The modulation index should not exceed 100% in order to avoid distortion in a demodulated signal whenever an envelope detector is employed.

Depending on the manner by which the instantaneous amplitude varies as a function of Δ_{AM}, assuming $|m(t)| = 1$, the following terminology is applicable:

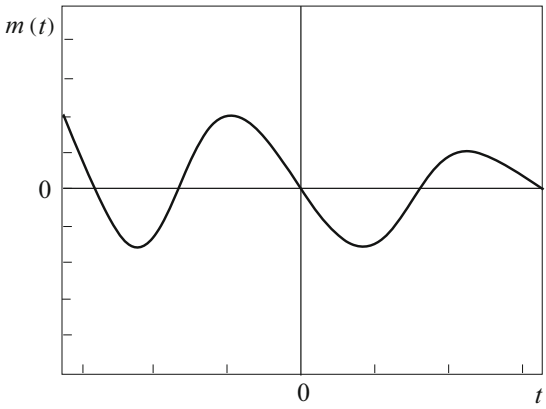


Fig. 7.3 An example of a modulating signal $m(t)$.

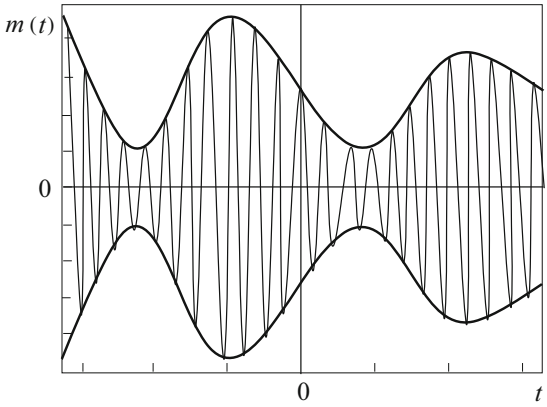


Fig. 7.4 Effect of the modulation index on the transmitted signal, for $\Delta_{AM} < 1$

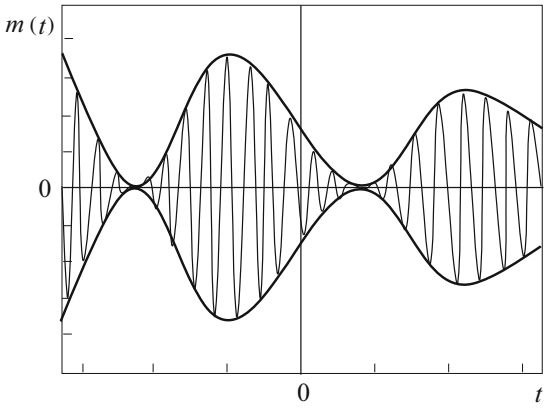


Fig. 7.5 Effect of the modulation index on the transmitted signal, for $\Delta_{AM} = 1$

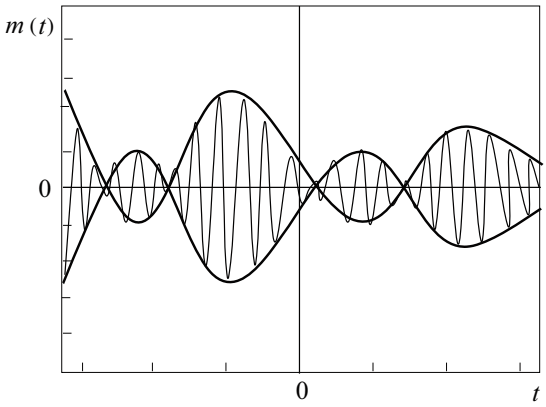


Fig. 7.6 Effect of the modulation index on the transmitted signal, for $\Delta_{AM} > 1$

- $\Delta_{AM} = 1$ implies 100% carrier modulation and allows envelope detection of the modulating signal;
- $\Delta_{AM} > 1$ means carrier overmodulation that causes phase rotation of the carrier and requires synchronous demodulation for recovering the modulating signal;
- $\Delta_{AM} < 1$, the carrier undermodulation case permits envelope detection of the modulating signal but does not make efficient use of the carrier power.

7.3 Amplitude Modulation by Random Signals

The mathematical treatment used in this section involves concepts of a random process, or random signal, which were developed previously. The results derived here by using an analysis based on random processes by far compensate for the extra effort required to understand the theory of stochastic processes. The theoretical development that follows, besides being particularly more elegant than the standard treatment based on deterministic signals, is closer to real-life signals of a true random nature.

Consider the modulated carrier given by $s(t) = a(t)\cos(\omega_c t + \phi)$, in which $a(t) = A[1 + \Delta_{AM} m(t)]$. Let the carrier phase ϕ be a random variable with a uniform distribution in the interval $[0, 2\pi]$. The signal $m(t)$ is assumed to be a stationary random process with zero mean and statistically independent of ϕ.

The carrier is a random process because it possesses a random phase and because it was modulated by a stochastic process. The stationarity of the carrier is thus guaranteed. The autocorrelation of a stationary random process, which was discussed previously, is given by

$$R_S(\tau) = E[s(t)s(t+\tau)]. \tag{7.6}$$

By plugging (7.4) into (7.6), it follows that

$$R_S(\tau) = E[(a(t)\cos(\omega_c t + \phi))(a(t+\tau)\cos(\omega_c(t+\tau) + \phi))], \tag{7.7}$$

or

$$R_S(\tau) = E[a(t)a(t+\tau)\cos(\omega_c t + \phi)\cos(\omega_c(t+\tau) + \phi)], \tag{7.8}$$

and by replacing a sum of cosines for a product of cosines it follows that

$$R_S(\tau) = \frac{1}{2}E[a(t)a(t+\tau)\,(\cos(\omega_c\tau) + \cos(2\omega_c t + \omega_c\tau + 2\phi))]. \tag{7.9}$$

By making use of properties of the mean, considering that $a(t)$ and ϕ are independent random variables and that the mean value of the carrier is zero, it follows that

$$R_S(\tau) = \frac{1}{2}R_A(\tau)\cos(\omega_c\tau), \tag{7.10}$$

in which the autocorrelation of the modulating signal $R_A(\tau)$ is defined as

$$R_A(\tau) = E[a(t)a(t+\tau)]. \tag{7.11}$$

Replacing the expression (7.3) for $a(t)$ in (7.11), it follows that

$$\begin{aligned} R_A(\tau) &= E[A(1 + \Delta_{AM}m(t))A(1 + \Delta_{AM}m(t+\tau))] \\ &= A^2E[1 + \Delta_{AM}m(t) + \Delta_{AM}m(t+\tau) + \Delta_{AM}{}^2m(t)m(t+\tau)]. \end{aligned} \tag{7.12}$$

Again using properties of the mean and recalling that $m(t)$ is stationary and zero mean, that is, that $E[m(t)] = E[m(t+\tau)] = 0$, it follows that

$$R_A(\tau) = A^2[1 + \Delta_{AM}^2 R_M(\tau)], \tag{7.13}$$

in which $R_M = E[m(t)m(t+\tau)]$ represents the autocorrelation of the message signal. Finally, the autocorrelation of the amplitude modulated carrier is given by

$$R_S(\tau) = \frac{A^2}{2}[1 + \Delta_{AM}^2 R_M(\tau)]\cos\omega_c\tau. \tag{7.14}$$

7.3.1 Total Power of an AM Carrier

The total power of an AM carrier is given by the value of its autocorrelation for $\tau = 0$, thus

$$P_S = R_S(0) = \frac{A^2}{2}(1 + \Delta_{AM}^2 P_M), \tag{7.15}$$

in which $P_M = R_M(0)$ represents the power in the message signal $m(t)$. The power in the unmodulated carrier is given by $\frac{A^2}{2}$, as can be easily checked, and represents a significant portion of the total transmitted power.

7.3.2 Power Spectral Density

The power spectral density of the AM-modulated carrier is obtained as the Fourier transform of the autocorrelation function $R_S(\tau)$. This result is known as the Wiener-Khinchin theorem, seen in previous chapters, that is,

$$S_S(\omega) = \mathcal{F}[R_S(\tau)],$$

in which $R_S(\tau) = \frac{1}{2} R_A(\tau) \cos(\omega_c \tau)$.

The function $R_S(\tau)$ can be seen as the product of two functions: $\frac{1}{2} R_A(\tau)$ and $\cos(\omega_c \tau)$. Using this line of reasoning, the Fourier transform of $R_s(\tau)$ is calculated with the application of the convolution theorem, that is,

$$\begin{aligned} S_S(\omega) = \mathcal{F}[R_S(\tau)] &= \frac{1}{2\pi}\left(\mathcal{F}[\frac{1}{2} R_A(\tau)] * \mathcal{F}[\cos(\omega_c \tau)]\right) \\ &= \frac{1}{2\pi}\left[\frac{1}{2} S_A(\omega) * (\pi\delta(\omega + \omega_c) + \pi\delta(\omega - \omega_c))\right], \end{aligned}$$

and applying the impulse filtering property it follows that

$$S_S(\omega) = \frac{1}{4}\left[S_A(\omega + \omega_c) + S_A(\omega - \omega_c)\right], \tag{7.16}$$

in which $S_A(\omega) = \mathcal{F}[R_A(\tau)]$.

The power spectral density of the modulating signal can be derived by writing the expression for $R_A(\tau)$ and then calculating its Fourier transform. Thus,

$$\begin{aligned} S_A(\omega) &= \mathcal{F}[A^2(1 + \Delta_{AM}{}^2 R_M(\tau)], \\ &= A^2[2\pi\delta(\omega) + \Delta_{AM}{}^2 S_M(\omega)]. \end{aligned}$$

in which $S_M(\omega)$ is the power spectral density of the message signal.

Finally, the power spectral density of the modulated AM carrier is given by

$$S_S(\omega) = \frac{A^2}{4}\left[2\pi(\delta(\omega + \omega_c) + \delta(\omega - \omega_c)) +\right.$$

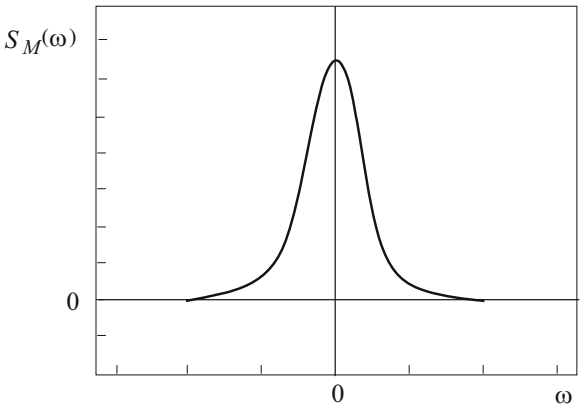


Fig. 7.7 Power spectral density of the message signal

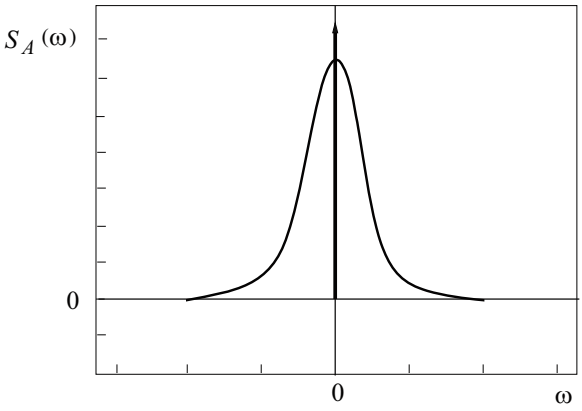


Fig. 7.8 Power spectral density the modulating signal

$$\Delta_{AM}{}^2(S_M(\omega+\omega_c)+S_M(\omega-\omega_c))\Big]. \tag{7.17}$$

The power spectral density of a message signal is depicted in Fig. 7.7, and the spectrum of the modulating signal is shown in Fig. 7.8, which includes an impulse at the origin due to the DC level.

Finally, the power spectral density of the modulated carrier is illustrated in Fig. 7.9, which shows two identical spectra centered at the carried frequency ω_c and its negative counterpart $-\omega_c$.

The bandwidth required for the transmission of an AM signal is precisely twice the bandwidth of the message signal. In AM radio broadcast, the maximum frequency of the message signal is limited to 5 kHz, and consequently the AM bandwidth for commercial radio broadcast is 10 kHz.

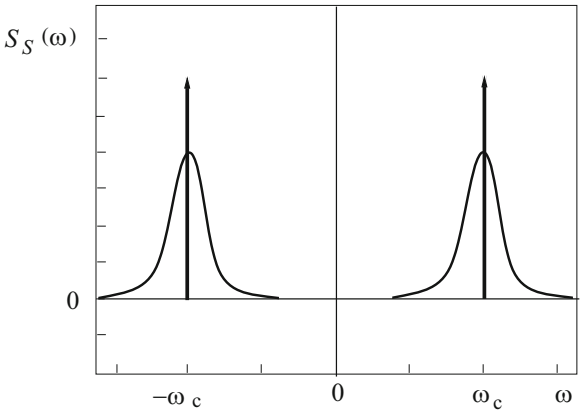


Fig. 7.9 Power spectral density of an AM signal

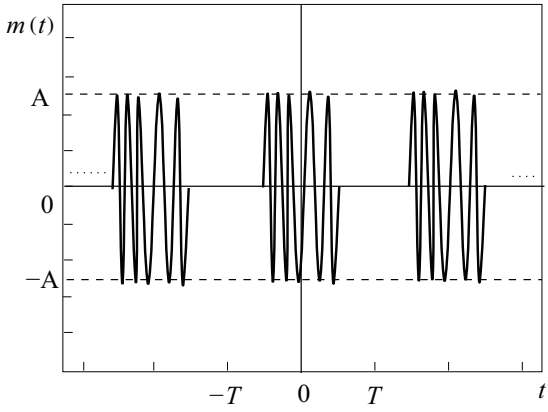


Fig. 7.10 A binary ASK signal

7.3.3 Digital AM Signal

The digital AM signal, also called amplitude-shift keying (ASK), can be generated by the process of multiplying the digital modulating signal by the carrier. An example of the ASK signal is shown in Fig. 7.10.

The ASK signal can also be represented in a different manner, using a phase diagram, which consists of representing the modulated signal in axes which are in phase (I axis) and in quadrature (with a phase lag of $\pi/2$, or Q axis) with respect to the carrier phase.

This diagram is also known as a constellation because it represents signal points as stars on a plane. The digital signal which amplitude modulates the carrier can be written as

$$m(t) = \sum_{k=-\infty}^{\infty} m_k p(t - kT_b), \tag{7.18}$$

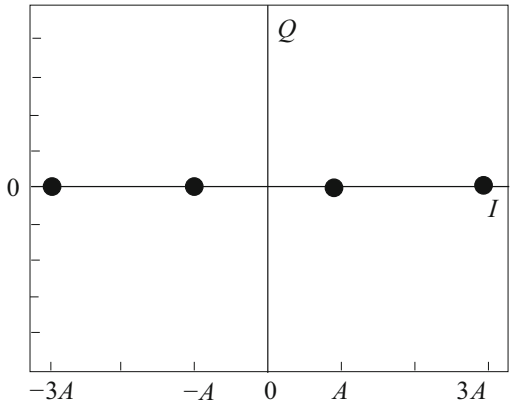


Fig. 7.11 Constellation diagram for a 4ASK signal

in which m_k represents the k-th randomly generated symbol, from a discrete alphabet, $p(t)$ is the pulse shape of the transmitted digital signal and T_b is the bit interval.

The modulated signal without carrier is then given by

$$s(t) = \sum_{k=-\infty}^{\infty} m_k p(t - kT_b) \cos(\omega_c t + \phi). \tag{7.19}$$

As an example, Fig. 7.11 shows the constellation diagram of a 4ASK signal, the symbols of which are $m_k \in \{-3A, -A, A, 3A\}$. All the signal points are on the cosine axis (in phase with the carrier), since there is no quadrature component in this case.

When the modulating signal is a digital signal, the transmitted power is calculated by considering the average power per symbol. In the 4-ASK case, by considering equiprobable symbols it follows that the transmitted power is given by

$$P_S = \frac{1}{2} \sum_{k=1}^{4} m_k^2 p(m_k) = \frac{1}{2} \left[\frac{(-3A)^2 + (-A)^2 + (A)^2 + (3A)^2}{4} \right] = \frac{5}{2} A^2. \tag{7.20}$$

The probability of error for the coherent binary ASK is (Haykin 1988)

$$P_e = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right), \tag{7.21}$$

in which E_b is the binary pulse energy, N_0 represents the noise power spectral density, and $\text{erfc}(\cdot)$ is the complementary error function

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt. \tag{7.22}$$

For a rectangular pulse, $E_b = A^2 T_b$, in which A is the pulse amplitude and T_b is the pulse duration.

7.4 Suppressed Carrier Amplitude Modulation

It has been seen that in standard AM systems, most of the power is spent in transmitting the carrier. In principle, this is a waste of power if one reasons that the carrier by itself conveys no information.

In a suppressed carrier AM system (AM-SC), the carrier is not sent as part of the AM signal, and all of the transmitter power is available for transmitting information over the two sidebands. Obviously, in terms of efficient use of transmitter power, the AM-SC system is more efficient than an AM system. However, a receiver for an AM-SC signal is significantly more complex than an AM receiver.

The AM-SC technique essentially translates the frequency spectrum of the modulating signal by multiplying it by a sinusoid, the frequency of which has a value equal to the desired frequency translation. In other words the original message, or modulating, signal $m(t)$ becomes $m(t)\cos(\omega_c t + \phi)$, after multiplication by the carrier.

7.5 Autocorrelation for the AM-SC Signal

Let $s(t) = m(t)\cos(\omega_c t + \phi)$, in which $m(t)$ is a stationary random process with zero mean and ϕ is a random variable uniformly distributed in the interval $[0, 2\pi]$ and statistically independent of $m(t)$:

$$R_S(\tau) = E[s(t)s(t+\tau)] \tag{7.23}$$

$$R_S(\tau) = E\left[m(t)m(t+\tau)\cos(\omega_c t + \phi)\cos(\omega_c(t+\tau) + \phi)\right] \tag{7.24}$$

$$R_S(\tau) = E\left[m(t)m(t+\tau)\right] \cdot E\left[\frac{1}{2}(\cos\omega_c\tau) + \cos(\omega_c\tau + 2\omega_c t + 2\phi)\right] \tag{7.25}$$

$$R_S(\tau) = E\left[m(t)m(t+\tau)\right]\left(\frac{1}{2}\cos\omega_c\tau\right) \tag{7.26}$$

$$R_S(\tau) = \frac{1}{2} R_M(\tau)\cos\omega_c(\tau) \tag{7.27}$$

The power of the AM-SC signal can be derived from the autocorrelation function computed for $\tau = 0$:

$$P_S = R_S(0) = \frac{1}{2} P_M. \tag{7.28}$$

7.5.1 Autocorrelation of a Digitally Modulated Signal

Consider the digital signal, with independent and equiprobable symbols from the set $\mathbb{M} = \{-A, A\}$, whose autocorrelation is defined by Formula 5.51.

The autocorrelation function for the amplitude modulated carrier, considering that the signal has zero mean, is given by

$$R_M(\tau) = \frac{A^2}{2} \left[1 - \frac{|\tau|}{T_b} \right] [u(\tau + T_b) - u(\tau - T_b)] \cos(\omega_c \tau), \tag{7.29}$$

in which T_b is the bit interval and A represents the pulse amplitude.

Figure 7.12 shows the modulated carrier autocorrelation. Its maximum, which is the modulated signal power, occurs at the origin and is equal to $A^2/2$. As can be observed from the figure, the autocorrelation decreases with an increase in the time interval and reaches zero at time T_b.

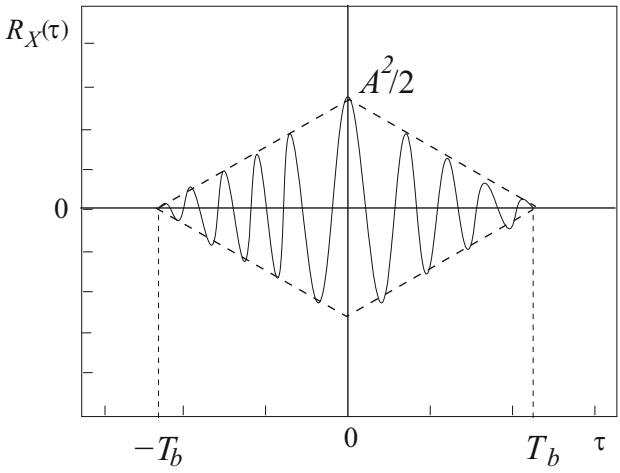


Fig. 7.12 Autocorrelation for the digitally modulated signal

7.6 AM-SC Power Spectral Density

The power spectral density is obtained by means of the Fourier transform of the autocorrelation function, as follows:

$$S_S(\omega) = \int_{-\infty}^{\infty} R_S(\tau) e^{-j\omega\tau} \mathrm{d}\tau, \tag{7.30}$$

$$S_S(\omega) = \frac{1}{4} S_M(\omega) * \delta[(\omega + \omega_c) + \delta(\omega - \omega_c)], \tag{7.31}$$

$$S_S(\omega) = \frac{1}{4}[S_M(\omega + \omega_c) + S_M(\omega - \omega_c)]. \tag{7.32}$$

Figure 7.13 represents the spectrum of the modulating signal, without the impulse associated to the DC level, and Fig. 7.14 is the spectrum of the modulated AM-SC signal, also called AM-DSB, without the impulses associated with the carrier. It is important to note that the modulated signal bandwidth is double the modulating signal bandwidth, which is a waste of the available frequency spectrum.

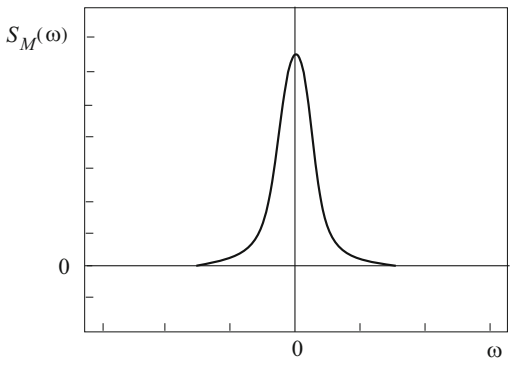


Fig. 7.13 Spectrum of the modulating signal $S_M(\omega)$

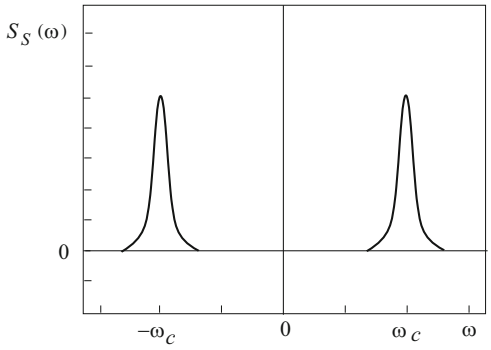


Fig. 7.14 Spectrum of the AM-DSB modulated signal $S_S(\omega)$

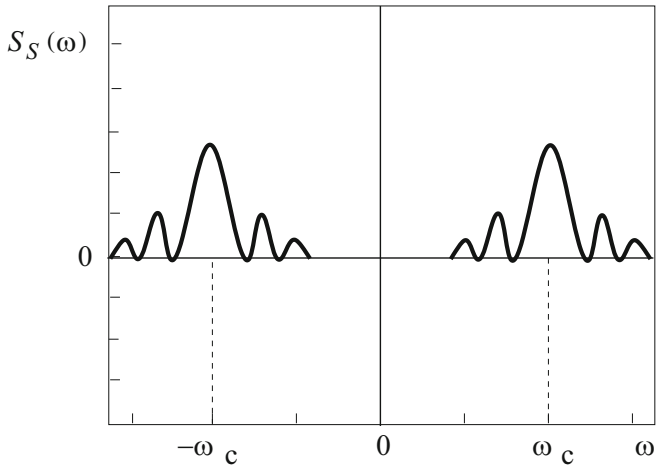


Fig. 7.15 Power spectral density for the digitally modulated signal

7.6.1 Spectrum of a Digitally Modulated Signal

For the digital signal, whose autocorrelation is given by Formula 5.51, the corresponding spectrum is

$$S_S(\omega) = \frac{A^2 T_b}{4}\left[\frac{\sin^2\left[(\omega+\omega_c)T_b/2\right]}{[(\omega+\omega_c)T_b/2]^2} + \frac{\sin^2\left[(\omega-\omega_c)T_b/2\right]}{[(\omega-\omega_c)T_b/2]^2}\right], \tag{7.33}$$

which represents two squared sampling functions and shows that the modulated digital signal has a continuous spectrum.

The power density spectrum is sketched in Fig. 7.15. The distance between the first nulls around the carrier frequency is a measure of the bandwidth.

7.7 Quadrature Amplitude Modulation

The quadrature modulation scheme (QAM) uses sine and cosine orthogonality properties to allow the transmission of two different signals in the same carrier, which occupies a bandwidth that is equivalent to the AM signal.

The QAM modulator can be assembled using two DSB-SC modulators. The information is transmitted by both the carrier amplitude and phase. The quadrature modulated signal $s(t)$ can be written as

$$s(t) = b(t)\cos(\omega_c t + \phi) + d(t)\sin(\omega_c t + \phi), \tag{7.34}$$

in which the random modulating signals $b(t)$ e $d(t)$ can be correlated or uncorrelated.

It is also possible to write the modulated signal as

$$s(t) = \sqrt{b^2(t) + d^2(t)} \cos\left(\omega_c t - \tan^{-1}\frac{d(t)}{b(t)} + \phi\right), \tag{7.35}$$

in which the modulating signal, or resultant amplitude, can be expressed as

$$a(t) = \sqrt{b^2(t) + d^2(t)}, \tag{7.36}$$

and the resultant phase is

$$\theta(t) = -\tan^{-1}\left[\frac{d(t)}{b(t)}\right]. \tag{7.37}$$

The autocorrelation function for the quadrature modulated carrier can be computed from the definition

$$R_S(\tau) = E[s(t) \cdot s(t+\tau)]. \tag{7.38}$$

Substituting Eq. 7.34 into 7.38 gives

$$\begin{aligned} R_S(\tau) = E\,[&[b(t)\cos(\omega_c t + \phi) + d(t)\sin(\omega_c t + \phi)] \\ &\cdot\ [b(t+\tau)\cos(\omega_c(t+\tau) + \phi) \\ &+\ d(t+\tau)\sin(\omega_c(t+\tau) + \phi)]]\,. \end{aligned} \tag{7.39}$$

Expanding the product gives

$$\begin{aligned} R_S(\tau) = E[&b(t)b(t+\tau)\cos(\omega_c t + \phi)\cos(\omega_c(t+\tau) + \phi) \\ &+ d(t)d(t+\tau)\sin(\omega_c t + \phi)\sin mega_c(t+\tau) + \phi) \\ &+ b(t)d(t+\tau)\cos(\omega_c t + \phi)\sin(\omega_c(t+\tau) + \phi) \\ &+ b(t+\tau)d(t)\cos(\omega_c(t+\tau) + \phi)\sin(\omega_c t + \phi)]. \end{aligned} \tag{7.40}$$

Using trigonometric properties and collecting terms which represent known autocorrelation functions, it follows that

$$\begin{aligned} R_S(\tau) = &\frac{R_B(\tau)}{2}\cos\omega_c\tau + \frac{R_D(\tau)}{2}\cos\omega_c\tau \\ &+ \frac{R_{DB}(\tau)}{2}\sin\omega_c\tau - \frac{R_{BD}(\tau)}{2}\sin\omega_c\tau. \end{aligned} \tag{7.41}$$

Which can be simplified to

$$R_S(\tau) = \left[\frac{R_B(\tau) + R_D(\tau)}{2}\right] \cos \omega_c \tau + \left[\frac{R_{DB}(\tau) - R_{BD}(\tau)}{2}\right] \sin \omega_c \tau. \quad (7.42)$$

It is observed that the QAM signal is subject to both amplitude and phase modulation, and its autocorrelation function can be written as

$$R_S(\tau) = \frac{R(\tau)}{2} \cos\left[\omega_c \tau + \theta(\tau)\right], \quad (7.43)$$

in which

$$R(\tau) = \sqrt{[R_B(\tau) + R_D(\tau)]^2 + [R_{DB}(\tau) - R_{BD}(\tau)]^2} \quad (7.44)$$

$$\theta(\tau) = -\tan^{-1}\left[\frac{R_{DB}(\tau) - R_{BD}(\tau)}{R_B(\tau) + R_D(\tau)}\right] \quad (7.45)$$

Considering zero mean uncorrelated modulating signals, $R_{BD}(\tau) = E[b(t)d(t+\tau)] = 0$ and $R_{DB}(\tau) = E[b(t+\tau)d(t)] = 0$. The resulting autocorrelation is then given by

$$R_S(\tau) = \frac{R_B(\tau)}{2} \cos \omega_c \tau + \frac{R_D(\tau)}{2} \cos \omega_c \tau. \quad (7.46)$$

The carrier power is given by the following formula

$$P_S = R_S(0) = \frac{P_B + P_D}{2}. \quad (7.47)$$

The power spectrum density is obtained by applying the Fourier transform to the autocorrelation function (Wiener-Khinchin theorem), which gives

$$\begin{aligned} S_S(\omega) &= \frac{1}{4}\left[S_B(\omega + \omega_c) + S_B(\omega - \omega_c) + S_D(\omega + \omega_c) + S_D(\omega - \omega_c)\right] \\ &= \frac{j}{4}\left[S_{BD}(\omega - \omega_c) + S_{BD}(\omega + \omega_c)\right. \\ &+ \left. S_{DB}(\omega + \omega_c) - S_{DB}(\omega - \omega_c)\right], \end{aligned} \quad (7.48)$$

in which $S_B(\omega)$ and $S_D(\omega)$ represent the respective power spectrum densities for $b(t)$ and $d(t)$; $S_{BD}(\omega)$ is the cross-spectrum density between $b(t)$ and $d(t)$; and $S_{DB}(\omega)$ is the cross-spectrum density between $d(t)$ and $b(t)$.

For uncorrelated signals, the previous formula can be simplified to

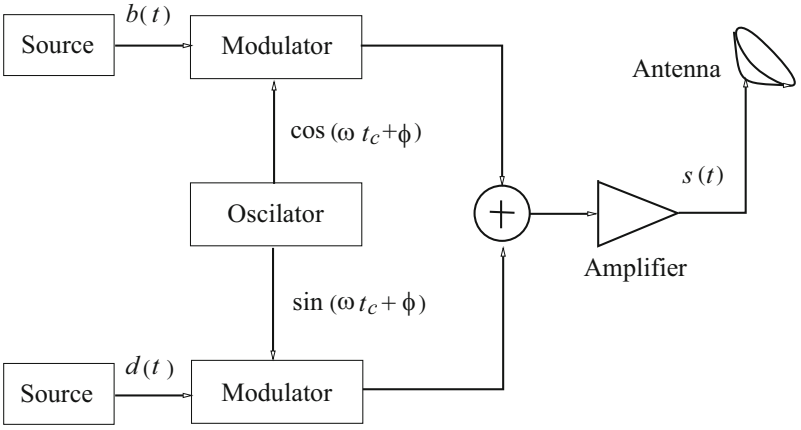


Fig. 7.16 Block diagram for the quadrature modulator

$$S_S(\omega) = \frac{1}{4}\left[S_B(\omega+\omega_c) + S_B(\omega-\omega_c) + S_D(\omega+\omega_c) + S_D(\omega-\omega_c)\right]. \tag{7.49}$$

The quadrature amplitude modulation scheme is shown in Fig. 7.16.

7.8 Single Sideband Amplitude Modulation

Single sideband amplitude modulation, or AM-SSB, is, in reality, a type of quadrature amplitude modulation which uses either the lower or upper AM side band for transmission. The AM-SSB signal can be obtained by filtering out the undesired side band of an AM-SC signal.

The SSB signal saves bandwidth and power, as compared to other systems, but needs frequency and phase synchronization to be recovered (Carlson 1975). In order to explain the process of generating the SSB signal, use is made of the Hilbert transform, which has been introduced in a previous chapter.

7.8.1 Producing the SSB Signal

The usual process to obtain the SSB signal is by filtering one of the AM-SC sidebands. Another method is to use the properties of the Hilbert transform, discussed previously. Consider a sinusoidal signal $m(t) = \cos(\omega_M t)$, which has a Fourier spectrum $S_M(\omega)$ represented by two impulses at $\pm\,\omega_M$.

The modulated signal is given by $\cos(\omega_M t)\cos(\omega_c t)$, whose Fourier spectrum is that of $S_M(\omega)$ shifted to $\pm\,\omega_c$. The carrier spectrum is formed by two impulses at

$\pm\omega_c$. Therefore, producing an SSB signal, for the special case, $m(t) = \cos(\omega_M t)$ is equivalent to generating the signal $\cos[(\omega_c - \omega_M)t]$ or the signal $\cos[(\omega_c + \omega_M)t]$.

By trigonometry, it follows that

$$\cos[(\omega_c - \omega_M)t] = \cos\omega_M t \cos\omega_c t + \sin\omega_M t \sin\omega_c t. \tag{7.50}$$

Thus, the desired SSB signal can be produced adding $\cos\omega_M t \cos\omega_c t$ and $\sin\omega_M t$ sin, $\omega_c t$. Signal $\cos\omega_M t \cos\omega_c t$ can be produced by a balanced modulator. Signal $\sin\omega_M t \sin\omega_c t$ can be written as $\cos(\omega_M t - \frac{\pi}{2})\cos(\omega_c t - \frac{\pi}{2})$.

This signal can also be generated by a balanced modulator, as long as $\cos\omega_M t$ and carrier $\cos\omega_c t$ be phase shifted by $\frac{\pi}{2}$. Although this result has been derived for the special case $m(t) = \cos\omega_M t$, it is valid to any waveform, because of the Fourier series properties, that allow the representation of any signal by a sum of sine and cosine functions.

The SSB signal associated with $m(t)$ is thus

$$s(t) = m(t)\cos(\omega_c t + \phi) + \hat{m}(t)\sin(\omega_c t + \phi), \tag{7.51}$$

in which $\hat{m}(t)$ is obtained by phase shifting all frequency components of $m(t)$ by $\frac{\pi}{2}$. The block diagram of the modulator is shown in Fig. 7.17.

7.8.2 *Lower Sideband SSB: Random Signal*

If the modulating signal $m(t)$ is a zero mean, stationary, stochastic process, the usual path to follow, in order to obtain the power spectrum density, is to compute its autocorrelation function. The modulated SSB signal is given by

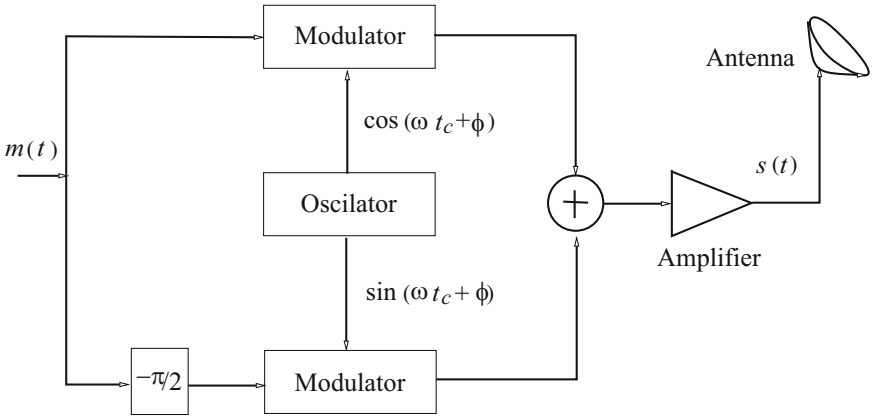


Fig. 7.17 Block diagram for the SSB modulator

$$s(t) = m(t)\cos(\omega_c t + \phi) + \hat{m}(t)\sin(\omega_c t + \phi) \tag{7.52}$$

The autocorrelation function is calculated, as usual, by the formula

$$R_S(\tau) = E[s(t)s(t+\tau)] \tag{7.53}$$

Substituting $s(t)$, given by Eq. 7.52

$$\begin{aligned} R_S(\tau) = E\big[&(m(t)\cos(\omega_c t + \phi) + \hat{m}(t)\sin(\omega_c t + \phi)) \\ &\cdot (m(t+\tau)\cos(\omega_c(t+\tau) + \phi) \\ &+ \hat{m}(t+\tau)\sin(\omega_c(t+\tau) + \phi))\big]. \end{aligned} \tag{7.54}$$

The previous equation can be split into

$$\begin{aligned} R_S(\tau) = &E\left[m(t)m(t+\tau)\cos(\omega_c t + \phi)\cos(\omega_c(t+\tau) + \phi)\right] \\ &+ E\left[m(t)\hat{m}(t+\tau)\cos(\omega_c t + \phi)\sin(\omega_c(t+\tau) + \phi)\right] \\ &+ E\left[\hat{m}(t)m(t+\tau)\sin(\omega_c t + \phi)\cos(\omega_c(t+\tau) + \phi)\right] \\ &+ E\left[\hat{m}(t)\hat{m}(t+\tau)\sin(\omega_c t + \phi)\sin(\omega_c(t+\tau) + \phi)\right]. \end{aligned} \tag{7.55}$$

After the corresponding simplifications, one obtains

$$\begin{aligned} R_S(\tau) = &\frac{1}{2}R_{MM}(\tau)\cos\omega_c\tau + \frac{1}{2}R_{M\hat{M}}(\tau)\sin\omega_c\tau \\ &- \frac{1}{2}R_{\hat{M}M}(\tau)\sin\omega_c\tau + \frac{1}{2}R_{\hat{M}\hat{M}}(\tau)\cos\omega_c\tau. \end{aligned} \tag{7.56}$$

It is known that $R_{MM}(\tau) = R_{\hat{M}\hat{M}}(\tau)$ and $R_{M\hat{M}}(\tau) = -R_{\hat{M}M}(\tau)$. Therefore, the power spectrum density can be computed as $S_S(\omega) = \mathcal{F}[R_S(\tau)]$, by the use of previous relations and using the equation for the Hilbert filter

$$H(\omega) = \begin{cases} -j \text{ se } \omega \geq 0 \\ +j \text{ se } \omega < 0 \end{cases} \tag{7.57}$$

which leads to

$$S_{\hat{M}\hat{M}}(\omega) = S_{MM}(\omega) \tag{7.58}$$

$$S_{M\hat{M}}(\omega) = j[u(-\omega) - u(\omega)]S_{MM}(\omega) \tag{7.59}$$

$$S_{\hat{M}M}(\omega) = j[u(\omega) - u(-\omega)]S_{MM}(\omega), \tag{7.60}$$

Fig. 7.18 illustrates the spectrum of the original signal, $S_{MM}(\omega)$ at the input of the modulator. The signals filtered by the Hilbert filter are seen in Figs. 7.19 and 7.20.

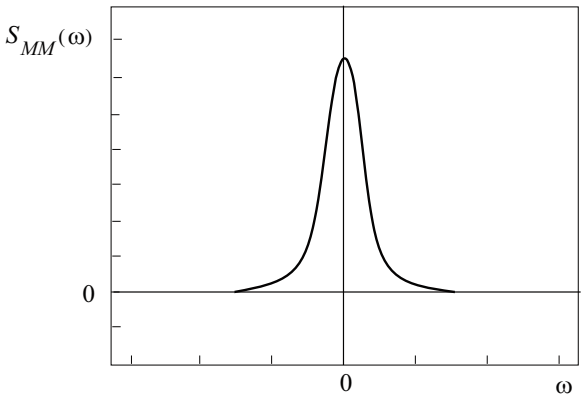


Fig. 7.18 Spectrum of the modulating signal $S_{MM}(\omega)$

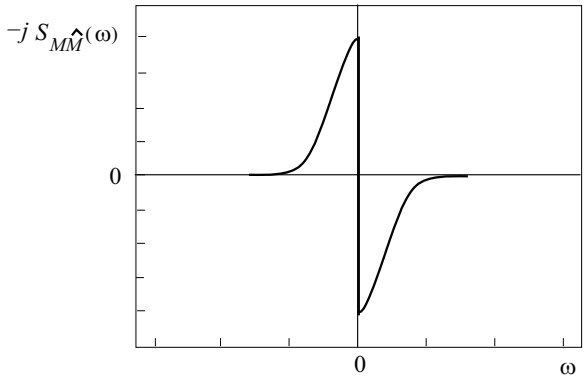


Fig. 7.19 Spectrum of the filtered modulating signal $-jS_{M\hat{M}}(\omega)$

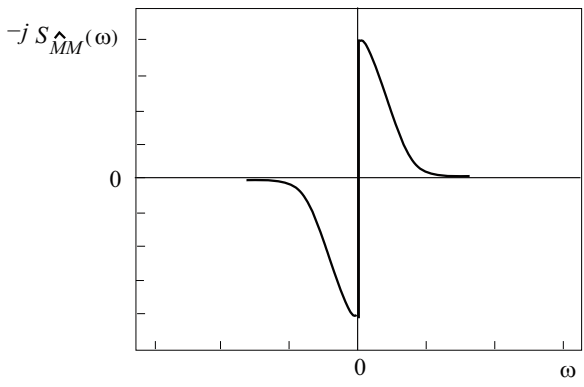


Fig. 7.20 Spectrum of the filtered modulating signal $-jS_{\hat{M}M}(\omega)$

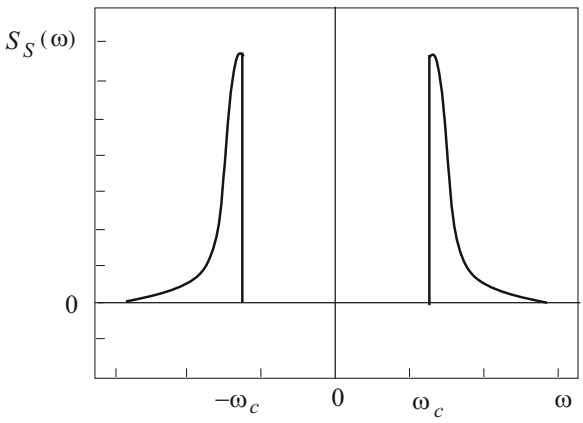


Fig. 7.21 Spectrum of the SSB modulated signal $S_S(\omega)$

It is possible to notice that the following power spectral density results for the SSB signal:

$$S(\omega)_{SSB} = S_M(\omega - \omega_c)u(-\omega + \omega_c) + S_M(\omega + \omega_c)u(\omega + \omega_c), \tag{7.61}$$

which represents the lower sideband SSB signal, obtained from the original spectrum $S_M(\omega)$.

The power spectral density for the SSB modulated carrier for the upper sideband SSB signal is given by

$$S(\omega)_{SSB} = S_M(\omega - \omega_c)u(\omega - \omega_c) + S_M(\omega + \omega_c)u(-\omega - \omega_c), \tag{7.62}$$

The spectrum for the upper sideband SSB modulated signal is shown in Fig. 7.21.

7.9 Angle Modulation

Angle modulation is a generic term for both frequency modulation (FM) and phase modulation (PM). The objective of this section is to derive a general mathematical model to analyze stochastic angle modulated signals. Several transmission systems use either frequency or phase modulation, including analog and digital mobile cellular communication systems, satellite transmission systems, television systems, and wireless telephones.

Nowadays, it is well established that a frequency-modulated signal cannot fit in a bandwidth that is narrower than the one occupied by the original, or modulating, signal. Ironically, frequency modulation was originally conceived as a means to reduce the bandwidth required for the transmission of a given signal.

It is not difficult to prove that this argument is false. John R. Carson was the first to recognize the fallacy of the reduction in bandwidth and clarified the subject in a

paper published in 1922 (Carson 1922). At the time, however, he could foresee the advantage of frequency modulation over amplitude modulation, and the subject was forgotten for some time, until Edwin H. Armstrong invented the first radio which modulated a carrier in frequency (Armstrong 1936).

Later on, Carson proved, in an unpublished memorandum of August 28, 1939, that the bandwidth of the frequency-modulated signal was equal to twice the sum of the peak frequency deviation and the highest frequency of the modulating signal (Carlson 1975).

7.10 Angle Modulation Using Random Signals

The use of stochastic processes allows the development of a general and elegant method to compute the power spectrum density of angle modulated signals, and also to determine the bandwidth associated with the resulting modulated carrier. The modulating signal is considered a stationary random process $m(t)$ that has autocorrelation $R_M(\tau)$.

7.10.1 Mathematical Model

Philip Mayne Woodward (1919–2018), a British engineer, mathematician, and horologist, published a paper establishing the concept that the spectrum of a high-index frequency-modulated (FM) waveform has a shape that is approximately that of the probability distribution of its instantaneous frequency (Woodward 1952; Alencar and Neto 1991).

The method described in this section estimates the spectrum of an angle-modulated signal, based on the first-order probability density function (pdf) of the random modulating signal to validate the previous theorem. A new proof is presented which includes the linear mean square estimator (Papoulis 1981a; Alencar 1998a).

The general angle modulated signal $s(t)$ is obtained from the following known equation:

$$s(t) = A\cos\left(\omega_c t + \theta(t) + \phi\right), \tag{7.63}$$

in which

$$\theta(t) = \Delta_{FM} \int_{-\infty}^{t} m(t)dt, \tag{7.64}$$

for frequency modulation and

$$\theta(t) = \Delta_{PM} m(t) dt, \tag{7.65}$$

for phase modulation, in which the constant parameters A, ω_c, Δ_{FM}, and Δ_{PM} represent, respectively, the carrier amplitude and angular frequency and the frequency and phase deviation indices.

The message signal is represented by $m(t)$, which is considered a zero mean random stationary process, which is a reasonable assumption. The carrier phase ϕ is random, assumed uniformly distributed in the range $(0, 2\pi]$ and statistically independent of $m(t)$.

The modulating signal, $\theta(t)$, represents the variation in the carrier phase, produced by the message signal. The frequency modulation index is defined as

$$\beta = \frac{\Delta_{FM}\sigma_M}{\omega_M}, \tag{7.66}$$

in which $\sigma_M = \sqrt{P_M}$ and P_M represents the power of the message signal, $m(t)$, and ω_M is the maximum angular frequency of the signal.

The frequency deviation $\sigma_F = \Delta_{FM}\sigma_M$ represents the shift from the original, or spectral, carrier frequency. The modulation index gives an idea of how many times the modulating signal bandwidth fits into the frequency deviation.

The phase modulation index, which indicates how the carrier phase deviates from the original phase, is defined as

$$\alpha = \Delta_{PM}\sigma_M. \tag{7.67}$$

In the evaluation of the spectrum of the angle modulated signal, subject to a stochastic modulating signal, the following steps are observed:

1 Compute the autocorrelation function of the modulated carrier $s(t)$ in Eq. 7.63.
2 Obtain an estimate of this autocorrelation, for the special cases of a low, medium, and high modulation indices, using the theory of stochastic processes and, in particular, the linear mean square estimator (Papoulis 1981a).
3 Compute the power spectrum density (PSD) of $s(t)$ as the Fourier transform of the autocorrelation estimate for each special case.

The first step involves the computation of the autocorrelation function of $s(t)$, defined by Eq. 7.63. It can be expressed as

$$R_S(\tau) = E[s(t)s(t+\tau)] \tag{7.68}$$

in which $E[\cdot]$ represents the expected value operator. It follows that

$$R_S(\tau) = \frac{A^2}{2} E[\cos(w_c\tau - \theta(t) + \theta(t+\tau))]. \tag{7.69}$$

The following sections demonstrated the computation of the power spectrum density of the modulated signal, for the two special cases of low modulation index and high modulation index.

7.10.2 Modulation with Low Index

For the case of a carrier modulated with a low modulation index, that is, $\beta < 0.5$, the autocorrelation function of the modulated carrier $s(t)$ can be obtained from Eq. 7.69 by expanding the cosine function

$$R_S(\tau) = \frac{A^2}{2}\cos(\omega_c\tau)E[\cos(-\theta(t)+\theta(t+\tau))] - \frac{A^2}{2}\sin(\omega_c\tau)E[\sin(-\theta(t)+\theta(t+\tau))]. \tag{7.70}$$

In this case, it is possible to expand the sine and cosine functions of Eq. 7.70 in Taylor series, neglecting the high-order terms, which are negligible, to obtain

$$R_S(\tau) = \frac{A^2}{2}\cos(\omega_c\tau)E\left[1-\frac{(-\theta(t)+\theta(t+\tau))^2}{2}\right] - \frac{A^2}{2}\sin(\omega_c\tau)E\left[-\theta(t)+\theta(t+\tau)\right]. \tag{7.71}$$

Considering that $m(t)$ is usually a zero mean, stationary process, it follows that

$$R_S(\tau) = \frac{A^2}{2}\cos(\omega_c\tau)[1-R_\Theta(0)+R_\Theta(\tau)], \tag{7.72}$$

in which

$$R_\Theta(\tau) = E[\theta(t)\theta(t+\tau)] \tag{7.73}$$

and $R_\Theta(0) = P_\Theta$ is the power of signal $\theta(t)$.

The power spectral density of the modulated carrier $s(t)$ is obtained by the use of the Wiener-Khinchin theorem or using the Fourier transform of Eq. 7.72 (Papoulis 1983b):

$$S_S(\omega) = \frac{\pi A^2(1-P_\Theta)}{2}[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)] + \frac{\Delta_{FM}^2 A^2}{4}\left[\frac{S_M(\omega+\omega_c)}{(\omega+\omega_c)^2}+\frac{S_M(\omega-\omega_c)}{(\omega-\omega_c)^2}\right] \tag{7.74}$$

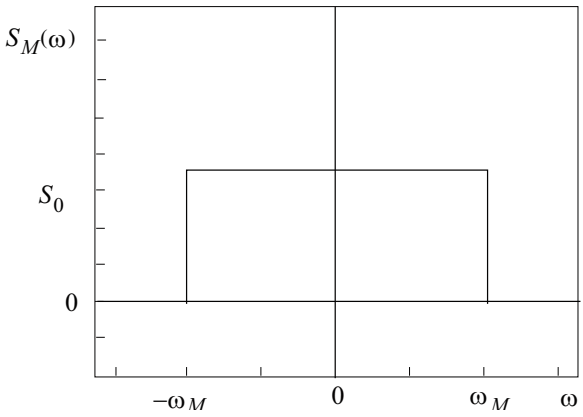


Fig. 7.22 Spectrum for the white Gaussian noise modulating signal

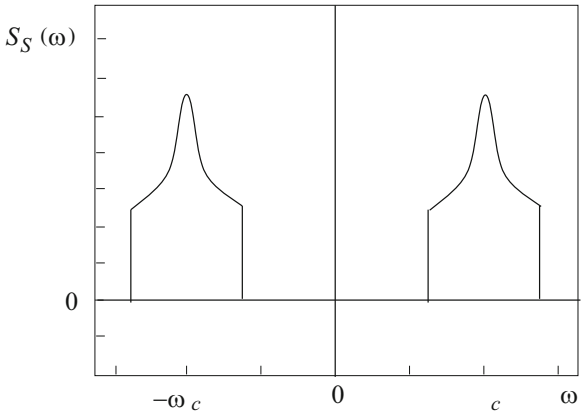


Fig. 7.23 Spectrum for a carrier modulated by white Gaussian noise, with low modulating index

in which $S_M(\omega)$ represents the PSD of the message signal $m(t)$, which has bandwidth ω_M. The modulated signal bandwidth is double the message signal bandwidth BW = $2\omega_M$.

Some observations are possible. From Eq. 7.74 one can notice that the FM spectrum has the shape of the message signal spectrum multiplied by a squared hyperbolic function.

Example Consider a uniform power spectrum density for the modulating signal, as depicted in Fig. 7.22.

The power spectral density for a carrier that is frequency modulated by white Gaussian noise, with low modulating index, is shown in Fig. 7.23, in which the power spectral density of the modulating signal is given by

$$S_M(\omega) = S_0 \left[\mathrm{u}(\omega + \omega_M) - \mathrm{u}(\omega - \omega_M) \right]. \tag{7.75}$$

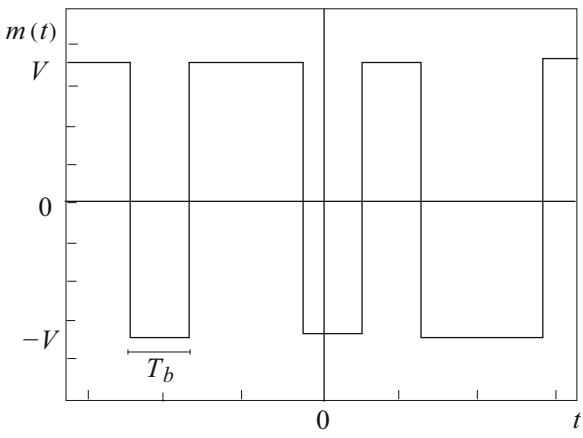


Fig. 7.24 Digital modulating signal

As can be observed, the main modulation effect is to distort the spectrum of the modulating signal. In fact, the power spectral density shows a peak at the carrier frequency and then decreases following the shape of a squared hyperbola.

7.10.3 Modulation with a Digital Signal

Consider that the carrier is modulated in frequency by a digital signal. The modulating signal, shown in Fig. 7.24, can be written as the following summation

$$m(t) = \sum_k m_k \, p(t - kT_b), \tag{7.76}$$

in which $m_k \in \{-V, V\}$, $k \in \mathbb{Z}$, $p(t)$ is a discrete unit pulse, and T_b is the bit interval. This digital modulating signal is shown in Fig. 7.24.

The frequency-modulated carrier by a digital signal can be written as

$$\begin{aligned} s(t) &= A\cos\left[\omega_c t + \Delta_{FM}\int_{-\infty}^{t} m(t)\mathrm{d}t + \phi\right] \\ &= A\cos\left[\omega_c t + \Delta_{FM}\int_{-\infty}^{t}\sum_k m_k \, p(t - kT_b)\mathrm{d}t + \phi\right] \\ &= A\cos\left[\omega_c t + \Delta_{FM}\sum_k m_k\int_{-\infty}^{t} p(t - kT_b)\mathrm{d}t + \phi\right] \\ &= A\cos\left[\omega_c t + \theta(t) + \phi\right], \end{aligned} \tag{7.77}$$

in which

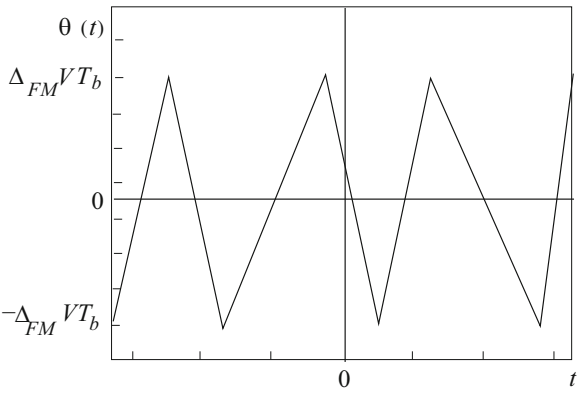


Fig. 7.25 Adjusted angle modulating signal

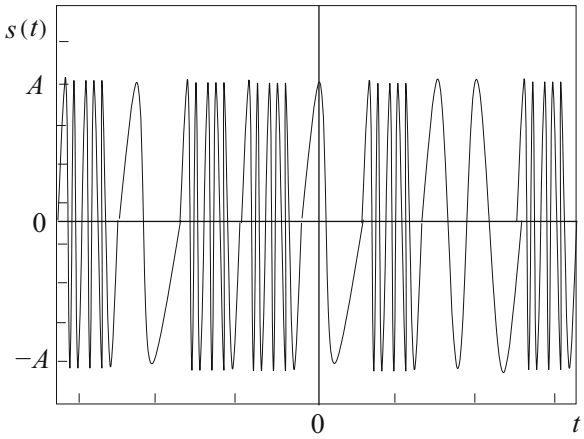


Fig. 7.26 Carrier modulated in frequency by a digital signal

$$\theta(t) = \Delta_{FM} \sum_k m_k \int_{-\infty}^{t} p(t - kT_b)\mathrm{d}t. \tag{7.78}$$

The adjusted modulating signal, $\theta(t)$, is represented in Fig. 7.25. Recall that the instantaneous angular frequency is the derivative of the angle function in time, as given by

$$\omega(t) = \frac{\mathrm{d}\theta(t)}{\mathrm{d}t}. \tag{7.79}$$

The carrier modulated in frequency by a digital signal is depicted in Fig. 7.26.

Again, for the digital signal, whose autocorrelation is given by Formula 5.51, the corresponding spectrum is depicted in Fig. 7.27. As can be observed, the spectrum shape is limited by the squared hyperbole.

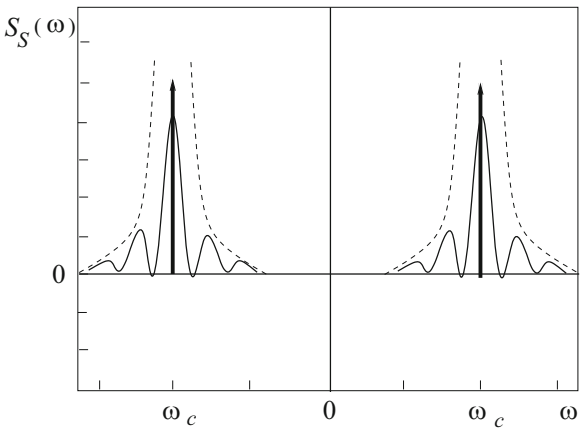


Fig. 7.27 Spectrum for a carrier modulated by a digital signal, with low modulating index

7.10.4 Modulation with High Index

As a result of Woodward's theorem, discussed in the following, the power spectrum density of the modulated carrier $s(t)$ approaches the probability density function of the modulating signal $m(t)$ in Eq. 7.64, as the modulation index is increased, that is, $\beta > 5$.

A high modulating index causes a wide spectrum broadening of the modulated signal, although the spectrum converges somehow for an infinite index. Actually, the modulated carrier power spectrum density converges to the probability density function of the modulating signal, as stated in Woodward (1952) and Blachman and McAlpine (1969).

As practical cases, the analog mobile cellular system, the commercial FM, and the angle modulated wireless telephone can be considered as high modulation index systems (Lee 1989). For the high modulation index case, it is more useful to use Euler's formula, named after the mathematician Leonhard Euler (1707–1783), and rewrite Eq. 7.69 as

$$R_S(\tau) = \frac{A^2}{4} e^{j\omega_c \tau} E[e^{j(-\theta(t)+\theta(t+\tau))}] + \frac{A^2}{4} e^{-j\omega_c \tau} E[e^{j(\theta(t)-\theta(t+\tau))}]. \tag{7.80}$$

Some initial considerations are in place. For instance, the second-order linear mean square estimate of the process $\theta(t+\tau)$ includes its current value $\theta(t)$ and its derivative $\theta'(t)$

$$\theta(t+\tau) \approx \alpha\,\theta(t) + \beta\,\theta'(t). \tag{7.81}$$

In this case, the mean square error is given by

$$e(t, \alpha, \beta) = E\left[\left(\theta(t+\tau) - \alpha\,\theta(t) - \beta\,\theta'(t)\right)^2\right]. \tag{7.82}$$

To obtain the values of the optimization parameters α and β, the derivative of the error must converge to zero. To begin with, the partial derivative of the error in terms of α gives

$$\frac{\partial e(t, \alpha, \beta)}{\partial \alpha} = E\left[\left(\theta(t+\tau) - \alpha\,\theta(t) - \beta\,\theta'(t)\right)\theta(t)\right] = 0, \tag{7.83}$$

which indicates that the minimum error is orthogonal to the random process.

It is useful to recognize the autocorrelations in the previous expression, which yields

$$R_\Theta(\tau) - \alpha R_\Theta(0) - \beta R_{\Theta'\Theta}(0) = 0.$$

Observe that the last term of the equation is zero, because the autocorrelation has a maximum at the origin. Thus,

$$\alpha = \frac{R_\Theta(\tau)}{R_\Theta(0)}. \tag{7.84}$$

The partial derivative of the error in terms of β gives

$$\frac{\partial e(t, \alpha, \beta)}{\partial \beta} = E\left[\left(\theta(t+\tau) - \alpha\,\theta(t) - \beta\,\theta'(t)\right)\theta'(t)\right] = 0 \tag{7.85}$$

or

$$R_{\Theta'\Theta}(\tau) - \alpha R_{\Theta'\Theta}(0) - \beta R_{\Theta'}(\tau) = 0.$$

The first term of the expression is the derivative of the autocorrelation function in relation to τ. The second term is zero, as explained previously, and the third term is the negative of the second derivative of the autocorrelation function in relation to τ. Thus,

$$\beta = \frac{R'_\Theta(\tau)}{R''_\Theta(0)}. \tag{7.86}$$

Therefore, the best approximation to the future value of the process, in the mean square sense, is

$$\theta(t+\tau) \approx \frac{R_\Theta(\tau)}{R_\Theta(0)}\theta(t) + \frac{R'_\Theta(\tau)}{R''_\Theta(0)}\theta'(t). \tag{7.87}$$

Considering that the random process is slowly varying, as compared to the spectral frequency of the modulated carrier, leads the approximation

$$R_\Theta(\tau) \approx R_\Theta(0).$$

Expanding the derivative of the autocorrelation in a Taylor series, named after the mathematician Brook Taylor (1685–1731), gives

$$R'_\Theta(\tau) \approx R'_\Theta(0) + \tau R''_\Theta(0). \tag{7.88}$$

Recalling that the autocorrelation has a maximum at the origin, simplifies the approximation to

$$R'_\Theta(\tau) \approx \tau R''_\Theta(0).$$

Finally, the future value of the random process can be approximated by

$$\theta(t+\tau) \approx \theta(t) + \tau\,\theta'(t). \tag{7.89}$$

Using the linear mean square estimator in Eq. 7.80 then gives (Alencar 1989; Papoulis 1983b)

$$R_S(\tau) = \frac{A^2}{4} e^{j\omega_c \tau} E[e^{j\tau\theta'(t)}] + \frac{A^2}{4} e^{-j\omega_c \tau} E[e^{-j\tau\theta'(t)}]. \tag{7.90}$$

By definition, $\theta'(t) = d\theta(t)/dt = \omega(t)$, in which $\omega(t)$ is the carrier angular frequency deviation, at the instantaneous carrier angular frequency, thus

$$R_S(\tau) = \frac{A^2}{4} e^{j\omega_c \tau} E[e^{j\tau\omega(t)}] + \frac{A^2}{4} e^{-j\omega_c \tau} E[e^{-j\tau\omega(t)}]. \tag{7.91}$$

Taking into account that

$$E[e^{j\tau\omega(t)}] = \int_{-\infty}^{\infty} p_\Omega(\omega(t)) e^{j\tau\omega(t)} d\omega(t) \tag{7.92}$$

represents the characteristic function of process $\omega(t) = \theta'(t)$ and $p_\Omega(\omega(t))$ is its probability density function, considered symmetrical here, without essential loss of generality.

Calculating the Fourier transform of Eq. 7.91, it follows that

$$S_S(\omega) = \frac{\pi A^2}{2}[p_\Omega(\omega + \omega_c) + p_\Omega(\omega - \omega_c)]. \tag{7.93}$$

Considering the definition of $\omega(t)$ from Eq. 7.64, it is noticed that $\omega(t) = \Delta_{FM}.m(t)$, thus

$$p_{\Omega}(\omega(t)) = \frac{1}{\Delta_{FM}} p_M\left(\frac{m}{\Delta_{FM}}\right) \tag{7.94}$$

in which $p_M(\cdot)$ is the probability density function of $m(t)$.

Substituting 7.94 into 7.93 gives, finally, the formula for the power spectrum density for the wideband frequency modulated signal (Alencar and Neto 1991):

$$S_S(\omega) = \frac{\pi A^2}{2\Delta_{FM}}\left[p_M\left(\frac{\omega+\omega_c}{\Delta_{FM}}\right) + p_M\left(\frac{\omega-\omega_c}{\Delta_{FM}}\right)\right]. \tag{7.95}$$

7.10.5 Phase Modulation

Following a similar line of thought, it is possible to derive a formula for the spectrum of the phase modulated signal (PM). It is instructive to recall the instantaneous angular frequency, given by

$$\omega(t) = \frac{\theta(t)}{dt} = \Delta_{PM} m(t). \tag{7.96}$$

Therefore,

$$S_S(\omega) = \frac{\pi A^2}{2\Delta_{PM}}\left[p_{M'}\left(\frac{\omega+\omega_c}{\Delta_{PM}}\right) + p_{M'}\left(\frac{\omega-\omega_c}{\Delta_{PM}}\right)\right], \tag{7.97}$$

in which $p_{M'}(\cdot)$ is the probability density function of the derivative of the message signal $m(t)$.

For a Gaussian modulating signal

$$p_M(m) = \frac{1}{\sqrt{2\pi P_M}} e^{-\frac{m^2}{2P_M}}, \tag{7.98}$$

in which $P_M = R_M(0)$ denotes the power of signal $m(t)$, which gives

$$p_M(m) = \frac{1}{\sqrt{2\pi R_M(0)}} e^{-\frac{m^2}{2R_M(0)}}. \tag{7.99}$$

From Eq. 7.99, and considering the property $R_{M'}(\tau) = -R''_M(\tau)$ (the autocorrelation of the signal derivative equals the negative of the second derivative of the autocorrelation function) and the fact that the response of a linear system to a Gaussian input is also Gaussian, one can determine the probability density function

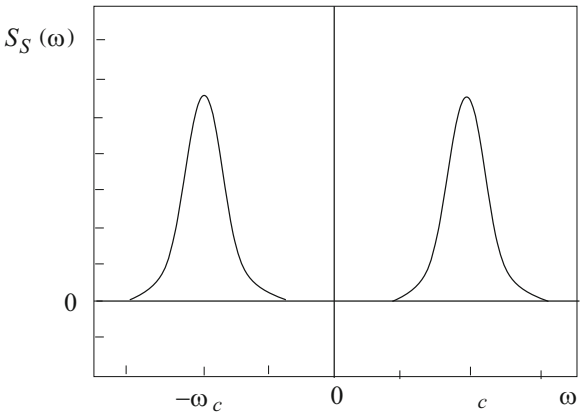


Fig. 7.28 Spectrum of FM signal for a Gaussian modulating signal and high modulation index

for the derivative of a Gaussian signal

$$p_{M'}(m) = \frac{1}{\sqrt{-2\pi R''_M(0)}} e^{-\frac{m^2}{(-2R''_M(0)}}). \tag{7.100}$$

That is, the spectrum of a wideband PM signal, when modulated by a Gaussian signal, presents similar characteristics to his FM counterpart. For a narrowband modulation, the PM spectrum approximates the spectrum of the modulating signal which is different from the FM spectrum for the same conditions (Fig. 7.28).

7.10.6 Modulated Carrier Bandwidth

It is noticeable that for a high modulation index, the frequency deviation is given by $\Delta_{FM}\sqrt{P_M}$ and the bandwidth approximated by the formula $\text{BW} = 2\Delta_{FM}\sqrt{P_M}$, in order to include most of the modulated carrier power. As previously derived, the bandwidth for a narrowband FM is $\text{BW} = 2\omega_M$. Interpolating between both values gives a formula that covers the whole range of modulation indices β

$$\begin{aligned} \text{BW} &= 2\omega_M + 2\Delta_{FM}\sqrt{P_M} \\ &= 2\left(\frac{\Delta_{FM}\sqrt{P_M}}{\omega_M} + 1\right)\omega_M \\ &= 2\left(\beta + 1\right)\omega_M, \end{aligned} \tag{7.101}$$

in which, as defined, the frequency modulation index is given by

$$\beta = \frac{\Delta_{FM}\sigma_M}{\omega_M}.$$

Equation 7.102 is the well-known Carson's rule, whose heuristic deduction first appeared in 1922 (Carson 1922). John Renshaw Carson (1886–1940) was an American electrical engineer who invented single-sideband modulation and developed the Carson bandwidth rule to estimate frequency modulation (FM) bandwidth.

A sinusoidal signal $m(t) = V\sin(\omega_M t + \varphi)$, in which φ is uniformly distributed in the interval $[0, 2\pi]$, has the following probability density function:

$$p_M(m) = \frac{1}{\pi\sqrt{V^2 - m^2}}, |m| < V. \tag{7.102}$$

Consequently, a carrier modulated by a sinusoidal signal, with high modulation index, has the following spectrum:

$$S_S(\omega) = \frac{1}{2\sqrt{(V\Delta_{FM})^2 - (\omega \pm \omega_c)^2}}, |\omega - \omega_c| < \Delta_{FM}V. \tag{7.103}$$

The modulated signal occupies, in this case, a bandwidth equivalent to $2\Delta_{FM}V$. Its PSD is shown in Fig. 7.29.

For a signal which is phase modulated, using the same sinusoid as before as the modulating signal, the derivative of the message signal is given by $m'(t) = \omega_M V\cos(\omega_M t + \varphi)$, which has the following probability density function:

$$p_{M'}(m) = \frac{1}{\pi\sqrt{(\omega_M V)^2 - m^2}}, |m| < \omega_M V. \tag{7.104}$$

The spectrum can be computed, resulting in

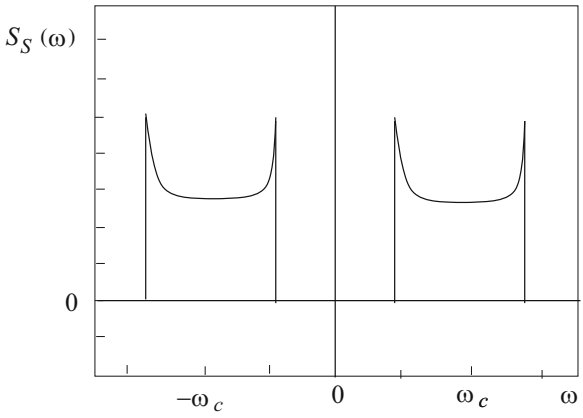


Fig. 7.29 Spectrum of an FM signal for a sinusoidal modulating signal and high modulation index

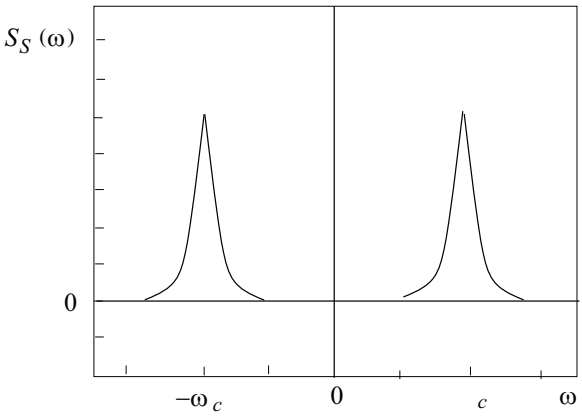


Fig. 7.30 Spectrum of an FM signal for a voice signal, modeled by a Laplace distribution, with high modulation index

$$S_S(\omega) = \frac{1}{2\sqrt{(\omega_M V \Delta_{PM})^2 - (\omega \pm w_c)^2}}, |\omega - \omega_c| < \omega_M \Delta_{PM} V. \tag{7.105}$$

The bandwidth is now $2\omega_M \Delta_{PM} V$, and its PSD is similar to the one shown in Fig. 7.29.

Example The probability density function of the speech signal can be modeled by a Laplace distribution (Gazor and Zhang 2003):

$$p_M(m) = \alpha e^{-\alpha|m|} \tag{7.106}$$

Therefore, using the previous development, the power spectral distribution can be expressed by the following formula, as shown in Fig. 7.30:

$$S_S(\omega) = \frac{\alpha A^2}{2\Delta_{FM}} \left[e^{\frac{\alpha}{\Delta_{FM}}(|\omega+\omega_c)|} + e^{-\frac{\alpha}{\Delta_{FM}}|(\omega-\omega_c)|} \right], \tag{7.107}$$

and its bandwidth is $\text{BW} = \frac{2\sqrt{2}\Delta_{FM}}{\alpha}$. Note that the bandwidth is an inverse function of the distribution parameter α, which controls the variance (power) of the probability density function:

$$\sigma_M^2 = \frac{2}{\alpha^2}. \tag{7.108}$$

Example A more precise model for the voice signal uses the bilateral Gamma distribution (Paez and Glisson 1972), defined by the following formula:

$$p_M(m) = \frac{\sqrt{k}}{2\sqrt{\pi}} \frac{e^{-k|m|}}{\sqrt{|m|}} \tag{7.109}$$

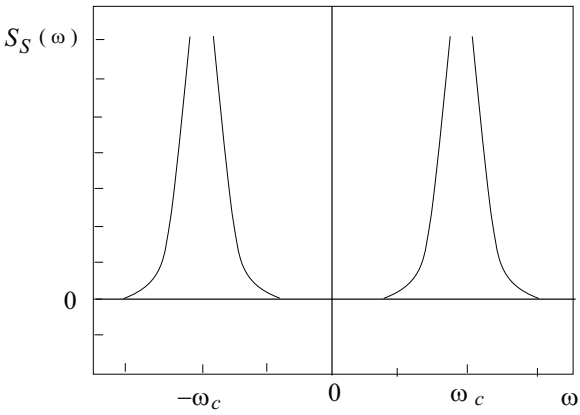


Fig. 7.31 Spectrum of an FM signal for a voice signal, modeled by a Gamma distribution, with high modulation index

Therefore, the corresponding FM spectrum, depicted in Fig. 7.31, is given by

$$S_S(\omega) = \frac{A^2\sqrt{\pi k}}{4\sqrt{\Delta_{FM}}} \frac{e^{-\frac{k}{\Delta_{FM}}|\omega \pm \omega_c|}}{\sqrt{|\omega \pm \omega_c|}} \tag{7.110}$$

and its bandwidth is $\text{BW} = \frac{2\Delta_{FM}\sqrt{0.75}}{k}$.

7.11 Digital Frequency Modulation

Because the digital signal has only two levels, from the set $\{-V, V\}$, it has a probability density function given by two impulses, as depicted in Fig. 7.32, and shown in the following formula that has been written considering equiprobable levels:

$$p_M(m) = \frac{1}{2}\left[\delta(m+V) + \delta(m-V)\right]. \tag{7.111}$$

When the digital signal modulates a carrier in frequency, with a high index of modulation, it produces a spectrum that is given by two impulses centered at ω_c and two additional impulses centered at $-\omega_c$.

Because the voltage levels of the modulating signal are V and $-V$, the voltage difference is $2V$. Therefore, the frequency range of the modulated carrier, or the modulated carrier bandwidth, is $\text{BW} = 2\Delta_{FM}V$ (Fig. 7.33).

Example Suppose that the modulating signal pulses are triangular in shape, as depicted in Fig. 7.34.

The probability density function for the digital modulating signal, which has a uniform probability density function, is given by the following formula:

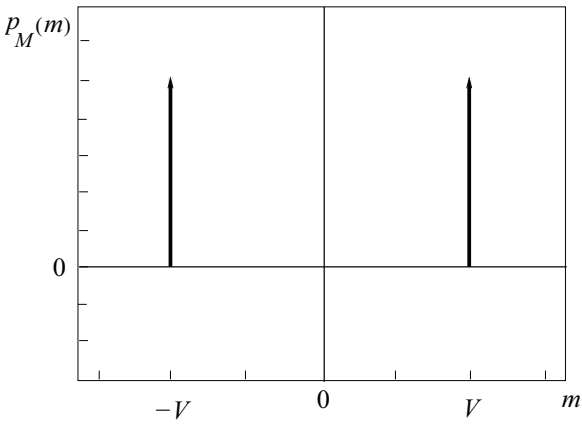


Fig. 7.32 Probability density function for a digital signal

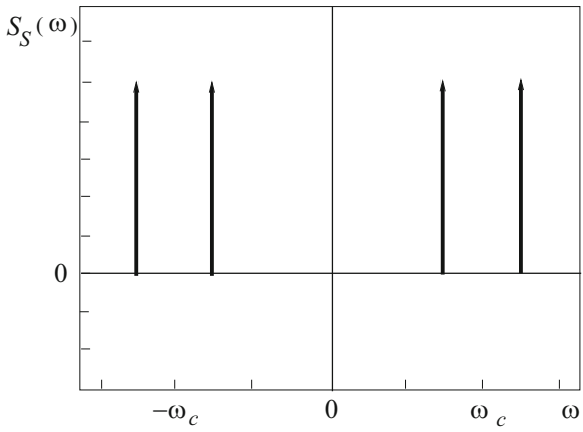


Fig. 7.33 Spectrum of an FM signal for a digital signal with high modulation index

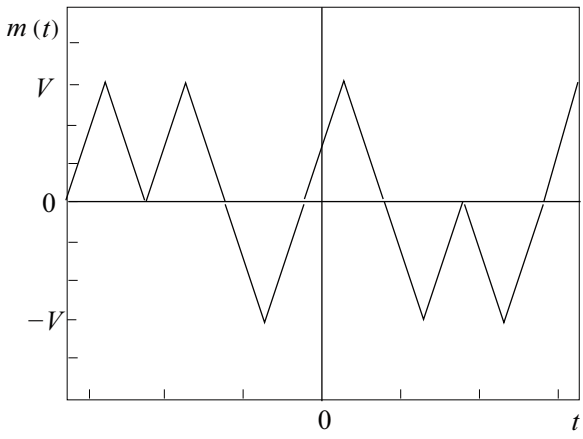


Fig. 7.34 Triangular-shaped digital modulating signal

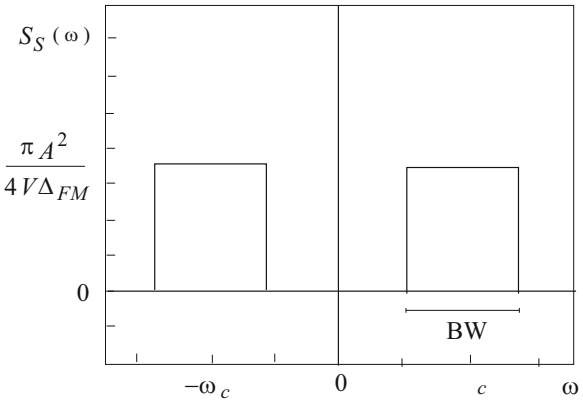


Fig. 7.35 Spectrum for carrier that is frequency modulated by a triangular-shaped digital signal

$$p_M(m) = \frac{1}{2V}\left[\mathrm{u}(m+V) + \mathrm{u}(m-V)\right]. \tag{7.112}$$

In this case, the modulated carrier power spectral density is given by the following formula and shown in Fig. 7.35:

$$S_S(\omega) = \frac{\pi A^2}{4V\Delta_{FM}}\left[\mathrm{u}\left(\frac{\omega+\omega_c+\Delta_{FM}V}{\Delta_{FM}}\right) - \mathrm{u}\left(\frac{\omega+\omega_c+\Delta_{FM}V}{\Delta_{FM}}\right) + \mathrm{u}\left(\frac{\omega-\omega_c-\Delta_{FM}V}{\Delta_{FM}}\right) - \mathrm{u}\left(\frac{\omega-\omega_c-\Delta_{FM}V}{\Delta_{FM}}\right)\right].$$

and the modulated carrier bandwidth is $\mathrm{BW} = 2\Delta_{FM}V$.

7.12 Problems

1. Describe the main modulation schemes and point their characteristics.
2. Is it possible to modulate, at the same time, the amplitude and the angle of a carrier? Give a practical example:

$$s(t) = a(t)\cos[\omega_c t + \theta(t) + \phi].$$

3. Explain why the radiation efficiency of radio signals depend on the antenna dimension.
4. Why the antenna dimensions must be compatible with the wavelength of the electromagnetic field and with the media characteristics?
5. Is it possible to use the sine function in place of the cosine in the formula for amplitude modulation? What would be the noticeable difference in the

modulated signal in time?

$$s(t) = a(t)\cos(\omega_c t + \phi).$$

6. In case the modulation index is greater than one ($\Delta_{AM} > 1$), that causes phase inversion in the modulated carrier, is it still possible to demodulate the signal? Explain!
7. Explain why the power spectral density of the modulating signal is always symmetrical.
8. Consider the power spectral density of the AM signal. Explain what happens to the spectrum for the cases $\Delta_{AM} \to 0$ and $\Delta_{AM} \to \infty$:

$$S_S(\omega) = \frac{A^2}{4}\left[2\pi(\delta(\omega+\omega_c)+\delta(\omega-\omega_c)) + \Delta_{AM}{}^2(S_M(\omega+\omega_c)+S_M(\omega-\omega_c))\right].$$

9. Consider the following digitally modulated signal. Why is it important to guarantee that the expected value of the random symbols be zero, that is, $\mathrm{E}[m_k] = 0$?

$$s(t) = \sum_{k=-\infty}^{\infty} m_k\, p(t - kT_b)\cos(\omega_c t + \phi).$$

10. The following equation gives the resultant amplitude for a QAM signal. Compute the expected power of the signal $\mathrm{E}[a(t)]$?

$$a(t) = \sqrt{b^2(t) + d^2(t)}.$$

11. The QAM resultant phase is given in the following. Plot the function $\theta(t)$ for the case: $b(t) = A\cos(\omega_M t)$ and $d(t) = A\sin(\omega_M t)$, in which $\omega_M = 2\pi f_M$ is a given angular frequency:

$$\theta(t) = -\tan^{-1}\left[\frac{d(t)}{b(t)}\right].$$

12. Determine the expected power of the resulting signal, $\mathrm{E}[a(t)]$, for the case $b(t) = A\cos(\omega_M t)$ and $d(t) = A\sin(\omega_M t)$.
13. The expression for the ISB signal is given in the following. Determine the expected power of the resulting signal, $\mathrm{E}[a(t)]$, for this modulation scheme.

$$s(t) = [l(t) + r(t)]\cos(\omega_c t + \phi) + [\hat{r}(t) - \hat{l}(t)]\sin(\omega_c t + \phi).$$

14. Compute the expected power of the modulated signal, $\mathrm{E}[s(t)]$, for the ISB scheme.
15. Determine a general expression for the power spectrum density of the ISB modulation scheme.
16. Give physical explanations for the frequency and phase modulation indices.
17. Considering that $m(t)$ is a zero mean, stationary process, compute the expected power of the modulated signal, $\mathrm{E}[s(t)]$, using the autocorrelation function for the case of low modulation index:

$$R_S(\tau) = \frac{A^2}{2}\cos(\omega_c\tau)[1 - R_\Theta(0) + R_\Theta(\tau)].$$

18. Derive the formula for the autocorrelation function of the modulated carrier, for the narrowband phase modulation (NBPM) scheme.
19. Compute the power of the angle-modulated carrier $s(t)$ using the autocorrelation function:

$$R_S(\tau) = \frac{A^2}{4}e^{j\omega_c\tau}E[e^{j\tau\omega(t)}] + \frac{A^2}{4}e^{-j\omega_c\tau}E[e^{-j\tau\omega(t)}].$$

20. Compute the power of the angle-modulated carrier $s(t)$ using the power spectral density function,

$$S_S(\omega) = \frac{\pi A^2}{2}[p_\Omega(\omega + \omega_c) + p_\Omega(\omega - \omega_c)].$$

21. Compute the bandwidth of the FM modulated carrier when the modulating signal has a Laplace distribution.

Chapter 8
Digital Modulation Theory*



It's fine to work on any problem, so long as it generates interesting mathematics along the way – even if you don't solve it at the end of the day.
– Andrew Wiles

8.1 Introduction

This chapter presents the general principles that are used to analyze digital communication schemes, in order to compute the effect of channel noise on the transmission of digital signals. Digital modulation can be considered as a mapping of symbols into waveforms, to transmit information through a communication channel.

Modulation schemes with varying carrier envelopes, such as QAM and ASK, are more susceptible to amplitude variations, because of nonlinear amplification, fading, and interference. Digital angle modulation signals occupy a wider bandwidth and can be used if resilience against noise is required.

Regarding the complexity of the modulation scheme, larger constellations can provide high transmission rates, but are more susceptible to noise and fading, while the small ones are more robust, but only support low bit transmission rates. Thus, there is always a trade-off between high transmission rates and greater error probabilities.

8.2 Signal Space

Digital modulation codes a finite bit sequence into a signal waveform, selected from a certain set of signals. Assuming equally probable messages, the receiver

* This chapter received contributions from Wamberto José Lira de Queiroz.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_8

minimizes error probability by decoding a received signal that is closer to the transmitted signal, under a given distance metric.

The usual procedure to represent the signal is to project it over a set of orthogonal basis functions, to obtain a vectorial representation of each waveform. This restricts the dimensionality of the signal space and permits the use of regular vectorial distance metrics.

8.2.1 Digital Signal Model

Consider the communication system shown in Fig. 8.1, which is subject to additive white Gaussian noise (AWGN), considered stationary, which has zero mean and a uniform power spectral density (PSD) $S_N(\omega) = N_0/2$.

The channel impulse response is $h(t) = \delta(t)$ and every T seconds the source selects for transmission a message m_i from a set $\mathcal{M} = \{m_1, \cdots, m_M\}$. Each message is chosen with probability p_i, such that $\sum_i p_i = 1$.

Because the message set $\mathcal{M}$ has cardinality M, each transmitted message carries, at most, $R = (\log_2 M)/T$ bits of information per second. Assuming M to be a power of two, the $\log_2 M$ bits corresponding to message m_i are coded as a signal $s_i(t) \in \mathcal{S} = \{s_1(t), \cdots, s_M(t)\}$ with energy

$$E_i = \int_T {s_i}^2(t)\mathrm{d}t, \quad i = 1, \cdots, M. \tag{8.1}$$

The transmitted signal is, therefore, given by

$$\sum_k s_i(t - kT), \tag{8.2}$$

in which $s_i(t)$ is an analog message that represents message m_i in the transmission interval $[kT, (k+1)T]$.

The received signal, corresponding to the original message m_i, in the interval $[kT, (k+1)T]$ is $x(t) = s_i(t - kT) + n(t)$. For each transmitted signal, $s_i(t - kT)$, the receiver must determine the best estimate of $s_i(t) \in \mathcal{S}$, which is equivalent to minimizing the error probability for each symbol



Fig. 8.1 Communication system model

$$P_e = \sum_{i=1}^{M} p_i P(\hat{m} \neq m_i | m_i \text{ sent}), \tag{8.3}$$

in each time interval $[kT, (k+1)T]$.

The geometrical representation of the set of signals is a way to solve the problem of finding the optimum receiver, subject to AWGN, based on a minimum distance criterion.

8.2.2 Representation by Basis Functions

The representation of signals by basis functions uses the Gram-Schmidt orthogonalization procedure (Wozencraft and Jacobs 1965), which states that it is possible to represent a set of M waveforms $\mathcal{S} = (s_1(t), \cdots, s_M(t))$ defined in the interval $[0, T]$ by a linear combination of $N \leq M$ orthonormal basis functions $\{\phi_1(t), \ldots, \phi_N(t)\}$.

Therefore, it is possible to write each waveform $s_i(t) \in \mathcal{S}$, as

$$s_i(t) = \sum_{i=1}^{N} s_{ij}\phi_j(t), \ 0 \leq t < T, \tag{8.4}$$

in which

$$s_{ij} = \int_0^T s_i(t)\phi_j(t)\mathrm{d}t \tag{8.5}$$

is a real coefficient that represents the projection of $s_i(t)$ over the basis function $\phi_j(t)$ and

$$\int_0^T \phi_i(t)\phi_j(t)\mathrm{d}t = \begin{cases} 1, i = j \\ 0, i \neq j. \end{cases} \tag{8.6}$$

If the functions $\{s_i(t)\}$ are linearly independent, then $N = M$; otherwise, $N < M$. For most modulation techniques, the basis set has only two functions ($N = 2$) that correspond to the in-phase and quadrature dimensions of the transmitted signal.

Coefficients $\{s_{ij}\}$ are denoted as $\boldsymbol{s}_i = (s_{i1}, \cdots, s_{iN})$, in the vector space. Given the basis functions, $\{\phi_1(t), \cdots, \phi_N(t)\}$, there is a one to one correspondence between the signal $s_i(t)$ and its vectorial representation $\boldsymbol{s}_i$.

Vector $\boldsymbol{s}_i$ defines a point corresponding to message m_i in the signal constellation. As discussed previously, signal constellations for the usual modulation schemes, such as M-PSK and M-QAM, are bidimensional. The noise signal is also projected

into the signal space, and this simplifies the performance analysis of the system, as well as the optimum receiver design.

Before converting the model described in Fig. 8.1 into a vectorial model, it is convenient to present some definitions to characterize a vector in the signal space $\mathbb{R}^N$.

The length of a vector in signal space is defined as

$$||\boldsymbol{s}_i||^2 = \boldsymbol{s}_i^T \boldsymbol{s}_i = \sum_{k=1}^{N} s_{ik}^2. \tag{8.7}$$

The positive square root of this length, $||\boldsymbol{s}_i||$, is the vector norm.

The distance between two vectors $\boldsymbol{s}_i$ and $\boldsymbol{s}_j$ is defined as

$$||\boldsymbol{s}_i - \boldsymbol{s}_j||^2 = \sum_{k=1}^{N} (s_{ik} - s_{jk})^2 = \int_0^T (s_i(t) - s_j(t))^2 \mathrm{d}t, \tag{8.8}$$

in which the second equality is obtained by writing $s_i(t)$ and $s_j(t)$ as basis functions in Eq. 8.4, and using the orthonormality property of the basis functions.

The inner product, $< s_i(t), s_j(t) >$, between two real signals $s_i(t)$ and $s_j(t)$ in the interval $[0, T]$ is defined as

$$< s_i(t), s_j(t) > = \int_0^T s_i(t) s_j(t) \mathrm{d}t. \tag{8.9}$$

In a similar manner, the inner product $< \boldsymbol{s}_i, \boldsymbol{s}_j >$ between two real vectors,

$$< \boldsymbol{s}_i, \boldsymbol{s}_j > = \boldsymbol{s}_i^T \boldsymbol{s}_j = \int_0^T s_i(t) s_j(t) \mathrm{d}t = < s_i(t), s_j(t) > . \tag{8.10}$$

Two signals are considered orthogonal, if their inner product is zero.

8.2.3 Receiver Design and Sufficient Statistics

The output of a channel is given by

$$x(t) = s_i(t) + n(t), 0 \leq t < T, \tag{8.11}$$

therefore, one can design a receiver to infer which m_i (or $s_i(t)$) has been sent in the time interval $[0, T]$, and a similar procedure is attempted for each time interval $[kT, (k+1)T]$.

Consider the receiver structure shown in Fig. 8.2, in which

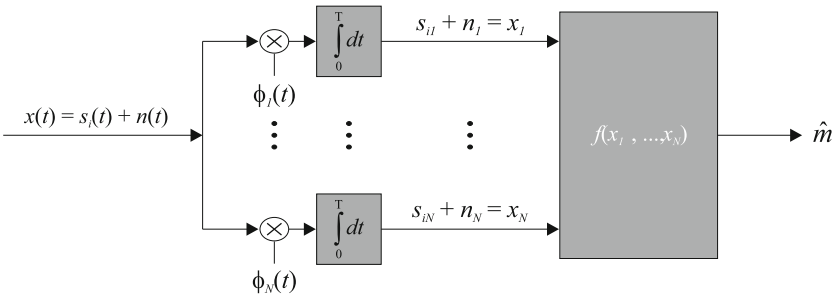


Fig. 8.2 Receiver structure to detect a signal subject to AWGN

$$s_{ij} = \int_0^T s_i(t)\phi_j(t)\mathrm{d}t \tag{8.12}$$

and

$$n_j = \int_0^T n(t)\phi_j(t)\mathrm{d}t. \tag{8.13}$$

It is possible to write $x(t)$ as

$$\sum_{j=1}^{N}(s_{ij} + n_j)\phi_j(t) + n_r(t) = \sum_{j=1}^{N} x_j\phi_j(t) + n_r(t), \tag{8.14}$$

in which $x_j = s_{ij} + n_j$ and

$$n_r(t) = n(t) - \sum_{j=1}^{N} n_j\phi_j(t)$$

denotes the remaining noise.

The receiver obtains an estimate $\hat{m}$ of the transmitted message m_i as a function of $x_1, \cdots, x_N$. Therefore, the receiver ignores any information contained in the remaining noise term $n_r(t)$, which is the noise component orthogonal to the signal space generated by the basis set $(\phi_1(t), \cdots, \phi_N(t))$.

If it is possible to prove that $n_r(t)$ is useless when deciding whether $s_i(t)$ has been sent, then it is only necessary to determine the optimum receiver for $(x_1, \ldots, x_N)$, once it contains all information needed to the optimum detection of $s_i(t)$ given $x(t)$. In other words, $(x_1, \cdots, x_N)$ constitutes a sufficient statistic to detect $s_i(t)$.

Because $n(t)$ is a Gaussian random process, the channel output $x(t) = s_i(t) + n(t)$ is also a Gaussian random process, and the random vector $(x_1, \cdots, x_N)$ is also Gaussian.

Recalling that $x_j = s_{ij} + n_j$, the mean of x_j is given by

$$\mu_{x_j} = E[x_j] = E[s_{ij} + n_j] = s_{ij}, \tag{8.15}$$

and because $n(t)$ has zero mean, the variance is

$$\sigma_{x_j} = E[x_j - \mu_{x_j}]^2 = E[s_{ij} + n_j - s_{ij}]^2 = E[n_j^2]. \tag{8.16}$$

Therefore, one obtains

$$\begin{aligned} \text{Cov}[x_j x_k] &= E[(x_j - \mu_{x_j})(x_k - \mu_{x_k})] = E[n_j n_k] \\ &= E\left[\int_0^T n(t)\phi_j(t)\mathrm{d}t \int_0^T n(\tau)\phi_k(\tau)\mathrm{d}\tau\right] \\ &= \int_0^T \int_0^T E[n(t)n(\tau)]\phi_j(t)\phi_k(\tau)\mathrm{d}t\mathrm{d}\tau \\ &= \int_0^T \int_0^T \frac{N_0}{2}\delta(t-\tau)\phi_j(t)\phi_k(\tau)\mathrm{d}t\mathrm{d}\tau \\ &= \frac{N_0}{2}\int_0^T \phi_j(t)\phi_k(\tau)\mathrm{d}t = \begin{cases} N_0/2, i = j \\ 0, i \neq j \end{cases} \end{aligned} \tag{8.17}$$

in which the last equality is a result of the basis functions orthogonality property. The terms x_i are uncorrelated, and independent, because they are Gaussian (Papoulis 1984). Furthermore, $E[n_j^2] = N_0/2$.

If the random vector corresponding to the correlator output is defined as $\boldsymbol{x} = [x_1, \cdots, x_N]$, given that message m_i is transmitted, then x_i has Gaussian distribution, with average s_{ij} and variance $N_0/2$.

Therefore, by the independence of the terms x_i, it follows that

$$p(\boldsymbol{x}|m_i) = \prod_{j=1}^{N} p(x_j|m_i) = \frac{1}{(\pi N_0)^{N/2}} \exp\left[-\frac{1}{N_0}\sum_{j=1}^{N}(x_j - s_{ij})^2\right]. \tag{8.18}$$

It can be shown that if $E[x_j n_r(t_k)] = 0$ for any $t_k : 0 \leq t_k < T$. Because every random process is completely characterized by its set of time samples, then x_j is independent of any function of the remaining noisy process $n_r(t)$. Besides, as the transmitted signal is independent of the channel noise, s_{ij} is independent of $n_r(t)$.

The main goal of the receiver design is to minimize the message symbol m_i detection error probability given the received signal $x(t)$. In order to minimize the error probability

$$P_e = p(\hat{m} \neq m_i|x(t)) = 1 - p(\hat{m} = m_i|x(t)), \tag{8.19}$$

one needs to maximize $p(\hat{m} = m_i|x(t))$.

Therefore, the receiver output, $\hat{m}$, given the received signal $x(t)$, must correspond to the message m_i that maximizes the conditional probability $p(m_i \text{ sent}|x(t))$.

Recall that $x(t)$ is completely described by $\boldsymbol{x} = (x_1, \cdots, x_N)$ and $n_r(t)$ and that the decision on which message has been transmitted must be a function of the remaining noisy process $f[n_r(t)]$, then

$$\begin{aligned} p(m_i \text{ sent}|x(t)) &= p((s_{i1}, \cdots, s_{iN}) \text{ sent}|(x_1, \cdots, x_N, f[n_r(t)])) \\ &= \frac{p((s_{i1}, \cdots, s_{iN}) \text{ sent}, (x_1, \cdots, x_N), f[n_r(t)]}{p((x_1, \cdots, x_N), f[n_r(t)])} \\ &= \frac{p((s_{i1}, \cdots, s_{iN}) \text{ sent}, (x_1, \cdots, x_N))p(f[n_r(t)])}{p((x_1, \cdots, x_N))p(f[n_r(t)])} \\ &= p((s_{i1}, \cdots, s_{iN}) \text{ sent}|(x_1, \cdots, x_N)), \end{aligned} \tag{8.20}$$

in which the third equality results from the fact that any function of the remaining noisy process is independent of $(x_1, \cdots, x_N)$ and $(s_{i1}, \cdots, s_{iN})$.

It is possible to show that $(x_1, \cdots, x_N)$ is a sufficient statistic for $x(t)$, regarding the detection of m_i, because the error detection probability is minimized.

8.2.4 Maximum Likelihood Decision

The previous section has demonstrated that the detection probability error is minimized if the detector output $\hat{m}$ is selected to maximize the probability of correct decision, $1 - P_e = p(\hat{m} \text{ sent}|\boldsymbol{x})$. Using Bayes rule,

$$p(\hat{m} \text{ sent}|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|\hat{m} \text{ sent})\, p(\hat{m} \text{ sent})}{p(\boldsymbol{x})}. \tag{8.21}$$

Assuming equiprobable messages ($p(m_i) = 1/M$), $\hat{m}$ is chosen as the argument that maximizes $p(\boldsymbol{x}|\hat{m})$,

$$\hat{m} = \arg\max_{m_i} p(\boldsymbol{x}|m_i), i = 1, \cdots, M. \tag{8.22}$$

The maximum likelihood function, associated with the receiver, is defined as

$$L(m_i) = p(\boldsymbol{x}|m_i \text{ sent}). \tag{8.23}$$

For a maximum likelihood receiver, $\hat{m}$ is chosen as the argument that maximizes $L(m_i)$. Because the logarithmic function is an increasing function of its positive argument, the maximization of $L(m_i)$ is equivalent to maximize the logarithm of the likelihood function, defined as

$$l(m_i) = \ln L(m_i) = -\frac{1}{N_0}\sum_{j=1}^{N}(x_j - s_{ij})^2, \tag{8.24}$$

in which the second equality is obtained by the substitution of $L(m_i)$ into Eq. 8.18.

Note that the log-likelihood function depends only on the distance between the received vector $\boldsymbol{x}$ and the set of points of the transmitted signal constellation, $\boldsymbol{s}_i$, $i = 1, \ldots, M$.

Therefore, the maximum likelihood receiver computes $\boldsymbol{x}$ from $x(t)$ using the structure presented in Fig. 8.2 and then decodes

$$\hat{m} = \arg\max_{m_i} l(m_i). \tag{8.25}$$

In a similar way, it is possible to find the signal constellation corresponding to the maximization of the message signal m_i in Eq. 8.25, as

$$\arg\max_{\boldsymbol{s}_i}\left(-\frac{1}{N_0}\sum_{j=1}^{N}(x_j - s_{ij})^2\right) = \arg\max_{\boldsymbol{s}_i}\left(-\frac{1}{N_0}||\boldsymbol{x} - \boldsymbol{s}_i||^2\right). \tag{8.26}$$

The maximum likelihood (ML) receiver decodes the message m_i corresponding to the signal constellation, $\boldsymbol{s}_i$, $i = 1, \ldots, M$, as the one which is nearest to the received vector $\boldsymbol{x}$. It is possible to divide the signal space into M decision regions

$$Z_i = (\boldsymbol{x} : ||\boldsymbol{x} - \boldsymbol{s}_i|| < ||\boldsymbol{x} - \boldsymbol{s}_j||, \forall j = 1, \cdots, M, j \neq i) \;\; i = 1, \cdots, M. \tag{8.27}$$

Then, the ML decoding implies the determination of the decision region Z_i in which the received vector $\boldsymbol{x}$ is located, that is, $\boldsymbol{x} \in Z_i \rightarrow \hat{m} = m_i$. Therefore, the function $f(\boldsymbol{x}) = f(x_1, \cdots, x_N)$ in Fig. 8.2 is given by $\hat{m} = m_i : \boldsymbol{x} \in Z_i$.

It can be shown that Fig. 8.2 is equivalent to the matched filter depicted in Fig. 8.3.

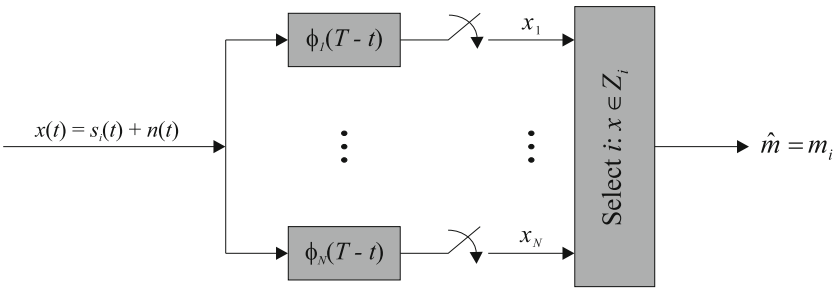


Fig. 8.3 Matched filter structure

8.2.5 Error Probability and the Union Bound

Consider the error probability associated with the ML receiver structure. For equiprobable messages, $p(m_i \text{ sent}) = 1/M$, one has

$$\begin{aligned} P_e &= \sum_{i=1}^{M} p(\boldsymbol{x} \notin Z_i | m_i \text{ sent}) p(m_i \text{ sent}) \\ &= \frac{1}{M} \sum_{i=1}^{M} p(\boldsymbol{x} \notin Z_i | m_i \text{ sent}) \\ &= 1 - \frac{1}{M} \sum_{i=1}^{M} p(\boldsymbol{x} \in Z_i | m_i \text{ sent}) \\ &= 1 - \frac{1}{M} \sum_{i=1}^{M} \int_{Z_i} p(\boldsymbol{x}|m_i) d\boldsymbol{x} \\ &= 1 - \frac{1}{M} \sum_{i=1}^{M} \int_{Z_i} p(\boldsymbol{x} = \boldsymbol{s}_i + \boldsymbol{n}|\boldsymbol{s}_i) d\boldsymbol{n} \\ &= 1 - \frac{1}{M} \sum_{i=1}^{M} \int_{Z_i - \boldsymbol{s}_i} p(\boldsymbol{n}) d\boldsymbol{n}. \end{aligned} \tag{8.28}$$

The integrals in Expression 8.28 are defined in the N-dimensional subset $Z_i \subset \mathbb{R}^N$. The method to compute the error probability is illustrated in Fig. 8.4, in which the constellation stars $s_1, \ldots, s_8$ are equally spaced around a circle, with minimum separation $d_{\min}$.

The probability of correct reception, assuming that the first symbol is sent, $p(\boldsymbol{x} \in Z_1 | m_1 \text{ sent})$, corresponds to the probability $p(\boldsymbol{x} = \boldsymbol{s}_1 + \boldsymbol{n}|\boldsymbol{s}_1)$. This implies that

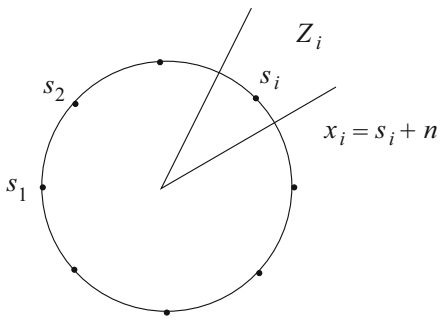


Fig. 8.4 Computation of the error probability

when noise is added to the transmitted signal $\boldsymbol{s}_1$, the resulting vector $\boldsymbol{x} = \boldsymbol{s}_1 + \boldsymbol{n}$ remains in the region Z_1 shown in Fig. 8.4.

The error probability is invariant to any rotation or translation of the signal constellation. If the signal undergoes a phase rotation θ and a translation P, then, $\boldsymbol{s}_i' = \boldsymbol{s}_i e^{j\theta} + P$.

Expression 8.28 provides an exact solution for the error probability, but, in general, it is not possible to determine the probability with a closed known function. Therefore, the union bound is used to find a solution to that case.

Let A_{ik} be the event $||\boldsymbol{x} - \boldsymbol{s}_k|| < ||\boldsymbol{x} - \boldsymbol{s}_i||$, given that the constellation point $\boldsymbol{s}_i$ has been sent. If event A_{ik} occurs, then the signal will be decoded with an error, since the received symbol $\boldsymbol{s}_i$ is not the nearest point to the received vector $\boldsymbol{x}$.

On the other hand, A_{ik} does not necessarily imply that $\boldsymbol{s}_k$ is decoded, instead of $\boldsymbol{s}_i$, because it is possible to exist another constellation point $\boldsymbol{s}_j$ with

$$||\boldsymbol{x} - \boldsymbol{s}_j|| < ||\boldsymbol{x} - \boldsymbol{s}_k|| < ||\boldsymbol{x} - \boldsymbol{s}_i||.$$

The constellation point can be correctly decoded if

$$||\boldsymbol{x} - \boldsymbol{s}_i|| < ||\boldsymbol{x} - \boldsymbol{s}_k|| \; \forall k \neq i.$$

Then,

$$P_e(m_i \text{ sent}) = p\left(\bigcup_{\substack{k=1 \\ k \neq i}}^{M} A_{ik}\right) \leq \sum_{k=1}^{M} p(A_{ik}), \tag{8.29}$$

in which the inequality follows from the union bound.

Consider $p(A_{ik})$ in detail. This gives,

$$\begin{aligned} p(A_{ik}) =& p(||\boldsymbol{s}_k - \boldsymbol{x}|| < ||\boldsymbol{s}_i - \boldsymbol{x}|| \; |\boldsymbol{s}_i \text{ sent}) \\ =& p(||\boldsymbol{s}_k - (\boldsymbol{s}_i + \boldsymbol{n})|| < ||\boldsymbol{s}_i - (\boldsymbol{s}_i + \boldsymbol{n})|| \\ &+ p(||\boldsymbol{n} + \boldsymbol{s}_i - \boldsymbol{s}_k|| < ||\boldsymbol{n}||), \end{aligned} \tag{8.30}$$

that is, the error probability equals the likelihood that the noise $\boldsymbol{n}$ is closer to the vector $\boldsymbol{s}_i - \boldsymbol{s}_k$ than to the origin.

The probability error depends only on the projection of the noise $\boldsymbol{n}$ on the line that connects the origin and the point $\boldsymbol{s}_i - \boldsymbol{s}_k$, as shown in Fig. 8.5. Given the properties of $\boldsymbol{n}$, that projection is a uni-dimensional Gaussian random variable $\boldsymbol{n}$, with zero mean and variance $N_0/2$.

Event A_{ik} occurs if the noise $\boldsymbol{n}$ is closer to $\boldsymbol{s}_i - \boldsymbol{s}_k$ than to zero, that is, if $n > d_{ik}/2$, in which $d_{ik} = ||\boldsymbol{s}_i - \boldsymbol{s}_k||$ is the distance between the constellation points $\boldsymbol{s}_i$ and $\boldsymbol{s}_k$. Therefore,

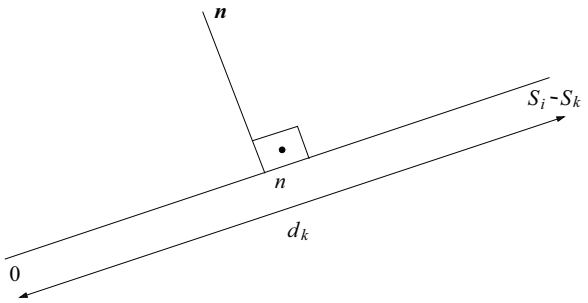


Fig. 8.5 Noise projection on signal coordinate

$$p(A_{ik}) = p(n > d_{ik}/2) = \int_{d_{ik}/2}^{\infty} \frac{1}{\sqrt{\pi N_0}} \exp\left[\frac{-v^2}{N_0}\right] \mathrm{d}v \tag{8.31}$$
$$= Q\left(\frac{d_{ik}}{\sqrt{2N_0}}\right).$$

Substitution into Eq. 8.29 gives

$$P_e(m_i \text{ sent}) \leq \sum_{\substack{k=1 \\ k \neq i}}^{M} Q\left(\frac{d_{ik}}{\sqrt{2N_0}}\right). \tag{8.32}$$

Adding all possible messages,

$$P_e = \sum_{i=1}^{M} p(m_i) P_e(m_i \text{ sent}) \leq \frac{1}{M} \sum_{i=1}^{M} \sum_{\substack{k=1 \\ k \neq i}}^{M} Q\left(\frac{d_{ik}}{\sqrt{2N_0}}\right). \tag{8.33}$$

Defining the minimum constellation distance as $d_{\min} = \min_{i,k} d_{ik}$, it is possible to simplify Expression 8.33 using the union bound

$$P_e \leq (M-1) Q\left(\frac{d_{\min}}{\sqrt{2N_0}}\right). \tag{8.34}$$

The use of a well-known bound

$$Q(x) \leq \frac{1}{2} e^{-x^2/2}$$

gives a closed formula for the error probability

$$P_e \leq \frac{M-1}{\sqrt{\pi}} \exp\left[\frac{-d_{\min}^2}{N_0}\right]. \tag{8.35}$$

For the binary case, $M = 2$, there is only one way to commit an error, and $d_{\min}$ is the distance between two points in the signal constellation; then the upper bound on the Expression 8.34 is met with equality,

$$P_b = Q\left(\frac{d_{\min}}{\sqrt{2N_0}}\right). \tag{8.36}$$

Note that P_e is the error probability of a symbol (message),

$$P_e = p(\hat{m} \neq m_i | m_i \text{ sent}),$$

in which m_i corresponds to a message with $\log_2 M$ bits, and it is possible to approximate the bit error probability as

$$P_b \approx \frac{P_e}{\log_2 M}, \tag{8.37}$$

that was obtained using the Gray code (Lee 1986).

8.3 Digital Modulation Schemes

The digital modulation schemes use the information bits to select the appropriate amplitude, phase, or frequency or a combination of them, to modulate the carrier. There are three principal modulation systems:

1 Amplitude shift keying (ASK)—The information signal modulates the carrier amplitude.
2 Phase shift keying (PSK)—The information signal bits modulate the carrier phase.
3 Quadrature amplitude modulation (QAM)—The information signal modulates both the carrier amplitude and phase.

Using the notation $s(t) = \Re\{u(t)e^{j2\pi f_c t}\}$ for the transmitted signal, then the modulating signal is given by

$$u(t) = \sum_n s_n g(t - nT_s), \qquad s_n = a_n + jb_n, \tag{8.38}$$

in which T_s denotes the symbol interval, $T_s \gg 1/f_c$, a_n and b_n are the real and imaginary parts of the symbol, n is the interval signaling index, $g(t)$ is the pulse format, and s_n is a complex number that represents $K = \log_2 M$ information bits that is invariant during the symbol interval.

The bit rate corresponds to K bits per symbol, or $R = K/T_s$ bits second. Because $s_n = a_n + jb_n$ is a complex number, both the signal envelope and phase are affected.

The information mapping on the complex number and the choice of the pulse format specify the digital modulation technique.

Therefore, the transmitted signal is

$$s(t) = \left[\sum_n a_n g(t - nT_s)\right] \cos(\omega_c t) - \left[\sum_n b_n g(t - nT_s)\right] \sin(\omega_c t), \tag{8.39}$$

in which the first term is the in-phase component, $s_I(t)$, and the second term represents the quadrature component, $s_Q(t)$, and $\omega_c = 2\pi f_c$.

The spectral properties of $s(t)$ and $u(t)$ are defined by the spectral characteristics of $g(t)$. The modulated signal bandwidth is, at least, twice the pulse bandwidth, and the design criteria, to minimize the main lobe bandwidth and the amplitude of the other lobes, to reduce co-channel interference, depend on the pulse format.

The mentioned modulation schemes can be represented by the general scheme of Fig. 8.6.

8.3.1 Amplitude Shift Keying

The simplest modulation system is the M-ASK that has no quadrature component ($b_n = 0$). The information is carried by the amplitude variation, A_m, of the transmitted signal

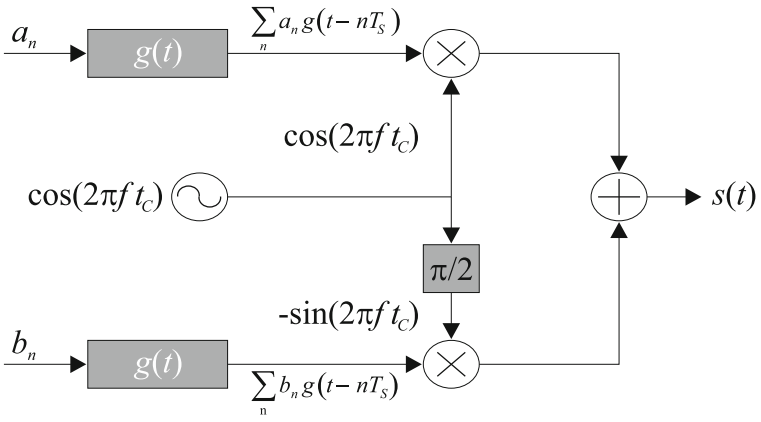


Fig. 8.6 General structure for a digital modulator

$$s_m(t) = \Re\left\{A_m g(t) e^{j\omega_c t}\right\} = A_m g(t)\cos(\omega_c t), \qquad 1 \le t \le T_s, \tag{8.40}$$

in which $A_m = (2m - 1 - M)d$, $m = 1, 2, \ldots, M$, and $\omega_c = 2\pi f_c$.

The transmitted signal amplitude is obtained from a set of $M = 2^K$ distinct values, which correspond to $\log_2 M = K$ bits per signaling interval T_s.

The transmitted signal energy, in a signaling interval $[0, T_s)$, is

$$\begin{aligned} E_{s_m} &= \int_0^{T_s} s_m^2(t)\mathrm{d}t = \int_0^{T_s} A_m^2 g^2(t) \cdot \frac{1}{2}[1 + \cos(2\omega_c t)]\mathrm{d}t \\ &\approx \frac{1}{2}A_m^2 \int_0^{T_s} g^2(t)\mathrm{d}t = \frac{1}{2}A_m^2 E_g, \end{aligned} \tag{8.41}$$

in which E_g is the pulse energy.

This is a good approximation, when $f_c T_s \gg 1$, because $g(t)$ is fairly constant in a cycle $T_c = 1/f_c$. The Euclidean distance between two symbols, which represent different information sequences, is given by

$$\begin{aligned} d_{mn} &= ||s_m(t) - s_n(t)|| = \sqrt{\int_0^{T_s} |s_m(t) - s_n(t)|^2 \mathrm{d}t} \\ &\approx \sqrt{0,5E_g}|A_m - A_n| \ge d\sqrt{2E_g} = d_{\min}, \end{aligned} \tag{8.42}$$

in which $d_{\min}$ is the smallest distance between any two constellation symbols.

It is usual to resort to the Gray code to associate the bit sequences with certain constellation symbols. For this mapping, the adjacent symbols, associated with certain bit sequences, differ in only one bit, as shown in Fig. 8.7. Therefore, an error that produces a change from a symbol to an adjacent one only changes a single bit in the sequence.

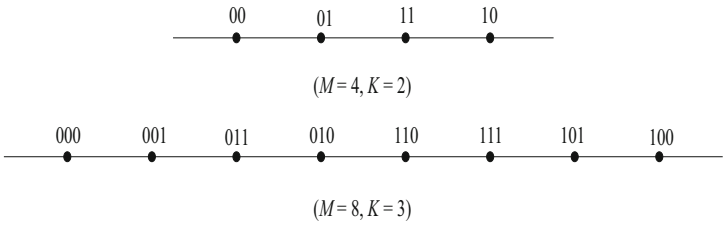


Fig. 8.7 Gray mapping for the M-ASK modulation scheme

8.3.2 Phase Shift Keying

For the M-PSK scheme, the information is coded in the transmitted signal phase, that is,

$$\begin{aligned} s_m(t) &= \Re\{g(t)e^{j2\pi(m-1)/M}e^{j\omega_c t}\}, \qquad 0 \le t \le T_s, \\ &= g(t)\cos\left[\omega_c t + \frac{2\pi}{M}(m-1)\right] \\ &= g(t)\cos\left[\frac{2\pi}{M}(m-1)\right]\cos(\omega_c t) \\ &\quad - g(t)\sin\left[\frac{2\pi}{M}(m-1)\right]\sin(\omega_c t), \end{aligned} \tag{8.43}$$

in which $\omega_c = 2\pi f_c$.

Thus, for $s_n = a_n + jb_n$, the coefficients a_n and b_n are given as $\cos\left[\frac{2\pi}{M}(m-1)\right]$ and $\sin\left[\frac{2\pi}{M}(m-1)\right]$, respectively. As in the M-ASK case, $g(t)$ is the pulse format, and $\theta_m = \frac{2\pi}{M}(m-1)$, $m = 1, 2, \ldots, M$ are the possible carrier phases that contain the information bits.

All the signals have equal energy:

$$E_{s_m} = \int_0^{T_s} s_m^2(t)\mathrm{d}t \approx \frac{1}{2}\int_0^{T_s} g^2(t)\mathrm{d}t = \frac{1}{2}E_g. \tag{8.44}$$

Observe that for a rectangular pulse, $g(t) = 1$ in the pulse interval, the transmitted signal has constant envelope, differing from the M-ASK and M-QAM schemes. However, rectangular pulses are spectrally inefficient, because they generate many sidelobes.

The distance d_{mn} between two M-PSK constellation symbols, $s_m - s_n$, is given by

$$d_{mn} = \sqrt{E_g\left(1 - \cos\left(\frac{2\pi}{M}(m-1)\right)\right)} \ge \sqrt{E_g(1-\cos(2\pi/M))} = d_{\min}. \tag{8.45}$$

As in the case of M-ASK, it is common to use the Gray code to map the signal, as illustrated in Fig. 8.8.

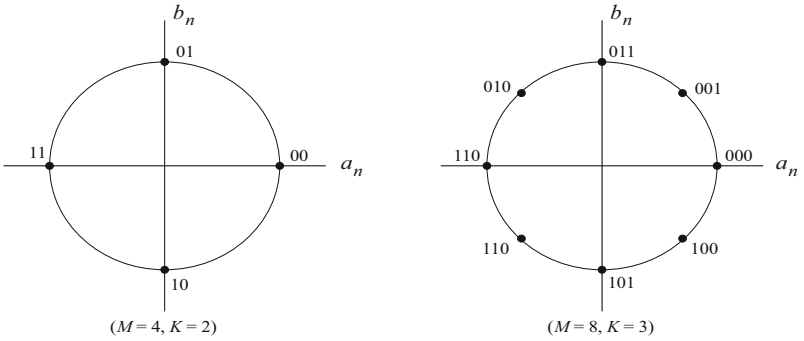


Fig. 8.8 Gray code mapping for the QPSK and 8-PSK modulation schemes

8.3.3 Quadrature Modulation

For the M-QAM scheme, the information bits modulate both the carrier phase and amplitude. There are, then, two degrees of freedom to code the information bits. As a result, the M-QAM scheme has better spectral efficiency, that is, for a given bandwidth and a certain average power, the M-QAM modulation scheme manages to code a larger number of bits per transmitted symbol.

The transmitted signal is

$$s_m(t) = \Re\{A_m e^{j\theta_m} g(t) e^{j\omega_c t}\} = A_m g(t)\cos(\omega_c t + \theta_m), \qquad 0 \le t \le T_s, \tag{8.46}$$

in which $\omega_c = 2\pi f_c$.

The energy of the signal $s_m(t)$ is

$$E_{s_m} = \int_0^{T_s} s_m^2(t)\mathrm{d}t \approx \frac{1}{2} A_m^2 E_g, \tag{8.47}$$

which is the same result for the M-ASK scheme.

The Euclidean distance d_{mn} between two constellation symbols, s_m and s_n, is given by

$$d_{mn} = \sqrt{\frac{1}{2} E_g[(a_m - a_n)^2 + (b_m - b_n)^2]}, \tag{8.48}$$

in which $a_k = A_k \cos\theta_k$ and $b_k = A_k \mathrm{sen}\,\theta_k$ for $k = m, n$.

Considering square M-QAM in which a_n and b_n assume values $(2m-1-L)d$ for $m = 1, 2, \ldots, L = 2^l$, the shortest distance between the symbols is $d_{\min} = d\sqrt{2E_g}$. This is equivalent to the distance obtained for the M-ASK constellation.

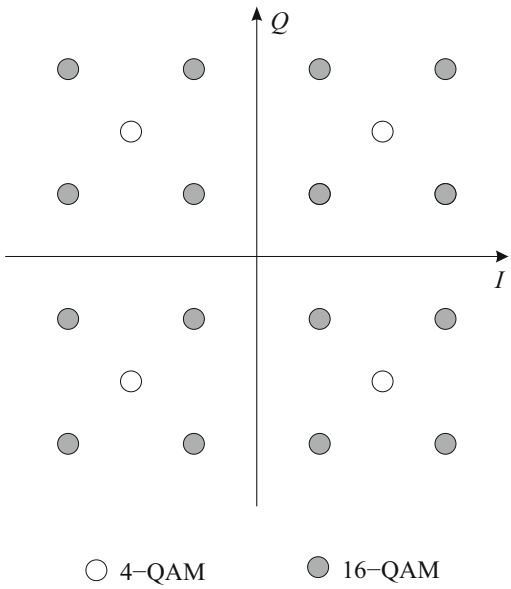


Fig. 8.9 Constellations 4-QAM and 16-QAM

On the other hand, the number of symbols in a square QAM constellation is $M = d^{2l}$. The transmission rate is $2l$ bits/symbol, that is, l bits per dimension. The usual square constellations are 4-QAM and 16-QAM, illustrated in Fig. 8.9.

8.4 Differential Coding

In the M-PSK and M-QAM schemes, the information modulates the carrier phase. Therefore, they require coherent demodulation to recover the original signal. This implies that the carrier phase must be known at the receiver.

The carrier phase can be recovered using a pilot signal, a low-amplitude subcarrier that is transmitted along with the signal or using a phase-locked loop (PLL) (Koufalas 1996). However, the additional complexity raises receiver costs and makes the detection more susceptible to carrier phase drift, for example. Besides, it is difficult to obtain a phase reference when dealing with fast fading channels.

Therefore, differential modulation schemes that do not require phase references are sometimes preferred in wireless applications.

The differential modulation techniques belong to a general modulation class known as modulation with memory, in which a symbol transmitted in a discrete time instant n depends either on current and on past bits,

Differential modulation can be used to adjust the transmitted signal spectrum format according to the channel spectral characteristics, as in the case of non-return to zero (NRZ) and non-return to zero inverted (NRZI).

As mentioned earlier, the main advantage of differential modulation is to avoid the need for a phase reference to demodulate a signal. The rationale is to use the previously sent symbol as the phase reference for the current symbol. That is, the information bits code the phase difference between the current symbol and the previous symbol.

Differential coding is less sensitive to carrier random phase shifts. However, if the channel is subject to Doppler shift, for instance, when the transmitter, or receiver, is moving, the phases in distinct time intervals can be uncorrelated, which renders the previous symbol a noisy phase reference. This implies that the error probability will have a floor, an irreducible level.

8.5 Offset Phase Modulation

The baseband signal, $s_n = a_n + jb_n$, that phase modulates a carrier assumes a certain initial phase in the four quadrants of the complex plane. At time nT_s, the transition to a new symbol can imply a 180° phase change, and this can make the signal amplitude to pass through zero.

Sudden phase changes, combined with large amplitude variations, produce sidelobes and the signal can be distorted by amplifiers or filters. The abrupt transitions can be avoided with the introduction of an offset in the quadrature pulse, which is accomplished by the introduction of a delay of a half symbol period, as shown in Fig. 8.10.

As can be observed, the offset quadrature phase shift modulation (OQPSK) has the same spectral characteristics of the QPSK system, for linear amplification, but has a better efficiency for nonlinear operation, because the maximum phase transition is 90°.

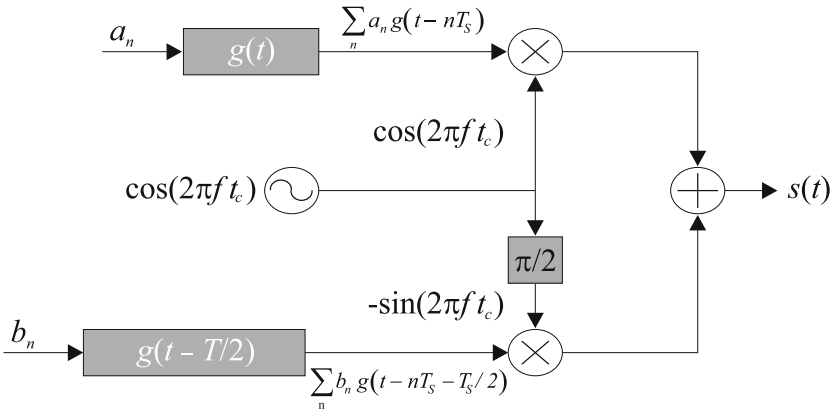


Fig. 8.10 Offset phase modulator

Another technique to minimize the phase transitions, used in some mobile cellular and digital television standards, is the $\pi/4$-QPSK, that rotates the QPSK by 45° and avoids switching the oscillators on and off. It can also be deferentially coded, eliminating the need for a phase reference, producing the $\pi/4$-DQPSK.

8.6 The Transmission Pulse

For the M-PSK modulation scheme, if $g(t)$ is a rectangular pulse of width T, the signal envelope is constant, which gives an advantage regarding nonlinear amplification, such as obtained from a satellite transponder amplifier. On the other hand, rectangular pulses produces high-energy sidelobes, which requires a large bandwidth, to minimize co-channel interference.

Of course, the pulse format can be modified, to reduce the out-of-band energy, but this must be done without introducing inter-symbol interference (ISI). Therefore, the pulse format $g(t)$ must satisfy the Nyquist criterion, which states that the ISI must be null at the sampling instants (Haykin 1989).

The following pulse formats satisfy the Nyquist criterion:

1. Rectangular pulse—That is, $g(t) = 1$, $0 \leq t \leq T_s$. This pulse format produces a constant envelope, for M-PSK schemes, but has poor spectral properties, with high-energy sidelobes.
2. Sine or cosine pulse—That is, $g(t) = \text{sin}(\pi t/T_s)$, $0 \leq t \leq T_s$. This pulse format is often used in the minimum shift keying (MSK) modulation scheme. The quadrature symbol is delayed by $T_s/2$, and this produces a constant amplitude modulation and also low-energy sidelobes. The Gaussian minimum shift keying (GMSK) modulation scheme, used in the Global System for Mobile Communications (GSM) standard, included this pulse format.
3. Raised cosine pulse—This format has been designed in the frequency domain, to obtain some desired spectral properties. Thus, the pulse $g(t)$ is specified as a function of its Fourier transform:

$$G(f) = \begin{cases} T_s, \ 0 \leq |f| \leq (1-\beta)/2T_s, \\ \frac{T_s}{2}\left[1 - \text{sin}\left(\frac{\pi T_s}{\beta}\left(f - \frac{1}{2T_s}\right)\right)\right], \\ (1-\beta)/2T_s < |f| \leq (1+\beta)/2T_s, \end{cases} \tag{8.49}$$

in which β is defined as the roll-off factor that determines the spreading percentage of the pulse.

The time domain representation of pulse $g(t)$, corresponding to $G(f)$, is given by

$$g(t) = \frac{\text{sen}(\pi t/T_s)}{\pi t/T_s} \frac{\cos(\beta \pi t/T_s)}{1 - 4\beta^2 t^2/T_s^2}. \tag{8.50}$$

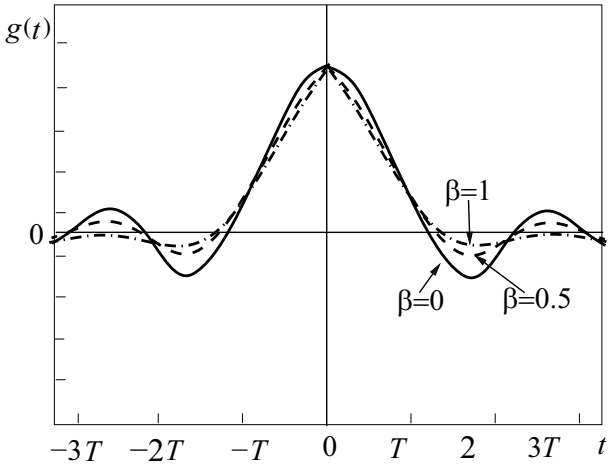


Fig. 8.11 Raised cosine pulse, in the time domain

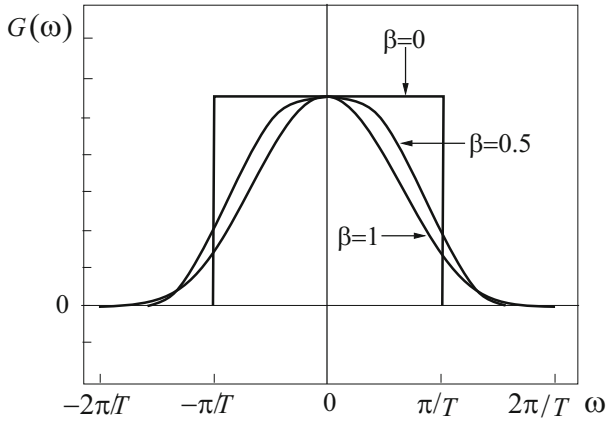


Fig. 8.12 Raised cosine pulse, in the frequency domain

Figures 8.11 and 8.12 present the raised cosine pulse in the time and frequency domains, respectively, for some values of β. In the limit, for $\beta = 0$, the pulse is rectangular. The pulse amplitude decay, in the time domain, is proportional to $1/t^3$, which is faster than the previous ones.

8.7 Constant Envelope Modulation

Constant envelope modulation permits the use of nonlinear amplification, which is common in satellite transmission, for example. That type of modulation is more efficient in terms of energy use. It is also more robust in the presence of channel fading and noise. However, this type of modulation has poor spectral efficiency and occupies a larger bandwidth, as compared to the ASK and QAM schemes. The FSK modulation scheme has a constant envelope and has the following available versions: M-FSK, MSK, and GMSK.

For the M-FSK scheme, the modulated signal has the format given by

$$s_m(t) = A\cos(\omega_c t + \alpha_m \Delta\omega_c t), \qquad 0 \leq t \leq T_s, \tag{8.51}$$

in which $\omega_c = 2\pi f_c$, $\alpha_m = (2m - 1 - M)$, $m = 1, 2, \ldots, M = 2^K$. This way, the shortest frequency distance between different symbols is $2\Delta f_c$.

The usual way to generate the M-FSK signal is to change the carrier frequency directly, as done in the analog FM case,

$$s_m(t) = A\cos\left[\omega_c t + 2\pi\beta\int_{-\infty}^{t} v(\tau)\mathrm{d}\tau\right] = A\cos\left[\omega_c t + \theta(t)\right], \tag{8.52}$$

in which

$$v(t) = \sum_n a_n g(t - nT_S)$$

is an M-ASK signal modulated by the information bits, as previously described, and

$$\theta(t) = 2\pi\beta\int_{-\infty}^{t} v(\tau)\mathrm{d}\tau.$$

The phase $\theta(t)$ is continuous, and the scheme is called continuous phase frequency shift keying (CPFSK).

Using Carson's rule, for small values of β, gives the following bandwidth: for $s(t)$

$$B_s \approx M\Delta f_c + 2B, \tag{8.53}$$

in which B is the M-ASK signal bandwidth, considering the modulating signal $v(t)$.

As discussed earlier, the MSK modulation is a special case of the FSK modulation scheme, in which the frequency separation is $\Delta f_c = 0,5/T_s$. This value of Δf_c is the minimum frequency separation, for which the inner product $< s_m(t), s_n(t) >$ is null in a symbol interval, for $m \neq n$. Therefore, the MSK signal occupies a smaller bandwidth in comparison with FSK.

Because FSK and MSK are nonlinear modulating techniques, their bandwidth is larger than that of the linear modulation schemes, for the same spectral efficiency. The bandwidth can be reduced by an adequate selection of the pulse format to convey the PAM signal. The Gaussian pulse shape is commonly used to improve the spectral efficiently, as is the case for the MSK modulation scheme.

The Gaussian pulse format is used to improve the MSK spectral efficiency because it has the minimum bandwidth by pulse width product:

$$g(t) = \frac{\sqrt{\pi}}{\alpha} e^{-\pi^2 t^2/\alpha^2}, \tag{8.54}$$

in which α is a parameter related to the spectral efficiency of the system.

The Gaussian pulse spectrum has a Gaussian format too, given by

$$G(f) = e^{-\alpha^2 f^2}, \tag{8.55}$$

in which the parameter α is related to the bandwidth

$$B = \frac{\sqrt{-\ln \sqrt{1/2}}}{\alpha}. \tag{8.56}$$

Therefore, an increase in the value of the parameter α has the potential to improve the spectral efficiency of the GMSK scheme. On the other hand, the Gaussian pulses do not satisfy the Nyquist criterion, and thus, there is an increasing intersymbol interference when α becomes larger.

The increase in the value of α to improve the spectral efficiently introduces an error floor that is caused by the high level of intersymbol interference. Therefore, Gaussian pulses are more common for voice transmission, which tolerates higher bit error rates.

Chapter 9
Information Theory



Mathematics is the science of what is clear by itself.
– Carl Jacobi

9.1 Uncertainty Measure

Information theory is the name of the scientific discipline created by Claude Shannon in 1948 in one of the most important articles in the history of engineering (Shannon 1948a). Shannon's pioneering work, entitled *A Mathematical Theory of Communication*, was published in the *Bell System Technical Journal*, vol. 27, July 1948, pp. 379–423 and October 1948, pp. 623–656, and established the scientific foundations of this new discipline. Information theory, in addition to dealing quantitatively with the concept of information, also deals with establishing performance limits for transmission or storage of information, among other issues. The word "information" is often used in situations where the correct term would be "uncertainty." Strictly speaking, as we will see later, *information* means the non-negative difference between two levels of uncertainty. In the case where one of the levels of uncertainty is equal to zero, it follows that the measure of information and the measure of uncertainty are equal, as is the case in the following sections.

The main objectives of information theory are the establishment of theoretical performance limits for information transmission systems, information storage systems, and cryptographic systems. Currently, we find applications of information theory in well-established areas, such as mathematics, in theorem proving.

Hartley's Measure of Information

The only work prior to Shannon's of which we are aware was developed by R. V. L. Hartley and is entitled *Transmission of Information*, Bell System Technical Journal, vol. 3, July 1928, pp. 535–564. Hartley clearly recognized the essential importance of some aspects of information. Perhaps the most important was his

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_9

recognition that the reception of a given symbol only provides information if it is part of a set of symbols with at least two symbols. Going a little further, in today's terms the occurrence of a symbol only provides information if it represents the value of a random variable. The idea proposed by Hartley represented a radical change for the time and took time to be assimilated by telecommunications engineers. In other words, the design of communication systems had to take into account the transmission of random quantities, rather than just reproducing static sinusoids. Hartley then set out to propose a measure of information based on the following reasoning.

Consider the occurrence of a symbol belonging to a set with L symbols. The information provided by the occurrence of n such symbols should be equal to n times the value of the information provided by the occurrence of a single symbol from this set, even though there are L^n possible different ways for n symbols to occur. This suggests that $\log_b(L^n) = n \log_b L$ is the appropriate measure of information in which the base b selected for the logarithm fixes the size of the unit of information, in Hartley's own words. We can therefore express Hartley's measure of the amount of information provided by the observation of a discrete random variable X as

$$I(X) = \log_b L \tag{9.1}$$

in which L denotes the number of possible values of X.

As noticed by Massey, when $b = 2$ in Eq. 9.1 Hartley's unit of information is called the *bit*, although the word *bit* was not used before Shannon's 1948 paper (Massey 1990). Hartley's information measure makes it possible to correctly interpret various technical problems. For example, 16 binary digits are required to access a given memory location in a semiconductor memory with 65,536 locations. In this manner the memory address provides 16 binary digits (bits) of information.

Why is Hartley's measure of information incomplete? The main issue with this measure of information is the fact that it ignores the probability of occurrence of the various values of X. In retrospect, Hartley's measure of information is correct only when X is uniformly distributed and fails when such values are not uniformly distributed. Hartley's pioneering work apparently had a small impact and his name is more frequently remembered in association with an oscillator rather than with his measure of information.

Shannon's Measure of Information

In 1948, 20 years after the publication of Hartley's article, Shannon published an article proposing a new measure of information, which caused a big turmoil of activities making use of Shannon's concepts, which lasts to the present day. Clearly Shannon perceived something that was not noticed by Hartley and that was essential to allow a general application of that theory.

As explained by Massey, a small modification to Hartley's original definition of information measure leads to the same measure as proposed by Shannon (Massey 1982). In general, if the ith value of X has probability $P_X(x_i)$, then Hartley's information measure $\log(1/P_X(x_i)) = -\log P_X(x_i)$ for this value should be weighted by $P_X(x_i)$ giving

$$-\sum_{i=1}^{L} P_X(x_i) \log P_X(x_i) = E[-\log P_X(X)] \tag{9.2}$$

as the amount of information provided by X. Expression 9.2 is precisely Shannon's measure of information, which can be interpreted as *average Hartley information*. Shannon referred to his measure of information as *entropy*.

An apparent difficulty appears in Eq. 9.2 when $p_i = 0$ for one or more choices of i. However, from a practical point of view, it can be argued that those corresponding values of X will never occur and that they would not contribute to the information provided by X. Therefore, it would be reasonable not to include such terms in Eq. 9.2. On the other hand, from a mathematical point of view, the following result can be used:

$$\lim_{p \to 0+} p \log p = 0$$

as a justification not to include terms with $p_i = 0$ in Eq. 9.2.

At this point it is convenient to introduce the terminology and notation that will be used in the sequel, aiming at a simplification in the presentation of some concepts. If f is any real valued function, then the *support* of f is defined as a subset of its domain in which f takes on non-zero values and is denoted as $\text{supp}(f)$. If P_X denotes the probability distribution of a random variable X, then $\text{supp}(P_X)$ represents the subset of values of X with non-zero probability of occurrence.

Fundamental Inequality

The inequality in Eq. 9.3 plays a key role in information theory, very often appearing in the proof of properties, theorems, etc., and for this reason is coined *Fundamental Inequality of Information Theory (IT Inequality)*

$$\ln x \leq x - 1, \tag{9.3}$$

where x denotes a positive real number and equality is verified if and only if $x = 1$. Multiplying by -1 both sides of the inequality in Eq. 9.3, the following equivalent form is obtained for the fundamental inequality of information theory:

$$\ln(1/x) \geq 1 - x, \tag{9.4}$$

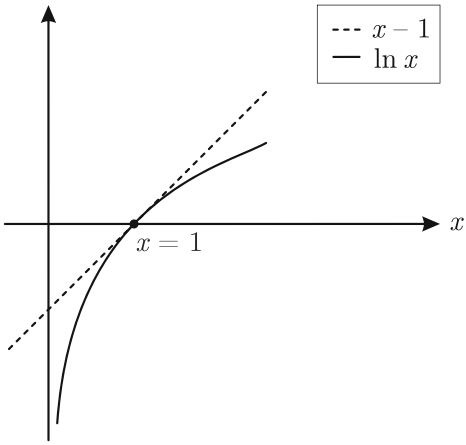


Fig. 9.1 Auxiliary graph to illustrate the proof of the IT inequality

where equality occurs if and only if $x = 1$.

Theorem 9.1 (IT Inequality) *For a given positive real number x,*

$$\log x \leq (x-1)\log e \tag{9.5}$$

with equality if and only if $x = 1$.

Proof The graphs for $\ln x$ and $x - 1$ coincide in $x = 1$ as illustrated in Fig. 9.1. However

$$\frac{d}{dx}(\ln x) = \frac{1}{x}\begin{cases} > 1 \text{ for } x < 1 \\ < 1 \text{ for } x > 1 \end{cases}$$

and therefore these graphs never cross each other. Thus, $\ln x \leq x - 1$ with equality if and only if $x = 1$. Multiplying by $\log e$ both sides of this inequality and recalling that $\log x = (\ln x)(\log e)$ then Eq. 9.5 results. □

9.2 Entropy

The concept of entropy is central to information theory and, according to Shannon, is associated with the idea of uncertainty. The approach followed by Shannon to measure entropy, or uncertainty, employs probability theory. Let X denote a random variable taking L possible values $x_1, x_2, \ldots, x_L$, which occur with probabilities $P_X(x_1), P_X(x_2), \ldots, P_X(x_L)$, respectively.

Definition 9.2 The entropy, or uncertainty, $H(X)$ of a random variable X is the quantity

$$H(X) = -\sum_{x \in \text{supp}(P_X)} P_X(x) \log_b P_X(x). \tag{9.6}$$

Due to its practical importance, this chapter will focus on digital information theory, i.e., will be concerned with discrete structures, for example, sources with a finite of countably infinite number of symbols, code dictionaries with a finite number of codewords, and so on.

Shannon associated the concept of entropy with *uncertainty*. Some people mistakenly refer to $H(X)$ as a measure of information. As we will see later on, information according to Shannon is always associated with a reduction in uncertainty, i.e., *information* is that which is received when *uncertainty* is reduced. Shannon used the word *entropy* because in statistical thermodynamics, this is the name given to the expression of $H(X)$ in the formula Eq. 9.6. Although the symbol H was also borrowed from thermodynamics, according to Massey (1990), there would be no harm in thinking of it as a late homage to Hartley.

Before presenting an example of calculating $H(X)$, it is necessary to define the base of the logarithms used. Defining the base b of logarithms implicitly defines a unit for measuring entropy. When $b = 2$ the unit of entropy is *bits*/symbol, the word *bit* having been suggested to Shannon by John Tukey to signify a contraction of *binary digit*. John Tukey is much more remembered, however, for his work with fast Fourier transforms. When $b = e$ (the basis of natural logarithms), the unit is *nepers*/symbol. There is also the base $b = 10$ the unit of which is *hartleys*/symbol, but almost never used. This text adopts the convention of not indicating the base of logarithms when base 2 is used.

Example 9.3 Suppose the random variable X takes the values $0, 1, 2$ and 3 with probabilities $1/2$, $1/4$, $1/8$ and $1/8$, respectively. The uncertainty $H(X)$ is calculated as

$$\begin{aligned} H(X) &= (1/2)\log 2 + (1/4)\log 4 + (1/8)\log 8 + (1/8)\log 8 \\ &= 1.75 \text{ bits/symbol.} \end{aligned}$$

Example 9.4 r-ary Entropy: If, instead of base 2 logarithms, base r logarithms are employed, the *r-ary entropy* is expressed as follows:

$$H_r(X) = -\sum_{x \in \text{supp}(P_X)} P_X(x) \log_r P_X(x) \ \ r\text{-ary units}$$

and is related to the entropy in base 2 as follows:

$$H_r(X) = \frac{H(X)}{\log r}.$$

Example 9.5 Suppose the random variable X takes only two values, x_1 and x_2, and that $P_X(x_1) = p$. Then, the uncertainty of X in bits, considering $0 < p < 1$, is given by

$$H(X) = -p \log p - (1 - p) \log(1 - p).$$

Due to its frequent occurrence in information theory, the expression $-p \log p - (1-p) \log(1-p)$ is given the name *binary entropy function* and is denoted as $h(p)$. A graph of $h(p)$ is shown in Fig. 9.2, and Table 9.1 contains a few useful values of $h(p)$. It is also specified that $h(0) = h(1) = 0$.

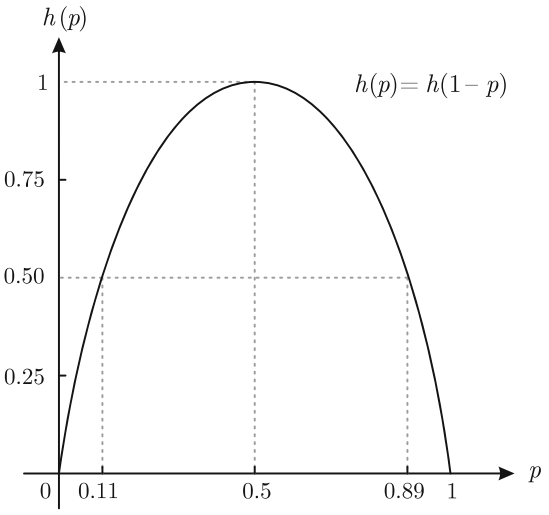


Fig. 9.2 The binary entropy function

Table 9.1 Some values of the binary entropy function $h(p)$

p	$-p \log_2 p$	$-(1-p) \log_2(1-p)$	$h(p)$ $(bits)$
0	0	0	0
0.05	0.216	0.070	0.286
0.10	0.332	0.137	0.469
0.11	0.350	0.150	0.500
0.15	0.411	0.199	0.610
0.20	0.464	0.258	0.722
0.25	0.500	0.311	0.811
0.30	0.521	0.360	0.881
1/3	0.528	0.390	0.918
0.35	0.530	0.404	0.934
0.40	0.529	0.442	0.971
0.45	0.518	0.474	0.993
0.50	1/2	1/2	1

Therefore we can write

$$h(p) = -p \log p - (1-p)\log(1-p),\ 0 \leq p \leq 1,$$

as long as the convention that

$$p \log p = 0 \text{ for } p = 0, \tag{9.7}$$

is adopted and that will be assumed in this chapter. In this manner, following the convention expressed in Eq. 9.7, we can write Eq. 9.6 as follows:

$$H(X) = -\sum_{x_i} P_X(x_i) \log P_X(x_i),$$

or, equivalently, in terms its mathematical expectation as

$$H(X) = E[-\log P_X(x_i)]. \tag{9.8}$$

It is worth observing that mathematically there is no distinction between *discrete random variables* and *discrete random vectors*. For example, it is possible to have $X = [X_1, X_2, \ldots, X_n]$, and thus the definition of entropy Eq. 9.8 also defines the uncertainty (or entropy) of random vectors. The uncertainty of $X = [X_1, X_2, \ldots, X_n]$ is denoted by $H(X_1 X_2 \ldots X_n)$, resulting in

$$H(X_1 X_2 \ldots X_n) = \\ -\sum_{x_1 x_2 \ldots x_n} P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n) \log P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n). \tag{9.9}$$

In order to simplify notation, in situations where there is no doubt about the origin of the variables in a given expression involving probabilities, it is used $P(x_1, x_2, \ldots, x_n)$ to mean $P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n)$.

Properties of the Entropy Function

Next, the first significative result about the measure of uncertainty is proved, which has the intuitive properties that the uncertainty about X attains a maximum when the values of X are equiprobable and attains a minimum when one of the values of X has probability 1. This result also shows that Shannon's measure of information coincides with the corresponding Hartley's information measure only when the values of the random variable considered are equiprobable.

Theorem 9.6 *If the discrete random variable X has L possible values, then the entropy $H(X)$ will satisfy the following inequalities:*

$$0 \leq H(X) \leq \log L, \tag{9.10}$$

with equality on the left if and only if $P(x_i) = 1$ *for some* x_i *or with equality on the right if and only if* $P(x_i) = 1/L$ *for all* x_i.

Proof

1. The inequality on the left in Eq. 9.10 is proved by using the definition of entropy Eq. 9.6 and expression 9.4, as follows:

$$\begin{aligned} H(X) &= -\sum_{x \in \text{supp}(P_X)} P(x_i) \log P(x_i) \\ &= (\log e) \sum_{x \in \text{supp}(P_X)} P(x_i) \ln(1/P(x_i)) \qquad (9.11) \\ &\geq (\log e) \sum_{i=1}^{L} P(x_i)(1 - P(x_i)) \qquad (9.12) \\ &= (\log e) \sum_{i=1}^{L} (P(x_i) - P^2(x_i)) \geq 0. \qquad (9.13) \end{aligned}$$

   Expression 9.12 results by applying Eq. 9.4 to each term in Eq. 9.11 and including those terms for which $P(x_i) = 0$. There will be equality simultaneously in Eqs. 9.12 and 9.13 if and only if $P(x_i) = 1$ for some value of i.
2. The expression $\log L - H(X)$ is considered next to prove the inequality on the right in Eq. 9.10, followed by the use of the inequality in Eq. 9.4, i.e.,

$$\begin{aligned} \log L - H(X) &= \log L + \sum_{x \in \text{supp}(P_X)} P(x_i) \log P(x_i) \\ &= \sum_{x \in \text{supp}(P_X)} P(x_i) \log L + \sum_{x \in \text{supp}(P_X)} P(x_i) \log P(x_i) \\ &= \sum_{x \in \text{supp}(P_X)} P(x_i) \log L P(x_i) \qquad (9.14) \\ &= (\log e) \sum_{x \in \text{supp}(P_X)} P(x_i) \ln L P(x_i) \\ &\geq (\log e) \sum_{x \in \text{supp}(P_X)} P(x_i) \left(1 - \frac{1}{L P(x_i)}\right) \qquad (9.15) \\ &= (\log e) \sum_{x \in \text{supp}(P_X)} \left(P(x_i) - \frac{1}{L}\right) \geq 0. \qquad (9.16) \end{aligned}$$

The inequality in Eq. 9.15 results by applying Eq. 9.4 to each term in Eq. 9.14. Equality occurs in Eq. 9.15 if and only if $P(x_i) = 1/L$ for all x_i. The inequality in Eq. 9.16 follows from the observation that the summation in x_i contains at most L terms.

□

9.3 Conditional Entropy

Very often there is interest in the behavior of a random variable when another random variable is specified. The following definition generalizes in a natural manner the concept of uncertainty in this situation.

Definition 9.7 The conditional uncertainty, or conditional entropy, of the discrete random variable X, given that the event $Y = y$ occurred is given by the expression

$$H(X|Y = y) = -\sum_{x \in \text{supp}(P(x|y))} P(x|y) \log P(x|y). \tag{9.17}$$

Alternatively, Eq. 9.17 can be written as the conditional expectation as follows:

$$H(X|Y = y) = E\left[-\log P_{X|Y}(X|Y = y)\right]. \tag{9.18}$$

Due to the similarity between expressions Eqs. 9.6 and 9.17, for $H(X)$ and $H(X|Y = y)$, respectively, the following result is immediate.

Corollary 9.8 *If the discrete random variable X has L possible values, then*

$$0 \leq H(X|Y = y) \leq \log L \tag{9.19}$$

with equality on the left if and only if $P(x|y) = 1$ for some value of x and with equality on the right if and only if $P(x|y) = 1/L$ for all x.

It should be kept in mind that y is a fixed value of Y in Corollary 9.8. When referring to the *uncertainty of X given Y*, it is meant the average conditional uncertainty of X given the event Y, taken over all possible values of Y.

Definition 9.9 The average conditional uncertainty, or average conditional entropy, of the discrete random variable X, given the discrete random variable Y, is given by the expression

$$H(X|Y) = \sum_{y \in \text{supp}(P(y))} P(y) H(X|Y = y). \tag{9.20}$$

Alternatively, Eq. 9.20 can be written as

$$H(X|Y) = E\left[-\log P_{X|Y}(X|Y)\right]. \tag{9.21}$$

The second most important result relating entropy is proved next, which has a pleasant intuitive interpretation that the knowledge of Y never increases the uncertainty about X but in general it can reduce such uncertainty or not to alter it in the particular case when X and Y are statistically independent random variables.

Theorem 9.10 *For any two random variables X and Y the expression*

$$H(X|Y) \leq H(X)$$

is satisfied with equality if and only if X and Y are statistically independent.

Proof Using expressions Eqs. 9.6 and 9.20, it follows that

$$\begin{aligned}
&H(X|Y) - H(X) = \qquad (9.22)\\
&= \sum_{y\in \text{supp}P(y)} P(y)H(X|Y=y) + \sum_{x\in\text{supp}(P(x))} P(x)\log P(x)\\
&= -\sum_{y\in\text{supp}(P(y))} P(y) \sum_{x\in\text{supp}(P_{X|Y}(.|y))} P(x|y)\log P(x|y)\\
&\quad + \sum_{y\in\text{supp}(P(y))}\sum_{x\in\text{supp}(P(x))} P(x,y)\log P(x)\\
&= -\sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\log P(x|y) + \sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\log P(x)\\
&= \sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\log\frac{P(x)}{P(x|y)}\\
&= \sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\log\frac{P(x)P(y)}{P(x,y)}.
\end{aligned}$$

However

$$\begin{aligned}
&\sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\log\frac{P(x)P(y)}{P(x,y)}\\
&\leq \sum_{(x,y)\in\text{supp}(P(x,y))} P(x,y)\left[\frac{P(x)P(y)}{P(x,y)} - 1\right]\log e \qquad (9.23)
\end{aligned}$$

$$
\begin{aligned}
&= \left[\sum_{x,y:P(x,y)\neq 0} P(x)P(y) - 1 \right] \log e \\
&\leq \left[\sum_{x,y} P(x)P(y) - 1 \right] \log e \qquad (9.24) \\
&= \left[\sum_{x} P(x) \sum_{y} P(y) - 1 \right] \log e = 0.
\end{aligned}
$$

Equality occurs in the inequality in Eq. 9.23 if and only if $P(x, y) = P(x)P(y)$, whenever $P(x, y) \neq 0$, and occurs with equality in the inequality in Eq. 9.24 if and only if $P(x)P(y) = 0$ whenever $P(x, y) = 0$. Therefore, equality occurs simultaneously in the inequalities in Eqs. 9.23 and 9.24 if and only if $P(x, y) = P(x)P(y)$ for all x and all y, which is just the definition of statistical independence. □

It follows immediately from expressions Eqs. 9.19 and 9.20 that

$$0 \leq H(X|Y) \leq \log L \qquad (9.25)$$

with equality on the right if and only if the two following conditions are satisfied:

(i) X and Y are statistically independent
(ii) $P(x) = 1/L$ for all x,

and with equality on the left if and only if for all y such that $P(y) \neq 0$ there is an x such that $P(x|y) = 1$.

It is possible to express Eq. 9.9 as a sum of entropies based on the following relation:

$$P(x_1, x_2, \ldots, x_n) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2) \ldots P(x_n|x_1, x_2, \ldots, x_{n-1},$$

obtaining

$$
\begin{aligned}
&H(X_1X_2 \ldots X_n) = \\
&\quad H(X_1) + H(X_2|X_1) + \cdots + H(X_n|X_1X_2 \ldots X_{n-1}. \qquad (9.26)
\end{aligned}
$$

Definition 9.11 The conditional uncertainty, or conditional entropy, of the discrete random variable X, given the discrete random variable Y and given that the event $Z = z$ occurs, is the quantity

$$H(X|Y, Z = z) = E\left[-\log P_{X|YZ}(X|Y, Z = z) \right]. \qquad (9.27)$$

Equivalently, Eq. 9.27 can be expressed as

$$H(X|Y, Z = z) = -\sum_{(x,y)\in \text{supp}(P_{XY|Z}(.,.|z)))} P_{XY|Z}(x, y|z) \log P_{X|YZ}(x|y, z). \tag{9.28}$$

Equations 9.18, 9.21, and 9.27 that

$$H(X|Y, Z) = \sum_{z\in \text{supp}(P_Z)} P_Z(z) H(X|Y, Z = z), \tag{9.29}$$

in which, by means of Eq. 9.21, it is observed that in the theory of discrete probability there is no essential difference between a single random variable Y and a random vector (Y, Z). Expression 9.29, together with expression 9.28, in many occasions represent the most adequate way to compute $H(X|Y, Z)$.

9.4 Informational Divergence

A very useful quantity when proving inequalities in information theory is introduced next. Basically this quantity represents a measure of *distance* between two probability distributions.

Definition 9.12 If X and $\tilde{X}$ are discrete random variables with the same domain, i.e., with the same set of possible values, $X(\Omega) = \tilde{X}(\Omega)$, then the informational divergence between P_X and $P_{\tilde{X}}$ is the quantity

$$D(P_X \parallel P_{\tilde{X}} = \sum_{x\in \text{supp}(P_X)} P_X(x) \log \left(\frac{P_X(x)}{P_{\tilde{X}}(x)} \right).$$

The informational divergence between P_X and $P_{\tilde{X}}$ is also known by various other names and that can be considered as an indirect proof of its utility. In some texts, it is called *relative entropy between X and* $\tilde{X}$ and is known in statistics as the *distance of Kullback-Leibler between* P_X *and* $P_{\tilde{X}}$.

Every time $D(P_X \parallel P_{\tilde{X}}$ is written, it is implicitly assumed that

$$X(\Omega) = \tilde{X}(\Omega),$$

because, without this assumption, the definition of informational divergence does not make sense. It follows from Definition 9.12 that, if there is an $x \in \text{supp}(P_X)$ but $x \notin \text{supp}(P_{\tilde{X}}$, i.e., for which $P_X(x) \neq 0$ but $P_{\tilde{X}}(x) = 0$, then

$$D(P_X \parallel P_{\tilde{X}} = \infty.$$

It is therefore apparent that, in general,

$$D(P_X \parallel P_{\tilde{X}} \neq D(P_{\tilde{X}} \parallel P_X)$$

and thus the informational divergence does not satisfy the symmetry property required to be a measure of *distance*.

In Definition 9.12, it is observed that the informational divergence can be written as an average or expected value, and that turns out to be useful, in the following manner:

$$D(P_X \parallel P_{\tilde{X}} = E\left[\log \frac{P_X(X)}{P_{\tilde{X}}(X)}\right].$$

The following inequality, simple and yet elegant, is the key to the usefulness of the informational divergence and serves as a justification for its interpretation as similar to a *distance*.

Theorem 9.13 *The informational divergence inequality*

$$D(P_X \parallel P_{\tilde{X}} \geq 0$$

occurs with equality if and only if $P_X = P_{\tilde{X}}$*, i.e., if and only if* $P_X(x) = P_{\tilde{X}}(x)$ *for all* $x \in X(\Omega) = \tilde{X}(\Omega)$.

Proof Consider

$$-D(P_X \parallel P_{\tilde{X}} = \sum_{x \in \text{supp}(P_X)} P_X(x) \log \frac{P_{\tilde{X}}(x)}{P_X(x)}$$

and by the IT inequality, we have

$$\begin{aligned} -D(P_X \parallel P_{\tilde{X}} &\leq \sum_{x \in \text{supp}(P_X)} P_X(x) \left[\frac{P_{\tilde{X}}(x)}{P_X(x)} - 1\right] \log e \\ &= \left[\sum_{x \in \text{supp}(P_X)} P_{\tilde{X}}(x) - \sum_{x \in \text{supp}(P_X)} P_X(x)\right] \log e \\ &\leq (1-1) \log e \\ &= 0. \end{aligned}$$

Equality occurs in the IT inequality if and only if $P_X(x) = P_{\tilde{X}}(x)$ for all $x \in \text{supp}(P_X)$, which is equivalent to $P_X(x) = P_{\tilde{X}}(x)$ for all $x \in X(\Omega) = \tilde{X}(\Omega)$ and that causes equality also in the second inequality. □

The following example serves the purpose of illustrating the usefulness of the informational divergence.

Example 9.14 Let X have L values, i.e., let $\#(X(\Omega)) = L$. Let $\tilde{X}$ be the random variable with probability distribution $P_{\tilde{X}}(x) = 1/L$ for all $x \in \tilde{X}(\Omega)$. Then

$$\begin{aligned} D(P_X \parallel P_{\tilde{X}} &= E\left[\log\left(\frac{P_X(x)}{P_{\tilde{X}(x)}}\right)\right] \\ &= E\left[\log\left(\frac{P_X(x)}{1/L}\right)\right] \\ &= \log L - E[-\log P_X(x)] \\ &= \log L - H(X). \end{aligned}$$

It follows from the informational divergence inequality that $H(X) \leq \log L$, with equality if and only if $P_X(x) = 1/L$ for all $x \in X(\Omega)$, which is the fundamental inequality on the right in Eq. 9.10, Theorem 9.6.

9.5 Uncertainty Reduction by Conditioning

The informational divergence inequality is used below to prove a very useful result which has an intuitively interesting interpretation, namely, that the knowledge of Y reduces, in general, the uncertainty about X.

Theorem 9.15 (Second Entropy Inequality) *For any two discrete random variables X and Y, the expression*

$$H(X|Y) \leq H(X) \tag{9.30}$$

occurs with equality if and only if X and Y are statistically independent random variables.

Proof Expressing $H(X)$ and $H(X|Y)$, respectively, as expected values, it follows that

$$\begin{aligned} H(X) - H(X|Y) &= E[-\log P_X(X)] - E[-\log P_{X|Y}(X|Y)] \\ &= E\left[\log \frac{P_{X|Y}(X|Y)}{P_X(X)}\right] \\ &= E\left[\log \frac{P_{X|Y}(X|Y)P_Y(Y)}{P_X(X)P_Y(Y)}\right] \end{aligned}$$

$$= E\left[\log \frac{P_{XY}(X, Y)}{P_X(X)P_Y(Y)}\right]$$
$$= D(P_{XY} \parallel P_{\tilde{X}\tilde{Y}},$$

in which $P_{\tilde{X}\tilde{Y}}(x, y) = P_X(x)P_Y(y)$ for all $(x, y) \in X(\Omega) \times Y(\Omega)$. It follows from the informational divergence inequality that $H(X|Y) \leq H(X)$, with equality if and only if $P_{XY}(x, y) = P_X(x)P_Y(y)$ for all $(x, y) \in X(\Omega) \times Y(\Omega)$, which is the definition of statistical independence for discrete random variables. □

Expressions 9.20 and 9.27 differ only by the fact that the probability distribution of the latter has an additional conditioning, on the event $Z = z$. In this manner, due to the mathematical similarity, the following result is stated.

Corollary 9.16 (From Theorem 9.15) *For any three discrete random variables X, Y and Z, the expression*

$$H(X|Y, Z = z) \leq H(X|Z = z) \tag{9.31}$$

occurs with equality if and only if $P_{XY|Z}(x, y|z) = P_{X|Z}(x|z)P_{Y|Z}(y|z)$ for all x and y. (Notice that z represents a fixed value of Z in this corollary.)

Multiplying by $P_Z(z)$ the two sides in the inequality in Eq. 9.31 and adding for all $z \in \text{supp}(P_Z)$, the following intuitive and useful result is obtained, which represents the last inequality to be presented relating various uncertainties and once more demonstrates that conditioning can never increase uncertainty.

Corollary 9.17 (From Theorem 9.15) *Third entropy inequality: For any three discrete random variables X, Y and Z, the expression*

$$H(X|YZ) \leq H(X|Z)$$

occurs with equality if and only if, for all $z \in \text{supp}(P_Z)$, the relation

$$P_{XY|Z}(x, y|z) = P_{X|Z}(x|z)P_{Y|Z}(y|z)$$

remains true for all x and y, or, equivalently, if and only if X and Y are independent when conditioned on the knowledge of Z.

Summarizing what has been presented about inequalities, involving the conditioning of random variables, it turns out that such conditioning never increases uncertainty about the conditioned variable. This is yet another intuitive property of the Shannon uncertainty measure. However, it is observed that conditioning a variable on an event can increase uncertainty, i.e., $H(X|Y = y)$ can be greater than $H(X)$. Example 9.18 presented below illustrates such a situation.

Example 9.18 Suppose that the random vector $[X, Y, Z]$ takes on any of the four values $[0, 0, 0]$, $[0, 1, 0]$, $[1, 0, 0]$, and $[1, 0, 1]$ with equal probability. Then

$P_X(0) = P_X(1) = 1/2$ and thus

$$H(X) = h(1/2) = 1\ bit.$$

Note that $P_{Y|X}(0|1) = 1$ and thus

$$H(Y|X = 1) = 0.$$

Analogously, $P_{Y|X}(0|0) = 1/2$ and therefore

$$H(Y|X = 0) = h(1/2) = 1\ bit.$$

Using Eq. 9.20 we obtain

$$H(Y|X) = \frac{1}{2}(1) = 1/2\ bit.$$

Because $P(z|xy)$ is equal to 1 for $(x, y, z) = (0, 0, 0)$, equal to 1 for $(x, y, z) = (0, 1, 0)$ and equal to $1/2$ for $(x, y, z) = (1, 0, 0)$, it follows that

$$H(Z|X = 0, Y = 0) = 0$$
$$H(Z|X = 0, Y = 1) = 0$$
$$H(Z|X = 1, Y = 0) = 1\ bit.$$

Noting that $P(x, y)$ is equal to $1/4$, $1/4$ and $1/2$ for (x, y) equal to $(0, 0)$, $(0, 1)$, and $(1, 0)$, respectively, it follows from Eq. 9.20 that

$$H(Z|XY) = \frac{1}{4}(0) + \frac{1}{4}(0) + \frac{1}{2}(1) = 1/2\ bit.$$

By making use of Eq. 9.26, we obtain

$$H(XYZ) = 1 + 1/2 + 1/2 = 2\ bits.$$

However there is a simpler way to compute $H(XYZ)$, based on the fact that $[X, Y, Z]$ is equally likely to assume any of the four values and so

$$H(XYZ) = \log 4 = 2\ bits.$$

Since $P_Y(1) = 1/4$, we obtain

$$H(Y) = h(1/4) = 0.811\ bits.$$

Therefore, it turns out that

$$H(Y|X) = 1/2 < H(Y) = 0.811$$

in agreement with Theorem 9.6. However, the following situation is observed:

$$H(Y|X = 0) = 1 > H(Y) = 0.811\ bits.$$

9.6 The Chain Rule for Uncertainty

One of the simplest, most intuitive, and useful information theory identities is presented below. Let $[X_1, X_2, \ldots, X_N]$ be a discrete random vector with N components that are discrete random variables. For the fact that discrete random vectors are also discrete random variables, Eq. 9.8 can be written as

$$H(X_1X_2 \ldots X_N) = E[-\log P_{X_1,X_2,\ldots,X_N}(X_1, X_2, \ldots, X_N)], \tag{9.32}$$

which can, equivalently, be written using the multiplication rule for probability distributions as

$$\begin{aligned} H(X_1X_2 \ldots X_N) \ = \ & E[-\log(P_{X_1}(X_1)P_{X_2|X_1}(X_2|X_1) \ldots \\ & \ldots P_{X_N|X_1,\ldots,X_{N-1}}(X_N|X_1, \ldots, X_{N-1})]. \end{aligned} \tag{9.33}$$

Expression 9.33 can be written in a more compact form as shown next.

$$\begin{aligned} H(X_1X_2 \ldots X_N) &= E\left[-\log \prod_{n=1}^{N} P_{X_n|X_1 \ldots X_{n-1}}(X_n|X_1, \ldots, X_{n-1}\right] \\ &= \sum_{n=1}^{N} E\left[-\log P_{X_n|X_1 \ldots X_{n-1}}(X_n|X_1, \ldots, X_{n-1}\right] \\ &= \sum_{n=1}^{N} H(X_n|X_1 \ldots X_{n-1}. \end{aligned} \tag{9.34}$$

By using a less compact notation, however easier to read, Eq. 9.34 can be written as

$$H(X_1X_2 \ldots X_N) = H(X_1) + H(X_2|X_1) + \cdots + H(X_N|X_1 \ldots X_{N-1}. \tag{9.35}$$

Expression 9.35, which is sometimes called *the chain rule for uncertainty*, can be interpreted as meaning that *the uncertainty of a random vector is equal to the uncertainty of its first coordinate, plus the uncertainty of its second coordinate when*

the first coordinate is known, . . ., plus the uncertainty of the last coordinate when all previous coordinates are known. This is such an intuitively pleasing property that, in Massey's words, can induce an individual to prematurely conclude that Shannon's measure of information is the correct measure of information (Massey 1990).

The reader should notice that in the derivation of Eq. 9.35 the choice of the order of the coordinates is arbitrary. Therefore it is possible to expand $H(XYZ)$, for example, in six distinct manners as shown next:

$$\begin{aligned} H(XYZ) &= H(X) + H(Y|X) + H(Z|XY) \\ &= H(X) + H(Z|X) + H(Y|XZ) \\ &= H(Y) + H(X|Y) + H(Z|XY) \\ &= H(Y) + H(Z|Y) + H(X|YZ) \\ &= H(Z) + H(X|Z) + H(Y|XZ) \\ &= H(Z) + H(Y|Z) + H(X|YZ). \end{aligned}$$

In general $H(X_1X_2 \ldots X_N)$ can be expanded in $N!$ distinct ways.

9.7 Mutual Information

As mentioned earlier, uncertainty is the central element in Shannon's information theory, who characterized *information* as a non-negative difference between two interrelated uncertainty values. Shannon's answer to the question "how much information the random variable Y provides about the random variable X?" was "it is the value by which Y reduces the uncertainty abour X," i.e., $H(X) - H(X|Y)$.

Definition 9.19 The mutual information between the discrete random variables X and Y is by definition the quantity

$$I(X;Y) = H(X) - H(X|Y). \tag{9.36}$$

As shown below, the denomination *mutual information* is very appropriate by the fact that $I(X;Y) = I(Y;X)$. This can be seen by expanding $H(XY)$ in two ways, namely,

$$\begin{aligned} H(XY) &= H(X) + H(Y|X) \\ &= H(Y) + H(X|Y), \end{aligned}$$

from which it follows that $H(X) - H(X|Y) = H(Y) - H(Y|X)$ or, equivalently,

$$I(X;Y) = I(Y;X). \tag{9.37}$$

The equality in Eq. 9.37 means that the amount of information provided by the random variable X about the random variable Y is exactly the same amount of information provided by the random variable Y about the random variable X. It is therefore a symmetric relationship between the random variables X and Y.

Example 9.20 (Continuing from Example 9.18. Recalling that $P_Y(1) = 1/4$ it follows that

$$H(Y) = h(1/4) = 0.811 \text{ bits}.$$

Therefore,

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= 0.811 - 0.500 = 0.311\, bits. \end{aligned}$$

This means that the component X of the random vector $[X, Y, Z]$ gives 0.311 bits of information about the component Y and vice versa.

It is interesting to note that Shannon, in 1948, neither referred to *mutual information* nor used a special symbol to denote it but instead always talked about difference between uncertainties (Massey 1990). It was Fano who later introduced the terminology *mutual information* or *average mutual information*, as it is frequently called, as well as the symbol $I(X;Y)$ (Fano 1961). Certainly, it is convenient to use the terminology introduced by Fano, but one should not lose sight of Shannon's idea that information means a change in uncertainty level. The following two definitions are natural consequences of the definition of mutual information.

Definition 9.21 The conditional mutual information between the discrete random variables X and Y, given that the event $Z = z$ occurred, is the quantity

$$I(X;Y|Z=z) = H(X|Z=z) - H(X|Y, Z=z). \tag{9.38}$$

Definition 9.22 The conditional mutual information between the discrete random variables X and Y, given the discrete random variable Z, is the quantity

$$I(X;Y|Z) = H(X|Z) - H(X|YZ). \tag{9.39}$$

It follows from Eqs. 9.20 and 9.32 that

$$I(X;Y|Z) = \sum_{z \in \text{supp}(P_Z)} P_Z(z) I(X;Y|Z=z). \tag{9.40}$$

By the fact that $H(XY|Z=z)$ and $H(XY|Z)$ can, each one of them, be expanded in two distinct manners, namely,

$$H(XY|Z=z) = H(X|Z=z) + H(Y|X,Z=z)$$
$$= H(Y|Z=z) + H(X|Y,Z=z)$$

and

$$H(XY|Z) = H(X|Z) + H(Y|XZ)$$
$$= H(Y|Z) + H(X|YZ),$$

it follows from Definition 9.21 and from Definition 9.22 that

$$I(X;Y|Z=z) = I(Y;X|Z=z)$$

and that

$$I(X;Y|Z) = I(Y;X|Z).$$

The fundamental inequalities satisfied by the mutual information are considered next. Expression 9.36, due to Eqs. 9.25 and 9.37, immediately implies the following result, expressed as a theorem.

Theorem 9.23 *For any two discrete random variables X and Y, we have*

$$0 \leq I(X;Y) \leq \min[H(X), H(Y)]$$

with equality on the left if and only if X and Y are statistically independent random variables and equality on the right if and only if Y essentially determines X, or X essentially determines Y, or both.

Similarly, Definition 9.21 and Definition 9.22 lead to the inequalities

$$0 \leq I(X;Y|Z=z) \leq \min[H(X|Z=z), H(Y|Z=z)] \tag{9.41}$$

and

$$0 \leq I(X;Y|Z) \leq \min[H(X|Z), H(Y|Z)], \tag{9.42}$$

respectively.

9.8 Discrete Information Sources

For our study of information theory, it is essential that we have a mathematical description of the information generation mechanism. Considering that information is generated by a message source, our next step is to mathematically characterize

message sources. Unless we note otherwise, we will from now on consider only discrete sources, that is, sources the set of generated symbols of which is finite or countably infinite.

Self-Information

Let E be an event that occurs with probability $P(E)$. If we are informed that event E occurred, then we say that we received

$$I(E) = \log \frac{1}{P(E)} \text{ bits}$$

of information. Recall that entropy can be interpreted as the average value of the self-information.

Discrete Memoryless Source

Let U denote the output of a discrete message source (DMS) S which emits sequences of symbols from a fixed alphabet $\{u_0, u_1, \ldots, u_L\}$, and those symbols are generated obeying a probability distribution $P(u_1), P(u_2), \ldots, P(u_L)$. We consider next the case where the symbols emitted by S are *statistically independent*.

In spite of its simplicity, the *discrete memoryless source* mathematical model of a message source is quite useful and is completely described by the source alphabet and by the probability distribution of occurrence of its symbols. The entropy (or average uncertainty) of a memoryless DMS is given by

$$H(U) = -\sum_{i=1}^{L} P(u_i) \log P(u_i) \text{ bits/symbol},$$

which can be interpreted as the uncertainty that an observer has, about which symbol will be emitted, before knowing which symbol was in fact emitted by the source. As we have already said, this source S is completely described by its alphabet and by the probability distribution of occurrence of its symbols $P(u_1), P(u_2), \ldots, P(u_L)$, which we will also denote by $P_1, P_2, \ldots, P_L$ when this does not cause ambiguity of interpretation.

Denoting the self-information per symbol by $I(u_i)$, i.e.,

$$I(u_i) = \log \frac{1}{P_i} = -\log P_i,$$

we can interpret the entropy $H(U)$ of a memoryless DMS as being the average value of the information obtained per observed symbol, that is,

$$H(U) = \sum_{i=1}^{L} P_i I(u_i).$$

Finite Extension of a Memoryless Source

In many practical situations as, for example, when enciphering or coding data, it is more convenient to work with blocks of source symbols rather than to work with individual symbols. For example, the ASCII representation of symbols on a computer keyboard uses binary 7 digit words, that is, binary blocks of length 7. For now we will consider blocks of fixed size equal to n symbols. The number n is called *order* of the source extension. We treat this case mathematically by considering a new source S^n the output of which, denoted by U^n, consists of all blocks formed by the concatenation of n symbols from the alphabet of S. We will denote the set of symbols from S^n by $\{\sigma_1, \sigma_2, \ldots, \sigma_{L^n}\}$. We present next the definition of extension of a memoryless source.

Definition 9.24 Consider a memoryless DMS, the output U of which emits symbols from the alphabet $\{u_1, u_2, \ldots, u_L\}$ and denote by P_i the probability of the DMS emitting the symbol u_i. The extension of order n of this DMS is the DMS with output denoted by U^n which emits symbols from the alphabet $\{\sigma_1, \sigma_2, \ldots, \sigma_{L^n}\}$, in which to each σ_i corresponds a specific sequence of n symbols from the alphabet $\{u_1, u_2, \ldots, u_L\}$. Supposing that σ_i corresponds to the sequence $u_{i_1}, u_{i_2}, \ldots, u_{i_n}$, the probability $P(\sigma_i)$ is given by

$$P(\sigma_i) = P(u_{i_1}, u_{i_2}, \ldots, u_{i_n} = P(u_{i_1} P(u_{i_2} \ldots P(u_{i_n} = P_{i_1} P_{i_2} \ldots P_{i_n}.$$

Example 9.25 Consider a memoryless binary source S with output alphabet $\{u_1, u_2\}$. The third-order extension of S, denoted by S^3, has the output alphabet $\{\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5, \sigma_6, \sigma_7, \sigma_8\}$, in which $\sigma_1 = u_1u_1u_1$, $\sigma_2 = u_1u_1u_2$, $\sigma_3 = u_1u_2u_1$, $\sigma_4 = u_1u_2u_2$, $\sigma_5 = u_2u_1u_1$, $\sigma_6 = u_2u_1u_2$, $\sigma_7 = u_2u_2u_1$, $\sigma_8 = u_2u_2u_2$.

Entropy of the Finite Extension of a Memoryless Source

Since each symbol of S^n, the extension of order n of the memoryless message source S, corresponds to n symbols from S, it would be reasonable to expect that the entropy per symbol of S^n, denoted by $H(U^n)$, be equal to n times the entropy per symbol of S. As we will show below, this result is indeed true. Using the definition of entropy and considering $\sigma_i = u_{i_1}, u_{i_2}, \ldots, u_{i_n}$, we have

$$H(U^n) = -\sum_{i=1}^{L^n} P(\sigma_i)\log P(\sigma_i) \tag{9.43}$$

$$= -\sum_{i_1=1}^{L}\cdots\sum_{i_n=1}^{L} P(u_{i_1},\ldots,u_{i_n}\log P(u_{i_1},\ldots,u_{i_n}$$

$$= -\sum_{i_1=1}^{L}\cdots\sum_{i_n=1}^{L} P(u_{i_1},\ldots,u_{i_n}\log(P_{i_1}\ldots P_{i_n}$$

$$= -\sum_{i_1=1}^{L}\cdots\sum_{i_n=1}^{L} P(u_{i_1},\ldots,u_{i_n}[\log P_{i_1}+\cdots+\log P_{i_n}]$$

$$= -\sum_{i_1=1}^{L} P_{i_1}\log P_{i_1} - \cdots - \sum_{i_n=1}^{L} P_{i_n}\log P_{i_n}$$

$$= nH(U). \tag{9.44}$$

In order to show the result in Eq. 9.44 and other related results later we used the following properties of the joint probability:

$$\sum_{u_{i_j}} P(u_{i_1},\ldots,u_{i_N} = P(u_{i_1},\ldots,u_{i_{j-1}},u_{i_{j+1}},\ldots,u_{i_n}. \tag{9.45}$$

$$\sum_{\sigma_i} P(\sigma_i) = \sum_{i_1}\cdots\sum_{i_n} P_{i_1}\ldots P_{i_n} = \sum_{i_1} P_{i_1}\cdots\sum_{i_n} P_{i_n} = 1 \tag{9.46}$$

$$-\sum_{\sigma_i} P(\sigma_i)\log P_{i_j} = -\sum_{i_1}\cdots\sum_{i_j}\cdots\sum_{i_n} P_{i_1}\ldots P_{i_j}\ldots P_{i_n}\log P_{i_j}$$

$$= -\sum_{i_j} P_{i_j}\log P_{i_j}\sum_{i_1} P_{i_1}\cdots\sum_{i_n} P_{i_n}$$

$$= -\sum_{i_j} P_{i_j}\log P_{i_j}$$

$$= H(U). \tag{9.47}$$

Example 9.26 Let S be the source in Example 9.25, with $P_1 = 1/3$ and $P_2 = 2/3$. The third-order extension of S, denoted as S^3, has the following symbol probability distribution:

$$
\begin{aligned}
P(\sigma_1) &= P(u_1u_1u_1) = P(u_1)P(u_1)P(u_1) = 1/27\\
P(\sigma_2) &= P(u_1u_1u_2) = P(u_1)P(u_1)P(u_2) = 2/27\\
P(\sigma_3) &= P(u_1u_2u_1) = P(u_1)P(u_2)P(u_1) = 2/27\\
P(\sigma_4) &= P(u_1u_2u_2) = P(u_1)P(u_2)P(u_2) = 4/27\\
P(\sigma_5) &= P(u_2u_1u_1) = P(u_2)P(u_1)P(u_1) = 2/27\\
P(\sigma_6) &= P(u_2u_1u_2) = P(u_2)P(u_1)P(u_2) = 4/27\\
P(\sigma_7) &= P(u_2u_2u_1) = P(u_2)P(u_2)P(u_1) = 4/27\\
P(\sigma_8) &= P(u_2u_2u_2) = P(u_2)P(u_2)P(u_2) = 8/27.
\end{aligned}
$$

For this source we obtain $H(U) = 0.918$ and $H(U^3) = 3 \times 0.918 = 2.754$.

Discrete Information Sources with Memory

Increasing a little the complexity of the message source model considered so far, which was the one in which the source has no memory, we will now consider sources where the occurrence of a symbol depends statistically on the m last emitted symbols, for a finite m. This type of source is called a *Markov source of order m* and is specified by its alphabet $\{u_1, u_2, \ldots, u_L\}$ and by its conditional probability distribution

$$P(u_i/u_{j_1}, u_{j_2}, \ldots, u_{j_l}, \ldots, u_{j_m},$$

in which $i \in \{1, 2, \ldots, L\}$, $j_l \in \{1, 2, \ldots, L\}$, $l \in \{1, 2, \ldots, m\}$ and the symbol sequence in time is $u_{j_1}, \ldots, u_{j_l}, \ldots, u_{j_m}, u_i$, i.e., u_i succeeds u_{j_m}.

For a Markov source of order m, the probability of it emitting a given symbol is known if the m preceding symbols are known. Due to the importance they have in characterizing a Markov source of order m, the preceding m symbols are called *state* of the source. Since the source alphabet consists of L symbols, the number of possible distinct states is L^m. As the source emits symbols, state changes occur. A useful way to represent the possible state changes of a Markov source is through the *state diagram*. In a state diagram, each node represents a state, and each possible transition between two states is represented by an oriented line (indicated by an arrow) connecting these two states.

Example 9.27 Consider the second-order ($m = 2$) Markov source with a binary alphabet ($L = 2$), shown in Fig. 9.3. The symbol conditional probabilities (transition probabilities) are the following:

$$
\begin{aligned}
P(0|00) &= P(1|11) = 0.8\\
P(1|00) &= P(0|11) = 0.2\\
P(0|01) &= P(0|10) = 0.5
\end{aligned}
$$

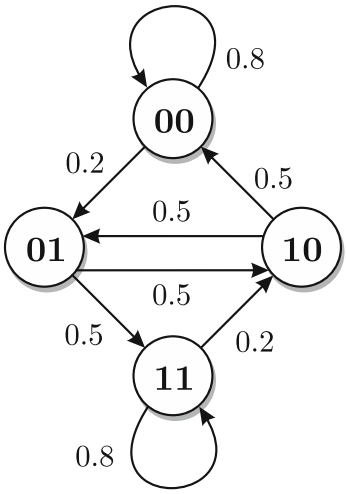


Fig. 9.3 State diagram of a second-order binary Markov source

$$P(1|01) = P(0|10) = 0.5.$$

Since this is a second-order binary source, i.e., $L = 2$ and $m = 2$, we have $L^m = 4$ and therefore four states: 00, 01, 10, 11. In Fig. 9.3, the four possible states are indicated by four circles. In general, the transitions between states are indicated by arrows, each arrow connecting two states, and the state transition probability is indicated by a number associated with each arrow. For example, from state 10, it is possible to go to state 00, with probability 0.5 if the source emits a 0, or to go to state 01, with probability 0.5 if the source emits a 1, but state 11 cannot be reached and the source is not allowed to remain in state 10. Notice however in this example that one of the arrows leaving state 00 and state 11, respectively, return to the same state with probability 0.8.

In our presentation of Markov sources, we shall consider only *ergodic sources*. It is beyond the scope of the current treatment to go into the nuances of the mathematical definition of what an ergodic source is. However, a purposefully imprecise conceptualization, which we introduce below, will serve our purposes well.

Definition 9.28 An ergodic source is one that, when observed over a long period of time, emits with probability 1 a typical sequence of its symbols.

In order to better characterize what is meant by a non-ergodic source, the following example is presented.

Example 9.29 Consider the second-order ($m = 2$) Markov source with a binary alphabet ($L = 2$) shown in Fig. 9.4. The symbol state transition probabilities (transition probabilities) are the following:

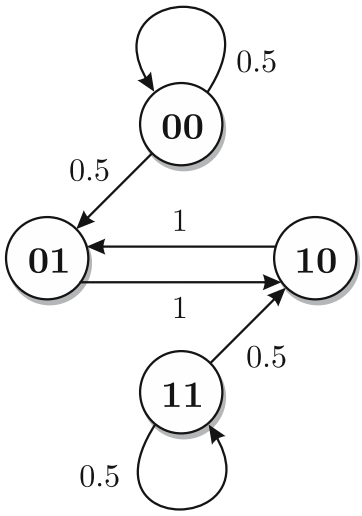


Fig. 9.4 State diagram of a non-ergodic second-order binary Markov source

$$
\begin{aligned}
P(0|00) &= P(1|11) = 0.5 \\
P(1|00) &= P(0|11) = 0.5 \\
P(0|01) &= P(1|10) = 1 \\
P(1|01) &= P(0|10) = 0.
\end{aligned}
$$

Similar to the case in Example 9.27, here we also have $L^m = 4$, i.e., four states denoted as 00, 01, 10, and 11. The state diagram for this source is shown in Fig. 9.4. Notice that, once states 01 or 10 are reached, from that point on, the source will permanently emit the sequence $\ldots 101010101010\ldots$ and will never reach state 00 or 11. Even if the initial state is randomly selected from a uniform probability distribution, that is, each state being selected in a statistically independent manner with probability $1/4$, after a sufficiently long number of state transitions either state 01 or state 10 will be reached with probability 1. Therefore, whatever the sequence of symbols emitted by this source, if we wait a sufficiently long time, with probability 1 we will see the sequence $\ldots 101010101010\ldots$ appear. Thus, with probability 1 we will not see a typical sequence emitted by this source, and for this reason, it is not ergodic.

An important property of ergodic Markov sources is that the probability distribution of states is unique, and the states in any long sequence of symbols emitted by the source will occur, with probability 1, according to this distribution. This probability distribution is called *stationary distribution* of the ergodic Markov process. Due to the fact that the stationary distribution of states does not depend on any initial distribution for choosing these states, it is possible to calculate it from the conditional probabilities of source symbols (Abramson 1963, Ch. 2).

Entropy of a Markov Source

Given that the Markov source of order m is in the state $u_{j_1}, u_{j_2}, \ldots, u_{j_m}$, its corresponding conditional uncertainty is given by

$$H(U|u_{j_1}, u_{j_2}, \ldots, u_{j_m} = \\ -\sum_{i=1}^{L} P(u_i|u_{j_1}, u_{j_2}, \ldots, u_{j_m} \log P(u_i|u_{j_1}, u_{j_2}, \ldots, u_{j_m} \tag{9.48}$$

The average value of $H(U|u_{j_1}, u_{j_2}, \ldots, u_{j_m}$, over all L^m possible states is defined as the entropy (or uncertainty) of U and is expressed as

$$H(U) = \sum_{j_1=1}^{L} \sum_{j_2=1}^{L} \ldots \sum_{j_m=1}^{L} P(u_{j_1}, u_{j_2}, \ldots, u_{j_m} H(U|u_{j_1}, u_{j_2}, \ldots, u_{j_m}. \tag{9.49}$$

The Adjoint Source

By employing the symbol first-order probability distribution of a Markov source, an auxiliary source is defined and called *adjoint source*.

Definition 9.30 Let $\{u_1, u_2, \ldots, u_L\}$ be the alphabet of a Markov source S of order m, and let $P_1, P_2, \ldots, P_L$ be the first-order probability distribution of the source symbols. The **adjoint source** of S, denoted by $\overline{S}$, is the memoryless source with the same alphabet as S and with symbol probability distribution $P_1, P_2, \ldots, P_L$.

We will prove below that the entropy of the adjoint source $\overline{S}$ is never less than the entropy of the source S. This important fact can be interpreted in the following way. The two sources, S and $\overline{S}$, have the same first-order symbol probability distribution; the essential difference among them comes from the fact that S has additional restrictions, expressed in the form of symbol conditional probabilities, which restrict the possible output sequences. Such restrictions reduce uncertainty about the source output.

In order to keep notation as simple as possible, we will write $H(U)$ for the entropy of S and will write $H(\overline{U})$ for the entropy of $\overline{S}$ and will prove that $H(\overline{U})$ is greater than or equal to $H(U)$ when S is a first-order Markov source. We leave as an exercise for the reader to prove the case in which S is a Markov source of order m (see Problem 16).

Let S be a first-order Markov source, the symbols of which are $u_1, u_2, \ldots, u_L$ and conditional probabilities are $P(u_i|u_j)$, $1 \leq i \leq L$, $1 \leq j \leq L$. Let us denote by $P_1, P_2, \ldots, P_L$ the symbol first-order probabilities of S and let $\overline{S}$ denote the adjoint source. Let us denote by $P(u_j, u_i)$ the joint probability of the source S being in the state specified by u_j and emit the symbol u_i. We can then write

$$P(u_j, u_i) = P(u_i|u_j)P_j. \tag{9.50}$$

Let us now consider the following double summation:

$$\sum_i \sum_j P(u_j, u_i) \log\left(\frac{P_j P_i}{P(u_j, u_i)}\right). \tag{9.51}$$

If follows from the fundamental inequality of information theory that Eq. 9.51 is less than or equal to zero, with equality if and only if

$$P(u_j, u_i) = P_j P_i \quad \text{for all} \quad i \text{ and } \quad j. \tag{9.52}$$

By combining Eqs. 9.50 and 9.51 and writing the inequality, we obtain

$$\sum_i \sum_j P(u_j, u_i) \log\left(\frac{P_i}{P(u_i|u_j)}\right) \leq 0 \tag{9.53}$$

or, equivalently,

$$\sum_i \sum_j P(u_j, u_i) \log\left(\frac{1}{P(u_i|u_j)}\right) \leq \sum_i \sum_j P(u_j, u_i) \log\left(\frac{1}{P_i}\right)$$

that is,

$$\begin{aligned} H(U) &\leq \sum_{i=1}^{L} \sum_{j=1}^{L} P(u_j, u_i) \log\left(\frac{1}{P_i}\right) \\ &= \sum_{i=1}^{L} \log\left(\frac{1}{P_i}\right) \sum_{u_j=1}^{L} P(u_j, u_i) \\ &= \sum_{i=1}^{L} P(u_i) \log\left(\frac{1}{P_i}\right) \\ &= H(\overline{U} \end{aligned}$$

and thus

$$H(U) \leq H(\overline{U},$$

recalling that the condition for equality Eq. 9.52 implies that u_i and u_j are statistically independent, or, that the source S is memoryless.

Finite Extension of a Markov Source

We take a look next at some properties of a Markov source considering blocks of n symbols from this source as forming a new symbol σ_i.

Definition 9.31 Let S be a Markov message source, of order m, with alphabet $u_1, u_2, \ldots, u_L$ and symbol conditional probability distribution $P(u_i|u_{j_1}, u_{j_2}, \ldots, u_{j_m}$. The extension of order n of S, denoted by S^n, is a Markov source of order μ with L^n symbols, $\sigma_1, \sigma_2, \ldots, \sigma_{L^n}$. Each σ_i corresponds to a certain sequence of n symbols $u_{i_1}, u_{i_1}, \ldots, u_{i_n}$, with conditional probability $P(\sigma_i|\sigma_{j_1}, \sigma_{j_1}, \ldots, \sigma_{j_\mu}$, where $\mu = \lceil m/n \rceil$.

Theorem 9.32 *The entropy $H(U^n)$ of the extension S^n of a Markov source S of order m is equal to n times the entropy $H(U)$ of the source S, that is,*

$$H(U^n) = nH(U).$$

Theorem 9.33 *Let $\overline{S}^n$ denote the adjoint source of S^n. The entropy $H(\overline{U}^n)$ of the adjoint source is greater than or equal to $H(U^n)$, that is,*

$$H(\overline{U}^n) \geq H(U^n) = nH(U).$$

Actually it is shown in Abramson (1963) that

$$H(\overline{U}^n) = nH(U) + \epsilon_n$$

in which ϵ_n is a positive constant and, for $n > m$, it depends only on the statistics of U, and that

$$\lim_{n\to\infty} \frac{\epsilon_n}{n} = 0.$$

Therefore

$$\lim_{n\to\infty} \frac{H(\overline{U}^n)}{n} = H(U).$$

In other words, for n sufficiently large, the Markov source constraints over symbols from U^n become less and less important.

9.9 Source Codes

In this section we investigate the representation of each symbol from a given source by a codeword of a given code. Only some special classes of such codes will be

considered, aiming at practical applications as well as keeping this presentation at a level more accessible to the reader. Initially, definitions and code properties will be presented and that is followed by some specific code constructions.

Frequently in practical applications like, for example, voice digital signal processing, there is a need to represent source symbols by codewords formed by a sequence of symbols from another alphabet, commonly a binary alphabet. A formal definition of a code is presented next.

Definition 9.34 Let $\{u_1, u_2, \ldots, u_L\}$ denote the set of symbols of a given alphabet U. A code is a mapping of all symbol sequences from U to symbol sequences from another alphabet $\{x_1, x_2, \ldots, x_D\}$ denoted by X. We call U the **source alphabet** and call X the **code alphabet**.

However, *Definition 9.34* is so general that it does not help us very much when constructing codes. Our next step is the introduction of restrictions in the definition of a code, in the form of properties required from a code. The first code property is to require it to be a **block code**.

Definition 9.35 A **block code** is a code which maps each symbol in the source alphabet U to a fixed sequence of symbols from the code alphabet X. These fixed sequences of symbols from X are called **codewords**.

Alternatively, the designation *a block code for* U is used to mean a list $(z_1, z_2, \ldots, z_L)$ of D-ary sequences for which z_i denotes the codeword for u_i, i.e., $U = u_i$ implies $Z = z_i$.

Example 9.36 Table 9.2 presents a binary block code.

The values of the random variable $Z = [X_1, X_2, \ldots, X_W]$ are D-ary sequences of variable length W, in which W is a random variable. The average codeword length $E[W]$ of the codewords will be used as an indication of the quality of the code, meaning that the shorter $E[W]$ the better the code.

If $z_i = [x_{i1}, x_{i2}, \ldots, x_{iw_i}]$ is the codeword for u_i and w_i is the length of this word, then it is possible to write the average length of words as

$$E[W] = \sum_{i=1}^{L} w_i P_U(u_i), \tag{9.54}$$

Table 9.2 A binary block code

Source Symbols	Codewords
u_1	0
u_2	00
u_3	01
u_4	10
u_5	01

Table 9.3 Non-singular block code

Source Symbols	Codewords
u_1	0
u_2	01
u_3	10
u_4	11
u_5	00

because W is a real function of the random variable U. Note that the values $Z = [X_1, X_2, \ldots, X_W]$ are, in general, random variables defined conditionally only, because X_i only takes on a value when $W \geq i$.

Uniquely Decodable Codes

Observing the code in Example 9.36 we notice the need to add a few restrictions to avoid, for example, that two distinct source symbols be represented by the same codeword, which is the case of u_3 and u_5 in Table 9.2, because that situation would lead to an ambiguity when decoding the codeword 01.

Definition 9.37 A block code is **non-singular** if all its codewords are distinct.

Example 9.38 Table 9.3 presents a non-singular block code.

Despite all the codewords in Table 9.3 being distinct, we will still have problems with ambiguity in decoding when, for example, the sequence 010 is received as it could represent u_1u_3 or u_2u_1. In other words, preventing against non-singularity considering only codewords alone is not enough, we need to prevent non-singularity from occurring when we consider the concatenation of codewords.

Definition 9.39 The nth extension of a block code, which maps the source symbols u_i to codewords Z_i, is the block code that maps the sequences $u_{i_1}, u_{i_2}, \ldots, u_{i_n}$ of source symbols to sequences of codewords $Z_{i_1}Z_{i_2}\ldots Z_{i_n}$.

It follows from Definition 9.39 that the nth extension of a block code is also a block code.

Example 9.40 Table 9.4 presents the second-order extension of the block code shown in Table 9.3.

Definition 9.41 A block code C is a **uniquely decodable code** if and only if the nth extension of C is non-singular for all finite n.

Definition 9.41 apparently guarantees only that any two sequences of source symbols containing the same number of symbols produce distinct codeword sequences. Obviously, it would be desirable to be more protected by requiring that sequences of source symbols of different lengths produce distinct sequences of codewords. This property, however, is contained in Definition 9.41 as we shall see shortly. On the contrary, suppose for a moment that code C fulfills Definition 9.41 but that $\mathcal{U}_1$ and $\mathcal{U}_2$ are two sequences of source symbols, of different length, which produce the same sequence $\mathcal{Z}_0$ of codewords. Consider now two new sequences of source

Table 9.4 Second-order extension of the block code in Example Eq. 9.38

Source symbols	Codewords	Source symbols	Codewords
u_1u_1	00	u_3u_4	1011
u_1u_2	001	u_3u_5	1000
u_1u_3	010	u_4u_1	110
u_1u_4	011	u_4u_2	1101
u_1u_5	000	u_4u_3	1110
u_2u_1	010	u_4u_4	1111
u_2u_2	0101	u_4u_5	1100
u_2u_3	0110	u_5u_1	000
u_2u_4	0111	u_5u_2	0001
u_2u_5	0100	u_5u_3	0010
u_3u_1	100	u_5u_4	0011
u_3u_2	1001	u_5u_5	0000
u_3u_3	1010	-	-

symbols, $\mathcal{U}'_1$ and $\mathcal{U}'_2$, and that $\mathcal{U}'_1$ is the sequence of source symbols formed by $\mathcal{U}_1$ followed by $\mathcal{U}_2$, while $\mathcal{U}'_2$ is the sequence of source symbols formed by $\mathcal{U}_2$ followed by $\mathcal{U}_1$. We notice that both $\mathcal{U}'_1$ and $\mathcal{U}'_2$ are sequences of equal length and that both produce the same sequence of codewords $\mathcal{Z}_0\mathcal{Z}_0$. In this case code C does not satisfy the condition to be uniquely decodable stated in Definition 9.41.

We now examine in more detail some typical uniquely decodable codes, illustrated in Example 9.42. Code $\mathcal{A}$ is an example of the simplest possible method of constructing a uniquely decodable code. Namely, all codewords in $\mathcal{A}$ have the same length and are distinct, i.e., $\mathcal{A}$ is non-singular. The two properties are sufficient to guarantee that a code is uniquely decodable (see Problem 1.Eq. 18). Code $\mathcal{B}$, in addition to being non-singular, employs the symbol 0 as a comma to separate its codewords and is thus uniquely decodable. Due to the comma, it is possible in code $\mathcal{B}$ to determine where one codeword ends and another begins. Code $\mathcal{C}$ is non-singular but differs from $\mathcal{A}$ and $\mathcal{B}$ in an aspect of practical importance. When decoding $\mathcal{C}$, on receiving a sequence of codewords, we are generally only able to detect the end of a codeword when we receive the first digit of the next codeword. The only exception to this rule occurs with the codeword 0111, the ending of which is easily identified since there is no codeword in $\mathcal{C}$ formed by a 0 followed by more than three 1's. We observe that in the codes $\mathcal{A}$ and $\mathcal{B}$, the end of each received codeword is identified immediately, that is, without the need to wait for the first digit of the next codeword. At this point in our discussion, it should be clear that the unique decodability depends essentially on our ability to determine the beginning and end of a codeword immersed in a finite sequence of codewords.

Example 9.42 In Table 9.5 three examples are presented of uniquely decodable codes.

Table 9.5 Uniquely decodable codes

Source symbols	Code $\mathcal{A}$	Code $\mathcal{B}$	Code $\mathcal{C}$
u_1	00	0	0
u_2	01	10	01
u_3	10	110	011
u_4	11	1110	0111

9.10 Prefix-Free Codes

We will begin this section by introducing the definition of a prefix-free code, so that we can then determine necessary and sufficient conditions for a code to be prefix-free.

Definition 9.43 A uniquely decodable code is also a **prefix-free code** if it is possible to decode each codeword, in a sequence of codewords, without needing to observe succeeding codeword symbols.

Example 9.44 Without going into details, we state (and the reader will be able to verify) that the codes $\mathcal{A}$ and $\mathcal{B}$ of Example 9.42 are prefix-free. Code $\mathcal{C}$, from this same example, despite being uniquely decodable, is not prefix-free.

Except for relatively simple codes, we generally need a practical criterion to test whether a given code is uniquely decodable or not. Such criterion exists and is quite simple as can be seen shortly.

Definition 9.45 Let $z_i = x_{i_1} x_{i_2} \dots x_{i_m}$ denote a codeword of a given code. The sequence of symbols $x_{i_1} x_{i_2} \dots x_{i_l}$, $1 \leq l \leq m$, in codeword z_i is called a prefix of z_i.

Example 9.46 The word 1110 has as prefixes 1, 11, 111 and 1110.

The test to verify whether a code is prefix-free is presented next in the form of a theorem.

Theorem 9.47 *A necessary and sufficient condition for a code C to be prefix-free is that none of its codewords be a prefix of another codeword in C.*

Proof The *sufficiency* condition is an immediate consequence of the definition of a prefix-free code. Supposing that no codeword is a prefix to another codeword, we proceed with decoding by scanning a received sequence of symbols from the code alphabet until we can form a subsequence which is a valid codeword. Such subsequence actually represents a single valid codeword because by hypothesis it cannot be the prefix to any other codeword. This way we can proceed with word-by-word decoding.

We will prove by contradiction the *necessary* part of the theorem. Let us then assume that, contrary to the hypothesis, there is a codeword Z_1 which is the prefix of another codeword Z_2. If we proceed with decoding by scanning a received sequence of symbols from the code alphabet we can form a subsequence Z_1, and such a

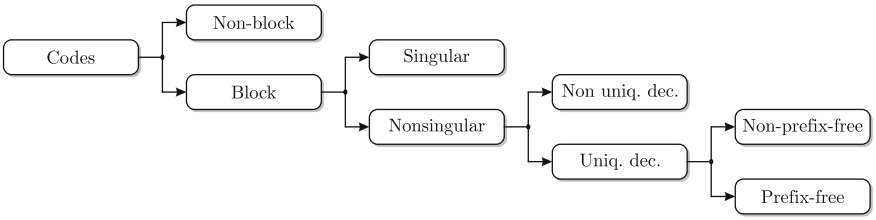


Fig. 9.5 Subclasses of source codes

Table 9.6 Construction of a prefix-free code

Source symbols	Step 1	Step 2	Step 3
u_1	0	0	0
u_2	*	10	10
u_3	*	*	110
u_4	*	*	111

subsequence may in fact be a valid codeword, or it may just be a prefix of Z_2. We will not be able to decide which of the two alternatives is true until we examine more symbols in the received sequence and this way the code is no longer prefix-free. □

Figure 9.5 schematically presents the sequence of code subclasses that takes us from a simple code to the subclass of prefix-free codes.

We will now discuss, through examples, the construction of prefix-free codes. Let us consider the construction of a prefix-free binary code for a source with four symbols. To begin with, let us allocate the word 0 to represent the symbol u_1. This choice leaves us with the option to choose only words that start with the digit 1, to represent u_2, u_3, and u_4. Obviously if we choose 1 to represent u_2, we will be unable to choose new words that do not have the 0 or 1 as a prefix. Therefore, we are forced (due to the way how u_1 was chosen) to choose a word with two digits, starting with 1, to represent u_2. Let our choice for u_2 be the word 10. We are then left with a single two-digit prefix that has not yet been used, that is, 11. As there are still two symbols left (u_3 and u_4) to be encoded and we have a single two-digit prefix not yet used, we are forced to choose a three-digit word for u_3, starting with 11. Let our choice for u_3 be word 110. Finally, our choice for u_4 is the word 111. Table 9.6 illustrates the allocation of words, and asterisks are employed to indicate positions yet void to be occupied by words yet to be selected.

9.11 Kraft Inequality

We concluded the previous section by discussing restrictions of a qualitative nature on the length of codewords in the construction of prefix-free codes. We will now deal with restrictions of a quantitative nature on the length of the codewords of such codes.

Theorem 9.48 *A necessary and sufficient condition for the existence of a prefix-free code, the codewords of which have lengths $w_1, w_2, \ldots, w_L$, respectively, is that these lengths satisfy the following condition:*

$$\sum_{i=1}^{L} D^{-w_i} \leq 1, \tag{9.55}$$

in which D denotes the number of distinct symbols in the code alphabet.

Proof We will first prove that the condition expressed in the inequality in Eq. 9.55 is enough to build a prefix-free code that satisfies it. Let us suppose we are given the set of codeword lengths $w_1, w_2, \ldots, w_L$ satisfying the inequality in Eq. 9.55 and we are asked to construct a prefix-free code with these word lengths. The length values, that is, $w_1, w_2, \ldots, w_L$, are not necessarily all distinct. Let us denote by n_i the number of words of length i. Therefore, n_1 denotes the number of words of length 1, n_2 denotes the number of words of length 2, etc. Denote by w the length of the longest codeword, that is, let $w = \max\ w_i,\ 1 \leq i \leq L$, and thus

$$\sum_{i=1}^{w} n_i = L.$$

Alternatively, we can write the inequality in Eq. 9.55 as

$$\sum_{i=1}^{w} n_i D^{-i} \leq 1. \tag{9.56}$$

Multiplying by D^w both sides in the inequality in Eq. 9.56, we have

$$\sum_{i=1}^{w} n_i D^{w-i} \leq D^w. \tag{9.57}$$

Expanding the inequality in Eq. 9.57 and placing all its terms on the right side, we obtain

$$0 \leq D^w - n_1 D^{w-1} - n_2 D^{w-2} - \cdots - n_{w-1} D - n_w. \tag{9.58}$$

We observe that, since $n_w > 0$, we can suppress it in the inequality in Eq. 9.58 and simplify the result dividing by D the two sides and obtain

$$0 \leq D^{w-1} - n_1 D^{w-2} - n_2 D^{w-3} - \cdots - n_{w-1}. \tag{9.59}$$

Continuing in an analogous manner, starting from the inequality in Eq. 9.59 we obtain

$$0 \leq D^{w-2} - n_1 D^{w-3} - n_2 D^{w-4} - \cdots - n_{w-2} \tag{9.60}$$

$$\vdots \; \vdots \; \vdots$$

$$0 \leq D^3 - n_1 D^2 - n_2 D - n_3 \tag{9.61}$$

$$0 \leq D^2 - n_1 D - n_2 \tag{9.62}$$

$$0 \leq D - n_1. \tag{9.63}$$

The proof follows by observing that from inequalities Eqs. 9.58–9.63, we can construct a prefix-free code in the following manner. We need to form n_1 codewords of length 1. We know that there are D distinct codewords of length 1, which we can form using a D-ary code. By the inequality in Eq. 9.63 $n_1 \leq D$, and therefore we can choose these n_1 codewords arbitrarily. In doing so, we will be left with $D - n_1$ prefixes that have not yet been used, to form codewords of length 2. Adding to the right of each of these $D - n_1$ symbols (possible prefixes), a second symbol, we will have $D(D - n_1) = D^2 - n_1 D$ possible codewords of length 2. By the inequality in Eq. 9.62 we only need n_2 codewords of length 2, where $n_2 \leq D^2 - n_1 D$. When using n_2 codewords of length 2, $D^2 - n_1 D - n_2$ prefixes will remain available to form codewords of length 3. By the inequality in Eq. 9.61, we will need at most this total of codewords of length 3. Continuing with this procedure, we build all the codewords. It is therefore proven that the condition expressed by the inequality in Eq. 9.55 is sufficient to construct a prefix-free code with the lengths of the codewords being $w_1, w_2, \ldots, w_L$. To prove that the condition in the inequality in Eq. 9.55 is also necessary, we reverse the steps used in proving the sufficiency condition and arrive at the desired result. □

Example 9.49 Verification of the prefix-free condition in source codes. It follows from Fig. 9.6 that the code in b) is non-prefix-free because 1, the codeword representing u_1, is a prefix of 11 which is the codeword representing u_3.

Rooted Trees

In order to facilitate the reader to penetrate the nuances of the nature of prefix-free codes, it is interesting to represent the digits in the codewords of a given code as branches of a rooted tree. In a rooted tree nodes, leaves and root are defined. The *nodes* are vertices from which branches of the tree stem outward in the direction away from the root and *leaves* are vertices with no outgoing branches. A tree *grows* by adding branches to a direction away from a vertex with special notation called *root*. The root is denoted by the *earth* symbol used in electrical circuits.

Figure 9.6 shows the *binary trees* associated with the binary codes of Example 9.49. Codewords are represented by dark circles at the end of branches marked with the last digit of the respective codeword. Note that all codewords of prefix-free codes match the leaves of the respective tree and that codes with a prefix also have one or more codewords that correspond to *nodes*. Some relevant definitions are presented below.

a) Prefix-free code

U	Z
u_1	0
u_2	10
u_3	11

b) Non-prefix-free code

U	Z
u_1	1
u_2	00
u_3	11

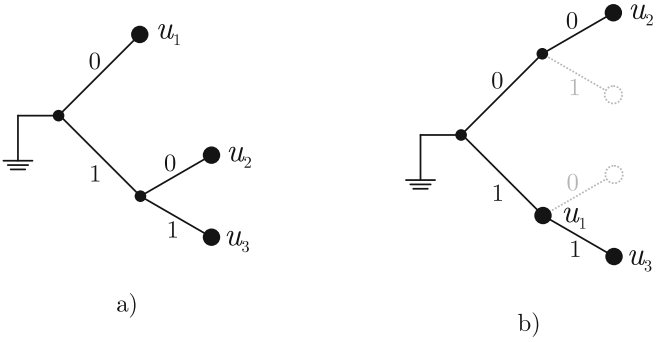


Fig. 9.6 A prefix-free code is shown in (**a**) and a non-prefix-free code is shown in (**b**)

Definition 9.50 A D-ary tree is a finite or semi-infinite rooted tree such that D branches emanate from each node of this tree in the direction away from the root.

Each of the D branches stemming outward from a node of a D-ary tree will be denoted by one of the D distinct D-ary letters, $0, 1, \ldots, D-1$ of the code alphabet.

Definition 9.51 A full D-ary tree of length N is the D-ary tree with D^N leaves, each leaf at depth N from the root.

It may now be clear to the reader that all D-ary prefix-free code can be identified with a set of leaves on a D-ary tree and that, conversely, any set of leaves in a D-ary tree defines a D-ary prefix-free code. In this text a D-ary tree is represented using the following convention: the leaves representing codewords are denoted by dark circles, and the leaves not used as codewords are denoted by hollow circles. The D-ary tree that represents a given D-ary prefix-free code is made unique by *pruning* the tree at each node from which no codewords stem.

Example 9.52 Construct a binary prefix-free code with codeword lengths $w_1 = 2$, $w_2 = 2$, $w_3 = 2$, $w_4 = 3$ and $w_5 = 4$.

Since $\sum_{i=1}^{5} 2^{-w_i} = 1/4 + 1/4 + 1/4 + 1/8 + 1/16 = 15/16 < 1$, by the Kraft inequality, it is concluded that a binary prefix-free code for this source exists. This code is constructed and shown in Fig. 9.7.

Example 9.53 Construct a binary prefix-free code with codeword lengths $w_1 = 1$, $w_2 = 2$, $w_3 = 2$, $w_4 = 3$ e $w_5 = 4$.

U	Z
u_1	00
u_2	01
u_3	10
u_4	110
u_5	1110

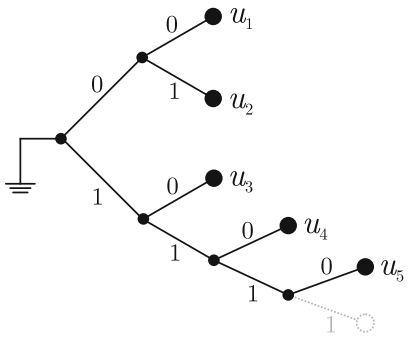


Fig. 9.7 Binary prefix-free code of Example 9.52

U	Z
u_1	0
u_2	10
u_3	11
u_4	120
u_5	1210

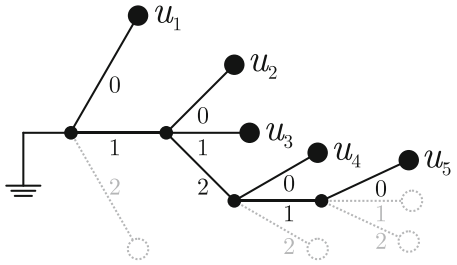


Fig. 9.8 Ternary prefix-free code of Example 9.54

Since $\sum_{i=1}^{5} 2^{-w_i} = 1/2 + 1/4 + 1/4 + 1/8 + 1/16 = 19/16 > 1$, the Kraft inequality establishes that this code cannot be constructed as specified.

Example 9.54 Construct a ternary prefix-free code with codeword lengths $w_1 = 1$, $w_2 = 2$, $w_3 = 2$, $w_4 = 3$ e $w_5 = 4$.

Since $\sum_{i=1}^{5} 3^{-w_i} = 1/3 + 1/9 + 1/9 + 1/27 + 1/81 = 49/81 < 1$, by the Kraft inequality, it is concluded that a ternary prefix-free code for this source exists. This code is constructed and shown in Fig. 9.8.

Rooted Trees with Probabilities

It is observed in a rooted tree with probabilities that the sum of the leaf probabilities is necessarily equal to one (see Example 9.57).

Lemma 9.55 (The Path Length Lemma) *In a rooted tree with probabilities, the average depth of the leaves is equal to the sum of the probabilities of the nodes, including the root node.*

Proof The probability of each node is equal to the sum of the probabilities of the leaves in the subtree stemming from that node. It turns out that a leaf at depth d is connected to the d subtrees that correspond to the d nodes, including the root node, on the path from the root to that leaf. In this way, the sum of the probabilities of

the nodes is equal to the sum of the products of the probability of each leaf by its respective depth, but this sum corresponds exactly to the average depth of the leaves according to the Eq. 9.54. □

Shannon-Fano Code

The *Shannon-Fano code* is a heuristic construction of a prefix-free source code which in general is not optimum. An intuitive way to approach the Shannon-Fano construction is obtained by initially considering the reasoning of the Hartley measure of information. In that situation, when the event $U = u_i$ occurs, everything happens as if one of $1/P_U(u_i)$ equally likely possibilities occurs. However the coding of L equally likely possibilities with D-ary words of the same length requires a codeword length of $\lceil \log_D L \rceil$ digits, where $\lceil x \rceil$ denotes the smallest integer number greater than or equal to x. This fact suggests that the length w_i, of the codeword for u_i, should be chosen as being

$$w_i = \left\lceil \log_D \frac{1}{P_U(u_i)} \right\rceil = \lceil -\log_D P_U(u_i) \rceil = \left\lceil \frac{-\log P_U(u_i)}{\log D} \right\rceil. \tag{9.64}$$

If $P_U(u_i) = 0$ occurs, the condition in Eq. 9.64 becomes meaningless. This apparent difficulty is resolved by agreeing that no codewords will be allocated to values of U that have zero probability of occurring. With this impasse resolved, two new questions arise. The first is whether there is a prefix-free code the codeword lengths of which are those indicated in Eq. 9.64. The answer to this question is obtained by observing the following inequality

$$x \leq \lceil x \rceil < x + 1. \tag{9.65}$$

It follows from Eq. 9.64 that

$$w_i \geq -\log_D P_U(u_i),$$

such that, if U is a L-ary random variable, we have

$$\sum_{i=1}^{L} D^{-w_i} \leq \sum_{i=1}^{L} D^{\log_D P_U(u_i)} = \sum_{i=1}^{L} P_U(u_i) = 1.$$

Therefore, the Kraft inequality guarantees that in fact it is possible to find a D-ary prefix-free code, the codeword lengths of which are given by Eq. 9.64. The quality of the codes thus generated can be evaluated by combining expressions Eqs. 9.64 and 9.65 resulting in this

$$\frac{-\log P_U(u_i)}{\log D} \leq w_i < \frac{-\log P_U(u_i)}{\log D} + 1. \tag{9.66}$$

Multiplying Eq. 9.66 by $P_U(u_i)$ and adding for all i, due to Eq. 9.54 it follows that

$$\frac{H(U)}{\log D} \leq E[W] < \frac{H(U)}{\log D} + 1. \tag{9.67}$$

The codes produced by this source coding method are called Shannon-Fano codes, because this technique appears implicit in Shannon's 1948 paper, but was first made explicit by Fano, and provides a prefix-free code the average codeword length of which lies within one digit of the lower bound in Eq. 9.67, satisfied by all prefix-free codes. The following theorem is thus proven.

Theorem 9.56 (Coding Theorem for a L-ary Random Variable.) *The average codeword length of an optimal prefix-free D-ary code for a L-ary random variable U satisfies the relation*

$$\frac{H(U)}{\log D} \leq E[W] < \frac{H(U)}{\log D} + 1 \tag{9.68}$$

with equality on the left if and only if the probability of each value of U is a negative integer power of D. Furthermore, the value $E[W]$ resulting from Shannon-Fano coding, which is generally not optimal, also satisfies inequalities Eq. 9.68 with equality on the left if and only if the probability of each value of U is an integer negative power of D.

Theorem 9.56 does not yet represent a complete justification for the Shannon information measure because it is not possible to make the upper bound arbitrarily close to the lower bound. However, since inequalities Eq. 9.68 are valid for any discrete memoryless source S, we may apply it to S^n, the nth extension of S, according to Eq. 9.44, i.e., $H(U^n) = nH(U)$, and obtain

$$\frac{H(U^n)}{\log D} \leq E_n[W] < \frac{H(U^n)}{\log D} + 1, \tag{9.69}$$

in which $E_n(W)$ denotes the average length of the codewords representing the symbols of S^n. Equivalently, we can write Eq. 9.69 as

$$\frac{H(U)}{\log D} \leq E_n[W]/n < \frac{H(U)}{\log D} + 1/n, \tag{9.70}$$

in which $E_n[W]/n$ represents the average number of code symbols per symbol of S. Thus it is possible to make $E_n[W]/n$ as close to $H(U)/\log D$ as we wish by coding the nth extension of S rather than coding individual symbols of S or, in mathematical terms

$$\lim_{n\to\infty} \frac{E_n[W]}{n} = \frac{H(U)}{\log D}. \tag{9.71}$$

The inequalities in Eq. 9.70 exhibit the result known as *Shannon's first theorem* or the *noiseless coding theorem*.

Example 9.57 Consider the binary Shannon-Fano code construction for the source with alphabet $U = \{u_1, u_2, u_3, u_4\}$ and symbol probability distribution as follows.

u	u_1	u_2	u_3	u_4
$P_U(u)$	0.4	0.3	0.2	0.1

Initially, Eq. 9.64 is used with $D = 2$ in order to determine the codeword lengths.

$$w_1 = \left\lceil \log \frac{1}{0.4} \right\rceil = 2$$

$$w_2 = \left\lceil \log \frac{1}{0.3} \right\rceil = 2$$

$$w_3 = \left\lceil \log \frac{1}{0.2} \right\rceil = 3$$

$$w_4 = \left\lceil \log \frac{1}{0.1} \right\rceil = 4.$$

The code is constructed by growing a tree as explained earlier in order to obtain the Shannon-Fano code with the tree illustrated in Fig. 9.9. By applying the path length lemma in Fig. 9.9, we obtain

$$E[W] = 1 + 0.7 + 0.3 + 0.3 + 0.1 = 2.4$$

and a direct calculation gives

$$H(U) = 1.846 \; bits.$$

Comment: It is observed that the condition in Eq. 9.68 is indeed satisfied. Although, clearly this code is not optimal, because it would simply be enough to choose the 4 possible codewords of length 2 which would result in a better code ($E[W] = 2$). Despite this, the coding theorem guarantees that no code can beat the average codeword length of the Shannon-Fano code by more than one digit.

Huffman Code

In the sequel, we show how to construct an optimum prefix-free D-ary code for a L-ary random variable U. We assume that $P_U(u_i) > 0$, $i = 1, 2, \ldots, L$, so that it is necessary to allocate a codeword for every possible value of U. Initially, binary coding is considered, i.e., $D = 2$. Two simple lemmas, presented next, provide the key to construct an optimum code.

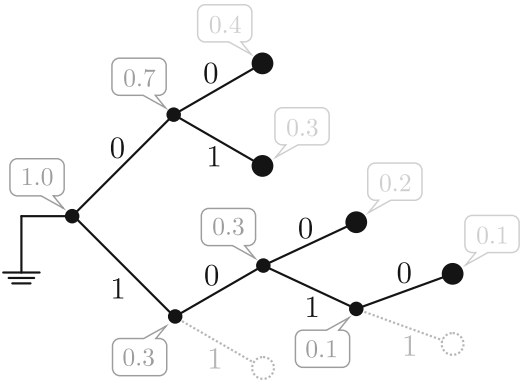


Fig. 9.9 Binary tree of the Shannon-Fano code of Example 9.57

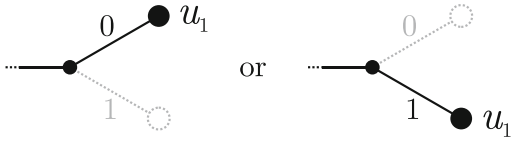


Fig. 9.10 Tree with leaf not used for a codeword

Lemma 9.58 *The binary tree of an optimum prefix-free binary code for U has no unused leaves.*

Proof Contrary to the hypothesis, suppose that the tree has unused leaves. Because the code is optimum, these unused leaves should be located at the maximum depth in the tree. Therefore, for at least one value u_i of U, the situation indicated in Fig. 9.10 will occur. In one or the other of the two cases indicated in Fig. 9.10, it is possible to suppress the last digit of the codeword for u_i, without changing the other codewords, and still have a prefix-free code. It turns out, however, that this new code has a smaller $E[W]$ value and therefore the original code could not be optimum.

□

Lemma 9.59 *There exists an optimum binary prefix-free code for U such that the two least likely codewords, without loss of essential generality, are associated with u_{L-1} and u_L and differ only in their last digit.*

Proof Suppose that $P_U(u_{L-1} \geq P_U(u_L)$. Let z_i be one of the longest codewords in an optimum code for U. So, because according to Lemma Eq. 9.58 there cannot be unused leaves in the code tree, the situation illustrated in Fig. 9.11 necessarily occurs, where u_j denotes some other value of the random variable U being encoded. So, if $j \neq L$, the leaves associated with u_j and u_L are swapped. This cannot increase $E[W]$ because $P_U(u_j) \geq P_U(u_L)$ and $w_j \geq w_L$. If $i \neq L-1$, the nodes associated with u_i and u_{L-1} are also swapped. This cannot increase $E[W]$ due to a similar argument. Because the original code is optimum, the new code is optimum too. However, the new optimum code has two less likely codewords, which differ only in the last digit.

□

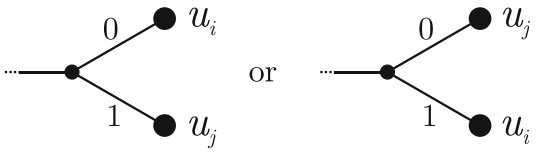


Fig. 9.11 Situation considered in the proof of Lemma Eq. 9.59, in which u_i and u_j denote values of the random variable U

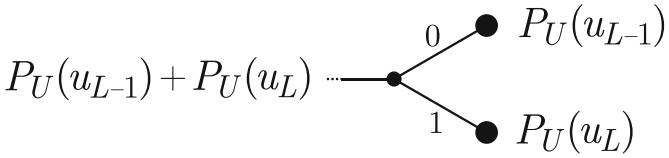


Fig. 9.12 Optimized choice of a node in a tree with probabilities

Due to Lemma Eq. 9.58 and the path length lemma, constructing an optimum prefix-free binary code for a L-ary random variable U is equivalent to constructing a binary tree with L leaves such that the sum of the node probabilities is minimum when the leaves have allocated probabilities $P_U(u_i)$ for $i = 1, 2, \ldots, L$. However, Lemma Eq. 9.59 suggests how a node can be chosen in an optimum tree of a code, specifically as indicated in Fig. 9.12, in which u_{L-1} and u_L represent the two least likely values of U. However, if the binary tree is pruned at this node, to make it a leaf with probability $P_U(u_{L-1} + P_U(u_L))$, this would become one of the $L - 1$ leaves in a new tree. Completing the code construction would be equivalent to building a binary tree with these $L - 1$ leaves such that the sum of the node probabilities is a minimum. Applying Lemma Eq. 9.59 allows you to choose the next specific node in this tree and so on. It is thus proven the validity of the following algorithm.

Huffman Algorithm
Purpose: Construction of an optimal prefix-free binary code for a L-ary random variable U, such that $P(u) \neq 0$ for all u.

Step 0: Designate L vertices (which will be the leaves in the final tree) as $u_1, u_2, \ldots, u_L$ and assign probability $P_U(u_i)$ to vertex u_i, for $i = 1, 2, \ldots, L$. Designate these L vertices as *active*.

Step 1: Create a node that interconnects the two least likely active vertices, using binary branches. Deactivate these two active vertices, activate the new node, and assign it a probability equal to the sum of the probabilities of the two vertices that have just been deactivated. This way, the number of active vertices decreases by one.

Step 2: If there is now a single active vertex, then make this vertex the root of the tree and stop. Otherwise, go to step 1.

Example 9.60 Consider the construction of a binary Huffman code for the random variable U such that

u	u_1	u_2	u_3	u_4	u_5	u_6
$P_U(u)$	0.05	0.10	0.15	0.20	0.23	0.27

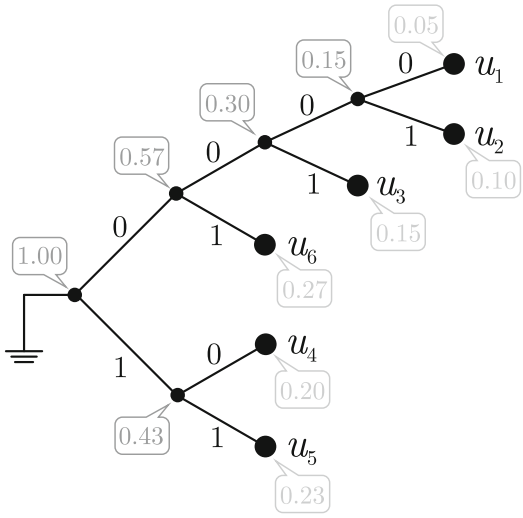


Fig. 9.13 Binary tree for the Huffman code of Example 9.60

Table 9.7 Huffman code constructed for the random variable U of Example 9.60

u	u_1	u_2	u_3	u_4	u_5	u_6
$P_U(u)$	0.05	0.10	0.15	0.20	0.23	0.27
Z	0000	0001	001	10	11	01

An optimal binary tree for U is shown in Fig. 9.13, and the corresponding Huffman code (optimum binary code) is given in Table 9.7.

It is noticed in this example that

$$E[W] = 2(0.20 + 0.23 + 0.27) + 3(0.15) + 4(0.10 + 0.05) = 2.45.$$

On the other hand,

$$H(U) = -\sum_{i=1}^{6} P_U(u) \log P_U(u) = 2.42 \; bits,$$

verifying that the inequality in Eq. 9.67 is satisfied.

9.12 Capacity of Discrete Noiseless Channels

Consider a message source U with a L symbol alphabet, for which we arbitrarily assign a uniquely decodable source code C. Consider that the L codewords of C have lengths w_i, $1 \leq i \leq L$, respectively. Following Shannon (1948a), we use the

expression *discrete channel*, in a generic manner, to refer to a system by which a *sequence of choices* of elements from a finite (or countably infinite) set of symbols $\{x_1, x_2, ..., x_D\}$ can be transmitted between two points. The electronic mail (e-mail) system is an example of a discrete channel for the transmission of information.

The capacity of a source code to transmit information in a discrete channel was defined in da Rocha Jr. (1999) as the maximum rate in bits per symbol of the code and turns out to be a function of the codeword lengths only. We assume that each of the symbols x_i, $1 \leq i \leq D$, has duration t_i seconds and that, in general, for any two symbols, x_i and x_j, $i \neq j$, we can have $t_i \neq t_j$. Telegraphic transmission gives us an example of a case where the symbols used, called *dot* and *dash*, have different durations. In general the system may have constraints, that is, the system may not allow the transmission of all possible symbol sequences from U. Just a few sequences may be allowed. These represent the possible *signals* for the channel.

In the telegraph channel, the following symbols are employed:

1 **DOT**, consisting of a pulse present during a time unit, and a pulse absent during a time unit.
2 **DASH**, consisting of a pulse present during three time units, and a pulse absent during a time unit.
3 **LETTER SPACE**, consisting of a pulse absent during three time units.
4 **WORD SPACE**, consisting of a pulse absent during six time units.

A natural restriction for telegraphy sequences is to forbid the transmission, for example, of two consecutive *letter spaces*, because they are identical to a *word space*.

We next consider the question of how we can measure the *capacity* of such channels to transmit information. For electronic mail, for example, which employs binary symbols of the same duration, and the messages of which consist of sequences of *words* of 7 binary digits, the response is immediate. If this system transmits R words per second, the natural response is that the channel has a capacity of $7R$ bits per second. This, however, does not mean that the system will be always transmitting information at this rate. The value is $7R$ represents the maximum information rate that the channel can transmit. In a more general case, where symbols have different durations and there is restriction on the sequences to be transmitted, Shannon (1948a) introduced the following definition.

Definition 9.61 The capacity $\mathcal{C}$ of a discrete noiseless channel is given by

$$\mathcal{C} = \lim_{T \to \infty} \frac{\log N(T)}{T} \tag{9.72}$$

in which $N(T)$ denotes the number of signals of duration T which are allowed.

By applying formula Eq. 9.72, it can be easily checked that for the case of electronic mail the result $\mathcal{C} = 7R$ bits per second coincides with that obtained earlier. Shannon (1948a) observed that the limit in formula Eq. 9.72 exists in the majority of cases of interest.

Capacity Without Restrictions

Let us now calculate the discrete noiseless channel capacity assuming a case in which all possible sequences of symbols $x_1, x_2, \ldots, x_D$ are allowed and that each of the symbols x_i has a certain duration in time t_i, assuming that for two distinct symbols x_i and x_j, we can have $t_i \neq t_j$. Representing by $N(t)$ the number of sequences of duration t, we have

$$N(t) = N(t - t_1) + N(t - t_2) + \cdots + N(t - t_D). \tag{9.73}$$

The total number of sequences $N(t)$ is equal to the sum of the numbers of sequences ending in x_1, or ending in x_2, ... etc., or ending in x_D, and these are, respectively, $N(t - t_1), N(t - t_2), \ldots, N(t - t_D)$.

Using the finite difference method, we find the following solution (Goldman 1968, pp. 311–317) for $N(t)$:

$$N(t) = A_1 a_1^t + A_2 a_2^t + \cdots + A_D a_D^t, \tag{9.74}$$

in which the A_i, $1 \leq i \leq D$, are arbitrary constants and the a_i, $1 \leq i \leq D$, are determined using Eq. 9.75. Notice that, in general,

$$N(t - t_i) = A_1 a_1^{t-t_i} + A_2 a_2^{t-t_i} + \cdots + A_r a_r^{t-t_i},$$

and substituting these values in Eq. 9.73, we obtain

$$\begin{aligned} & A_1 a_1^t [\, a_1^{-t_1} + a_1^{-t_2} + \cdots + a_1^{-t_D} - 1] \\ + \; & A_2 a_2^t [\, a_2^{-t_1} + a_2^{-t_2} + \cdots + a_2^{-t_D} - 1] \\ \vdots \quad & \qquad\qquad \vdots \\ + \; & A_D a_D^t [\, a_D^{-t_1} + a_D^{-t_2} + \cdots + a_D^{-t_D} - 1] = 0. \end{aligned}$$

The general solution of Eq. 9.73 is given by Eq. 9.74, in which the a_i, $1 \leq i \leq D$, are the roots of the characteristic equation

$$X^{-t_1} + X^{-t_2} + \cdots + X^{-t_D} - 1 = 0. \tag{9.75}$$

The value of $N(t)$ is necessarily real and non-negative for all positive values of t. By making t sufficiently large, Eq. 9.74 reduces to

$$N(t) = A_l a_l^t, \tag{9.76}$$

in which a_l is the largest real root in Eq. 9.75 since, asymptotically, the contribution of all other roots becomes negligible. Shannon used the notation $X_0 = a_l$. Substituting the asymptotic value of $N(t)$, given in Eq. 9.76, in Eq. 9.72 of the capacity definition, we obtain the value

$$\mathcal{C} = \log X_0.$$

By examining Eq. 9.76 it is noticed that the number of possible sequences of duration t grows exponentially for sufficiently large values of t.

Capacity with Restrictions

A very general type of restriction that can be imposed on permitted sequences is the following. Initially we will define *state* as the consecutive symbols of a segment of arbitrary but fixed length, l, formed by the symbols $x_{j_1}, x_{j_2}, ..., x_{j_l}$ of a given allowed sequence. For each state, just certain symbols from the set $x_1, x_2, ..., x_D$ can be transmitted. We assume that different states are associated with different subsets of permitted symbols.

When one of the allowed symbols is transmitted, the status changes to a new state depending on the previous state and the transmitted symbol. If the restrictions on the allowed sequences can be described by a *linear graph*, or *state diagram*, $\mathcal{C}$ exists and is calculated from the result of the following theorem (Shannon 1948a).

Theorem 9.62 *Let $b_{ij}^{(l)}$ be the duration of the lth symbol which is allowed in state i and leads to state j. Then the capacity of the discrete noiseless channel $\mathcal{C}$ is equal to* $\log W$*, being W the largest real root of the equation defined by the determinant*

$$\left| \sum_l W^{-b_{ij}^l} - \delta_{ij} \right| = 0, \tag{9.77}$$

in which $\delta_{ij} = 1$ if $i = j$ and is equal to 0 otherwise.

The code alphabet for the telegraph channel employs the following symbols: **dot** $= +-$, **dash** $= +++-$, **letter space** $= ---$ and **word space** $= -----$. Table 9.8 illustrates the states and respective state transitions for the telegraph channel.

By using Eq. 9.77 we obtain the following determinant equation for the telegraph channel which embeds the restriction on allowable sequences that no spaces follow each other; otherwise, if two letter spaces are consecutive, they are identical to a word space.

Table 9.8 Constraints on the use of telegraph symbols

	TO STATE 1	TO STATE 2
FROM STATE 1	-1	$W^{-2} + W^{-4}$
FROM STATE 2	$W^{-3} + W^{-6}$	$W^{-2} + W^{-4} - 1$

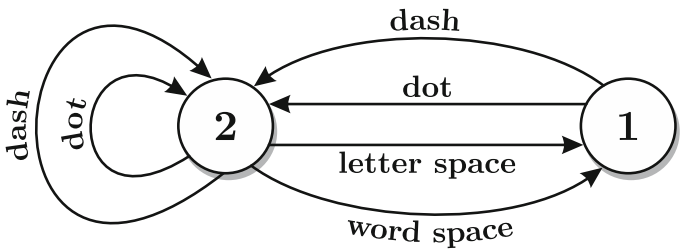


Fig. 9.14 Constraints on the use of telegraph symbols

$$\begin{vmatrix} -1 & (W^{-2}+W^{-4} \\ (W^{-3}+W^{-6} & (W^{-2}+W^{-4}-1) \end{vmatrix} = 0,$$

or, equivalently,

$$W^{-10}+W^{-8}+W^{-7}+W^{-5}+W^{-4}+W^{-2}-1=0. \tag{9.78}$$

Figure 9.14 illustrates the constraints on the use of telegraph symbols.

9.13 Capacity of Discrete Noisy Channels

The basic objective of the communications engineer is to transmit information in an efficient and reliable way, from the source, through a channel, to its destination. Having already dealt with the fundamental aspects of information sources earlier in this chapter, the next step is to address fundamental aspects of communication channels. The first question that arises naturally is: what is a *channel*?

The authors in (Wozencraft and Reiffen 1961, p. 6) attribute to John L. Kelly [1] a provocative description of a channel as being that part of a communications system that one is either *unwilling to change* or is *unable to change*. For example, if it is mandatory to use radio signals in a certain band of the frequency spectrum, then this restriction becomes part of the channel. When an individual is restricted from using communications equipment, consisting of a transceiver and antenna, such restriction also becomes part of the channel. After all restrictions have been specified, thus characterizing the channel, the freedom that remains is what can be used to transmit information.

If one is allowed the unaltered transmission of only a single signal, for example, a sinusoid with invariant amplitude, frequency, and phase, then it is not possible to transmit information to a recipient. In order for information transmission to occur,

[1] John L. Kelly was a colleague of Shannon's at Bell Laboratories, but worked in another department. Kelly worked on the *asset allocation problem* in investment markets.

freedom of choice between at least two different signal options is required. Once the choice of a signal for transmission has been exercised, from then on, the channel governs what happens with that signal during the journey to the destination point. Said in mathematical terms, the channel specifies the conditional probabilities of the various signals that can be received; however the *a priori* probabilities of the input signals are chosen by the sender using the channel.

Here only discrete time channels are considered, such that both the input and output of the channel can both be described as sequences of random variables. The input sequence $X_1, X_2, X_3, \ldots$ is chosen by the sender, but it is the channel that determines the conditional probabilities of the corresponding output sequence $Y_1, Y_2, Y_3, \ldots$. The *discrete memoryless channel* (DMC) is mathematically the simplest channel model and is the one that receives the most attention here. The characterization of a DMC requires the following three specified quantities:

1. The *input alphabet*, A, which is a finite or countably infinite set $\{a_1, a_2, a_3, \ldots\}$, each element of which represents one of the signals that a sender can choose each instant of time when using the channel;
2. The *output alphabet*, B, which is also a finite or countably infinite set $\{b_1, b_2, b_3, \ldots\}$, each element of which represents one of the output signals that can result at each instant of time in which the channel is used;
3. The *conditional probability distributions* $P_{Y|X}(.|x)$ over B for each $x \in A$, which govern channel behavior in such a way that

$$P(y_n|x_1, \ldots, x_{n-1}, x_n, y_1, \ldots, y_{n-1} = P_{Y|X}(y_n|x_n), \tag{9.79}$$

for $n = 1, 2, 3, \ldots$.

Equation 9.79 represents the mathematical condition that corresponds to the *memoryless* nature of the DMC, that is, what happens to the signal sent in the nth use of the channel is statistically independent of what happens in the previous $n - 1$ uses. Using the Eq. 9.79, it can also be observed that the DMC is invariant in the sense that the probability distribution $P_{Y_n|X_n}$ does not depend on n. For this reason $P(y_n|x_n)$ cannot be written on the right side of Eq. 9.79, because this is the abbreviated notation, used to mean $P_{Y_n|X_n}(y_n|x_n)$, and which would not imply that such probability distribution does not depend on n.

Discrete memoryless channels are generally specified using diagrams as illustrated in Fig. 9.15 in which:

1. the nodes on the left are associated with the input alphabet A;
2. nodes on the right are associated with the output alphabet B; and
3. the directed branch from a_i to b_j is associated with the conditional probability $P_{Y|X}(b_j|a_i)$, unless this probability is equal to zero, in which case this branch would be omitted.

The channel illustrated in Fig. 9.15a is called *binary symmetric channel* (BSC), while the channel illustrated in Fig. 9.15b is called *binary channel with erasure*

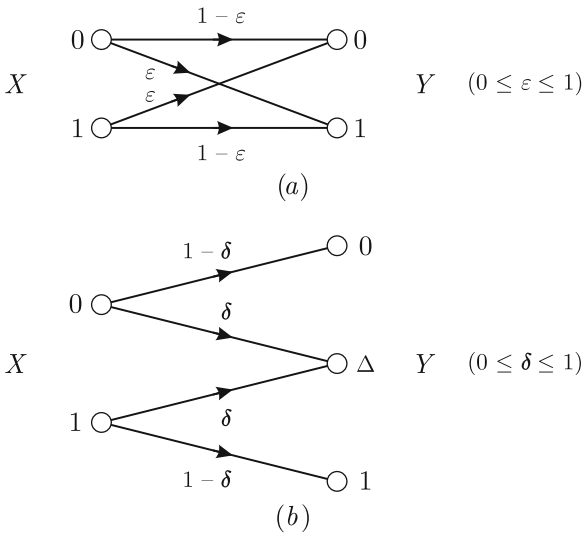


Fig. 9.15 (**a**) Binary symmetric channel (BSC). (**b**) Binary erasure channel (BEC)

(BEC). We shall give special attention to these two, apparently very simple, discrete memoryless channel models.

The following discussion is made with the purpose of illustrating the modeling of a real channel by a BSC channel. Suppose a communications system that allows transmitting a voltage pulse lasting t_0 microseconds with two possible amplitudes, for example, $-V$ or $+V$ volts, which can represent 0 or 1, respectively. Suppose that these signals are transmitted over a broadband medium, such as a coaxial cable, and are processed by a receiver. The receiver examines the received signal during consecutive intervals of t_0 microseconds and makes a *hard decision* about which voltage pulse is most likely in each interval of time considered. The use of a receiver with *soft decision*, which additionally provides information about the reliability of the decision, is generally preferable. As a consequence, receivers with *hard decision*, in general, perform worse than receivers with *soft decision*.

Due to the presence of thermal noise in the transmission medium and at the receiver input, there is a probability that the decision made is wrong. If the bandwidth of the communications system is wide enough, the errors in each time slot will be statistically independent, in such a way that the *complete system* formed by the pulse transmitter, the medium of transmission, and the receiver with hard decision becomes a BSC that is used every t_0 microseconds. Other examples could be presented to illustrate practical situations in which the BSC is the appropriate model from the information theory point of view. The beauty and strength of information theory comes from the fact that it is possible to ignore many of the technical details in practical systems and focus attention only on what really matters, related to the ability to transmit information, which are the conditional probability distributions that are created.

When working with probabilities, it is not uncommon for a student to ask how to specify a conditional probability for which the conditioning event has a probability of occurrence equal to zero. The teacher responds that, in this case, the value specified for the conditional probability is arbitrary. In general, such an answer does not leave the student completely convinced. For this reason, this situation is illustrated next using a DMC as an example. Suppose the BSC in Fig. 9.15a is being used and it has been decided that only 0's will be transmitted. Said equivalently, we choose $P_{X_n}(0) = 1$ and $P_{X_n}(1) = 0$ for all n. Even so, the channel specifies $P_{Y_n|X_n}(1|1) = 1 - \varepsilon$, even though the conditioning event has probability equal to zero by the formula $P_{X_nY_n}(1, 1)/P_{X_n}(1)$. In this channel, although the conditional probability distribution can be chosen arbitrarily when the conditioning event has a probability equal to zero, the channel is left to do such an arbitrary choice as this avoids having to treat as special cases those situations in which it is decided not to transmit one or more symbols.

The case in which DMC is used without feedback is discussed next, that is, when the inputs are selected in such a way as to satisfy the following expression:

$$P(x_n|x_1, \ldots, x_{n-1}, y_1, \ldots, y_{n-1} = P(x_n|x_1, \ldots, x_{n-1}), \tag{9.80}$$

for $n = 1, 2, 3, \ldots$. When successive input digits are chosen, note that Eq. 9.80 does not imply that each input digit is chosen independently of the previous input digits, just that the previous output digits are not being used at all, as could be the case when a feedback channel is available from the output to the input of the DMC. Next, an important result is demonstrated using Theorem Eq. 9.63.

Theorem 9.63 *When a DMC is used without feedback, then*

$$P(y_1, \ldots, y_n|x_1, \ldots, x_n) = \prod_{i=1}^{n} P_{Y|X}(y_i|x_i) \tag{9.81}$$

for $n = 1, 2, 3, \ldots$.

Proof It follows from the conditional probability multiplication rule that

$$P(x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^{n} P(x_i|x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1} P(y_i|x_1, \ldots, x_i, y_1, \ldots, y_{i-1}. \tag{9.82}$$

Applying Eqs. 9.79 and 9.80 to the right side of Eq. 9.82, we have

$$P(x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^{n} P(x_i|x_1, \ldots, x_{i-1} P_{Y|X}(y_i|x_i) =$$

$$\prod_{j=1}^{n} P(x_j|x_1, \ldots, x_{j-1} \prod_{i=1}^{n} P_{Y|X}(y_i|x_i) = P(x_1, \ldots, x_n) \prod_{i=1}^{n} P_{Y|X}(y_i|x_i),$$

and after dividing both sides by $P(x_1, \ldots, x_n)$ we obtain Eq. 9.81. □

It is observed here that the relationship Eq. 9.81 is so fundamental that it is often presented as the definition of DMC. The reader is cautioned to remember this fact when you are consulting the literature on information theory. When Eq. 9.81 is used, it is implied that DMC is being used without feedback.

Example 9.64 It is observed for the BSC that

$$P_{Y|X}(y|x) = \begin{cases} (1-\varepsilon) & \text{if } y = x \\ \varepsilon & \text{if } y \neq x. \end{cases}$$

Thus, when a BSC is used without feedback to transmit a block of N binary digits, it results

$$\begin{aligned} P(\mathbf{y}|\mathbf{x} &= \varepsilon^{d(\mathbf{x},\mathbf{y}}(1-\varepsilon)^{N-d(\mathbf{x},\mathbf{y} \\ &= (1-\varepsilon)^N \left(\frac{\varepsilon}{1-\varepsilon}\right)^{d(\mathbf{x},\mathbf{y}}, \end{aligned} \tag{9.83}$$

in which $\mathbf{x} = [x_1, x_2, \ldots, x_N]$ denotes the transmitted block, $\mathbf{y} = [y_1, y_2, \ldots, y_N]$ denotes the received block, and $d(\mathbf{x}, \mathbf{y}$ denotes the Hamming distance between $\mathbf{x}$ and $\mathbf{y}$, i.e., the number of positions at which the vectors $\mathbf{x}$ and $\mathbf{y}$ differ.

Channel Capacity

Recall that a DMC specifies the conditional probability distribution $P(y|x)$, but also recall that the sender is free to choose the input probability distribution $P(x)$. Therefore, it is natural to define the capacity $\mathcal{C}$ of a DMC as the maximum value of the average mutual information $I(X;Y)$ which can be obtained by choosing $P(x)$, that is,

$$\mathcal{C} = \max_{P_X} I(X;Y) \tag{9.84}$$

or, equivalently,

$$\mathcal{C} = \max_{P_X}[H(Y) - H(Y|X)]. \tag{9.85}$$

However, we now face the more challenging task of showing that this definition is useful, that is, showing that $\mathcal{C}$ provides the answer to some important practical questions regarding transmission of information. However, before demonstrating this fact, the channel capacity of some relatively simple channel models is calculated below, which nevertheless include most channels of practical interest.

Fano's Lemma

Next, the notion of *errors* in the study of information theory is introduced for the first time in this chapter. Consider that the random variable $\hat{U}$ represents an estimate of the random variable U. It is required that $\hat{U}$ takes on values in the same alphabet as U. Therefore, an error is the event in which $\hat{U} \neq U$ and the error probability, denoted by P_e, is therefore

$$P_e = P(\hat{U} \neq U). \tag{9.86}$$

After this brief characterization of the notion of error, one of the most important and interesting results in information theory is presented next, which relates P_e with the conditional uncertainty $H(U|\hat{U}$.

Lemma 9.65 (Fano's Lemma) *Suppose that U and $\hat{U}$ are L-ary random variables with the same alphabet and that the error probability P_e is defined by the Eq. 9.86, then*

$$h(P_e) + P_e \log_2(L-1) \geq H(U|\hat{U}, \tag{9.87}$$

in which the uncertainty $H(U|\hat{U})$ is expressed in bits.

Converse of the Noisy Coding Theorem

The following theorem states that it is impossible to transmit *information reliably* through a DMC at a *rate* above its capacity. Without loss of essential generality, suppose that the information to be transmitted comes from the output of a binary symmetric source (BSS), represented by the DMS with $P_U(0) = P_U(1) = 1/2$, having a rate of $H(U) = 1$ bit per source letter. Usually the letters in the binary output sequence $U_1, U_2, U_3, \ldots$ of the BSS are called *bits of information* because each letter carries a bit of information, although care should be taken in use of this abuse of terminology. Assume that L information bits are sent to their destination via N uses of the DMC without feedback, leading to a rate of transmission R defined as

$$R = L/N \; bits \; per \; use \; of \; the \; DMC. \tag{9.88}$$

Our real interest is in the fraction of these digits that are in error, i.e., in the *bit error probability*

$$P_b = \frac{1}{L} \sum_{i=1}^{L} P_{e_i} \tag{9.89}$$

in which we have used the notation

$$P_{e_i} = P(\hat{U}_i \neq U_i). \tag{9.90}$$

Theorem 9.66 (Converse of the Noisy Coding Theorem) *If information bits from a BSS are sent to the receiver at a rate R (in bits per channel use) through a DMC with capacity C (in bits per use) without feedback, then the error probability at the destination satisfies the condition*

$$P_b \geq h^{-1}\left(1 - C/R\right), \text{ if } R > C, \tag{9.91}$$

in which h^{-1} denotes the inverse binary entropy function, defined as $h^{-1}(x) = \min\{p : h(p) = x\}$, in a manner that the minimum is selected so as to make the inverse unique.

Noisy Coding Theorem for a DMC

Theorem 9.66 establishes what is impossible to do and now we turn to see what is possible when one wishes to transmit a BSS over a DMC. We consider here block coding in which each *block* of N channel digits is determined by a corresponding *block* of L information bits. The decoder makes its decisions on a block of L information bits by an examination of only the corresponding received block of N digits. The rate R, and the bit error probability, P_b, are already defined by Eqs. 9.88 and 9.89, respectively. We now introduce the block error probability, P_B, which is defined as

$$P_B = P(\hat{U}_1 \ldots \hat{U}_L \neq U_1 \ldots U_L). \tag{9.92}$$

Theorem 9.67 *Consider the transmission of information bits from a BSS to a receiver at a rate $R = L/N$, using block coding with block length N, via a DMC with capacity C (in bits per use) operating without feedback. Then, it is always possible to achieve $P_B < \varepsilon$, for any given $\varepsilon > 0$, provided that $R < C$, and that N is chosen sufficiently large and that suitable encoders and decoders are designed.*

Theorem 9.67 was the most surprising result of Shannon's article Shannon (1948a). Before this result appeared, it was believed that the only ways to make communication more reliable were:

- reduce the rate of information transmission, or
- increase the signal-to-noise ratio,

as all good telecommunications engineers knew in 1947. Theoretically, Shannon put an end to these myths forever, which however in practice proved hard to die. Under the condition of $R < C$, Shannon showed that it is possible to obtain greater reliability just by increasing the complexity of the coding system, without the need to change the signal-to-noise ratio.

Even though this is not a complex matter, the proof of Theorem 9.67 is not presented here for the following reasons:

1. The theorem lacks an important practical consideration, which does not exist when it lacks an indication of how large N must be for a given ε. That is, without considering how complex the encoder and decoder need to be in order to achieve a given reliability.
2. The proof of the theorem will be much more significant after having deepened the knowledge about channel encoder and channel decoder.

Among the breakthroughs in coding theory with great impact in practical applications, it is worth noting the discovery of turbo codes (Berrou and Glavieux 1996), the re-discovery of LDPC codes (MacKay and Neal 1996), and the discovery of polar codes (Arikan 2009).

9.14 Problems

1. Derive Eq. 9.26.
2. You are given nine apparently identical coins and a scale, with which it is possible to compare two groups of coins. Using your knowledge of information theory, prove that it is possible, through three weighings, to determine the existence or not of a fake coin among the nine coins, and determine whether the eventual fake coin is lighter or heavier than the others.
3. Suppose the random vector $[X_1, X_2, X_3]$ takes the values $[0, 0, 0]$, $[0, 1, 1]$, $[1, 0, 1]$ and $[1, 1, 0]$, each of them with probability 1/4.

   (i) Compute $H(X_1X_2X_3)$

   (ii) Compute the following uncertainties:

   i. $H(X_1)$, $H(X_2)$ and $H(X_3)$
   ii. $H(X_2|X_1 = 0)$, $H(X_2|X_1 = 1)$ and $H(X_2|X_1)$
   iii. $H(X_3|X_1X_2 = 00)$, $H(X_3|X_1X_2 = 01)$

   (iii) Verify, using (a) and (b), that the expression

   $$H(X_1X_2X_3) = H(X_1) + H(X_2|X_1) + H(X_3|X_1X_2)$$

   is satisfied.

4. Show that conditional uncertainties also have expansions analogous to the Eq. 9.35, i.e., show that

   $$(a)\quad H(X_1X_2\ldots X_N|Y = y) = H(X_1|Y = y) + H(X_2|X_1, Y = y) + \cdots + H(X_N|X_1\ldots X_{N-1}, Y = y),$$

   $$(b)\quad H(X_1X_2\ldots X_N|Y) = H(X_1|Y) + H(X_2|X_1Y) + \cdots + H(X_N|X_1\ldots X_{N-1}Y),$$

$$(c)\quad H(X_1X_2\ldots X_N|Y,Z=z)=H(X_1|Y,Z=z)+$$
$$H(X_2|X_1Y,Z=z)+\cdots+H(X_N|X_1\ldots X_{N-1}Y,Z=z).$$

5. State the conditions for equality in the inequalities Eqs. 9.41 and 9.42.
6. A memoryless binary source U outputs the symbol u_1 with probability 0.8 and symbol u_2 with probability 0.2. Successive symbols are output independently at a rate of 80 symbols per minute. A binary channel is available, that is, it transmits 0's and 1's, without restrictions, at a rate of 1 symbol per second.

   (i) What is the information rate of U, measured in bits per second?
   (ii) Is it possible to transmit the information generated by U through the given binary channel? Justify your answer.
   (iii) If the answer to the previous question is yes, construct a source code which allows the transmission of the information generated by U.

7. For the ergodic Markov source in Example 9.27, calculate the following:

   (i) The probability distribution for the states of the Markov source.
   (ii) The probability of 0's and 1's being emitted by the ergodic Markov source from Example 9.27.

8. Compute the entropy of the ergodic Markov source from Example 9.27. Hint: use the state probabilities from Problem 7.
9. Define a memoryless (or zero-memory) message source.
10. Define the extension of order n of a memoryless source U and show that its entropy, $H(U^n)$, is equal to $nH(U)$, in which $H(U)$ denotes the entropy of the memoryless source.
11. Define a Markov message source.
12. Define adjoint source and explain what is meant by an extension of a Markov source.
13. Show that it is also valid for the nth extension of a Markov source with entropy $H(U)$ a result analogous to that for memoryless sources (see Problem 10), i.e., $H(U^n)=nH(U)$.
14. A second-order binary Markov source has the following transition probabilities: $P(0/00)=P(1/11)=p,\ P(0/01)=P(1/10)=q$.

    (a) Calculate the stationary probability distribution of the states of this source.
    (b) Calculate the source entropy $H(U)$.
    (c) Calculate the entropy of the adjoint source $H(\overline{U})$.

15. Let U_0 be a binary memoryless source, for which the probability of the symbol 0 is equal to p (almost equal to 1). In view of the high occurrence of zeros a source code is proposed for transmitting the length of the sequences of consecutive zeros. That is, a new source U is considered with symbols $u_1, u_2, u_3, \ldots$, for which u_1 corresponds to the symbol 1 from U_0, u_2 corresponds to the sequence 01, u_3 corresponds to the sequence 001, etc.

    a. Calculate the entropy $H(U)$ as a function of p.

b. Calculate the ratio $H(U)/H(U_0)$.

16. Prove that the entropy $H(\overline{U})$ of the adjoint source $\overline{U}$ is never less than the entropy $H(U)$ of the source U, when U is a Markov source of order m.
17. Define a uniquely decodable code.
18. Prove that code $\mathcal{A}$ in Example 9.42 is uniquely decodable. Generalize this result by proving that the following properties are sufficient for a block code to be uniquely decodable.

    (a) All codewords have the same length.
    (b) The code is non-singular.

19. Show how to construct a prefix-free code.
20. Prove Kraft inequality.
21. Find the largest real solution of the characteristic equation as presented in Eq. 9.78 and determine the capacity of the telegraph channel.
22. It is common in several cryptographic systems to eliminate the spaces between words in the plain text in order to make the enemy cryptanalyst's work more difficult. Calculate the information transmission capacity, measured in bits per unit of time, of the code that uses the symbols: **dot** $= +-$, **dash** $= +++-$, and **letter space** $= ---$, without any other additional restriction.
23. Mutual information.
    A code C, consisting of just two equiprobable codewords, 000 and 111, is used to transmit messages through a BSC with transition probability ε. Calculate the mutual information between the input and the output of this BSC, when the codeword 000 is sent and the sequence 100 is received.
24. Capacity of discrete memoryless channels (DMC).
    Determine the capacity of the DMC channels shown in Fig. 9.16.
25. In a binary (0, 1) transmission, suppose that 1 occurs with probability p, $0 < p < 1$, and that the digits of this sequence are statistically independent. The probability of a binary sequence of length R, in which the first $R-1$ digits are 0 and the Rth digit is 1 is given by

$$P_R(r) = p(1-p)^{r-1},\ r = 1, 2, 3, \ldots.$$

    (a) Show that the expected value of the length of this sequence is $E[R] = 1/p$ and its variance is $\mathrm{Var}(R) = (1-p)/p^2$.

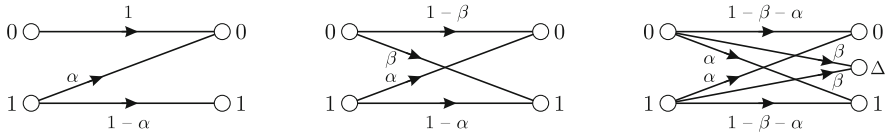


Fig. 9.16 Discrete memoryless channels of Problem 24

(b) Find the uncertainty $H(R)$ in terms of both the binary entropy function and $E[R]$.

26. Coding for the binary erasure channel (BEC).
Suppose we have a binary symmetric memoryless source U and a BEC. Suppose the encoder for U can observe the outputs of this BEC and constructs a variable length code described as follows. When the information symbol produced by U is a zero, the encoder keeps sending successive zeros through the channel until the channel output is a zero. If the information symbol is a 1, then the encoder keeps sending 1's through the BEC until the channel output is a 1. For each information symbol the corresponding number of channel symbols employed is a random variable.

(a) Calculate the average length of a codeword for each bit of information transmitted.
(b) What is the information transmission rate of this coding scheme, measured in information bits per unit of channel use?
(c) What is the rate of information symbols (bits) not correctly recovered after decoding?

Chapter 10
Error-Correcting Codes



If I were again beginning my studies, I would follow the advice of Plato and start with mathematics.
– Galileo Galilei

10.1 Block Codes

One way to characterize block codes is by their encoding process, which consists of segmenting a stream of message digits, to be transmitted or stored, into blocks with a fixed length and attaching to each block a fixed number of redundant digits. These redundant digits are calculated for each corresponding block of message digits, with the purpose of detecting or correcting errors or correcting erasures caused by noise on the channel during transmission or by faults in the storage media. Block codes can be classified as linear or nonlinear. In a linear block code, redundant digits result from a calculation that performs a linear combination of information digits. Linear block codes, as a consequence, benefit from mathematical tools such as linear algebra and finite field theory, or *Galois field* theory, with significant relevance in the development of algebraic decoding. In order to benefit from a Galois field structure, the code alphabet is q-ary, where q denotes a power of a prime, with predominance of $q = 2$ for binary codes. Unless otherwise noted, in the following, we will mainly consider binary linear block codes due to their importance in practical applications. A formal definition of what is meant by a *linear block code* is presented next.

Definition 10.1 A q-ary (n, k, d) linear block code consists of a set of q^k codewords of length n forming a subspace of the vector space of all q^n q-ary n-tuples, in which any two distinct codewords differ in at least d corresponding positions.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_10

10.1.1 Linear Algebra Representation

The codewords of a linear block code can be represented by vectors with n components. The codeword components are in general elements of a q-ary finite field, denoted by GF(q) and called a *Galois field*. For $q = 2$, i.e., for binary fields, the codeword elements are represented by zero and 1. Because a linear code constitutes a q-ary subspace of the space of all q-ary n-tuples, it follows that any codeword can be represented by a linear combination of the basis vectors of the subspace, i.e., by a linear combination of linearly independent vectors. A convenient representation for the basis vectors consists in writing them as rows of a matrix, called the code *generator matrix* (Lin and Costello 2004, p. 67). The code generator matrix $\mathbf{G}$ has k rows and n columns, and from $\mathbf{G}$, a matrix $\mathbf{H}$ is derived, with $n - k$ rows and n columns, which is orthogonal to the row space of $\mathbf{G}$. Equivalently, if $\mathbf{v}_i$ is a codeword (or vector) in the row space of $\mathbf{G}$, then

$$\mathbf{v}_i \mathbf{H}^T = \mathbf{0},\ 0 \leq i \leq q^k - 1,$$

in which $\mathbf{H}^T$ denotes the transpose of $\mathbf{H}$ and $\mathbf{0}$ denotes the all-zero $(n - k)$-tuple. The $\mathbf{H}$ matrix, also called the code *parity-check matrix*, can be represented as

$$\mathbf{H} = [\mathbf{h} : I_{n-k}],$$

in which $\mathbf{h}$ denotes an $(n - k) \times k$ matrix and I_{n-k} is the $(n - k) \times (n - k)$ identity matrix. The $\mathbf{G}$ matrix can be written as

$$\mathbf{G} = [I_k : \mathbf{g}], \tag{10.1}$$

called reduced echelon form of $\mathbf{G}$ (Peterson and Weldon Jr. 1972, pp. 45-46), in which $\mathbf{g}$ denotes a $k \times (n - k)$ matrix and I_k denotes the $k \times k$ identity matrix. The $\mathbf{g}$ and $\mathbf{h}$ matrices are related by the expression $\mathbf{g} = \mathbf{h}^T$. Since the rows of $\mathbf{H}$ are linearly independent, they generate a $(n, n - k, d')$ linear code called the *dual code* of the (n, k, d) code generated by $\mathbf{G}$. The code generated by $\mathbf{H}$ constitutes the dual subspace of the code generated by $\mathbf{G}$. Thus, it follows that an encoder for a linear block code has the function of performing the product $\mathbf{mG}$ of a row matrix $\mathbf{m}$, with k elements which represent the information digits, by the code generator $\mathbf{G}$ matrix. The product $\mathbf{mG}$ gives as a result a linear combination of the rows of $\mathbf{G}$ and is thus a codeword.

10.1.2 Error Detection and Error Correction

In order to characterize the error detection or the error correction power of a code, we need first define *Hamming weight* of an n-tuple and the *Hamming distance* between two n-tuples.

Definition 10.2 The Hamming weight $W_H(\mathbf{v})$ of an n-tuple $\mathbf{v}$ is defined as the number of nonzero coordinates in $\mathbf{v}$.

Definition 10.3 The Hamming distance between two n-tuples, $\mathbf{v}_1$ and $\mathbf{v}_2$, denoted as $d_H(\mathbf{v}_1, \mathbf{v}_2)$, is defined as the number of coordinate positions is which these two n-tuples differ.

The modulo-q sum of any two codewords of a linear code over GF(q) produces a codeword, as a consequence of the code linearity. Equivalently, if $\mathbf{v}_i$ and $\mathbf{v}_j$ are codewords then $\mathbf{v}_i + \mathbf{v}_j = \mathbf{v}_l$, where $\mathbf{v}_l$ is a codeword. It follows from the definition of Hamming distance and the definition of Hamming weight that $d_H(\mathbf{v}_i, \mathbf{v}_j) = W_H(\mathbf{v}_i - \mathbf{v}_j) = W_H(\mathbf{v}_s)$, where $\mathbf{v}_s$ is a codeword.

Definition 10.4 The minimum distance of a code is defined as the smallest Hamming distance between pairs of distinct codewords.

It follows from the linearity property of a code that to determine its minimum distance means to find the minimum nonzero Hamming weight among the codewords. So, for a linear code with M codewords, instead of making $M(M-1)/2$ operations of addition modulo-q and the corresponding Hamming weight calculation, it is sufficient to calculate the Hamming weight of the $M-1$ nonzero codewords to obtain the code minimum distance. If a linear code satisfies additional mathematical properties, the calculation of the code minimum distance or the determination of upper or lower bounds for the minimum distance can be further simplified. Additionally, the minimum distance d of a code can be interpreted as the minimum number of coordinate changes necessary to convert a codeword into another codeword. Therefore, the occurrence of up to $d-1$ errors in a codeword can be detected, because the resulting n-tuple does not match any valid codeword. When performing error correction, after errors are detected in a codeword, the decoder must decide which codeword is more likely be the correct one. Very often, it is assumed that the codewords are equiprobable, and in this case, the decoder decides in favor of the codeword nearest to the received n-tuple, i.e., the codeword at minimum Hamming distance from the received n-tuple. This approach to decoding will always make the correct decision as long as up to t errors occur in a codeword and t satisfies the condition $2t + 1 \leq d$.

10.1.3 Standard Array Decoding

Suppose a codeword $\mathbf{v}$ of a linear block code is transmitted through a noisy channel or stored in a noisy media. At the receiver, the signal associated with $\mathbf{v}$ is processed on arrival to produce an n-tuple $\mathbf{r}$ defined over the code alphabet, and $\mathbf{r}$ may differ from $\mathbf{v}$ due to the noise added during transmission or storage. The decoder has the task of recovering $\mathbf{v}$ from $\mathbf{r}$, and as a first step, it checks whether $\mathbf{r}$ is a codeword. For a code with generator matrix $\mathbf{G}$ and parity-check matrix $\mathbf{H}$, this checking process can be implemented as follows:

$$\mathbf{r}\mathbf{H}^T = \mathbf{s},$$

in which $\mathbf{s}$ denotes a vector with $n - k$ components, called *syndrome*. If the $n - k$ components in $\mathbf{s}$ are equal to zero, denoted as $\mathbf{s} = \mathbf{0}$, we assume that no errors occurred, which implies $\mathbf{r} = \mathbf{v}$.

However, $\mathbf{s} \neq \mathbf{0}$ implies $\mathbf{r} \neq \mathbf{v}$, and more than that, $\mathbf{r}$ does not match any codeword in the row space of $\mathbf{G}$. The decoder can then use this nonzero syndrome for detection or for detection and correction. Let the received n-tuple $\mathbf{r}$ be written as

$$\mathbf{r} = \mathbf{v} + \mathbf{e},$$

where $+$ denotes modular addition of corresponding components with $\mathbf{e}$ defined over the code alphabet and denoting the error pattern. Notice that an undetectable error will occur if $\mathbf{e} = \mathbf{v}'$, in which $\mathbf{v}'$ denotes a codeword because then $\mathbf{s} = \mathbf{0}$. However, the code is designed such that $\mathbf{e} = \mathbf{v}'$ is a rare event and in practice has a negligible impact on the error rate. The decoder task is to perform a process leading to a decision on which codeword was sent to the receiver. A systematic way for the decoder to implement the decision process, assuming a binary code, is to distribute the 2^n distinct n-tuples into 2^k disjoint sets, each set having cardinality 2^{n-k}, in a manner that each one of the disjoint sets contains only one codeword. The decoder decides correctly if the received n-tuple $\mathbf{r}$ is in the subset of the codeword sent to the receiver. We now describe one way of doing this. The 2^n binary n-tuples are separated into cosets as follows. The 2^k codewords are written in the first row then, below the all-zero codeword, place an n-tuple $\mathbf{e}_1$ which is not present in the first row. Form the second row by adding modulo-2 to $\mathbf{e}_1$ the codewords of the first row as follows:

$$\begin{array}{cccccc} \mathbf{0} & \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_{2^k-1} \\ \mathbf{e}_1 & \mathbf{e}_1 \oplus \mathbf{v}_1 & \mathbf{e}_1 \oplus \mathbf{v}_2 & \cdots & \mathbf{e}_1 \oplus \mathbf{v}_{2^k-1}, \end{array}$$

in which $\oplus$ denotes modulo-2 addition of corresponding coordinates. Subsequent rows are formed similarly, and each new row begins with an element not present in the previous rows. In this manner, the array presented in Table 10.1 is constructed, which is called the *standard array*.

The standard array rows are called *cosets*, and the element furthest left in each coset is called a *coset leader*. The procedure used to construct the standard array for a given linear code is called the *coset decomposition* of the vector space of n-tuples over GF(q). When using the standard array, it is necessary to find the row and, therefore, the associated coset leader, to which the incoming n-tuple belongs. There are situations in which this procedure is not practical to implement, for example, when 2^{n-k} is large. Consequently, the concept of standard array proves more useful as a way to understand the structure of linear codes, rather than as a practical decoding algorithm. A few decoding procedures which are potentially more practical are presented next.

Table 10.1 A binary linear code of block length n having codewords $\mathbf{v}_i$, $0 \leq i \leq 2^k - 1$, with $\mathbf{v}_0 = \mathbf{0}$, used to construct the standard array decomposition of an n-dimensional vector space over GF(2)

$\mathbf{0}$	$\mathbf{v}_1$	$\mathbf{v}_2$	$\cdots$	$\mathbf{v}_{2^k-1}$
$\mathbf{e}_1$	$\mathbf{e}_1 \oplus \mathbf{v}_1$	$\mathbf{e}_1 \oplus \mathbf{v}_2$	$\cdots$	$\mathbf{e}_1 \oplus \mathbf{v}_{2^k-1}$
$\mathbf{e}_2$	$\mathbf{e}_2 \oplus \mathbf{v}_2$	$\mathbf{e}_2 \oplus \mathbf{v}_2$	$\cdots$	$\mathbf{e}_2 \oplus \mathbf{v}_{2^k-1}$
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$
$\mathbf{e}_{2^{n-k}-1}$	$\mathbf{e}_{2^{n-k}-1} \oplus \mathbf{v}_1$	$\mathbf{e}_{2^{n-k}-1} \oplus \mathbf{v}_2$	$\cdots$	$\mathbf{e}_{2^{n-k}-1} \oplus \mathbf{v}_{2^k-1}$

10.1.4 Maximum Likelihood Decoding

Let $\mathbf{v}$ denote a codeword that is transmitted through a noisy channel and let $\mathbf{r}$ denote the corresponding word arriving at the receiver. The conditional probability of $\mathbf{r}$ being received when $\mathbf{v}$ is the transmitted codeword is denoted as $P(\mathbf{r}|\mathbf{v})$. In statistics, the conditional probability $P(\mathbf{r}|\mathbf{v})$ is called the *likelihood function*. If all codewords have the same probability of being selected for transmission, then it follows from the probability formula:

$$P(\mathbf{v}, \mathbf{r}) = P(\mathbf{v})P(\mathbf{r}|\mathbf{v})$$

that $P(\mathbf{v}, \mathbf{r})$ is maximized when the decoder selects that codeword $\mathbf{v}$ which maximizes $P(\mathbf{r}|\mathbf{v})$. Therefore, if the codewords of a (n, k, d) code are equally likely to occur and are independently selected to be sent through a noisy channel, an optimum way to decode them is by deciding for that codeword which maximizes the likelihood function. Let $\mathbf{r}$ denote a received n-tuple. The task of the decoder is to compare $\mathbf{r}$ with all possible codewords. For a binary (n, k, d) code, this means comparing $\mathbf{r}$ with 2^k distinct codewords. The decoder then decides for the codeword nearest to $\mathbf{r}$ in terms of the Hamming distance, i.e., decoding is completed when the decoder selects the codeword that differs from $\mathbf{r}$ in the least number of positions. The decoder implementation operates by comparing $\mathbf{r}$ with 2^k codewords during a time interval corresponding to the duration of n channel digits. This fact sets a practical limit to the use of this decoding process, i.e., it is not appropriate in many practical cases even for moderate values of k. Again, a practical limit for the use of this decoding process occurs due to high decoder complexity if one chooses to trade search time by a parallel decoder implementation.

10.1.5 Probabilistic Decoding

Important probabilistic decoding algorithms have appeared in the literature since the early 1960s. A probabilistic decoding algorithm can operate on non-quantized values of the received n-tuple; however, for practical reasons, channel output

quantization is employed. *Hard-decision* decoding is the name given to a decoding technique in which the channel output is quantized to a number of levels equal to that of the code alphabet; otherwise, it is called a *soft-decision* decoding technique. For example, when a binary code is employed and the channel output is quantized to two levels, the decoding technique is called hard-decision, and if instead the quantization employs more than two levels, the decoding technique is called a *soft-decision* decoding technique. It was introduced in Hartmann and Rudolph (1976) an optimal probabilistic decoding algorithm, in the sense that it minimizes the probability of error per digit, with the assumption that the codewords are equiprobable and are transmitted in the presence of additive noise in a memoryless channel. The Hartmann-Rudolph (HR) decoding algorithm is exhaustive in that every codeword of the dual code is used in the decoding process. This feature makes the HR decoding algorithm practical for use with high rate codes, differing in this manner to what happens with some other decoding techniques. In Wolf (1978), an efficient maximum likelihood decoding algorithm was introduced which consists of a rule to traverse a trellis type structure which is dependent on the code parity-check $\mathbf{H}$ matrix. Based on the received n-tuple, the decoder determines the most probable path through the trellis, i.e., the most probable codeword. In Honary and Markarian (1998), the authors address trellis decoders for practical applications employing block codes. Later in this chapter, a brief view of the probabilistic decoding of low-density parity-check (LDPC) codes is presented (MacKay and Neal 1996).

10.2 Simple Block Codes

Codes constructed with a relatively simple structure are described in the sequel to help the reader become familiar with this approach and later understand more sophisticated code constructions.

10.2.1 Repetition Codes

A binary (n, k, d) repetition code has just two codewords, namely, the all-0 codeword and the all-1 codeword, the block length n is arbitrary and the other code parameters are: $k = 1$, $d = n$. The parity-check digits are a repetition of the information digit. A simple decoding rule for this type of code consists in counting which binary digit occurs more often in a received word and declare it as the information digit transmitted. This decoding rule assumes that the probability of a bit being in error is less than $1/2$, obviously, and works fine for n odd. If n is even, a tie may occur when counting of occurrence of zeros and 1's, in which case the decoder declares that errors have been detected. The rate or efficiency of a repetition code is given by $R = k/n = 1/n$, and the number t of correctable errors must satisfy the condition $t \leq \lfloor (n-1)/2 \rfloor$.

10.2.2 Single Parity-Check Codes

A binary (n, k, d) single parity-check code has parameters $k = n-1$, $d = 2$, and the block length n is arbitrary. The single parity-check digit is calculated so as to make even the number of 1's in the codeword, i.e., make the parity-check digit equal to 1 when the number of information digits in a codeword is odd, otherwise make it equal to 0. The rate or efficiency of a single parity-check code is given by $R = k/n = (n-1)/n$, and its Hamming distance is $d = 2$. A simple decoding rule for single parity-check codes consists in counting the number of 1's in the received word. If the resulting count is an even number, the received word is assumed to be free of errors and is delivered to the data sink. Otherwise, the received word is declared not free from error, and the receiver declares *error detected* and notifies the data recipient. Single parity-check codes detect an odd number of errors in a received word and are more effective when decoded with a soft-decision algorithm. In general, codes which detect errors can be efficiently employed when an ideally error-free feedback channel is available to request retransmission of messages detected to contain error.

10.2.3 Hamming Codes

The first nontrivial codes for correcting errors were proposed in Hamming (1950) and are called Hamming codes. Hamming codes are linear (n, k, d) codes with a minimum distance $d = 3$, block length $n \leq 2^{n-k} - 1$, where $n - k$ is the number of redundant digits, and are thus capable of correcting one error per codeword. The condition on n guarantees enough redundancy to detect and correct an error in a codeword, because the number of nonzero syndromes, $2^{n-k} - 1$, is always greater than or equal to the number of positions where an error can be.

Example 10.5 Consider the construction of the $(n, k, d) = (7, 4, 3)$ Hamming code. The number of parity-check digits in this code is $n - k = 3$. Numbers representing powers of 2, i.e., 2^j , $0 \leq j \leq 2$, are associated with parity-check positions c_i, $1 \leq i \leq 3$, in a codeword. Other codeword positions are associated with information digits k_i, $1 \leq i \leq 4$, as indicated in Table 10.2. Now, looking downward at the columns in Table 10.2, the parity-check equations denoted as c_i, $1 \leq i \leq 3$, are written as modulo-2 sums of information positions where a 1 appears in the particular column considered. That is,

$$c_1 = k_1 \oplus k_2 \oplus k_4$$

$$c_2 = k_1 \oplus k_3 \oplus k_4$$

$$c_3 = k_2 \oplus k_3 \oplus k_4.$$

Table 10.2 Nonzero numbers that can be written with $n - k = 3$ binary digits and associated codeword positions in the $(7, 4, 3)$ Hamming code

0	0	1	c_1
0	1	0	c_2
0	1	1	k_1
1	0	0	c_3
1	0	1	k_2
1	1	0	k_3
1	1	1	k_4

When a word is received, its parity-check digits are recalculated by the decoder and are then added modulo-2 to their corresponding parity-check digits in the received word in order to obtain the syndrome. If, for example, an error has altered the digit k_2, the syndrome digits in positions c_1 and c_3 will be 1 indicating failure, while in position c_2, no failure is indicated because c_2 does not check k_2. This syndrome is represented as

$$(c_3, c_2, c_1) = (1, 0, 1),$$

which corresponds to the row for k_2 in Table 10.2. The error in k_2 has been identified and can then be corrected. Obviously, this construction and decoding procedure can be applied to Hamming codes of any length $n = 2^k - 1$ and to their shortened versions.

In an important sense, Hamming codes are special as no other class of nontrivial codes can be so easily decoded and, furthermore, Hamming codes are perfect codes as defined next.

Definition 10.6 An error-correcting code over GF(q) with parameters (n, k, d) which corrects t errors per codeword is defined as perfect if and only if the following condition is satisfied:

$$\sum_{i=0}^{t} (q-1)^i C_n^i = q^{n-k}.$$

The only nontrivial linear perfect codes are the Hamming codes, the binary $(23, 12, 7)$ Golay code, and the $(11, 6, 5)$ ternary Golay code. There are no other nontrivial linear perfect codes (Pless 1982, p.20). A class of nonlinear single error-correcting codes with the same parameters as the Hamming codes was published in Vasil'ev (1962).

10.2.4 Low-Density Parity-Check Codes

Turbo codes (Berrou et al. 1993), publicly presented in 1993, were the first capacity approaching practical codes. Three years after the discovery of turbo codes, a class of linear codes called low-density parity-check (LDPC) codes were rediscovered (MacKay and Neal 1996) and turned out to be the strongest competitor to turbo codes at that time. Later, LDPC codes were adopted in the 4G and 5G generations of mobile communications. Originally, LDPC codes were discovered by Gallager in 1960 (Gallager 1963), who presented a decoding algorithm with a complexity that increases linearly with block length. When LDPC codes were discovered, the means to perform computer simulation or practical implementation were not available. In 1981, a graphical representation of LDPC codes was introduced (Tanner 1981) which significantly contributed to their rediscovery, followed by further theoretical advances, namely, in MacKay (1999) it was shown that long LDPC codes achieve a performance close to the Shannon capacity with a low error rate, when iterative decoding is employed. Among the advantages of LDPC codes with respect to turbo codes, it is worth mentioning that LDPC codes do not require a long interleaver in order to achieve lower error rates, and their error floor occurs at lower bit error rates for a decoder complexity comparable to that of turbo codes. An LDPC code is characterized by its parity-check matrix as follows.

Definition 10.7 A binary LDPC code is defined as the set of codewords that satisfy a parity-check matrix $\mathbf{H}$, where $\mathbf{H}$ has ρ 1's per row and γ 1's per column. The number of 1's in common between any two columns in $\mathbf{H}$, denoted by λ, is at most 1, i.e., $\lambda \leq 1$.

In Definition 10.7, the parameters ρ and γ denote positive integers, where ρ is small in comparison with the code block length and γ is small in comparison with the number of rows in $\mathbf{H}$. A number of good LDPC codes were constructed by computer search after their rediscovery (MacKay and Neal 1996), which were mainly non-systematic codes, causing them to lack in further mathematical structure and consequently to have more complex encoding than naturally systematic LDPC codes. A more practical construction using systematic algebraic LDPC codes was introduced in Kou et al. (2001) based on finite geometries.

10.3 Cyclic Codes

From the point of view of practical engineering applications, cyclic codes are the most important (Clark Jr. and Cain 1981, p.333) among block codes. Cyclic codes are used in magnetic recording (Immink 1994), communication protocols (Nguyen et al. 1992), music (Compact Disc), video (Digital Versatile Disc), coding schemes for space application, etc. Basically, this is due to the fact that cyclic codes have a mathematical structure based on finite fields and that allows lower complexity

encoders and decoders for these codes. A formal treatment of cyclic codes employs polynomial rings, with polynomials over a Galois field GF(q), modulo $x^n - 1$, in which n denotes the code block length and q is a power of a prime (Berlekamp 1968, p.119). A simpler way to characterize and define a cyclic code is as follows.

Definition 10.8 A cyclic code is a block code that is invariant to cyclic shifts of its codewords, i.e., a cyclic shift applied to any codeword produces a codeword in the same code. Namely, if $\mathbf{v} = (v_0, v_1, v_2, \ldots, v_{n-1})$ is a codeword, then a cyclic shift by i places to the right applied to $\mathbf{v}$ produces a codeword in the same code, i.e.,

$$\mathbf{v}^i = (v_{n-i}, v_{n-i+1}, \ldots, v_0, v_1, \ldots, v_{n-i-1})$$

in which indices are reduced modulo n.

We call the reader's attention to the fact that a cyclic code in Definition 10.8 is not necessarily a linear code. In order to be a linear code, a cyclic code must satisfy the same property as linear block codes, i.e., a linear combination of codewords always produces a codeword belonging to the codebook of the cyclic code considered. A useful representation of an n-tuple $\mathbf{v} = (v_0, v_1, v_2, \ldots, v_{n-1})$ results by employing a polynomial of degree at most $n - 1$ as follows:

$$v(x) = v_0 + v_1 x + v_2 x^2 + \cdots + v_{n-1} x^{n-1}.$$

Based on properties of finite fields, it can be shown that the codewords of a (n, k, d) cyclic code are multiples of a polynomial $g(x)$, of degree $n - k$, called the code *generator polynomial* and conversely that all polynomials of degree at most $n - 1$ which are divisible by $g(x)$ are codewords of this code (Lin and Costello 2004, p.140). Furthermore, $g(x)$ is a factor of $x^n - 1$.

10.3.1 Matrix Representation of a Cyclic Code

Because each codeword of a cyclic code is a multiple of the code generator polynomial $g(x)$, as mentioned earlier, it follows that the polynomials $g(x)$, $xg(x)$, $x^2 g(x), \ldots,$ $x^{k-1} g(x)$ are codewords and are also linearly independent. Therefore, the coefficients of the polynomials $g(x)$, $xg(x)$, $x^2 g(x)$, $\ldots$, $x^{k-1} g(x)$ can be used to construct a generator matrix $\mathbf{G}$ for the cyclic code which has $g(x)$ as its generator polynomial. Each row of $\mathbf{G}$ is associated one to one with one of the k polynomials $g(x)$, $xg(x)$, $x^2 g(x)$, $\ldots$, $x^{k-1} g(x)$ and contains n elements, consisting of the coefficients of the corresponding polynomial, while the remaining empty positions in a row are filled with zeros. The cyclic shift code preserving property of cyclic codes is useful for encoding purposes, allowing a sequential implementation of the $\mathbf{G}$ matrix as described next.

10.3.2 Shift-Register Encoder with $n - k$ Delay Cells

The $n - k$ delay cell shift-register encoder exploits the property that each codeword in a cyclic code is a multiple of the code generator polynomial $g(x)$. The implementation of this encoder performs the multiplication of the polynomial $I(x)$ by x^{n-k} to produce $I(x)x^{n-k}$, i.e., a polynomial of degree at most $n - 1$ with its nonzero terms confined to the higher order k terms. Division of $I(x)x^{n-k}$ by $g(x)$ produces as a result

$$I(x)x^{n-k} = Q(x)g(x) + R(x),$$

in which $Q(x)$ and $R(x)$ denote the quotient polynomial and the remainder polynomial, respectively. The remainder polynomial $R(x)$ has degree at most $n - k - 1$. By subtracting $R(x)$ from $I(x)x^{n-k}$, the result is a multiple of $g(x)$, and a codeword is thus generated, i.e.,

$$I(x)x^{n-k} - R(x) = Q(x)g(x).$$

In this manner, the parity-check digits are represented by $R(x)$ which has no terms overlapping with $I(x)x^{n-k}$. The operations involved can be implemented with the circuit illustrated in Fig. 10.1. The circuit in Fig. 10.1 employs $n - k$ delay cells of a shift register with feedback connections specified by the code generator polynomial as explained next, performing a pre-multiplication of the information polynomial $I(x)$ by x^{n-k}. Let $g(x) = x^{n-k} + g_{n-k-1}x^{n-k-1} + \cdots + g_1x + 1$ denote the generator polynomial. In Fig. 10.1, a switch is associated with each coefficient g_i, $1 \leq i \leq n - k - 1$, which is closed if $g_i = 1$; otherwise, it is kept open. The shift-register cells are initially filled with zeros, with switch S_1 closed and switch S_2 staying in position 1. The information digits are then fed simultaneously to the output and into the division circuit. After being fed with the k information digits, the shift-register contents are the remainder, i.e., the parity-check digits. In the next step, switch S_1 is left open, switch S_2 is moved to position 2, and during the next $n - k$ clock pulses, the parity-check digits are sent to the output. This encoding

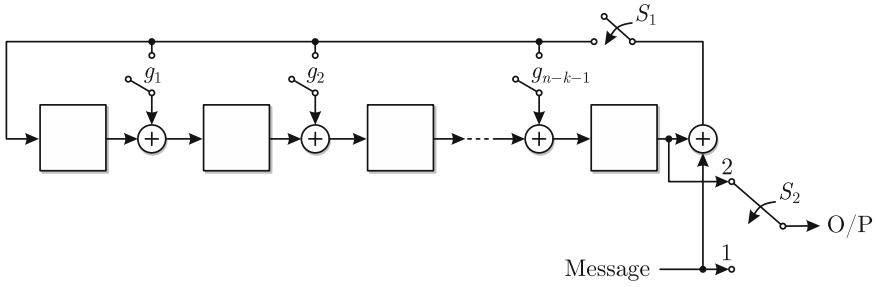


Fig. 10.1 Encoder for a binary cyclic code employing $n - k$ shift-register delay cells

procedure is performed in the same manner for all subsequent k-digit information blocks.

Another sequential encoding procedure exists for cyclic codes based on the polynomial $h(x) = (x^n - 1)/g(x)$, which employs k shift-register delay cells and is described elsewhere in the coding literature, in Clark Jr. and Cain (1981, p.73), Lin and Costello (2004, p.148), for example. In the sequel, a few classes of cyclic codes are presented which benefit from their cyclic structure regarding encoding and decoding operations.

10.3.3 Cyclic Hamming Codes

The Hamming codes presented earlier in Sect. 10.1 admit also a cyclic representation. The binary cyclic (n, k, d) Hamming codes have as their generator polynomial a primitive polynomial $p(x)$ of degree m (Peterson and Weldon Jr. 1972, p. 161) and have the following parameters:

$$n = 2^m - 1,\ k = 2^m - m - 1,\ d = 3.$$

As examples of easily implementable decoders for cyclic Hamming codes, one can cite the Megitt decoder and the error-trapping decoder, which are described later. Due to the fact that Hamming codes are perfect codes (see Definition 10.6), very often, they appear in the literature in most varied applications as, for example, subsets of their codewords being used as protocol sequences for the collision channel without feedback (da Rocha Jr. 1993) and for a low-power, wide-area (LPWA) networking protocol designed to wirelessly connect battery operated devices to the Internet (Seller and Sornin 2014).

10.3.4 Maximum-Length-Sequence Codes

The length of a q-ary sequence generated by a shift register with m delay cells, employing linear feedback, is at most $q^m - 1$. By considering $q = 2$, a binary sequence of length $2^m - 1$ generated in this manner, together with its cyclic shifts and the all zero $(2^m - 1)$-tuple constitute a linear binary cyclic code called an m-sequence code. The parameters of a (n, k, d) binary cyclic m-sequence code are the following:

$$n = 2^m - 1,\ k = m,\ d = 2^{m-1},\ \text{for } m \geq 2,$$

the generator polynomial of which has the form $g(x) = (x^n - 1)/p(x)$, in which $p(x)$ denotes a degree m primitive polynomial. All nonzero codewords of an m-sequence code have the same Hamming weight and have the property of

being equidistant. The m-sequence codes are also called simplex codes. Regarding their decoding, m-sequence codes are completely orthogonalizable in one step (Massey 1963) which makes them easily decodable by majority logic. The practical applications of an m-sequence code include direct sequence spread spectrum, ranging, and location techniques.

10.3.5 Bose-Chaudhuri-Hocquenghem Codes

The Bose-Chaudhuri-Hocquenghem codes appeared in the literature first in France (Hocquenghem 1959) and about a year later in the USA (Bose and Ray-Chaudhuri 1960), but due to a possible lack of good communications, it is considered that they were independently discovered. The respective authors of both articles are considered codiscovers of these codes which became known as BCH codes. Among cyclic codes, the BCH codes are considered one of the most important classes of block codes having an algebraic decoding algorithm. Given two arbitrary positive integers, m and t, a (n, k, d) BCH code can be constructed with the following parameters:

$$n = q^m - 1,\ n - k \leq mt,\ d \geq 2t + 1.$$

The lower bound $d \geq 2t+1$ for a BCH code minimum distance is known as the *BCH bound theorem* (da Rocha Jr. 2014, p. 23). The BCH codes can be interpreted as a generalization of Hamming codes, which can correct multiple errors in a codeword. Since a BCH code is a cyclic code, it can be defined in terms of the roots of its generator polynomial as follows.

Definition 10.9 Let α denote a primitive element of GF(q^m). A primitive (n, k, d) BCH code over GF(q), with block length $n = q^m - 1$, which can correct t errors per codeword, has $\alpha^{h_0},\ \alpha^{h_0+1},\ \ldots,\ \alpha^{h_0+2t-1}$ as roots of its generator polynomial, for any integer h_0.

It follows from Definition 10.9 that the generator polynomial $g(x)$ of a BCH code can be written as the least common multiple (LCM) of minimal polynomials of the specified roots (Berlekamp 1968, p.101), that is,

$$g(x) = \text{LCM}\{m_0(x), m_1(x), ..., m_{2t-1}(x)\},$$

in which $m_i(x)$ is the minimal polynomial of $\alpha^{h_0+i},\ 0 \leq i \leq 2t-1$. Non-primitive BCH codes are those BCH codes for which α is not a primitive element of GF(q^m), and, as a consequence, non-primitive BCH codes have a block length equal to the multiplicative order of α. In case $h_0 = 1$, the corresponding codes are called *narrow sense BCH codes*. Alternatively, BCH codes can be defined in terms of the finite field Fourier transform (da Rocha Jr. 2014, Appendix C) of the generator polynomial $g(x)$ (Blahut 1983, p.207).

10.3.6 Reed-Solomon Codes

Reed-Solomon (RS) codes (Blahut 1983, p.174) are an important class of nonbinary BCH (n, k, d) codes due to their algebraic structure and the fact that they constitute a class of *maximum distance separable* codes (MacWilliams and Sloane 1977, p.323), i.e., $d = n - k + 1$. An RS (n, k, d) code defined over GF(q) has the following parameters:

$$n = q - 1,\ n - k = 2t,\ d = 2t + 1.$$

A generator polynomial $g(x)$ of an RS code over GF(q) has $n - k = 2t$ consecutive roots, and if the roots are $\alpha, \alpha^2, \alpha^3, \ldots, \alpha^{2t}$, then $g(x)$ can be written as

$$g(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^3) \ldots (x - \alpha^{2t}),$$

in which α denotes a primitive element in GF(q). Very often, practical applications require binary digits to be employed, and that makes $q = 2^r$ symbols a natural choice for the alphabet size of an RS code since then each 2^r-ary symbol can be represented by r binary digits. An RS code defined over GF(2^r) when mapped to binary produces a binary code of block length nr which can correct any combination of t erroneous binary r-tuples; in other words, this binary code is capable of correcting both random errors and burst errors. RS codes have been successfully employed as outer codes in coding schemes known as serial concatenated coding (Clark Jr. and Cain 1981, p.333).

10.3.7 Golay Codes

There are two Golay codes, one is a binary $(n, k, d) = (23, 12, 7)$ code while the other is a ternary $(n, k, d) = (11, 6, 5)$ code. Those are the only perfect codes with $t > 1$ (van Lint 1982, p.102).

10.3.8 The Binary (23, 12, 7) Golay Code

The binary (23, 12, 7) Golay code is a linear code which can be algebraically treated as a non-primitive BCH code and can be generated as follows. Let α be a primitive element of GF(2^{11}), and since $2^{11} - 1 = 89 \times 23$, it follows that $\beta = \alpha^{89}$ is a non-primitive element of order 23 because $\beta^{23} = 1$ and $\beta^i \neq 1$, for $1 \leq i \leq 22$. The factoring of $x^{23} + 1$ over GF(2) produces

$$x^{23}+1 = (x+1)(x^{11}+x^{10}+x^6+x^5+x^4+x^2+1)(x^{11}+x^9+x^7+x^6+x^5+x+1),$$

in which one of the degree 11 factors has β, β^2, β^3, and β^4 as the longest string of consecutive roots, while the other degree 11 factor has β^{19}, β^{20}, β^{21}, and β^{22} as the longest string of consecutive roots. Any of the two degree 11 factors of $x^{23}+1$ can be the code generator polynomial. We call the reader's attention to the fact that by the BCH bound (Lin and Costello 2004, p.205) this code has a designed distance $\delta = 5$, while its true minimum distance is 7. Since the minimum distance of the binary Golay code is an odd number, namely, $d = 7$, it follows that we can extend the binary Golay code to become a (24, 12, 8) code by appending an overall parity-check digit to each codeword in a manner to make even the total number of 1s (see Problem 2.10.11).

10.3.9 The Ternary (11, 6, 5) *Golay Code*

The only nontrivial perfect ternary code is the (11, 6, 5) Golay code. By employing finite field tools, the ternary Golay code can be classified as a non-primitive BCH code over GF(3). Let α denote a primitive element in GF(3^5), i.e., $\alpha^{3^5-1} = 1$ and $\alpha^i \neq 1$, for $1 \leq i \leq 3^5 - 2$. Since $3^5 - 1 = 242 = 22 \times 11$, it follows that α^{22} is an element of order 11 in GF(3^5), i.e., $(\alpha^{22})^{11} = \alpha^{242} = 1$ and $(\alpha^{22})^i \neq 1$, for $1 \leq i \leq 10$. Letting $\beta = \alpha^{22}$, the powers $\beta^i = (\alpha^{22})^i$, $0 \leq i \leq 10$, are the roots of $x^{11} - 1$ and can be separated into the cyclotomic classes $\{1\}$, $\{\beta, \beta^3, \beta^9, \beta^5, \beta^4\}$, $\{\beta^2, \beta^6, \beta^7, \beta^{10}, \beta^8\}$. Therefore, $x^{11} - 1$ can be factored over GF(3) into three irreducible factors as

$$x^{11} - 1 = (x-1)(x^5 + x^4 + 2x^3 + x^2 + 2)(x^5 + 2x^3 + x^2 + 2x + 2),$$

in which $\{\beta, \beta^3, \beta^9, \beta^5, \beta^4\}$ are the roots of $x^5 + x^4 + 2x^3 + x^2 + 2$ and $\{\beta^2, \beta^6, \beta^7, \beta^{10}, \beta^8\}$ are the roots of $x^5 + 2x^3 + x^2 + 2x + 2$. Either degree 5 factor of $x^{11} - 1$ can be used as a generator polynomial for the cyclic (11, 6, 5) ternary Golay code. Both degree 5 irreducible factors of $x^{11} - 1$ have a maximum of three consecutive roots, and, by the BCH bound, the codes they generate have a designed distance $\delta = 4$ while the code true minimum Hamming distance is $d = 5$, required for double-error correction. By extending the (11, 6, 5) ternary Golay code with an overall parity-check digit added to each codeword, the (12, 6, 6) extended ternary Golay code is obtained (MacWilliams and Sloane 1977, p. 635).

10.3.10 Reed-Muller Codes

Reed-Muller (RM) codes are binary block codes equivalent to cyclic codes with an overall parity-check digit appended to each codeword. The RM codes are a large class of binary codes which in general have a lower error correcting power in

comparison with BCH codes of equivalent rate. One important aspect of RM codes is that they are easily decodable by majority logic, and another relevant aspect is that they form a subclass of codes constructed from Euclidean geometry (da Rocha Jr. 2014, pp. 142 - 160). In order to define RM codes, it is necessary first to define the Hadamard product of vectors.

Definition 10.10 Given m binary vectors $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m}$ of length n, their Hadamard product, denoted as $\mathbf{v_1 v_2} \ldots \mathbf{v_m}$, is the binary vector of length n the coordinates of which are obtained by multiplying the corresponding coordinates of the factors.

Example 10.11 Given the binary vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$, $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, and $\mathbf{c} = (c_1, c_2, \ldots, c_n)$, their Hadamard product denoted by **abc** is given by

$$\mathbf{abc} = (a_1 b_1 c_1, a_2 b_2 c_2, \ldots, a_n b_n c_n).$$

Let $\mathbf{v_0}$ denote the all-1 vector of length $n = 2^m$, i.e., having all n coordinates equal to 1. Let $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m}$ denote binary vectors of length $n = 2^m$ forming the rows of a matrix having $n = 2^m$ columns consisting of all the distinct binary m-tuples.

Definition 10.12 The Reed-Muller code of order r is the set of codewords produced by the generator matrix, the rows of which are the vectors $\mathbf{v_0}, \mathbf{v_1}, \ldots, \mathbf{v_m}$ and their respective Hadamard products two at a time, three at a time, ..., r at a time. For any given positive integer m, the resulting Reed-Muller code of order r has the following parameters:

$$n = 2^m, \; k = \sum_{i=0}^{r} C_m^i, \; d = 2^{m-r}.$$

10.3.11 Quadratic Residue Codes

Quadratic residue (n, k, d) codes constitute a well-defined class of cyclic codes with prime block length $n = p$, rate $k/n \approx 1/2$, and minimum Hamming distance $d \geq \sqrt{p}$. The name *quadratic residue code* reflects the connection of these codes with *quadratic residues* studied in number theory as explained next.

Definition 10.13 Given a prime number p, an integer r is defined as a quadratic residue of p if and only if there exists an integer s such that $s^2 \equiv r \mod p$, i.e., the square of s reduced modulo p equals r.

For a given prime n, let R_0 denote the set of quadratic residue modulo n, and let $\overline{R_0}$ denote the set of quadratic non-residue modulo n. If $n = 8m \pm 1$ is a prime number, then 2 is a quadratic residue of n (MacWilliams and Sloane 1977, p.481),

and letting α denote an element of multiplicative order n in an extension field of GF(2), it follows that

$$x^n + 1 = (x + 1)G_r(x)G_{\overline{r}}(x),$$

in which

$$G_r(x) = \prod_{r \in R_0} (x + \alpha^r) \text{ and } G_{\overline{r}}(x) = \prod_{\overline{r} \in \overline{R_0}} (x + \alpha^{\overline{r}}).$$

Definition 10.14 A quadratic residue code is a cyclic code with generator polynomial $g(x)$, such that $g(x) \in \{G_r(x),\ (1 + x)G_r(x),\ G_{\overline{r}}(x), (x + 1)G_{\overline{r}}(x)\}$.

Permutation decoding represents an efficient way of decoding quadratic residue codes (MacWilliams and Sloane 1977, p. 513). However, permutation decoders are usually more complex than algebraic BCH decoders. The relevance of quadratic residue codes comes from the fact that some of these codes of moderate length have a minimum Hamming distance greater than that of BCH codes of comparable block length.

10.3.12 Alternant Codes

An important class of algebraic codes which includes as special cases the BCH codes, Goppa codes, Srivastava codes, and Chien-Choy codes is the class of alternant codes (MacWilliams and Sloane 1977, p.332). By a relatively simple, but not obvious, modification of the parity-check matrix of a BCH code, an alternant code is constructed. Consider a (n, k, d) BCH code with designed distance δ over GF(q) and parity-check matrix $\mathbf{H} = [h_{ij}]$, in which $h_{ij} = \alpha^{ij}$, $1 \leq i \leq \delta - 1, 0 \leq j \leq n - 1$, and α is a primitive nth root of unity in GF(q^m) (MacWilliams and Sloane 1977, p.196). By making $h_{ij} = x_j^{i-1} y_j$, an alternant code results, in which $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is a vector over GF(q^m) with distinct components and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ is a vector with nonzero components also over GF(q^m) (MacWilliams and Sloane 1977, Ch. 12).

10.4 Decoding Cyclic Codes

The decoding procedures for linear block codes are also applicable to cyclic codes. However, the algebraic properties associated with the cyclic structure allow important simplifications when implementing a decoder for a cyclic code, both for calculating the syndrome and for correcting errors. The syndrome computation consists in dividing by the generator polynomial $g(x)$ the polynomial representing

the word received by the decoder. The remainder of this division is the syndrome, denoted by $s(x)$. If $s(x) = 0$, the received word is accepted as being a codeword. Otherwise, i.e., if $s(x) \neq 0$, we declare that errors have occurred. In this manner, it is clear that a circuit to detect errors with a cyclic code is rather simple. The problem of locating error positions in a received word for correction is a different matter and in general requires more elaborate techniques in order to be implemented in practice. So far, the most important algebraic decoding techniques for cyclic codes are those based on the Berlekamp-Massey algorithm Massey (1969) and on the Euclidean algorithm (MacWilliams and Sloane 1977, pp. 362-365), (Clark Jr. and Cain 1981, pp. 216-218). In the sequel, we give a brief presentation of the more relevant decoding procedures for cyclic codes.

10.4.1 Meggitt Decoder

The Meggitt decoding algorithm (Lin and Costello 2004, p.156) consists in employing a circuit to identify those syndromes that correspond to error patterns containing an error in the highest-order position of the received word, i.e., an error at position x^{n-1}. Thus, when the digit at position x^{n-1} is being delivered to the sink, it can be altered or not, depending on the decision determined by the circuit that identifies errors in that position. If we cyclically shift the received word, digit by digit with a shift register, all the digits in this word will necessarily occupy position x^{n-1} after a certain number of shifts and will be examined by the circuit that identifies errors in that position. The decision for choosing the Meggitt decoder depends on the complexity of the circuit that identifies errors at position x^{n-1} of the received word. A straightforward way of implementing a Meggitt decoder is by the use of programmable memories.

10.4.2 Error-Trapping Decoder

An error-trapping decoder (Lin and Costello 2004, p.166) for a (n, k, d) cyclic code is able to locate and correct an error which cyclically spreads to at most $n - k$ positions in a received word and has Hamming weight at most t, $t \leq \lfloor (d-1) \rfloor / 2$. Suppose the Hamming weight of the syndrome of a received word is nonzero, i.e., the received word is not error-free. If the Hamming weight of the syndrome is at most t, then the errors are confined to the parity-check section of the received word, the information digits are error-free, and no further action is required from the decoder. Otherwise, an error-trapping decoder then operates by cyclically shifting the syndrome shift register, bit by bit, and observing its Hamming weight at each step. If, at some step, a syndrome of Hamming weight at most t is detected, then

the corresponding segment of length $n - k$ of the error vector, containing all the errors, coincides with this shifted syndrome. After an error pattern is identified, or trapped, then error correction is immediate. However, if after n cyclic shifts, the syndrome Hamming weight was never equal to or less than t the decoder declares the occurrence of an error pattern which cyclically spreads over a length greater than $n - k$ of cyclically consecutive word positions and decoding stops. Error-trapping decoding is more appropriate for use with low rate codes, since in this case it will be easier to satisfy the requirement $n/k = 1/R > t$ to correct t errors in a received word by employing this technique.

10.4.3 Information Set Decoding

An *information set* (Clark Jr. and Cain 1981, p.102) defined for a given a linear (n, k, d) block code contains k elements corresponding to k positions which can be independently specified in a codeword. Therefore, the k symbols in an information set can be used to construct a codeword. It follows that an information set free from errors in a received n-tuple can be used to reconstitute the transmitted codeword. Furthermore, information set decoding is applicable to linear codes in general, not just to cyclic codes. A conceivable decoding algorithm based on a number of information sets consists of the following steps:

1. First, construct several information sets for the given code.
2. Use each information set from the previous step to form an estimate of the transmitted codeword.
3. The received word is now compared with each one of the estimates obtained in the previous step and decide for the codeword nearest to the received word in terms of Hamming distance.

10.4.4 Threshold Decoding

Threshold decoding (Massey 1963) is a technique that has been employed with success in a number of applications using cyclic codes and is also known by the name of *majority logic decoding*. The implementation of threshold decoders is based on the use of *parity-check sums* or just *parity sums* as described in the sequel. Given a (n, k, d) binary linear code with parity-check matrix $\mathbf{H} = [h_{ij}]$, the syndrome $\mathbf{s} = (s_0, s_1, \ldots, s_{n-k-1})$ is a vector of length $n - k$, obtained by multiplying an error word $\mathbf{e} = (e_0, e_1, \ldots, e_{n-1})$ by the transpose of the code parity-check matrix, i.e.,

$$\mathbf{s} = \mathbf{e}\mathbf{H}^T,$$

in which each syndrome component s_j, $0 \leq j \leq n-k-1$, is computed as

$$s_j = \sum_{i=0}^{n-1} e_i h_{ij}. \tag{10.2}$$

Consider the linear combination of syndrome digits:

$$A = \sum_{i=0}^{n-k-1} a_i s_i,$$

in which a_i is either 0 or 1, and by employing Eq. 10.2, it can be written as

$$A = \sum_{i=0}^{n-1} b_i e_i, \tag{10.3}$$

in which b_i is either 1 or 0. Expression 10.3 is called a *parity-check sum* which checks an error in position e_i if $b_i = 1$.

Definition 10.15 A set of J parity-check sums $A_1, A_2, \ldots, A_J$ is defined as orthogonal in a position e_l in the error vector $\mathbf{e} = (e_0, e_1, \ldots, e_{n-1})$ if e_l is checked by each one of the J parity-check sums and all other error vector positions $e_i, i \neq l$, are checked by at most one of the J parity-check sums.

Consider that t errors occurred in a received word, $t \leq \lfloor J/2 \rfloor$. The threshold decoding procedure works as follows:

i. Suppose $e_l = 1$. This hypothesis implies that the other $t-1$ errors can affect at most $t-1 \leq \lfloor J/2 \rfloor - 1$ of the J parity sums, leaving at least $J - (\lfloor J/2 \rfloor - 1) = \lceil J/2 \rceil + 1$ of the parity sums equal to 1, i.e., the indication provided by the majority of the parity sums allows one to correctly decide for $e_l = 1$.
ii. Suppose $e_l = 0$. This hypothesis implies that the t errors can affect at most $\lfloor J/2 \rfloor$ parity sums, i.e., at most half of the parity sums fail in a worst case situation, and one can safely assume that $e_l = 0$ in this case.

If the binary linear code considered is a cyclic code, the threshold decoding operation should employ J parity sums orthogonal on position e_{n-1} and start by checking for error in that position. After that is done, the received word is cyclically shifted by one position to the right and the same set of J parity sums is used to decode position e_{n-2} and continue in a similar manner for decoding the remaining received word positions until the complete word is decoded. Completely orthogonalizable codes are those codes for which $J = d - 1$. There are cases when it is not possible to construct J parity sums orthogonal on a given position of a codeword. For various classes of codes (Lin and Costello 2004, p.296), orthogonal parity sums can be obtained in L steps, where at each step one uses parity sums

orthogonal on a sum of codeword positions. The topic *threshold decoding* is treated in greater detail in da Rocha Jr. (2014, Ch. 11).

10.5 Hard-Decision Algebraic Decoding

It is well known that in practice the decoding process for a given (n, k, d) code has a complexity higher than the corresponding encoding process. Consequently, the choice of the most suitable code for a given application depends on the budget available. In some cases, the budget constraint forces the use of a suboptimum solution by opting for a decoder with an amenable implementation cost. However, computationally efficient decoding algorithms (Berlekamp 1968, p.178), Blahut (1983) have been discovered for some classes of algebraic codes. The decoding of algebraic codes involves solving a system of nonlinear equations, the direct solution of which is not obvious in general.

10.5.1 Berlekamp-Massey Time Domain Decoding

The Berlekamp-Massey (BM) time domain decoding algorithm is widely applicable for decoding algebraic codes, including RS codes and BCH codes (Berlekamp 1968; Massey 1969). Obviously, for binary algebraic codes, there is no need to calculate error magnitude values, since in GF(2) it is sufficient to determine the positions of the errors to perform error correction. For nonbinary algebraic codes, including RS codes and BCH codes, both error locations and error magnitudes have to be determined in order to perform error correction.

Let (n, k, d) denote a cyclic code with generator polynomial $g(x)$ having roots $\alpha, \alpha^2, \ldots, \alpha^{2t}$ and coefficients both in some finite field GF(q). Let $c(x) = \sum_{i=0}^{n-1} c_i x^i$ denote a codeword, let $e(x) = \sum_{i=0}^{n-1} e_i x^i$ denote the error polynomial with coefficients in GF(q), and let $r(x) = c(x) + e(x)$, with addition defined in GF(q), denote the received word, i.e., an n-tuple in polynomial form. The algebraic decoding procedure contains, in general, the following steps:

1. Calculate the first $2t$ coefficients $s_0, s_1, \ldots, s_{2t-1}$ of the syndrome polynomial $s(x) = s_0 + s_1 x + \cdots + s_{2t-1} x^{2t-1} + \cdots$, in which $s_0 = r(\alpha), s_1 = r(\alpha^2), \ldots, s_{2t-1} = r(\alpha^{2t})$.
2. The sequence $s_0, s_1, \ldots, s_{2t-1}$ is fed as an input to the BM algorithm to compute the error-locator polynomial $\sigma(x)$, of degree τ, $\tau \leq t$, expressed as

$$\sigma(x) = 1 + \sigma_1 x + \sigma_2 x^2 + \cdots + \sigma_\tau x^\tau .$$

3. The roots $\beta_1, \beta_2, \ldots, \beta_\tau$ of $\sigma(x)$ are found in this step and their multiplicative inverses give the error locations.

4. If the code considered is nonbinary, then calculate the error magnitudes.

A procedure for computing error magnitudes for nonbinary cyclic codes was introduced in Berlekamp (1968, p.220) and defined the polynomial

$$Z(x) = 1 + (s_0 + \sigma_1)x + (s_1 + \sigma_1 s_0 + \sigma_2)x^2 + \cdots \\ \cdots + (s_{\tau-1} + \sigma_1 s_{\tau-2} + \sigma_2 s_{\tau-3} + \cdots + \sigma_\tau)x^\tau. \tag{10.4}$$

An error magnitude at position β_i^{-1}, $1 \leq i \leq \tau$, is calculated as

$$e_i = \frac{Z(\beta_i)}{\prod_{j=1, j\neq i}^{\tau}(1 - \beta_j^{-1}\beta_i)}. \tag{10.5}$$

The BM algorithm gives as a result an estimate of the error pattern of minimum Hamming weight that satisfies all the syndrome equations. Whether or not such an estimate represents the true error pattern that hit the transmitted codeword will depend on its Hamming weight τ in comparison to t. The true error pattern is found by the BM algorithm whenever the condition $\tau \leq t$ is satisfied. A flowchart of the BM time domain decoding algorithm is presented in da Rocha Jr. (2014, p. 35).

10.5.2 Euclidean Frequency Domain Decoding

Reference (Blahut 1983, p.193) brings an authoritative presentation of the decoding procedure known as *algebraic decoding in the frequency domain* which is applicable to any cyclic code and, in particular, is more efficient with (n, k, d) BCH codes. We assume $\mathbf{r}$ is a received word representing the sum over GF(q) of a codeword $\mathbf{v}$ and an error word $\mathbf{e}$, i.e., $\mathbf{r} = \mathbf{v} + \mathbf{e}$. The Euclidean frequency domain decoding employs the following steps:

1. **Received word finite field Fourier transform**
   First, the received vector $\mathbf{r} = (r_0, r_1, \ldots, r_{n-1})$ is used to compute its finite field Fourier transform $\mathbf{R}$. The linearity of the Fourier transform $\mathbf{R}$ can be expressed as a sum of the corresponding Fourier transforms of $\mathbf{v}$ and $\mathbf{e}$, i.e., as $\mathbf{R} = \mathbf{V} + \mathbf{E}$.
2. **Syndrome**

$$S(z) = \sum_{j=0}^{2t-1} S_j z^j,$$

in which

$$S_j = R_{j+h_0} = \sum_{i=0}^{n-1} r_i \alpha^{i(j+h_0)},\ j = 0, 1, \ldots, 2t-1,$$

for a code having $2t$ consecutive roots α^{j+h_0}, $0 \leq j \leq 2t-1$, in its generator polynomial and h_0 is a fixed integer satisfying $0 \leq h_0 \leq n-1$.

3. **Error-locator polynomial**
   The error-locator polynomial, denoted by $L(z)$, can be expressed as

$$L(z) = \sum_{k=0}^{\nu} L_z z^k = \prod_{k=1}^{\nu} (1 - z\alpha^{i_k}),$$

   where $\nu \leq t$ and $i_1, i_2, \ldots, i_\nu$, correspond to error locations. The Euclidean algorithm (da Rocha Jr. 2014, pp. 176 - 177) is then applied with $a(z) = z^{2t}$ and $b(z) = S(z)$, stopping when the degree of $r_i(z)$ becomes less than t. Finally, we make $L(z)$ equal to $g_i(z)$. This step is also known as the *solution of the key equation* (Clark Jr. and Cain 1981, p.189).
4. **Error word finite field Fourier transform**
   The error word $\mathbf{e}$ has its finite field Fourier transform denoted by $\mathbf{E}$, with the following polynomial representation:

$$E(z) = \sum_{j=0}^{n-1} E_j z^j,$$

   in which the coefficients E_j, $j = h_0, h_0+1, \ldots, h_0+2t-1$ are known since the code is assumed to have $2t$ consecutive roots in its generator polynomial. The following recursive formula is employed to calculate the remaining unknown coefficients of $E(z)$:

$$\sum_{k=0}^{n-1} L_k E_{j-k} = 0,$$

   which can be slightly simplified to be expressed as

$$E_j = -\sum_{k=1}^{\nu} L_k E_{j-k},$$

   because $L_0 = 1$ and $L(z)$ has degree $\nu \leq t$.
5. **Error correction**
   The inverse finite field Fourier transform of the n-tuple $\mathbf{E}$ gives the error word $\mathbf{e}$, and, finally, the decoded codeword $\mathbf{v}$ is given by

$$\mathbf{v} = \mathbf{r} - \mathbf{e}.$$

10.6 Soft-Decision Decoding

Probabilistic decoding algorithms, as mentioned earlier, are used to implement soft-decision decoding. Received symbol reliability information supplied by the demodulator is used by a soft-decision decoder when deciding which codeword was transmitted. A typical gain of at least 2.0 dB results when soft-decision decoding is employed instead of hard-decision decoding.

10.6.1 Decoding Low-Density Parity-Check Codes

The decoding algorithm called sum-product algorithm (SPA) (Moreira and Farrell 2006, p.282) is a symbol-by-symbol soft-in soft-out iterative decoding algorithm particularly efficient for decoding low density parity-check (LDPC) codes. The SPA uses the code parity-check matrix $\mathbf{H}$ and makes iterations starting with the received symbols fed to the decoder, in order to improve the reliability of the decoded symbols. Symbol reliability values after each iteration are used to make hard decisions and to output a decoded binary n-tuple $\mathbf{z}$. The operation $\mathbf{zH}^T$ is performed and decoding stops if $\mathbf{zH}^T = \mathbf{0}$ because $\mathbf{z}$ is a codeword. However, if $\mathbf{zH}^T \neq \mathbf{0}$, then $\mathbf{z}$ is not a codeword, and the iteration process is continued to produce fresh symbol reliability values and test again whether a codeword is obtained. A maximum number of decoder iterations is specified which is employed to halt the decoding process when the decoder does not find a codeword, and a decoding failure is then declared. Let $\mathbf{y}$ denote a received n-tuple with real-valued coordinates when a codeword $\mathbf{v} = (v_1, v_2, \ldots, v_i, \ldots, v_n)$ is transmitted. The SPA operates by computing the marginal probabilities $P(v_i|\mathbf{y})$, for $1 \leq i \leq n$. In MacKay (1999), the reader will find a detailed description of SPA decoding of LDPC codes.

10.7 Convolutional Codes

Convolutional codes were introduced in the literature in 1954 (Elias 1954), and since then, many researchers have dedicated time to study their structure (Forney Jr. 1970). In 1961, the first practical decoding algorithm for convolutional codes was described (Wozencraft and Reiffen 1961). In 1967, another way of decoding convolutional codes was discovered (Viterbi 1967) which at first was thought to be asymptotically optimum. A few years later, in 1973 (Forney Jr. 1973), it was shown that the Viterbi algorithm was a maximum likelihood decoding algorithm for convolutional codes. In 1993, a turbo decoding algorithm was introduced for a code construction employing convolutional codes (Berrou et al. 1993). At the time, this was by far the most remarkable result in coding theory since Shannon's papers. A turbo decoder allows performance very close to the Shannon limit (channel

capacity) (Berrou and Glavieux 1996), in the presence of additive white Gaussian noise. Turbo decoding was also shown to work equally well with block codes (Pyndiah et al. 1994). Convolutional codes have led the code race in space research (Massey 1992) in combination with Reed-Solomon codes (Hagenauer et al. 1993; Wicker and Bhargava 1994). Convolutional codes offer an alternative for error control which is quite different from that offered by block codes. Some relevant concepts which are needed for implementing turbo codes (Heegard and Wicker 1999; Johanneson and Zigangirov 1999; Lin and Costello 2004) are presented in this chapter. In the sequel, the basic theory of convolutional codes is revisited.

10.7.1 Representation and Basic Concepts

As mentioned in Viterbi and Omura (1979, p. 227), convolutional codes are related to a special class of linear block codes; however, the additional convolutional structure aggregates further properties which are relevant to reduce decoder complexity at the cost of increased memory and higher computational speed requirements. A linear convolutional encoder can be treated as a finite state machine, and from this point of view, a state diagram and a trellis appear as important tools both to describe the code and to analyze its behavior.

10.7.2 Shift-Register Encoder

A *binary convolutional encoder* (BCE) is a device, the output of which consists of blocks of n binary symbols, which are dependent on the current k binary input symbols as well as on m consecutive k-bit blocks of the previous m messages. A convolutional encoder with parameters (n, k, m) can be implemented as a sequential logic circuit with k inputs, n outputs, and a memory order of m bits (see Definition 10.19). Usually, n and k are small positive integers, with $k < n$. Larger values of the memory order m allow the construction of convolutional codes that can achieve lower probability of error after decoding. The ratio k/n is called the asymptotic code rate. A *binary convolutional code* (BCC) is the set of all codewords which can be produced at the output of a binary convolutional encoder.

Example 10.16 The BCE illustrated in Fig. 10.2 employs a shift register with three memory elements which accepts message blocks of length $k = 1$ symbol at the input, denoted by the vector $\mathbf{u}$, and has output blocks of length $n = 2$ symbols denoted by the vectors $\mathbf{v}^{(0)}$ and $\mathbf{v}^{(1)}$, respectively, i.e., it implements a convolutional code with parameters $(n, k, m) = (2, 1, 3)$, and asymptotic code rate $k/n = 1/2$.

The information sequence $\mathbf{u} = (u_0, u_1, u_2, \ldots)$ is fed bit by bit to the encoder. Since the encoder implements a linear system, its two output sequences, namely,

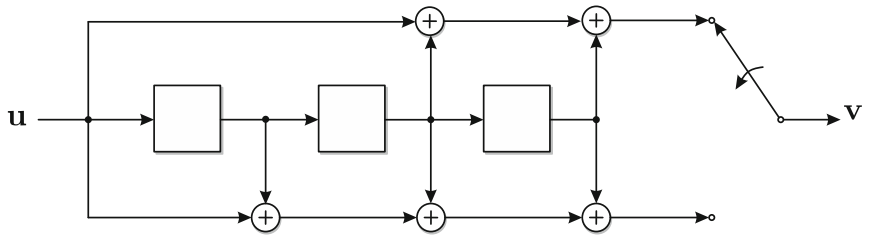


Fig. 10.2 BCE with parameters $(n, k, m) = (2, 1, 3)$ implemented with a shift register and exclusive or gates

$$\mathbf{v}^{(0)} = (v_0^{(0)}, v_1^{(0)}, v_2^{(0)}, \ldots)$$

and

$$\mathbf{v}^{(1)} = (v_0^{(1)}, v_1^{(1)}, v_2^{(1)}, \ldots)$$

can be obtained by the convolution of the input sequence $\mathbf{u}$ with the two impulse responses of the encoder circuit. The two impulse responses are obtained when the encoder input is fed with $\mathbf{u} = (1\ 0\ 0 \ldots)$ and the corresponding two output sequences are observed at the encoder output. Since the encoder memory is m, the impulse responses are $\mathbf{g}^{(0)} = (g_0^{(0)}, g_1^{(0)}, g_2^{(0)}, \ldots, g_m^{(0)})$ and $\mathbf{g}^{(1)} = (g_0^{(1)}, g_1^{(1)}, g_2^{(1)}, \ldots, g_m^{(1)})$. For the encoder in Fig. 10.2, it follows that the impulse responses are $\mathbf{g}^{(0)} = (1\ 0\ 1\ 1)$ and $\mathbf{g}^{(1)} = (1\ 1\ 1\ 1)$ and are called code *generator sequences*.

The encoding equations can be written as

$$\mathbf{v}^{(0)} = \mathbf{u} * \mathbf{g}^{(0)},$$

$$\mathbf{v}^{(1)} = \mathbf{u} * \mathbf{g}^{(1)},$$

in which $*$ denotes the discrete convolution and all operations of addition and multiplication are reduced modulo 2. The convolution implies that for all $l \geq 0$

$$v_l^{(j)} = \sum_{i=0}^{m} u_{l-i} g_i^{(j)} = u_l g_0^{(j)} + u_{l-1} g_1^{(j)} + \cdots + u_{l-m} g_m^{(j)}, \quad j \in \{0, 1\},$$

in which $u_{l-i} = 0$ for all $l < i$. Therefore, for the encoder in Fig. 10.2, we have

$$v_l^{(0)} = u_l + u_{l-2} + u_{l-3},$$

$$v_l^{(1)} = u_l + u_{l-1} + u_{l-2} + u_{l-3},$$

which can be immediately obtained by direct inspection of the encoding circuit. The two output encoded sequences, $\mathbf{v}^{(0)}$ and $\mathbf{v}^{(1)}$, are then multiplexed to form a single sequence called a *codeword*, ready for storage or transmission through a communication channel. The codeword is represented as

$$\mathbf{v} = (v_0^{(0)} v_0^{(1)}, v_1^{(0)} v_1^{(1)}, v_2^{(0)} v_2^{(1)}, \ldots).$$

Convolutional Encoder Types

Digital filters can be implemented with shift registers, and because of that, a BCE can be neatly described using the terminology for digital filters, i.e., we may classify a BCE from the point of view of a digital filter. A BCE may exhibit a finite impulse response (FIR), as the BCE in Fig. 10.2, also called a *non-recursive*, *feedback-free*, or *feed-forward* BCE (Heegard and Wicker 1999). A BCE may also exhibit an infinite impulse response (IIR) and implement *recursive* convolutional codes or convolutional encoders with *feedback* (Heegard and Wicker 1999). Recursive BCEs played an important role in the development of turbo codes and are treated in more detail later in this chapter. Figure 10.3 shows an example of a binary convolutional encoder with feedback, i.e., a recursive BCE. Finally, a BCE can also be classified as *systematic* or *non-systematic*, thus leading to a systematic BCC or a non-systematic BCC, respectively. A BCC is called systematic if each block of length n at the output of the BCE contains the same k information symbols present at each corresponding k-bit input block, for example, the BCE in Fig. 10.3, in which $\mathbf{v}^{(0)} = \mathbf{u}$, is systematic. Figures 10.4 and 10.5 illustrate other types of BCE.

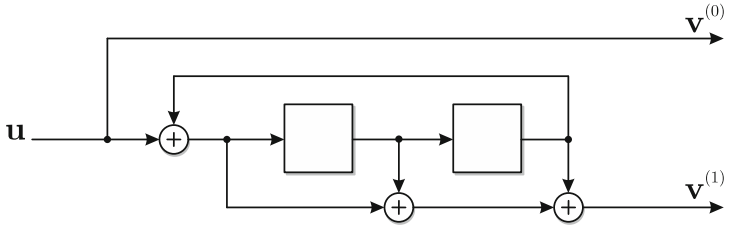
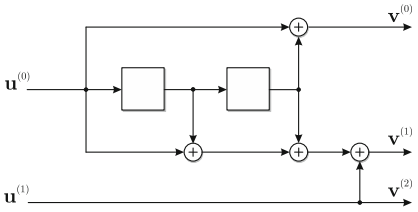


Fig. 10.3 Recursive BCE implemented with shift registers and exclusive or gates

Fig. 10.4 Non-systematic and non-recursive BCE



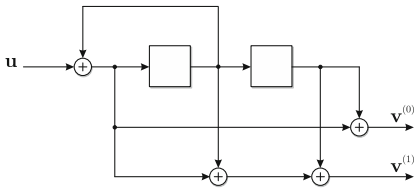


Fig. 10.5 Non-systematic and recursive BCE

10.7.3 Polynomial Matrix

For any linear system, operations in the time domain containing convolutions can be replaced by operations with polynomials. Since a convolutional encoder is a linear system, by a linear correspondence, each sequence in its encoding equations can be replaced by a polynomial, and each convolution operation can be replaced by a polynomial multiplication. For example, in a $(2, 1, m)$ convolutional code, the encoding equations can be expressed as functions of a delay operator D as follows:

$$\mathbf{v}^{(0)}(D) = \mathbf{u}(D)\mathbf{g}^{(0)}(D),$$

and

$$\mathbf{v}^{(1)}(D) = \mathbf{u}(D)\mathbf{g}^{(1)}(D),$$

in which

$$\mathbf{u}(D) = u_0 + u_1 D + u_2 D^2 + \cdots$$

denotes the information sequence, and

$$\mathbf{v}^{(0)}(D) = v_0^{(0)} + v_1^{(0)} D + v_2^{(0)} D^2 + \cdots$$

and

$$\mathbf{v}^{(1)}(D) = v_0^{(1)} + v_1^{(1)} D + v_2^{(1)} D^2 + \cdots$$

are the encoded sequences, and

$$\mathbf{g}^{(0)}(D) = g_0^{(0)} + g_1^{(0)} D + g_2^{(0)} D^2 + \cdots + g_m^{(0)} D^m,$$

and

$$\mathbf{g}^{(1)}(D) = g_0^{(1)} + g_1^{(1)} D + g_2^{(1)} D^2 + \cdots + g_m^{(1)} D^m,$$

are the respective code generator polynomials.

Recalling that the encoder is a linear system and that $\mathbf{u}^{(i-1)}(D)$ represents the ith input sequence and $\mathbf{v}^{(j-1)}(D)$ represents the jth output sequence, the generator polynomial $g_{i-1}^{(j-1)}(D)$ can be interpreted as the encoder transfer function relating the ith input to the jth output. Considering a linear system with k inputs and n outputs, there are a total of kn transfer functions which can be represented by a $k \times n$ transfer function matrix, also called polynomial generator matrix $\mathbf{G}(D)$, i.e.,

$$\mathbf{G}(D) = \begin{bmatrix} \mathbf{g}_0^{(0)}(D) & \mathbf{g}_0^{(1)}(D) & \cdots & \mathbf{g}_0^{(n-1)}(D) \\ \mathbf{g}_1^{(0)}(D) & \mathbf{g}_1^{(1)}(D) & \cdots & \mathbf{g}_1^{(n-1)}(D) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{g}_{k-1}^{(0)}(D) & \mathbf{g}_{k-1}^{(1)}(D) & \cdots & \mathbf{g}_{k-1}^{(n-1)}(D) \end{bmatrix}, \tag{10.6}$$

in which $\mathbf{g}_{i-1}^{(j-1)}(D)$ is the generator polynomial relating the ith input to the jth output.

By employing the polynomial transfer function matrix, the encoding equations for a (n, k, m) code can be written as

$$\mathbf{V}(D) = \mathbf{U}(D)\mathbf{G}(D), \tag{10.7}$$

in which $\mathbf{U}(D) = [\mathbf{u}^{(0)}(D), \mathbf{u}^{(1)}(D), \ldots, \mathbf{u}^{(k-1)}(D)]$ is a k-tuple of input sequences and $\mathbf{V}(D) = [\mathbf{v}^{(0)}(D), \mathbf{v}^{(1)}(D), \ldots, \mathbf{v}^{(n-1)}(D)]$ is an n-tuple of output sequences.

Example 10.17

(a) The polynomial generator matrix $G(D)$ for the BCE in Fig. 10.2 is

$$\mathbf{G}(D) = \left[1 + D^2 + D^3 \;\; 1 + D + D^2 + D^3 \right].$$

and

(b) The polynomial generator matrix $G(D)$ for the BCE in Fig. 10.4 is

$$\mathbf{G}(D) = \begin{bmatrix} 1 + D^2 & 1 + D + D^2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

10.7.4 State Diagram

Consider a (n, k, m) convolutional encoder with polynomial generator matrix as presented in Eq. 10.6.

Definition 10.18 The constraint length for the ith input in a polynomial generator matrix is defined as

$$\nu_i = \max_{0 \le j \le n-1} \{\deg \ \mathbf{g}_i^{(j)}(D)\}, \qquad 0 \le i \le k-1.$$

Definition 10.19 The encoder memory order m is defined as the maximum constraint length, i.e.,

$$m = \max_{0 \le i \le k-1} \{\nu_i\}.$$

Definition 10.20 The total encoder memory ν is defined as the sum of all constraint lengths for all k inputs, i.e.,

$$\nu = \sum_{i=0}^{k-1} \nu_i.$$

Therefore, for a (n, k, m) code with $k > 1$, the ith encoder shift register contains the previous ν_i information symbols. Thus, the elements in the vector $(\nu_1, \nu_2, \ldots, \nu_k)$ represent the respective lengths of the k encoder shift registers, i.e., the ith shift register contains ν_i memory elements.

The state of the encoder at time instant l, when the encoder inputs are

$$u_l^{(0)} u_l^{(1)} \ldots u_l^{(k-1)},$$

is the binary m-tuple

$$(u_{l-1}^{(0)} u_{l-2}^{(0)} \ldots u_{l-m_1}^{(0)} u_{l-1}^{(1)} u_{l-2}^{(1)} \ldots u_{l-m_2}^{(1)} \ldots u_{l-1}^{(k-1)} u_{l-2}^{(k-1)} \ldots u_{l-m_k}^{(k-1)}).$$

It follows that a binary convolutional encoder with memory ν has a total of 2^ν states.

Example 10.21 For a $(n, 1, m)$ code, we have $\nu = \nu_1$, and the encoder state at time instant l is

$$(u_{l-1} u_{l-2} \ldots u_{l-m}).$$

The state diagram is a graph consisting of nodes, representing the states of the encoder, and lines with arrows representing the transitions between states. Each line is labelled with a pair (input k-bits)/(output n-bits). There are 2^k branches leaving each state, corresponding to each distinct input block. For a $(n, 1, m)$ code, there are just two branches leaving each state, and each branch is labelled with a pair (input 1-bit)/(output n-bits).

Example 10.22 Consider the $(n = 2,\ k = 1,\ m = 3)$ encoder shown in Fig. 10.2. The corresponding state diagram is shown in Fig. 10.6. The encoder has eight states labelled as 000, 001, 010, 011, 100, 101, 110, and 111, in which each state represents the contents of the shift register. There are two possible branches leaving each state, corresponding to the two possible values of the message symbol.

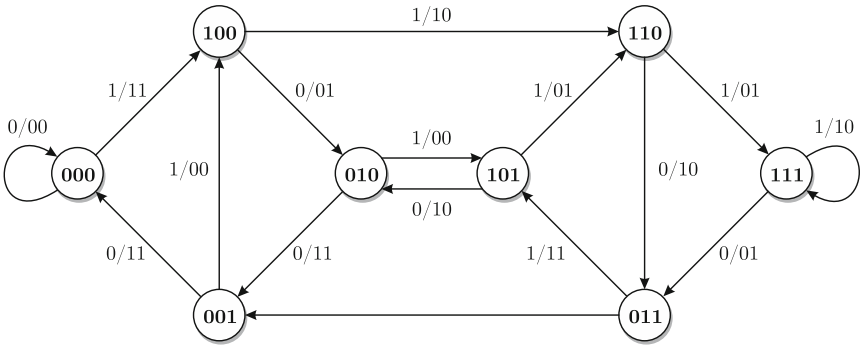


Fig. 10.6 State diagram of the encoder illustrated in Fig. 10.2, in which the labels indicated on the branches represent pairs (input 1-bit/output 2-bits)

Given the current encoder state, the input information sequence determines both the path through the state diagram and the associated output sequence. Some new blocks of k inputs cause a transition to a new state; however, for some blocks of k inputs, a state transition does not occur, and the state remains the same. For example, in Fig. 10.6, each new input block causes a transition to a new state, except for state 000 when the input is a 0 and for state 111 when the input is a 1.

10.7.5 Trellis Diagram

A trellis diagram is obtained from a state diagram by drawing all possible state transitions and writing the respective input/output sequences expanded in time (Viterbi and Omura 1979; Lin and Costello 2004). As an example, consider the trellis diagram in Fig. 10.8 for the $(2, 1, 2)$ code illustrated in Fig. 10.7 with polynomial generator matrix:

$$\mathbf{G}(D) = \left[1 + D^2 \quad 1 + D + D^2\right],$$

and an information sequence of length $L = 5$. The trellis diagram contains $L+m+1$ time units or levels which are labelled from 0 to $L + m$ as indicated in Fig. 10.8. Considering that the encoder initial state is 00 and that the final state is also 00, the encoder uses the first m time units to be able to reach any state in its trellis diagram and uses the last m time units to return to the state 00. Consequently, not all states can be reached neither in the first m time units nor in the last m time units. However, all states are reachable in the center part of the trellis, where each time unit contains a replica of the state diagram. There are two branches leaving and two branches entering each state. The dotted branches represent input symbols equal to 1 and the continuous branches represent input symbols equal to 0. Each branch is labelled

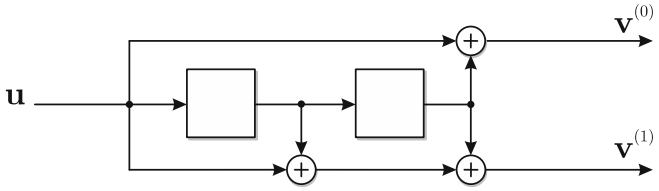


Fig. 10.7 FIR non-systematic code used for constructing the trellis in Fig. 10.8

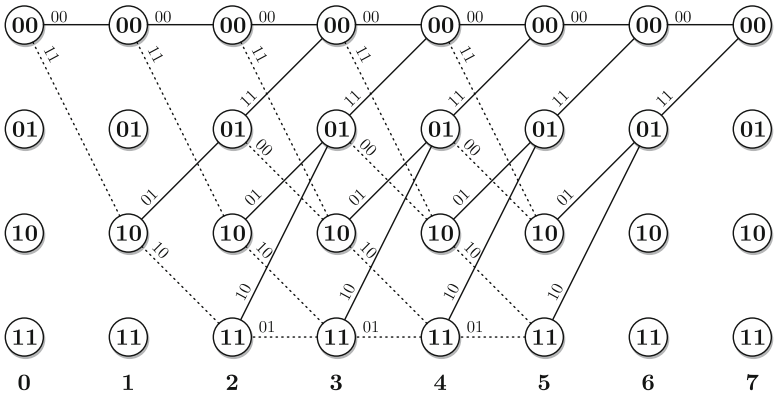


Fig. 10.8 Trellis diagram for the encoder in Fig. 10.7. The states are represented by 00, 01, 10, and 11. The labels indicated on the branches represent the encoder outputs. The dotted branches and the continuous branches represent input symbols equal to 1 and equal to 0, respectively. The numbers in boldface below the trellis represent the time intervals considered, also referred to as the trellis depth

with a corresponding encoder output block $\mathbf{v}_i$, and each one of the 2^L codewords of length $N = n(L+m)$ is represented by a unique path in the trellis. For example, the codeword corresponding to the information sequence $\mathbf{u} = (1\ 1\ 0\ 0\ 1)$ has its path highlighted in the trellis in Fig. 10.9.

Considering the general case of a (n, k, m) code and an information sequence of length kL, there are 2^k branches leaving and 2^k branches entering each state and 2^{kL} distinct paths in the trellis corresponding to 2^{kL} codewords.

10.8 Recursive Systematic Convolutional Codes

The recursive systematic convolutional (RSC) codes (Heegard and Wicker 1999; Vucetic and Yuan 2001; Berrou and Glavieux 1996), also called infinite impulse response (IIR) codes, can be obtained from finite impulse response (FIR) codes as shown in the sequel.

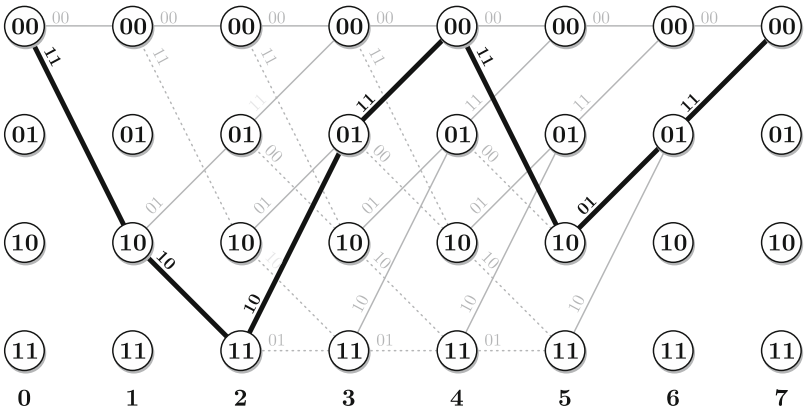


Fig. 10.9 The sequence of labels (11 10 10 11 11 01 11) on the highlighted path corresponds to the sequence of information subblocks (11001)

In a systematic (n, k, m) convolutional code, the first k output sequences are exact replicas of the input information sequences. The polynomial generator matrix of a systematic convolutional code has the following form:

$$\mathbf{G}(D) = \left[\mathbf{I}\ \mathbf{P}(D)\right],$$

in which $\mathbf{I}$ denotes the $k \times k$ identity matrix and $\mathbf{P}(D)$ is a $k \times (n-k)$ matrix. In order to obtain the systematic form for a given polynomial generator matrix of a convolutional code, it is necessary first to define an equivalent polynomial generator matrix.

Definition 10.23 Two polynomial generator matrices are equivalent when they generate the same convolutional code.

Example 10.24 Consider the polynomial generator matrix for the encoder in Fig. 10.7, i.e.,

$$\mathbf{G}(D) = \left[1 + D^2\ 1 + D + D^2\right].$$

The code sequence for this encoder is given by

$$\begin{aligned}\mathbf{V}(D) &= \mathbf{U}(D)\mathbf{G}(D)\\ &= \mathbf{U}(D)\left[1 + D^2 \quad 1 + D + D^2\right]\\ &= \mathbf{U}(D)(1 + D^2)\left[1 \quad \frac{1 + D + D^2}{1 + D^2}\right]\end{aligned} \tag{10.8}$$

$$= \mathbf{U}'(D)\left[1 \quad \frac{1+D+D^2}{1+D^2}\right]$$
$$= \mathbf{U}'(D)\mathbf{G}_1(D),$$

in which

$$\mathbf{G}_1(D) = \left[1 \quad \frac{1+D+D^2}{1+D^2}\right], \quad \mathbf{U}'(D) = \mathbf{U}(D)\mathbf{T}(D)$$

and

$$\mathbf{T}(D) = 1 + D^2.$$

The set $\mathbf{V}(D)$ of output sequences is produced by the product of the set $\mathbf{U}'(D)$ of input sequences by the generator matrix $\mathbf{G}_1(D)$, as well as by the product of the set $\mathbf{U}(D)$ of input sequences by the original generator matrix $\mathbf{G}(D)$, in which

$$\mathbf{G}(D) = \mathbf{T}(D)\mathbf{G}_1(D).$$

The two generator matrices $\mathbf{G}(D)$ and $\mathbf{G}_1(D)$ are said to be equivalent if $\mathbf{T}(D)$ is invertible. Matrix $\mathbf{G}_1(D)$ in Example 10.24 is in systematic form; however, its parity entry is not a polynomial but rather a rational function. The parity sequence output in Eq. 10.8 is obtained by the multiplication of its input sequence by the polynomial $1+D+D^2$ and division by the polynomial $1+D^2$. The multiplication operations can be performed with a shift register without feedback, while the division operations can be performed with a shift register with feedback. The entries in the parity submatrix of the generator matrix of a RSC encoder are rational functions in the variable D with binary coefficients; in other words, they are a ratio of two polynomials with binary coefficients. A rational transfer function like this

$$\frac{f_0 + f_1 D + \cdots + f_m D^m}{1 + q_1 D + \cdots + q_m D^m},$$

can be implemented with the circuit shown in Fig. 10.10 (Johanneson and Zigangirov 1999) as part of an encoder. The output v_j in Fig. 10.10 is a linear function of both the input data and the shift-register contents. The shift-register input is a function of both its input data and its contents. At time instant j the output v_j is given by

$$v_j = \sum_{i=0}^{m} f_i w_{j-i}.$$

By employing the polynomial representation, it follows that

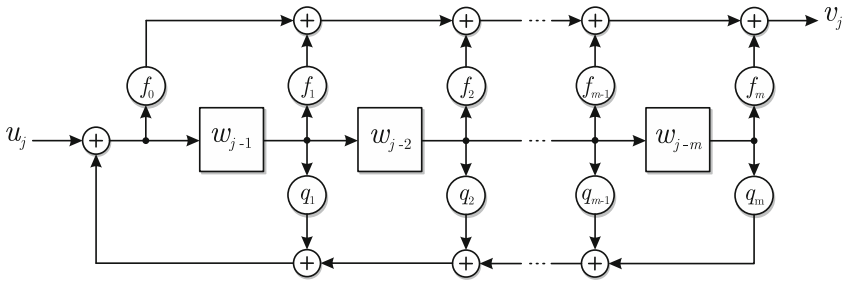


Fig. 10.10 Circuit for implementing the $\frac{f_0+f_1D+\cdots+f_mD^m}{1+q_1D+\cdots+q_mD^m}$ rational transfer function

$$v(D) = \sum_{j=-\infty}^{\infty} v_j D^j = \sum_{j=-\infty}^{\infty} \sum_{i=0}^{m} f_i w_{j-i} D^j$$
$$= \sum_{s=-\infty}^{\infty} \left(\sum_{i=0}^{m} f_i D^i \right) w_s D^s = f(D)w(D), \tag{10.9}$$

in which $j - i$ was replaced by s and $f(D)$ and $w(D)$ are compact representations for

$$f(D) = f_0 + f_1 D + \cdots + f_m D^m,$$

and

$$w(D) = \sum_{s=-\infty}^{\infty} w_s D^s.$$

From Fig. 10.10, it follows that

$$w_j = u_j + \sum_{i=1}^{m} q_i w_{j-i},$$

or, equivalently, supposing $q_0 = 1$, we have

$$u_j = \sum_{i=0}^{m} q_i w_{j-i}.$$

By repeating the steps employed to obtain Eq. 10.9, we have

$$u(D) = q(D)w(D), \tag{10.10}$$

in which

$$u(D) = \sum_{j=-\infty}^{\infty} u_j D^j,$$

and

$$q(D) = 1 + q_1 D + \cdots + q_m D^m.$$

By combining Eq. 10.9 and Eq. 10.10, it follows that

$$v(D) = u(D)\frac{f(D)}{q(D)} = u(D)\frac{f_0 + f_1 D + \cdots + f_m D^m}{1 + q_1 D + \cdots + q_m D^m}. \tag{10.11}$$

Let

$$g(D) = \frac{f(D)}{q(D)}$$

in Eq. 10.11, then

$$v(D) = u(D)g(D).$$

The function $g(D)$ is called a rational transfer function. In general, a matrix $\mathbf{G}(D)$ the inputs of which are rational functions is called a matrix of rational transfer functions. The matrices of rational transfer functions associated with the encoders in Figs. 10.3 and 10.5, respectively, are

$$\mathbf{G}(D) = \left[\, 1 \;\; \tfrac{1+D+D^2}{1+D^2} \,\right],$$

$$\mathbf{G}(D) = \left[\, \tfrac{1+D^2}{1+D} \;\; \tfrac{1+D+D^2}{1+D} \,\right].$$

10.9 Punctured Convolutional Codes

In certain applications, there is a need to vary the rate of a convolutional code without altering the structure of the encoder. In this situation, the code rate is altered by not transmitting certain parity symbols in the codeword or, in other words, by puncturing the original code (Vucetic and Yuan 2001; Clark Jr. and Cain 1981). The main reason for constructing punctured codes is the fact that their respective trellises possess a structure which is simpler than the corresponding non-punctured codes, thus allowing a simplification when implementing the Viterbi algorithm at the cost of a small increase in the bit error probability.

Punctured codes are convolutional codes with parameters (n, k, m), obtained in general from a convolutional code with parameters $(n, 1, m)$. It is possible to explain the construction of a punctured convolutional code by means of an example.

Example 10.25 Consider the encoder shown in Fig. 10.7, with polynomial generator matrix:

$$\mathbf{G}(D) = \left[1 + D^2 \; 1 + D + D^2\right],$$

and trellis diagram illustrated in Fig. 10.8. For each four output symbols, one output symbol is punctured (or extracted), and so the encoder will output three symbols for every two input information symbols. The resulting punctured code has asymptotic rate $2/3$. The trellis diagram for this code is shown in Fig. 10.11, in which $\times$ indicates a punctured symbol. The set of code sequences produced by this punctured code is identical to the set of code sequences generated by the code with asymptotic rate $2/3$ and polynomial generator matrix:

$$\mathbf{G}(D) = \begin{bmatrix} 1 + D & 1 + D & 1 \\ 0 & D & 1 + D \end{bmatrix},$$

for which the encoder and trellis diagram are illustrated in Figs. 10.12 and 10.13, respectively. It is noticed that the trellis illustrated in Fig. 10.13 is more complex than that for the punctured code illustrated in Fig. 10.11, since a maximum of four branches enter each state instead of only two branches as in the punctured code trellis. As a result, the encoding and decoding operations are more complex. The

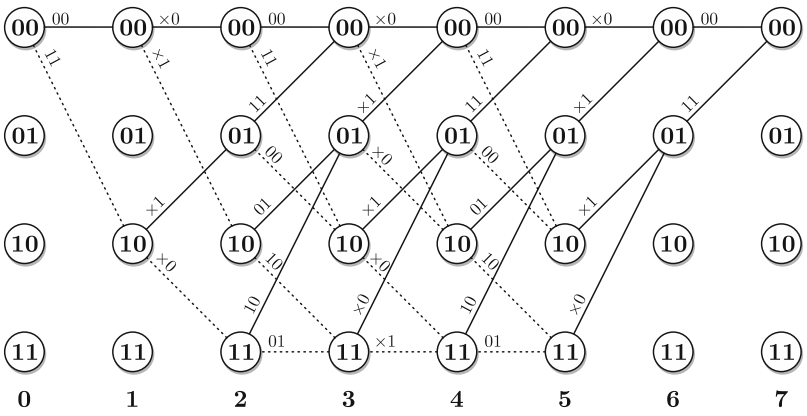


Fig. 10.11 Trellis diagram for the punctured code obtained from the code with polynomial generator matrix $G(D) = \left[1 + D^2 \;\; 1 + D + D^2\right]$. For each four output symbols, one output symbol is punctured (or extracted), and in this manner, the encoder will output three symbols for every two input information symbols. The resulting punctured code has asymptotic rate $2/3$

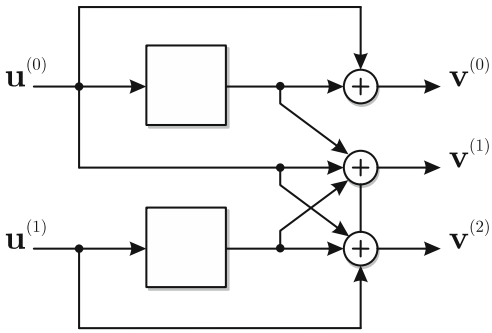


Fig. 10.12 Encoder for a rate 2/3 code

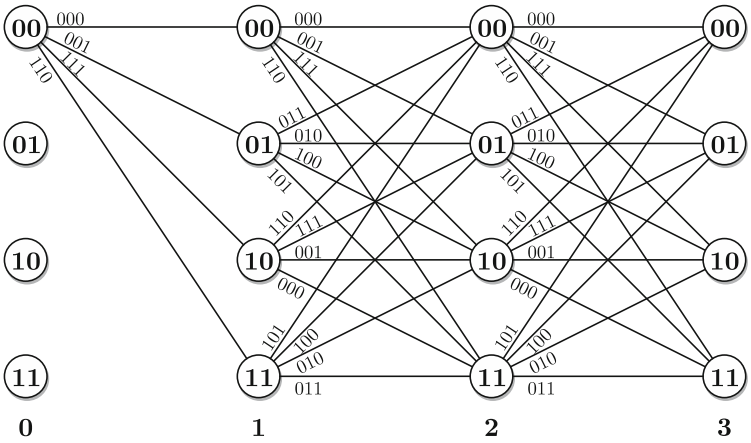


Fig. 10.13 Trellis for the encoder illustrated in Fig. 10.12

symbols to be extracted from the code are indicated in the trellis in Fig. 10.11 and can be described by the following puncturing table:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

in which a "0″ in the puncturing table means that the corresponding code symbol is not transmitted. In Example 10.27, the first symbol of the label in every second branch is not transmitted. In general, a punctured convolutional code with asymptotic rate p/q can be constructed from a convolutional code with parameters $(n, 1, m)$ after puncturing $np - q$ symbols from each np symbols corresponding to p input information symbols. The convolutional code with parameters $(n, 1, m)$ is called the mother code and is specified by the following generator matrix:

$$\left[\mathbf{g}_1^{(1)} \mathbf{g}_1^{(2)} \cdots \mathbf{g}_1^{(n)}\right].$$

The symbols to be punctured are specified in a puncturing table $\mathbf{P}$, i.e.,

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1j} & \cdots & p_{1p} \\ p_{21} & p_{22} & \cdots & p_{2j} & \cdots & p_{2p} \\ \vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\ p_{i1} & p_{i2} & \cdots & p_{ij} & \cdots & p_{ip} \\ \vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nj} & \cdots & p_{np} \end{bmatrix},$$

in which $p_{ij} \in \{0, 1\}$, $1 \leq i \leq n$, and $1 \leq j \leq p$. The puncturing operation is performed periodically at each np code symbols and p is called the puncturing period.

10.10 Decoding Convolutional Codes

At present, in most practical applications of convolutional codes, the Viterbi algorithm (VA) (Viterbi 1967; Viterbi and Omura 1979) is employed for decoding. This preference comes from the fact that the bit error rate at the output of a Viterbi decoder is considered adequate in such applications. Lower bit error rates however may call for a sequential decoder. As mentioned earlier, the VA performs maximum likelihood decoding and benefits from the trellis structure of convolutional codes. On receiving a sequence of convolutionally encoded symbols contaminated by noise, at the decoder, the VA estimates the most likely sequence path through the corresponding code trellis.

The scope of this chapter and the limited space available do not allow us to address the VA decoding of convolutional codes at its deserved depth. However, the interested reader now has available a wide choice of publications covering the Viterbi algorithm as, for example, references Viterbi and Omura (1979), Lin and Costello (2004), and Johanneson and Zigangirov (1999), to name a few.

In the following, we present the original BCJR algorithm (Bahl et al. 1974) and the modified BCJR algorithm (Berrou and Glavieux 1996), both of which minimize the bit error probability, recalling that the VA algorithm minimizes the block error probability.

10.10.1 The BCJR Algorithm

The Viterbi algorithm (Lin and Costello 2004; Michelson and Levesque 1985; Johanneson and Zigangirov 1999) provides an optimum decoding method in the sense that it minimizes the probability of error per block for convolutional codes

contaminated by additive white Gaussian noise. The Viterbi algorithm does not necessarily minimizes the probability of error per symbol. An algorithm which minimizes the probability of error per symbol was proposed in 1974 by Bahl, Cocke, Jelinek, and Raviv and became known as the BCJR decoding algorithm (Bahl et al. 1974). The BCJR algorithm was later modified (Berrou and Glavieux 1996) to be used with recursive convolutional encoders and was used for decoding turbo codes in mobile communication (3G systems[1]). Both the original BCJR algorithm and the modified BCJR algorithm (Berrou and Glavieux 1996; Thitimajshima 1993) are addressed in the sequel.

10.10.2 The Original BCJR Algorithm

Without loss of essential generality, consider a systematic convolutional encoder with asymptotic code rate of $1/2$ and M states. The corresponding sequence of input information symbols is denoted as

$$\mathbf{u} = \mathbf{u}_1^N = \{u_1, u_2, \ldots, u_k, \ldots, u_N\}.$$

The associated encoded output sequence is denoted as

$$\mathbf{v} = \mathbf{v}_1^N = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \ldots, \mathbf{v}_N\},$$

in which $\mathbf{v}_k = (v_k^{(1)}, v_k^{(2)}) = (u_k, v_k^{(2)})$ is the output block associated with an information symbol, $v_k^{(2)}$ is the non-systematic encoder output, and $\mathbf{v}_1^N$ is fed as input to a memoryless channel contaminated by additive white Gaussian noise the output of which is the received sequence

$$\mathbf{r} = \mathbf{r}_1^N = \{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_k, \ldots, \mathbf{r}_N\},$$

in which $\mathbf{r}_k = (x_k, y_k)$. The random variables x_k and y_k at time instant k are defined by the following equations:

$$x_k = (2u_k - 1) + i_k \tag{10.12}$$

$$y_k = (2v_k^{(2)} - 1) + q_k, \tag{10.13}$$

in which i_k and q_k are two independent noise samples with identical variance σ^2.

[1] The evolution of technology for mobile communication has moved from turbo codes (3G systems), the decoding of which is based on the BCJR algorithm (Berrou and Glavieux 1996), to LDPC (4G systems) (Gallager 1963; Tanner 1981; MacKay 1999), and now, 5G systems, which employ LDPC codes on the coding scheme for the data channel and Polar Codes (Arikan 2009), are used on the coding scheme for the control channel.

The BCJR algorithm starts by computing the log-likelihood ratio $\Lambda(u_k)$ associated with each information symbol as follows:

$$\Lambda(u_k) = \log\left\{\frac{P\{u_k = 1|\mathbf{r}\}}{P\{u_k = 0|\mathbf{r}\}}\right\}, \tag{10.14}$$

in which $P\{u_k = i|\mathbf{r}\}$, $i \in \{0, 1\}$, is the *a posteriori* probability of the information symbol u_k. Considering $1 \leq k \leq N$, where N denotes the length of the received sequence $\mathbf{r}$, the decoder decides that $\hat{u}_k = 1$, if $P\{u_k = 1|\text{observation}\} \geq P\{u_k = 0|\text{observation}\}$; otherwise, it decides that $\hat{u}_k = 0$. This decision rule can be expressed as follows:

$$\Lambda(u_k) \begin{cases} \geq 0 : \hat{u}_k = 1, \\ < 0 : \hat{u}_k = 0. \end{cases}$$

The modulo of $\Lambda(u_k)$ represents the soft-decision information associated with the hard-decision value (0 or 1) estimated for u_k. The state of the encoder at the kth time interval is given by S_k. The *a posteriori* probability of each information symbol can be extracted from the joint probability $\lambda_k^i(m)$ defined as

$$\begin{aligned} \lambda_k^i(m) &= P\{u_k = i, S_k = m|\mathbf{r}_1^N\} \\ &= \frac{p\{u_k = i, S_k = m, \mathbf{r}_1^N\}}{p\{\mathbf{r}_1^N\}}, \end{aligned} \tag{10.15}$$

in which $p\{\mathbf{r}_1^N\}$ represents the probability density function of $\mathbf{r}_1^N$. Therefore, the *a posteriori* probability of the information symbol u_k is given by

$$P\{u_k = i|\mathbf{r}_1^N\} = \sum_{m=0}^{M-1} \lambda_k^i(m), \quad i \in \{0, 1\}.$$

Thus, the log-likelihood ratio in Eq. 10.14 can be written as

$$\Lambda(u_k) = \log\left\{\frac{\sum_{m=0}^{M-1} \lambda_k^1(m)}{\sum_{m=0}^{M-1} \lambda_k^0(m)}\right\}. \tag{10.16}$$

From Eq. 10.15 and Eq. 10.16, it follows that

$$\Lambda(u_k) = \log\left\{\frac{\sum_m \sum_{m'} p\{u_k = 1, S_k = m, S_{k-1} = m', \mathbf{r}_1^{k-1}, \mathbf{r}_k, \mathbf{r}_{k+1}^N\}}{\sum_m \sum_{m'} p\{u_k = 0, S_k = m, S_{k-1} = m', \mathbf{r}_1^{k-1}, \mathbf{r}_k, \mathbf{r}_{k+1}^N\}}\right\}. \tag{10.17}$$

By taking into account the fact that events after time instant k are not influenced by the observation $\mathbf{r}_1^k$ and by bit u_k whenever state S_k is known, it follows that

$$\Lambda(u_k) = \log\left\{\frac{\sum_m \sum_{m'} p\{\mathbf{r}_{k+1}^N | S_k = m\} p\{S_{k-1} = m', \mathbf{r}_1^{k-1}\}}{\sum_m \sum_{m'} p\{\mathbf{r}_{k+1}^N | S_k = m\} p\{S_{k-1} = m', \mathbf{r}_1^{k-1}\}} \times \frac{p\{u_k = 1, S_k = m, \mathbf{r}_k | S_{k-1} = m'\}}{p\{u_k = 0, S_k = m, \mathbf{r}_k | S_{k-1} = m'\}}\right\}. \tag{10.18}$$

By introducing the following probability functions

$$\begin{aligned} \alpha_k(m) &= p\{S_k = m, \mathbf{r}_1^k\}, \\ \beta_k(m) &= p\{\mathbf{r}_{k+1}^N | S_k = m\}, \\ \gamma_i(\mathbf{r}_k, m', m) &= p\{u_k = i, S_k = m, \mathbf{r}_k | S_{k-1} = m'\}, \end{aligned}$$

it follows that

$$\Lambda(u_k) = \log\left\{\frac{\sum_m \sum_{m'} \gamma_1(\mathbf{r}_k, m', m)\alpha_{k-1}(m')\beta_k(m)}{\sum_m \sum_{m'} \gamma_0(\mathbf{r}_k, m', m)\alpha_{k-1}(m')\beta_k(m)}\right\}, \tag{10.19}$$

in which $\alpha_k(m)$, for $k = 1, 2, \ldots, N$, can be computed in a recursive manner by means of

$$\begin{aligned} \alpha_k(m) &= \sum_{m'=0}^{M-1} p\{S_{k-1} = m', S_k = m, \mathbf{r}_1^k\} \\ &= \sum_{m'=0}^{M-1} p\{S_{k-1} = m', \mathbf{r}_1^{k-1}\} p\{S_k = m, \mathbf{r}_k | S_{k-1} = m', \mathbf{r}_1^{k-1}\} \\ &= \sum_{m'=0}^{M-1} p\{S_{k-1} = m', \mathbf{r}_1^{k-1}\} p\{S_k = m, \mathbf{r}_k | S_{k-1} = m'\} \qquad (10.20) \\ &= \sum_{i=0}^{1} \sum_{m'=0}^{M-1} p\{S_{k-1} = m', \mathbf{r}_1^{k-1}\} p\{u_k = i, S_k = m, \mathbf{r}_k | S_{k-1} = m'\} \\ &= \sum_{i=0}^{1} \sum_{m'=0}^{M-1} \alpha_{k-1}(m')\gamma_i(\mathbf{r}_k, m', m). \qquad (10.21) \end{aligned}$$

Equality in Eq. 10.20 follows from the fact that events after time $k-1$ are not influenced by $\mathbf{r}_1^{k-1}$ whenever S_{k-1} is known. Considering that the code trellis is initialized at state $S_0 = \mathbf{0}$, the boundary conditions are as follows:

$$\alpha_0(\mathbf{0}) = 1 \text{ and } \alpha_0(m) = 0, \text{ for } m \neq 0. \tag{10.22}$$

Similarly, for $k = 1, 2, \ldots, N-1$, $\beta_k(m)$ can also be calculated in a recursive manner as

$$\begin{aligned}
\beta_k(m) &= \sum_{m'=0}^{M-1} p\{S_{k+1} = m', \mathbf{r}_{k+1}^N | S_k = m\} \\
&= \sum_{m'=0}^{M-1} p\{S_{k+1} = m', \mathbf{r}_{k+1} | S_k = m\} p\{\mathbf{r}_{k+2}^N | S_k = m, S_{k+1} = m', \mathbf{r}_{k+1}\} \\
&= \sum_{m'=0}^{M-1} p\{S_{k+1} = m', \mathbf{r}_{k+1} | S_k = m\} p\{\mathbf{r}_{k+2}^N | S_{k+1} = m'\} \\
&= \sum_{i=0}^{1} \sum_{m'=0}^{M-1} p\{u_{k+1} = i, S_{k+1} = m', \mathbf{r}_{k+1} | S_k = m\} p\{\mathbf{r}_{k+2}^N | S_{k+1} = m'\} \\
&= \sum_{i=0}^{1} \sum_{m'=0}^{M-1} \gamma_i(\mathbf{r}_{k+1}, m, m') \beta_{k+1}(m').
\end{aligned} \tag{10.23}$$

The appropriate boundary conditions when the decoder is guided to terminate at state $\mathbf{0}$, i.e., $S_N = \mathbf{0}$, are the following:

$$\beta_N(\mathbf{0}) = 1 \text{ and } \beta_N(m) = 0, \text{ for } m \neq 0. \tag{10.24}$$

When the final decoder state in unknown, the appropriate boundary conditions (Berrou and Glavieux 1996) are

$$\beta_N(\mathbf{0}) = 1/M \text{ and } \beta_N(m) = 0, \text{ for } m \neq 0. \tag{10.25}$$

The probabilities $\gamma_i(\mathbf{r}_k, m', m)$ can be determined from the transition probabilities of the channel contaminated with additive white Gaussian noise and the transition probabilities of the encoder trellis as

$$\begin{aligned}
\gamma_i(\mathbf{r}_k, m', m) = P\{S_k = m | S_{k-1} = m'\} p\{\mathbf{r}_k | u_k = i, S_k = m, S_{k-1} = m'\} \\
P\{u_k = i | S_k = m, S_{k-1} = m'\}.
\end{aligned} \tag{10.26}$$

The transition probabilities $P\{S_k = m | S_{k-1} = m'\}$ are defined by the *a priori* probabilities of the input bits. When the input bits are equiprobable, i.e., $P\{u_k = 1\} = P\{u_k = 0\} = 1/2$, then $P\{S_k = m | S_{k-1} = m'\} = 1/2$. The probability $P\{\mathbf{r}_k | u_k = i, S_k = m, S_{k-1} = m'\}$ is the transition probability of the channel contaminated with additive white Gaussian noise and can be written as

$$p\{\mathbf{r}_k|u_k = i, S_k = m, S_{k-1} = m'\} = p\{\mathbf{r}_k|\mathbf{v}_k\}, \tag{10.27}$$

in which

$$p\{\mathbf{r}_k|\mathbf{v}_k\} = \prod_{j=0}^{n-1} p\{r_k^{(j)}|v_k^{(j)}\},$$

and

$$p\{r_k^{(j)}|v_k^{(j)}\} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(r_k^{(j)} - v_k^{(j)})^2}{2\sigma^2}\right),$$

thus

$$p\{r_k^{(j)}|v_k^{(j)} = -1\} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(r_k^{(j)} + 1)^2}{2\sigma^2}\right),$$

$$p\{r_k^{(j)}|v_k^{(j)} = +1\} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(r_k^{(j)} - 1)^2}{2\sigma^2}\right).$$

The probability $P\{u_k = i|S_k = m, S_{k-1} = m'\}$ is either equal to 0 or 1 since the convolutional encoder is a deterministic machine. The expression for $\gamma_i(\mathbf{r}_k, m', m)$ can be written as

$$\gamma_i(\mathbf{r}_k, m', m) = \begin{cases} P\{u_k = i\} \exp\left(-\frac{\sum_{j=0}^{n-1}(r_k^{(j)} - v_{k,i}^{(j)})^2}{2\sigma^2}\right) & \text{for } m, m' \in B_k^i \\ 0, & \text{otherwise,} \end{cases} \tag{10.28}$$

in which $v_{k,i}^{(j)}$ is the encoder output associated with the transition of state $S_{k-1} = m'$ to state $S_k = m$ and input $u_k = i$ and B_k^i denotes the set of state transitions from $S_{k-1} = m'$ to $S_k = m$ caused by the input bit $u_k = i$. Substituting $\gamma_i(\mathbf{r}_k, m', m)$ from Eq. 10.28 into Eq. 10.19, we have for $\Lambda(u_k)$ the following:

$$\Lambda(u_k) = \log\left\{\frac{\sum_m \sum_{m'} P\{u_k = 1\} \exp\left(-\frac{\sum_{j=0}^{n-1}(r_k^{(j)} - v_{k,1}^{(j)})^2}{2\sigma^2}\right)\alpha_{k-1}(m')\beta_k(m)}{\sum_m \sum_{m'} P\{u_k = 0\} \exp\left(-\frac{\sum_{j=0}^{n-1}(r_k^{(j)} - v_{k,0}^{(j)})^2}{2\sigma^2}\right)\alpha_{k-1}(m')\beta_k(m)}\right\}. \tag{10.29}$$

In order to simplify notation, we make $P\{u_k = i\} = p_k(i)$ in Eq. 10.29 and write

$$\Lambda(u_k) = \log \left\{ \frac{\sum_m \sum_{m'} p_k(1) \exp\left(-\frac{\sum_{j=0}^{n-1} (r_k^{(j)} - v_{k,1}^{(j)})^2}{2\sigma^2} \right) \alpha_{k-1}(m') \beta_k(m)}{\sum_m \sum_{m'} p_k(0) \exp\left(-\frac{\sum_{j=0}^{n-1} (r_k^{(j)} - v_{k,0}^{(j)})^2}{2\sigma^2} \right) \alpha_{k-1}(m') \beta_k(m)} \right\}, \tag{10.30}$$

for $(m, m') \in B_k^i$, where it is assumed that in Eq. 10.29 and 10.30 that the numerator contemplates all state transitions from $S_{k-1} = m'$ to $S_k = m$ which are caused by the input bit $u_k = 0$, $(m, m' \in B_k^0)$ and the denominator contemplates all state transitions from $S_{k-1} = m'$ to $S_k = m$ which are caused by the input bit $u_k = 1$ $(m, m' \in B_k^1)$.

Summarizing, the steps for the BCJR algorithm to be executed are as follows:

1. Initial conditions for $\alpha_0(m)$ and $\beta_N(m)$ are presented in Eq. 10.22 and Eq. 10.24 or Eq. 10.25.
2. When $\mathbf{r}_k$ is received, the decoder computes $\gamma_i(\mathbf{r}_k, m', m)$ using Eq. 10.28 and computes $\alpha_k(m)$ using Eq. 10.21. The values of $\alpha_k(m)$ are stored, for all k and m.
3. After the complete sequence $\mathbf{r}_1^N$ is received, the decoder recursively computes $\beta_k(m)$ using Eq. 10.23. The value computed for $\beta_k(m)$ is then multiplied by the appropriate $\alpha_k(m)$ and $\gamma_i(\mathbf{r}_k, m', m)$ to obtain Eq. 10.30.

Example 10.26 Consider a coding system employing a RSC (2, 1, 1) with polynomial generator matrix:

$$\mathbf{G}(D) = \left[1 \ \ \frac{1}{1+D} \right].$$

The corresponding trellis is illustrated in Fig. 10.14. The BPSK modulation is assumed and $\frac{E_b}{N_0} = 2$ dB, where E_b denotes the signal energy per bit and N_0 denotes the noise power. The transmitted information sequence is

$$\mathbf{u} = (001101),$$

and the received code sequence is

$$\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}_5, \mathbf{r}_6, \mathbf{r}_7),$$

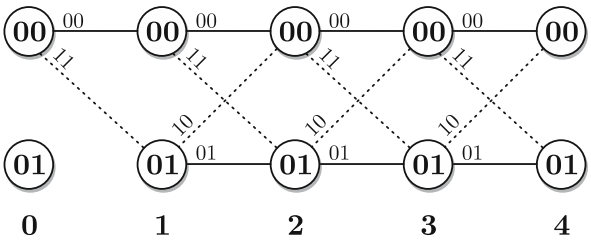


Fig. 10.14 Trellis constructed with the polynomial generator matrix $G(D) = \left[1 \ \ \frac{1}{1+D} \right]$

Table 10.3 Parameters α, β, γ, and Λ after the BCJR algorithm is applied to decode codewords from a RSC (2,1,1) with polynomial generator matrix $G(D) = \left[1 \ \frac{1}{1+D}\right]$. Values are rounded off to four decimal places due to space limitations

k	1	2	3	4	5	6	7
u_k	0	0	1	1	0	1	–
x_k	−0.0236	−2.1084	0.7163	1.4271	−1.4689	0.7887	0.2671
y_k	−1.5870	−1.0626	0.2762	0.2771	−1.7265	1.5689	−2.0307
$\gamma_0(\mathbf{r}_k, 0, 0)$	0.2071	0.2165	0.0224	0.0030	0.3009	0.0006	0.0816
$\gamma_0(\mathbf{r}_k, 1, 1)$	0.0278	0.0121	0.0474	0.0064	0.0028	0.0457	0.0003
$\gamma_1(\mathbf{r}_k, 0, 1)$	0.0026	0	0.3317	0.3097	0	0.3893	0.0007
$\gamma_1(\mathbf{r}_k, 1, 0)$	0.1942	0.0007	0.1566	0.1459	0.0056	0.0055	0.1687
$\alpha_k(0)$	0.2071	0.0448	0.0010	0.0022	0.0007	0	0
$\alpha_k(1)$	0.0026	0	0.0149	0.0004	0	0.0002	0
$\beta_k(0)$	0.0001	0.0002	0.0001	0.0099	0.0329	0.0408	0.5
$\beta_k(1)$	0	0.0001	0.0014	0.0002	0.0041	0.0843	0
$\Lambda_k(u_k)$	−8.7332	−9.5391	5.4428	6.0920	−5.6578	6.8868	–
$\hat{u}_k$	0	0	1	1	0	1	–

in which the values of x_k and y_k obtained from Eq. 10.12 and Eq. 10.13, respectively, are presented in Table 10.3. Our goal here is to compute both the hard output and the soft output from the BCJR decoder. After feeding the complete input sequence, one more bit is fed to the encoder so as to have an information sequence of length 6 bits and an output sequence of length 14 bits. The last bit appended to the information sequence was not intended to reach state **0** in the trellis; for this reason, the boundary conditions used for $\beta_N(m)$ were

$$\beta_7(\mathbf{0}) = 0.5 \quad and \quad \beta_7(m) = 0, \quad \text{for} \quad m \neq 0.$$

The asymptotic code rate is

$$R = \frac{5}{10+2} = 0.416.$$

The white Gaussian noise variance is

$$\sigma^2 = \frac{N_0}{2}.$$

The signal to noise ratio is

$$SNR = \frac{E_s}{N_0},$$

in which E_s denotes the signal energy per block

$$E_s = RE_b.$$

The variance can be written as follows:

$$\sigma^2 = \frac{E_s}{2SNR} = \frac{0.5}{R\left(\frac{E_b}{N_0}\right)},$$

and substituting the value for E_s and expressing $\frac{E_b}{N_0}$ in dB, we have

$$\sigma^2 = \frac{0.5}{0.416(10^{0.2})}.$$

The values computed at time instants $k = 1, 2, \ldots, 7$ for $\gamma_i(\mathbf{r}_k, m', m)$, $\alpha_k(m)$, $\beta_k(m)$, and $\Lambda_k(u_k)$ are illustrated in Table 10.3.

10.10.3 The Modified BCJR Algorithm

When the original BCJR algorithm is used with a RSC, the value of $\alpha_k(m)$ decreases rapidly as the value of k increases, and the value of $\beta_k(m)$ decreases rapidly as the value of k diminishes (see Table 10.3), consequently making it difficult to compute the value of $\Lambda(u_k)$ in Eq. 10.19. In order to avoid this difficulty, the modified BCJR algorithm was introduced Berrou et al. (1993), Berrou and Glavieux (1996), and Thitimajshima (1993).

Starting from (10.18) and introducing the probability functions

$$\alpha_k(m) = P\{S_k = m|\mathbf{r}_1^k\}, \tag{10.31}$$

$$\beta_k(m) = \frac{p\{\mathbf{r}_{k+1}^N|S_k = m\}}{p\{\mathbf{r}_{k+1}^N|\mathbf{r}_1^k\}}, \tag{10.32}$$

$$\gamma_i(\mathbf{r}_k, m', m) = p\{u_k = i, S_k = m, \mathbf{r}_k|S_{k-1} = m'\}, \tag{10.33}$$

the value of $\Lambda(u_k)$ in Eq. 10.19 can be obtained by recursively computing Eq. 10.31 and Eq. 10.32 as explained next. From Eq. 10.31, it follows that

$$\begin{aligned}\alpha_k(m) &= P\{S_k = m|\mathbf{r}_1^k\} \\ &= \frac{p\{S_k = m, \mathbf{r}_1^k\}}{p\{\mathbf{r}_1^k\}} = \frac{p\{S_k = m, \mathbf{r}_1^{k-1}, \mathbf{r}_k\}}{p\{\mathbf{r}_1^{k-1}, \mathbf{r}_k\}} \\ &= \frac{p\{S_k = m, \mathbf{r}_k|\mathbf{r}_1^{k-1}\}}{p\{\mathbf{r}_k|\mathbf{r}_1^{k-1}\}}.\end{aligned} \tag{10.34}$$

The numerator in Eq. 10.34 can be written as a function of bit u_k and the state S_{k-1} as follows:

$$\begin{aligned} p\{S_k = m, \mathbf{r}_k | \mathbf{r}_1^{k-1}\} &= \sum_{m'} \sum_{i=0}^{1} p\{u_k = i, S_{k-1} = m', S_k = m, \mathbf{r}_k | \mathbf{r}_1^{k-1}\} \\ &= \sum_{m'} \sum_{i=0}^{1} p\{u_k = i, S_k = m, \mathbf{r}_k | S_{k-1} = m', \mathbf{r}_1^{k-1}\} \\ &\quad P\{S_{k-1} = m' | \mathbf{r}_1^{k-1}\} \\ &= \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_k, m', m) \alpha_{k-1}(m'). \end{aligned} \tag{10.35}$$

Equation 10.35 results from the fact that events after time $(k-1)$ are independent of the observation $\mathbf{r}_1^{k-1}$, if S_{k-1} is known. The denominator in (10.34) can be written as

$$\begin{aligned} p\{\mathbf{r}_k | \mathbf{r}_1^{k-1}\} &= \sum_{m} \sum_{m'} \sum_{i=0}^{1} p\{u_k = i, S_k = m, S_{k-1} = m', \mathbf{r}_k | \mathbf{r}_1^{k-1}\} \\ &= \sum_{m} \sum_{m'} \sum_{i=0}^{1} p\{u_k = i, S_k = m, \mathbf{r}_k | S_{k-1} = m', \mathbf{r}_1^{k-1}\} \\ &\quad P\{S_{k-1} = m' | \mathbf{r}_1^{k-1}\} \\ &= \sum_{m} \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_k, m', m) \alpha_{k-1}(m'). \end{aligned} \tag{10.36}$$

Finally, the probability $\alpha_k(m)$ can be written as a function of $\alpha_{k-1}(m)$ by employing Eq. 10.35 and Eq. 10.36, i.e.,

$$\alpha_k(m) = \frac{\sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_k, m', m) \alpha_{k-1}(m')}{\sum_{m} \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_k, m', m) \alpha_{k-1}(m')}. \tag{10.37}$$

In a similar manner, it follows that $\beta_k(m)$ can be computed recursively, and from Eq. 10.32, it follows that

$$\beta_k(m) = \frac{p\{\mathbf{r}_{k+1}^N | S_k = m\}}{p\{\mathbf{r}_{k+1}^N | \mathbf{r}_1^k\}}$$

$$= \frac{\sum_{m'} \sum_{i=0}^{1} p\{u_{k+1} = i, S_{k+1} = m', \mathbf{r}_{k+2}^{N}, \mathbf{r}_{k+1} | S_k = m\}}{p\{\mathbf{r}_{k+1}^{N} | \mathbf{r}_1^k\}}. \tag{10.38}$$

The numerator in Eq. 10.38 can equivalently be written as

$$\begin{aligned} p\{\mathbf{r}_{k+1}^{N} | S_k = m\} &= \sum_{m'} \sum_{i=0}^{1} p\{\mathbf{r}_{k+2}^{N} | S_{k+1} = m'\} \\ &\quad \times p\{u_{k+1} = i, S_{k+1} = m', \mathbf{r}_{k+1} | S_k = m\} \\ &= \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_{k+1}, m, m') \beta_{k+1}(m') p\{\mathbf{r}_{k+2}^{N} | \mathbf{r}_1^{k+1}\}. \end{aligned}$$

Also, $\beta_k(m)$ can equivalently be written as

$$\beta_k(m) = \frac{\sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_{k+1}, m, m') \beta_{k+1}(m')}{P\{\mathbf{r}_{k+1} | \mathbf{r}_1^k\}}. \tag{10.39}$$

Substituting k by $k + 1$ in Eq. 10.36, the denominator in Eq. 10.39 becomes

$$p\{\mathbf{r}_{k+1} | \mathbf{r}_1^k\} = \sum_{m} \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_{k+1}, m', m) \alpha_k(m').$$

Finally, the probability $\beta_k(m)$ can equivalently be written as a function of $\beta_{k+1}(m')$ by employing the following relation:

$$\beta_k(m) = \frac{\sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_{k+1}, m, m') \beta_{k+1}(m')}{\sum_{m} \sum_{m'} \sum_{i=0}^{1} \gamma_i(\mathbf{r}_{k+1}, m', m) \alpha_k(m')}. \tag{10.40}$$

The modified BCJR algorithm follows steps similar to the original BCJR algorithm:

1. Initial conditions $\alpha_0(m)$ and $\beta_N(m)$ are given in Eq. 10.22 and Eq. 10.24 or Eq. 10.25.
2. When $\mathbf{r}_k$ is received, the decoder computes $\gamma_i(\mathbf{r}_k, m', m)$ using Eq. 10.26 and computes $\alpha_k(m)$ using Eq. 10.37. The values of $\alpha_k(m)$ are stored for all k and m.
3. After the complete $\mathbf{r}_1^N$ sequence is received, the decoder recursively computes $\beta_k(m)$ using Eq. 10.40. The value computed for $\beta_k(m)$ is multiplied by $\alpha_k(m)$ and the appropriate $\gamma_i(\mathbf{r}_k, m', m)$ to obtain $\Lambda(u_k)$ as given in Eq. 10.19.

Example 10.27 We now solve Example 10.26 by employing the modified BCJR algorithm. The code considered is a RSC (2, 1, 1) with polynomial generator matrix:

Table 10.4 Parameters α, β, γ, and Λ resulting from the application of the modified BCJR algorithm to a system encoded with a RSC $(2, 1, 1)$ having polynomial generator matrix $G(D) = \left[1 \ \frac{1}{1+D}\right]$. Values are rounded off to four decimal places due to space limitations

k	1	2	3	4	5	6	7
u_k	0	0	1	1	0	1	–
x_k	− 0.0236	− 2.1084	0.7163	1.4271	− 1.4689	0.7887	0.2671
y_k	− 1.5870	− 1.0626	0.2762	0.2771	− 1.7265	1.5689	− 2.0307
$\gamma_0(\mathbf{r}_k, 0, 0)$	0.2071	0.2164	0.0224	0.0030	0.3009	0.0006	0.0816
$\gamma_0(\mathbf{r}_k, 1, 1)$	0.0278	0.0121	0.0474	0.0064	0.0028	0.0457	0.00033
$\gamma_1(\mathbf{r}_k, 0, 1)$	0.0026	0	0.3316	0.3097	0	0.3893	0.0007
$\gamma_1(\mathbf{r}_k, 1, 0)$	0.1942	0.0007	0.1566	0.1459	0.0056	0.0055	0.1687
$\alpha_k(0)$	0.9876	0.9991	0.0635	0.8419	0.9981	0.0017	0.9980
$\alpha_k(1)$	0.0124	0.0009	0.9365	0.1580	0.0019	0.9983	0.0019
$\beta_k(0)$	1.0104	0.9988	0.0661	1.1810	0.9997	0.4834	0.5000
$\beta_k(1)$	0.0129	0.1713	1.0613	0.0232	0.1240	0.9989	0
$\Lambda_k(u_k)$	− 8.7332	− 9.5391	5.4428	6.0920	− 5.6579	6.8868	–
$\hat{u}_k$	0	0	1	1	0	1	–

$$\mathbf{G}(D) = \left[1 \ \frac{1}{1+D}\right].$$

The received value of $\mathbf{r}$ is the same as that in Table 10.3. It is observed in Table 10.4 that the values of $\gamma_i(\mathbf{r}_k, m', m)$ and $\Lambda(u_k)$ are the same as in Table 10.3. The values of $\alpha_k(m)$, computed according Eq. 10.37, do not decrease monotonically as k increases. The values of $\beta_k(m)$, computed according to Eq. 10.40, do not decrease monotonically as the values of k become smaller and smaller.

10.11 Problems

1. Calculate the probability that an error pattern will not be detected at the receiver when a $(7, 3, 4)$ m-sequence code is employed for transmission through a BSC with probability of error p, $p < 1/2$.
2. Calculate the block error rate after decoding is performed on a received word when a $(7, 3, 4)$ m-sequence code is employed for transmission through a BSC with probability of error p, $p < 1/2$.
3. Compute the Hamming weight of the vectors $\mathbf{v}_1 = (1, 0, 1, 0, 2)$ and $\mathbf{v}_2 = (1, 1, 3, 0, 2)$ and the Hamming distance $d_H(\mathbf{v}_1, \mathbf{v}_2)$.
4. Consider a binary (n, k, d) block code for which d is an odd number. Prove that an extended $(n + 1, k, d + 1)$ code results when an overall parity-check digit is appended to each codeword of the given (n, k, d) code.
5. Construct the set of codewords for the linear binary $(7, 3, 4)$ code the parity-check digits of which satisfy the following expressions:

$$c_1 = k_1 \oplus k_2,\ c_2 = k_1 \oplus k_3,$$
$$c_3 = k_2 \oplus k_3,\ c_4 = k_1 \oplus k_2 \oplus k_3.$$

6. Consider the code specified in Problem 2.10.11. Construct the generator matrix and the parity-check matrix for this code.
7. Show that the $(2^m - 1, 2^m - m - 1, 3)$ binary Hamming code is capable of correcting any two erasures per codeword.
8. Consider the primitive polynomial $p(x) = x^3 + x^2 + 1$ having α as a root. Construct a table with the elements of the Galois field GF(8) expressed as powers of α.
9. Consider the primitive polynomial $p(x) = x^4 + x^3 + 1$ having α as a root. Construct a table with the elements of the Galois field GF(16) expressed as powers of α.
10. Determine the generator polynomial $g(x)$ of a Reed-Solomon code over GF(8) with minimum distance 5, having the binary primitive polynomial $p(x) = x^3 + x + 1$ as a factor of $g(x)$.
11. Suppose a (15, 5, 7) binary BCH code is required to have the roots α^i, $1 \leq i \leq 6$, in its generator polynomial $g(x)$. Determine the polynomial $g(x)$ satisfying this root composition requirement.
12. Specify three parity-check sums, orthogonal on position e_6 for the (7, 3, 4) binary cyclic m-sequence code with generator polynomial $g(x) = x^4 + x^3 + x^2 + 1$.
13. The (7, 4, 3) binary cyclic Hamming code is threshold decodable in two steps. Consider $g(x) = x^3 + x^2 + 1$ as the code generator polynomial and describe a corresponding two-step decoding algorithm. Hint: Look at the code parity-check matrix $\mathbf{H}$.
14. Consider the (15, 7, 5) binary BCH code, with generator polynomial $g(x) = m_1(x)m_3(x) = (x^4 + x + 1)(x^4 + x^3 + x^2 + x + 1)$, and decode the received word $r(x) = x + x^9$ with the BM algorithm.
15. Decode the received word $r(x) = \alpha^2 x^3$ employing the BM algorithm with the (7, 5, 3) RS code over GF(8) having generator polynomial $g(x) = (x - \alpha)(x - \alpha^2)$.
16. Decode the received word $r(x) = \alpha^2 x + \alpha^3 x^9$ employing the BM algorithm with the (15, 11, 5) RS code over GF(16) having generator polynomial $g(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^3)(x - \alpha^4)$, and consider α to be a primitive element of GF(16).
17. Decode the received word $r(x) = x^6 + x^3$ employing the Euclidean frequency domain decoder with the binary (7, 4, 3) BCH code having α, α^2, and α^4, as roots of the generator polynomial, and consider α to be a primitive element of GF(2^3).
18. Draw the state diagram and trellis for the $(n, k, m) = (2, 1, 1)$ binary convolutional code with polynomial generator matrix:

$$\mathbf{G} = [1 \quad 1 + D].$$

Suppose this code is used on a BSC and the received sequence for the first seven branches is 11 10 11 01 11 01 00. Sketch the decisions on a trellis diagram and label the survivor's Hamming distance metric at each node level. In case a tie occurs in the metrics of two paths converging to the same state in the trellis, randomly decide between the upper and lower paths to choose the survivor.

19. Draw the state diagram for the $(n, k, m) = (2, 1, 3)$ binary convolutional code with polynomial generator matrix:

$$\mathbf{G} = \left[1 + D^3 \; 1 + D^2 + D^3 \right]$$

and write the codeword corresponding to the input sequence 1 1 1 0 0 1 0.

20. Apply the original BCJR algorithm to the encoder considered in Example 10.26 with the input data given in Table 10.3 and compute $\Lambda_k(u_k)$, $1 \leq k \leq 6$.
21. Apply the modified BCJR algorithm to the encoder considered in Example 10.27 with the input data given in Table 10.4, and compute $\Lambda_k(u_k)$, $1 \leq k \leq 6$.

Chapter 11
Computer Networks



If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is.
– John von Neumann

Computer networks have become ubiquitous and permeate the society. They are the main means of social communication, which includes voice, data and video communication, and services, including electronic commerce, information search, distance learning, academic consultation, events, business, and malicious activities.

A typical computer network, with the servers of network and users' machines, is illustrated in Fig. 11.1.

Computer networks differ from telephone networks, which dominated the second half of the twentieth century, in the switching technique used and in the transmission rates, because they employ packet switching, variable topology, and high transmission rates. Information routing is performed at each node in the system.

The telephone networks, in turn, use exchanges with circuit switching; they have a transmission rate limited to 64 kbits/s, for fixed-line telephony, and use a star topology, with the switches located in the center of the star (Hammond and O'Reilly 1986).

The purpose of this chapter is to present the concepts of data flow, queue models, structure and topology, performance measures, traffic, channel capacity, latency, protocols, and more representative architectures of computer networks.

These concepts are quickly incorporated into the communication systems, in a way that its intrinsic data traffic characteristics need to be taken into account in the design of the systems. In particular, the information presented in this chapter is essential to understand how common activities take place, such as sending files, searching for information, and providing telephony via the computer network, known as IP telephony.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5_11

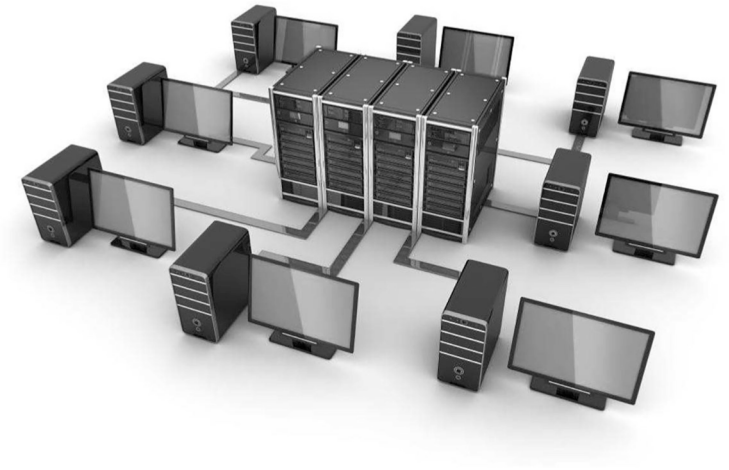


Fig. 11.1 Example of a computer network (Marsi 2023)

11.1 Data Flow in Computer Networks

The performance analysis of computer networks is related to the nature and characteristics of the data stream, which in computer networks is typically not uniform and random. The arrival times of messages, packets, or characters are random.

The time to process a message on the channel depends on the number of bits of the message, which can be random, so that the time also becomes a random variable. Efficiency, throughput, delay, and other parameters of interest are measures of how the network processes messages.

Queuing theory, which encompasses the theory of telephone traffic, provides an important model for the analysis of quantitative performance, correlating the most interesting computer networks (Hayes 1986).

A computer network can have thousands of servers, interconnected by thousands of communication channels, usually with some kind of cooperation between the servers and users. This makes capacity analysis a complex task. Therefore, designers use simpler parameters to be specified, such as flow (throughput) and latency.

11.1.1 Stationary Data Flow

Several properties related to stationary flow of packets or messages can be derived regardless of the distributions of the parameters that define the flow. In the analysis of the data flow, it is assumed that the network stores the messages, in the sense that they cannot be created, destroyed, or modified by the network.

Messages can only flow in or out of the network perimeter or can remain stored for certain of time on the network. The time spent in the network corresponds to the sum of the processing times of the packets by the servers. On each server, the message needs to be examined, eventually corrected, and then routed to the next destination.

If the average entry rate into the network exceeds the exit rate, the number of stored messages is constantly increasing. On the other hand, if the average rate is higher than the input rate, the number of stored messages may decrease to zero.

From these considerations, it was concluded that, for stable operation of network in steady state, the input and output rates must be equivalent.

Consider $\alpha(t)$ the number of incoming packets and $\delta(t)$ the number of network outgoing packets in a certain time interval $(0, t)$. The difference between these quantities, denoted by $N(t)$, represents the increase in the number of messages stored on the network in the given time interval:

$$N(t) = \alpha(t) - \delta(t). \tag{11.1}$$

The input rate, in a time interval t, is defined as

$$\lambda_t = \frac{\alpha(t)}{t}. \tag{11.2}$$

Another measure of interest is the total time that all messages spend on the network:

$$\gamma(t) = \int_0^t N(x)\mathrm{d}x = \int_0^t \alpha(x) - \delta(x)\mathrm{d}x. \tag{11.3}$$

Proceeding in this way, it is possible to find the average number of messages, N_t, on the network in the range $(0, t)$:

$$N_t = \frac{1}{t}\int_0^t N(x)\mathrm{d}x = \frac{\gamma(t)}{t}. \tag{11.4}$$

The average time spent by any message that has entered in the range $(0, t)$ is given by

$$T_t = \frac{\gamma(t)}{\alpha(t)}. \tag{11.5}$$

The previous formulas can be combined, taking the limit of each term, assuming that they remain finite, providing

$$N = \lambda T, \tag{11.6}$$

which is known as Little's law, in honor of Professor John Little, and establishes that the number of messages stored in a network is equal to the product of the rate of incoming messages to the network by the average time that these messages remain on the network, for a stationary system (Little 1961).

Considering the network illustrated in Fig. 11.2, equipped with a unit of storage or buffered and an output channel with rate C bits per second, it is assumed that there are λ messages per second and that the average length of messages $1/\mu$ bits per message and the traffic intensity, or time the channel is busy, is given by ρ (Alencar 2012).

The average transmission or processing time is $1/\mu C$ seconds. This implies μC messages processed per second, so that the output rate can be expressed by

$$\lambda = \rho \mu C \tag{11.7}$$

or

$$\rho = \frac{\lambda}{\mu C}, \tag{11.8}$$

indicating that the traffic intensity equals the arrival rate, divided by the channel transmission rate. So that there are no messages left in the network, which would lead to unstable behavior, the arrival rate should be less or equal the channel average processing rate, then $0 \leq \rho \leq 1$.

Little's law can be applied to a region which includes the buffer and the channel, providing

$$N = \lambda T. \tag{11.9}$$

If the region involves only the buffer, which stores N_q messages on average for a period of W seconds, one has

$$N_q = \lambda W. \tag{11.10}$$

As T is the sum of the average delays in the buffer and in the channel, one obtains

$$T = W + \frac{1}{\mu C}. \tag{11.11}$$

A relationship can be obtained between N and N_q from the previous equation, multiplying the equality by λ, which results in

$$N = N_q + \rho. \tag{11.12}$$

Equation 11.12 expresses the number of messages stored in the network as the sum of the average quantity stored in the buffer, N_q, plus ρ, which must represent the number of messages stored in the channel.

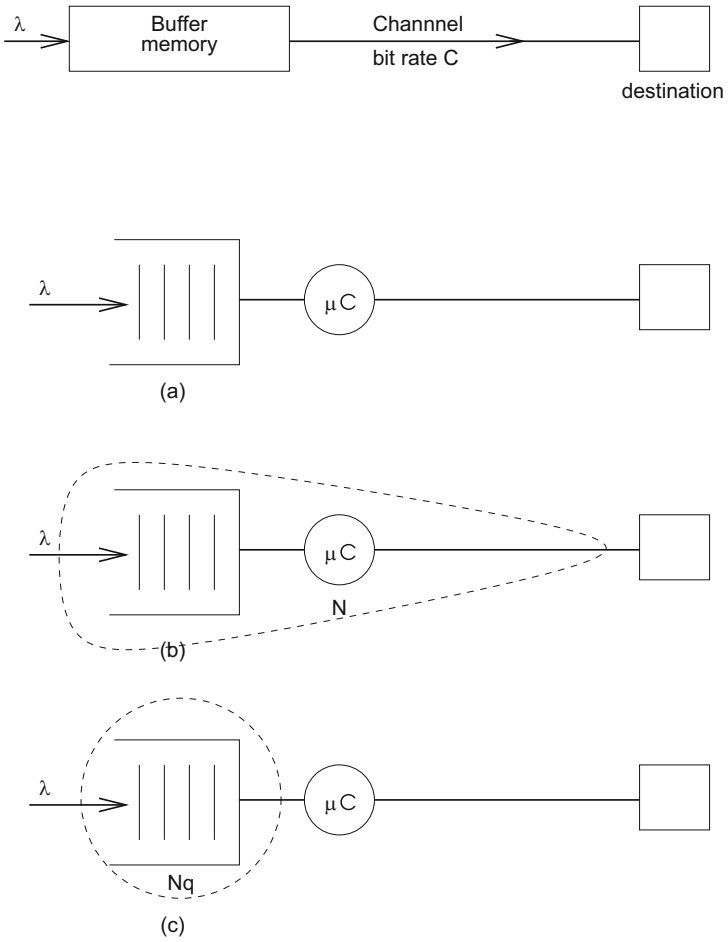


Fig. 11.2 The network's theoretical model. The top figure indicates the network as the composition of a buffer memory and a communication channel with capacity C. The lower figures represent, respectively, (**a**) the network model as described earlier, (**b**) the network model with a region including the memory buffer and the channel and (**c**) the network model with the region that includes only the memory. (Adapted from Alencar 2012)

11.2 Queueing Models

Computer networks generally start out structured but typically evolve in an amorphous way, as computational demands appear. The network manager, in addition to dealing with keeping the system running and its security, needs to take care of network optimization, which involves knowledge of queuing theory (Whittle 2007; Leon-Garcia and Widjaja 2000).

Queuing theory provides adequate models for the quantitative analysis of system and network performance, or even user behavior, and allows you to correlate important parameters. The usual notation indicates the premises for the input process, the server process, and the number of output channels (Kleinrock 1975).

There are four different types of queue that can be listed as follows:

Simple queue or linear queue For this kind of queue, an insertion takes place from one end, while the deletion occurs from another end.

Circular queue In this queue, all the nodes are represented in a circular format. It is similar to the linear queue except that the last element of the queue is connected to the first element. It is also known as ring buffer.

Double-ended queue A kind of queue in which the insertion and deletion can be done from both ends of the queue either from the front or rear. There are two types of priority queue that are presented in the following:

- **Input restricted double-ended queue** In input restricted queue, the insertion operation can be performed at only one end, while deletion can be performed from both ends.
- **Output restricted double-ended queue** In the output restricted queue, the deletion operation can be performed at only one end, while insertion can be performed from both ends.

Priority queue A type of queue in which the elements are arranged based on a predefined set of priorities. There are two types of priority queue that are discussed as follows:

- **Ascending priority queue** In this case, the elements can be inserted in arbitrary order, but only the one with smallest priority assgined can be deleted first.
- **Descending priority queue** In this case, the elements can be inserted in arbitrary order, but only element with the largest priority can be deleted first.

A queue is a non-primitive linear data structure that allows insertion of an element at one end and deletion of an element at the other end. The characteristics that define a queue, for simplicity, are described using the notation of the English mathematician and statistician David George Kendall (1918–2007), proposed in 1953 (Kendall 1953), and composed of a series of symbols of the form A/B/C/D/E/Q, in which:

A Distribution of times between successive arrivals
B Distribution of service times or attendance
C Number of servers or service stations
D Physical capacity of the system
E Size of the population
F Service discipline

The most common probability distributions that are used in queuing theory are:

- Exponential (M)
- Uniform (U)
- Arbitrary or general (G)
- Erlang type k (E_k)
- Hyperexponential (H_k)

Some of the most common service disciplines in queuing theory are:

FIFO First in, first out, that is, the first to enter is the first to leave.
FCFS First come, first served is identical do the previous one, that is, the first to arrive is the first to be served.
LCFS Last come, first served, that is, the last to arrive is the first to be served.
LIFO Last in, first out, that is, the last to arrive is the first to leave.
SIRO Serve in random order, that is, the service is done according to a statistical procedure.

The following examples illustrate the use of Kendall's notation, known for his work on probability, statistical shape analysis, and queueing theory.

$M/G/1$	In the case of an entry process being Poisson, without restriction on the server process and only one output channel.
$M/M/20$	If the entry process follows a Poisson sistribution, as well as the server process, with 20 output channels.
$M/D/1$	If the entry process is Poisson, but the server is deterministic, with only one output channel.
$M/E_k/30/200$	If the entry process is Poisson, the server is Erlang type k, with 30 channels of output and capacity for 200 clients.
$M/M/2/\infty/\infty$/FIFO	If the entry process is Poisson, as well as the server process, with two output channels, unlimited capacity, infinite population, and service discipline in which the first to enter is the first to leave.

11.2.1 The Markov Model

The Markov model shown in Fig. 11.3 serves as a basis for the traffic analysis on computer networks. Andrey Andreyevich Markov (1856–1922) was a Russian mathematician well known for his work on stochastic processes. He was elected to the Saint Petersburg Academy of Sciences, in 1886, the same year he was appointed extraordinary professor of the Saint Petersburg University. In 1896, Markov was elected an ordinary member of the academy as the successor of Pafnuty Lvovich Chebyshev (1821–1894), a Russian mathematician who made fundamental contributions to the fields of probability, statistics, mechanics, and number theory, and is considered the founding father of Russian mathematics.

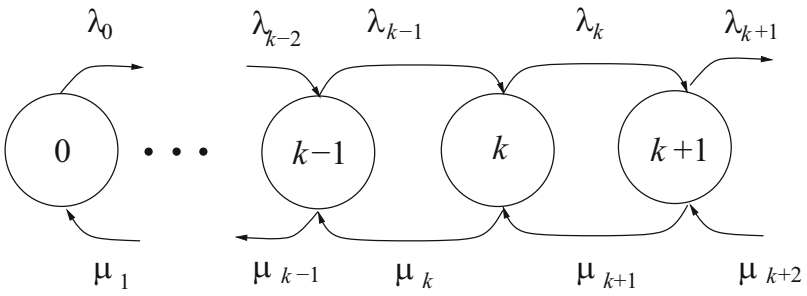


Fig. 11.3 Markov model of birth and death

A discrete-time Markov chain is used to model a stochastic process in honor of mathematician Andrei Andreyevich Markov. For this chain, the states prior to the current one are not relevant to the prediction of future states, as long as the current state is known.

The stochastic process is known as the birth and death model, because the transitions occur only between adjacent states. For example, from the state k, it is possible to go only to state $k+1$ or state $k-1$, with some assigned probability. This reflects the fact that it is negligible the likelihood of more than one user entering the system at the same time. Using this model, it is possible to calculate the steady-state probabilities of the Markov chain (Kleinrock 1975).

The transition matrix probabilities $\mathbf{P} = \{p_{ij}\} = \{p(y_j|x_i)\}$ define the dynamics of the model. The transition probabilities are obtained from the Markov model, in which λ_k and μ_k are the birth and death parameters.

$$\mathbf{P} = \begin{bmatrix} p(y_1|x_1) & p(y_2|x_1) & \dots & p(y_N|x_1) \\ p(y_1|x_2) & p(y_2|x_2) & \dots & p(y_N|x_2) \\ & \dots\dots\dots\dots & & \\ p(y_1|x_M) & p(y_2|x_M) & \dots & p(y_N|x_M) \end{bmatrix}. \tag{11.13}$$

In a stable Markov chain, it is possible to reach a steady state after a certain number of iterations. The probabilities of the steady state, $\Pi = \{ pi_k|k = 1, 2, 3 \dots\}$, can be calculated using one of the known techniques (Kleinrock 1975; Adke and Manjunath 1984). Each state k defines the number of users, packets, or other objects in the system.

Two cases of application of the Markov chain are presented in the following. Initially, one can define a problem in which the birth and death parameters are constant for any state. It can be determined that $\lambda_k = \lambda$ and $\mu_k = \mu$.

The first case, which illustrates the operation of a traditional computer network, connected by wires or cables, which operates with defined flow parameters, produces a geometric distribution for the probabilities:

$$p_k = (1-\rho)\rho^k \quad k = 0, 1, 2, \dots, \text{ for } \rho < 1, \tag{11.14}$$

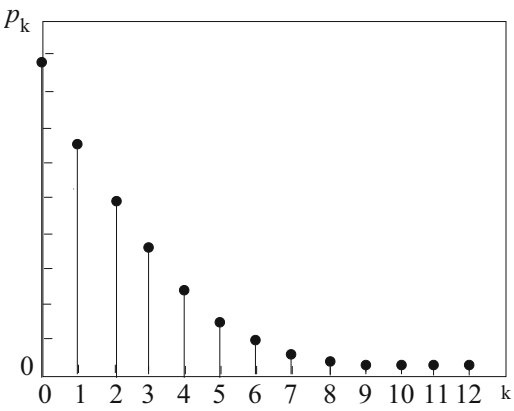


Fig. 11.4 Geometric probability distribution, as a function of the state k of the system

in which $\rho = \lambda/\mu$ is usually called the use of the system. Figure 11.4 illustrates the geometric probability distribution as a function of the state k of the system.

For the geometric distribution, the statistical average is given by $\rho/(1-\rho)$ and the variance is $\rho/(1-\rho)^2$. The probability of finding more than L users at a given time in the system is ρ^{L+1}.

Then, the problem related to the efficiency of the system server is calculated, and it reacts to the increased data flow in the system. This is equivalent to the case when users drop out as a function of, for example, the delay in completing a connection.

This Markov model exemplifies the operation of a wireless computer network, for example, in the case of the interconnection of notebooks, tablets, or cell phones, in which there is some competition to access the channel. For this case, one obtains $\lambda_k = \lambda/(k+1)$ and $\mu_k = \mu$, in which λ and μ are fixed probabilities.

The solution for the second case generates a Poisson distribution:

$$p_k = \frac{\rho^k}{k!} e^{-\rho} \quad k = 0, 1, 2, \ldots \tag{11.15}$$

with mean and variance given by ρ. Figure 11.5 illustrates the geometric probability distribution as a function of the state k of the system.

11.2.2 Markov Solution for a Stable Network

The formula that relates the probability of k existence packets at a given time on a network is given by the solution of the following equations, which depends on the configuration assumed for the system (Kleinrock 1975):

$$p_k = p_o \prod_{i=0}^{k-1} \frac{\lambda_i}{\mu_{i+1}},$$

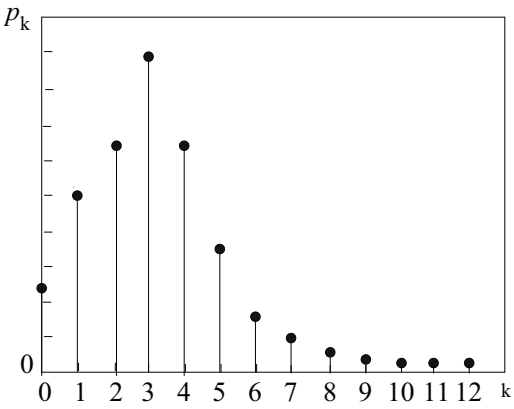


Fig. 11.5 Poisson probability distribution, depending on the state k of the system

$$p_o = \frac{1}{1 + \sum_{k=1}^{\infty} \prod_{i=0}^{k-1} \frac{\lambda_i}{\mu_{i+1}}},$$

$$N = \sum_{k=0}^{\infty} k p_k,$$

$$\sigma_N^2 = \sum_{k=0}^{\infty} (k - N)^2 p_k,$$

$$N = \lambda T,$$

in which:

- p_o—probability that there are no packets in the system
- p_k—probability of having k packets in the system
- λ_i—packet arrival rate to state i
- μ_{i+1}—departure rate for state packet $i + 1$
- N—average number of packets in the system
- σ_N^2—variance of the number of packets in the system
- T—average waiting time in line

These equations serve as a starting point for solving several queuing theory problems, as well as, for other solutions shown below in this chapter.

11.2.3 A System with Constant Arrival Rate

In this type of system, the arrival and departure rates of packets can be described, in terms of the coefficients, as follows:

$$\lambda_k = \lambda, k = 1, 2, 3, 4, \ldots$$
$$\mu_k = \mu, k = 1, 2, 3, 4, \ldots$$

The probabilities associated to k packets and none packet are given by

$$p_k = p_o \prod_{i=0}^{k-1} \frac{\lambda}{\mu},$$

$$p_k = p_o \left(\frac{\lambda}{\mu}\right)^k,$$

$$p_o = \frac{1}{1 + \sum_{k=1}^{\infty} \prod_{i=0}^{k-1} \frac{\lambda_i}{\mu_{i+1}}},$$

$$p_o = 1 - \frac{\lambda}{\mu},$$

With the restriction that $\frac{\lambda}{\mu}$ must be less than 1. Making $\rho = \frac{\lambda}{\mu}$, one obtains

$$p_k = (1 - \rho)\rho^k$$

and

$$p_o = (1 - \rho).$$

The number of packets in the system is given by a summation of terms consisting of the product of the number k of packets by the corresponding probability of occurrence p_k. Mathematically, this can be represented by

$$N = \sum_{k=0}^{\infty} k p_k,$$

after some mathematical manipulations, by recalling that

$$\frac{d}{d\rho} \sum_{k=1}^{\infty} \rho^k = \sum_{k=1}^{\infty} k\rho^{k-1}$$

and that

$$\sum_{k=1}^{\infty} \rho^k = \frac{1}{1 - \rho},$$

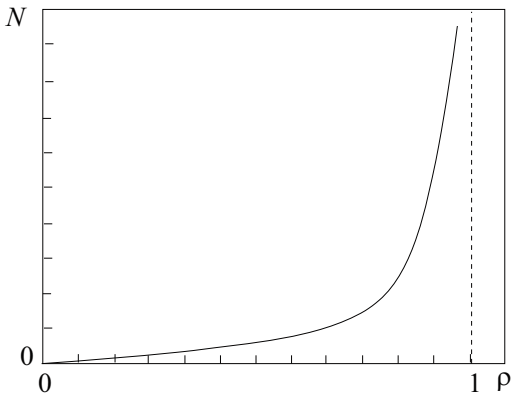


Fig. 11.6 Growth in the number of N packets on the network with ρ

one has

$$N = \frac{\rho}{(1-\rho)}.$$

Figure 11.6 illustrates the growth in the number of packets on a network, as a function of the coefficient ρ.

The variance in the number of packets in the system is given by

$$\sigma_N^2 = \sum_{k=0}^{\infty} (k-N)^2 p_k,$$

$$\sigma_N^2 = \frac{\rho}{(1-\rho)^2}.$$

The described results do not depend on the the individual parameters λ and μ but only of the ratio between them. This ratio relates the arrival and processing parameters.

The average waiting time in the queue is given by the quotient between the average number of packets and the average rate of arrival of packets to the system. Therefore

$$T = \frac{N}{\lambda} = \frac{1/\mu}{(1-\rho)}.$$

Figure 11.7 illustrates the increase in latency in a network, as the coefficient ρ increases.

When $\rho = 0$, the latency T corresponds to the expected service time for the first packet, being equal to $1/\mu$. This average time depends on the parameter μ individually, contrary to the previous ones.

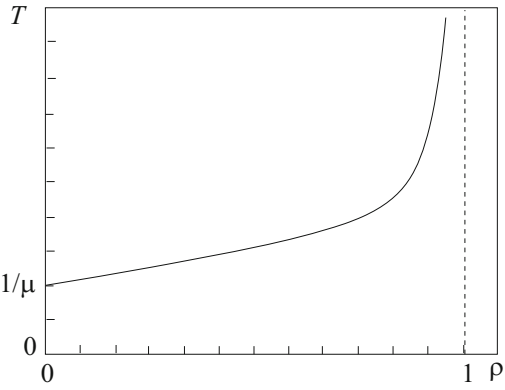


Fig. 11.7 Increased network latency due to ρ

Note that, as ρ tends to unity, that is, as the rate of packets that arrives on the network approaches the processing rate of the network's servers, the average number of packets in the system and the average waiting time in the queue tend to grow in an unlimited manner. This type of behavior when ρ tends to one is characteristic of almost all queuing systems encountered.

In general, this feature is exploited by hackers for an attack known as denial of service (denial of service—DoS), in which the network receives a flood of packets, so that the arrival rate approaches the processing rate, and the number of packets on the network tends to infinity.

Another interesting quantity to be analyzed is the probability that there must be at least k packets in the system. This probability is given by

$$P[x \geq k \text{in the system}] = \sum_{i=k}^{\infty} p_i$$

$$= \sum_{i=k}^{\infty} (1-\rho)\rho^i = \rho^k,$$

for ρ less than 1, since $\sum_{i=k}^{\infty} \rho^i$ only converges if $\rho < 1$.

11.2.4 Model for the Efficient Server

This type of system can be interpreted in two ways: as a server responsible for increasing the service rate, as the the number of packets in the queue increases, or as a new server available for each incoming packets. The model can be described by the following parameters:

$$\lambda_k = \lambda \quad k = 1, 2, 3, 4, \ldots$$
$$\mu_k = k\mu \; k = 1, 2, 3, 4, \ldots$$

The results are given as

$$p_k = \frac{(\lambda/\mu)^k}{k!} e^{-\frac{\lambda}{\mu}},$$

$$p_o = e^{-\frac{\lambda}{\mu}},$$

$$N = \frac{\lambda}{\mu},$$

$$T = \frac{1}{\mu},$$

from Little's equation.

It can be noted that the system with an efficient server is equivalent to a system with discouraged arrivals, as described next, when $\alpha = \lambda$.

11.2.5 A System with Discouraged Arrivals

In general, it is common to consider only a fixed network of computers interconnected by wire cables, or optical cables, to solve problems related to latency and the number of stored packets. However, the access network, typically wireless, is also important and includes the equipment used for the connection, such as cell phones, notebooks, and tablets. In this type of network, competition is established between users to gain access to available channels. This competition changes the scenario used for mathematical modeling. The fundamental change is to make the arrival rate dependent on the number of users (or packets) on the network.

Consider now an access system in which the entry of users tends to be discouraged, as the number of packets increases in system. To model this system, the parameter values must be assigned as follows:

$$\lambda_k = \tfrac{\alpha}{k+1} \; k = 1, 2, 3, 4, \ldots$$
$$\mu_k = \mu \quad k = 1, 2, 3, 4, \ldots$$

It is assumed that there is a lack of incentive for new arrivals, as the number of packets in the system increases. For this system, one obtains

$$p_k = p_o \prod_{i-0}^{k-1} \frac{\frac{\alpha}{i+1}}{\mu},$$

$$p_k = p_o \left(\frac{\alpha}{\mu}\right)^k \frac{1}{k!},$$

$$p_o = \frac{1}{1 + \sum_{k=1}^{\infty} \prod_{i-0}^{k-1} \frac{\lambda_i}{\mu_{i+1}}},$$

$$p_o = e^{-\frac{\alpha}{\mu}},$$

$$\rho = 1 - e^{-\frac{\alpha}{\mu}},$$

$$N = \sum_{k=0}^{\infty} k p_k,$$

$$N = \frac{\alpha}{\mu},$$

$$T = \frac{N}{\lambda},$$

$$T = \frac{\alpha}{\mu^2 \left(1 - e^{-\frac{\alpha}{\mu}}\right)},$$

$$P[x \geq k \text{in the system}] = \sum_{i=k}^{\infty} p_i.$$

This solution is equivalent to the one obtained for the efficient server, that is, different assumptions can lead to similar mathematical models.

11.2.6 Models of Queues M/G/1, M/M/1 and M/D/1

The $M/G/1$ model is used as a starting point for analysis. With respect to the service process, it is convenient to consider the message processing as a sequence of variables that are independent and identically distributed.

For the Poisson entry process, the number of arrivals at any time is statistically independent of the number of messages arriving at any other time interval that is not superimposed.

The probability of k messages arriving in a time interval T is given by

$$P\{k\} = \frac{(\lambda T)^k e^{-\lambda T}}{k!},\ k = 0,\ 1,\ 2, \ldots \tag{11.16}$$

The message arrival rate, that is, the average number of message arrivals at $T = 1$ second, is λ messages per second. Analyzing the $M/G/1$ model, it is concluded that the average number of messages on the network is given by

$$N = \frac{2\rho - \rho^2 + \lambda^2\sigma_Y^2}{2(1-\rho)} = \rho + \frac{\rho^2 + \lambda^2\sigma_Y^2}{2(1-\rho)}. \tag{11.17}$$

in which σ_Y^2 is the variance of message Y processing time.

The application of Little's law permits to obtain an expression for the message average time in the system (Alencar 2012):

$$T = \frac{1}{\mu C} + \frac{\rho + \lambda\mu C\sigma_Y^2}{2\mu C(1-\rho)}. \tag{11.18}$$

The Pollaczek-Khinchine formulas 11.16, 11.17, and 11.18 show the growth in the number N of packets in the network, for typical values of network rates and capacity. The model $M/M/1$ is a special case of the previous model. In this case, the message processing time is described by a Poisson process.

Therefore, it can be demonstrated that the variance of the processing time is given by $\sigma_Y^2 = (1/\mu C)^2$ and, thus, the number of messages in the system becomes

$$N = \frac{\rho}{1-\rho}. \tag{11.19}$$

As a result, the average delay time on the route between the buffer and the channel is given by

$$T = \frac{\rho}{\lambda(1-\rho)} = \frac{1}{\mu C(1-\rho)}. \tag{11.20}$$

The formulas for N_q, the average number of messages stored in the buffer, and W, the average time in the buffer, are, respectively

$$N_q = \frac{\rho^2}{1-\rho} \tag{11.21}$$

and

$$W = \frac{\rho}{\mu C(1-\rho)}. \tag{11.22}$$

It should be noted that the model $M/D/1$ is also a special case of the model $M/G/1$, in which the message processing time is set as invariant and denoted by $1/\mu C$.

As the variance in this case is zero, one can write

$$N = \rho + \frac{\rho^2}{2(1-\rho)} = \frac{\rho(2-\rho)}{2(1-\rho)} \tag{11.23}$$

and

$$T = \frac{1}{\mu C} + \frac{\rho}{2\mu C(1-\rho)} = \frac{2-\rho}{2\mu C(1-\rho)}. \tag{11.24}$$

As a consequence

$$N_q = \frac{\rho^2}{2(1-\rho)} \tag{11.25}$$

and

$$W = \frac{\rho}{2\mu C(1-\rho)}. \tag{11.26}$$

The curves in Fig. 11.8 show the number of packets, N, in the network versus ρ, for a $M/D/1$ queue, compared with similar results for a $M/M/1$ queue (Alencar 2012).

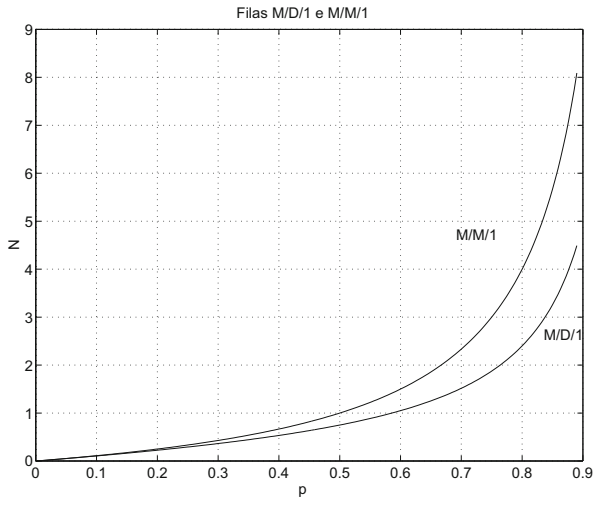


Fig. 11.8 Curves for N versus ρ, for an $M/D/1$ queue, compared to the $M/M/1$ queue

11.3 Local Area Networks

Computer networks often make use of efficient communication channels and take advantage of their potential for resource sharing among users (Moura et al. 1986; Giozza et al. 1986). The communication of processes takes place through the network, between its edge nodes.

A generic computer network can be defined as a set of autonomous computers interconnected, so that no machine completely controls the others. Computer networks have been developed to enable the sharing of facilities and the distribution of resources among users. The function of the network is to provide a means by which the computers connected to the network can exchange data.

11.3.1 Network Definitions

From the user's point of view, computer networks have become the basis for the exchange of information, for the purchase of products and services, and for carrying out works of various kinds. There are several types of computer networks. A geographic classification is generally used (Tanenbaum 1989):

- Local area networks
- Wide area networks

Wide area networks, such as the Internet, cover large geographic areas, often extending nationally or globally. Local area networks, such as Fibernet, Ethernet, or Novell, are characterized by restricted physical dimensions, usually covering a few hundred meters.

The difference in physical dimensions has a pronounced effect on the structure of networks. The local network, due to its limited physical extension, may be owned by network users. In this way, it is possible the use of dedicated high-speed and reliable channels.

The typical means of transmission on local computer networks are twisted pair, optical fiber link, 50 Ω coaxial cable (for baseband transmission), and 75 Ω coaxial cable (for broadband transmission). Wireless networks are also used, such as Wi-Fi (IEEE 802.11), whose IEEE 802.11n standard employs transfer rates from 65 Mbit/s to 600 Mbit/s, with transmission based on MIMO-OFDM, operating in the frequency range of 2.4 GHz or 5 GHz (Gast 2005).

Long-distance networks, for regulatory and economic reasons, used telephone links, which resulted in lower transmission rates. With the deployment of long-haul optical networks, including submarine cables, underground cables, and transmission over high voltage lines (OPGW), transmission rates have increased considerably.

Wireless networks are also formed with WiMAX technology (IEEE 802.16), stand for Worldwide Interoperability for Microwave Access, and offer fixed, nomadic, portable, or mobile connectivity, without the need for line of sight with

a base station. For a typical scenario, with distances from 3 to 10 km, WiMAX presents a transmission rate of 40 Mbit/s per channel (IEEE 2004).

Four physical layer standards have been specified for the metropolitan WiMAX network: SC (Single Carrier), SCa (Single Carrier a), OFDM (Orthogonal Frequency-Division Multiplexing), and OFDMA (Orthogonal Frequency-Division Multiple Access). However, cellular mobile telephony with long-term evolution technology (LTE) eclipsed WiMAX.

11.3.2 Applications for Local Area Networks

One of the most versatile applications for local networks is the office automation, in which the local network interconnects workstations, computing equipment and printers (Soares et al. 1995).

In an academic or business environment, the local network can provide the interconnection of workstations for development of computer-aided programs or projects.

Point-of-sale systems can have their equipment, such as cash registers and optical readers, in addition to consultation of files interconnected in network.

Local area networks can also be used in banking systems, libraries, laboratories, computational facilities in universities, and several other applications that require communication between computers. Figure 11.9 illustrates a generic point-to-point (P2P) local area network.

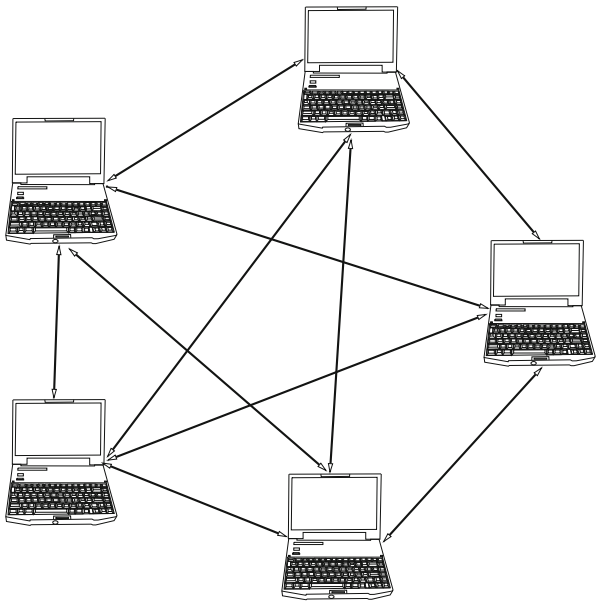


Fig. 11.9 Generic local peer-to-peer network

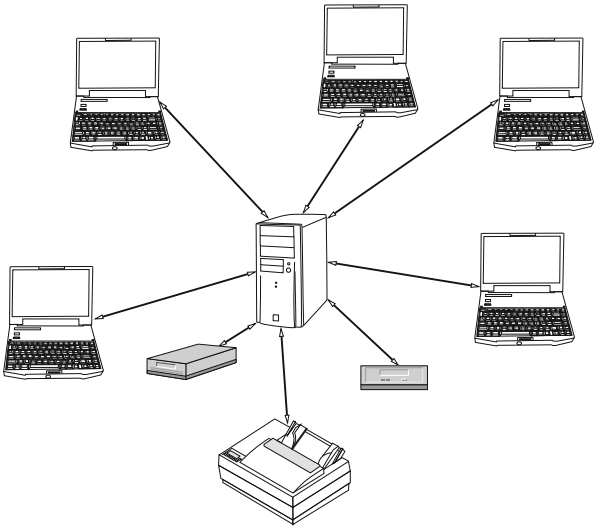


Fig. 11.10 Generic local client-server network

Local networks implement data transfer in discrete units called packets. A packet can vary from a few bits to thousands of bits according to each particular protocol. Eventually, there is a general server on the network, as in the case of the client-server model of Fig. 11.10.

11.3.3 Network Structure and Topology

The nodes of a network can be connected in several ways. The unstructured or amorphous topology is common for wide area networks, which grow due to the disorderly demand for services, but uncommon for local networks.

The storage and retransmission operation is typical of unstructured networks, which allows flexibilization in the positioning of the nodes and in the communication between them. Local area networks have a combination of requirements that make the storage and retransmission operation unnecessary.

In local area networks, efficient communication between devices is the main goal. Therefore, the delay and cost associated with routing messages between nodes are considered excessive.

Unstructured networks generally need to perform the error control at each link, due to the insertion of noise in the path. Error control between source and recipient is enough for local networks. Local area networks frequently use the broadcasting of messages, which is unusual in long-distance networks.

Some common network topologies have characteristics that are appropriate for the provided services, such as the star, bus, and ring. The tree structure is

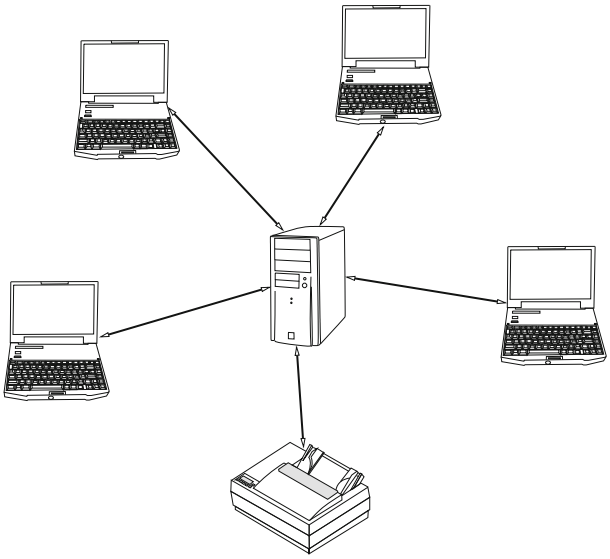


Fig. 11.11 Structures for typical networks: star network

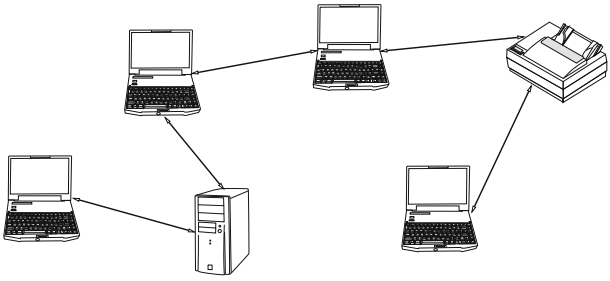


Fig. 11.12 Structures for typical networks: bus network

a generalization of the bus network. Typical network structures are shown in Figs. 11.11, 11.12, 11.13, and 11.14 (Alencar 2012).

For the bus topology, the station sends the message in both directions. Each station, or node, has the ability to read the packet heading, and there are buffers to store messages for which the node is the destiny.

Communication is typically in one direction in the ring topology. Therefore, routing is usually not necessary. The ring topology uses the central node as a traffic redirector, as in a telephone exchange. The links must be of the full-duplex type.

The tree network structure is common for pay-TV (CATV), which uses coaxial cable, optical cable, or a hybrid optical-coaxial system, and access by frequency division. Local networks can use the CATV cabling or use the power distribution network, with the Power Line Communications (PLC) technology (Marcelo et al. 2006).

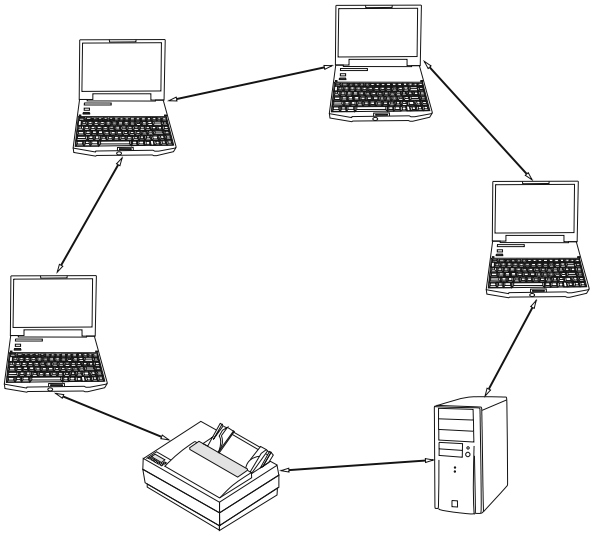


Fig. 11.13 Structures for typical networks: ring network

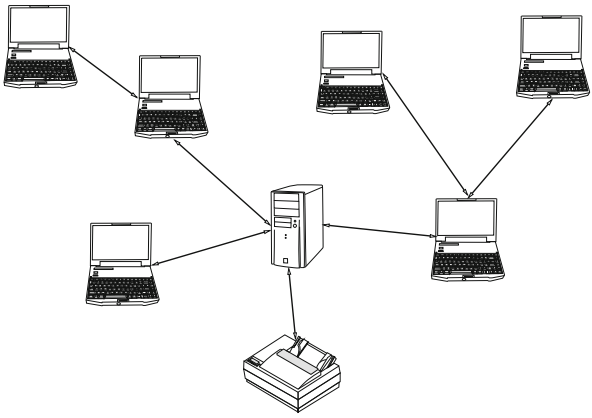


Fig. 11.14 Structures for typical networks: tree network

11.3.4 Access Techniques

There is a number of different access techniques that are used for different topologies. The choice of the technique of access is a determining factor in network performance. Techniques can be classified into four major categories: fixed allocation, random allocation, allocation on demand, and adaptive allocation.

In the fixed allocation, resources are placed at disposition of each station in a predetermined way. Two common schemes in this category are multiple access by multiple division into frequency (FDMA) and division into time (TDMA). At the opposite end to the fixed allocation strategy, there are the random allocation methods. The simplest method is known as pure ALOHA, in which each station transmits as soon as it is ready.

The adaptive or on-demand allocation methods require a control mechanism, operating in real time, to allocate the channel capacity to all stations in an optimum or quasi-optimum manner.

The on-demand allocation methods are still classified as central or distributed control. Algorithms to monitor systems illustrate central control, while ring networks use distributed control.

The adaptive allocation methods try to refine the random on-demand allocation protocols. Adaptive protocols usually perform contention resolution, provided that collisions are allowed, under low loading, to minimize access times. For high loading, however, the procedures adjust for some form of polling or TDMA. A characteristic of these methods is that some estimate of the load on the network is required by all stations.

11.3.5 Type of Network Traffic

In a computer network, the nature of the traffic offered by the user devices is an important factor in evaluating performance (Alencar 2012).

However, the traffic offered to the network is variable and random, because it depends on many factors. Peak data rates range from 100 bits/s for security systems, telemetry, and networks wireless sensors, at 100 Mbits/s for video transmission.

The establishment of a probability distribution for the stationary state is a difficult problem, because of the significant differences between averages taken at distinct time intervals.

The data in Fig. 11.15, for example, suggest an exponential distribution for the arrival times, and this distribution is often used as a basic model for research and development of networks:

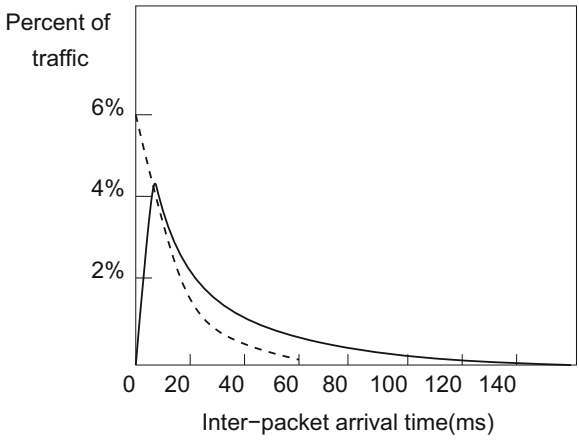


Fig. 11.15 Distribution of arrival time between packets on an Ethernet network. (Adapted from Hammond and O'Reilly 1986)

$$p_T(t) = \beta e^{-\beta t} u(t), \tag{11.27}$$

in which $u(\cdot)$ represents the unit step function and α is a parameter to be estimated.

The distribution of packet lengths over an experimental Ethernet network has a bimodal character. Smaller packets occur depending on the interactive traffic (bursty), while larger packets represent the result of the transfer of large volumes of data.

11.3.6 Measuring the Performance of the Network

An important performance measure, from the user point of view, is the network response time, that is, the time to correctly transmit a packet from one user to another, with the confirmation. Note that both network delays and user/station link delays can dominate the response time between users. If the network delays are significant, the choice of the best access technique is very important.

At the network level, a specific measure is the throughput, which measures the number of bits/s or packets/s, which can, on average, be processed by the network. If the average input rate for one part of the network is λ packets/s, the channel transmission rate is R bits/s, and there is $\bar{X}$ bits/packet on average; therefore, the normalized throughput is given by

$$S = \frac{\lambda \bar{X}}{R}. \tag{11.28}$$

Throughput is used in the design and performance evaluation of computer networks because the capacity, based on information theory, is not amenable to calculation for a complex network. But it must be kept in mind that the upper limit for the transmission rate in a communication system is its capacity, calculated as the maximum of the mutual information for all probability distributions of the input alphabet.

The average transfer delay, T, is defined as the time between the arrival of the last bit of a packet to a network station and the delivery of the last bit of that packet to the destination. The normalized average delay, in relation to the average time of transmission in the channel, is given by

$$\hat{T} = \frac{T}{\bar{X}/R} = \frac{RT}{\bar{X}}. \tag{11.29}$$

The delay, or latency, of the network is an important parameter when dealing with voice transmission in real time, as in the case of Voice over IP (VoIP), which is the technology used in applications like WhatsApp and Skype.

The computation of the total delay T, or network latency, which is a random variable, is done using properties of probability theory. Consider that each partial

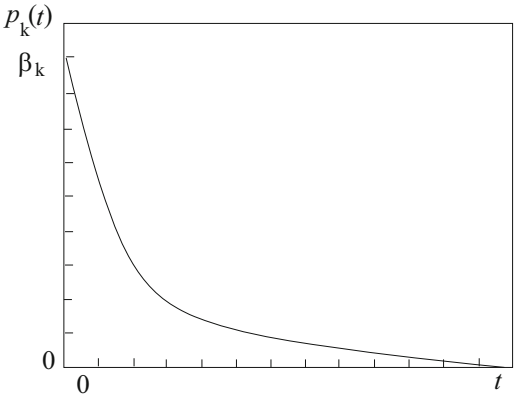


Fig. 11.16 Probabilistic model for the time delay in a computer network

delay time, typically represented by the processing time of the message by each individual server, plus the traffic time on the corresponding link, has exponential probability distribution:

$$T = \sum_{k=1}^{M} t_k, \tag{11.30}$$

in which M is the number of servers in the message path, and each particular time delay has distribution:

$$p_k(t) = \beta_k e^{-\beta_k t} u(t), \tag{11.31}$$

in which $u(\cdot)$ is the unit step function and β_k represents the parameter it regulates the probability distribution for the time t_k. The distribution is illustrated in Fig. 11.16.

The sum of independent random variables leads to the convolution of the two respective probability distributions. Therefore, the probability distribution of the total time is given by

$$p_T(t) = p_1(t) * p_2(t) * \cdots * p_M(t), \tag{11.32}$$

in which the convolution operation between two functions $f(t)$ and $g(t)$ is defined as

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) \mathrm{d}\tau. \tag{11.33}$$

Example For two servers, or two subnets, $M = 2$, the calculation can be done using Formula 11.33:

$$
\begin{aligned}
p_T(t) &= \int_{-\infty}^{\infty} p_1(\tau) p_2(t-\tau) \mathrm{d}\tau \\
&= \int_{-\infty}^{\infty} \beta_1 e^{-\beta_1 \tau} \mathrm{u}(\tau) \beta_2 e^{-\beta_2 (t-\tau)} \mathrm{u}(t-\tau) \mathrm{d}\tau \\
&= \beta_1 \beta_2 e^{-\beta_1 t} \int_0^t e^{-(\beta_2 - \beta_1)\tau} \mathrm{d}\tau \\
&= \frac{\beta_1 \beta_2}{\beta_1 - \beta_2} \left(e^{-\beta_2 t} - e^{-\beta_1 t} \right) \mathrm{u}(t).
\end{aligned}
$$

If the processing times are the same for both servers, that is, the parameters $\beta_1 = \beta_2 = \beta$, the solution converges to

$$p_T(t) = \beta^2 t e^{-\beta t} \mathrm{u}(t).$$

This is a special case of the Erlang distribution, a family of continuous probability distributions with support $t \in [0, \infty)$. The Erlang probability density function represents the distribution of a sum of k independent exponential variables with mean $1/\beta$ each, in which k is the shape and β is the rate parameter.

The Erlang probability density function is given by

$$p_T(t) = \frac{\beta^k}{(k-1)!} t^{k-1} e^{-\beta t} \mathrm{u}(t), \tag{11.34}$$

and the Erlang cumulative density function is

$$P_T(t) = \left[1 - \sum_{n=0}^{k-1} \frac{1}{n!} (\beta t)^n e^{-\beta t} \right] \mathrm{u}(t). \tag{11.35}$$

The random variable expected value is k/β, and the variance is k/β^2.

The Erlang distribution was proposed by Agner Krarup Erlang (1878–1929), a Danish mathematician, statistician, and engineer, who developed the fields of traffic engineering and queueing theory, to examine the number of telephone calls which might be placed to the telephone exchange. Erlang worked for the Copenhagen Telephone Company, from 1908 until his death in Copenhagen after an abdominal surgery.

The Erlang distribution can be used to model the time between incoming calls, in a telephone network, or packets, in a computer network. It is used in conjunction with the expected duration of incoming calls, or packets, to produce information about the traffic load measured in erlangs (Erl).

This information is used to determine the probability of packet loss or delay, according to certain assumptions made about whether blocked calls are aborted, which results in the Erlang B formula, or queued until served, giving the solution in terms of the Erlang C formula.

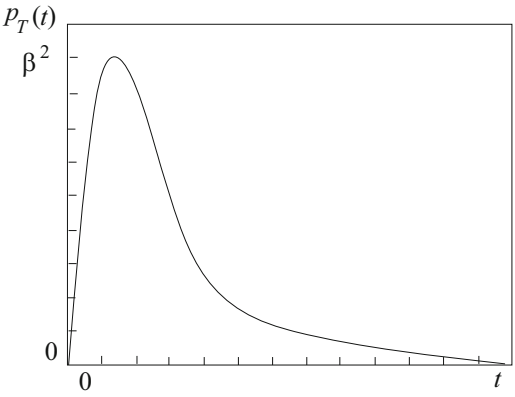


Fig. 11.17 Probabilistic model for combined latency in two subnets with same delay time

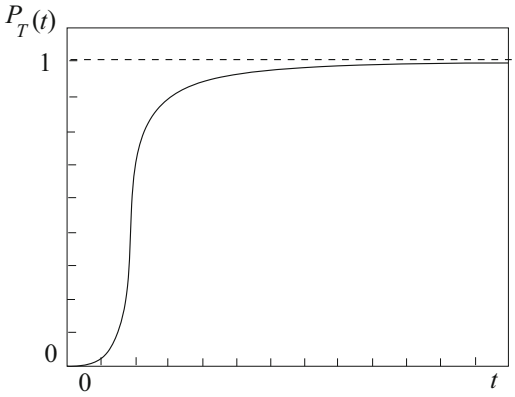


Fig. 11.18 Cumulative distribution function for the combined latency in two subnets with same delay

The solution for the combined latency of two servers, or two subnets, is represented in Fig. 11.17. As can be noticed, it resembles the distribution of the arrival time between packets on an Ethernet network, illustrated in Fig. 11.15.

The cumulative distribution function for the combined latency of two servers is represented in Fig. 11.18. The general case, for M servers with different processing times, produces the following probability distribution for the total time (Akkouchi 2008):

$$p_T(t) = \sum_{k=1}^{M} \frac{\beta_1 \cdots \beta_M}{\prod_{i=1, i\neq k}^{M} (\beta_i - \beta_k)} e^{-\beta_k t} u(t). \tag{11.36}$$

The probability distribution allows to calculate the average total network delay, latency, and also the deviation around this delay, which indicates the jitter. Latency is a measure of time that a packet takes to get from the source to the destination.

The jitter is the stochastic variation of delay in data delivery, that is, the temporal variation between successive data packets. The jitter limits the minimum separation

between bits, which affects the baud rate on the network. For the case of the Erlang distribution, the jitter is just the standard deviation, $\sigma_T = \sqrt{k}/\beta$.

11.4 Network Protocols and Architecture

In informatics, computer science, and engineering, the term protocol means the set of rules that makes possible the execution of a program efficiently and without errors or the exchange of information between servers of a computer network, for sending information. Nowadays, the protocols in evidence are those associated with the Internet, the worldwide network that connects most of Earth's population. But it was not always so Alencar (2010).

The word protocol comes from the Greek *protókollon*, meaning leaf placed in front, referring to the content description, usually placed on first page of a book. It came to English in medieval Latin *protocollu*, with the usual meaning of the word protocol, that is, a set of rules or criteria fulfilled in a given activity, whether in the execution, evaluation, or acceptance of services and materials.

11.5 Network Layer Architecture

Computer networks use layered architectures, to organize the project and minimize its complexity. Each layer represents a logical entity that performs certain functions. The layers switch information through primitives, simple commands that define simple operations.

The services provided by the highest level of the network, known as application layer, are sent directly to users. Each layer provides services for the next layer above it Alencar (2012).

When users in different nodes communicate, corresponding layers also exchange information using appropriate rules. A set of rules, called a protocol, is necessary so that each level can communicate in a structured way. A layered architecture, with the indicated protocols, is shown in Fig. 11.19.

Protocols are hierarchically organized, corresponding to the layers of the network. Protocols control exchange of information through a single layer, and all the information layers communicate by logical or virtual paths. The lowest layer protocol is the only one to control the data flow in a physical connection channel.

The flow of information from one level to another is done through an interface that is present between each pair of adjacent layers. The purpose of the interface is to interpret formats and other characteristics of the protocols.

The number of layers, the name of each layer, and its function differ from one network to another. The most extreme layers are usually the physical and application layers. The application layer, as the name implies, handles user-specific applications such as access to files or graphic output from a computer assisted project. The

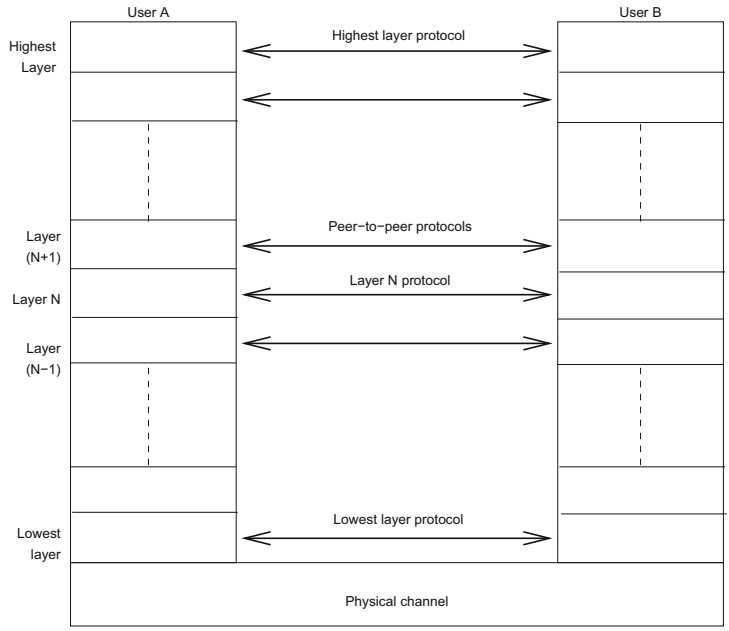


Fig. 11.19 A layered architecture with the indicated protocols

physical layer is always the lowest level and is responsible for the transmission of signals between two nodes, providing an electrical or logical connection between processes.

When designing a network architecture, it must be taken into account that lower-level protocols must be transparent to higher-level protocols. Each layer must perform a defined function. The main advantages of layer architecture are flexibility to adapt lower-level protocols without affecting those of the highest level and the ability to isolate well-defined functions (Alencar 2012).

Some major network architectures have become classic, including the IBM System Network Architecture (SNA), the Distributed Network Architecture (DNA) by Digital Equipment, and the Department of Defense Advanced Research Projects Agency Network (ARPANET), from the USA.

11.5.1 The ISO Reference Model

The International Standards Organization (ISO) developed a reference model to compare different architectures and to construct new networks. This model is called the Open Systems Reference Model for Interconnection (OSI). The layers of the ISO reference model are illustrated in Fig. 11.20 Marksteiner et al. (2017), as compared to the TCP/IP model.

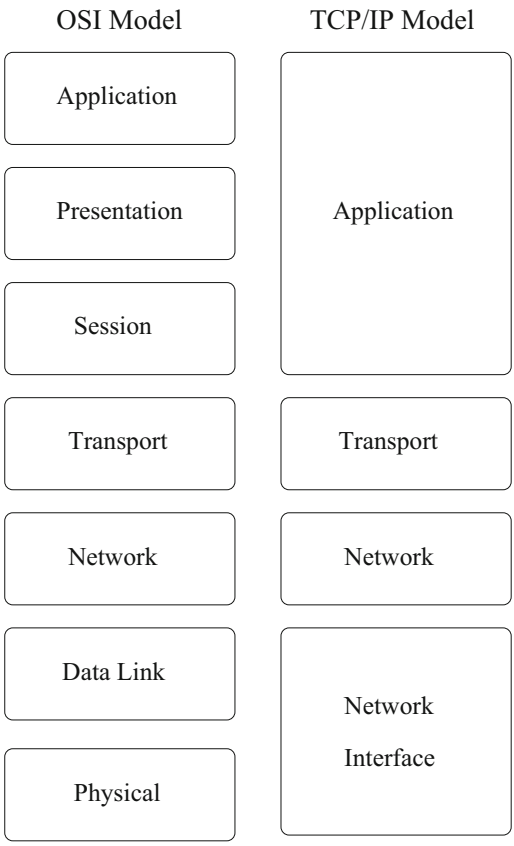


Fig. 11.20 Layer structure of the ISO reference model compared to the TCP/IP model

It is important to point out that the terminology for system, in this context, can cover from one simple terminal to a complete computer network. The term open is used to indicate that the model applies to the transfer of information between interconnected systems, without concern with the internal operation of the systems themselves.

Generally, all layers of the reference model are used at the origin and destination nodes, but only three are necessary in each intermediary node that just serves to transport packets to the destination nodes.

11.5.2 Layers of the Reference Model

The layers of the ISO reference model are described in detail in this section. Concisely, the physical layer coordinates the necessary functions for the transmission of information in a physical connection, and the link layer synchronizes the characters and messages and makes the transmission more reliable.

For the transfer of packets between the source and the destination, there is the network layer, for this purpose, the transport layer divides messages into packets, the session layer serves as an interface between the user and the network, the presentation layer simplifies communication between end users, and the application layer allows user access to the network.

Physical Layer
The electrical, mechanical, and functional characteristics, necessary for the transmission of the sequence of bits, are specified in the physical layer. These project specifications include signal features, such as amplitude, period, frequency spectrum (e.g., baseband or modulated transmission), and the digital modulation scheme that is adopted Farouzan (2008).

Some details regarding physical connections, such as number of pins, are also considered, as well as the mode of operation (full or half-duplex). The physical layer synchronizes the bits, establishing, maintaining, and releasing the connection between the nodes.

Data Link Layer
The data link layer provides the synchronizations of character and message (frame) and ensures reliable transmission of data blocks or frames between physically connected nodes. For this, the link layer creates and recognizes the limits (with the encapsulation) of the frames and uses redundant bits for the detection of errors.

The data link layer retransmits frames with errors, producing error messages. It does confirmations and manages other necessary details. The data link layer also controls the flow of packets to prevent the receiver from being overloaded and controls access, to determine which device takes control of the link at every moment.

Network Layer
The network layer performs the transfer of a data packet between the origin and the destination, typically through several links. It provides the services required to establish and maintain the flow of messages between users connected to the network.

The basic data unit of the network layer is the packet, and one of the functions of that layer is to ensure that packets are forwarded to their destinations. Considering that users are not necessarily connected by a direct physical link, this layer is responsible for routing and switching messages.

For forwarding messages, the network layer adds a header to the packet, which includes the logical addresses of the transmitter and receiver. The type of service provided to the transport layer by the network layer is usually specified as datagram service or virtual circuit. In addition, the network layer controls flow and congestion on the network, preventing many packets from being routed to the same connection.

Transport Layer
The transport layer subdivides messages passed through the session layer into units, called packets, if necessary, and allows the transmission of these units to the intended destination. It performs segmentation of the message at the source and its assembly at the destination, identifying and replacing damaged packets during the transmission.

The transport layer also provides associated services to the application processes, which run on hosts or user devices. Its function is to transfer data transparently between application programs in a way that is as independent as possible of the network. Some of these services involve the delivery of e-mail messages, with appropriate ordering, the diffusion of messages for multiple destinations, and the delivery of isolated messages.

Session Layer
The session layer establishes and maintains a connection between application processes, serving as an interface for the user with the network. In addition, the layer can verify the authenticity of the user, provide charging, and decide the type of communication, for example, full-duplex or half-duplex.

It also allows for the exchange of information between two systems and makes it possible for the process to verify a data stream. Therefore, there is no need to transmit the entire message in case of loss of part of the packets.

Presentation Layer
The presentation layer has the task of providing as many general functions as possible to simplify the communication between end users and to protect the information that travels on the network. It is responsible for the syntax and semantics of the information exchanged between two systems.

The layer deals with the interoperability of different methods of encryption, encryption of information, to guarantee privacy, and data compression. Examples of services provided by the of presentation layer are the conversion of codes, the text compression, and the use of layout standards for terminals and printers.

Application Layer
The application layer allows the user to access the computer network, such as the Internet. For example, it provides the right interface and support for certain services, such as message exchange, file transfer, establishment of a virtual terminal which emulates a remote system terminal, and shared database management.

This layer has a configuration and includes functions that depend on the user. Therefore, at this level, the software must contain the specific application programs, as a provision for the management of the required resources.

11.5.3 IEEE 802 Standards for Computer Networks

The Institute of Electrical and Electronics Engineers (IEEE) has committees that produce standards for all areas related to their professional activities, including those relating to computer networks, communication protocols, and modulation, among others.

The IEEE 802 standard, for example, defines the standards for local area and metropolitan area network physical and link layers of the OSI model for computer

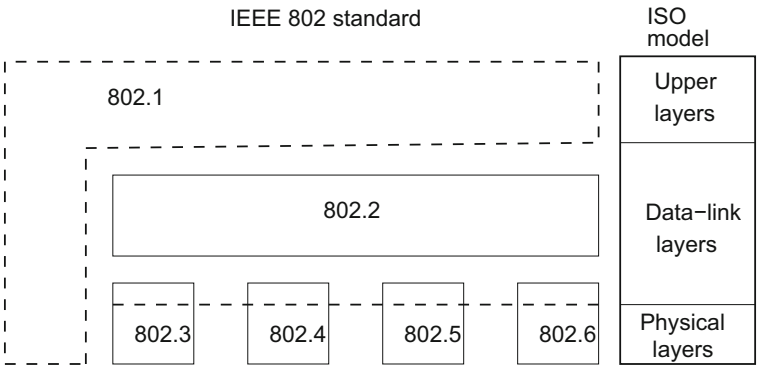


Fig. 11.21 Components of the IEEE 802 standard and the ISO model

networks. The standards specify different types of networks, such as Ethernet, wireless network, and fiber optics among others.

The IEEE 802 standards are divided into six basic parts, according to Fig. 11.21, plus nine specific parts. The standards specify the physical and link layers of the ISO reference, with a unique logical link protocol and four types of media access technologies (Alencar 2012):

1. Standard IEEE 802.1, a document describing network management, the relationship between the various parts of the pattern, and the relationship with the ISO reference model and with protocols for the highest level.
2. Standard IEEE 802.2, a common protocol for the control of logical link (logical link control—LLC).
3. Standard IEEE 802.3 specifies the syntax and semantics for access control in the middle (media access control—MAC). A bus that uses CSMA/CD as access method.
4. Standard IEEE 802.4, a bus that uses token passing as access method (Token Bus—TB).
5. Standard IEEE 802.5, a ring that uses token passing as access method (Token Ring—TR).
6. Standard IEEE 802.6, a metropolitan network (metropolitan area network—MAN).
7. Standard 802.7 defines broadband MAN.
8. Standard 802.8 defines the transmission in fiber optic network.
9. Standard 802.9 specifies the integration of local networks.
10. Standard 802.10 deals with security on local networks.
11. Standard 802.11, protocol for wireless local area network (LAN).
12. Standard 802.12 defines the demand priority access method, physical layer, and repeater specifications.
13. Standard 802.15 defines the personal wireless network (wireless personal area network—WPAN) (Bluetooth).

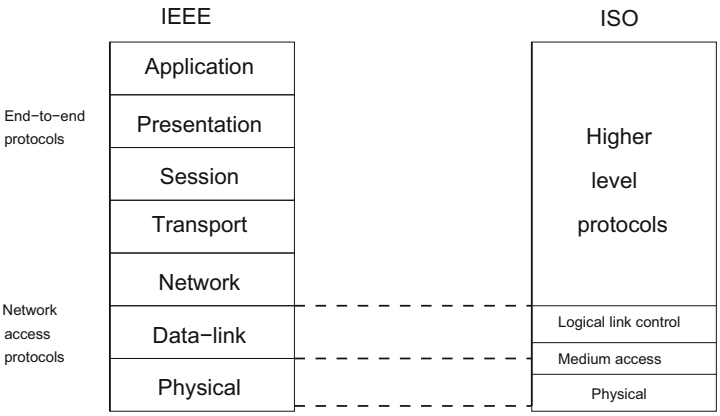


Fig. 11.22 Comparison between the reference model for local IEEE networks and the ISO model

14. Standard 802.16 defines broadband wireless access (BWA) (Wimax).
15. Standard IEEE 802.17 defines the Resilient Packet Ring (RPR) access.
16. Standard 802.20 specifies mobile wireless access (WMA) (Mobile-fi).
17. Standard 802.22 describes the wireless regional area network (WRAN).
18. Standard 802.24, defined by the attributions of the Vertical Applications Technical Advisory Group that determines what enhancements to the horizontal technologies would improve their ability to support the applications.

The IEEE standard for local area networks presents, essentially, the same characteristics for transmission and reception of bits that are described in the ISO model, as can be seen in Fig. 11.22.

11.6 The TCP/IP History

In 1975, the Defense Advanced Research Projects Agency (DARPA), which originated the ARPA network, started to develop the TCP/IP stack. In 1979, a committee was formed to lead the development of these protocols. This committee was called Internet Configuration Control Board (ICCB).

DARPA has assigned the rights to the TCP/IP code to the University of California for it to be distributed in UNIX version, in 1983, and requested that all computers connected to ARPANET use the TCP/IP. Protocols spread quickly over the network.

It has been defined that information can travel on communications networks in two different ways:

- Packet switching—In which a line on the network can be shared by multiple information packets from different sources. Internet traffic is based on packet switching.

- Circuit switching—In which there is an exclusive line between the source and the destination. The transmission of information is done as a telephone call, in which a circuit is established between the parties throughout the communication.

11.7 The Internet Coordination Structure

The Internet is an articulated set of computer networks, which uses the TCP/IP to provide packet communication, and contains many servers that generally work with operating systems UNIX and Linux. The Internet is coordinated by some entities, which take care of the organization of the network, distribution of IP addresses, creation of protocols, documentation, intermediation with government institutions, and the implementation of new technologies. The main bodies are:

- The Internet Advisory Board (IAB) is made up of several organizations and aims to coordinate the general organization of the Internet.
- The Internet Network Information Center (InterNIC) was created by NSF to distribute IP addresses.
- The Internet Research Task Force (IRTF) is one of the committees that make up the IAB, responsible for research activities, such as the development of protocols.
- The Requests for Comments (RFC) are technical documents related to Internet protocols. They can contain standards for the protocol or can become standards. These documents form the Internet documentation.
- The Federal Networking Council (FNC) is a committee that produces information on the Internet. The FNC performs the intermediation between the IAB and government institutions, in addition to supporting agencies in the use of the Internet.
- The Internet Engineering Task Force (IETF) is a subcommittee of the IAB that deals with constructive problems of network and also with the implementation of new technologies.

The Internet works as follows. Suppose a machine on a local network wants to send a packet of information to a distant network. Initially, the packet is sent to the local network, where it is received by all machines and the router. The router checks the destination address of the packet, queries its route table, and sends it forward.

The packet travels through routers of various networks, in which it is sent to routers closer to the final address, until it reaches the destination machine. If the packet reaches the final address, the originating machine receives an acknowledgment message. If this message does not arrive within a certain time, the packet is retransmitted.

11.8 Types of Computer Networks

The networks that make up the Internet can have several topologies, according to the coverage area, or coverage, and also depending on the technology used.

- Local area network (LAN) is a network located in a building or on campus. Its coverage is restricted to a few kilometers and its objective is to interconnect computers, workstations, and peripherals; this shares resources such as printers.
- Metropolitan area network (MAN) is a network that covers from some offices and neighborhoods to an entire city.
- Wide area network (WAN) is a network that covers a large geographic area, which can be a country or continent, usually formed by the interconnection of LAN networks.

11.9 Data Transmission Protocols

As described earlier, protocols are the transmission agreements established between network servers, to enable communication. They can be connection oriented or not.

- Connection-oriented protocols—In this type of transmission, packets do not need to have overheads, as occurs in nonconnection-oriented transmissions. At the beginning of the connection, the source and destination exchange all information needed for transmission. It has mechanisms for forwarding packets, as well as acknowledgment messages.
- Nonconnection-oriented protocols—The origin and the destination need to have a prior agreement on the communication and on the characteristics of quality of service (QoS). This mode of transmission associates each packet with a global address, which identifies the source and destination of the packet. Successive packets transferred are considered independent.

For nonconnection-oriented transmission mode, flow control is not important, nor is there recognizing or resending packets. In this way, one can communicate with any machine without having to make a connection. On the other hand, there is no guarantee of the success of the transmission, and the monitoring of the process needs to be more effective.

11.10 Interconnection Equipment

Physical Internet connections rely on certain basic components:

- Router—A circuit or computer with routing software, which has ports connected to different networks. When the router receives an information packet, it checks the destination address and looks for the equivalent port, based on network level addressing, which in the case of TCP/IP architecture is IP addressing. It tries to send the packet to networks that are closer to the destination network, decreasing the number of networks on which the packet travels.
- Gateway—It is a component of the network that aims to connect different networks, converting different levels of protocols, or in the case of IP, in order to perform the routing.
- Bridge—This device is used to connect similar networks. It forwards messages based on the level two addressing, known as medium access control (MAC). It segments network traffic, creating a single collision domain if necessary.

11.11 Interconnection Protocols

The Transmission Control Protocol (TCP) performs communication between applications from two different hosts. The TCP is a transport-level protocol that works with acknowledgment messages, specification of the information format, and security mechanisms. It ensures that all protocol data units (PDUs) are sent successfully, as it performs connection-oriented transmissions.

The TCP allows the use of various applications aimed at conversation. When executed, it uses the IP, not connection oriented. The TCP is then responsible for control of secure data transfer procedures. For greater efficiency in communications, TCP encompasses several functions that could be in the applications, such as word processor, database, and electronic mail. It was created to be a universal software that contains these functions.

The TCP performs additional services, such as:

- Flow control—Flow control assigns a transmission window to the source host. This window limits the number of bytes transmitted at a time interval.
- Transmission security—Reliability in transmissions via TCP in the connection orientation of the protocol and works with sequential and positive recognition numbers.

The TCP of the originating host transfers the data in the form of octets (bytes). Each octet is assigned numbers in sequence. The TCP of the target host analyzes these numbers to ensure the order and integrity of the message sent. If the transfer is perfect, the TCP of the destination host sends an acknowledgment message to the origin. Otherwise, a numeric string is sent to the TCP from the host source that informs the type of the problem, as well as order a new transmission.

For the TCP, sequential numbers can also be used to eliminate duplicate octets, which can occur on account of nonconnection-oriented transmission. The source TCP has a timer to ensure that time is not wasted between the transmission of wrong

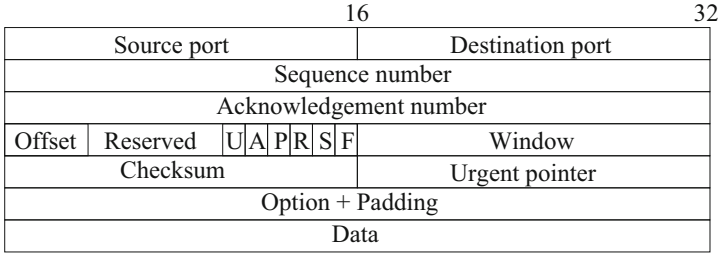


Fig. 11.23 Structure of a TCP header

messages and their correction. When the originating TCP receives an error message, a time-out occurs and the message is resent.

11.11.1 Some Additional TCP Services

The TCP performs additional services, which include:

- Open/close commands—Through physical devices, the TCP can establish a virtual connection, with the open command. At that moment, the TCP performs the three-way handshake, which is a process in which the source and destination TCP exchange acknowledgment messages to make the connection possible.

  By the time the transfer of information ends, any host (source or destination) can close the virtual connection with the close command.
- Information management in connection-oriented transmissions—The TCP can control all aspects of the information being transmitted, as it is a connection-oriented transmission protocol.
- Priority and security—The TCP allows the host administrator to control the levels of security and access permission, as well as connection priorities. These characteristics are not present in all versions, although they are defined in the TCP standards.
- Flow-oriented transfer stream—Interface applications send data to TCP in a targeted manner to stream. When the information arrives at TCP, it is grouped into packets and sent to the other transmission levels.

  The TCP can use several ways of sending messages, including telephone lines, local networks, or high-speed fiber optic networks (Fig. 11.23).

11.11.2 TCP Header Structure

The structure of the TCP header includes:

- Source port—Contains the source port number.
- Destination port—Contains the destination port number.
- Sequence number—The sequence number of the segment's first data octet (except when SYN is present). If SYN is present, the sequence number is the starting sequence number (ISN), and the first data octet is ISN + 1.
- Acknowledgment number—If the ACK control bit is activated, the field contains the value of the next sequence number that the recipient of the segment is waiting to receive.
- Data offset—Indicates where the data field starts within the TCP header.
- Reserved—Six bits reserved for future use.
- Control bits—The control bit can be (from left to right):
  - U (URG) Informs the application of the arrival of urgent data, which must be processed in the buffer.
  - A (ACK) Indicates that the acknowledgment field is important.
  - P (PSH) Function push.
  - R (RST) Reinitializes the connection.
  - S (SYN) Synchronize sequence numbers.
  - F (FIN) Indicates the end of data transmission.
- Window—16 bits. The number of data octets that the recipient is waiting to receive, starting with the octet that indicates the acknowledgment field.

11.12 The Internet Protocol

The main function of the IP is to transport datagrams from one network to another on the Internet. It is a nonconnection-oriented transmission protocol, with the following characteristics:

- It has no retransmission mechanisms.
- There is no guarantee of full or orderly transmission.
- Uses IP addresses as a basis for targeting datagrams.
- Discards a datagram if it is not delivered or if it has spent a long of time traveling on the Internet.
- Its operations and standards are described in Request for Comments (RFCs) and Internet Engineering Notes (IENs).

Although the IP does not have these characteristics, they are important so that data transmitted integrity remains with the TCP. Sending datagrams via IP goes through some basic steps that are explained in the following.

11.12.1 IP Addresses

Each computer connected to the Internet has one or more IP numbers, each IP number on the Internet being unique, which prevents datagrams from being sent to the wrong place. A computer can play the role of host or gateway or both. IP addresses are made up of four bytes, separated by periods and divided into network address and local address.

The network address, provided by InterNIC, contains the part that refers to the main network and the subnets. It is represented in the first three bytes of the IP number. The local address is assigned by the host administrator. It occupies the last byte and is used to identify local machines.

To facilitate the identification of addresses, the Domain Name System (DNS) was created, which associates a name to each IP number, with some characteristics:

- Names (domains) are separated by periods.
- There are no spaces between domains.
- The number of names can vary from one address to another, and they identify a single machine.
- There is no difference between uppercase and lowercase letters.
- Unlike the IP number, they are read from right to left.

11.12.2 How to Define a DNS

DNS example: `dee.ufcg.edu.br`, in which:

- `dee` is the name of the machine (host).
- `ufcg` identifies the organization to which the host is connected.
- `edu` is part of the organization's identification.
- `br` is the field that represents the country (Brazil) or the type of the organization.

In addition to the common DNS hierarchy (country name, organization name, machine name), there may also be a special hierarchy, such as the names ufcg and edu, where edu indicates an educational institution and ufcg is a university.

Associated with this address in the form of names can be the user's name (at the far left), separated by the symbol `@` from the rest of the address, as in the example: `malencar@iecom.org.br`, on what:

- `malencar` is the part that identifies the user (also called login).
- `@` is the character that separates the user's name from the rest of the address.
- `iecom.org.br` constitutes the remainder of the domain that goes from the name of the organization to the name of the country.

11.12.3 Datagram Fragmentation

When exchanging packets, Internet applications find a difference in size messages on different networks. The IP allows for fragmentation, and datagrams are divided into smaller units.

The fragmentation procedure is performed by a gateway, and the messages are broken into units and identified. The destination host regroups the instructions based on the identification (ID) of the gateway.

When identifying fragments, the gateway creates a header for each fragment, which contains the initial addresses of the networks (source address) and a message ID.

Upon regrouping, the destination host, upon receiving the first fragment, triggers a timer (time to live). If the standard time is exceeded and the message is not reconnected, the destination host discards fragments received and send an error message to the source.

11.12.4 IP Routing

A datagram sent by a host is not able to reach its destination without being guided by one or more routers.

A router receives the sent datagram and verifies its destination IP address, comparing it to its table. After this analysis, the datagram is sent to the gateway closest to the destination or even to the destination itself.

The table of a router can be static or dynamic. The dynamics are more flexible to adapt to the network modifications.

These tables determine the forwarding of datagrams to:

- Another subnet connected to the same router as the source network.
- A gateway, in case the network has access through one or more gateways (indirect routing).
- A default route if the address is not identified in the local table. In this case, the usual default is to send the datagram to an external circuit.

Router tables are made up of four numeric fields, as shown in Table 11.1.

- Field 1: IP address field. It contains all addresses sent by a given network.
- Field 2: This field contains a control called a metric, which uses algorithms to analyze the most efficient for a given address (including distance, equipment, and time).
- Field 3: The field that contains the IP address of the destination, be it a host from another subnet, a gateway, or an address default.
- Field 4: Field four contains the flags. A flag controls the frequency of use of a given address, so that, in the case of pivot tables, the address can be deleted if it is little used.

Table 11.1 Example of router tables

Router fields			
145.55.0.0	134	145.55.0.1	1
200.19.162.00	122	145.55.0.3	0
200.133.0.0	59	145.44.0.2	1
200.112.0.0	171	145.54.1.22	0
Default	173	821.27.8.11	1

Routers use a protocol to communicate or to communicate to hosts connected to them the information of status. This protocol can be the Routing Information Protocol (RIP) or Open Shortest Path First (OSPF).

There is yet another protocol, the Internet Control Message Protocol (ICMP), which informs about possible errors in the routing process.

11.12.5 IP Address Resolution

Sometimes, when a host wants to send a message, it may not know the full address of the host destination, that is, to know only the IP number. For these cases, the IP has a protocol called Address Resolution Protocol (ARP).

Two situations can occur in address resolution:

- The destination host is on the same network. Then, the originating host sends an ARP packet containing the destination IP number of the message to all other stations of your network. Only the station that identifies itself with the IP number of that packet will respond to the message, sending back your MAC address.

  In addition to sending the message, the originating host stores the MAC address received from the host from destination in a temporary table, which relates IP numbers to the MAC addresses of the supposed local network.
- The destination host is on a distant network. In this case, the originating host will send an ARP packet containing the IP number to the gateway, which then will guide this packet through several connections until it reaches the desired local network.

  Then, there will be the same as in the previous situation, with the difference that gateway itself will send ARP packets for all local stations.

The Reverse Address Resolution Protocol (RARP) is a protocol that allows a station to discover the user's own IP address. It is used by workstations without a hard disk, since they cannot permanently store an IP address.

Each time a station of this type is going to send a message, it needs to discover its IP address. To do so, it sends a message to your local network's RARP server, containing its MAC address. The server returns its IP number to the station, thus enabling the sending of messages.

The ARP Proxy is a variation of the ARP. It allows an organization to have only one IP address for its various networks.

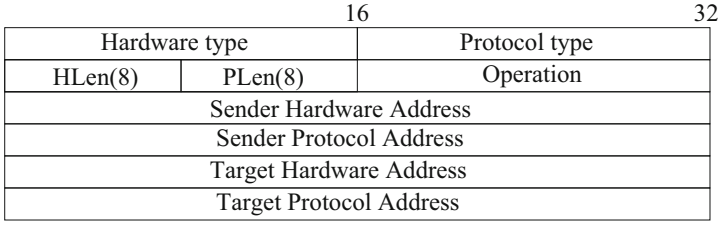


Fig. 11.24 Structure of the ARP/RARP header

In that case, all networks are connected to a router. When a host wants to communicate with a host from another network (without knowing its MAC address), it will dump a packet with the destination host IP number.

However, the packet is intercepted first by the router, which returns to the host destination MAC address itself.

Subsequent information is directed to the router, which rewrites it to the host destination, according to its own address table.

The structure of the header ARP/RARP is shown in Fig. 11.24.

The structure of the ARP/RARP header is composed as follows:

- *Hardware Type*—Specifies the type of hardware interface from which the issuer of the request expects a response from the request.

  Protocol Type—Specifies the type of high-level protocol the issuer of the request has.

  HLen—Hardware address length.

  PLen—Protocol address length.

  Operation—Values are as follows:

1. ARP request
2. ARP response
3. RARP request
4. RARP response
5. Dynamic RARP request
6. Dynamic RARP reply
7. Dynamic RARP error
8. InARP request
9. InARP reply

The structure of the ARP/RARP header includes:

- Sender hardware address—Hardware address of the request sender
- Sender protocol address—Address of the third-level (top level) protocol of the request sender
- Target hardware address—Hardware address of the request recipient
- Target protocol address—Recipient third-level (top level) protocol address

11.12.6 User Datagram Protocol

The UDP is restricted to ports and sockets and transmits data in a nonconnection-oriented manner. It is just an interface to the IP.

The basic function of UDP is to serve as a multiplexer or demultiplexer for IP information traffic. Like TCP, it works with ports that properly guide the information traffic to each application. These ports are:

- Destination Port—It is a part of the datagram (one end) that indicates the application to which whether to send the incoming information.
- Originating Port—It is located at the other end of the datagram and indicates the application that sent the message. It can be used for a resend, or when not used, it is filled with zeros.

11.12.7 Applications Based on TCP and UDP

There are a number of top-level Internet applications such as e-mail that use TCP or UDP services. These applications are widely used Internet standards, mainly by its versatility.

Some TCP application protocols:

- TELNET—The telecommunications network protocol allows a user to work on a host distant. It emulates a special terminal that does the necessary conversions between two different terminals, allowing you to remotely act on a host without the need for both hosts to have a similar terminal.
- FTP—The File Transfer Protocol is a tool for transmitting files over the Internet. It defines the procedures for managing the exchange of information between TCP hosts. An FTP connection goes through two processes: control connection and data transfer.

  The control connection is the first step in the FTP connection process. It serves to hold the host and define security and file manipulation levels.

  Data transfer is the stage at which files are transmitted. It depends on the success of control connection to be made.

11.12.8 Applications Based on TCP and UDP

The TCP application protocols are:

- SMTP—Simple Mail Transfer Protocol is a top-level application that is connected to the transmission e-mail via the Internet.

It is one of the most used top-level protocols on the Internet, which works as follows. SMTP is made up of two parts, which are the source and the destination, each of which has access to a storage server.

When the source sends a message to the destination, this message is stored on the source's storage server. The server then tries to send messages, and, if there is a problem with the destination, the server will later try to resend the message. If not, the message will be sent back to the source or to postmaster.

- SNMP—The Simple Network Management Protocol is the most expressive standard in terms of network management. It is a protocol that is used to deal with eventual network or equipment failures. SNMP is still used for monitoring networks, mainly in networks that use TCP/IP.
- RPC—Remote Procedure Call is a protocol that allows a host to use a function located on a remote host. RPC allows the exchange of messages, in which the source sends parameters to a server and is waiting for a return that provides the result of the remote function.
- TFTP—Trivial File Transfer Protocol is one of the most elementary of all, as it has few utilities, such as time and integrity control. It is a simpler model of FTP, as it has no mechanism security, and as it is based on the UDP, the integrity of your transmissions cannot be trusted. This protocol is not widely used today, but some vendors still distribute it to their customers' equipment to avoid possible incompatibilities.

11.13 The TCP/IP

A protocol is a set of rules that allow two or more computers to communicate, and TCP/IP is the most widely used network protocol today. It is, actually, a set or stack of protocols. The acronym refers to two different protocols, Transmission Control Protocol (TCP) and the Internet Protocol (IP).

In fact, there are many other protocols that compose the TCP/IP stack, including FTP, HTTP, SMTP, and UDP. TCP/IP has four layers.

The programs communicate with the application layer. In the application layer, it is possible to find application protocols such as SMTP (for email), FTP (for file transfer), and HTTP (for network browsing).

After processing the program request, the protocol in the application layer communicates with another protocol in the transport layer, usually TCP. This layer is responsible for taking the data sent by the upper layer, divide them into packets, and send them to the immediately lower layer, the Internet layer. In addition, during data reception, this layer is responsible for placing packets received from the network in order (since they may arrive out of order) and also check that the contents of the packets are intact.

In the Internet layer, there is the IP, which takes the packets received from the transport layer and adds information address, also known as IP address (the address of the computer sending the data and the address the computer receiving the data).

Then, the data packets are sent to the next lower layer, the network interface layer. In this layer, the packets are called datagrams, a reference to the time-honored telegram.

The network interface layer receives the packets sent by the Internet layer and sends them to the network (or receives data from the network, if the computer is receiving data). What is in that layer depends on the type of network the computer is connected to.

Currently, most computers use the Ethernet network and, therefore, the layers of the network interface must be found in the Ethernet, which are logical link control (LLC), media access control (MAC), and physics. Packets transmitted over the network are called frames.

11.13.1 Application Layer

This layer communicates between programs and transport protocols. There are several protocols that operate at the application layer. The most known are:

- The HyperText Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML.
- The Simple Mail Transfer Protocol Mail Transfer (SMTP) is an Internet standard communication protocol for electronic mail transmission. Mail servers and other message transfer agents use SMTP to send and receive mail messages.
- The File Transfer Protocol (FTP) is a standard to download, upload, and transfer files from one location to another on the Internet and between computer systems.
- The Simple Network Management Protocol (SNMP) is an Internet Standard protocol for collecting and organizing information about managed devices on IP networks.
- The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet.
- Telnet is a protocol for communication between remote terminals. It is a client/server application protocol that provides access to virtual terminals of remote systems on local area networks or the Internet.

When an email client program wants to download messages that are stored on the email server, it makes this request to the application layer of TCP/IP, being served by the SMTP.

When entering a www address to view a page on the Internet, the communication is executed with the layer application of TCP/IP, being served by the HTTP.

The application layer communicates with the transport layer through a door. The doors are numbered and the doors standard applications always use the same ports.

For example, the SMTP always uses port 25, the HTTP always uses port 80, and FTP uses ports 20 (for data transmission) and 21 (for transmission of control information).

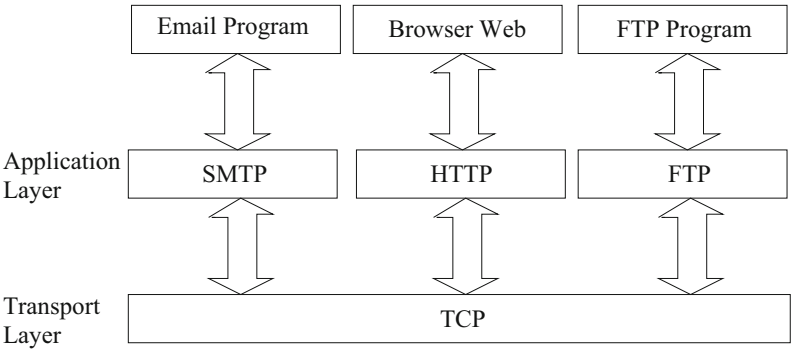


Fig. 11.25 How the application layer works

The use of a port number allows the Transmission Control Protocol (TCP) to know what type of content of the data packet (e.g., if it is an email) and receiver, to know which application protocol it should deliver the data packet to.

Upon receiving a packet destined for port 25, the TCP delivers it to the protocol that is connected to this port, typically SMTP, which in turn delivers the data to the application that requested it (the email program).

Figure 11.25 illustrates how the application layer works.

11.13.2 Transport Layer

In data transmission, the transport layer is responsible for taking the data passed through the data layer application and transform them into packets.

The TCP is the most used protocol in the transport layer. At the reception of data, the TCP takes the packets passed through the Internet layer and puts them in order, since that the packets can reach the destination out of order, check if the data inside the packets are intact and sends a confirmation signal called acknowledge (ACK) to the transmitter, warning that the packet has been received correctly and that the data is healthy.

If no acknowledgment signal (ACK) is received, either because the data did not reach its destination or because TCP discovered that the data was corrupted, the transmitter will resend the lost packet.

While TCP reorders packets and uses an acknowledgment mechanism, which is desirable in data transmission, there is another protocol that operates in this layer that does not have these resources. This protocol is UDP (User Datagram Protocol).

For this reason, TCP is considered a reliable protocol, while UDP is considered an unreliable protocol. UDP is typically used when no important data is being transmitted, such as DNS requests. Since UDP does not reorder packets or use an acknowledgment mechanism, it is faster than TCP.

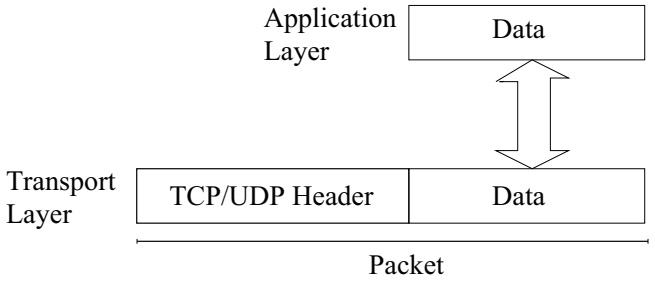


Fig. 11.26 Data packet generated at the transport layer

When UDP is used, the application that requests the transmission is responsible for verifying that the data received they are intact or not and also to reorder the received packets, that is, the application does the work of TCP.

During data transmission, both UDP and TCP receive data passed from the application layer and add a header to that data. Upon receipt of data, the header will be removed before the data is sent to the appropriate port.

This header contains control information, in particular the number of the source port, destination port number, sequence number (for acknowledgment of receipt and reordering mechanisms used by TCP), and a checksum (called checksum or CRC, which is a calculation used to verify that the data was received intact at the destination).

The UDP header is eight bytes long, while the TCP header has between 20 and 24 bytes, depending on whether the options field is being used or not.

Figure 11.26 illustrates the data packet generated at the transport layer. This data packet is sent to the Internet layer, for data transmission, or is received from the Internet layer, when receiving data.

11.13.3 Internet Layer

In TCP/IP networks, each computer is identified with a unique virtual address, called an IP address. The Internet layer is responsible for adding a header to the data packet received from the transport layer where, among other control data, the source IP address and the destination IP address are added, that is, the IP address of the computer sending the data and the IP address of the computer that should receive it.

Each computer network card has a physical address. This address is recorded in the card's ROM memory network and is called MAC address. That way, on a local network, if computer A wants to send data to the computer B, it needs to know the MAC address of computer B. In a small local network computers are easily found. The task is not so simple on a global network such as the Internet.

If no virtual addressing scheme is used, you need to know the MAC address of the target computer, which is not only a complicated task but also does not help in the routing of packets, as this address does not use a tree structure.

Routing is the path that the data must use to reach the destination. When someone requests data from a Internet server, for example, this data goes through several locations (called routers) before it arrives to the computer.

In all networks connected to the Internet, there is a device called a router, which bridges between computers on the local network and the Internet. Every router has a table containing the known networks and also a configuration called the default gateway pointing to another router on the Internet.

When the computer sends a data packet to the Internet, the router connected to the network first checks if it knows the destination computer; in other words, the router checks whether the destination computer is located on the same network or on a network that it knows the route. If it does not know the route to the destination computer, it will send the packet to its default gateway, which is another router. This process is repeated until the data packet reaches its destination.

There are several protocols that operate at the Internet layer:

- The Internet Protocol (IP) is the network layer communications protocol in the Internet Protocol suite for relaying datagrams across network boundaries.
- The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet Protocol suite that is used by network devices, including routers, to send error messages and operational information indicating success or failure when communicating with another IP address.
- The Address Resolution Protocol (ARP) is a procedure that connects a dynamic Internet Protocol (IP) address to a fixed physical machine address, also known as a media access control (MAC) address, in a local area network (LAN).
- The Reverse Address Resolution Protocol (RARP) is a procedure used to link a MAC address to an Internet Protocol (IP) address.

Data packets are transmitted using the IP. The IP takes the data packets received from the transport layer and divides them into datagrams.

The datagram is a packet that does not contain any type of acknowledgment of receipt (ACK), which means that IP does not implement a receipt acknowledgment mechanism, that is, it is an untrusted protocol.

It should be noted that during data transfer the TCP will be used over the Internet layer (i.e., over IP) and TCP implements the receipt acknowledgment mechanism.

Therefore, although the IP does not verify if the datagram has arrived at the destination, the TCP will do this verification. The connection will then be reliable, although IP alone is an unreliable protocol. Figure 11.27 illustrates a datagram in the Internet layer.

Each IP datagram can have a maximum size of 65,535 bytes, including its header, which can use 20 or 24 bytes, depending on whether a field called options is used or not. That way, IP datagrams can carry up to 65,515, which means 65,511 bytes of data.

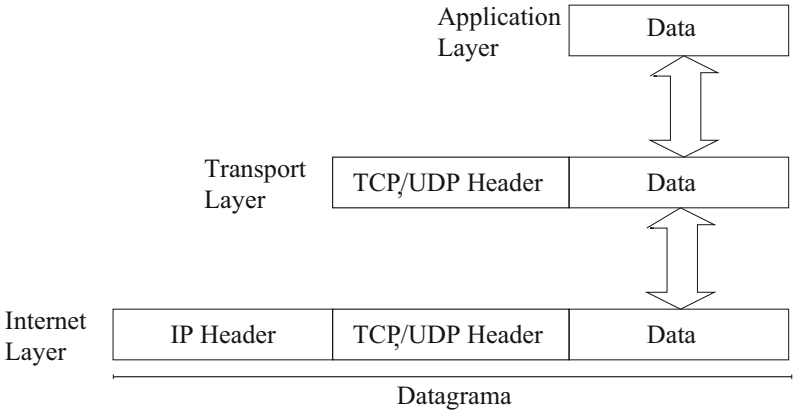


Fig. 11.27 Datagram in the Internet layer

If the data packet received from the transport layer is larger than than 65,515, or 65,511 bytes, the IP will fragment packets into as many datagrams as needed The header added by the IP includes the source IP address, the destination IP address, and several other control information.

11.14 Security Protocols for TCP/IP

The protocol architecture that governs data transmission on computer networks that make up the Internet, designated by the acronym TCP/IP, initially foresaw an open network environment with few security mechanisms. This provided the public nature of the references (Request For Comments—RFC) and standard implementations of the TCP/IP suite.

This public and open nature of the Internet Protocol, without protection mechanisms, made the security one of the main structural problems of the TCP/IP architecture. This opening was responsible for the success and popularization of the protocol. However, the deficiency in the security aspect represents one of the limitations to the use of technology, especially in commercial applications.

Among the deficiencies, in the security aspect, of the IP is its inability to authenticate a machine on the network. Based on the source IP address of a received packet, it is not possible to determine with certainty the identity of the machine that has it originated. There is also no guarantee that the contents of a received packet are unchanged or that data privacy has been preserved (Pouw 1999).

The initial solution to provide network security was to implement individual algorithms, for each application, when the needs arose. In this way, the PGP was created for communication by e-mail, SSL/TLS for Internet browsing, and SSH for login secure remote. However, although implementation at the application layer is

simpler, since it does not involve the operating system and is restricted to a specific application, a service implemented in this layer makes it specific, creating the need for a new development for each new application.

The most appropriate solution was to implement the security service in the internetwork layer, which delivers packets without connection, which allows control by flow or connection. Some protocols were developed for this layer for this purpose, some run only on IP (as in the case of GRE and PPTP), and others, more versatile, are able to handle not only IP packets but also IPX and NetBEUI (as in the case of L2F or L2TP).

The guarantee of security on the Internet requires that the IP itself offers this service, without relying on other protocols. The integration of this functionality with the new version of IP (IPv6) was proposed, and its development was in charge of the working group IP Security (IPSec), the Internet Engineering Task Force (IETF).

Considering the delay in the migration of the Internet to the new version of IP, several adaptations were made for IPSec to run over IPv4. Its implementation was made with special headers, with header extensions in IPv6 and an additional protocol header in IPv4, typically placed between the original IP header and its payload.

11.14.1 The IP Security

The IP Security is an open platform formed by a set of protocols that provide the following security services:

- Access control
- Packet integrity
- Origin authentication
- Packet privacy
- Privacy in packet flow
- Protection against replays

The open architecture makes it possible to include other authentication and encryption algorithms, in addition to the necessary HMAC-MD5 and HMAC-SHA-1, for authentication and encryption, and DES-CBC, for encryption only.

11.14.2 Vulnerabilities and Safety Measures Regarding TCP/IP

Vulnerabilities in the TCP/IP stack are exploited through the following attacks.

SYN Flooding This attack consists of sending several packets to open a data exchange connection (SYN packets), containing a forged IP address of a machine

that is not operational and, therefore, that is not able to complete the connection opening process.

IP Spoofing When the attacker masks his computer as a legitimate network machine. This is done by forging the source address of IP datagrams. Thus, the attacker associates his or her datagrams with a legitimate IP address of the target network.

Sequence Number Attack To ensure that the packets arrive at the destination in the order in which they were sent, the TCP uses a sequence of numbers associated with the packets. The TCP Sequence Number Prediction attack consists of predicting the next sequence number of the packet that travels and thus forging packets with the numbering expected by the target machine.

TCP Session Hijacking This is a variation of the attack by sequence number, which also uses a forged IP address. This attack can be performed against any application based on the TCP, such as Telnet, `rlogin`, and FTP. It consists of hijacking the connection between two machines: a client machine and a server.

RST and FIN attacks Because the TCP has flags that are used to control the connection, an attacker can use the RST and FIN flags to generate a denial of service attack. Normally, the RST flag is used to restart a connection, and the FIN flag is used to indicate that there is no more data to be sent.

Ping O'Death The Ping program is used to test whether a machine is operational, by sending a packet of the type "ICMP echo" and waiting for the receipt of this packet to be received by the target machine. Older versions of this program make it possible to send a packet with a larger size than allowed. When the target machine receives this packet and starts processing, system variables overflow may occur.

Possible measures against these attacks on the TCP/IP stack are:

- Block the transmission of packets, the source address of which does not belong to the set of addresses that make up the local network
- Block the reception of packets that originate on the Internet and that contain a source address equal to an IP address on the local network
- Use encryption and authentication in communication between machines via the Internet and in all TCP services
- Periodically update the operating systems (Ping O'Death)

11.14.3 Vulnerabilities and Safety Measures Regarding TCP/IP Services

Some services are vulnerable to attack by hackers. The vulnerabilities of the services offered by TCP/IP are typically associated with certain stack protocols:

Telnet This is a program used for communication between hosts. The vulnerability of Telnet is that it does not protect the user's login and password during transmission to the remote machine.

Finger It is used to obtain information about the users of a given host, such as the user's name and the last time he had access to the system. An attacker could use information provided by the finger command to perform an attack, for example, using the user's login and name to try to find out their password.

File Transfer Protocol (FTP) FTP is vulnerable because, when establishing a connection between a client and an FTP server, the user's login and password are sent without protection, and an attacker who is monitoring packet traffic on the network can discover the user's password.

The usual solutions to these problems are listed next:

- Periodically update the network's software.
- Limit the services offered by the network's servers.
- Inspect the system's logs files.
- Check running processes and strange files.
- Validate system settings.
- Educate users about network security.
- Limit users' access to the system.
- Disable the Telnet and Finger commands.
- Run frequent backups.
- Constant updating of administrators and those responsible for security in relation to methods of systems invasion.
- Encourage the use of cryptography on the network.
- Install firewalls to filter incoming packets.

11.15 Video Transmission Over the Internet

The Society of Motion Picture and Television Engineers (SMPTE) developed the SMPTE ST 2110 standards suites that specifies the carriage, synchronization, and description of separate elementary nature streams over IP for real-time production, reproduction, and other professional media applications. Each stream is individually timed by the ST 2110 system and can take different routes over the network to reach the destination via unicast or multicast at one or more receivers (SMPTE 2024).

The SMPTE ST 2110 suite is based on VSF TR-03, a standard designed for professional broadcast applications that uses Real-time Transport Protocol (RTP) as its underlying transport protocol. The audio-video-data synchronization using Precision Time Protocol (PTP) clocks guarantees that the accurate synchronization of all streams regardless of how the packets were routed.

The purpose of SMPTE ST 2110 is to create a single standard for the professional media industries to transmit and receive media. The ST 2110 suite of standards include audio encapsulation, background and system overview, data encapsulation,

synchronization and identification, traffic shaping and delivery timing, and video encapsulation (Mason 2024):

The standards are split into four main components: 2110-10 for timing and synchronization, 2110-20 for video, 2110-30 for audio, and 2110-40 for ancillary data (Matrox 2024; SMPTE 2024), as follows:

- ST 2110-10 System Timing and Definitions—Defines the timing and requirements for all the ST 2110 series essence streams. This family of engineering documents defines an extensible system of RTP-based essence streams referenced to a common reference clock, in a manner which specifies their timing relationships.
- ST 2110-20 Uncompressed Active Video—Standardizes the transport of uncompressed video over IP networks. An SDP-based signaling method is defined for image technical metadata necessary to receive and interpret the stream.
- ST 2110-21 Video Stream Packet Shaping—Specifies the traffic shaping and delivery timing for uncompressed video. A timing model for SMPTE ST 2110-10 video RTP streams as measured leaving the RTP sender and defines the sender SDP parameters used to signal the timing properties of such streams.
- ST 2110-22 Constant Bit-Rate Compressed Video—Describes the payload for constant rate compression video. The real-time, RTP-based transport of constant bit rate compressed video over IP networks, referenced to a common reference clock.
- ST 2110-30 Professional Media Over Managed IP Networks—Standardizes the transport of uncompressed audio streams, PDCM digital audio, over IP networks based on AES67. The real-time, RTP-based transport of PCM digital audio streams over IP networks by reference to AES67. An SDP-based signaling method is defined for metadata necessary to receive and interpret the stream. Non-PCM digital audio signals including compressed audio signals are outside the scope of this standard.
- ST 2110-31 AES3 Audio Streams—Specifies the real-time, RTP-based transport of AES3 signals over IP networks, referenced to a network reference clock, that is, the transport of uncompressed audio streams over IP networks using AES3.
- ST 2110-40 Ancillary Data—Standardizes how ancillary data is encapsulated and transported using RTP. It maps ancillary data packets, as defined in SMPTE ST 291-1, into Real-time Transport Protocol (RTP) packets that are transported via User Data Protocol/Internet Protocol (UDP/IP) and enables those packets to be moved synchronously with associated video and audio essence streams.
- SMPTE ST 2110-41—Defines a flexible RTP payload framework for data items. The framework can be used to transport data items which are tightly time associated with video or audio RTP Streams or those that are independent of any video or audio RTP streams.
- SMPTE ST 2110-43—Adds support for Timed Text Markup Language (TTML) for captions and subtitles in systems conforming to SMPTE ST 2110-10.

The SMTPE ST 2110 suite is an improvement over previous IP video transport protocols, because it is based on openly available and easily accessible standards,

the technology is also openly available, which means that the users are not tied to a particular vendor, it is flexible to adapt to different use cases and workflows, it supports high-quality, high-resolution audio and video formats, and it can also work on gigabyte IP networks with compressed video.

The SMPTE ST 2110 suite transports media by splitting the video, audio, and ancillary components into independent essence streams. This allows each component to be processed, transported, and stored separately from the others. Essence streams in ST 2110 are carried in a UDP/RTP packet and are self-contained, with each stream having all the information necessary for that media type, including timing information (Matrox 2024).

Some Recommended Practices (RPs) support and extend the core standards:

- SMPTE RP 2110-23—Specifies how a single video essence, with a high rate such as the UHD video, can be transported using multiple ST 2110-20 streams.
- SMPTE RP 2110-24—Addresses special considerations for standard definition (SD) video in SMPTE ST 2110 systems.
- SMPTE RP 2110-25—Recommends nomenclature for measurements on SMPTE 2110 systems, together with their associated formulas.

Appendix A
Fourier Series and Fourier Transform

Mathematics consists of proving the most obvious thing in the least obvious way.

– George Pólya

This appendix presents a summary of properties of the Fourier series and also a collection of Fourier transforms to help the interested reader in solving the problems. The Parseval's theorem is also presented (Hsu 1973; Lathi 1989).

Trigonometric Series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_o t + b_n \sin n\omega_0 t)$$

Cosine Series

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

Complex Exponential Series

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

It is important to note, for all the former series, that

$$f(t+T) = f(t), \qquad \omega_0 = \frac{2\pi}{T}$$

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

Conversion Formulas

$$F_0 = a_0, \qquad F_n = \frac{1}{2}(a_n - jb_n), \qquad F_{-n} = \frac{1}{2}(a_n + jb_n);$$

$$F_n = |F_n|e^{j\phi_n}, \qquad |F_n| = \frac{1}{2}\sqrt{a_n^2 + b_n^2};$$

$$a_0 = F_0, \qquad a_n = F_n + F_{-n}, \qquad b_n = j(F_n - F_{-n}),$$

$$C_0 = a_0, \qquad C_n = 2|F_n| = \sqrt{a_n^2 + b_n^2}, \qquad \theta_n = -\arctan\left(\frac{b_n}{a_n}\right).$$

Symmetry Conditions for the Periodical Waveforms

Symmetry type	Conditions	Fourier series format
Even	$f(t) = f(-t)$	$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$
Odd	$f(t) = -f(-t)$	$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$
Half-wave	$f(t) = -f(t + \frac{T}{2})$	$f(t) = \sum_{n=1}^{\infty}[a_{2n-1}\cos(2n-1)\omega_0 t + b_{2n-1}\sin(2n-1)\omega_0 t]$
Quarter wave, even	$f(t) = f(-t)$ and $f(t) = -f(t + \frac{T}{2})$	$f(t) = \sum_{n=1}^{\infty} a_{2n-1}\cos(2n-1)\omega_0 t$
Quarter wave, odd	$f(t) = -f(-t)$ and $f(t) = -f(t + \frac{T}{2})$	$f(t) = \sum_{n=1}^{\infty} b_{2n-1}\sin(2n-1)\omega_0 t$

Symmetry type	Conditions	Formula for the Fourier coefficients
Even	$f(t) = f(-t)$	$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\omega_0 t)\mathrm{d}t$
Odd	$f(t) = -f(-t)$	$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\omega_0 t)\mathrm{d}t$
Half-wave	$f(t) = -f(t + \frac{T}{2})$	$a_{2n-1} = \frac{4}{T}\int_0^{T/2} f(t)\cos[(2n-1)\omega_0 t]\mathrm{d}t$ $b_{2n-1} = \frac{4}{T}\int_o^{T/2} f(t)\sin[(2n-1)\omega_0 t]\mathrm{d}t$
Quarter wave, even	$f(t) = f(-t)$ $f(t) = -f(t + \frac{T}{2})$	$a_{2n-1} = \frac{8}{T}\int_0^{T/4} f(t)\cos[(2n-1)\omega_0 t]\mathrm{d}t$
Quarter wave, odd	$f(t) = -f(-t)$ $f(t) = -f(t + \frac{T}{2})$	$b_{2n-1} = \frac{8}{T}\int_0^{T/4} f(t)\sin[(2n-1)\omega_0 t]\mathrm{d}t$

Definition of the Fourier Transform and Some Properties

- Definition: $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\mathrm{d}t$

- Inverse: $f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\mathrm{d}\omega$
- Magnitude and phase: $F(\omega) = |F(\omega)|e^{j\theta(\omega)}$
- Even $f(t)$ function: $F(\omega) = 2\int_{0}^{\infty} f(t)\cos\omega t\mathrm{d}t$
- Odd $f(t)$ function: $F(\omega) = -2j\int_{0}^{\infty} f(t)\sin\omega t\mathrm{d}t$
- Area under the curve in time: $F(0) = \int_{-\infty}^{\infty} f(t)\mathrm{d}t$
- Area under the transform: $f(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\mathrm{d}\omega$
- Linearity: $\alpha f(t) + \beta g(t) \leftrightarrow \alpha F(\omega) + \beta G(\omega)$

Parseval's Theorem

$$\int_{-\infty}^{\infty} f(t)g(t)\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)G^*(\omega)\mathrm{d}\omega,$$

$$\int_{-\infty}^{\infty} |f(t)|^2\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\mathrm{d}\omega,$$

$$\int_{-\infty}^{\infty} f(\omega)G(\omega)\mathrm{d}\omega = \int_{-\infty}^{\infty} F(\omega)g(\omega)\mathrm{d}\omega.$$

A.1 Table of Fourier Transforms

This section presents a set of Fourier transforms that are necessary to solve some of the proposed problems. Some properties are also presented, which help computing the transforms (Hsu 1973; Spiegel 1976; Lathi 1989; Gradshteyn and Ryzhik 1990; Oberhettinger 1990).

$f(t)$	$F(\omega)$
$f(at)$	$\frac{1}{\lvert a\rvert}F(\frac{\omega}{a})$
$f(-t)$	$F(-\omega)$
$f^*(t)$	$F^*(-\omega)$
$f(t-\tau)$	$F(\omega)e^{-j\omega\tau}$
$f(t)e^{j\omega_0 t}$	$F(\omega-\omega_0)$
$f(t)\cos\omega_0 t$	$\frac{1}{2}F(\omega-\omega_0)+\frac{1}{2}F(\omega+\omega_0)$
$f(t)\sin\omega_0 t$	$\frac{1}{2j}F(\omega-\omega_o)-\frac{1}{2j}F(\omega+\omega_0)$
$F(t)$	$2\pi f(-\omega)$
$f'(t)$	$j\omega F(\omega)$
$f^{(n)}(t)$	$(j\omega)^n F(\omega)$
$\int_{-\infty}^{t} f(x)dx$	$\frac{1}{j\omega}F(\omega)+\pi F(0)\delta(\omega)$
$-jtf(t)$	$F'(\omega)$
$(-jt)^n f(t)$	$F^{(n)}(\omega)$

(continued)

$f(t)$	$F(\omega)$
$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)dx$	$F(\omega)G(\omega)$
$\delta(t)$	1
$\delta(t-\tau)$	$e^{-j\omega\tau}$
$\delta'(t)$	$j\omega$
$\delta^{(n)}(t)$	$(j\omega)^n$
$f(t)g(t)$	$\frac{1}{2\pi}F(\omega) * G(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\phi)G(\omega-\phi)d\phi$
$e^{-at}u(t)$	$\frac{1}{a+j\omega}$
$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$
e^{-at^2}	$\sqrt{\frac{\pi}{a}}e^{-\omega^2/(4a)}$
te^{-at^2}	$j\sqrt{\frac{\pi}{4a^3}}\omega e^{-\omega^2/(4a)}$
$p_T(t) = \begin{cases} 0 & \text{para } \lvert t\rvert > T/2 \\ A & \text{para } \lvert t\rvert \le T/2 \end{cases}$	$AT\frac{\sin(\frac{\omega T}{2})}{(\frac{\omega T}{2})}$
$\frac{\sin at}{\pi t}$	$p_{2a}(\omega)$
$te^{-at}u(t)$	$\frac{1}{(a+j\omega)^2}$
$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$	$\frac{1}{(a+j\omega)^n}$
$e^{-at}\sin bt\, u(t)$	$\frac{b}{(a+j\omega)^2+b^2}$
$e^{-at}\cos bt\, u(t)$	$\frac{a+j\omega}{(a+j\omega)^2+b^2}$
$\frac{1}{a^2+t^2}$	$\frac{\pi}{a}e^{-a\lvert\omega\rvert}$
$\frac{t}{a^2+t^2}$	$j\pi e^{-a\lvert\omega\rvert}[u(-\omega)-u(\omega)]$
$\frac{\cos bt}{a^2+t^2}$	$\frac{\pi}{2a}[e^{-a\lvert\omega-b\rvert}+e^{-a\lvert\omega+b\rvert}]$
$\frac{\sin bt}{a^2+t^2}$	$\frac{\pi}{2aj}[e^{-a\lvert\omega-b\rvert}-e^{-a\lvert\omega+b\rvert}]$
$\sin bt^2$	$\frac{\pi}{2b}\left[\cos\frac{\omega^2}{4b}-\sin\frac{\omega^2}{4b}\right]$
$\cos bt^2$	$\frac{\pi}{2b}\left[\cos\frac{\omega^2}{4b}+\sin\frac{\omega^2}{4b}\right]$
$\operatorname{sech} bt$	$\frac{\pi}{b}\operatorname{sech}\frac{\pi\omega}{2b}$
$\ln\left[\frac{x^2+a^2}{x^2+b^2}\right]$	$\frac{2e^{-b\omega}-2e^{-a\omega}}{\pi\omega}$
$f_P(t) = \frac{1}{2}[f(t)+f(-t)]$	$\mathrm{Re}\,(\omega)$
$f_I(t) = \frac{1}{2}[f(t)-f(-t)]$	$j\mathrm{Im}\,(\omega)$
$f(t) = f_P(t)+f_I(t)$	$F(\omega) = \mathrm{Re}\,(\omega)+j\mathrm{Im}\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
$\sin\omega_0 t$	$-j\pi[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$
$\sin\omega_0 t u(t)$	$\frac{\omega_0}{\omega_0^2-\omega^2}+\frac{\pi}{2j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$

(continued)

$f(t)$	$F(\omega)$
$\cos\omega_0 t u(t)$	$\frac{j\omega}{\omega_0^2-\omega^2}+\frac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$
$u(t-\tau)$	$\pi\delta(\omega)+\frac{1}{j\omega}e^{-j\omega\tau}$
$tu(t)$	$j\pi\delta'(\omega)-\frac{1}{\omega^2}$
1	$2\pi\delta(\omega)$
t	$2\pi j\delta'(\omega)$
t^n	$2\pi j^n\delta^{(n)}(\omega)$
$\lvert t\rvert$	$\frac{-2}{\omega^2}$
$\frac{1}{t}$	$\pi j-2\pi j u(\omega)$
$\frac{1}{t^n}$	$\frac{(-j\omega)^{n-1}}{(n-1)!}[\pi j-2\pi j u(\omega)]$
$u(t)-u(-t)$	$\frac{2}{j\omega}$
$\frac{1}{e^{2t}-1}$	$\frac{-j\pi}{2}\coth\frac{\pi\omega}{2}+\frac{j}{\omega}$
$\delta_T(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT)$	$\omega_0\delta_{\omega_0}(\omega)=\omega_0\sum_{n=-\infty}^{\infty}\delta(\omega-n\omega_0)$
$\cos\left(\frac{t^2}{4a}-\frac{1}{4}\pi\right)$	$2\sqrt{\pi a}\cos(a\omega^2)$
$\sin\left(\frac{t^2}{4a}+\frac{1}{4}\pi\right)$	$2\sqrt{\pi a}\sin(a\omega^2)$
$\frac{\Gamma(1-s)\sin\left(\frac{1}{2}s\pi\right)}{\lvert t\rvert^{1-s}}$	$\pi\lvert\omega\rvert^{-s}\ 0<\mathrm{Re}\,(s)<1$
$\frac{1}{\lvert t\rvert}$	$\frac{\sqrt{2\pi}}{\lvert\omega\rvert}$
$\frac{\sqrt{\sqrt{\alpha^2+t^2}+\alpha}}{\sqrt{\alpha^2+t^2}}$	$\sqrt{\frac{2\pi}{\lvert\omega\rvert}}e^{-\alpha\lvert\omega\rvert}$
$\frac{\cos(\frac{1}{2}\alpha)\cosh(\frac{1}{2}t)}{\cosh(t)+\cos(\alpha)}$	$\frac{\pi\cosh(\alpha\omega)}{\cosh(\pi\omega)}\quad-\pi<\alpha<\pi$
$\frac{\sin(\alpha)}{\cosh(t)+\cos(\alpha)}$	$\frac{\pi\sinh(\alpha\omega)}{\sinh(\pi\omega)}\quad-\pi<\alpha<\pi$
$J_0(\alpha t)$	$\frac{2}{\sqrt{\alpha^2-\omega^2}}\quad\lvert\omega\rvert<\alpha$
	$0\qquad\lvert\omega\rvert>\alpha$
$J_0(\alpha\sqrt{b^2-t^2})\lvert t\rvert<b$	$2\frac{\sin[b(\alpha^2+\omega^2)^{\frac{1}{2}}]}{\sqrt{\alpha^2+\omega^2}}$
	$0\qquad\lvert t\rvert>b$
$j^n J_{n+\frac{1}{2}}(t)$	$\sqrt{2}P_n(\omega)\lvert\omega\rvert<1$
	$0\qquad\lvert\omega\rvert>1$
$J_0(\alpha\sqrt{t^2+b^2})$	$\frac{2\cos(b\sqrt{\alpha^2-\omega^2})}{\sqrt{\alpha^2-\omega^2}}\ \lvert\omega\rvert<\alpha$
	$0\qquad\lvert\omega\rvert>\alpha$
$J_0(\alpha\sqrt{t^2-b^2})$	$\frac{2\cosh(b\sqrt{\alpha^2-\omega^2})}{\sqrt{\alpha^2-\omega^2}}\ \lvert\omega\rvert<\alpha$
	$0\qquad\lvert\omega\rvert>a$
$\frac{1}{(\alpha-jt)^\nu}$	$2\pi\omega^{\nu-1}e^{-\alpha\omega}/\Gamma(\nu),\quad\omega>0$
$\mathrm{Re}\,\alpha>0,\quad\mathrm{Re}\,\nu>0$	$0,\qquad\omega<0$

(continued)

$f(t)$	$F(\omega)$
$\frac{1}{(\alpha+jt)^{\nu}}$	$-2\pi(-\omega)^{\nu-1}e^{\alpha\omega}/\Gamma(\nu), \quad \omega<0$
$\mathrm{Re}\,\nu>0, \quad \mathrm{Re}\,\alpha>0$	$0, \qquad \omega>0$
$\frac{1}{(t^2+\alpha^2)(jt)^{\nu}}$	$\pi\alpha^{-\nu-1}e^{-\lvert\omega\rvert\alpha}$
$\lvert\nu\rvert<1, \quad \mathrm{Re}\,\alpha>0$	$\arg(jt)=1/2\pi, (t>0)$
	$\arg(jt)=-1/2\pi, (t<0)$
$\frac{1}{(t^2+\alpha^2)(\beta+jt)^{\nu}} \quad \mathrm{Re}\,\nu>-1$	$\frac{\pi e^{-\alpha\omega}}{\alpha(\alpha+\beta)^{\nu}}, \quad \omega>0$
	$\mathrm{Re}\,\alpha>0, \quad \mathrm{Re}\,\beta>0$
$\frac{1}{(x^2+\alpha^2)(\beta-jt)^{\nu}}$	$\frac{\pi(\beta-\alpha)^{\nu}e^{\alpha\omega}}{\alpha}, \quad \omega>0$
$\mathrm{Re}\,\nu>-1, \quad \mathrm{Re}\,\alpha>0$	$\mathrm{Re}\,\beta>0, \quad \alpha\neq\beta$
$\frac{1}{(\alpha-e^{-t})e^{\lambda t}}$	$\pi\alpha^{\lambda-1+j\omega}\cot(\pi\lambda+j\pi\omega)$
	$0<\mathrm{Re}\,\lambda<1, \quad \alpha>0$
$\frac{1}{(\alpha+e^{-t})e^{\lambda t}}$	$\pi\alpha^{\lambda-1+j\omega}\mathrm{cosec}(\pi\lambda+j\pi\omega)$
	$0<\mathrm{Re}\,\lambda<1, \quad -\pi<\arg\alpha<\pi$
$\frac{t}{(\alpha+e^{-t})e^{\lambda t}}$	$\pi\alpha^{\lambda-1+j\omega}\mathrm{cosec}(\pi\lambda+j\pi\omega)$
$0<\mathrm{Re}\,\lambda<1, \quad -\pi<\arg\alpha<\pi$	$\times[\log\alpha-\pi\cot(\pi\lambda+j\pi\omega)]$
$\frac{t^2}{(1+e^{-t})e^{\lambda t}}$	$\pi^3\mathrm{cosec}^3(\pi\lambda+j\omega\pi)[2-\sin^2(\pi\lambda+j\omega\pi)]$
$0<\mathrm{Re}\,\lambda<1$	
$\frac{1}{(\alpha+e^{-t})(\beta+e^{-t})e^{\lambda t}}$	$\pi(\beta-\alpha)^{-1}(\alpha^{\lambda-1+j\omega}-\beta^{\lambda-1+j\omega})$
$0<\mathrm{Re}\,\lambda<2, \quad \beta\neq\alpha$	$\times\,\mathrm{cosec}\,(\pi\lambda+j\omega\pi)$
$\lvert\arg\alpha\rvert<\pi, \quad \lvert\arg\beta\rvert<\pi$	
$\frac{t}{(\alpha+e^{-t})(\beta+e^{-t})e^{\lambda t}}$	$\frac{\pi(\alpha^{\lambda-1+j\omega}\log\alpha-\beta^{\lambda-1+j\omega}\log\beta)}{(\alpha-\beta)\sin(\lambda\pi+j\omega\pi)}$
$0<\mathrm{Re}\,\lambda<2, \quad \alpha\neq\beta$	$+\frac{\pi^2(\alpha^{\lambda-1+j\omega}-\beta^{\lambda-1+j\omega})\cos(\lambda\pi+j\omega\pi)}{(\beta-\alpha)\sin^2(\lambda\pi+j\omega\pi)}$
$\lvert\arg\alpha\rvert<\pi, \quad \lvert\arg\beta\rvert<\pi$	
$\frac{1}{(1+e^{-t})^n e^{\lambda t}}$	$\pi\,\mathrm{cosec}\,(\pi\lambda+j\omega\pi)\prod_{j=1}^{n-1}(j-\lambda-j\omega)/(n-1)!$
$n=1,2,3,\cdots, \quad 0<\mathrm{Re}\,\alpha<n$	
$e^{-\lambda t}\log\lvert1-e^{-t}\rvert$	$\pi(\lambda+j\omega)^{-1}\cot(\pi\lambda+j\omega\pi)$
$-1<\mathrm{Re}\,\lambda<0$	
$e^{-\lambda t}\log(1+e^{-t})$	$\pi(\lambda+j\omega)^{-1}\mathrm{cosec}\,(\pi\lambda+j\omega\pi)$
$-1<\mathrm{Re}\,\lambda<0$	
$e^{-\lambda t}\log\left(\frac{\lvert1+e^{-t}\rvert}{\lvert1-e^{-t}\rvert}\right)$	$\pi(\lambda+j\omega)^{-1}\tan(\frac{1}{2}\pi\lambda+\frac{1}{2}j\omega\pi)$
$\lvert\mathrm{Re}\,\lambda\rvert<1$	
$\frac{1}{(\sinh t+\sinh\alpha)} \quad \alpha>0$	$-\pi j e^{j\alpha\omega}\mathrm{sech}\,\alpha\,\mathrm{cosech}\,(\pi\omega)\times[\cosh(\pi\omega)-e^{-2j\alpha\omega}]$
$\frac{1}{[\Gamma(\nu-t)\Gamma(\mu+t)]}$	$[2\cos(\frac{1}{2}\omega)]^{\mu+\nu-2}e^{\frac{1}{2}j\omega(\mu-\nu)}\times[\Gamma(\mu+\nu-1)]^{-1}, \lvert\omega\rvert<\pi$
	$0, \quad \lvert\omega\rvert>\pi$

A.2 Fourier Transform in Two Dimensions

This section contains some interesting properties of the Fourier transform in two dimensions (Bracewell 1965; Hsu 1973; Spiegel 1976; Lathi 1989; Gradshteyn and Ryzhik 1990; Oberhettinger 1990).

$f(t,\tau)$	$F(\omega,\phi)$
$\alpha f(t,\tau)+\beta g(t,\tau)$	$\alpha F(\omega,\phi)+\beta G(\omega,\phi)$
$f(at,b\tau)$	$\frac{1}{\lvert ab\rvert}F(\frac{\omega}{a},\frac{\phi}{b})$
$f(-t,-\tau)$	$F(-\omega,-\phi)$
$f^*(t,\tau)$	$F^*(-\omega,-\phi)$
$f(t-\sigma,\tau-\zeta)$	$F(\omega,\phi)e^{-j\omega\sigma-j\phi\zeta}$
$f(t,\tau)e^{j\omega_0 t}$	$F(\omega-\omega_0,\phi)$
$f(t,\tau)\cos\omega_0 t$	$\frac{1}{2}F(\omega-\omega_0,\phi)+\frac{1}{2}F(\omega+\omega_0,\phi)$
$f(t,\tau)\,\mathrm{sen}\,\omega_0 t$	$\frac{1}{2j}F(\omega-\omega_o,\phi)-\frac{1}{2j}F(\omega+\omega_0,\phi)$
$F(t,\tau)$	$(2\pi)^2 f(-\omega,-\phi)$
$\frac{\partial}{\partial t}f(t,\tau)$	$j\omega F(\omega,\phi)$
$\frac{\partial}{\partial \tau}f(t,\tau)$	$j\phi F(\omega,\phi)$
$\frac{\partial^2}{\partial t\,\partial\tau}f(t,\tau)$	$-\omega\phi F(\omega,\phi)$
$\frac{\partial^n}{\partial t^n}f(t,\tau)$	$(j\omega)^n F(\omega,\phi)$
$\frac{\partial^n}{\partial \tau^n}f(t,\tau)$	$(j\phi)^n F(\omega,\phi)$
$\int_{-\infty}^{t} f(t,\tau)dt$	$\frac{1}{j\omega}F(\omega,\phi)+\pi F(0,\phi)\delta(\omega,\phi)$
$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t,\tau)dtd\tau$	$F(0,0)$
$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \lvert f(t,\tau)\rvert^2 dtd\tau$	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \lvert F(\omega,\phi)\rvert^2 d\omega d\phi$
$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t,\tau)g^*(t,\tau)dtd\tau$	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(\omega,\phi)G^*(\omega,\phi)d\omega d\phi$
$f(t,\tau)*g(t,\tau)$	$F(\omega,\phi)G(\omega,\phi)$
$f(t,\tau)*f^*(-t,-\tau)$	$\lvert F(\omega,\phi)\rvert^2$

Appendix B
Discrete Fourier Transforms

Anyone who attempts to generate random numbers by deterministic means is, of course, living in a state of sin.

– John von Neumann

This appendix presents discrete Fourier transforms and a table of discrete Fourier transforms.

Some properties are also presented, which help computing the transforms (Lourtie 2007; Mandal and Asif 2007; Körner 2002; Oppenheim et al. 2002).

Plancherel and Parseval's Theorems

If X_k and Y_k are the discrete Fourier transforms of x_n and y_n, respectively, then Parseval's theorem states that

$$\sum_{n=0}^{N-1} x_n y_n^* = \frac{1}{N} \sum_{k=0}^{N-1} X_k Y_k^*,$$

in which the asterisk denotes complex conjugation.

Plancherel theorem is a special case of the Parseval's theorem and states that

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2.$$

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

Periodicity

The periodicity of the discrete Fourier transform can be demonstrated directly from the definition:

$$X_{k+N} \triangleq \sum_{n=0}^{N-1} x_n e^{-\frac{i2\pi}{N}(k+N)n} = \sum_{n=0}^{N-1} x_n e^{-\frac{i2\pi}{N}kn} e^{-i2\pi n}$$

$$X_{k+N} = \sum_{n=0}^{N-1} x_n e^{-\frac{i2\pi}{N}kn} = X_k,$$

because $e^{-i2\pi n} = 1$.

In a similar way, it can be shown that the inverse discrete Fourier transform formula leads to a periodic extension.

Shift Theorem

Multiplying x_n by the complex exponential $e^{\frac{i2\pi(n-1)}{N}m}$, for some integer m, that leads to a linear phase, corresponds to a circular shift of the output X_k, that is, X_k is replaced by X_{k-m}, in which the subscript is interpreted modulo N.

Similarly, a circular shift of the input x_n corresponds to introducing a linear phase shift in the output X_k. Mathematically, if $\{x_n\}$ represents the vector x, then, if

$$\mathcal{F}(\{x_n\})_k = X_k,$$

then

$$\mathcal{F}\left(\left\{x_n \cdot e^{\frac{i2\pi}{N}nm}\right\}\right)_k = X_{k-m}$$

and

$$\mathcal{F}\left(\{x_{n-m}\}\right)_k = X_k \cdot e^{-\frac{i2\pi}{N}km}.$$

B.1 Table of Discrete Fourier Transforms

$x(n)$	$X(k)$
$\alpha x_1(n) + \beta x_2(n)$	$\alpha X_1(k) + \beta X_2(k)$
$x(n-l)$	$e^{-j2\pi lk/N} X(k)$
$x(n)e^{j2\pi ln}$	$X(k+l)$
$x(n)e^{-j2\pi ln}$	$X(k-l)$
$x(N-n)$	$X(N-k)$
$x(-n)$	$X(-k)$
$x_1(n) * x_2(n)$	$X_1(k) \cdot X_2(k)$
$x_1(n) \cdot x_2(n)$	$\frac{1}{N}[X_1(k) * X_2(k)]$
$x^*(n)$	$X^*(N-k)$
$x^*(N-n)$	$X^*(k)$
$\binom{N-1}{n}$	$\left[1 + e^{-j2\pi k/N}\right]^{N-1}$
α^n	$\begin{cases} N & \text{if } \alpha = e^{j2\pi kl/N} \\ \frac{1-\alpha^N}{1-\alpha e^{-j2\pi k/N}} & \text{otherwise} \end{cases}$
$x(n)$ real and even	$\sum_{n=0}^{N-1} x(n) \cos\left[\frac{2\pi kn}{N}\right]$
$x(n)$ real and odd	$-j \sum_{n=0}^{N-1} x(n) \sin\left[\frac{2\pi kn}{N}\right]$
$x_n \in \mathbb{R}$	$X_k = X^*_{N-k}$
$\Re(x_n)$	$\frac{1}{2}\left(X_k + X^*_{N-k}\right)$
$\Im(x_n)$	$\frac{1}{2i}\left(X_k - X^*_{N-k}\right)$
$\frac{1}{2}\left(x_n + x^*_{N-n}\right)$	$\Re(X_k)$
$\frac{1}{2i}\left(x_n - x^*_{N-n}\right)$	$\Im(X_k)$

Appendix C
Table of Hilbert Transforms

It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.

– Carl Friedrich Gauss

The appendix presents a table of Hilbert transform formulas and some important inequalities.

This appendix presents a table of Hilbert transform, named after the German mathematician David Hilbert (1862–1943) Hilbert (1912). For the third and fourth properties, the signal is assumed bandpass. In the following, consider that $\text{sinc}\,(t) = \frac{\sin(\pi t)}{\pi t}$ (Haykin 1987; Poularikas 1954; Erdélyi 1954; Baskakov 1986).

Recall that the Hilbert transform of a time signal $f(t)$ is denoted by $\mathcal{H}[f(t)]$, or $\hat{f}(t)$, and defined by the following integral (Gagliardi 1978):

$$\mathcal{H}[f(t)] = \hat{f}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f(\tau)}{t-\tau}\mathrm{d}\tau. \tag{C.1}$$

The functions $f(t)$ and $\hat{f}(t)$ form a pair of Hilbert transforms, and the transform is the Cauchy principal value of the integral (Tables C.1, C.2, C.3, and C.4). The Hilbert transform is a linear operation, and its inverse is given by

$$\mathcal{H}^{-1}[\hat{f}(t)] = f(t) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\hat{f}(t)}{t-\tau}\mathrm{d}\tau. \tag{C.2}$$

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

Table C.1 Useful relations between functions and their Hilbert transforms

Property	Relation
Orthogonality	$\int_{-\infty}^{\infty} f(t)\hat{f}(t)\mathrm{d}t = 0$
Autocorrelation	$\int_{-\infty}^{\infty} f(t)f(t-\tau)\mathrm{d}t = \int_{-\infty}^{\infty} \hat{f}(t)\hat{f}(t-\tau)\mathrm{d}t$
Energy	$\int_{-\infty}^{\infty} \lvert f(t)\rvert^2\mathrm{d}t = \int_{-\infty}^{\infty} \lvert\hat{f}(t)\rvert^2\mathrm{d}t$
Integration	$\int_{-\infty}^{\infty} f(t)\hat{g}(t)\mathrm{d}t = \int_{-\infty}^{\infty} \hat{f}(t)g(t)\mathrm{d}t$
	$\int_{-\infty}^{\infty} f(t)g(t)\mathrm{d}t = \int_{-\infty}^{\infty} \hat{f}(t)\hat{g}(t)\mathrm{d}t$
	$\int_{-\infty}^{\infty} f^2(t)\mathrm{d}t = \int_{-\infty}^{\infty} \hat{f}^2(t)\mathrm{d}t$

Table C.2 Table of Hilbert transforms

$f(t)$	$\hat{f}(t)$
$f(t)$	$\hat{f}(t) = \frac{1}{\pi t} * f(t)$
$\alpha f(t) + \beta g(t)$	$\alpha\hat{f}(t) + \beta\hat{g}(t)$
$f(t)\cos\omega_c t$	$f(t)\sin\omega_c t$
$f(t)\sin\omega_c t$	$-f(t)\cos\omega_c t$
1	0
$\frac{\sin t}{t}$	$\frac{1-\cos t}{t}$

Table C.3 Table of Hilbert transforms

$f(t)$	$\hat{f}(t)$
$f(t-\sigma)$	$\hat{f}(t-\sigma)$
$f(\alpha t)$	$\mathrm{sgn}(\alpha)\hat{f}(\alpha t)$
$\frac{\mathrm{d}}{\mathrm{d}t}f(t)$	$\frac{\mathrm{d}}{\mathrm{d}t}\hat{f}(t)$
e^{jt}	je^{jt}
$\frac{1}{t}$	$-\pi\delta(t)$
$\frac{1}{1+t^2}$	$\frac{1}{1+t^2}$
$\frac{\alpha}{\alpha^2+t^2},\ \alpha > 0$	$\frac{t}{\alpha^2+t^2}$
$\delta(t-\alpha) - \delta(t+\alpha)$	$\frac{2}{\pi}\left[\frac{\alpha}{t^2-\alpha^2}\right]$
$\delta(t-\alpha) + \delta(t+\alpha)$	$\frac{2}{\pi}\left[\frac{t}{t^2-\alpha^2}\right]$
$u(t+1/2) - u(t-1/2)$	$-\frac{1}{\pi}\ln\left\lvert\frac{t-1/2}{t+1/2}\right\rvert$

(continued)

Table C.3 (continued)

$f(t)$	$\hat{f}(t)$
$e^{-t},\ t > 0$	$\frac{1}{\pi} e^{-t}\mathrm{Ei}(t)$
$e^{-\lvert t\rvert}$	$\frac{1}{\pi}\left[e^{-t}\,\mathrm{Ei}(t) + e^{t}\,\mathrm{Ei}(-t)\right]$
$\mathrm{sgn}(t)\, e^{-\lvert t\rvert}$	$\frac{1}{\pi}\left[e^{-t}\,\mathrm{Ei}(t) - e^{t}\,\mathrm{Ei}(-t)\right]$
$\mathrm{sgn}(t)$	$-\infty$
$\mathrm{u}(t+\tau) - \mathrm{u}(t-\tau)$, even rectangular pulse	$\frac{1}{\pi}\ln\left\lvert\frac{t+\tau}{t-\tau}\right\rvert$
$2\mathrm{u}(t) - \mathrm{u}(t+\tau) - \mathrm{u}(t-\tau)$, odd rectangular pulse	$-\frac{1}{\pi}\ln\left\lvert\frac{t+\tau}{t}\right\rvert + \frac{1}{\pi}\ln\left\lvert\frac{t-\tau}{t}\right\rvert$
$\frac{1}{\tau}(\tau - t)\,[\mathrm{u}(t+\tau) - \mathrm{u}(t-\tau)]$, ramp pulse	$\frac{1}{\pi}\left[\frac{t-\tau}{\tau}\ln\left\lvert\frac{t-\tau}{t}\right\rvert + 1\right]$
$\frac{1}{\tau}(\tau - \lvert t\rvert)\,[\mathrm{u}(t+\tau) - \mathrm{u}(t-\tau)]$, triangular pulse	$\frac{1}{\pi}\left[\frac{t-\tau}{\tau}\ln\left\lvert\frac{t-\tau}{t}\right\rvert + \frac{t+\tau}{\tau}\ln\left\lvert\frac{t+\tau}{t}\right\rvert\right]$
$\frac{1}{\tau}(\tau - \lvert t\rvert)\,[2\mathrm{u}(t) - \mathrm{u}(t+\tau) - \mathrm{u}(t-\tau)]$, sawtooth pulse	$\frac{1}{\pi}\left[\frac{t-\tau}{\tau}\ln\left\lvert\frac{t-\tau}{t}\right\rvert - \frac{t+\tau}{\tau}\ln\left\lvert\frac{t+\tau}{t}\right\rvert + 2\right]$
$\cos(\alpha t) J_n(\beta t),\ 0 < \beta < \alpha$	$-\sin(\alpha t) J_n(\beta t)$

Table C.4 Table of Hilbert transforms

$f(t)$	$\hat{f}(t)$
$\delta(t)$	$\frac{1}{\pi t}$
c	0
$\frac{1}{\sqrt{t}}$	$\frac{1}{\sqrt{t}}$
$\mathrm{sinc}\, t$	$\mathrm{sinc}\,(t/2)\sin(\pi t/2)$
$\mathrm{sinc}'\, t$	$\frac{\cos t + t\sin t - 1}{t^2}$
$\delta'(t)$	$-\frac{1}{\pi t^2}$
$\frac{1}{\pi t^2}$	$\delta'(t)$
$\delta''(t)$	$\frac{2}{\pi t^3}$
$-\frac{2}{\pi t^3}$	$\delta''(t)$
$\frac{\alpha\beta - t^2}{t^2(\alpha+\beta)^2 + (\alpha\beta - t^2)^2}$	$\frac{(\alpha+\beta)t}{t^2(\alpha+\beta)^2 + (\alpha\beta - t^2)^2}$
$\lvert t\rvert^{z-1},\ 0 < \mathrm{Re}\, z < 1$	$\cot\left[\frac{\pi z}{2}\right]\lvert t\rvert^{z-1}\mathrm{sgn}(t)$
$\mathrm{sgn}(t)\,\lvert t\rvert^{z-1},\ 0 < \mathrm{Re}\, z < 1$	$-\tan\left[\frac{\pi z}{2}\right]\lvert t\rvert^{z-1}$
$e^{j\omega t}$	$-j\,\mathrm{sgn}(\omega) e^{j\omega t}$
$e^{-j\omega t}$	$j\,\mathrm{sgn}(\omega) e^{-j\omega t}$
e^{-t^2}	$-je^{-t^2}\mathrm{erf}(jt)$
$\cos\omega_c t$	$\sin\omega_c t$
$\sin\omega_c t$	$-\cos\omega_c t$
$\cos^2 t$	$\frac{\cos(2t)}{2}$
$\sin^2 t$	$-\frac{\sin(2t)}{2}$

Appendix D
Formulas and Important Inequalities

To not know math is a severe limitation to understanding the world.

– Richard Feynman

The appendix presents some important formulas and inequalities used in the text.

Schwartz Inequality

$$\left|\text{Re}\int_{-\infty}^{\infty} x(t)y^*(t)\text{d}t\right| \leq \left[\int_{-\infty}^{\infty} |x^2(t)|\text{d}t \int_{-\infty}^{\infty} |y^2(t)|\text{d}t\right]^{1/2}$$

the equality verifies for $x(t) = ky(t)$, in which k is a constant.

Holder Inequality

$$\left|\sum_{i=0}^{\infty} a_i b_i\right| \leq \left[\sum_{i=0}^{\infty} \left|a_i^2\right| \sum_{i=0}^{\infty} \left|b_i^2\right|\right]^{1/2}$$

the equality is verified for $a_i = kb_i$.

Other Inequalities

1. Consider that $w_1, w_2, \ldots, w_N$ are arbitrary positive numbers and let $q_1, q_2, \ldots, q_N$ be positive numbers, such that $\sum_{i=1}^{N} p_i = 1$. Then

$$\sum_{i=1}^{k} q_i w_i \log \sum_{i=1}^{k} q_i w_i \leq \sum_{i=1}^{k} q_i w_i \log q_i,$$

this equality is verified because of the continuity and convexity of the function $x \log x$ in the closed interval $(0, 1)$. The equality occurs if $q_i = q$, for all $i = 1, \ldots, N$ (Nedoma 1957).

2. Let $p_1, p_2, \ldots, p_N$ be arbitrary positive numbers and let $q_1, q_2, \ldots, q_N$ be positive numbers, such that their sum is unit, that is, $\sum_{i=1}^{N} p_i = 1$. Then

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

$$q_1^{p_1} \cdots q_k^{p_k} \leq \sum_{i=1}^{k} p_i q_i, \ \sum_{i=1}^{N} q_i = 1, \ q_i \geq 0,$$

the equality occurs for $k = N$ and $q_i = q$ (Ash 1990).

3. If $p_1 \geq p_2 \geq \cdots \geq p_k$ and $q_1 \geq q_2 \geq \cdots \geq q_k$, then (Chebyshev Inequality)

$$\left(\frac{p_1 + p_2 \cdots p_k}{k}\right)\left(\frac{q_1 + q_2 \cdots q_k}{k}\right) \leq \frac{1}{k}\sum_{i=1}^{k} p_i q_i,$$

or

$$\sum_{i=1}^{k} q_i \sum_{j=1}^{k} p_j \leq k \sum_{i=1}^{k} p_i q_i,$$

the equality is verified only and only if $p_i = p$ or $q_i = q$, for $1 \leq i \leq k$ (Gradshteyn and Ryzhik 1990).

4. Consider $w_1, w_2, \ldots, w_N$ positive arbitrary numbers and let $p_1, p_2, \ldots, p_N$ be positive numbers, such that $\sum_{i=1}^{N} p_i = 1$. Then

$$w_1^{p_1} \cdots w_k^{p_k} \leq \sum_{i=1}^{k} p_i w_i, \ \sum_{i=1}^{N} p_i = 1, \ p_i \geq 0,$$

the equality is obtained for $k = N$ and $w_i = w$. The inequality works even if any p_i equals zero, if $\sum_{i=1}^{N} p_i = 1$ (Ash 1990).

Final Value Theorem

$$\lim_{t \to \infty} x(t) = \lim_{\omega \to 0} [j\omega X(\omega)]$$

MacLaurin Series

$$f(x) = f(0) + f^{'}(0)x + \frac{1}{2!}f^{''}(0)x^2 + \frac{1}{3!}f^{'''}(0)x^3 + \cdots$$

Bessel Identities

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-x^2/4)^k}{k!(n+k)!}$$

$$= \frac{j^{-n}}{\pi} \int_0^{\pi} e^{jx\cos\theta} \cos(n\theta) d\theta$$

$$J_n(xe^{jm\pi}) = e^{jnm\pi} J_n(x)$$

$$I_n(x) = j^n J_n(x/j)$$

$$= \frac{1}{\pi} \int_0^{\pi} e^{x\cos\theta} \cos(n\theta) d\theta$$

Complex Identities

$$z = x + jy = \sqrt{x^2 + y^2} e^{j \tan^{-1}(y/x)}$$

$$z^* = x - jy = \sqrt{x^2 + y^2} e^{-j \tan^{-1}(y/x)}$$

$$|z|^2 = zz^* = x^2 + y^2$$

$$\mathrm{Re}\{z\} = \frac{1}{2}[z + z^*]$$

$$\mathrm{Im}\{z\} = \frac{1}{2j}[z - z^*]$$

$$\mathrm{Re}\{z_1 z_2\} = \mathrm{Re}\{z_1\}\mathrm{Re}\{z_2\} - \mathrm{Im}\{z_1\}\mathrm{Im}\{z_2\}$$

$$\mathrm{Im}\{z_1 z_2\} = \mathrm{Re}\{z_1\}\mathrm{Im}\{z_2\} + \mathrm{Im}\{z_1\}\mathrm{Re}\{z_2\}$$

Trigonometric Identities

$$\mathrm{sen}\,\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

$$\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$e^{\pm j\theta} = \cos\theta \pm j\,\mathrm{sen}\,\theta$$

$$\mathrm{sen}\,(\alpha \pm \beta) = \mathrm{sen}\,\alpha\cos\beta \pm \cos\alpha\,\mathrm{sen}\,\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \mathrm{sen}\,\alpha\,\mathrm{sen}\,\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

$$\mathrm{sen}\,\alpha\,\mathrm{sen}\,\beta = \frac{1}{2}\cos(\alpha - \beta) - \frac{1}{2}\cos(\alpha + \beta)$$

$$\cos\alpha\cos\beta = \frac{1}{2}\cos(\alpha-\beta) + \frac{1}{2}\cos(\alpha+\beta)$$

$$\operatorname{sen}\alpha\cos\beta = \frac{1}{2}\operatorname{sen}(\alpha-\beta) + \frac{1}{2}\operatorname{sen}(\alpha+\beta)$$

$$e^{\pm j\theta} = \cos\theta \pm j\operatorname{sen}\theta$$

$$\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$\operatorname{sen}\theta = (e^{j\theta} - e^{-j\theta})/2j$$

$$\operatorname{sen}^2\theta + \cos^2\theta = 1$$

$$\cos^2\theta - \operatorname{sen}^2\theta = \cos 2\theta$$

$$\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$$

$$\cos^3\theta = \frac{1}{4}(3\cos\theta + \cos 3\theta)$$

$$\operatorname{sen}^2\theta = \frac{1}{2}(1-\cos 2\theta)$$

$$\operatorname{sen}^3\theta = \frac{1}{4}(3\operatorname{sen}\theta - \operatorname{sen}3\theta)$$

Series Expansions

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots |nx| < 1$$

$$e^x = 1 + x + \frac{1}{2!}x^2 + \cdots$$

$$a^x = 1 + x\ln a + \frac{1}{2!}(x\ln a)^2 + \cdots$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots$$

$$\operatorname{sen} x = x - \frac{1}{3}!x^3 + \frac{1}{5}!x^5 - \cdots$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots$$

$$\tan x = x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \cdots$$

$$\sum_{m=0}^{M} x^m = \frac{(x^M - 1)}{(x - 1)}$$

$$e^{a\cos b} = \sum_{i=0}^{\infty} \epsilon_i I_i(a)\cos(ib), \quad \epsilon_0 = 1,\ \epsilon_i = 2,\ i \geq 1$$

$$\cos(x\,\mathrm{sen}\,\theta) = J_0(x) + 2\sum_{k=1}^{\infty} J_{2k}(x)\cos(2k\theta)$$

$$\mathrm{sen}\,(x\,\mathrm{sen}\,\theta) = 2\sum_{k=0}^{\infty} J_{2k+1}(x)\mathrm{sen}\,[(2k+1)\theta]$$

$$\cos(x\cos\theta) = J_0(x) + 2\sum_{k=0}^{\infty}(-1)^k J_{2k}(x)\cos(2k\theta)$$

$$\mathrm{sen}\,(x\cos\theta) = 2\sum_{k=0}^{\infty}(-1)^k J_{2k+1}(x)\cos[(2k+1)\theta]$$

$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^n} = \frac{(\pi/n)}{\mathrm{sen}\,(\pi/n)}, \quad n > 1 \qquad \int_0^{\infty} \frac{\mathrm{d}x}{(a^2+x^2)^2} = \frac{\pi}{4a^3}$$

$$\int_0^{\infty} \frac{x^u \mathrm{d}x}{1+x^n} = \left(\frac{\pi}{n}\right)\mathrm{cossec}\left[\frac{(u+1)\pi)}{n}\right] \qquad \int_0^{\infty} \frac{\mathrm{d}x}{1+x^2} = \tan^{-1}(b)$$

Indefinite Integrals

$$\int \mathrm{sen}\,ax\,\mathrm{d}x = -\frac{1}{a}\cos ax$$

$$\int \cos ax\,\mathrm{d}x = \frac{1}{a}\mathrm{sen}\,ax$$

$$\int \mathrm{sen}^2\,ax\,\mathrm{d}x = \frac{x}{2} - \frac{\mathrm{sen}\,2ax}{4a}$$

$$\int \cos^2 ax \, \mathrm{d}x = \frac{x}{2} + \frac{\operatorname{sen} 2ax}{4a}$$

$$\int \operatorname{sen} ax \cos ax \, \mathrm{d}x = \frac{1}{2a}\operatorname{sen}^2(ax)$$

$$\int x \operatorname{sen} ax \, \mathrm{d}x = \frac{1}{a^2}(\operatorname{sen} ax - ax \cos ax)$$

$$\int x \cos ax \, \mathrm{d}x = \frac{1}{a^2}(\cos ax + ax \operatorname{sen} ax)$$

$$\int x^2 \operatorname{sen} ax \, \mathrm{d}x = \frac{1}{a^3}(2ax \operatorname{sen} ax + 2\cos ax - a^2x^2 \cos ax)$$

$$\int x^2 \cos ax \, \mathrm{d}x = \frac{1}{a^3}(2ax \cos ax - 2\operatorname{sen} ax + a^2x^2 \operatorname{sen} ax)$$

$$\int \operatorname{sen} ax \operatorname{sen} bx \, \mathrm{d}x = \frac{\operatorname{sen}(a-b)x}{2(a-b)} - \frac{\operatorname{sen}(a+b)x}{2(a+b)} \qquad a^2 \neq b^2$$

$$\int \cos ax \cos bx \, \mathrm{d}x = \frac{\operatorname{sen}(a-b)x}{2(a-b)} + \frac{\operatorname{sen}(a+b)x}{2(a+b)}x \qquad a^2 \neq b^2$$

$$\int \operatorname{sen} ax \cos bx \, \mathrm{d}x = \frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} \qquad a^2 \neq b^2$$

$$\int e^{ax} \mathrm{d}x = \frac{1}{a}e^{ax}$$

$$\int x e^{ax} \mathrm{d}x = \frac{1}{a^2}e^{ax}(ax - 1)$$

$$\int x^2 e^{ax} \mathrm{d}x = \frac{1}{a^3}e^{ax}(a^2x^2 - 2ax + 2)$$

$$\int e^{ax} \operatorname{sen} bx \, \mathrm{d}x = \frac{1}{a^2 + b^2}e^{ax}(a \operatorname{sen} bx - b \cos bx)$$

$$\int e^{ax} \cos bx \, \mathrm{d}x = \frac{1}{a^2 + b^2}e^{ax}(a \cos bx + b \operatorname{sen} bx)$$

$$\int \left[\frac{\operatorname{sen} ax}{x}\right]^2 \mathrm{d}x = a \int \frac{\operatorname{sen} 2ax}{x} \mathrm{d}x - \frac{\operatorname{sen}^2 ax}{x}$$

$$\int \frac{\mathrm{d}x}{a^2+b^2x^2} = \frac{1}{ab}\tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{x^2\mathrm{d}x}{a^2+b^2x^2} = \frac{x}{b^2} - \frac{a}{b^3}\tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{\mathrm{d}x}{(a^2+b^2x^2)^2} = \frac{x}{2a^2(a^2+b^2x^2)} + \frac{1}{2ab^3}\tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{x^2\mathrm{d}x}{(a^2+b^2x^2)^2} = \frac{-x}{2b^2(a^2+b^2x^2)} + \frac{1}{2ab^3}\tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{\mathrm{d}x}{(a^2+b^2x^2)^3} = \frac{x}{4a^2(a^2+b^2x^2)^2} + \frac{3x}{8a^4(a^2+b^2x^2)} + \frac{3}{8a^5b}\tan^{-1}\left(\frac{bx}{a}\right)$$

Definite Integrals

$$\int_0^\infty \frac{\operatorname{sen} ax}{x}\mathrm{d}x = \begin{cases} \pi/2 & a > 0 \\ 0 & a = 0 \\ -\pi/2 & a < 0 \end{cases}$$

$$\int_0^x \frac{\operatorname{sen} u}{u}du \triangleq \mathrm{Si}(x) \quad \text{the integral is a function of } x$$

$$\int_0^\infty \frac{\operatorname{sen}^2 ax}{x^2}\mathrm{d}x = |a|\pi/2$$

$$\int_0^\infty e^{-ax^2}\mathrm{d}x = \frac{1}{2}\sqrt{\pi/a}$$

$$\int_0^\infty xe^{-ax^2}\mathrm{d}x = \frac{1}{2a}$$

$$\int_0^\infty x^2e^{-ax^2}\mathrm{d}x = \frac{1}{4a}\sqrt{\pi/a}$$

$$\int_0^\infty \frac{\mathrm{d}x}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab(a+b)} \qquad a > 0,\ b > 0$$

$$\int_0^\infty \frac{\mathrm{d}x}{ax^4+b} = \frac{\pi}{2\sqrt{2b}}\left(\frac{b}{a}\right)^{1/4} \qquad ab > 0$$

$$\int_0^\infty \frac{\mathrm{d}x}{ax^6+b} = \frac{\pi}{3b}\left(\frac{b}{a}\right)^{1/6} \qquad ab > 0$$

Useful Summations

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1-\alpha},\ |\alpha| < 1$$

$$\sum_{n=k}^{\infty} \alpha^n = \frac{\alpha^k}{1-\alpha},\ |\alpha| < 1$$

$$\sum_{n=k}^{N} \alpha^n = \frac{\alpha^k - \alpha^{N+1}}{1-\alpha},\ \alpha \neq 1$$

$$\sum_{n=0}^{N-1} \alpha^n = \begin{cases} \frac{1-\alpha^N}{1-\alpha}, & \text{if } |\alpha| \neq 1 \\ N, & \text{if } \alpha = 1 \end{cases}$$

Appendix E
Probability Theory

Logic is the anatomy of thought.

– John Locke

The appendix presents a review of probability theory covering set theory, functions and measure probability theory, and random variables. It also includes Bayes' theorem, moments, variance, the characteristic function and function of a random variable, and the concept of joint random variables.

E.1 Set Theory, Functions, and Measure

The general theory of sets was developed by Georg Cantor, in the nineteenth century. Georg Ferdinand Ludwig Philipp Cantor (1845–1918), a mathematician who played an important role in the development of set theory, was born in Saint Petersburg, Russian Empire. Cantor established the basis for this theory and demonstrated some of its most important results, including the concept of set cardinality.

Cantor lived most of his life in Germany Boyer (1974), where he defined the concept of cardinal and ordinal numbers and also their arithmetic. The ideas relative to the notions of universal set, empty set, set partition, discrete systems, continuous systems, and infinity are, in reality, as old as philosophy itself.

In the time of Zenon, one of the most famous pre-Socratic philosophers, the notion of infinity was already discussed. Zenon, considered the creator of the dialectic, was born in Elea, Italy, around 504 B.C. and was the defendant of Parmenides, his master, against criticism from the followers of Pythagoras. Pythagoras was born in Samos around 580 B.C. and created a philosophic current based on the quantification of the universe.

For the Pythagoreans, the unity is itself the result of existence and of nonexistence. It can be noticed that the concept of emptiness is expressed in the previous sentence, as well as the concept of a universal set.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

The Pythagoreans established an association between the number one and the point, between the number two and the line, between the number three and the surface, and between the number four and the volume (de Souza 1996). In spite of dominating the notion of emptiness, the Greeks still did not have the concept of zero.

Zenon, by his turn, defended the idea of a unique being continuous and indivisible, of Parmenides, against the multiple being, discontinuous and divisible of Pythagoras. Aristotle presents various of Zenon's arguments relative to movement, with the objective of establishing the concept of a continuum.

Aristotle was born in Estagira, Macedonia, in the year 384 B.C. The first of Aristotle's arguments suggests the idea of an infinite set, to be discussed later (Durant 1996). This argument is as follows: If a mobile object has to cover a given distance, in a continuous space, it must then cover first the first half of the distance before covering the whole distance.

Reasoning in this manner Zenon's argument implies that infinite segments of distance cannot be successively covered in a finite time. The expositive logic in Aristotle's counter argument is impressive(de Souza 1996):

> In effect, length and time and in general all contents are called infinite in two senses, whether meaning division or whether with respect to extremes. No doubt, infinities in quantity cannot be touched in a finite time, but infinities in division, yes, since time itself is also infinite in this manner. As a consequence, it is in an infinite time and not in a finite time that one can cross infinite, and, if infinities are touched, they are touched by infinities and not by finite.

Despite the reflections of pre-Socratic philosophers and of others that followed, no one had yet managed to characterize infinite until 1872. In that year, Julius Wilhelm Richard Dedekind (1831–1916) pointed to the universal property of infinite sets, which has found applications as far as in the study of fractals (Boyer 1974): "A system S is called infinite when it is similar to a part of itself. On the contrary, S is said to be finite."

Cantor also recognized the fundamental property of sets, but, differing from the observations of Dedekind, he noticed that not all infinite sets are equal. This notion originated the cardinal numbers, which will be covered later, in order to establish a hierarchy of infinite sets in accordance with their respective powers.

The results obtained by Cantor led him to establish set theory as a fully developed subject. As a consequence of his results on transfinite arithmetic, too advanced for his time, Cantor suffered attacks of mathematicians like Leopold Kronecker (1823–1891), who disregarded him for a position at the University of Berlin.

Cantor spent most of his carrier life in the smaller University of Halle, in a city of medieval aspect with the same name in Germany, famous for its mines of rock salt, and died there in an institution for persons with mental health problems .

However, his theory received from David Hilbert (1862–1943), one of the greatest mathematicians of the twentieth century, the following citation Boyer (1974): "The new transfinite arithmetic is the most extraordinary product of human thought and one of the most beautiful achievements of human activity in the domain of the purely intelligible."

E.1.1 Set Theory

The definition of a set, a concept of fundamental importance in Mathematics, is axiomatic—which means that a set does not admit a nonrecursive definition, that is, a definition which will not resort to the original notion of a set. On the other hand, the concept of a set is fundamental for all branches of Mathematics.

Set theory is usually developed based on a set of axioms, called fundamental axioms: axiom of extension, axiom of specification, Peano axioms, axiom of choice, besides Zorn's lemma, and Schröder-Bernstein's theorem (Halmos 1960).

This section presents the theory of sets in an informal manner, eventually quoting the fundamental axioms, since this theory is used as a basis to establish the notion of probability measure. Some examples of common sets are given next.

- The set of faces of a coin: $A = \{H, T\}$
- The binary set: $B = \{-1, 1\}$
- The set of natural numbers: $N = \{1, 2, 3, 4, \dots\}$
- The set of integer numbers: $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

The most important relations in set theory are the belonging relation, denoted as $a \in A$, in which a is an element of the set A, and the inclusion relation, $A \subset B$, which is read "A is a subset of the set B" or B is a superset of the set A.

Sets may be specified by means of examples or using propositions as, for instance, "the set of students that do not complain about the exams" or a little more formally $A = \{x \mid x \text{ do not complain about exams }\}$. This is, in a few cases, a way of denoting the empty set! Alias, the empty set can be written formally as $\emptyset = \{a \mid a \neq a\}$, that is, the set the elements that are not equal to themselves.

A fundamental concept is the notion of a universal set. It is understood as that set which contains all other sets of interest. An example of a universal set is provided by the sample space in probability theory, usually denoted as S or Ω.

On the other hand, the empty set is that set which contains no element and which is usually denoted as $\emptyset$ or $\{\ \}$. It is implicit that the empty set is contained in any set, that is, that $\emptyset \subset A$, for any given set A. However, the empty set is not in general an element of any other set.

The Venn diagram is a practical way of representing the logical relation between sets, which was popularized by John Venn in the 1880s, as illustrated in Fig. E.1. John Venn (1834–1923) was an English mathematician, logician, and philosopher, who published the book The Logic of Chance, in which he explained the frequency theory of probability.

Two sets are said to be disjoint if they have no element in common, as illustrated in Fig. E.2. Thus, for example, the set of even natural numbers and the set of odd natural numbers are disjoint.

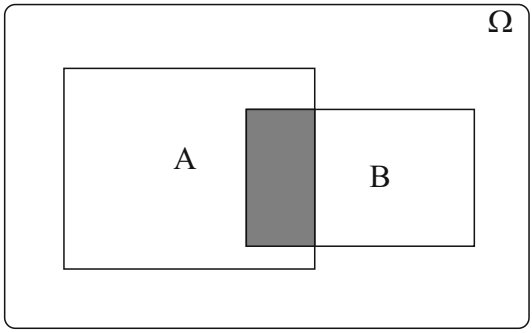


Fig. E.1 Venn diagram representing sets that intercept

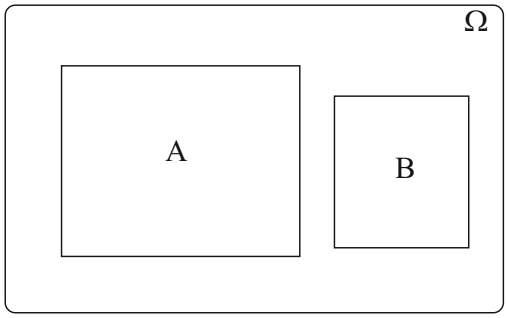


Fig. E.2 Disjoint sets

E.1.2 Operations on Sets

- The operation $\overline{A}$ represents the complement of A with respect to the sample space Ω.
- The subtraction of sets, denoted $C = A - B$, gives as a result the set the elements of which belong to A and do not belong to B.
  Note: If B is completely contained in $A : A - B = A \cap \overline{B}$.
- The set of elements belonging to A and to B, but not belonging to $(A \cap B)$ é, is specified by $A \Delta B = A \cup B - A \cap B$.

The generalization of these concepts to families of sets, for example, $\cup_{i=1}^{N} A_i$ and $\cap_{i=1}^{N} A_i$, is immediate. The following properties are usually employed as axioms in developing the theory of sets (Lipschutz 1968).

- **Idempotent**
  $A \cup A = A, \qquad A \cap A = A$
- **Associative**
  $(A \cup B) \cup C = A \cup (B \cup C), \qquad (A \cap B) \cap C = A \cap (B \cap C)$
- **Commutative**
  $A \cup B = B \cup A, \qquad A \cap B = B \cap A$
- **Distributive**
  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- **Identity**

$$A \cup \emptyset = A, \qquad A \cap U = A$$
$$A \cup U = U, \qquad A \cap \emptyset = \emptyset$$

- **Complementary**

$$A \cup \overline{A} = U, \qquad A \cap \overline{A} = \emptyset \qquad \overline{(\overline{A})} = A$$
$$U = \emptyset, \qquad \overline{\emptyset} = U$$

- **de Morgan laws**

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \qquad \overline{A \cap B} = \overline{A} \cup \overline{B}$$

E.1.3 Families of Sets

Among the most interesting families of sets, it is worth mentioning an increasing sequence of sets, such that $\lim_{i\to\infty} \cup A_i = A$, as shown in Fig. E.3. This sequence is used in proofs of limits over sets.

A decreasing sequence of sets is defined in a similar manner with $\lim_{i\to\infty} \cap A_i = A$, as shown in Fig. E.4.

E.1.4 Indexing of Sets

The Cartesian product is a way of expressing the idea of indexing of sets. The indexing of sets expands the possibilities for the use of sets, allowing to produce eventually entities known as vectors and signals.

Example Consider the set of amplitudes $B_i = \{0, 1\}$. Starting from this set, it is possible to construct an indexed sequence of sets by defining its indexing: $\{B_{i\epsilon I}\}$, $I = \{0, \cdots, 7\}$. This family of indexed sets B_i constitutes a finite discrete sequence, that is, a vector. The ASCII set of binary symbols is an example.

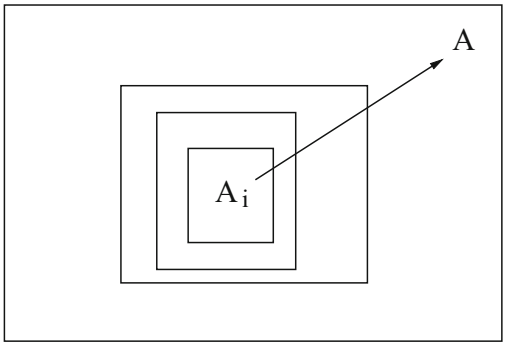


Fig. E.3 Increasing sequence of sets

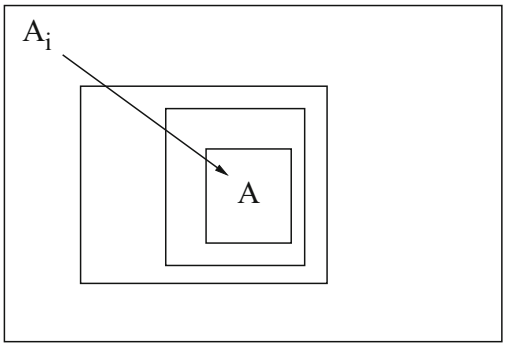


Fig. E.4 Decreasing sequence of sets

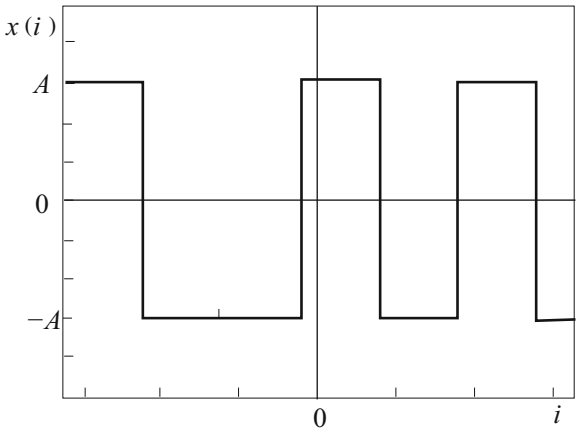


Fig. E.5 Signal discrete in time and in amplitude

Example Consider the set of amplitudes given by $B_i = \{-A, A\}$, and let the indexing set $I = Z$ (the set of positive and negative integers plus zero). It follows that $\{B_{i\epsilon Z}\}$, which represents an infinite series of $-A$'s and A's, that is, it represents a binary digital signal. For example, Fig. E.5 represents a signal, $x(i)$, that is discrete both in amplitude and time.

Example Letting the set $B_i = \{-A, A\}$, but considering the indexing over the set of real numbers, $\{B_{i\epsilon I}\}$, in which $I = R$, a signal is displayed which is discrete in amplitude but continuous in time, such as the telegraph signal in Fig. E.6.

Example Consider now the sets $B = R$ and $I = R$. In this manner, the result is an analog signal, that is, continuous in time and in amplitude, as shown in Fig. E.7.

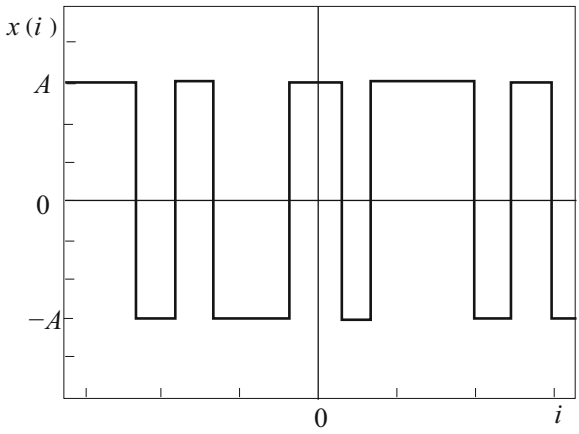


Fig. E.6 Telegraphic signal, discrete in amplitude, and continuous in time

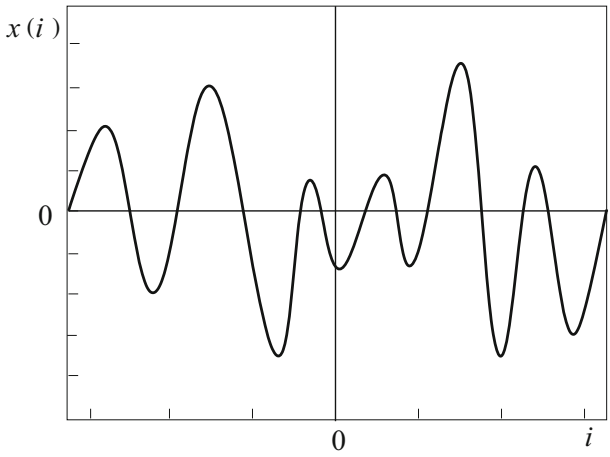


Fig. E.7 Analog signal

E.1.5 Algebra of Sets

In order to construct an algebra of sets or, equivalently, to construct a field over which operations involving sets make sense, a few properties have to be obeyed.

1. If $A \in \mathcal{F}$, then $\overline{A} \in \mathcal{F}$. A is the set containing desired results or over which one wants to operate.
2. If $A \in \mathcal{F}$ and $B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$.

The above properties guarantee the closure of the algebra with respect to finite operations over sets. It is noticed that the universal set Ω always belongs to the algebra, that is, $\Omega \in \mathcal{F}$, because $\Omega = A \cup \overline{A}$. The empty set also belongs to the algebra, that is, $\emptyset \in \mathcal{F}$, since $\emptyset = \overline{\Omega}$, follows by property 1.

Example The family $\{\emptyset, \Omega\}$ complies with the above properties and therefore represents an algebra. In this case, $\emptyset = \{\}$ and $\overline{\emptyset} = \Omega$. The union is also represented, as can be easily checked.

Example Given the sets $\{C_H\}$ and $\{C_T\}$, representing the faces of a coin, respectively, if $\{C_H\} \in \mathcal{F}$, then $\{\overline{C_H}\} = \{C_T\} \in \mathcal{F}$. It follows that $\{C_H, C_T\} \in \mathcal{F} \Rightarrow \Omega \in \mathcal{F} \Rightarrow \emptyset \in \mathcal{F}$.

The previous example can be translated by the following expression. If there is a measure for heads, then there must be also a measure for tails, in order for the algebra to be properly defined. Whenever a probability is assigned to an event, then a probability must also be assigned to the complementary event.

The cardinality of a finite set is defined as the number of elements belonging to this set. Sets with an infinite number of elements are said to have the same cardinality if they are equivalent, that is, $A \sim B$ if $\sharp A = \sharp B$. Some examples of sets and their respective cardinals are presented next.

- $I = \{1, \cdots, k\} \Rightarrow C_I = k$.
- $N = \{0, 1, \cdots\} \Rightarrow C_N$ or $\aleph_0$.
- $Z = \{\cdots, -2, -1, 0, 1, 2, \cdots\} \Rightarrow C_Z$.
- $Q = \{\cdots, -1/3, 0, 1/3, 1/2, \cdots\} \Rightarrow C_Q$.
- $R = (-\infty, \infty) \Rightarrow C_R$ or $\aleph$.

For the above examples, the following relations are verified: $C_R > C_Q = C_Z = C_N > C_I$. The notation $\aleph_0$, for the cardinality of the set of natural numbers, was employed by Cantor.

The cardinality of the power set, that is, of the family of sets consisting of all subsets of a given set I, $\mathcal{F} = 2^I$, is 2^{C_I}.

E.1.6 Borel Algebra

The Borel algebra $\mathcal{B}$, or σ-algebra, is an extension of the algebra so far discussed to operate with limits at infinity. The following properties are required from a σ-algebra:

1. $A \in \mathcal{B} \Rightarrow \overline{A} \in \mathcal{B}$
2. $A_i \in \mathcal{B} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$

The above properties guarantee the closure of the σ-algebra with respect to enumerable operations over sets. These properties allow the definition of limits in the Borel field.

Examples Considering the above properties, it can be verified that $A_1 \cap A_2 \cap A_3 \cdots \in \mathcal{B}$. In effect, it is sufficient to notice that

$$A \in \mathcal{B} \text{ and } B \in \mathcal{B} \Rightarrow A \cup B \in \mathcal{B},$$

and

$$\overline{A} \in \mathcal{B} \text{ and } \overline{B} \in \mathcal{B} \Rightarrow \overline{A} \cup \overline{B} \in \mathcal{B},$$

and finally

$$\overline{\overline{A} \cup \overline{B}} \in \mathcal{B} \Rightarrow A \cap B \in \mathcal{B}.$$

In summary, any combination of unions and intersections of sets belongs to the Borel algebra. In other words, operations of union or intersection of sets, or a combination of these operations, produce a set that belongs to the σ-algebra.

E.2 Probability Theory

This section summarizes the more basic definitions related to probability theory, random variables, and stochastic processes, the main results and conclusions of which will be used in subsequent chapters.

Probability theory began in France with studies about games of chance. Antoine Gombaud (1607–1684), known as *Chevalier de Méré*, was very keen on card games and would discuss with Blaise Pascal (1623–1662) about the probabilities of success in this game. Pascal, also interested on the subject, began a correspondence with Pierre de Fermat (1601–1665) in 1654, which originated the theory of finite probability (Zumpano and de Lima 2004).

However, the first known work about probability is *De Ludo Aleae* (About Games of Chance), by the Italian medical doctor and mathematician Gerolamo Cardano (1501–1576), published in 1663, almost 90 years after his death. This book was a handbook for players, containing some discussion about probability.

The first published treatise about the probability theory, dated 1657, was written by the Dutch scientist Christiaan Huygens (1629–1695), a folder titled *De Ratiociniis in Ludo Aleae* (About Reasoning in Games of Chance).

Another Italian, the physicist and astronomer Galileo Galilei (1564–1642), was also concerned with random events. In a fragment probably written between 1613 and 1623, entitled *Sopra le Scorpete dei Dadi* (About Dice Games), Galileu answers a question asked; it is believed, by the Grand Duke of Tuscany.

Abraham de Moivre (1667–1754) was another important mathematician for the development of probability theory. He wrote a book of great influence at the time, called *Doctrine of Chances*. The law of large numbers was discussed by Jacques Bernoulli (1654–1705), Swiss mathematician, in his work Ars Conjectandi (The Art of Conjecturing).

The study of probability was deepened in the eighteentn and nineteenth centuries, being worth of mentioning the works of French mathematicians Pierre-Simon de Laplace (1749–1827) and Siméon Poisson (1781–1840), as well as the German mathematician Carl Friedrich Gauss (1777–1855).

E.2.1 Axiomatic Approach to Probability

Probability theory is usually presented in one of the following manners: the classical approach, the relative frequency approach, and the axiomatic approach. The classical approach is based on the symmetry of an experiment but employs the concept of probability in a cyclic manner, because it is defined only for equiprobable events. The relative frequency approach to probability is more recent and relies on experiments. The following example illustrates the classical approach.

Considering the theoretical difficulties found in the two previous approaches to probability, respectively, the cyclic definition in the first case and the problem of convergence in a series of experiments for the second case, henceforth only the axiomatic approach will be followed in this text. Those readers interested in the classical or in the relative frequency approach are referred to the literature (Papoulis 1981a).

The axioms of probability were established by Andrey Nikolaevich Kolmogorov (1903–1987), a Russian mathematician whose work influenced many branches of modern mathematics. Just three statements, as follows, permitted the development of the complete theory (Papoulis 1983a):

Axiom 1 $P(\Omega) = 1$, in which Ω denotes the sample space or universal set and $P(\cdot)$ denotes the associated probability measure.

Axiom 2 $P(A) \geq 0$, in which A denotes an event belonging to the sample space.

Axiom 3 $P(A \cup B) = P(A) + P(B)$, in which A and B are mutually exclusive events and $A \cup B$ denotes the union of events A and B.

Using his axiomatic approach to probability theory, Kolmogorov established a firm mathematical basis on which other theories rely, for example, the theory of stochastic processes, communications theory, and information theory.

Kolmogorov's fundamental work was published in 1933, in Russia, and soon afterward was published in German with the title *Grundbegriffe der Wahrscheinlichkeits Rechnung* (Fundamentals of Probability Theory) (James 1981). In this work, Kolmogorov managed to combine advanced set theory, of Cantor, with measure theory, of Lebesgue, in order to produce what to this date is the modern approach to probability theory.

By applying the above axioms, it is possible to deduce all results relative to probability theory. For example, the probability of the empty set, $\emptyset = \{\}$, is easily calculated as follows. First, it is noticed that

$$\emptyset \cup \Omega = \Omega,$$

since the sets $\emptyset$ and Ω are disjoint. Thus, it follows that

$$P(\emptyset \cup \Omega) = P(\Omega) = P(\emptyset) + P(\Omega) = 1 \Rightarrow P(\emptyset) = 0.$$

In the case of sets A and B which are not disjoint, it follows that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

E.2.2 Bayes' Rule

In probability theory and statistics, Bayes' rule, or Bayes' theorem, concerns the computation of conditional probabilities. Thomas Bayes (1701–1761) was the English statistician, philosopher, and Presbyterian minister who formulated the theorem.

It can be expressed as

$$P(A|B) = \frac{P(A \cap B)}{P(B)},$$

assuming $P(B) \neq 0$. An equivalent manner of expressing the same result is the following:

$$P(A \cap B) = P(A|B) \cdot P(B) \ , \quad P(B) \neq 0.$$

Curiously, Bayes never published the theorem that bears his name. In fact, his notes were edited and published posthumously by Richard Price (1723–1791), a Welsh moral philosopher, minister, and mathematician.

Some important properties of measure of sets are presented next, in which A and B denote events from a given sample space:

- If A is independent of B, then $P(A|B) = P(A)$. It then follows that $P(B|A) = P(B)$ and that B is independent of A.
- If $B \subset A$, then: $P(A|B) = 1$.
- If $A \subset B$, then: $P(A|B) = \frac{P(A)}{P(B)} \geq P(A)$.
- If A and B are independent events, then $P(A \cap B) = P(A) \cdot P(B)$.
- If $P(A) = 0$ or $P(A) = 1$, then event A is independent of itself.
- If $P(B) = 0$, then $P(A|B)$ can assume any arbitrary value. Usually, in this case, one assumes $P(A|B) = P(A)$.
- If events A and B are disjoint and non-empty, then they are dependent.

A partition is a possible splitting of the sample space into a family of subsets, in a manner that the subsets in this family are disjoint and their union coincides with the sample space. It follows that any set in the sample space can be expressed by using a partition of that sample space and thus be written as a union of disjoint events.

The following property can be illustrated by means of a Venn diagram, as illustrated in Fig. E.8:

$$B = B \cap \Omega = B \cap \cup_{i=1}^{M} A_i = \cup_{i=1}^{N} B \cap A_i.$$

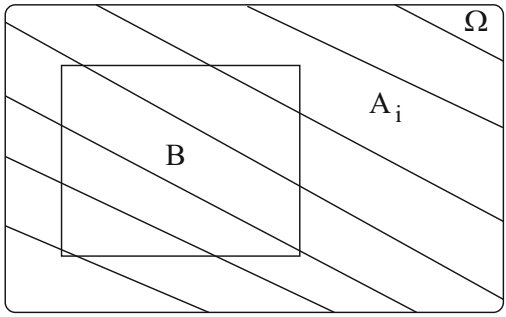


Fig. E.8 Partition of a set

It now follows that

$$P(B) = P(\cup_{i=1}^{N} B \cap A_i) = \sum_{i=1}^{N} P(B \cap A_i),$$

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{i=1}^{N} P(B \cap A_i)} = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{i=1}^{N} P(B|A_i) \cdot P(A_i)}.$$

Therefore, the probability of a set B can be computed using the partition of this set. Also, the conditional probability of a set A_i, given the set B, can be calculated using the *a priori* probabilities of the set B, given the set A_i.

E.3 Random Variables

A random variable (r.v.) X represents a mapping of the sample space on the line (the set of real numbers). A random variable is usually characterized by a cumulative probability function (CPF) $P_X(x)$ or by a probability density function (pdf) $p_X(x)$.

Example A random variable X, defined in the interval $[0, 1]$, with a uniform probability density function $p_X(x)$ is described by the equation $p_X(x) = \mathrm{u}(x) - \mathrm{u}(x-1)$. It follows by Axiom 1, and from the definition of the unit step function, that

$$\int_{-\infty}^{+\infty} p_X(x)\mathrm{d}x = \int_{0}^{1} [\mathrm{u}(x) - \mathrm{u}(x-1)]\mathrm{d}x = 1. \tag{E.1}$$

For a given probability distribution, $p_X(x)$ the probability that X is in the interval $(a, b]$ is given by

$$P(a < x \leq b) = \int_{a}^{b} p_X(x)\mathrm{d}x. \tag{E.2}$$

The cumulative probability function $P_X(x)$, of a random variable X, is defined as the integral of the probability density function, $p_X(x)$, that is,

$$P_X(x) = \int_{-\infty}^{x} p_X(t)\mathrm{d}t. \tag{E.3}$$

The cumulative probability is a measure, defined in the real set, $\mathbb{R}$. It measures the probability that a given random variable spans the real line, from minus infinity to the value x.

E.3.1 The Mean Value of a Random Variable

Let $f(X)$ denote a function of a random variable X. The average value (or expected value) of $f(X)$ with respect to X is defined as

$$E[f(X)] = \int_{-\infty}^{+\infty} f(x)p_X(x)\mathrm{d}x. \tag{E.4}$$

The following properties of the expected value follow from (E.4):

$$E[\alpha X] = \alpha E[X], \tag{E.5}$$

$$E[X + Y] = E[X] + E[Y] \tag{E.6}$$

and if X and Y are statistically independent random variables, then

$$E[XY] = E[X]E[Y]. \tag{E.7}$$

E.3.2 Moments of a Random Variable

The nth moment of a random variable X is defined as

$$m_n = E[X^n] = \int_{-\infty}^{+\infty} x^n p_X(x)\mathrm{d}x. \tag{E.8}$$

Some of the moments of X have special importance in probability, statistics and random processes, and also physical interpretation in engineering.

- $m_1 = E[X]$ is the arithmetic mean, average value, average voltage, or statistical mean.
- $m_2 = E[X^2]$ represents the quadratic mean or total power.
- $m_3 = E[X^3]$ is a measure of asymmetry of the probability density function.
- $m_4 = E[X^4]$ conveys a measure of flatness of the probability density function.

E.3.3 The Variance of a Random Variable

The variance of a random variable X is an important quantity in signal analysis and communication theory, usually associated to the AC power, and is defined as follows:

$$V[X] = E[(X - m_1)^2] = m_2 - m_1^2. \tag{E.9}$$

The standard deviation of a random variable, σ_X, is defined as the square root of the variance of X. It is a measure of the deviation from the statistical of the random variable:

$$\sigma_X = \sqrt{E[(X - m_1)^2]}. \tag{E.10}$$

E.3.4 The Characteristic Function of a Random Variable

The characteristic function $P_X(\omega)$, or moment generating function, of a random variable X is usually defined from the Fourier transform of the probability density function (pdf) of X, which is equivalent to making $f(x) = e^{-j\omega x}$ in (E.4), that is,

$$P_X(\omega) = E[e^{-j\omega x}] = \int_{-\infty}^{+\infty} e^{-j\omega x} p_X(x)\mathrm{d}x, \text{ in which } j = \sqrt{-1}. \tag{E.11}$$

The moments of a random variable X can be obtained directly from the characteristic function, by taking the partial derivatives of the function for $\omega = 0$, as follows:

$$m_i = \frac{1}{(-j)^i} \frac{\partial^i P_X(\omega)}{\partial \omega^i}\bigg|_{\omega=0}. \tag{E.12}$$

E.3.5 Function of a Random Variable

Given that X is a random variable, it follows that a given function of X, $Y = f(X)$ is also a random variable, obtained by the application of the transformation $f(\cdot)$. The probability density function of Y is related to that of X by the formula (Blake 1987):

$$p_Y(y) = \frac{p_X(x)}{|\mathrm{d}y/\mathrm{d}x|}\bigg|_{x=f^{-1}(y)}, \tag{E.13}$$

in which $f^{-1}(\cdot)$ denotes the inverse function of $f(\cdot)$. This formula assumes the existence of the inverse function of $f(\cdot)$ as well as its derivative at all points.

Some Important Random Variables

Some random variables appear in several mathematical applications to model physical phenomena and deserve special mention.

1. **Gaussian random variable**
   The random variable X with pdf

$$p_X(x) = \frac{1}{\sigma_X\sqrt{2\pi}} e^{-\frac{(x-m_X)^2}{2\sigma_X^2}} \tag{E.14}$$

   is called a Gaussian (or Normal) random variable. The Gaussian random variable plays an extremely important role in engineering, considering that many well-known processes can be described or approximated by this pdf.

   The noise present in either analog or digital signal processing or communications systems usually can be considered Gaussian as a consequence of the influence of many factors (Leon-Garcia 1989). In (E.14) m_X represents the average value and σ_X^2 represents the variance of X.

   Figure E.9 illustrates the Gaussian pdf, and its corresponding cumulative probability function is depicted in Fig. E.10i.

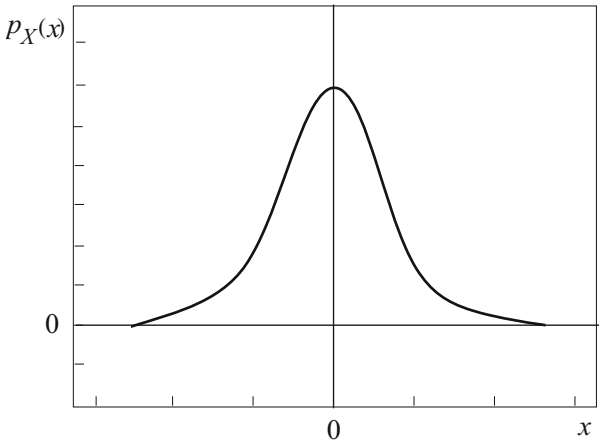


Fig. E.9 Gaussian probability density function

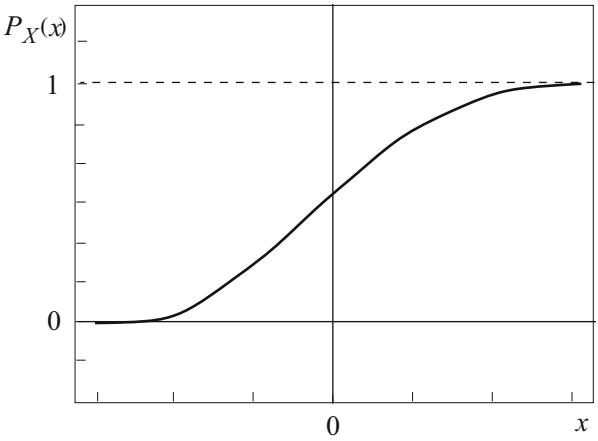


Fig. E.10 Gaussian cumulative probability function

2. **Rayleigh random variable**
   An often used model to represent the behavior of the amplitudes of signals subjected to fading employs the following pdf Kennedy (1969), Proakis (1990):

$$p_X(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} u(x) \tag{E.15}$$

   known as the Rayleigh pdf, with average $E[X] = \sigma\sqrt{\pi/2}$ and variance $V[X] = (2-\pi)\frac{\sigma^2}{2}$.

   The Rayleigh pdf represents the effect of multiple signals, reflected or refracted, which are captured by a receiver, in a situation in which there is no main signal component or main direction of propagation (Lecours et al. 1988). In this situation, the phase distribution of the received signal can be considered uniform in the interval $(0, 2\pi)$. It is noticed that it is possible to closely approximate a Rayleigh pdf by considering only six waveforms with independently distributed phases (Schwartz et al. 1966).
3. **Sinusoidal random variable**
   A sinusoidal tone X has the following pdf:

$$p_X(x) = \frac{1}{\pi\sqrt{V^2 - x^2}}, \ |x| < V. \tag{E.16}$$

   The pdf and the CPF of X are illustrated in Figs. E.11 and E.12.

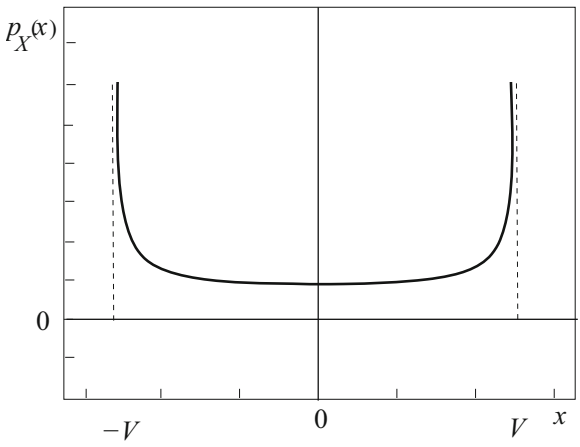


Fig. E.11 Probability density function of a sinusoidal random variable

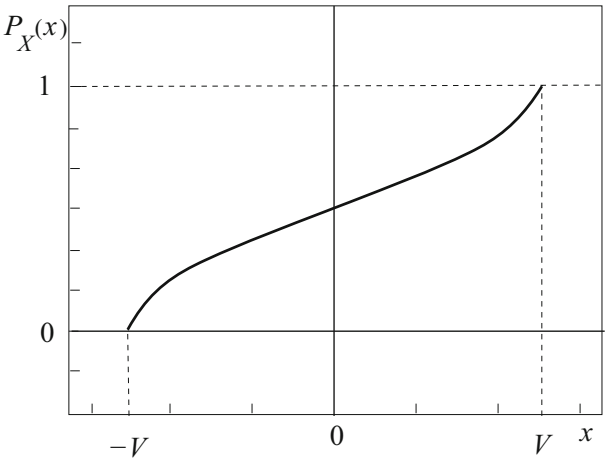


Fig. E.12 Cumulative probability function of a sinusoidal random variable

E.3.6 Joint Random Variables

Considering that X and Y represent a pair of real random variables, with joint pdf $p_{XY}(x, y)$, as illustrated in Fig. E.13, then the probability of x and y being simultaneously in the region defined by the polygon [abcd] is given by the joint integral:

$$\text{Prob}\{a < x < b, c < y < d\} = \int_a^b \int_c^d p_{XY}(x, y)\mathrm{d}x\mathrm{d}y, \tag{E.17}$$

in which a, b, c, d are given constants that define the limits of the polygon.

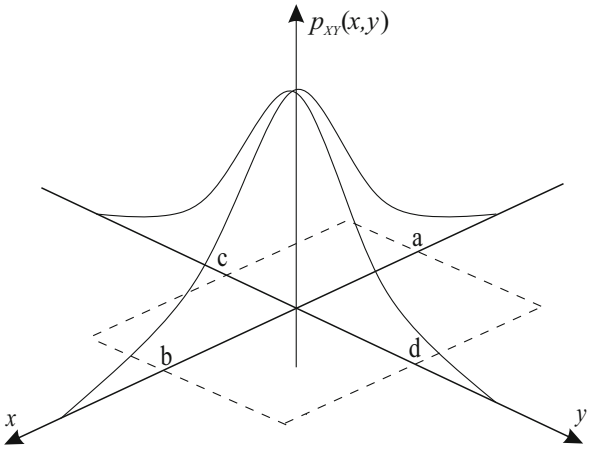


Fig. E.13 Joint probability density function

The individual distributions of X and Y, also called marginal pdfs, result from the integration of the joint pdf as follows:

$$p_X(x) = \int_{-\infty}^{+\infty} p_{XY}(x, y)\mathrm{d}y, \tag{E.18}$$

and

$$p_Y(y) = \int_{-\infty}^{+\infty} p_{XY}(x, y)\mathrm{d}x. \tag{E.19}$$

The joint average $\mathrm{E}[f(X, Y)]$ is calculated as

$$\mathrm{E}[f(X, Y)] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x, y)p_{XY}(x, y)\mathrm{d}x\mathrm{d}y, \tag{E.20}$$

for an arbitrary function $f(X, Y)$ of X and Y.

The joint moments m_{ik}, of order ik, are calculated as

$$m_{ik} = \mathrm{E}[X^i, Y^k] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} x^i y^k p_{XY}(xy)\mathrm{d}x\mathrm{d}y. \tag{E.21}$$

The two-dimensional characteristic function is defined as the two-dimensional Fourier transform of the joint probability density $p_{XY}(x, y)$:

$$P_{XY}(\omega, \nu) = \mathrm{E}[e^{-j\omega X - j\nu Y}]. \tag{E.22}$$

When the sum $Z = X + Y$ of two statistically independent random variables is considered, it is noticed that the characteristic function of Z turns out to be

$$P_Z(\omega) = \mathrm{E}[e^{-j\omega Z}] = \mathrm{E}[e^{-j\omega(X+Y)}] = P_X(\omega)P_Y(\omega). \tag{E.23}$$

As far as the pdf of Z is concerned, it can be said that

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(\rho)p_Y(z-\rho)\mathrm{d}\rho, \tag{E.24}$$

or

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(z-\rho)p_Y(\rho)\mathrm{d}\rho. \tag{E.25}$$

Equivalently, the sum of two statistically independent random variables has a pdf given by the convolution of the respective pdfs of the random variables involved in the sum.

The random variables X and Y are uncorrelated if $\mathrm{E}[XY] = \mathrm{E}[X]\mathrm{E}[Y]$. The criterion of statistical independence of random variables, which is stronger than that for the random variables being uncorrelated, is satisfied if $p_{XY}(x, y) = p_X(x).p_Y(y)$.

Glossary

The union of the mathematician with the poet, fervor with measure, passion with correctness, this surely is the ideal.

– William James

This glossary presents common acronyms and terminology used in digital modulation theory, computer networks, and digital communications (Furiati 1998; Alencar 1999; Design 2001; MobileWord 2001; Skycell 2001; TIAB2B.com 2001; Alencar and da Rocha Jr. 2005; Alencar 2011c, 2009; Alencar et al. 2018).

ADPCM—Adaptive Differential Pulse Code Modulation ADPCM is a speech coding method that achieves bit rate reduction using adaptive prediction quantization.

ADSL—Asymmetric Digital Subscriber Line A modem that permits the transmission of digital information in a twisted pair.

A/D—Analog-to-Digital Converter A device that converts from the analog domain to the discrete domain.

AES3—Audio Engineering Society 3 A standard for the exchange of digital audio signals between professional audio devices.

AF Audio Frequency. The range of frequencies, approximately 20 Hz to 20 kHz, that can be heard by the normal human ear, when transmitted as acoustic waves.

AGC—Automatic Gain Control A function of the receiver that produces a constant power output, even if the input is varying, for instance, because of fading.

AM—Amplitude Modulation In a modulation system, in which the modulating signal controls the carrier amplitude.

AMPS—Advanced Mobile Phone Service The first-generation analog cellular phone system that originated in the USA.

AM-SC—Amplitude Modulation Suppressed Carrier A modulation scheme in which the unmodulated carrier is eliminated to save energy.

AM-VSB—AM-Vestigial Sideband Modulation A method used to modulate carriers in analog video.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

ANSI—American National Standards Institute American National Standards Institute, a US-based organization which develops standards and defines interfaces for telecommunications.

ARQ—Automatic ReQuest for retransmission A type of communication link, in which the receiver asks the transmitter to resend a block of data when errors are detected.

ASCII—American Standard Code for Information Interchange ASCII data is a standard seven-bit code with one parity bit. Otherwise referred to as text. ASCII data can be interchanged between almost every type of computer.

ASK—Amplitude Shift Keying A modulation scheme in which the modulating signal changes the carrier amplitude.

ATSC—Advanced Television System Committee The American forum to discuss digital television.

AWGN—Additive White Gaussian Noise The usual type of noise that affects signal transmission in a channel, usually produced by thermal effect.

BCH—Bose-Chaudhuri-Hocquenghem A code used in the AMPS standard.

BCJR Bahl-Cocke-Jelinek-Raviv Decoding Algorithm.

BEC—Binary Erasure Channel A discrete channel model.

BER—Bit Error Rate A measure of the error introduced in the signal, computed at the receiver end.

BPSK—Binary Phase Shift Keying A binary phase modulation scheme.

BSC—Binary Symmetric Channel A discrete channel model.

BSS Broadcasting Satellite Service.

BSS Binary Symmetric Source.

BTA Broadcasting Technology Association (Japanese industrial organization).

BW—Bandwidth The difference between the limiting frequencies within which performance of a device, regarding some characteristic, falls within specified limits.

C Channel Capacity.

CIF—Common Interchange Format A standardized format for the picture resolution, frame rate, color space, and color subsampling of digital video sequences, used in video systems.

C-QUAM Compatible QUAM.

CATV—Community Antenna Television Often called community access television or cable TV, it is television distribution method in which signals from distant stations are received, amplified, and then transmitted by (coaxial or fiber) cable or microwave links to users.

Cable Modem Modem used with optical fiber or coaxial cable.

CCI—Cochannel Interference Superposition of adjacent signals in the frequency spectrum.

CCITT—Commité Consultatif International de Telegraphique et Telephonique Consultative Committee International Telephone and Telegraph is an international organization which develops standards and defines interfaces for telecommunications (now ITU-T)

CDMA—Code Division Multiple Access A multiple access system that uses spread spectrum.

CELP—Code-Excited Linear Prediction An analog-to-digital voice coding scheme.

CENELEC Comité Européen de Normalisation Électrotechnique.

C/I—Carrier-to-Interference Ratio The ratio of the desired unmodulated signal power to the interfering signal power.

CNR—Carrier-to-Noise Ratio The ratio of the level of the carrier to that of the noise in the intermediate frequency (IF) band before any nonlinear process, such as amplitude limitation and detection, takes place.

Codec—Coder/Decoder Device that converts analog signals to digital signals and vice versa.

COFDM—Coded Orthogonal Frequency Division Multiplex A frequency multiplexing system that uses coded signals as inputs.

CRC—Cyclic Redundancy Check A method of detecting errors in the serial transmission of data. A CRC for a block of data is calculated before it is sent and is then sent along with the data. A new CRC is calculated on the received data. If the new CRC does not match the one that has been sent along with the data, then an error has occurred.

D Cardinality of a Source Code Alphabet.

$D(P_X||P_{\tilde{X}})$ Informational Divergence between P_X **and** $P_{\tilde{X}}$.

D/A—Digital-to-Analog Converter A device to convert from the discrete to the analog domain.

dB—Decibel Abbreviation for decibel(s). One-tenth of the common logarithm of the ratio of relative powers, equal to 0.1 B (bel).

dBi Unit used to express the gain related to the isotropic antenna.

dBm Decibel related to 1 milliwatt that is used as a reference.

DCT—Discrete Cosine Transform A transform that is used in video compression for digital television.

DEMUX—Demultiplexer The separation of two or more channels previously multiplexed, **i.e.**, the reverse of multiplexing.

DFE—Decision Feedback Equalizer An equalizer that uses a decision over the received data do adjust the frequency response of the reception filter.

DM—Delta Modulation A simple analog-to-digital and digital-to-analog signal conversion technique used for transmission of voice information.

DMC Discrete Memoryless Channel.

DMS Discrete Memoryless Source.

DSS Discrete Stationary Source.

DPCM—Differential Pulse Code Modulation Form of pulse code modulation for which efficiency is enhanced by transmitting the difference between the current signal strength and the previous pulse signal strength rather than the absolute values.

DQPSK Differential Quadrature Phase Shift Keying.

DS-SS—Direct Sequence Spread Spectrum A type of spread spectrum technique that uses a pseudorandom sequence.

DSL—Digital Subscriber Line A dedicated link using leased line or wireless for subscriber connection. In Integrated Services Digital Networks (ISDN), it is an equipment that provides full-duplex service on a single twisted metallic pair at a rate sufficient to support ISDN basic access and additional framing, timing recovery, and operational functions.

DTMF—Dual-Tone Multifrequency The signaling scheme used in "touch tone" telephones. Each depressed key generates two audio tones in this scheme. In telephone systems, multifrequency signaling in which standard set combinations of two specific voice band frequencies, one from a group of four low frequencies and the other from a group of four higher frequencies, are used.

DTV—Digital Television An acronym for any digital television standard.

DVB—Digital Video Broadcasting An acronym for the European digital television standard.

DVD—Digital Video Disk An acronym for the digital video standard.

$E(X)$ Average Value or Expectation of the Random Variable X.

Eb/No—Energy-Per-Bit to Noise Density Ratio A ratio, or figure of merit, that is used to compare digital modulation schemes.

EDTV—Enhanced Definition Television The quality of EDTV is between SDTV and HDTV.

ETSI—European Telecommunications Standards Institute The board that has the authority to produce international communications standards in Europe.

FCC—Federal Communications Commission The US government board that has the authority to regulate all nonfederal government interstate telecommunications (including radio and television broadcasting) as well as all international communications that originate or terminate in the USA.

FDMA Frequency Division Multiple Access.

FEC—Forward Error Correction System of error control for data transmission for which the receiving device has the capability to detect and correct any character or code block that contains fewer than a predetermined number of symbols in error.

FFT—Fast Fourier Transform An efficient algorithm devised by John Tukey and J. W. Cooley to compute the Fourier transform of a discrete signal.

FH-SS—Frequency-Hopping Spread Spectrum A type of spread spectrum technique in which the frequency of the carrier changes pseudorandomly.

FM—Frequency Modulation A modulation scheme in which the carrier frequency is a linear function of the modulating signal.

FSK—Frequency Shift Keying A digital modulation technique in which the modulating signal changes the carrier frequency.

FTP—File Transfer Protocol File transfer protocol is the TCP/IP standard for remote file transfer.

FVC—Forward Voice Channel The radio channel used for communication of voice and user data from the base station to a cellular device.

GEO—Geostationary Earth Orbit Communications system with satellites in geosynchronous orbits 40,000 km miles above the Earth, on the Equator plane.

GHz—Gigahertz A unit of frequency denoting 10^9 Hz.

GMSK—Gaussian Minimum Shift Keying A modulation scheme that uses a Gaussian pulse.

GSM—Global System for Mobile Communication GSM originally stood for Groupe Speciale Mobile but has been renamed to Global System for Mobile Communications, an international digital cellular standard.

$h(p)$ Binary Entropy Function.

$H(X|Y = y)$ Conditional Uncertainty of the Discrete Random Variable X Given That the Event $Y = y$ Occurs.

$H(X|Y)$ Conditional Uncertainty of the Discrete Random Variable X Given the Discrete Random Variable Y.

$H(X)$ Entropy or Uncertainty of the Random Variable X.

$H_r(X)$ r-ary Entropy or Uncertainty of the Random Variable X.

$H(X_1X_2 \ldots X_n)$ Joint Entropy of Mutual Entropy of the Random Variables $X_1X_2 \ldots X_n$.

HDSL High Data Rate Digital Subscriber Line.

HDTV—High-Definition Television Television that has approximately twice the horizontal and twice the vertical emitted resolution specified by the NTSC standard.

HEO—Highly Elliptical Orbit This class of satellites covers orbits which have large eccentricities (are highly elliptical).

HF—High Frequency From 3 to 30 MHz.

HFC Hybrid network, including fiber and coaxial cable.

Hz—Hertz Frequency unit equivalent to 1 cycle per second.

$I(X; Y)$ Mutual Information Between the Discrete Random Variables X and Y.

$I(X; Y|Z)$ Mutual Information Between the Discrete Random Variables X and Y, Given the Random Variable Z.

IEEE Institute of Electrical and Electronic Engineers

IEC International Electrotechnical Commission.

IF—Intermediate Frequency A frequency to which a carrier frequency is shifted as an intermediate phase in transmission or reception.

IFFT Inverse Fast Fourier Transform.

IP—Internet Protocol The Internet Protocol is the network layer communications protocol in the Internet Protocol suite for relaying datagrams across network boundaries.

ISB—Independent Sideband Double-sideband transmission in which the information carried by each sideband is different.

ISDB Integrated Services Digital Broadcasting.

ISDN—Integrated Services Digital Network A digital phone service which provides two data channels, each with its own phone number

ISDTV—International System for Digital Television HDTV standard developed in Brazil and based on the ISDB standard.

ISI—Intersymbol Interference Digital communication system impairment in which adjacent symbols in a sequence are distorted, creating dispersion that interferes in the time domain with neighboring symbols.

ISO International Organization for Standardization.

ITU—International Telecommunications Union A United Nations (UN) agency, headquartered in Geneva, Switzerland, specialized in producing recommendations for communication systems.

ITU-R International Telecommunications Union, Radiocommunication Sector.

ITU-T International Telecommunications Union, Telecommunication Standardisation Sector.

JPEG—Joint Photographic Experts Group A method of lossy compression for digital images.

LAN—Local Area Network A computer network limited to the immediate area, usually the same building or floor of a building.

LDPC Low Density Parity Check.

LEO—Low Earth Orbit Mobile communications satellite between 700 and 2000 kilometers above the Earth.

LF—Low Frequency Frequency band between 30 kHz and 300 kHz.

$\lim$ Limit.

$\ln$ Natural Logarithm.

$\log$ Base 2 Logarithm.

$\log_b$ Base b Logarithm.

LMDS Local Multipoint Distribution System.

MCM—Multicarrier Modulation A technique of transmitting data by dividing it into several interleaved bit streams and using these to modulate several carriers.

MEO—Medium Earth Orbit MEO satellites orbit about 10,000 km above the earth.

ML Maximum Likelihood.

Modem—Modulator/Demodulator A device that can encode digital signals from a computer into analog signals that can be transmitted over analog phone lines and vice versa.

MPEG Motion Picture Experts Group.

MPSK—M-ary Phase Shift Keying A modulation scheme in which the carrier phase is a function of the information signal.

MSE Mean Square Error.

MSK—Minimum Shift Keying A modulation scheme in which the carrier frequency is a function of the information signal and the pulse format has the sine shape.

MUX—Multiplexer A device that combines multiple inputs into an aggregate signal to be transported via a single transmission channel.

n Length of a Block of Symbols.

N Length of a Codeword.

NBFM—Narrowband Frequency Modulation A scheme with a low modulation index.

NEXT—Near-End Crosstalk Impairment typically associated with twisted-pair transmission, in which a local transmitter interferes with a local receiver.

NF—Noise Figure A measure of how much noise is produced by a device.

NMT—Nordic Mobile Telephony One of the earliest cellular network developed jointly in Denmark, Finland, Iceland, Norway, and Sweden. Originally operated in the 450 MHz band and later the 900 MHz was used as well.

NTT Nippon Telephone and Telegraph Co.

NRZ—Not-Return-to-Zero Data encoding format in which each bit is represented for the entire duration of the bit period as a logic high or a logic low. A transmission encoding scheme in which two separate states are used to define binary ones and zeros, neither state is the electrical zero state.

OFDM—Orthogonal Frequency Division Multiplexing Multi-carrier signaling technique designed to maximize throughput in channels with potentially poor frequency response. It is the technology used in discrete multitone (DMT), in the ADSL standard, and also in television standards.

OQPSK—Offset Quadrature Phase Shift Keying QPSK system in which the two bits that compose a QPSK symbol are offset in time by a half-bit period for nonlinear amplification.

PABX—Private Automatic Branch eXchange An exchange linked to the PSTN that handles calls automatically.

PAL—Phase Alternated Line A television signal standard (625 lines, 50 Hz, 220 V primary power) used in the UK, much of the rest of western Europe, several South American countries, some Middle East and Asian countries, several African countries, Australia, New Zealand, and other Pacific Island countries.

PAL-M A modified version of the phase-alternation-by-line (PAL) television signal standard (525 lines, 50 Hz, 220 V primary power), used in Brazil.

PAM—Pulse Amplitude Modulation Amplitude modulation of a carrier which uses pulses of varying amplitude to transmit information from source to destination.

P_B Block Error Probability.

PCM—Pulse Code Modulation A common way of converting an analog signal to a digital signal. This is done by sampling the signal and coding the sample. The usual sampling rate is 8000 times per second, and each sampling represents 8 bits. This produces a transmission rate of 64 kbit/s.

PCS Personal Communications Service.

P_e Error Probability.

PLL—Phase-Locked Loop A tracking system that is used to recover a modulated carrier.

PM—Phase Modulation A modulation system in which the carrier phase is a linear function of the modulating signal.

PMD—Principle of Majority Decision The most frequent symbol is chosen in a detection system.

PN—Pseudorandom Noise An interfering signal produced by a shift register that emulates noise.

PSK—Phase Shift Keying A modulation system in which the carrier phase is a function of the discrete modulating signal.

PSTN—Public Switched Telephone Network The traditional voice network infrastructure, including both local service and long distance service.

PTP—Precision Time Protocol A protocol used to synchronize clocks throughout a computer network.

P_X Probability Distribution of the Discrete Random Variable X.

QAM Quadrature Amplitude Modulation.

QCIF—Quarter Common Interchange Format A common interchange format, in which the height and width of the frame are halved.

QoS—Quality of Service A parameter used to describe the attributes of a variety of network functions.

QPSK Quadrature Phase Shift Keying.

QUAM Quadrature Amplitude Modulation.

R**—Code Rate** Rate of a block code.

RACE—Research and Development for Advanced Communications A European Community endeavor aimed at creating advanced communications networks.

RF Radio Frequency.

RGB—Red-Blue-Green Related to the use of three separate signals to carry the red, green, and blue components, respectively, of a color video image.

RLE Run-Length Encoding.

RS—Reed-Solomon Error correcting code.

RTP—Real-Time Protocol A network protocol for delivering audio and video over IP networks.

Rx Receiver.

RZ Return to Zero. Data encoding format in which each bit is represented for only a portion of the bit period as a logic high or a logic low and what remains of the duration of the bit returns to logic zero.

SAN—Small Area Network A network, generally limited to tens of meters, which uses specialized communications methods and is applied in such areas as process control and other specific real time computer applications.

S-CDMA Synchronous Code Division Multiple Access.

SDH—Synchronous Digital Hierarchy The ITU standard for synchronous transmission.

SDMA—Space Division Multiple Access A multiple access scheme that relies on intelligent antennas to separate the transmitted signals.

SDTV—Standard Definition Television A digital television standard whose resolution is equivalent to the analog television.

SHF—Super High Frequency Frequency band from 3 to 30 GHz.

σ^i ith Symbol of SourceU^n.

supp(f) Support of f] If f is any real-valued function, then the *support* of f is defined as a subset of its domain in which f takes on nonzero values and is denoted as supp(f).

SMPTE—Society of Motion Picture and Television Engineers An internationally recognized standards organization that provides a technical framework to the industry, including standards, recommended practices, and engineering guidelines that deal with motion-imaging content.

SNR—Signal-to-Noise Ratio A metric used to compare modulation schemes.

SONET—Synchronous Optical Network The US standard for digital optical networks.

SQNR—Signal-to-Quantization Noise Ratio A metric used to compare quantization schemes.

SSB—Single Sideband A modulation scheme that transmits only of the two sidebands and saves bandwidth.

TACS—Total Access Communication System A British analog mobile telephone standard based on the US AMPS system.

TCM Trellis-Coded Modulation.

TCP/IP—Transmission Control Protocol/Internet Protocol This is the suite of protocols that defines the Internet. Originally designed for the UNIX operating system, TCP/IP software is now available for every major kind of computer operating system.

TDM Time Division Multiplexing.

TDMA Time Division Multiple Access.

TIA—Telecommunications Industry Association A membership organization concerned with various standards aspects of the telecommunications industry.

TWTA—Travelling Wave Tube Amplifier An amplifier technology designed for generation of very high-power microwave signals, such as those used in satellite communication applications.

Tx Transmitter.

U Discrete Information Source.

U^n Order n Extension of a Discrete Information Source.

UDP/IP—User Datagram Protocol/Internet Protocol The User Datagram Protocol (UDP) is one of the core communication protocols of the Internet protocol suite used to send messages, transported as datagrams in packets, to other hosts on an Internet Protocol (IP) network.

UHF—Ultra-High Frequency Frequency band from 300 MHz to 3 GHz.

ULF—Ultra-Low Frequency Frequency band from 300 Hz to 3000 Hz.

VA Viterbi Algorithm.

VCO—Voltage-Controlled Oscillator A device in which the output frequency is a function of the input voltage.

VDSL—Very High-Speed Digital Subscriber Line The next-generation twisted-pair technology (after ADSL) that targets higher data transmission rates than ADSL in exchange for shorter guaranteed distances.

VHF—Very-High Frequency Frequency band from 30 Hz to 300 Hz.

VLF—Very-Low Frequency Frequency band from 3 Hz to 30 Hz.

VSWR—Voltage Standing Wave Ratio In a transmission line, the ratio of maximum to minimum voltage in a standing wave pattern.

Var$X$ Variance of the Discrete Random Variable X.

W Length of Word of a Source Code.

WAN—Wide Area Network A network which covers a larger geographical area than a LAN and in which telecommunications links are implemented.

WAP—Wireless Application Protocol A global protocol for wireless systems to permit interaction between data services.

WBFM—Wide-Band Frequency Modulation A modulation that uses a high deviation index.

WDM—Wavelength Division Multiplexing A multiplexing scheme for the light frequency range.

WLL—Wireless Local Loop A wireless system meant to bypass a local landline telephone system.

X Discrete Random Variable X.

$\#(X(\Omega))$ Cardinality of set $X(\Omega)$.

$\lceil X \rceil$ Least Integer That Is Greater Than or Equal to X.

$\lfloor X \rfloor$ Greatest Integer That Is Less Than or Equal to X.

X.25—A Standard for Packet Transmission A set of ITU protocols for long distance networks.

X.400—A Standard for Packet Transmission A suite of ITU recommendations that define standards for data communication networks for message handling systems.

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

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M. Sampaio de Alencar, V. C. da Rocha, Jr., *Digital Communications*,
https://doi.org/10.1007/978-3-031-92397-5

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